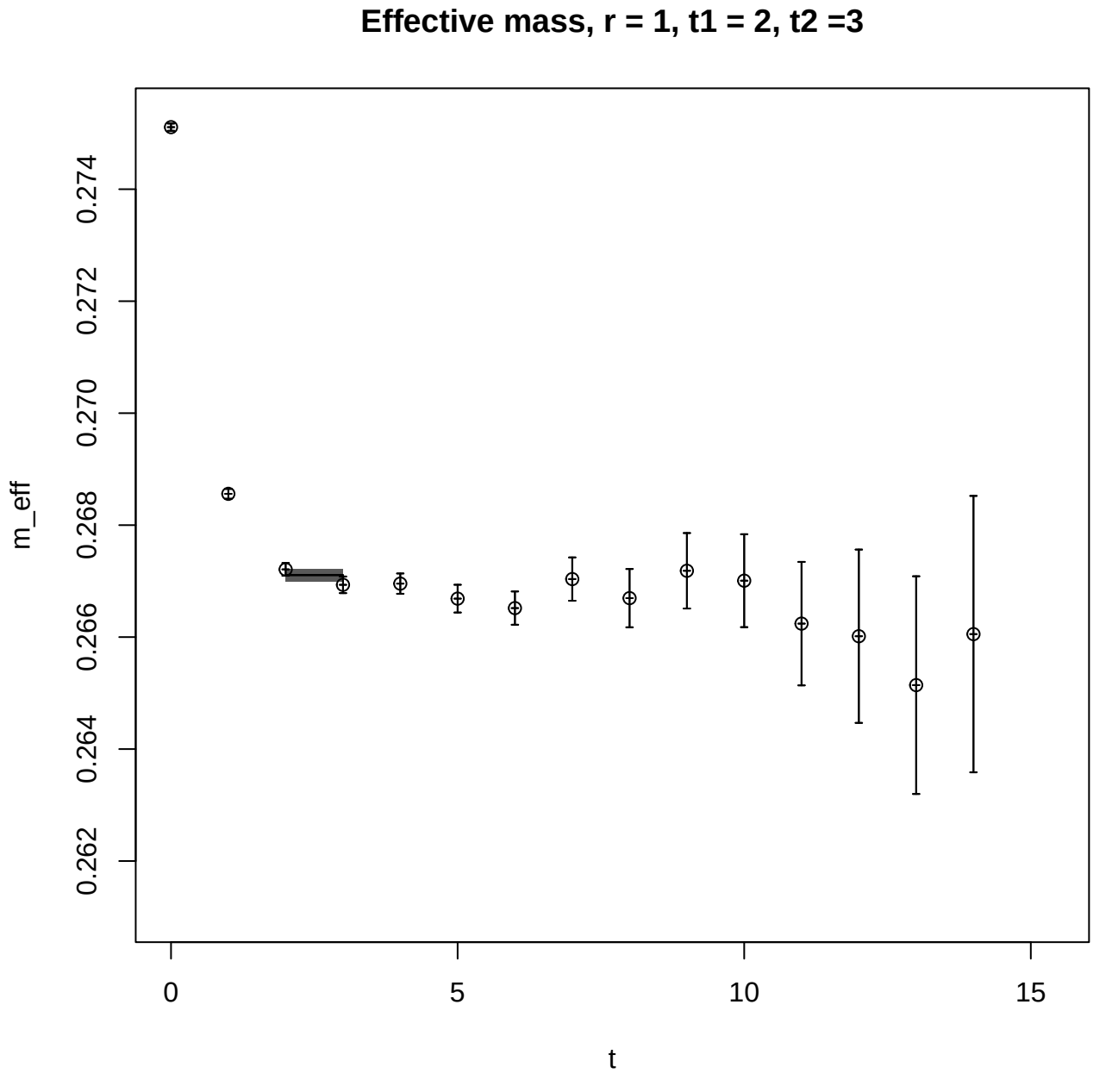
Effective mass, r = 1, t1 = 2, t2 =3
m_eff
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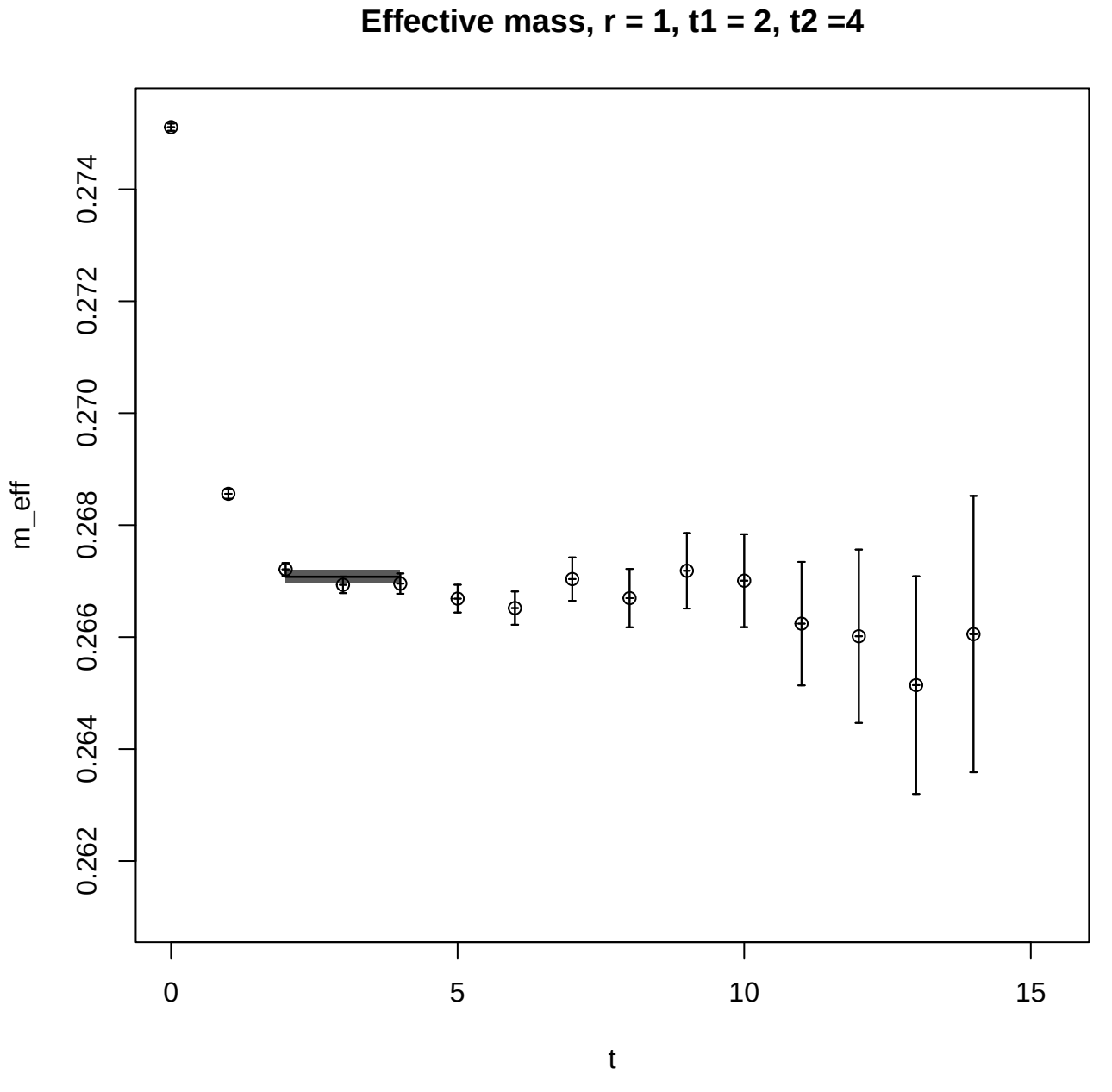
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =4
m_eff
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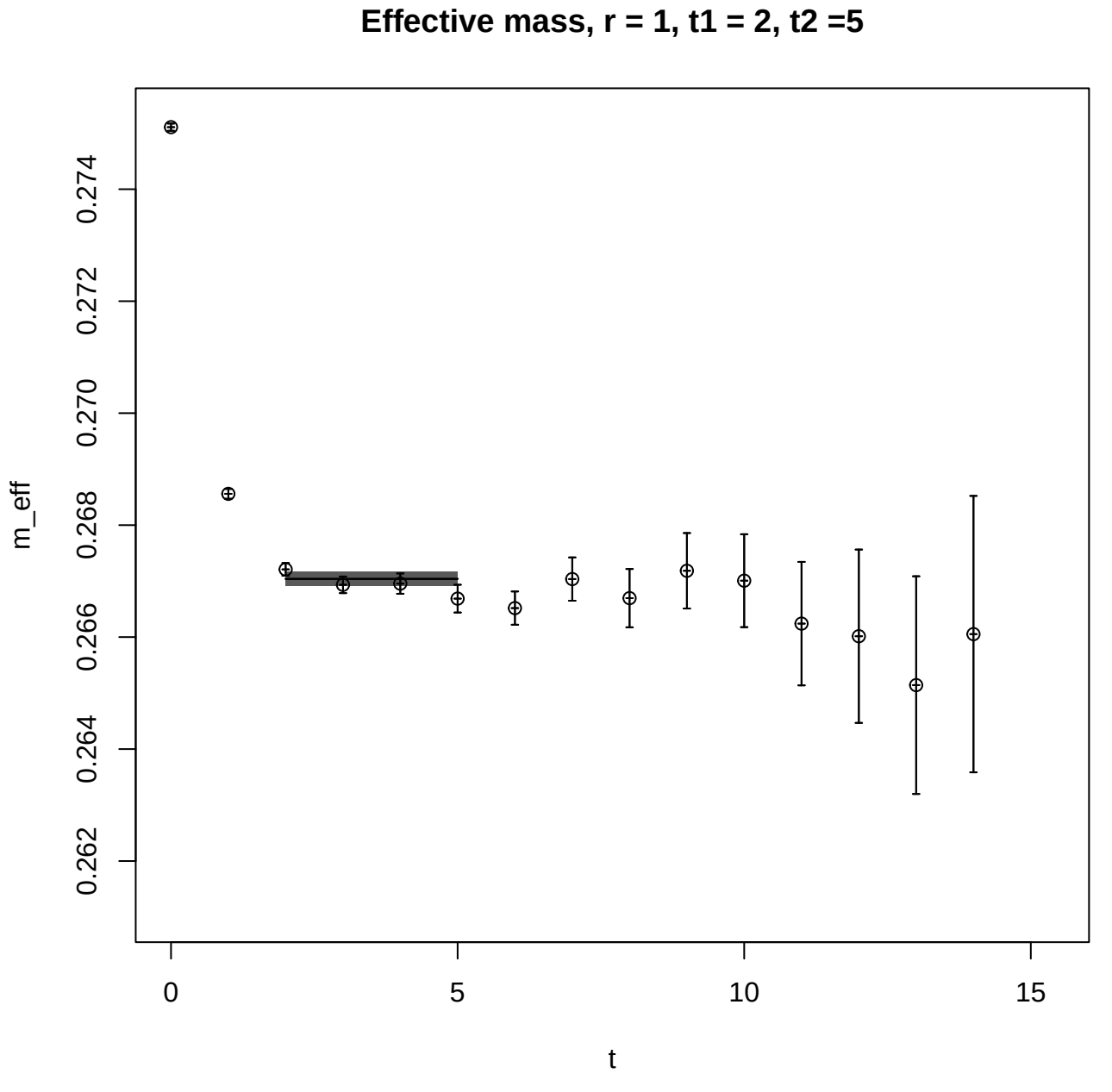
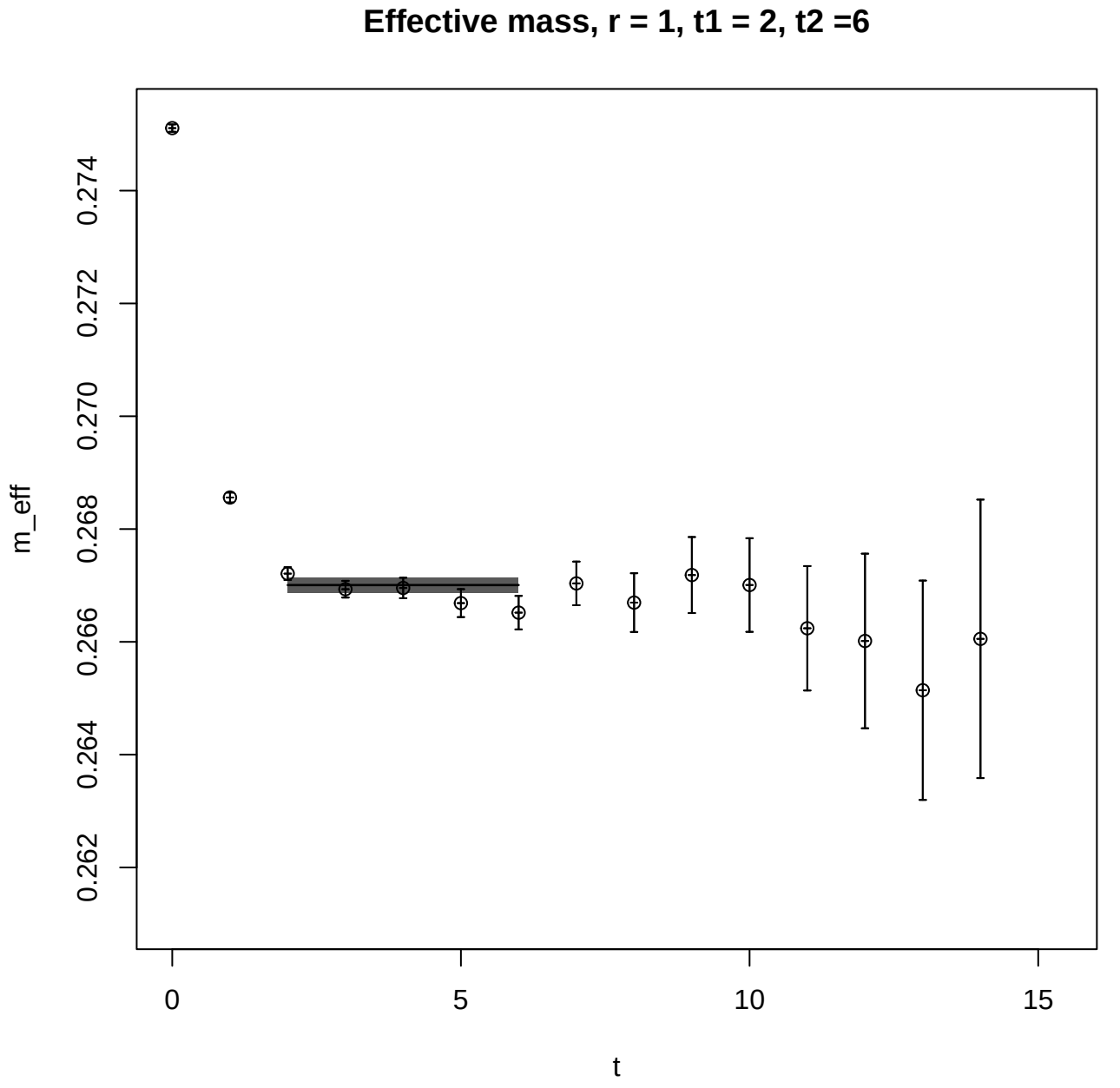
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =5
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =6
m_eff
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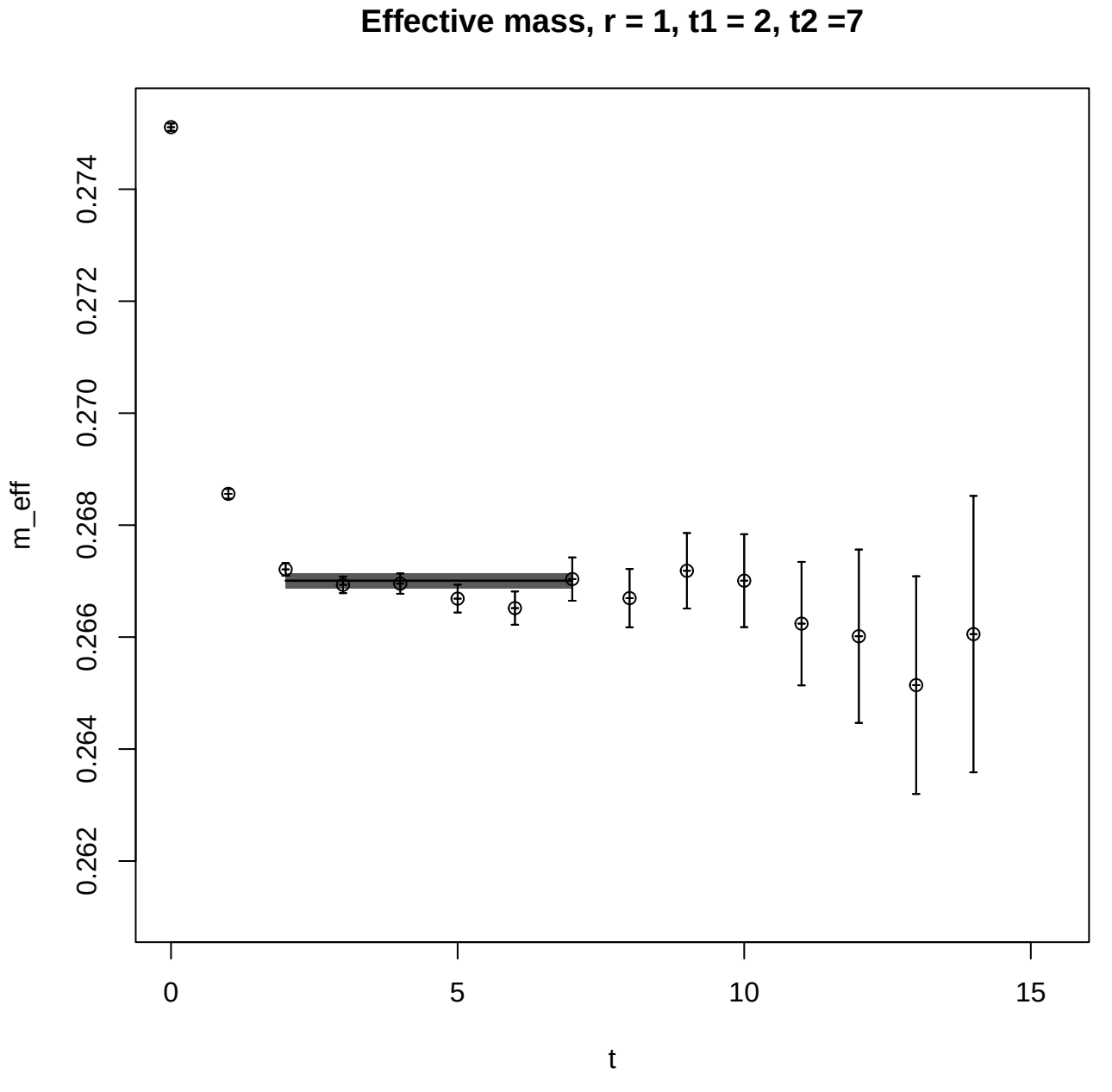
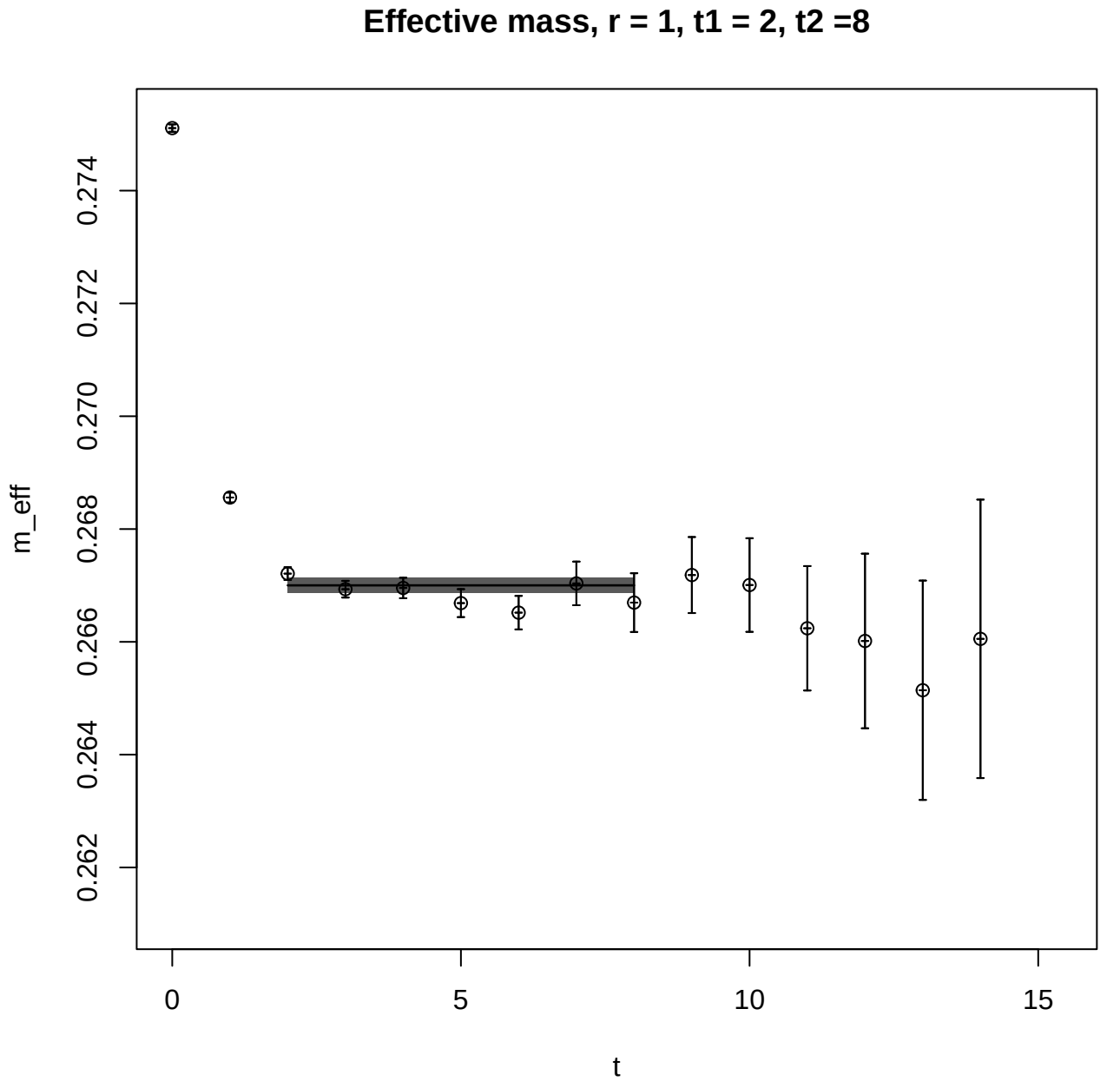
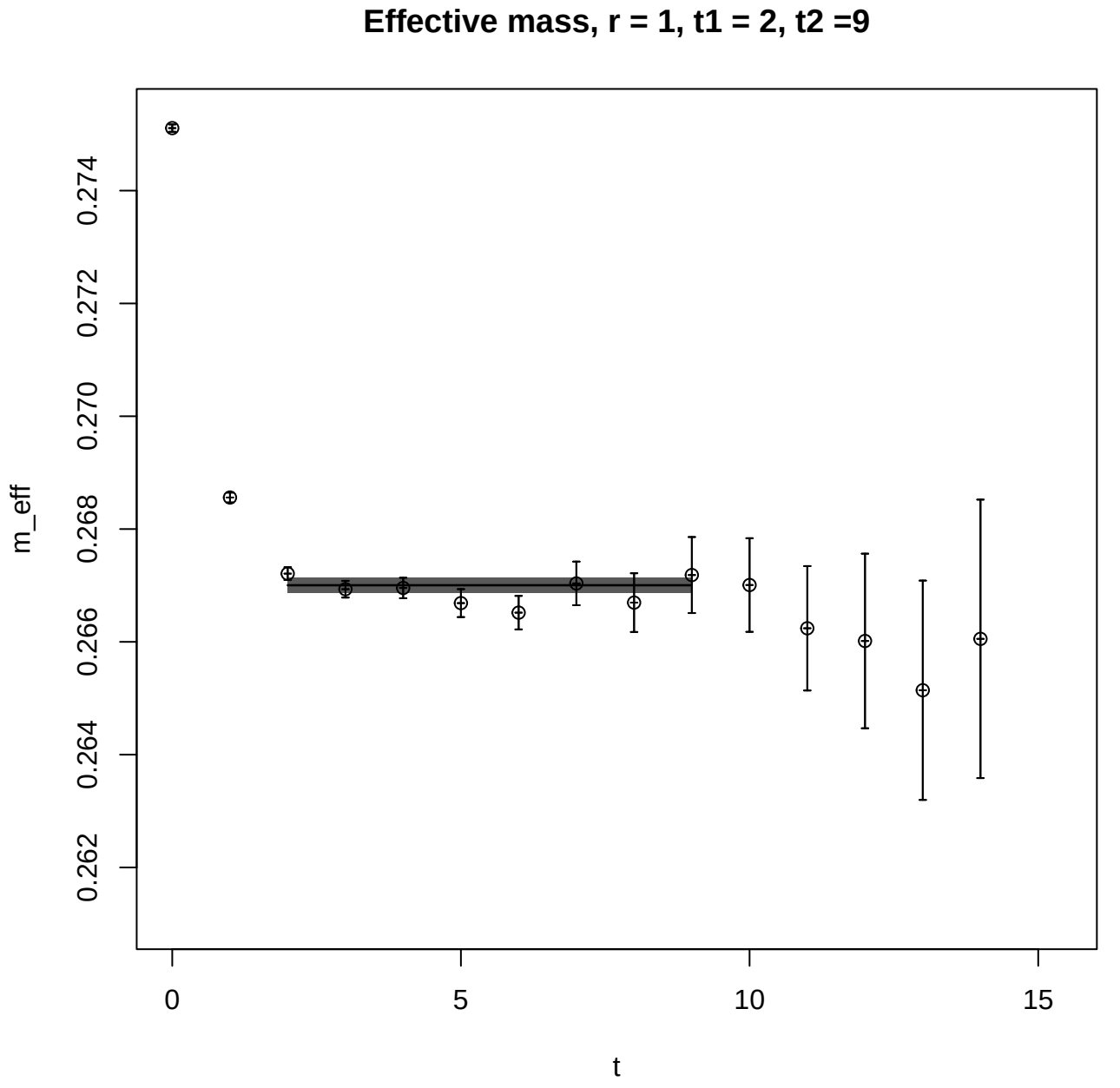
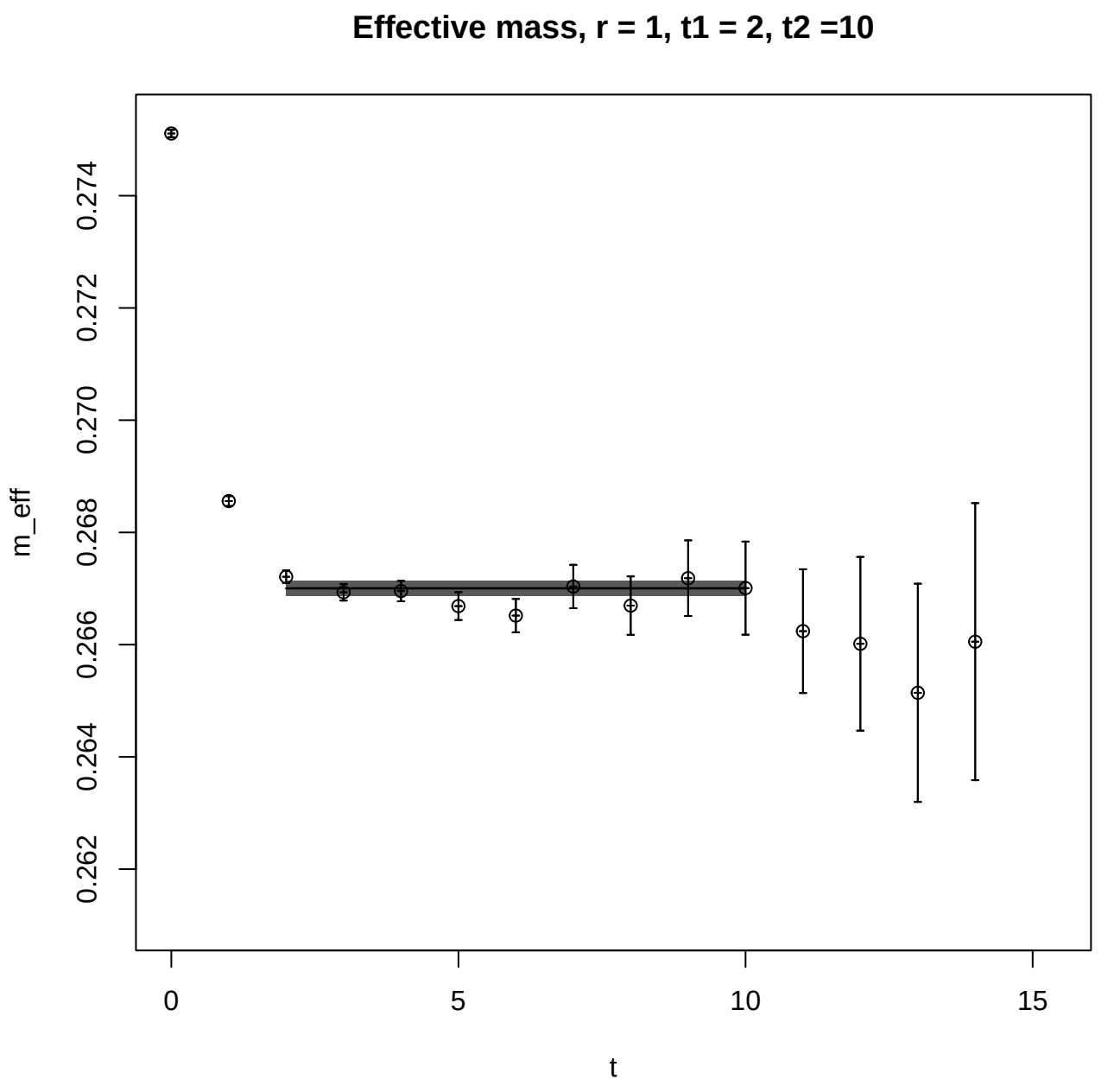
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =7
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =8
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =9
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =10
m_eff
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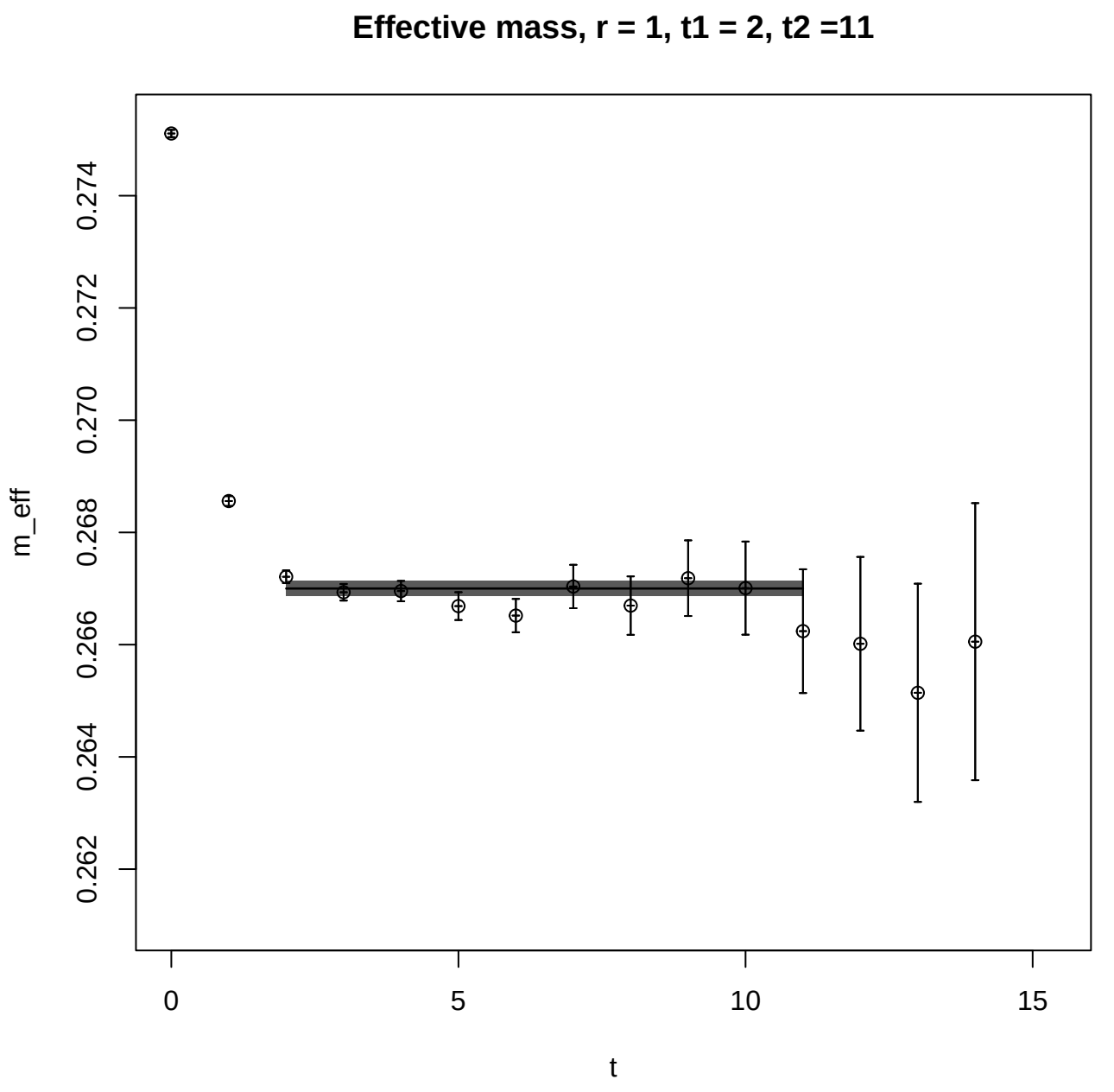
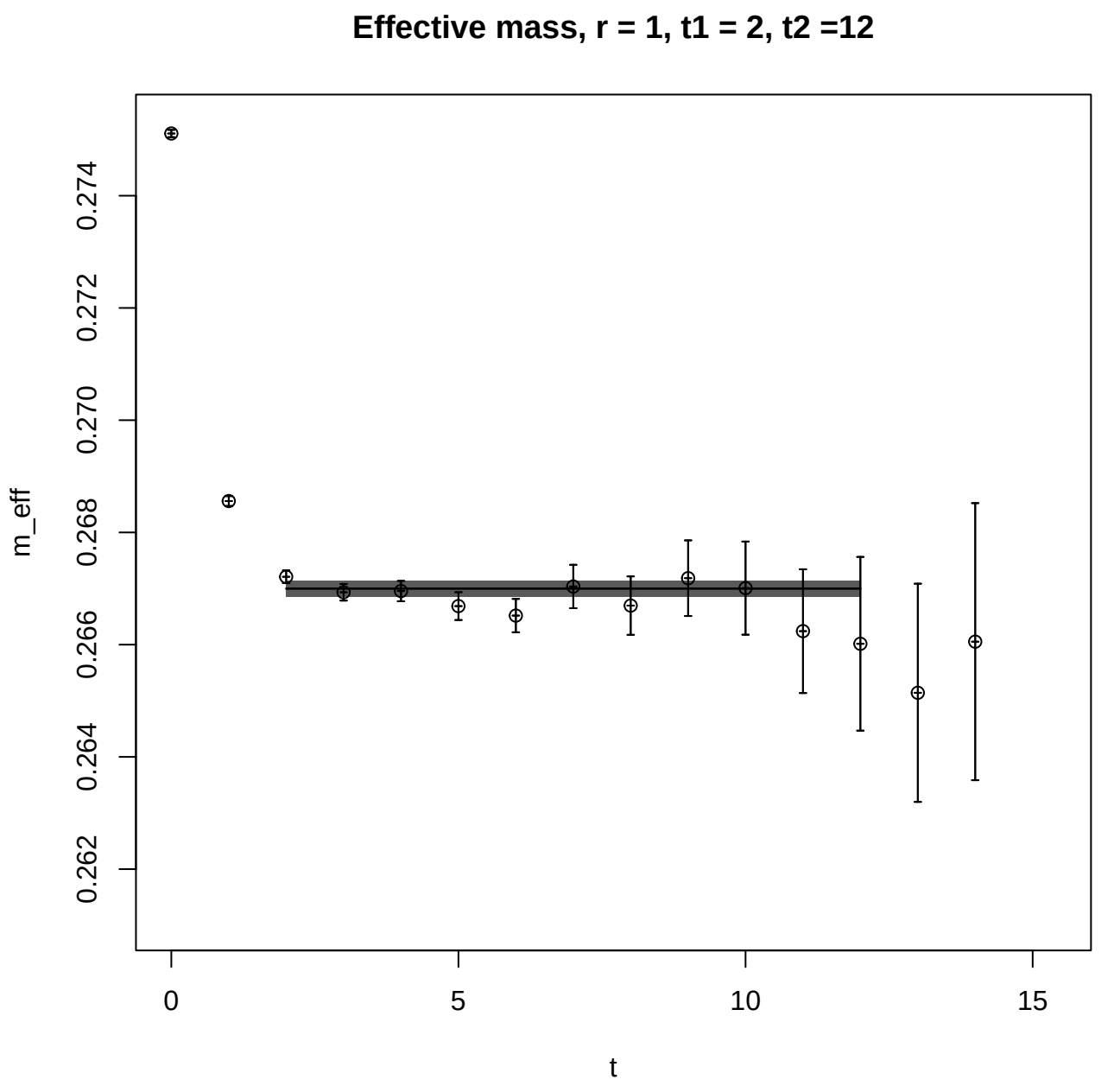
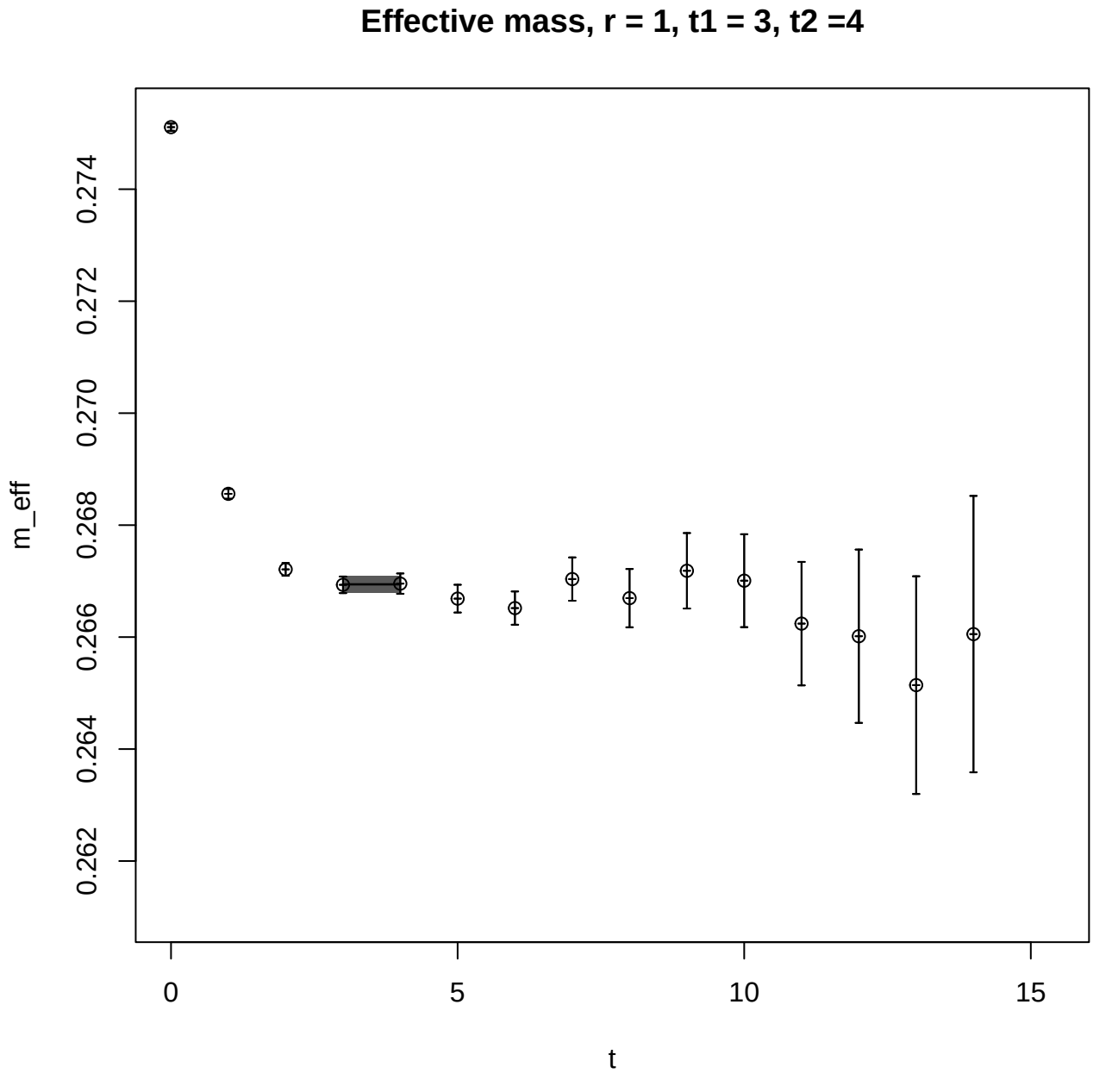
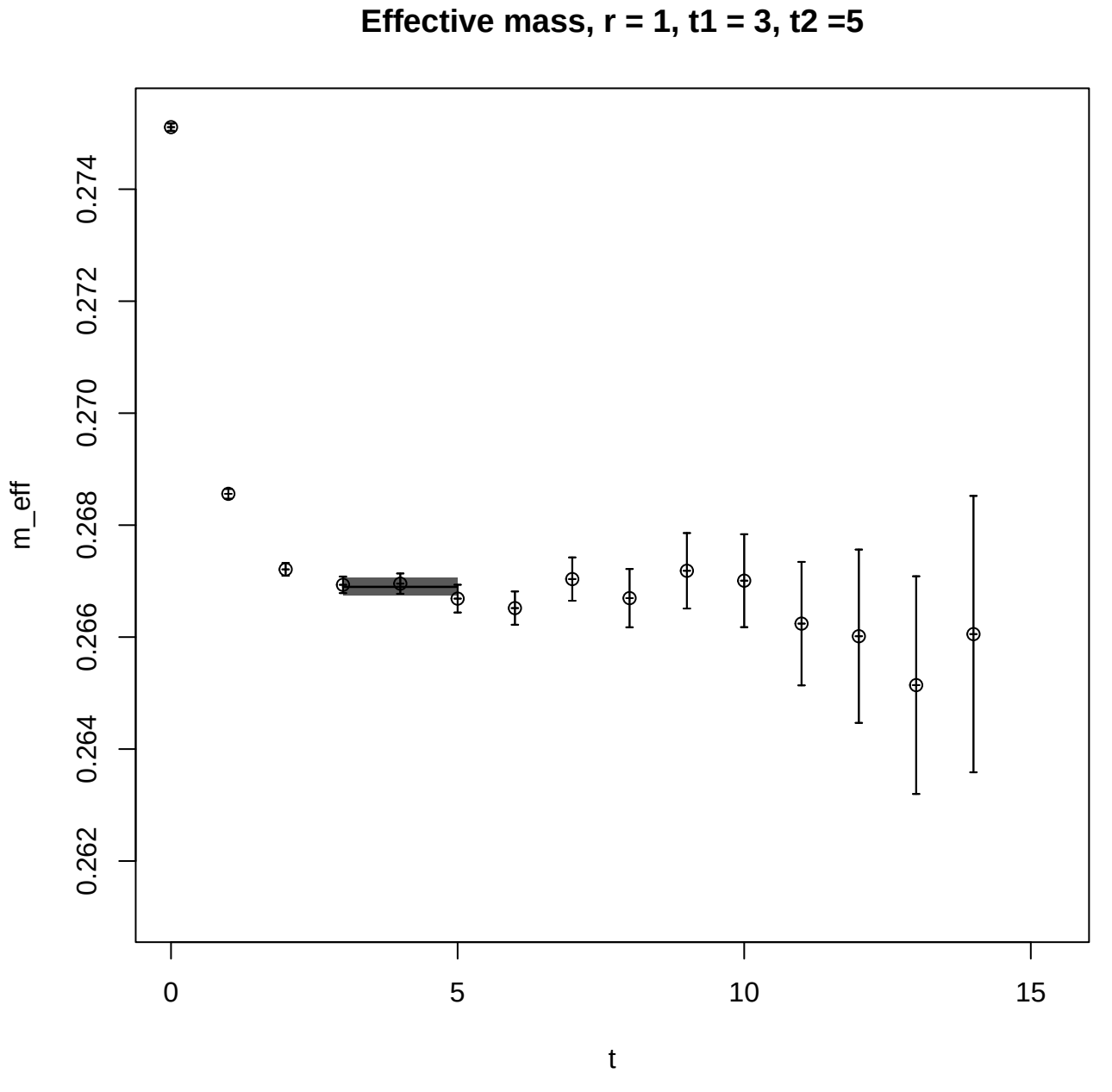
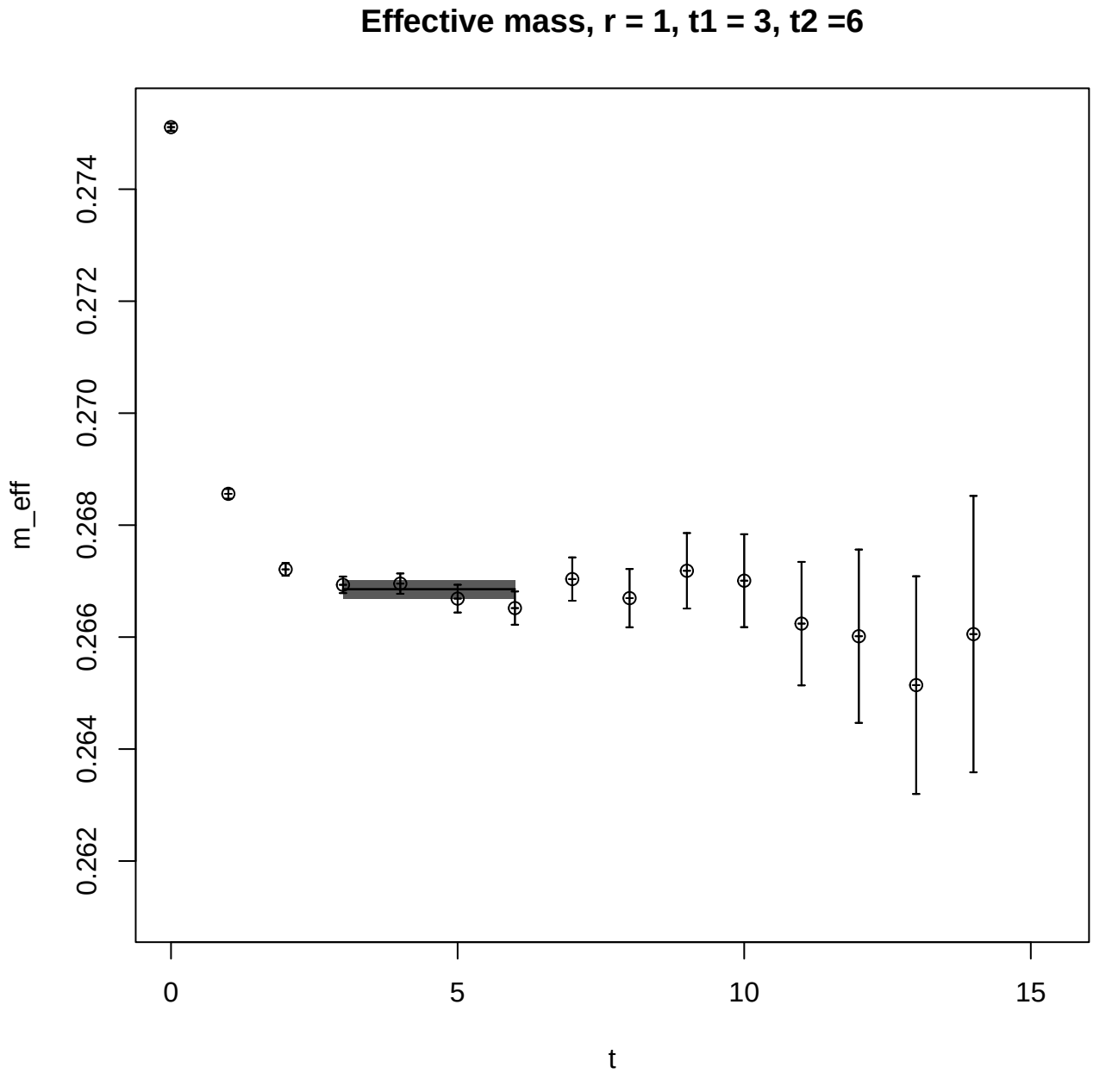
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =12
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =4
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =5
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =6
m_eff
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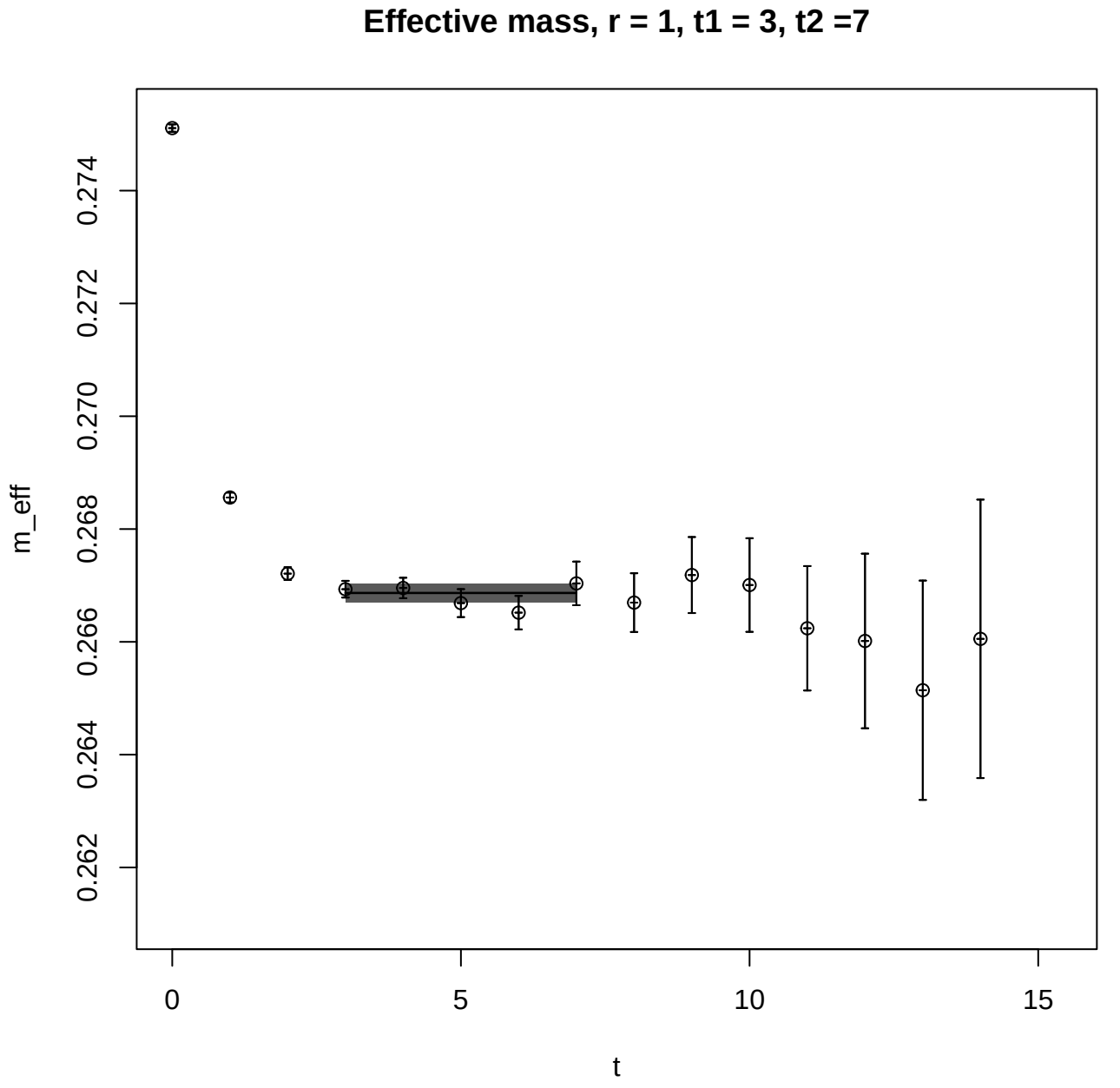
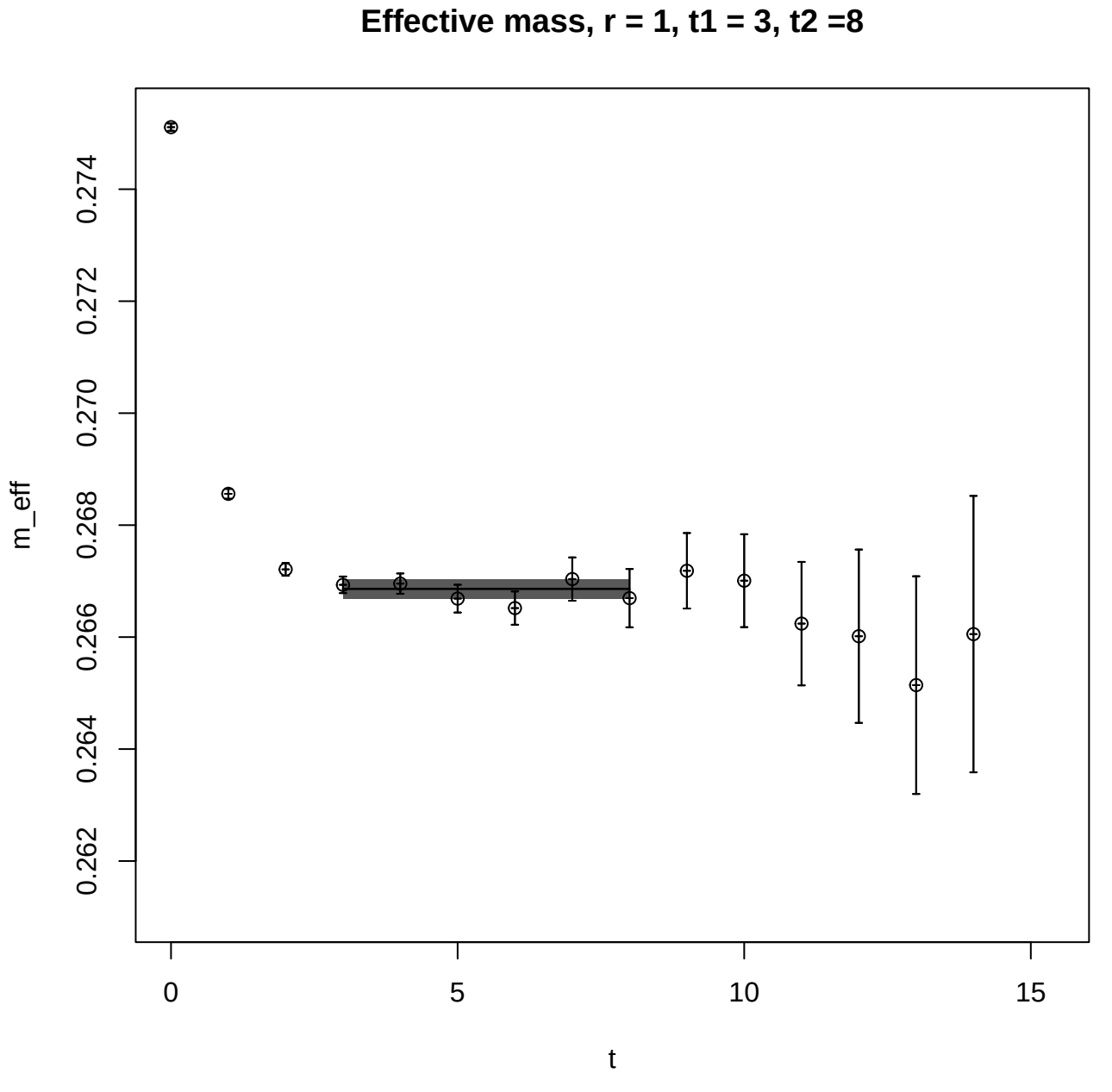
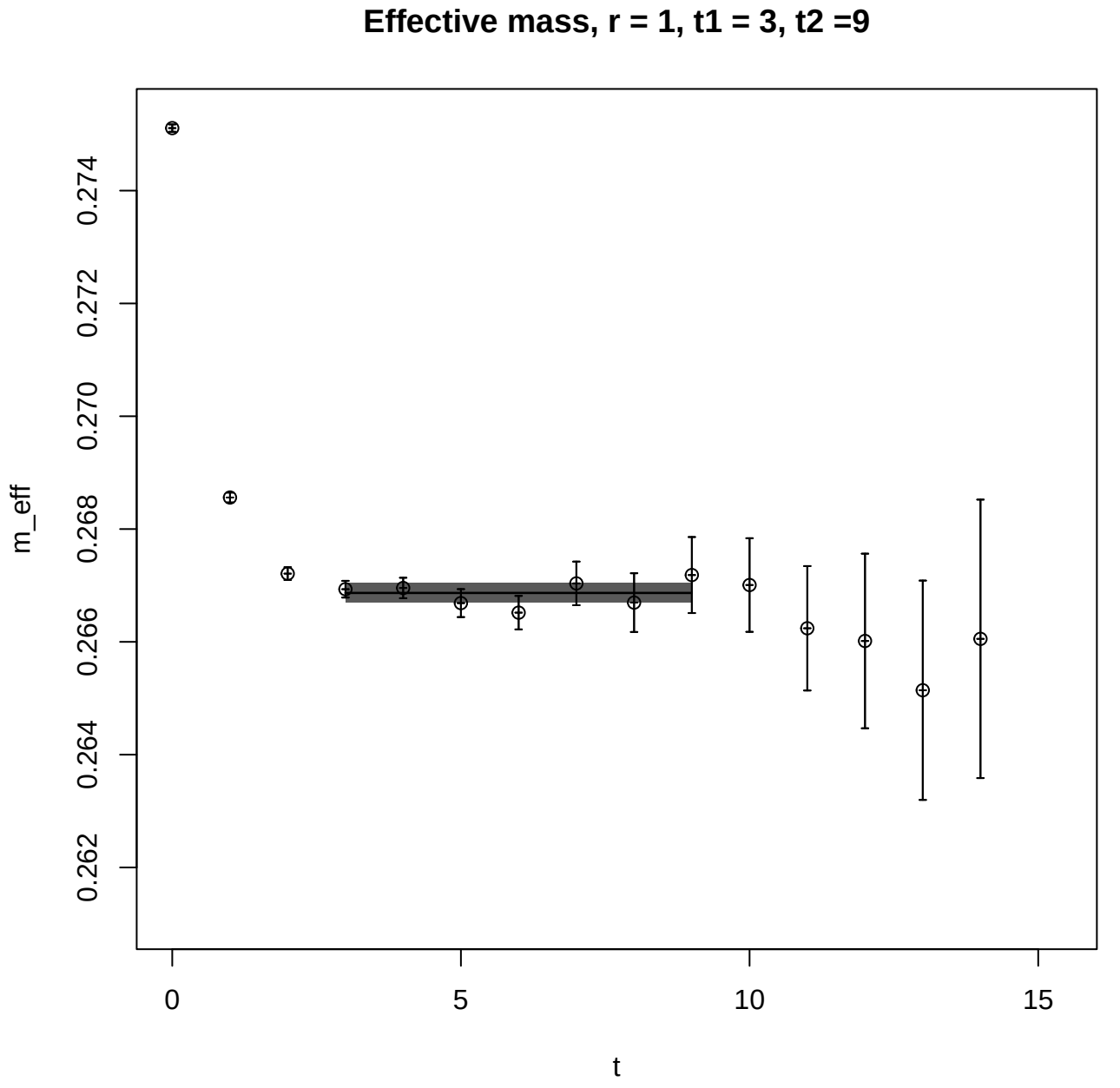
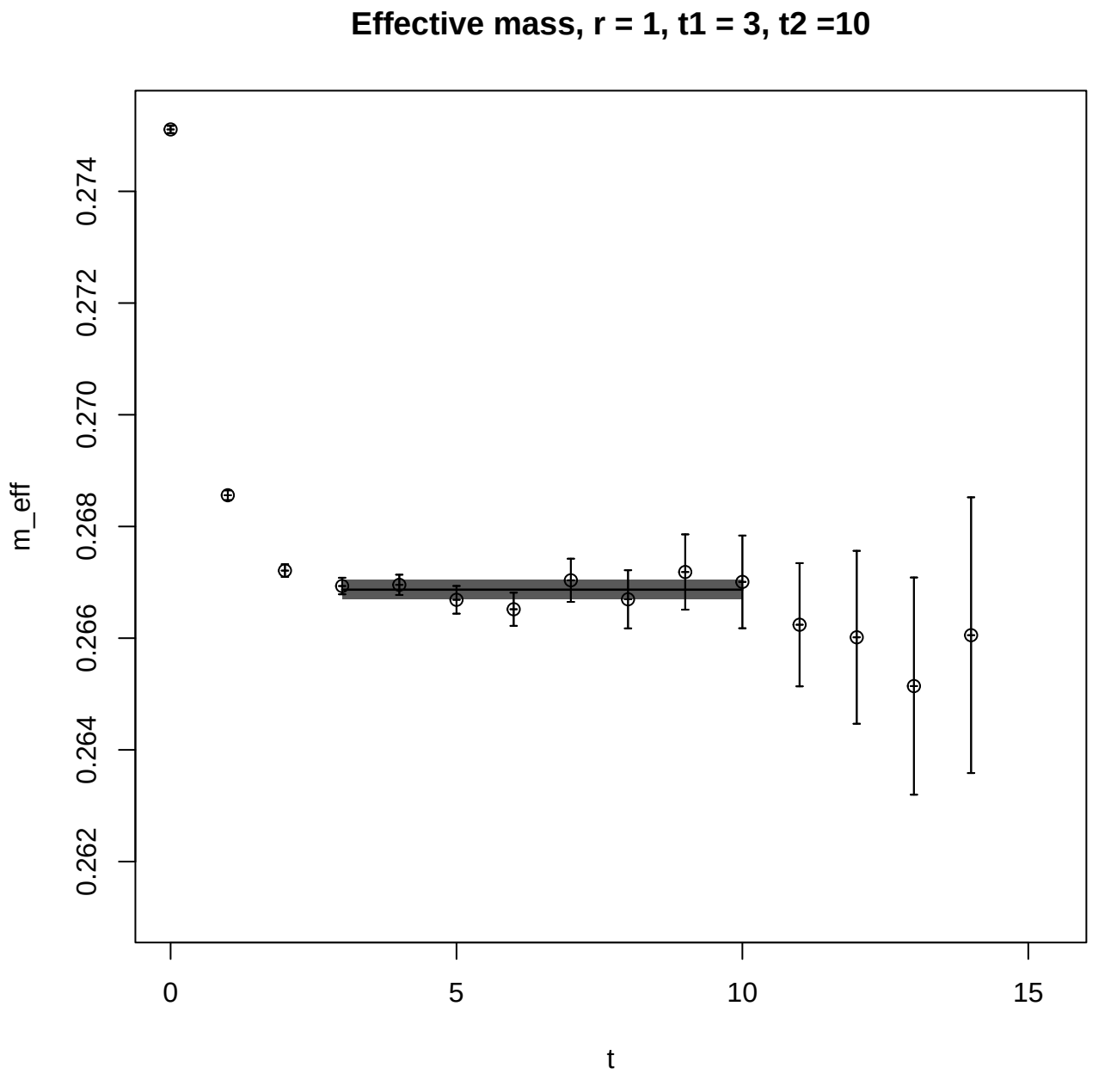
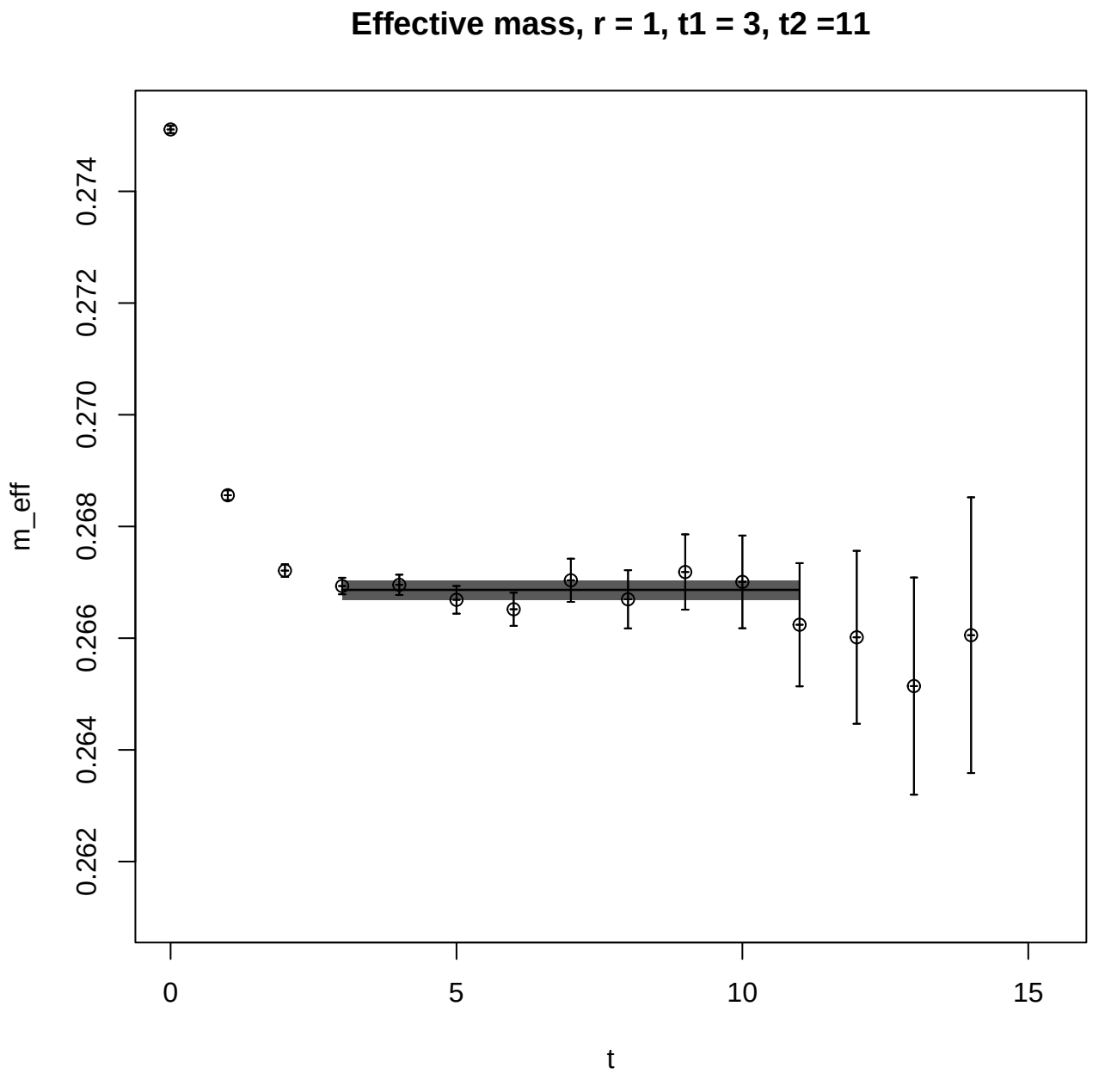
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

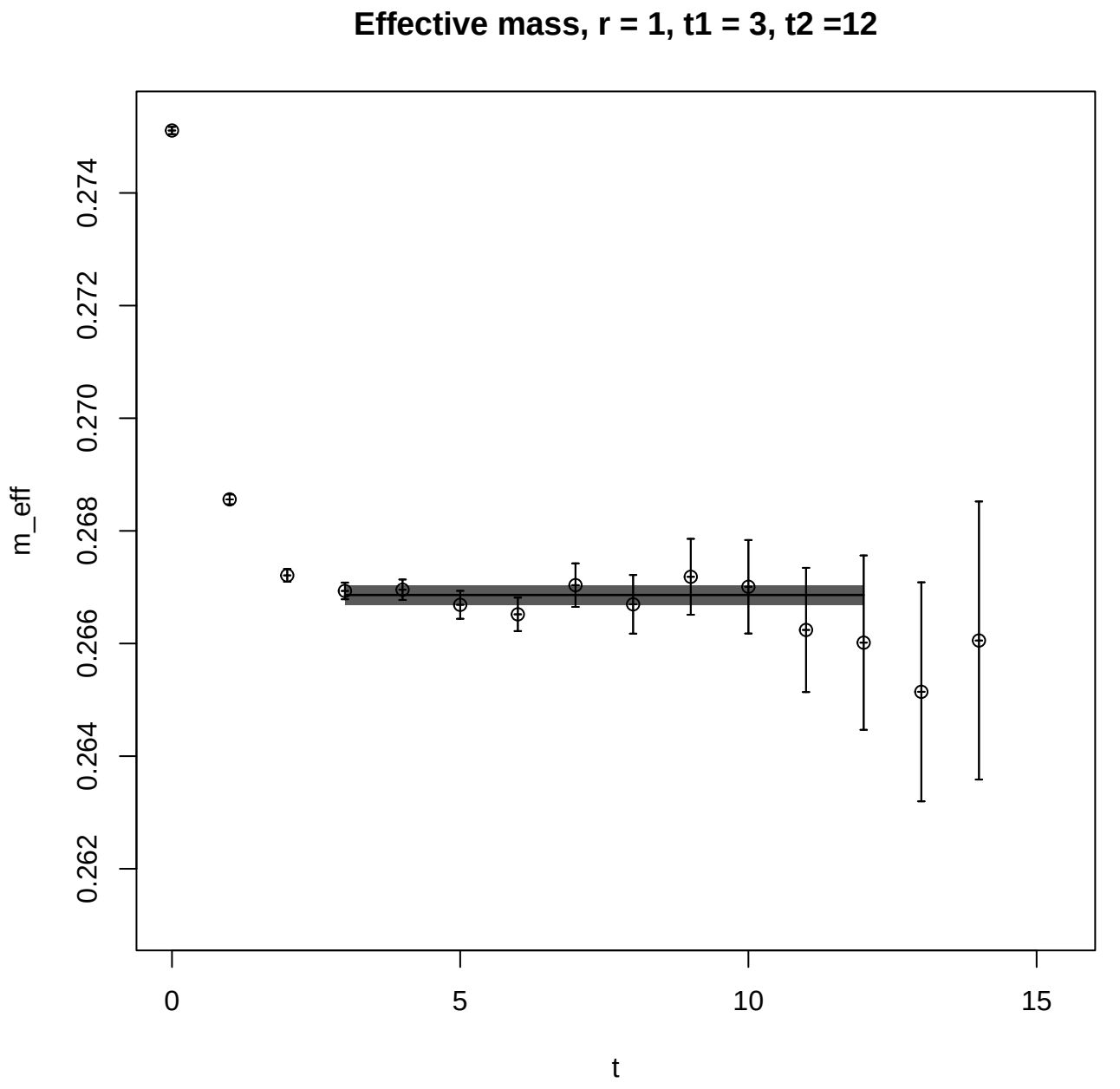

Effective mass, r = 1, t1 = 3, t2 =7
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

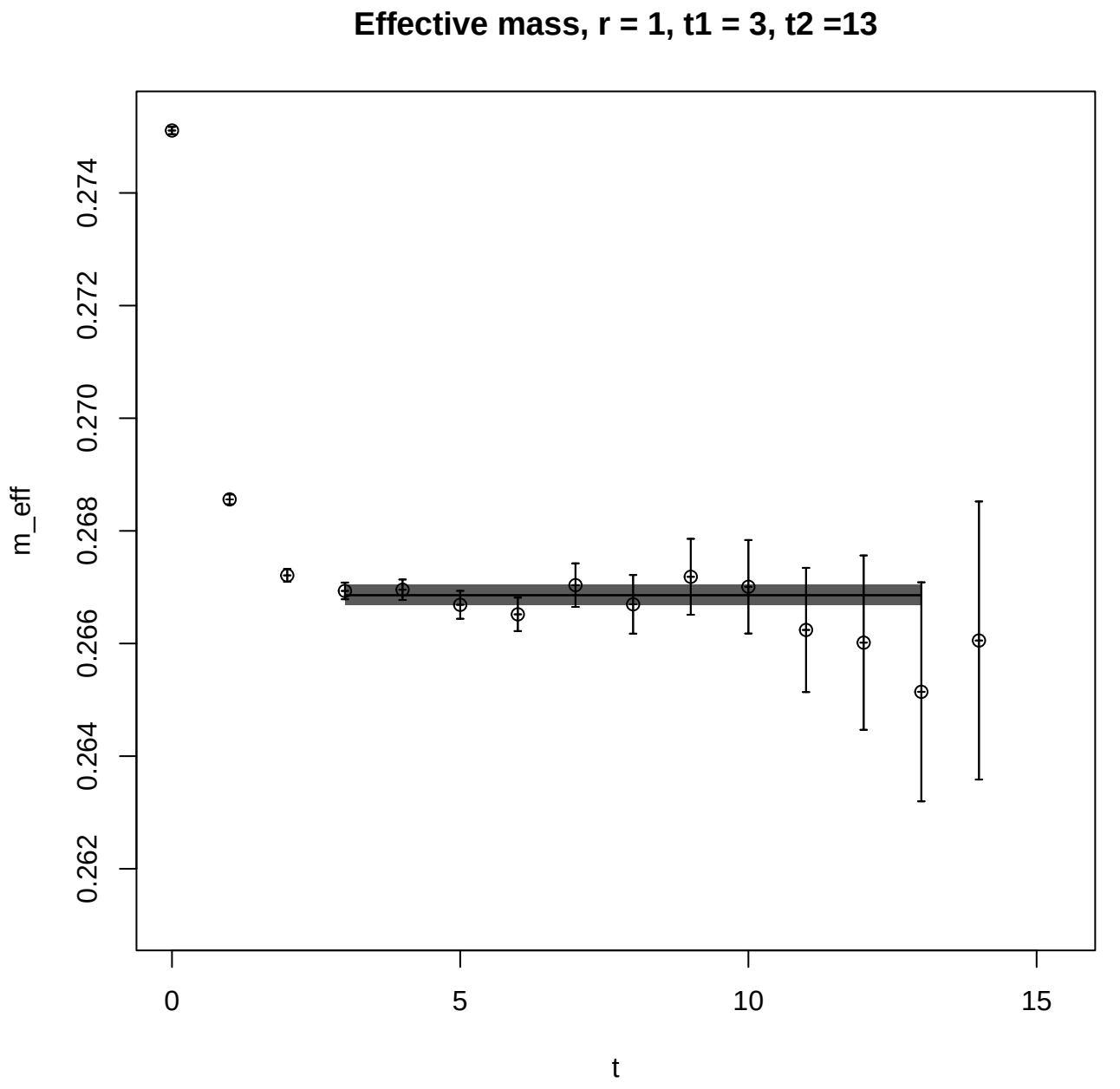

Effective mass, r = 1, t1 = 3, t2 =8
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

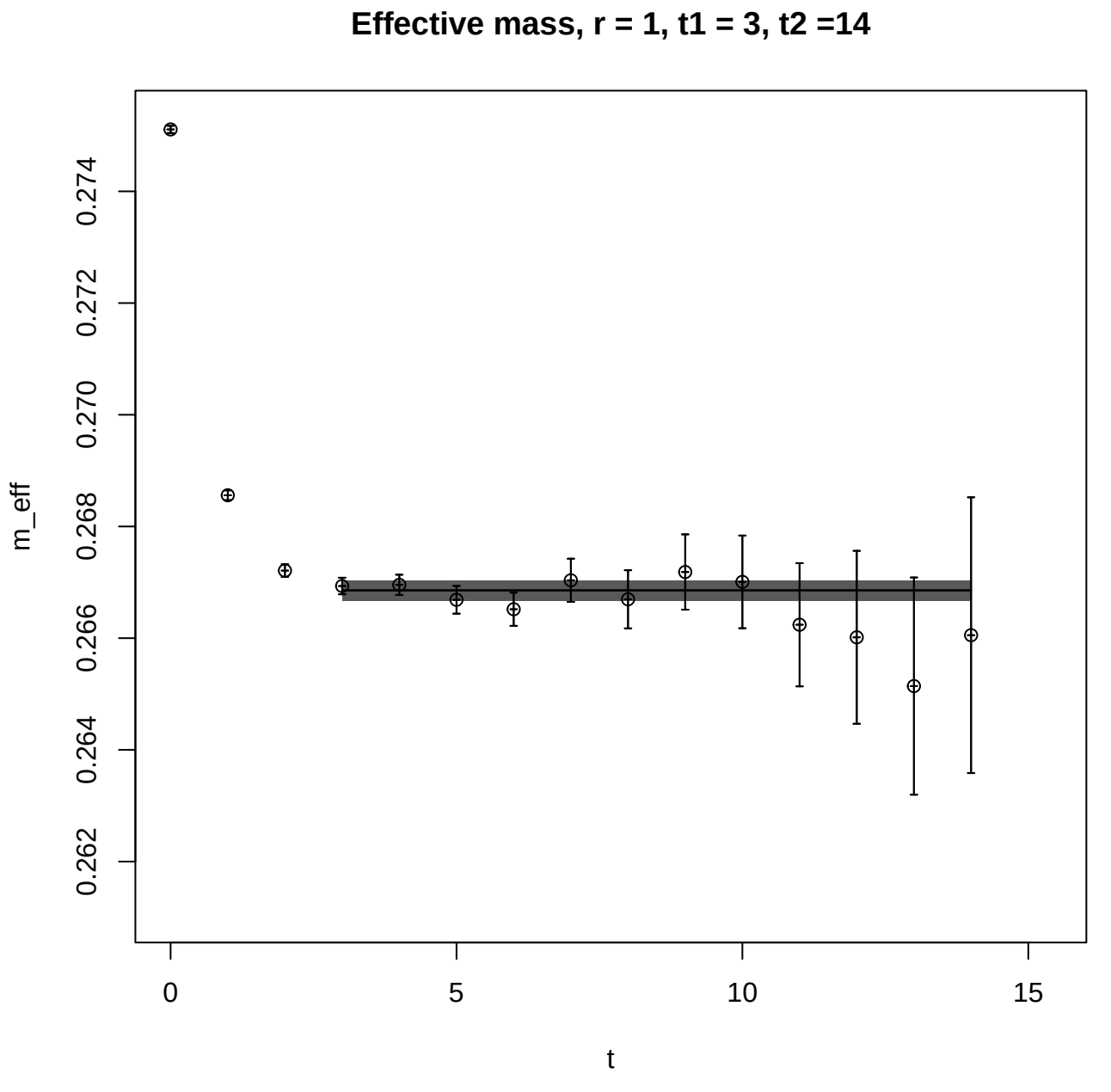

Effective mass, r = 1, t1 = 3, t2 =9
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

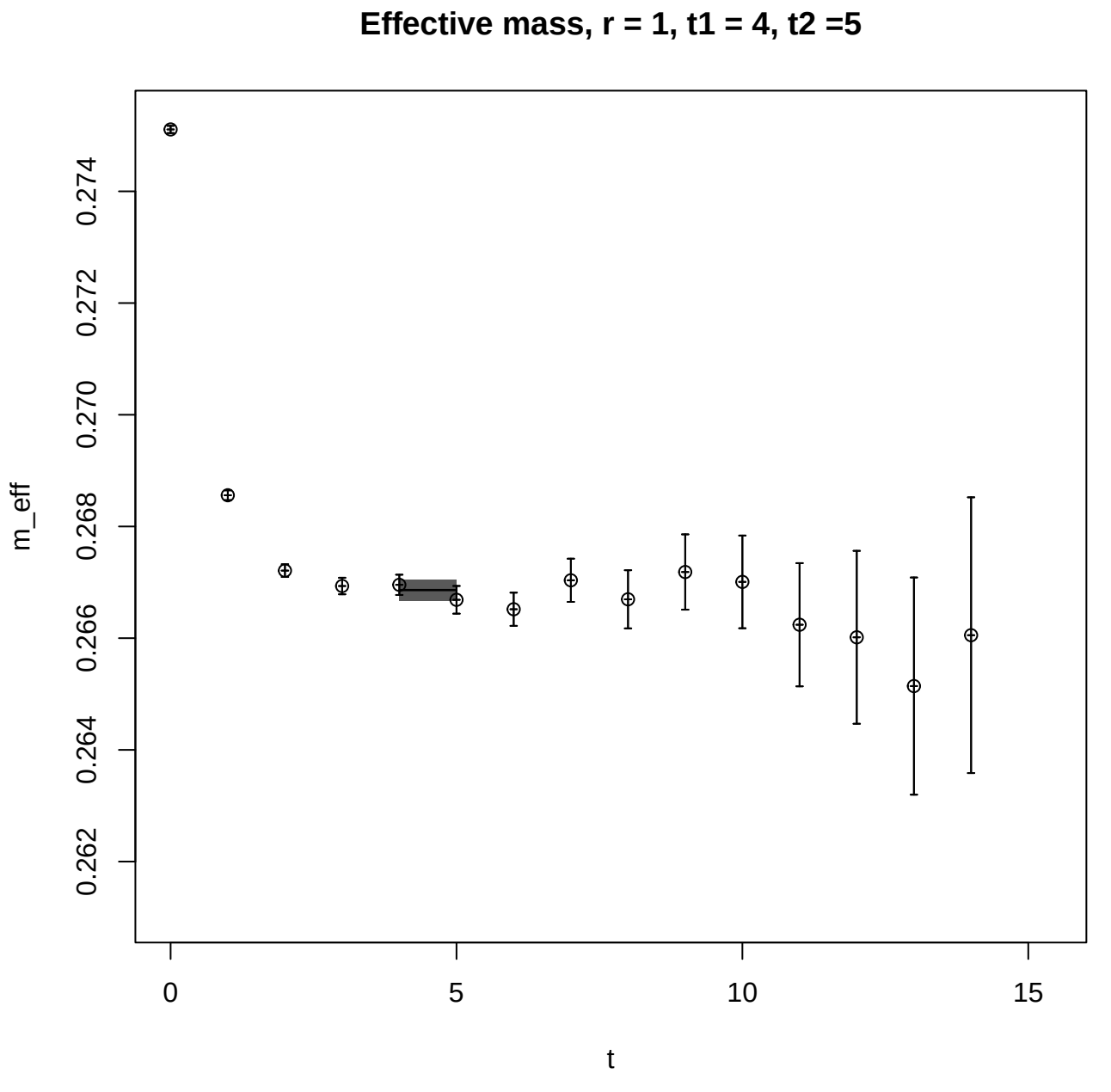

Effective mass, r = 1, t1 = 3, t2 =10
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =12
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =13
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =5
m_eff
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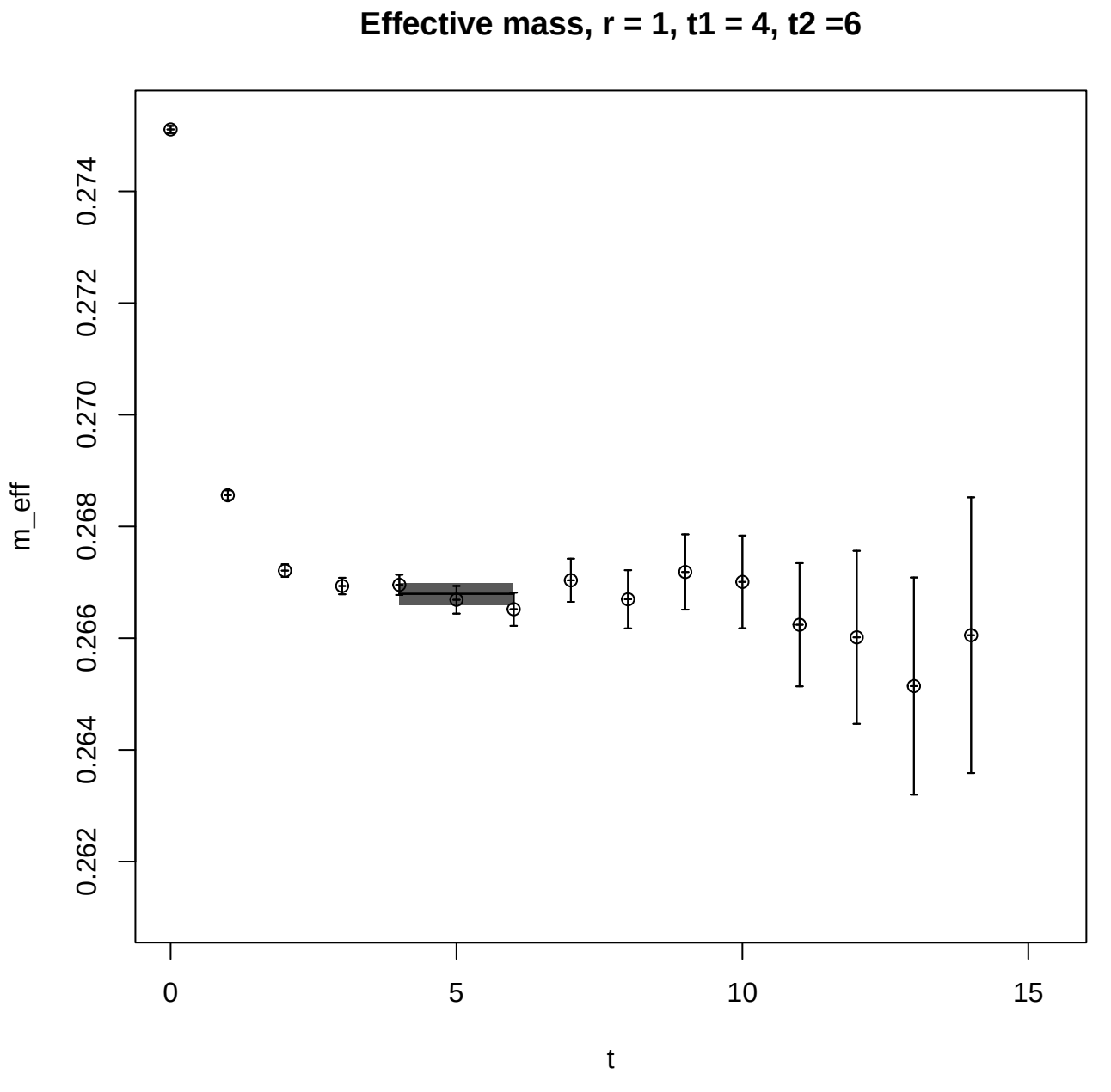
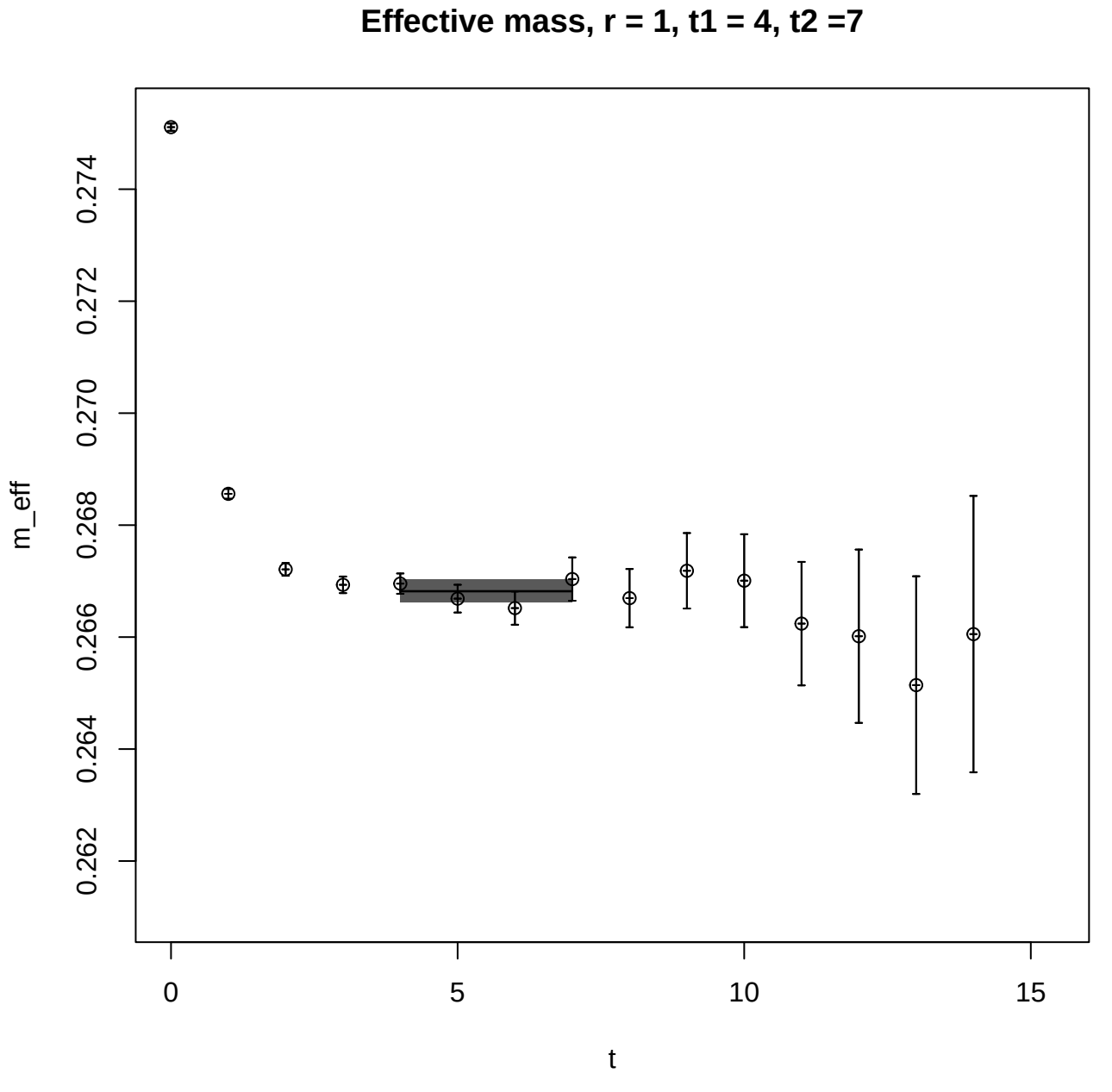
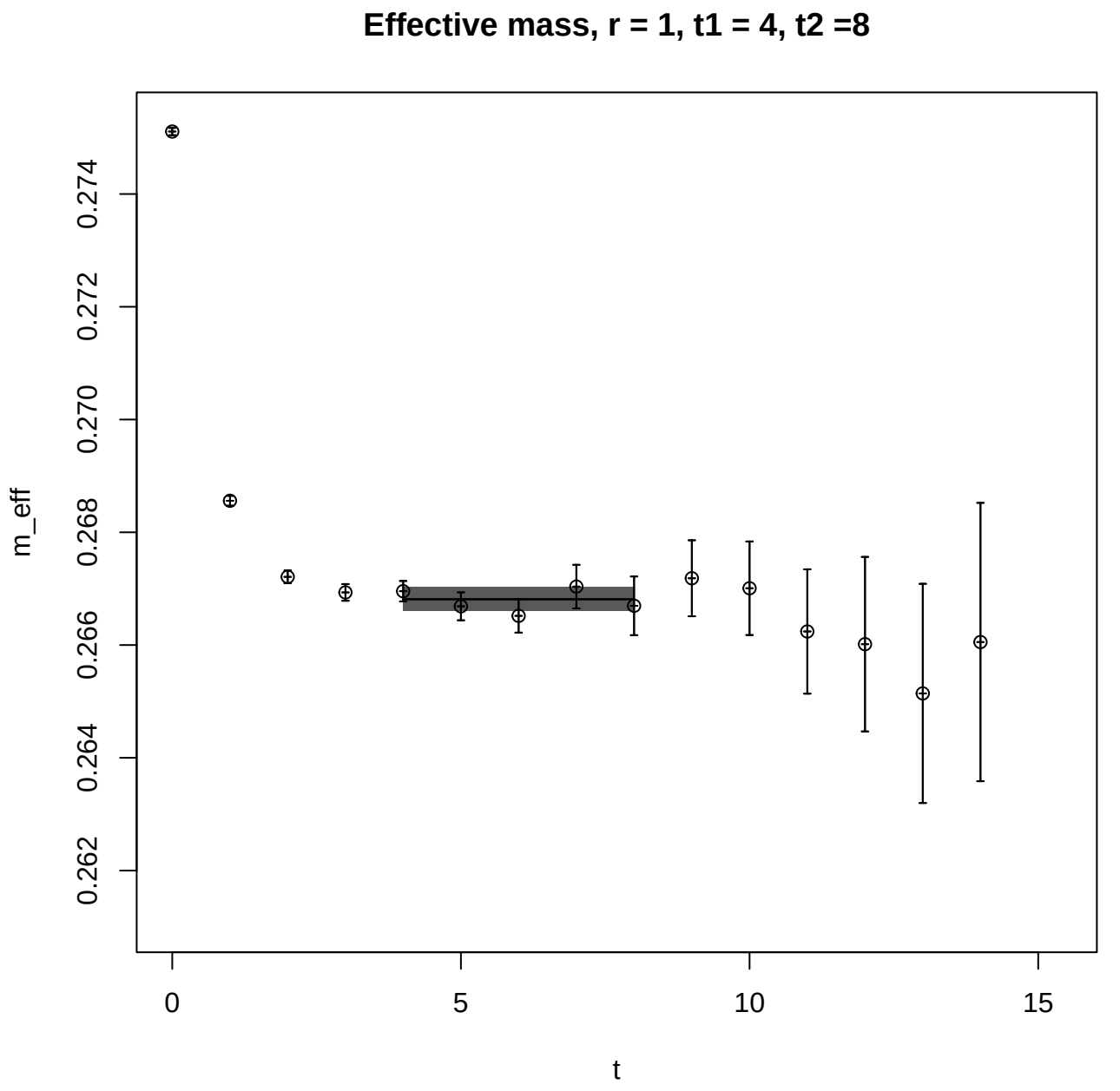
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =6
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =7
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =8
m_eff
0.262
0.264
0.266
0.268
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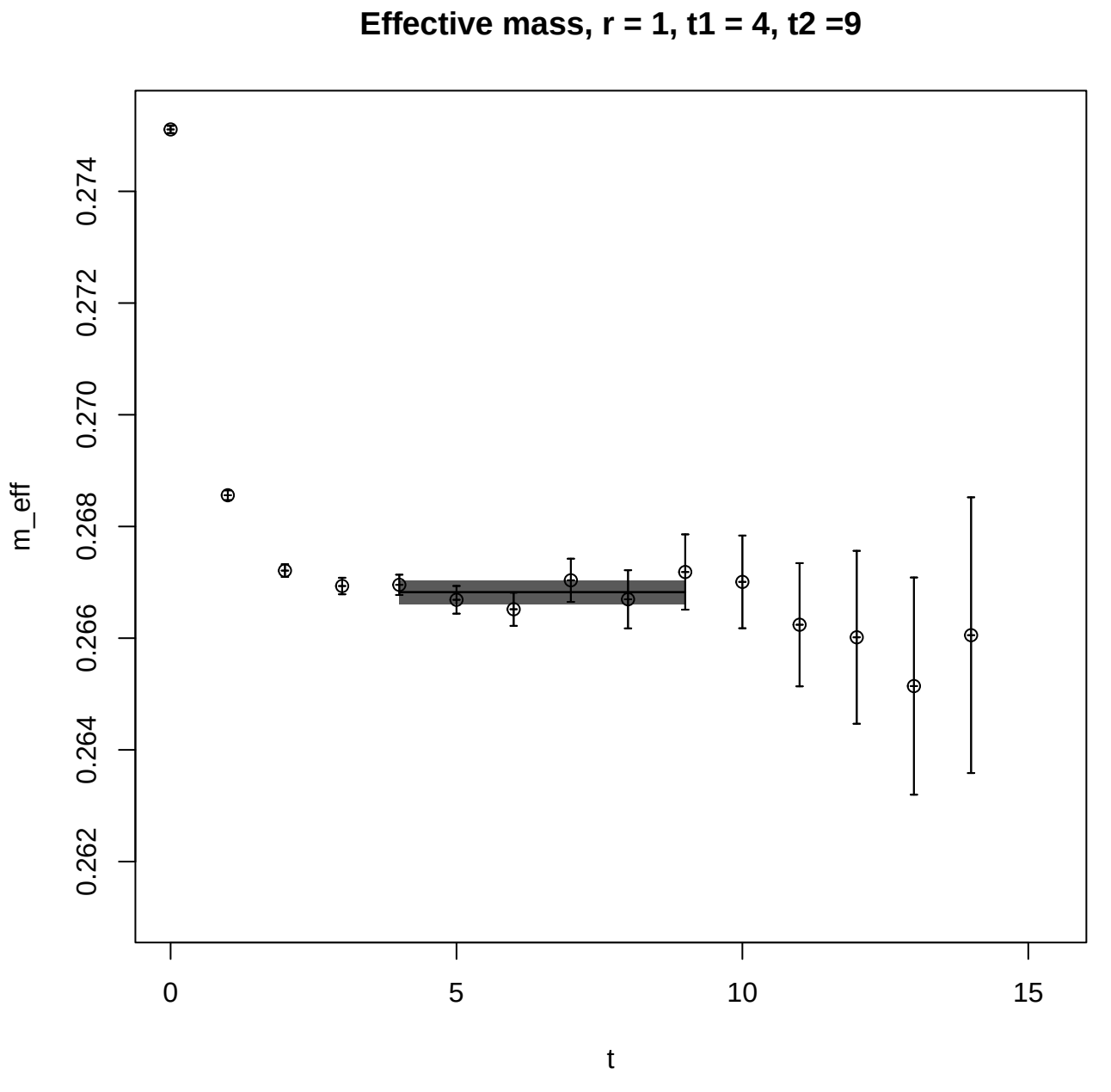
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =9
m_eff
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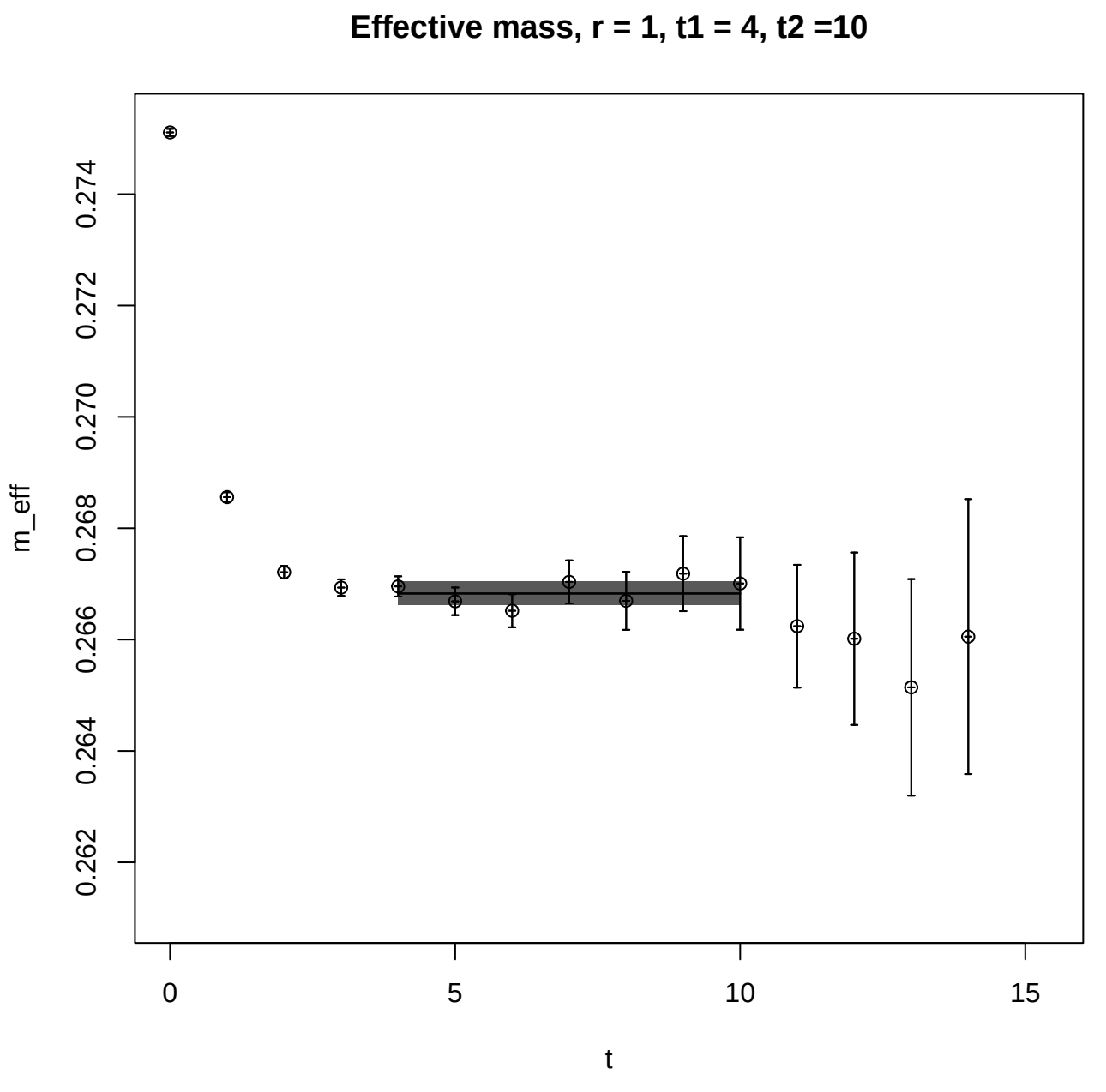
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =10
m_eff
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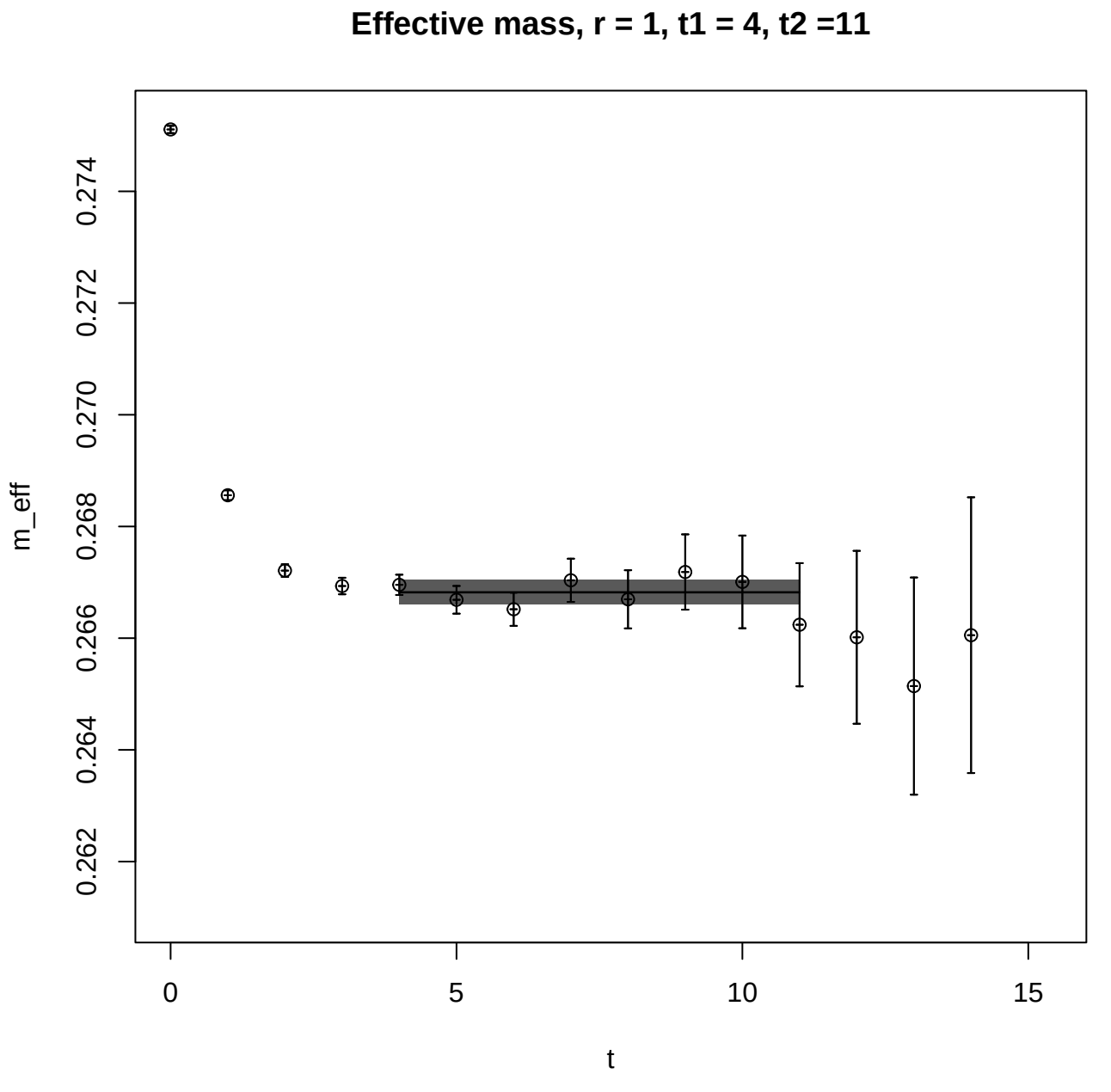
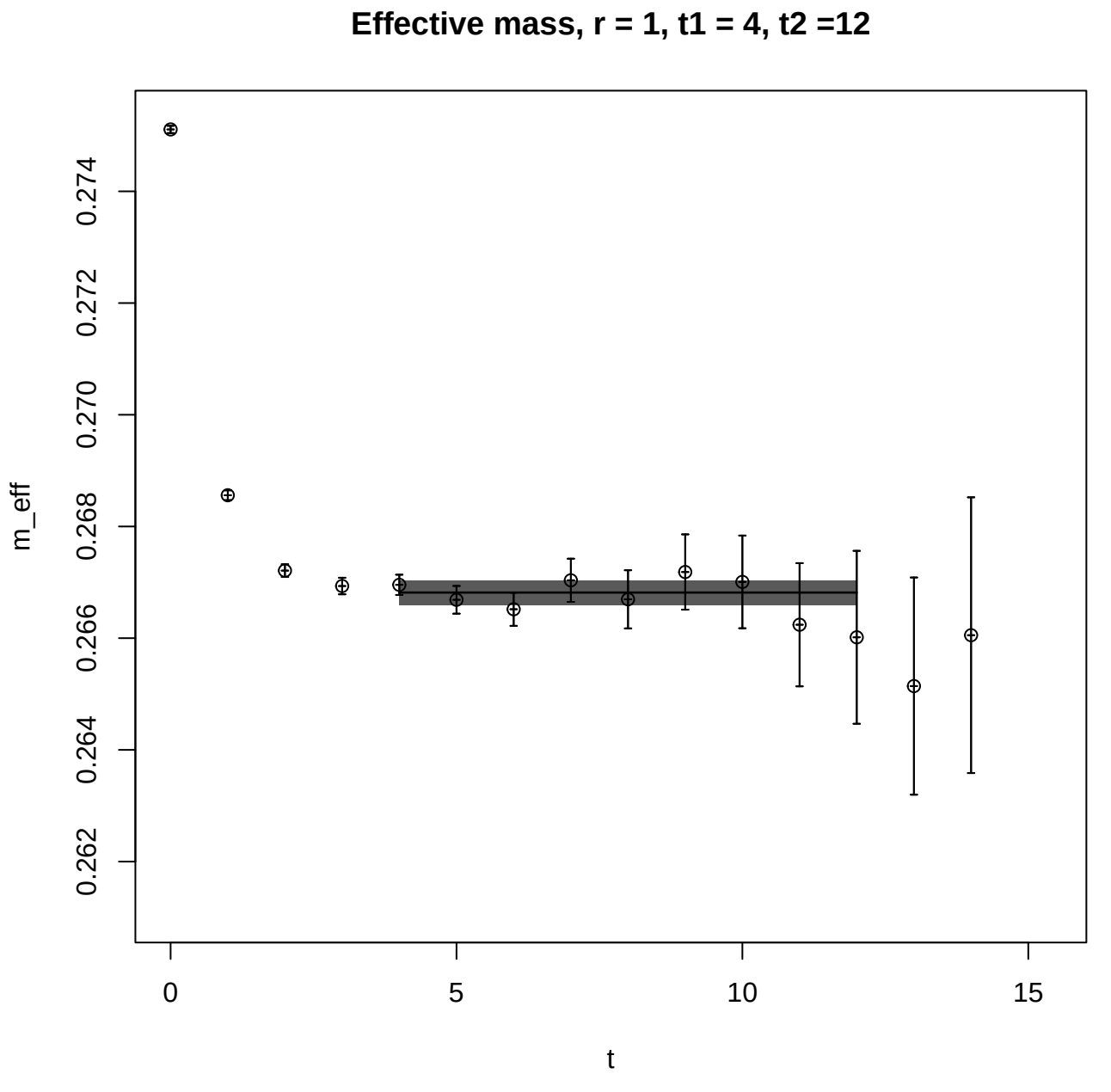
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =12
m_eff
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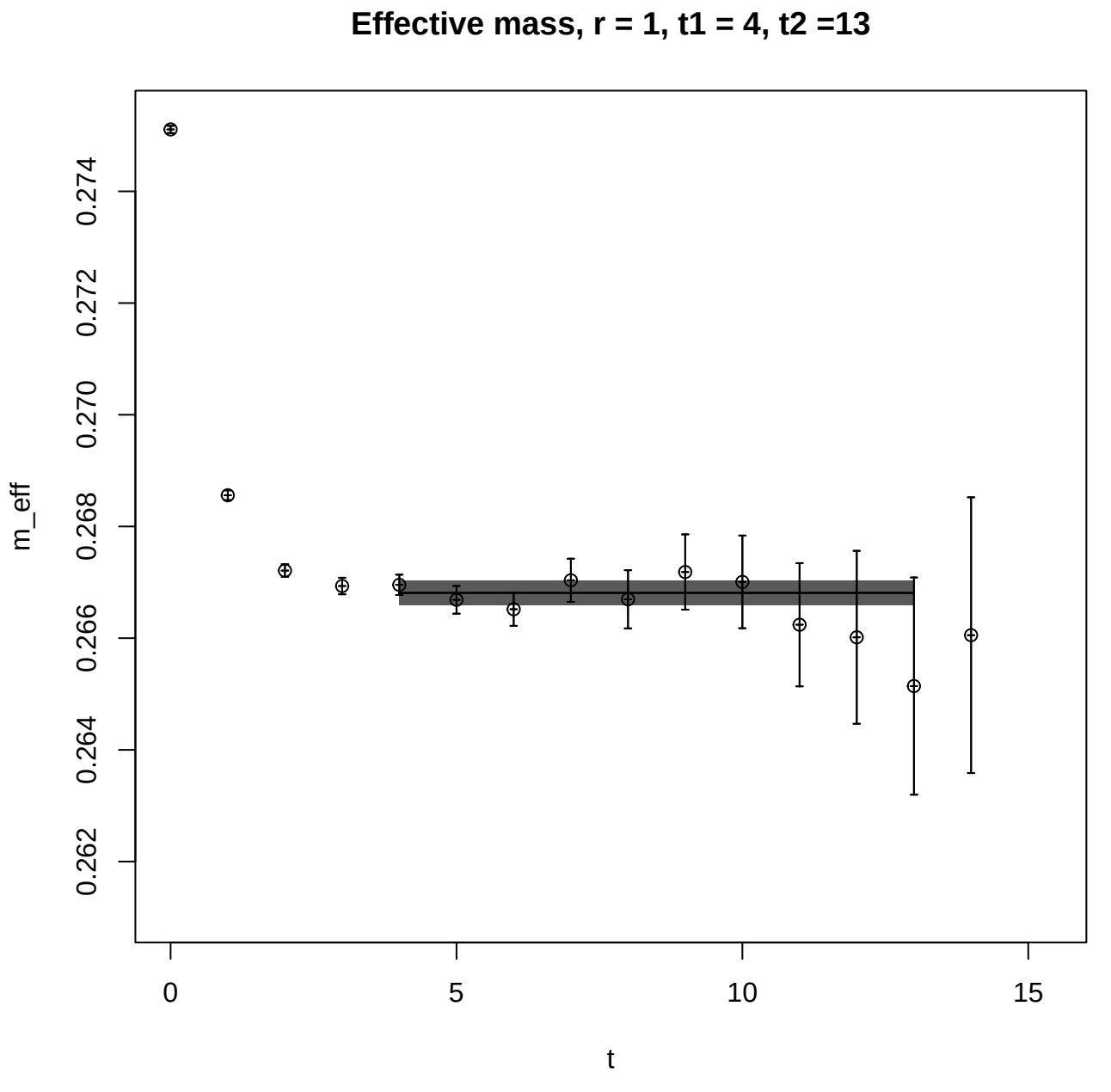
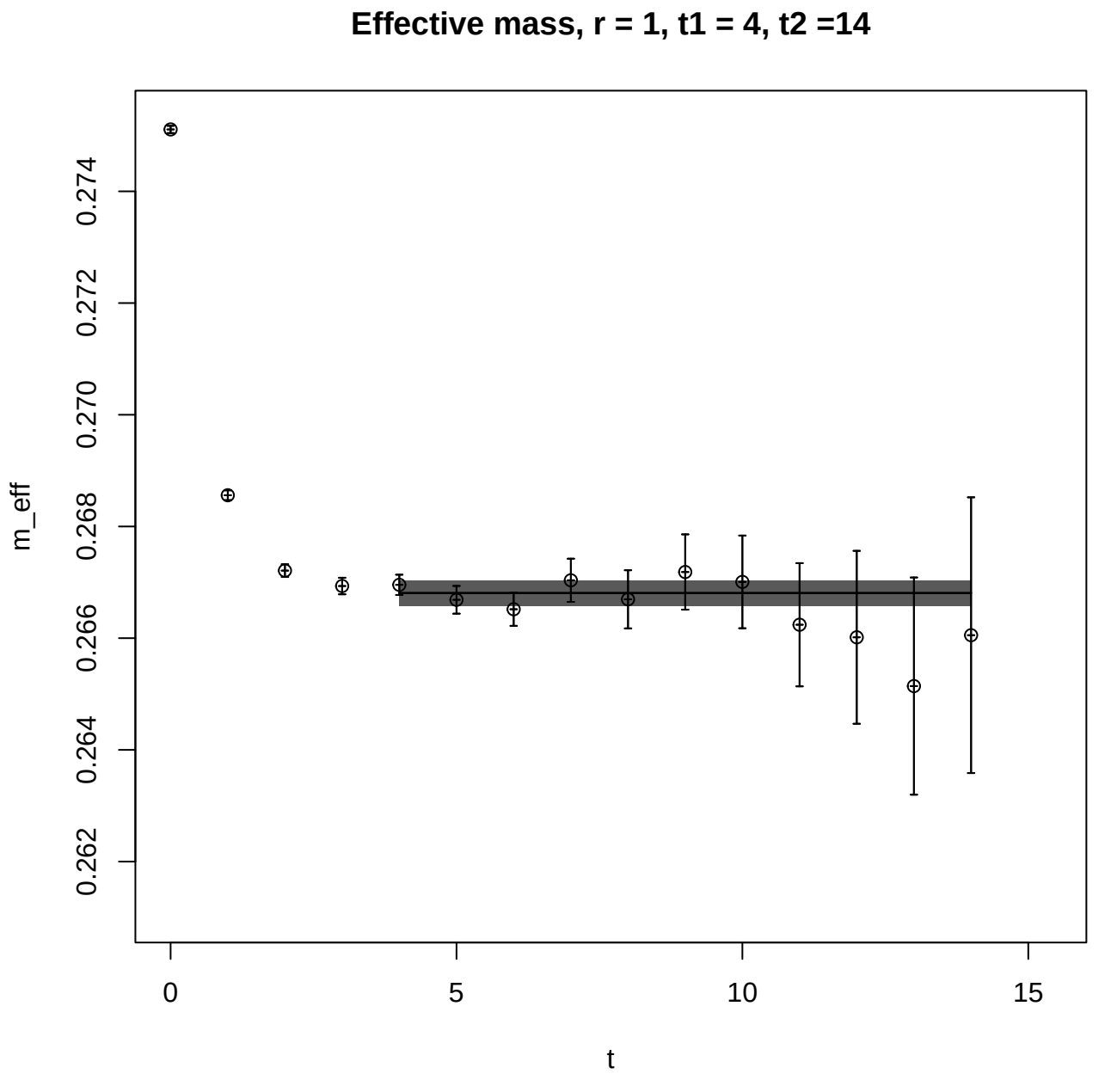
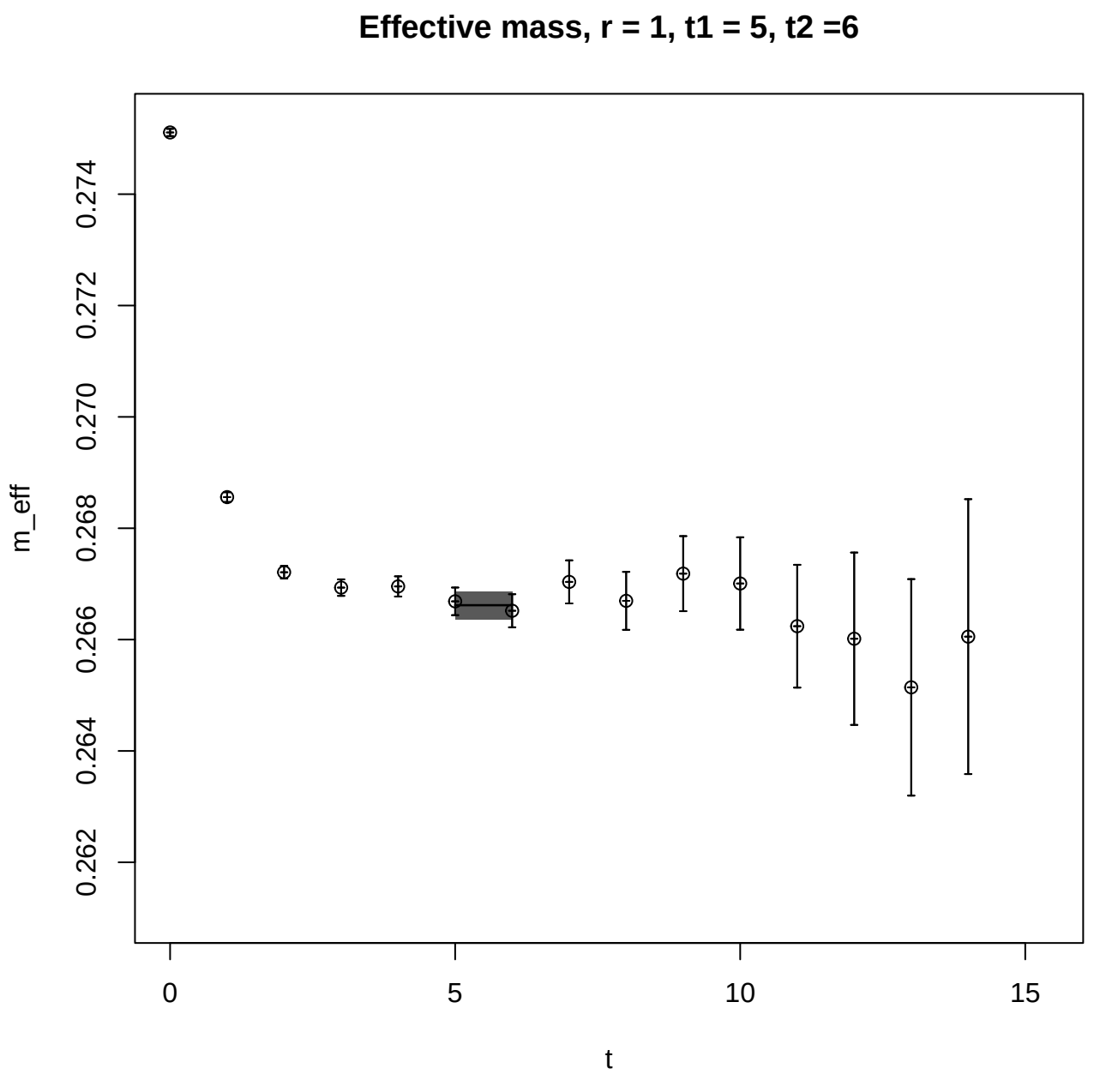
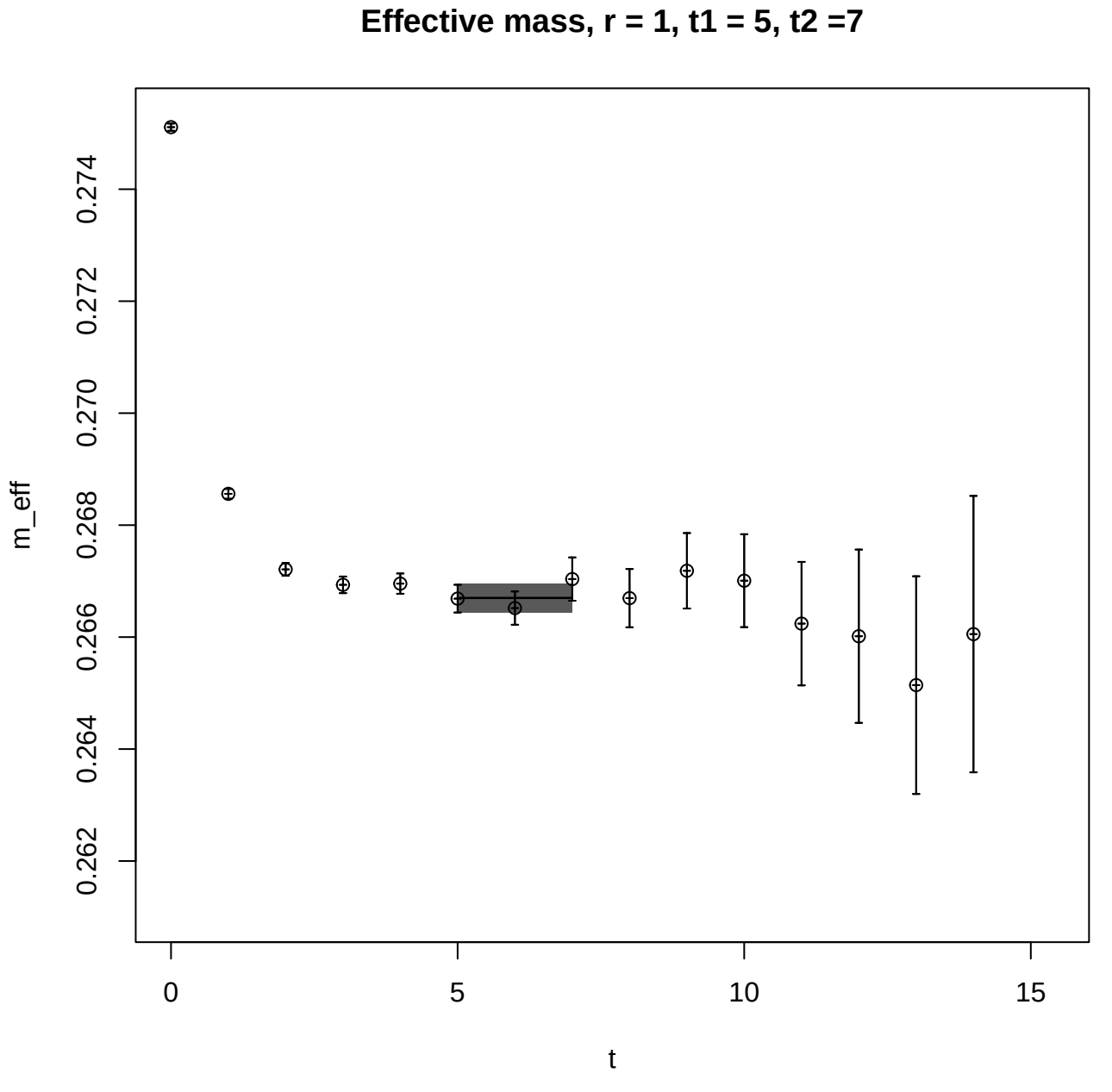
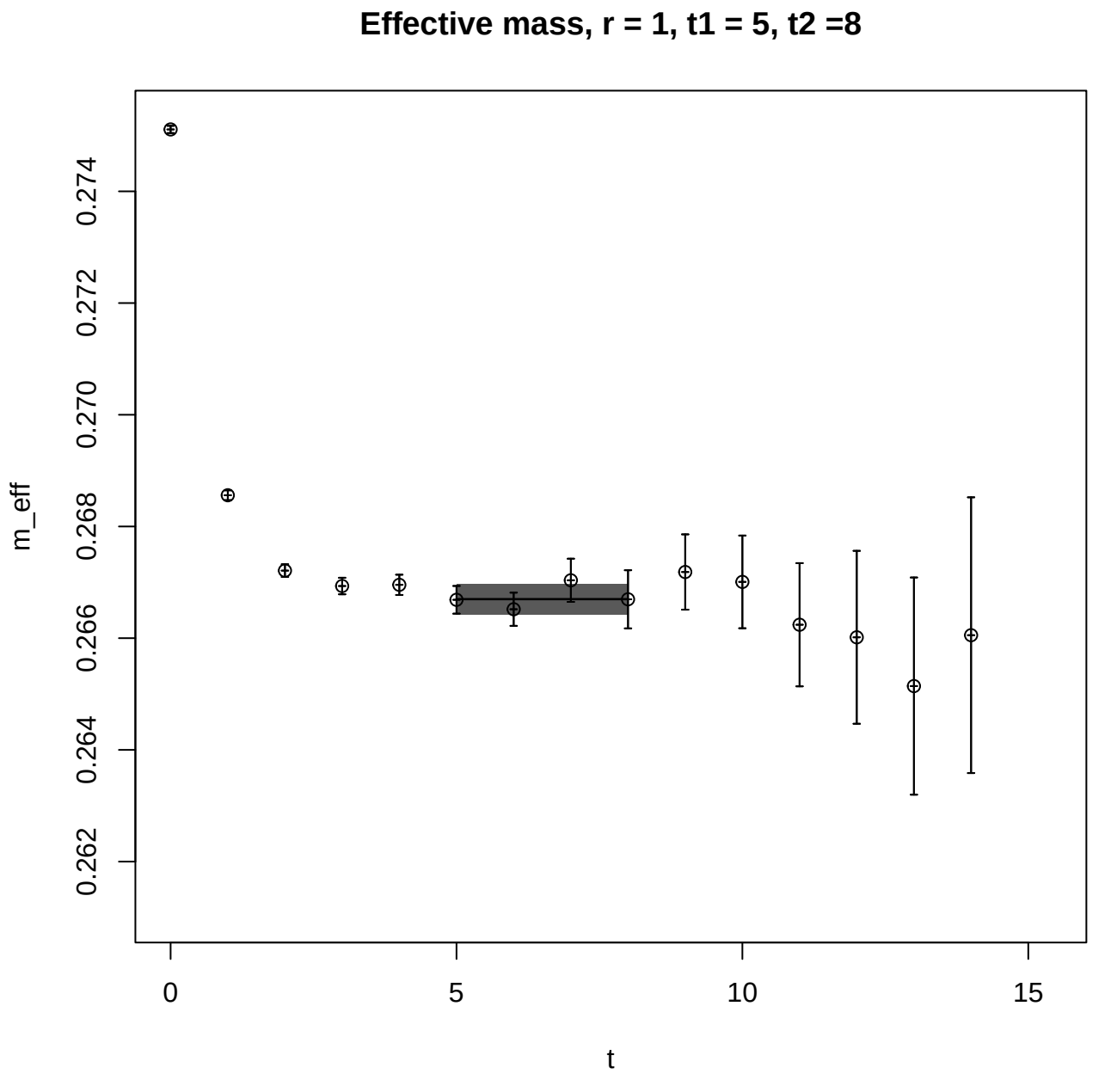
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

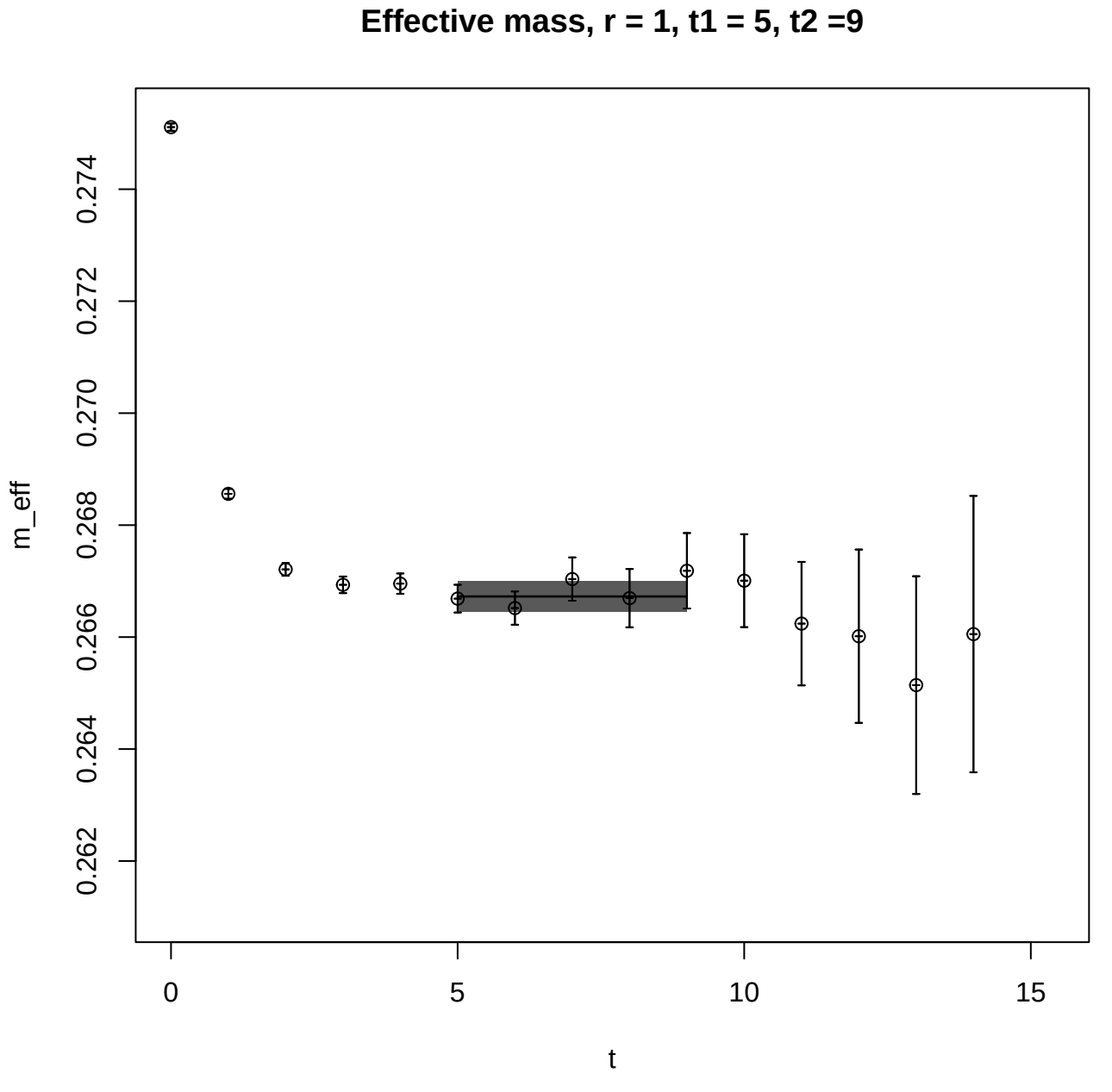

Effective mass, r = 1, t1 = 4, t2 =13
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =6
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =7
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =8
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =9
m_eff
0.262
0.264
0.266
0.268
0.270
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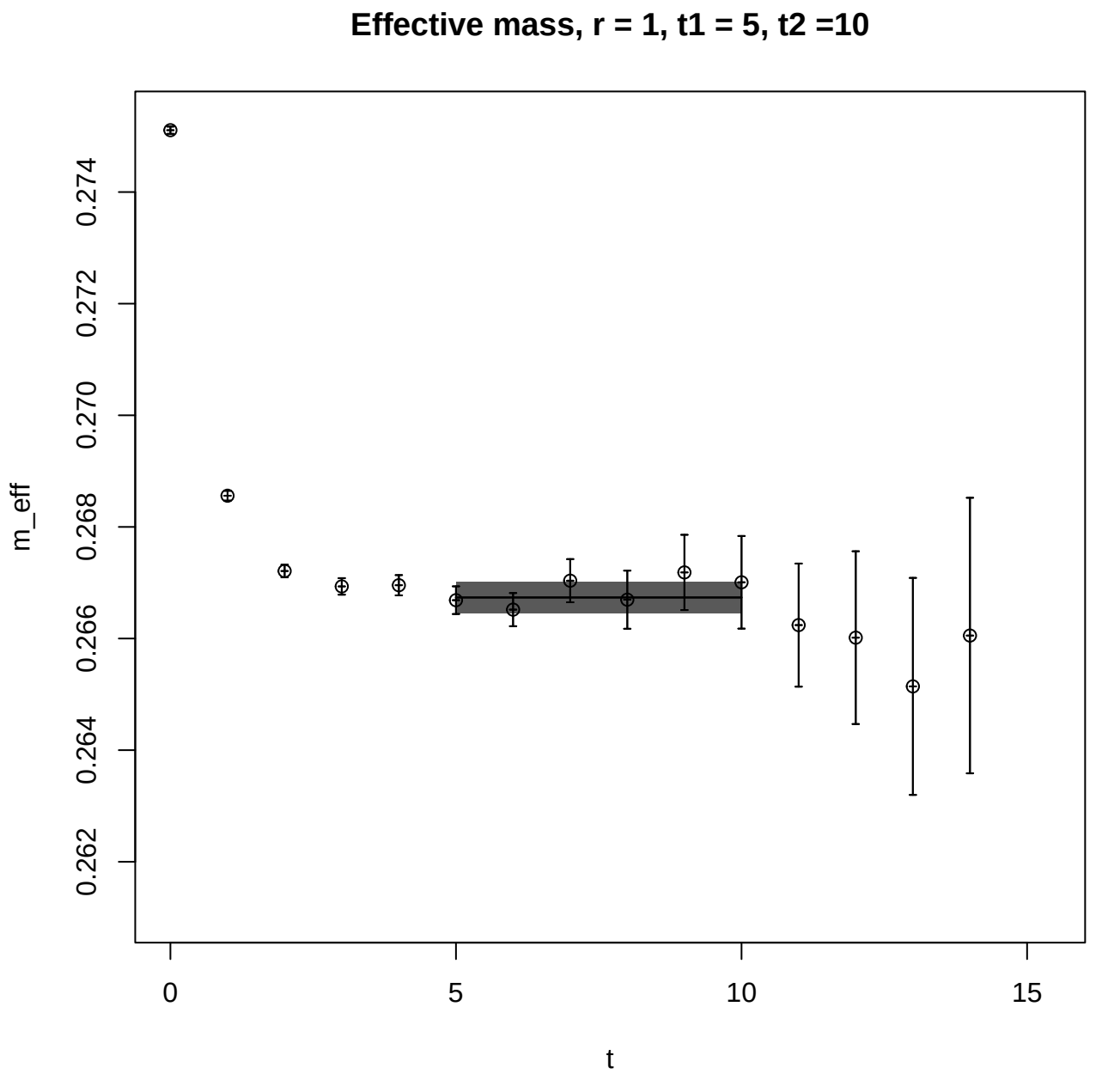
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =10
m_eff
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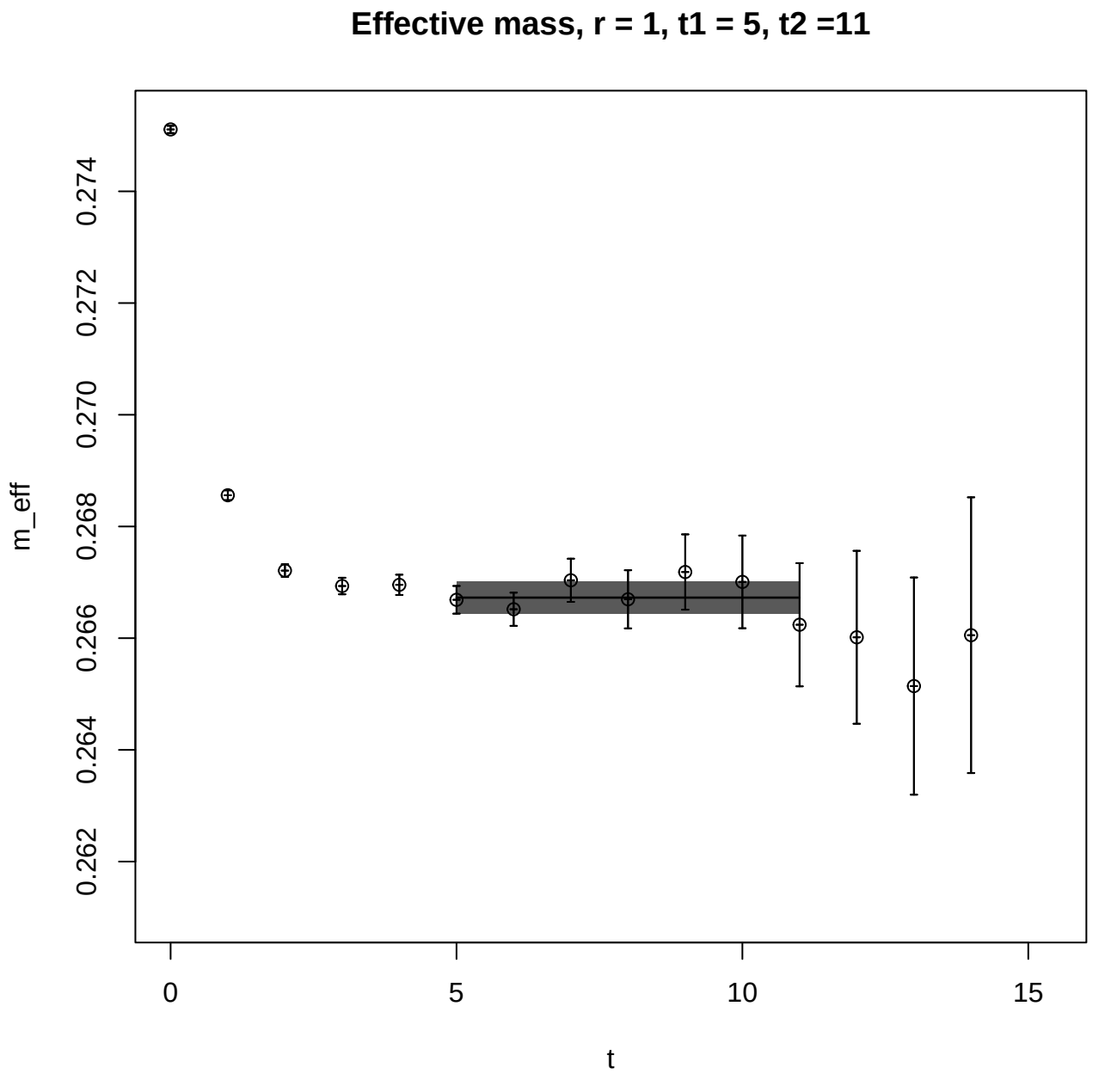
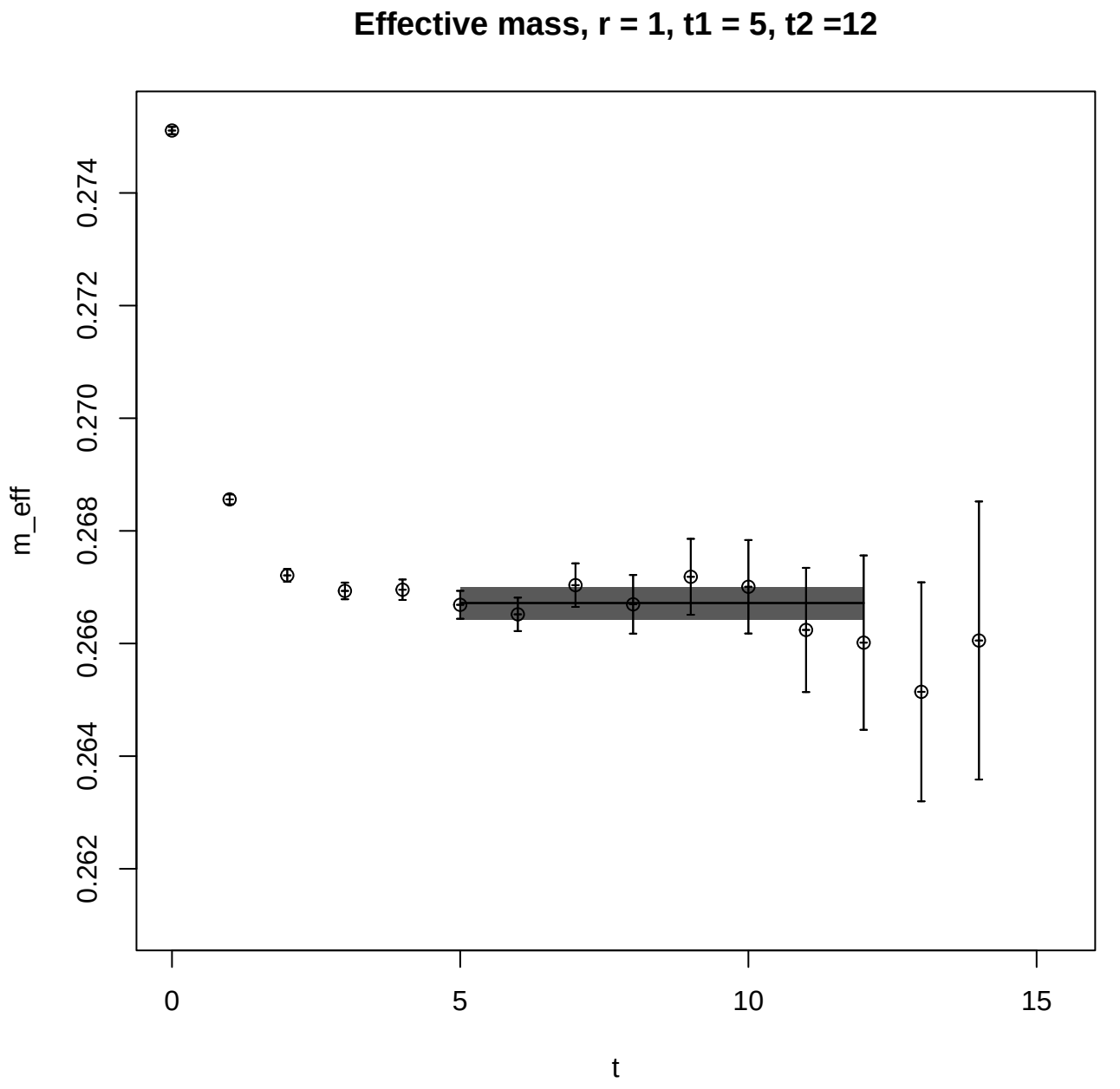
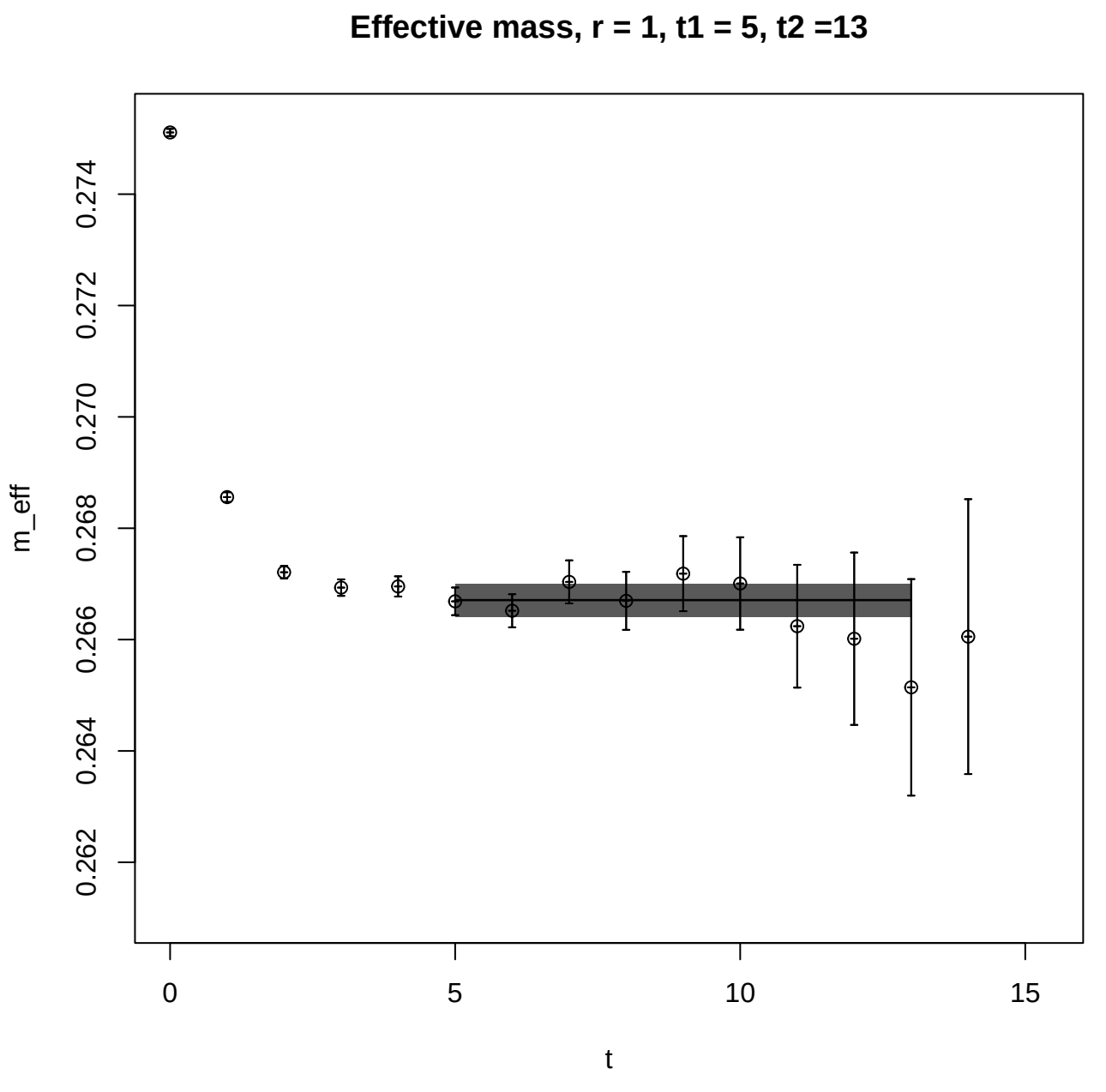
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =12
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =13
m_eff
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0.266
0.268
0.270
0.272
0.274
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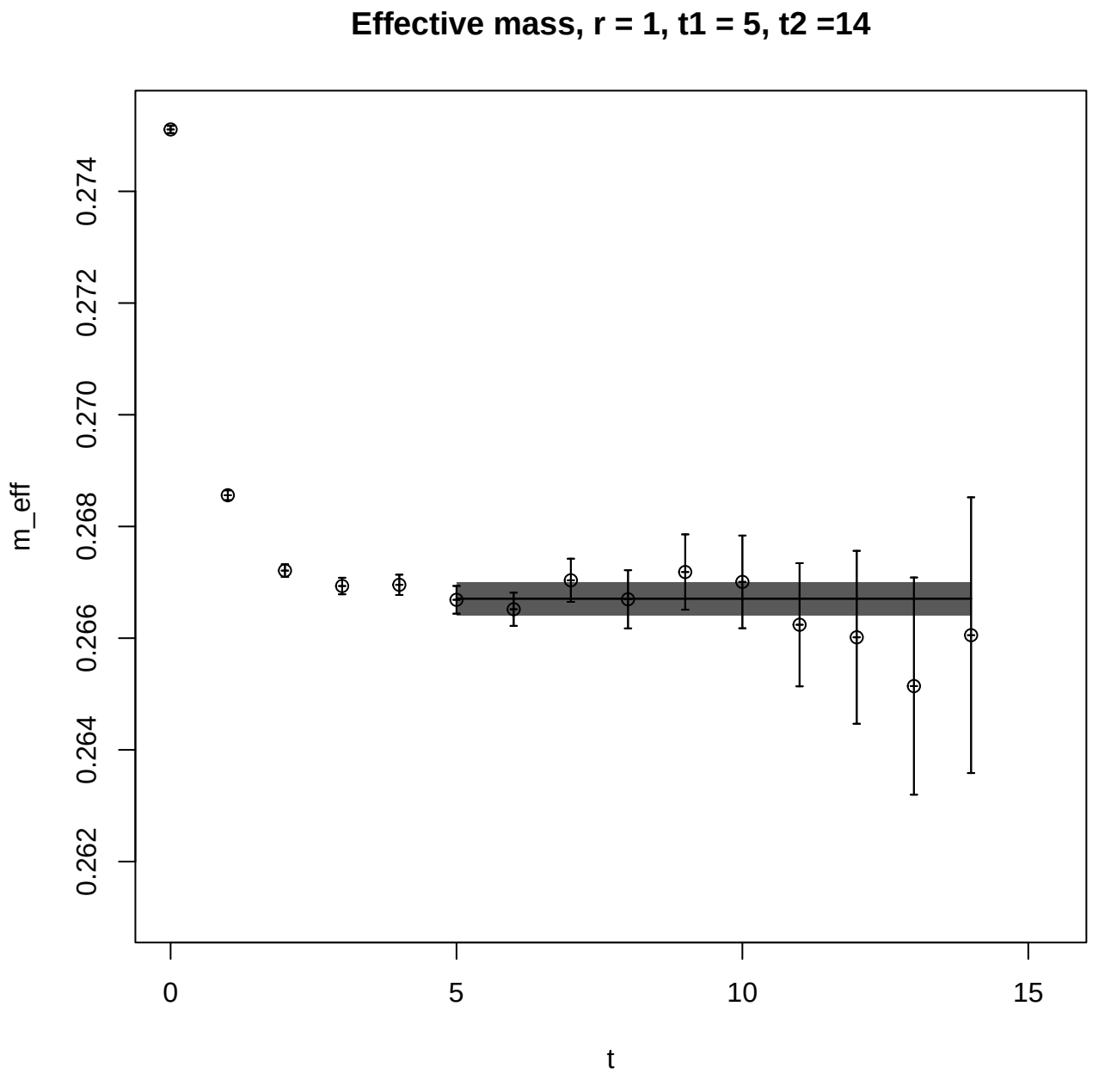
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =14
m_eff
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0.264
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0.268
0.270
0.272
0.274
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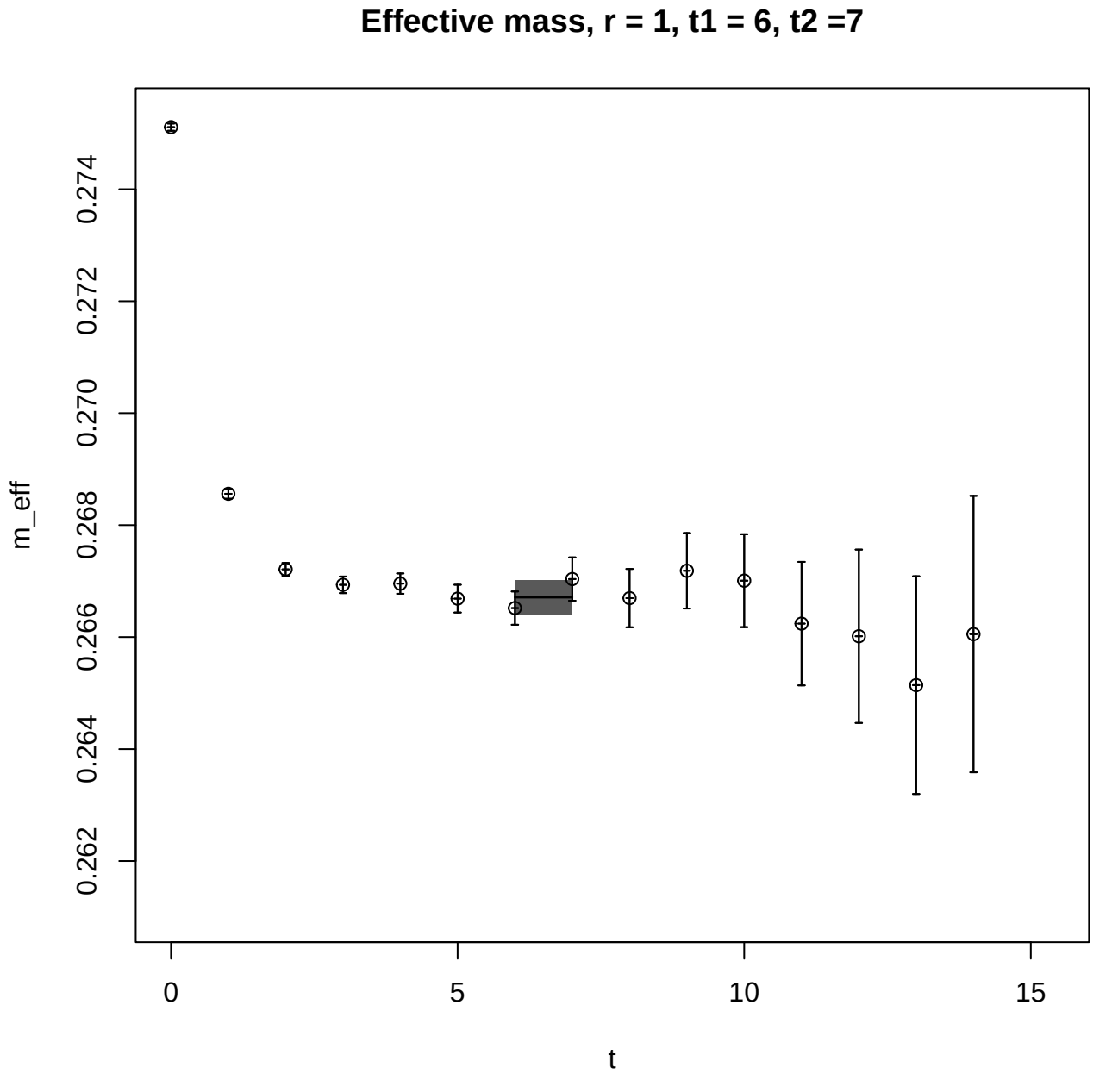
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =7
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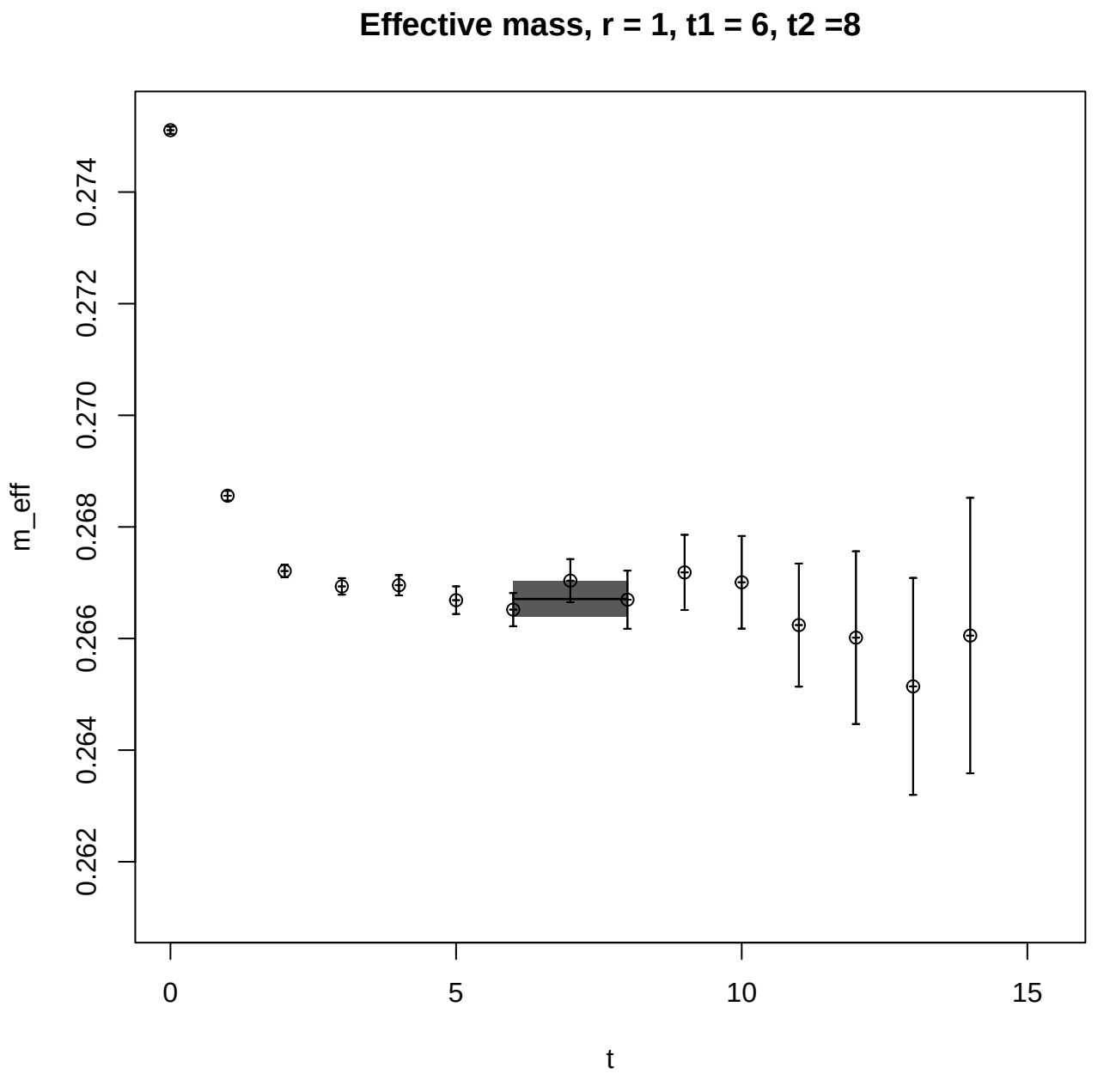
15
t


Effective mass, r = 1, t1 = 6, t2 =8
m_eff
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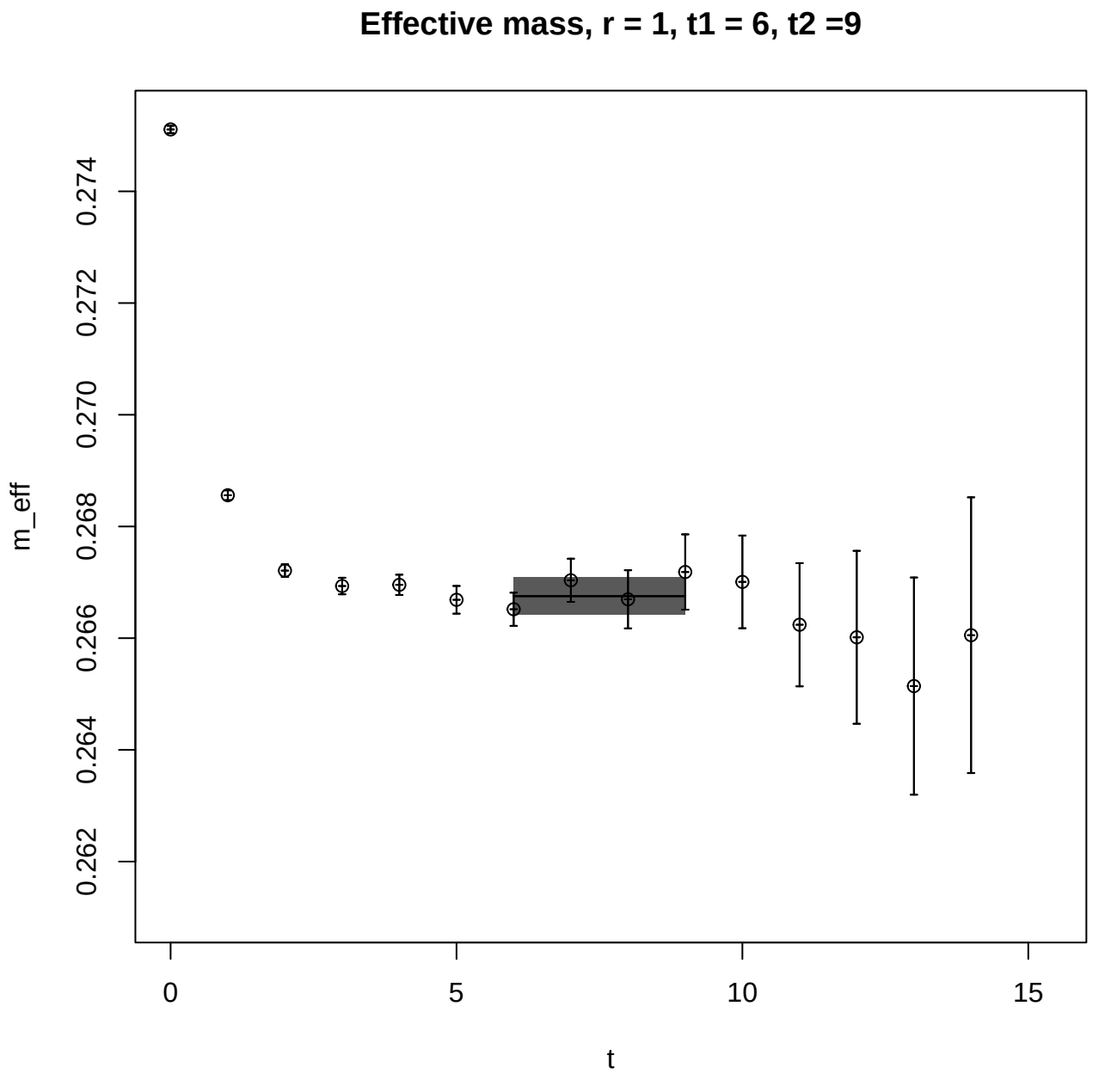
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =9
m_eff
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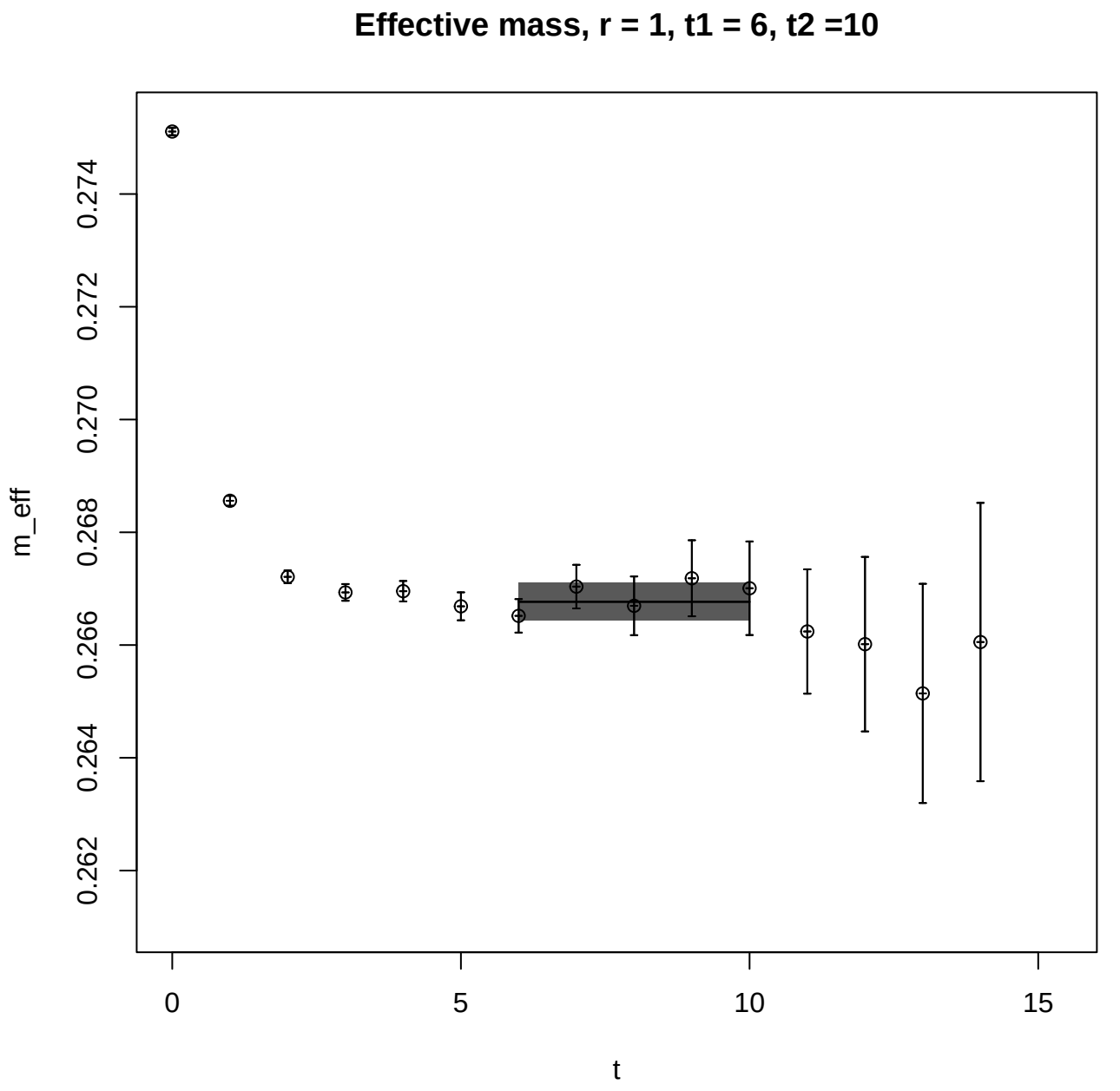
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =10
m_eff
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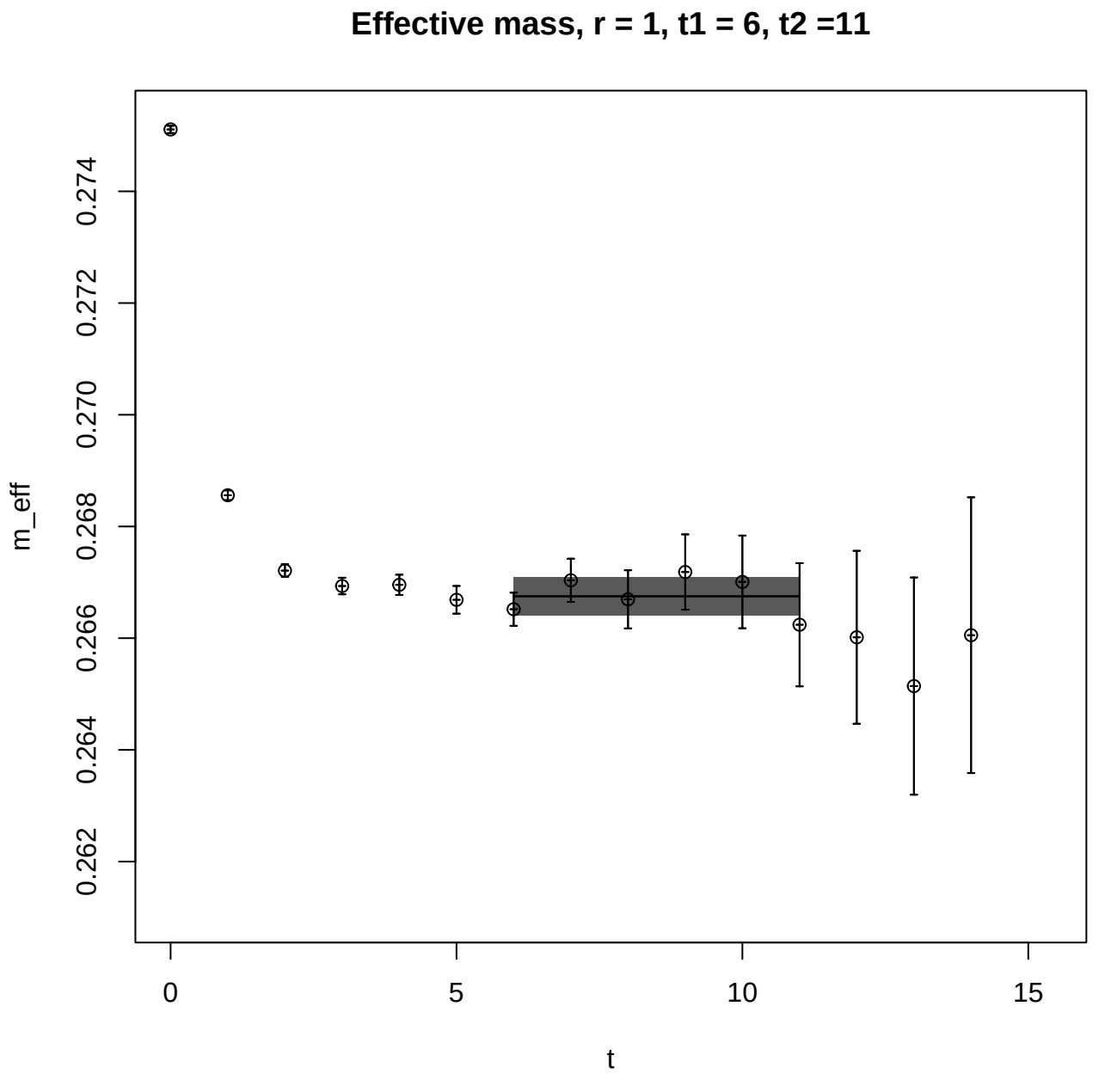
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =11
m_eff
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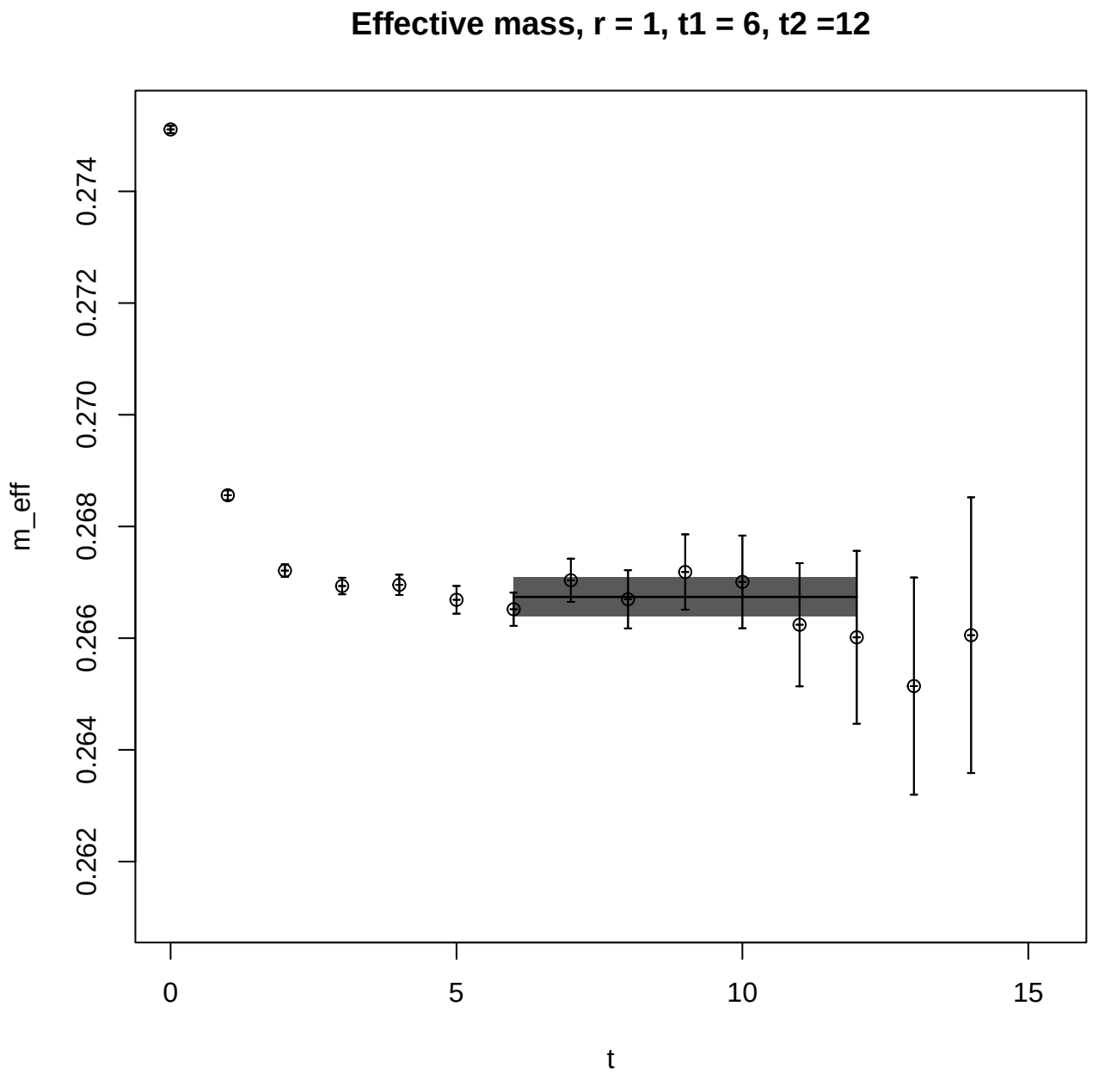
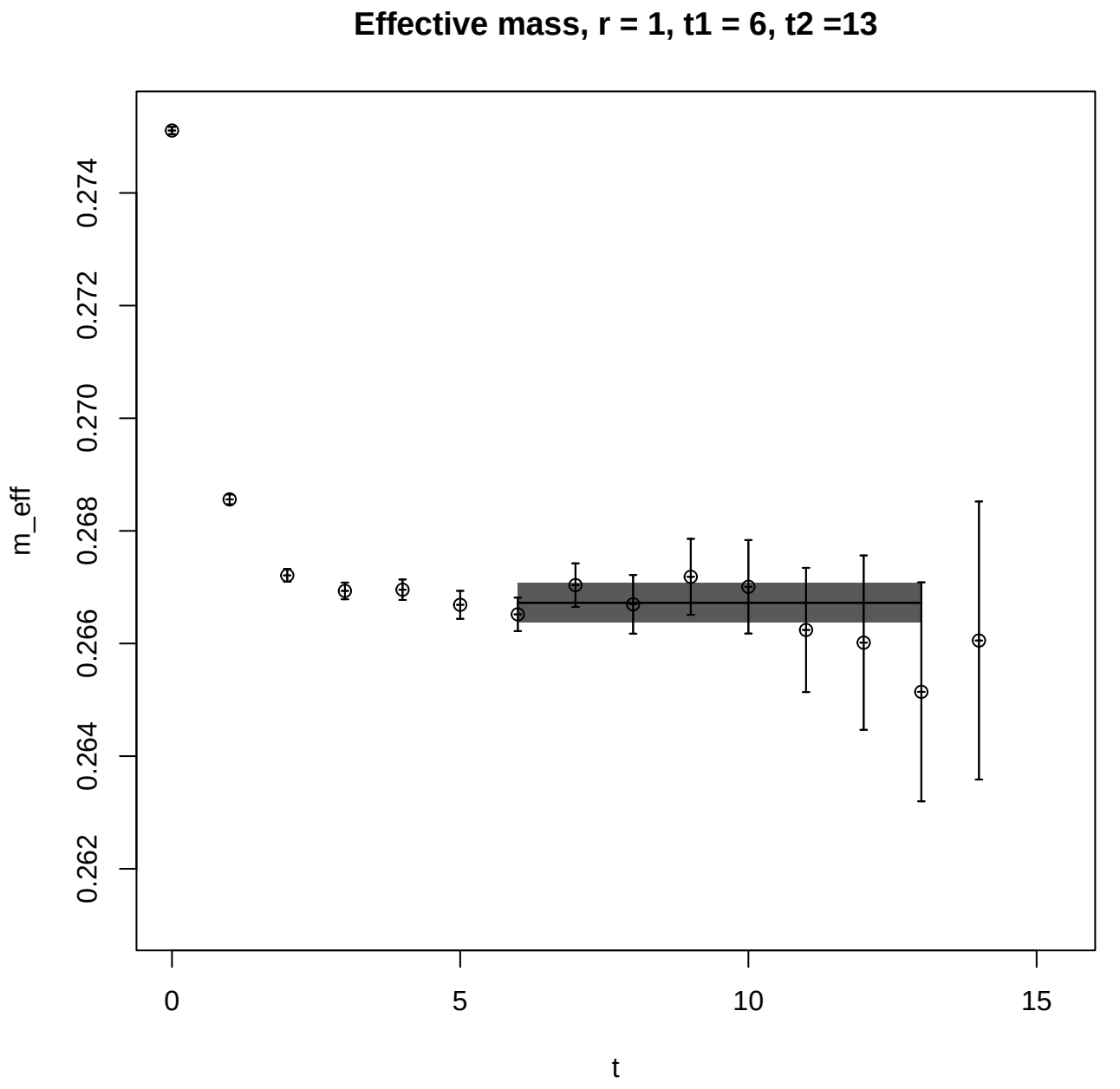
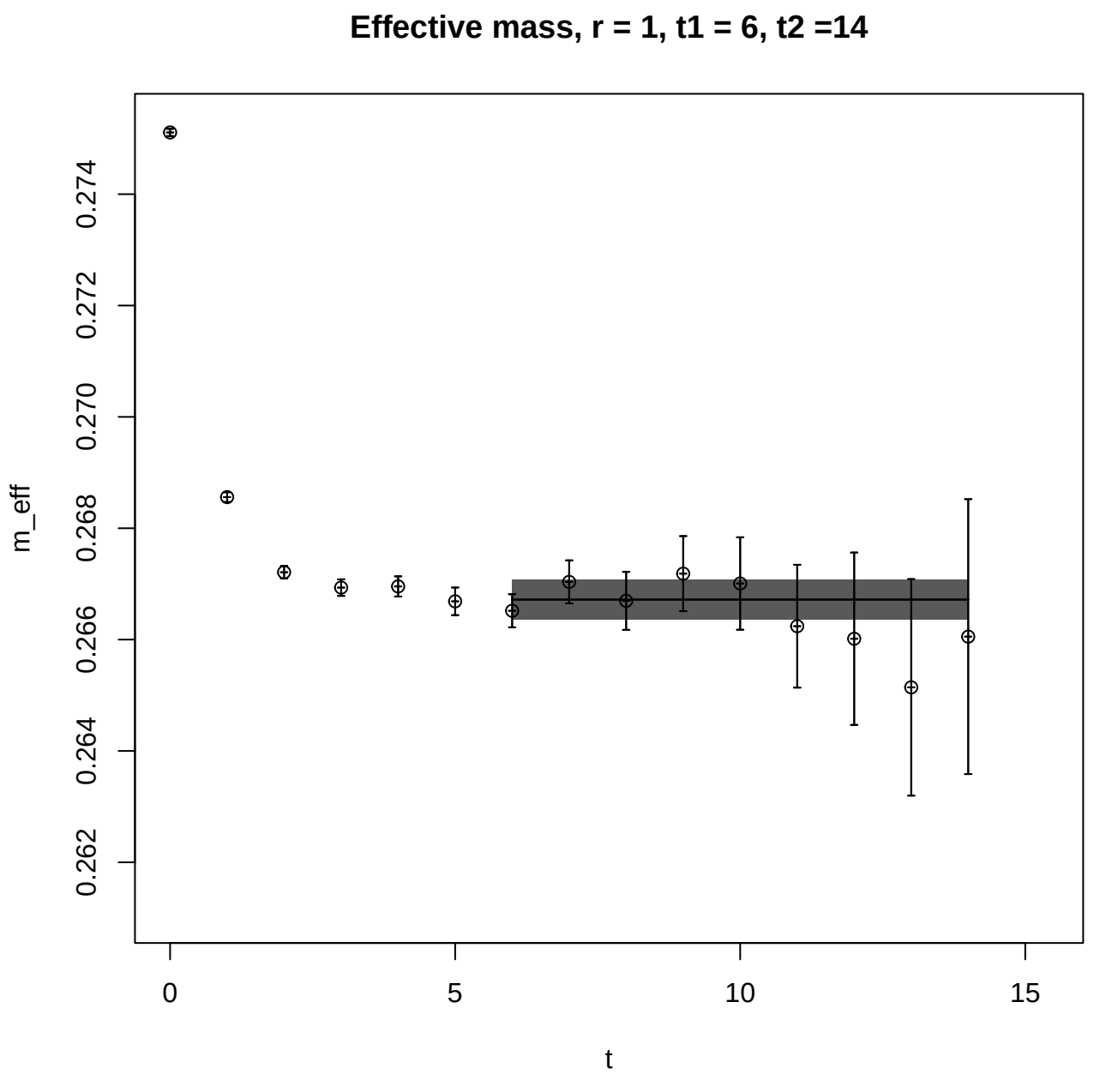
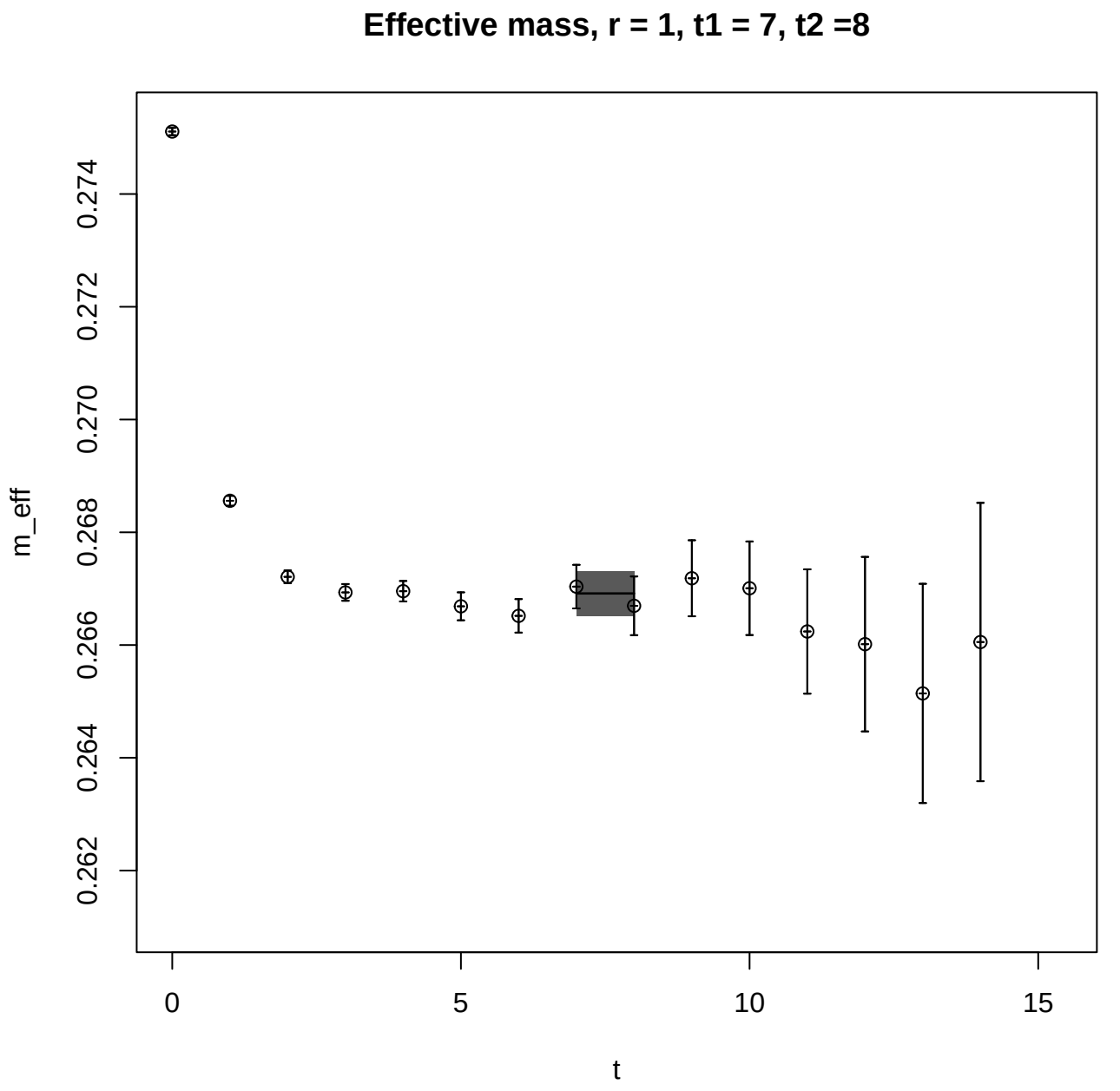
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =12
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =13
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 6, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =8
m_eff
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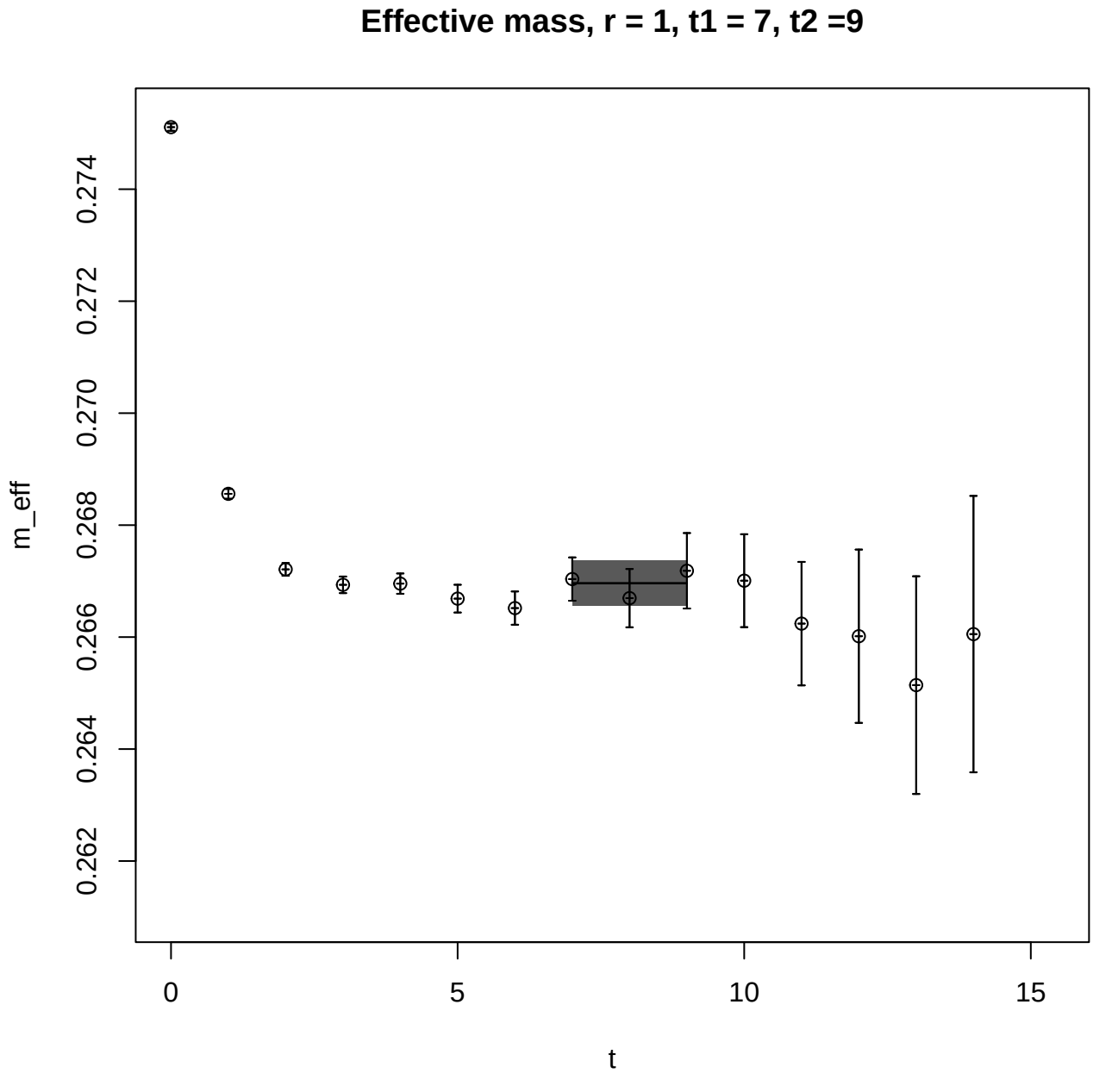
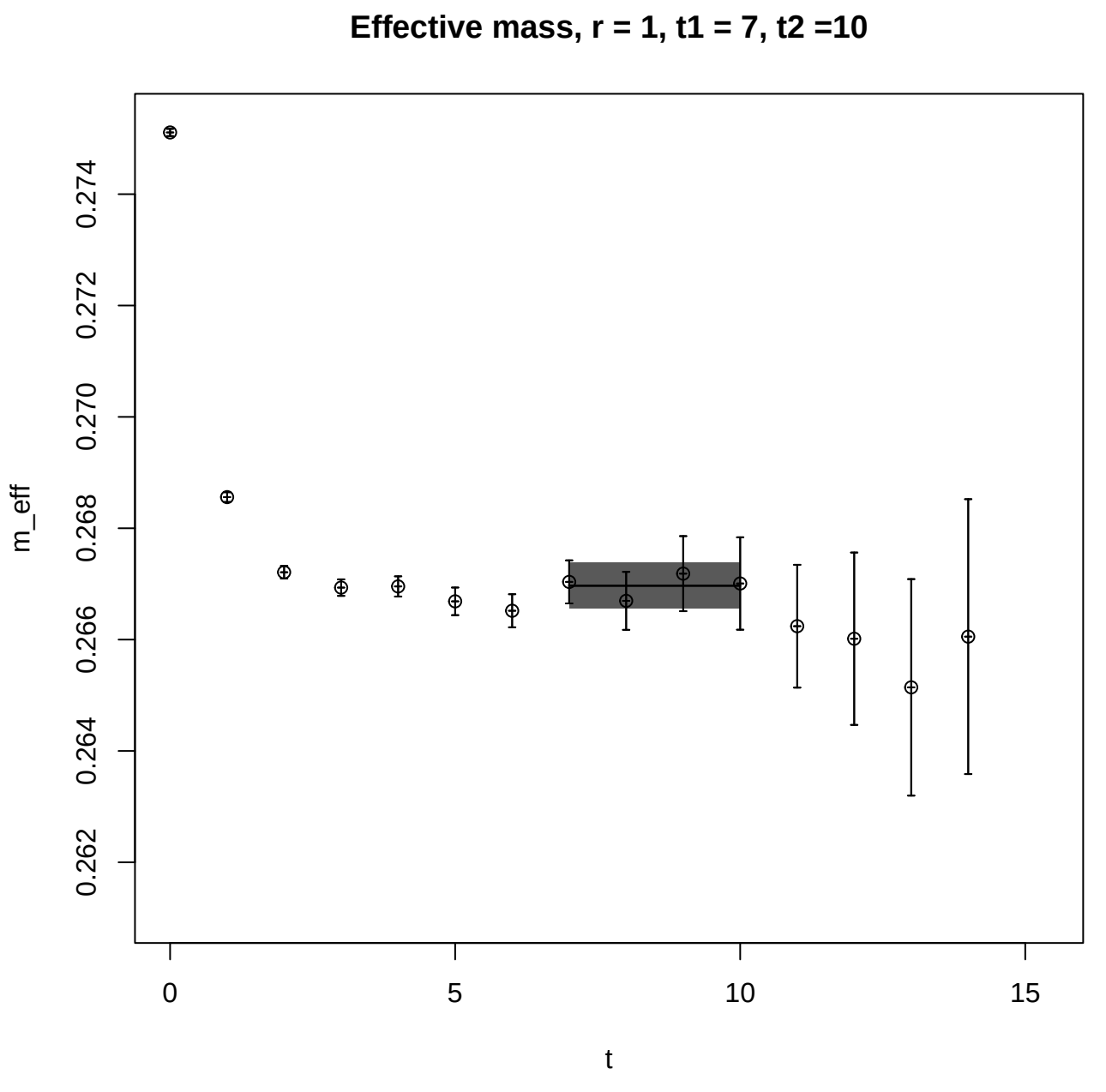
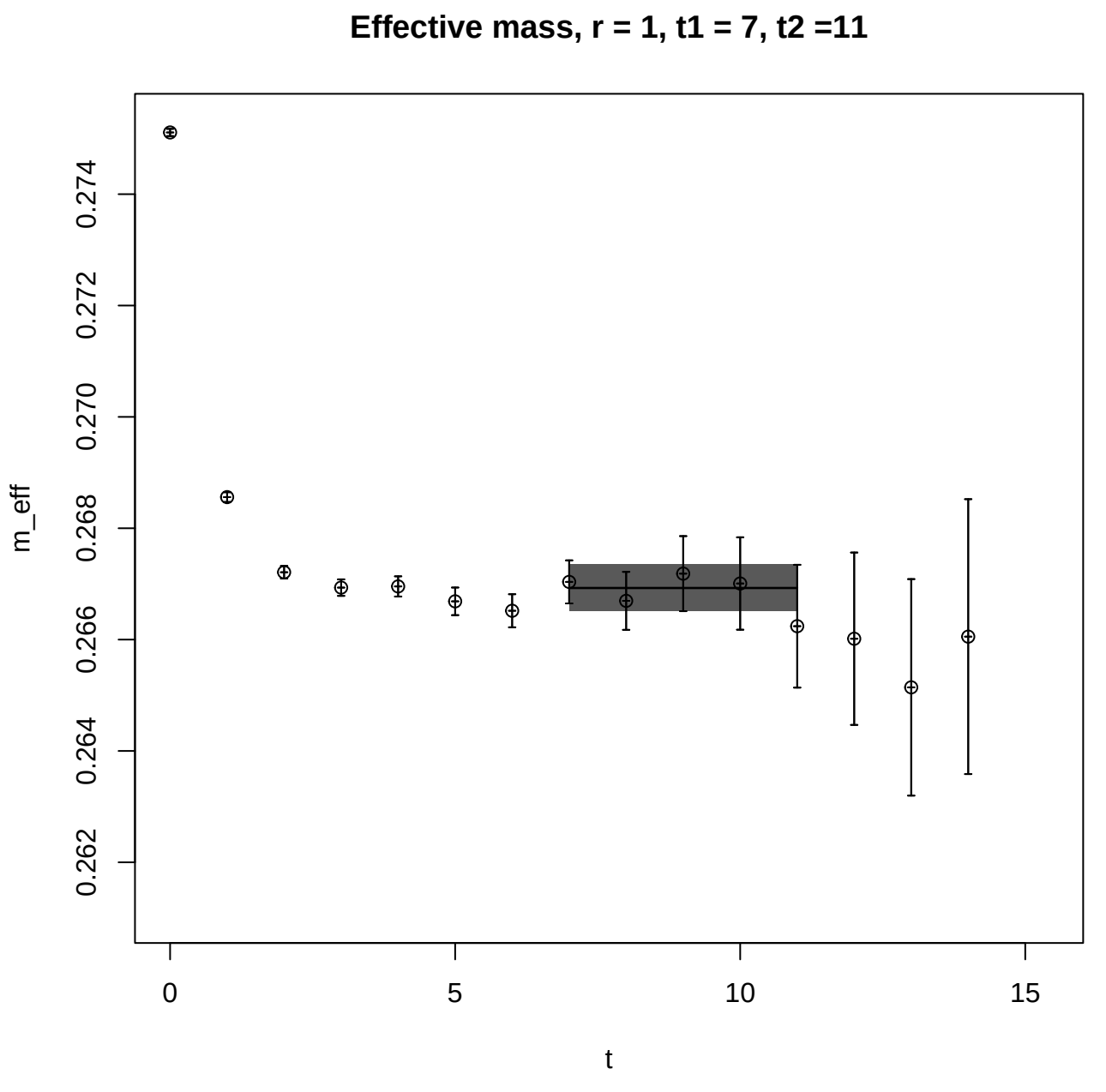
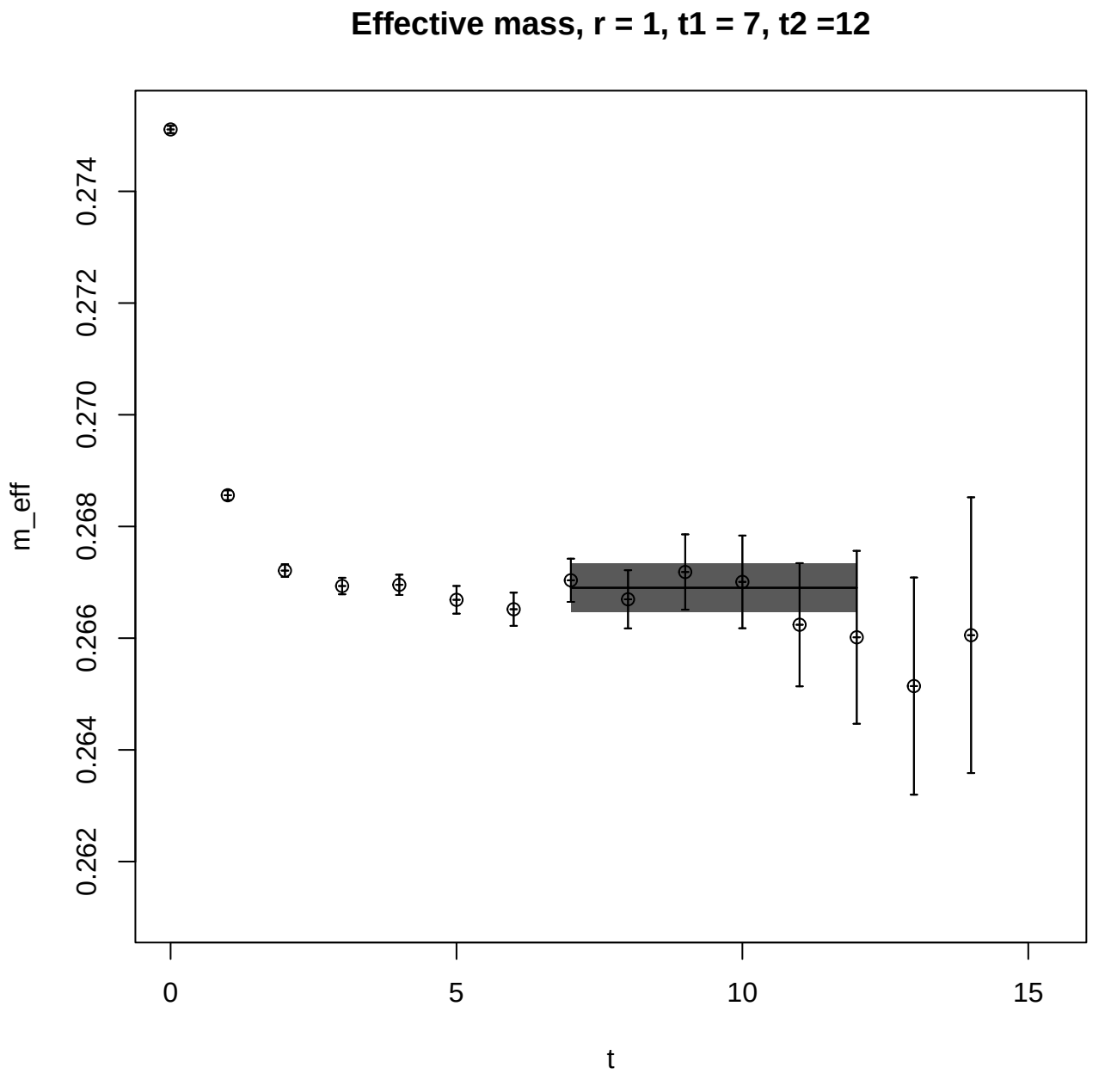
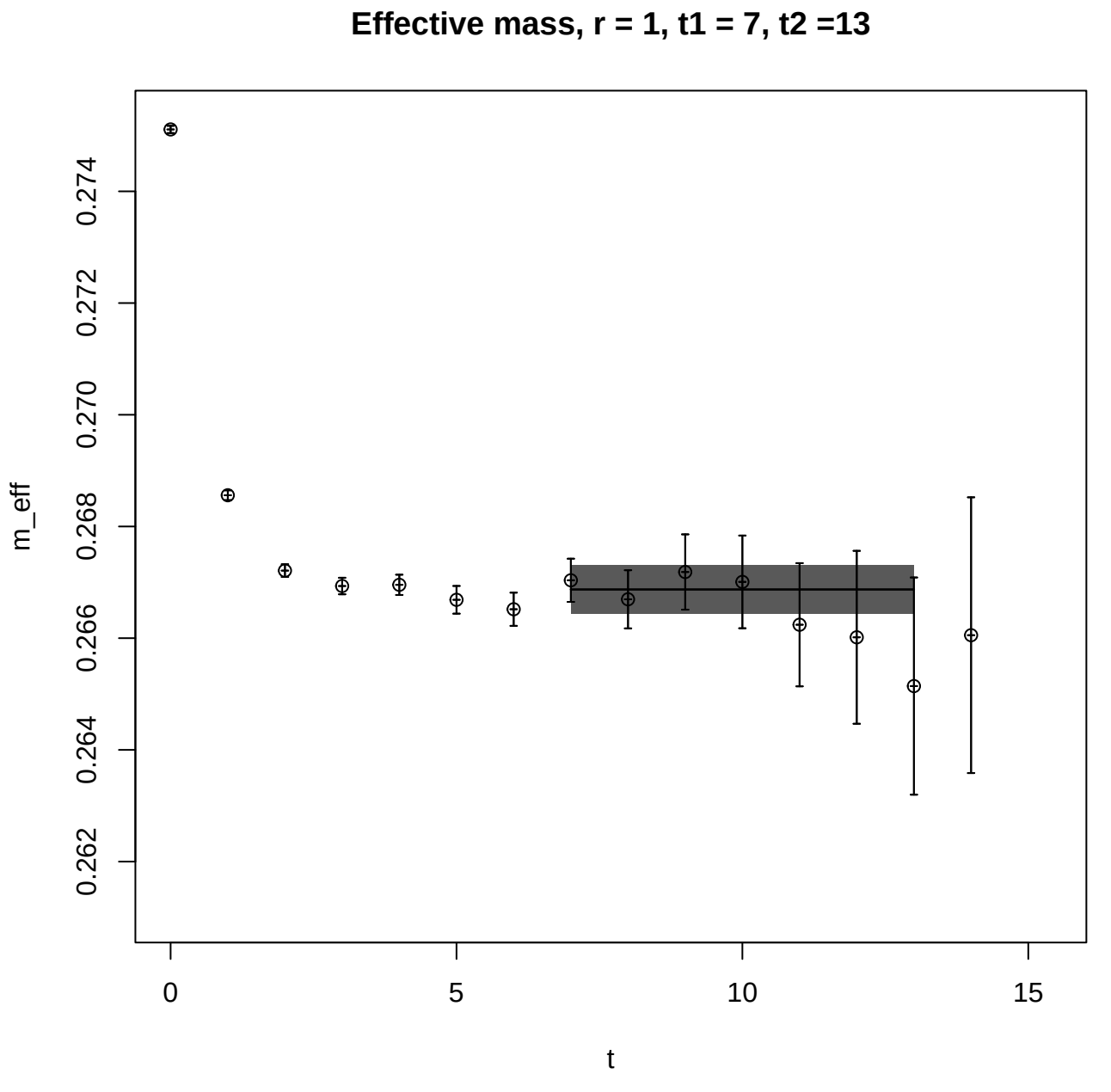
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =9
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =10
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =12
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =13
m_eff
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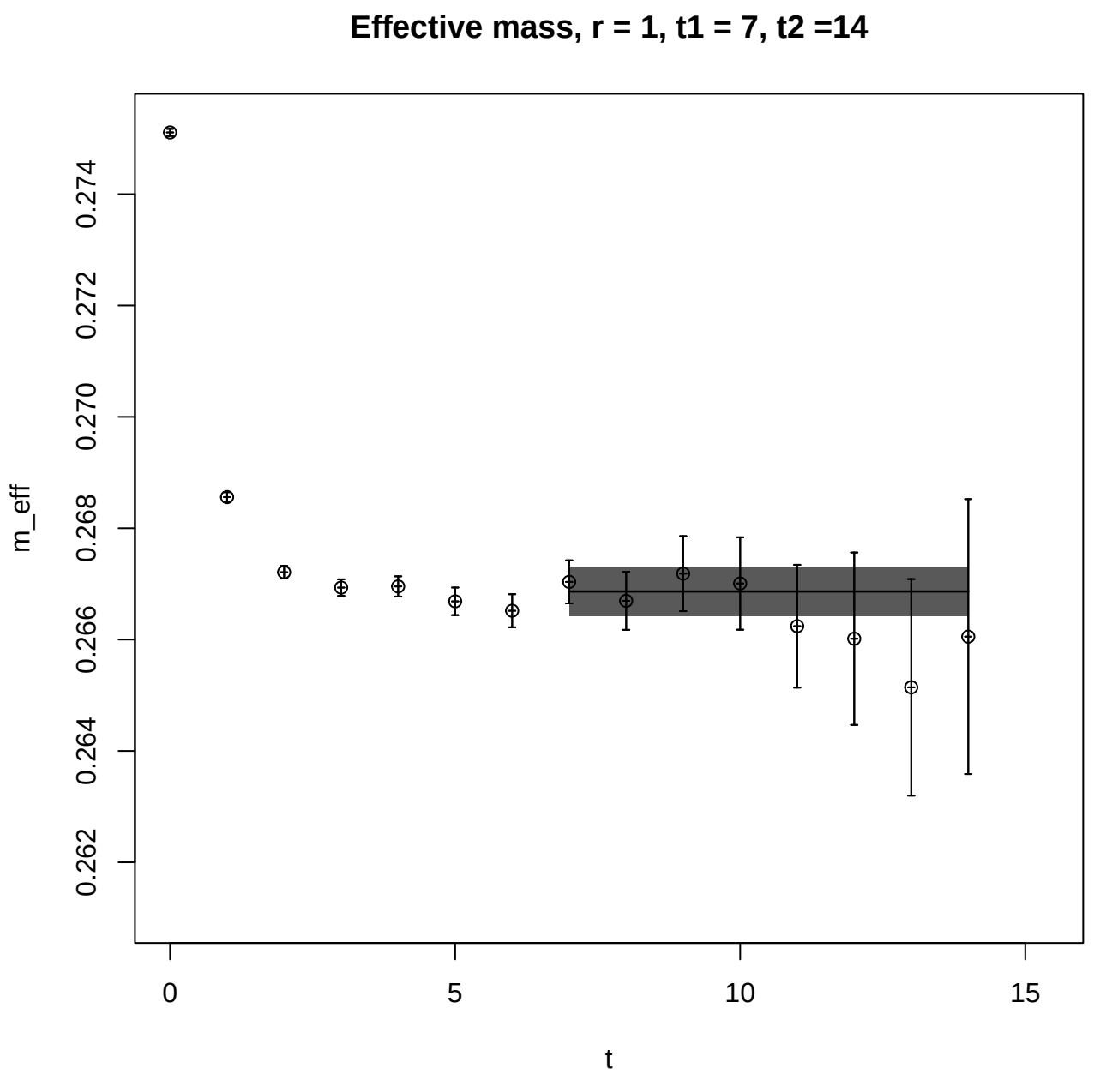
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =14
m_eff
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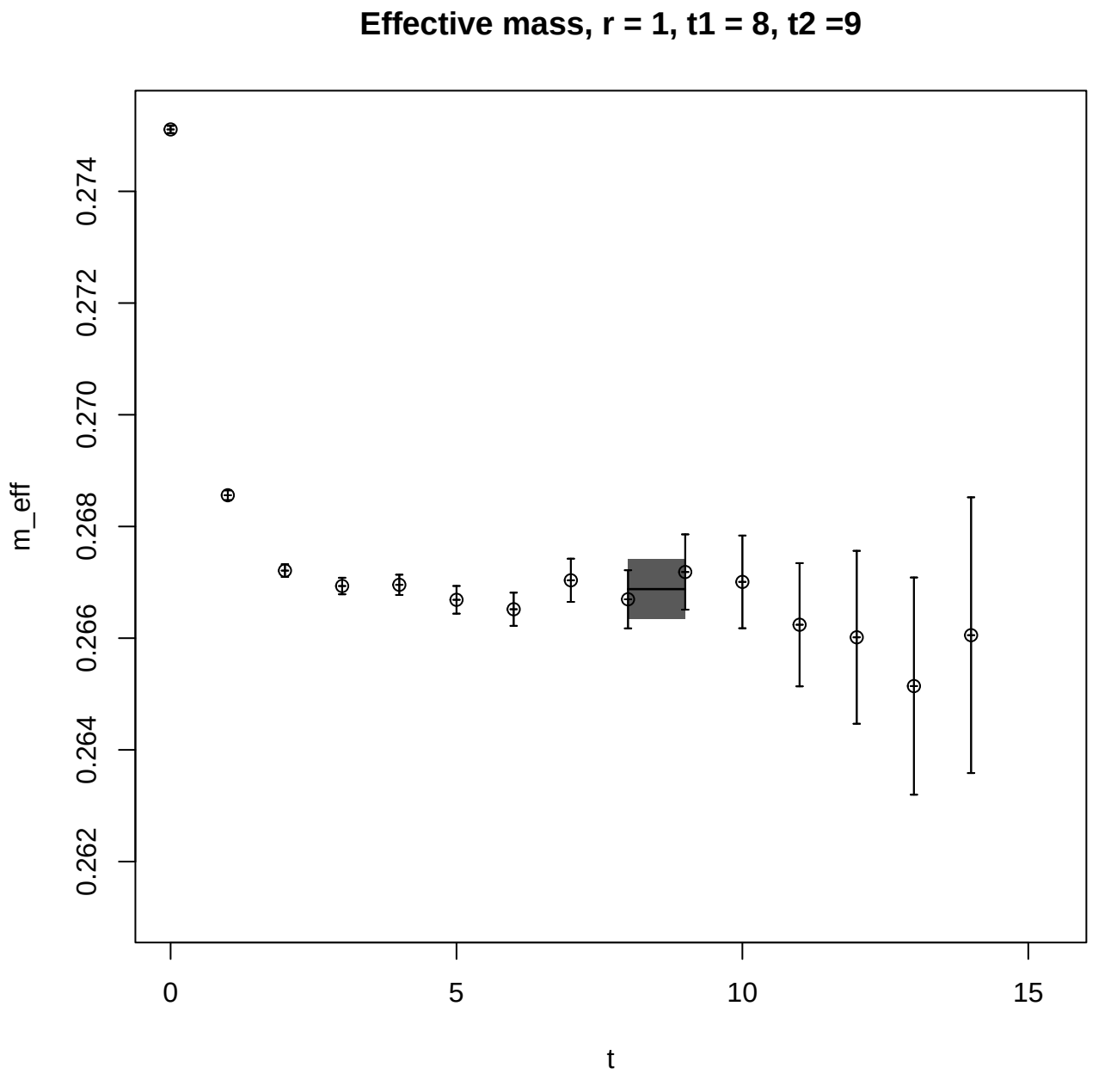
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =9
m_eff
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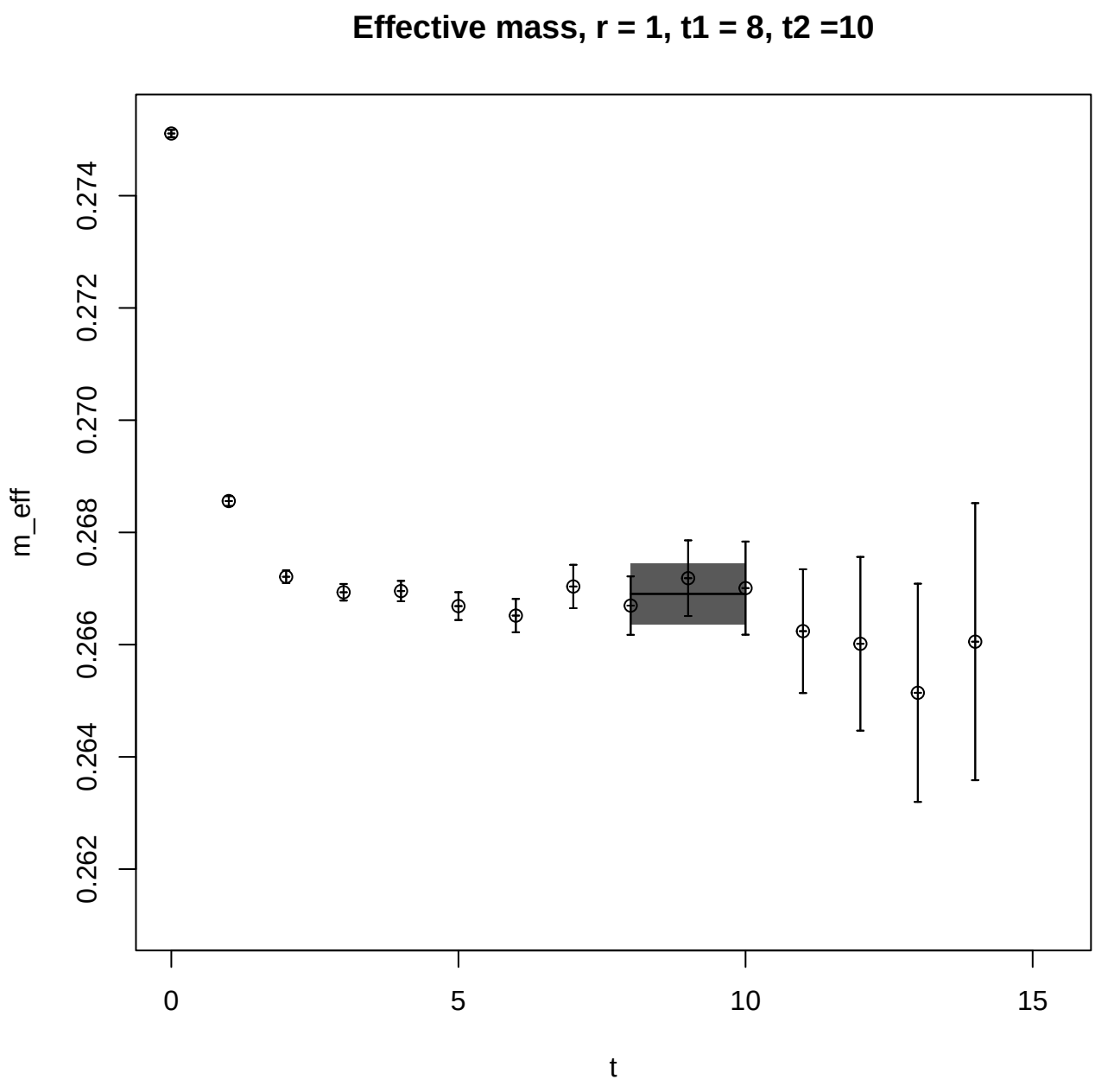
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =10
m_eff
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0.264
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0.268
0.270
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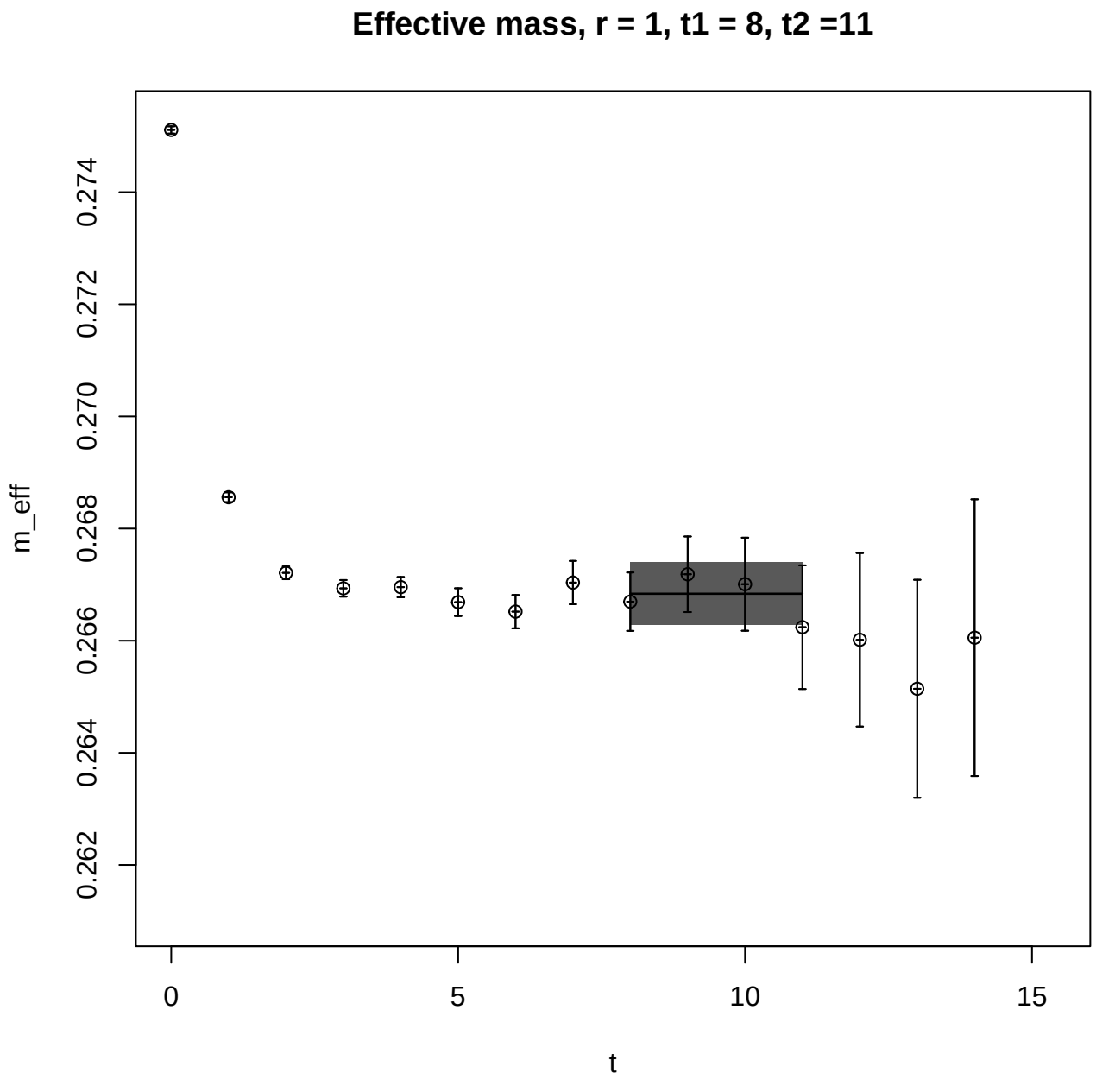
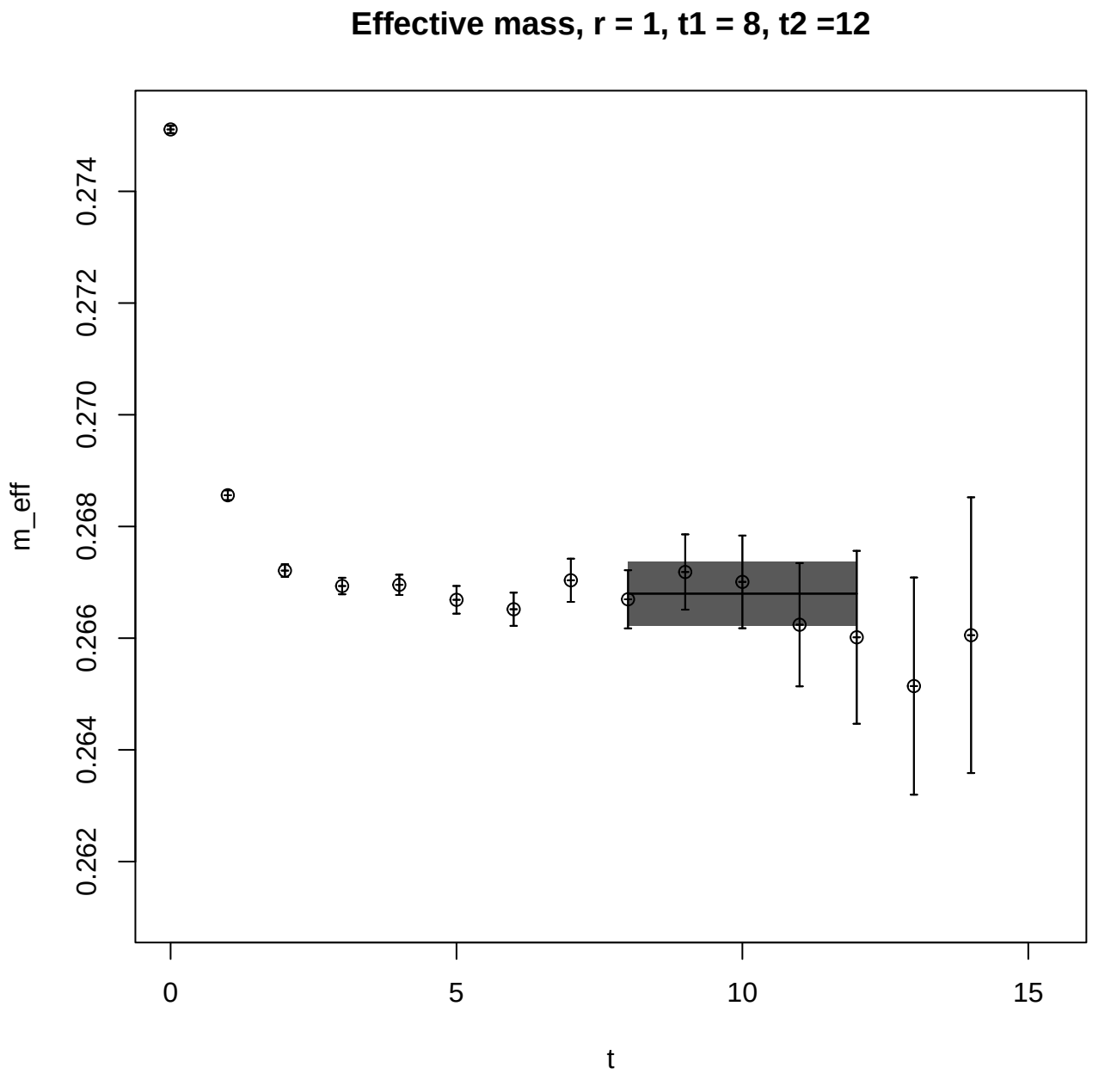
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =11
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =12
m_eff
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0.264
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0.268
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0.272
0.274
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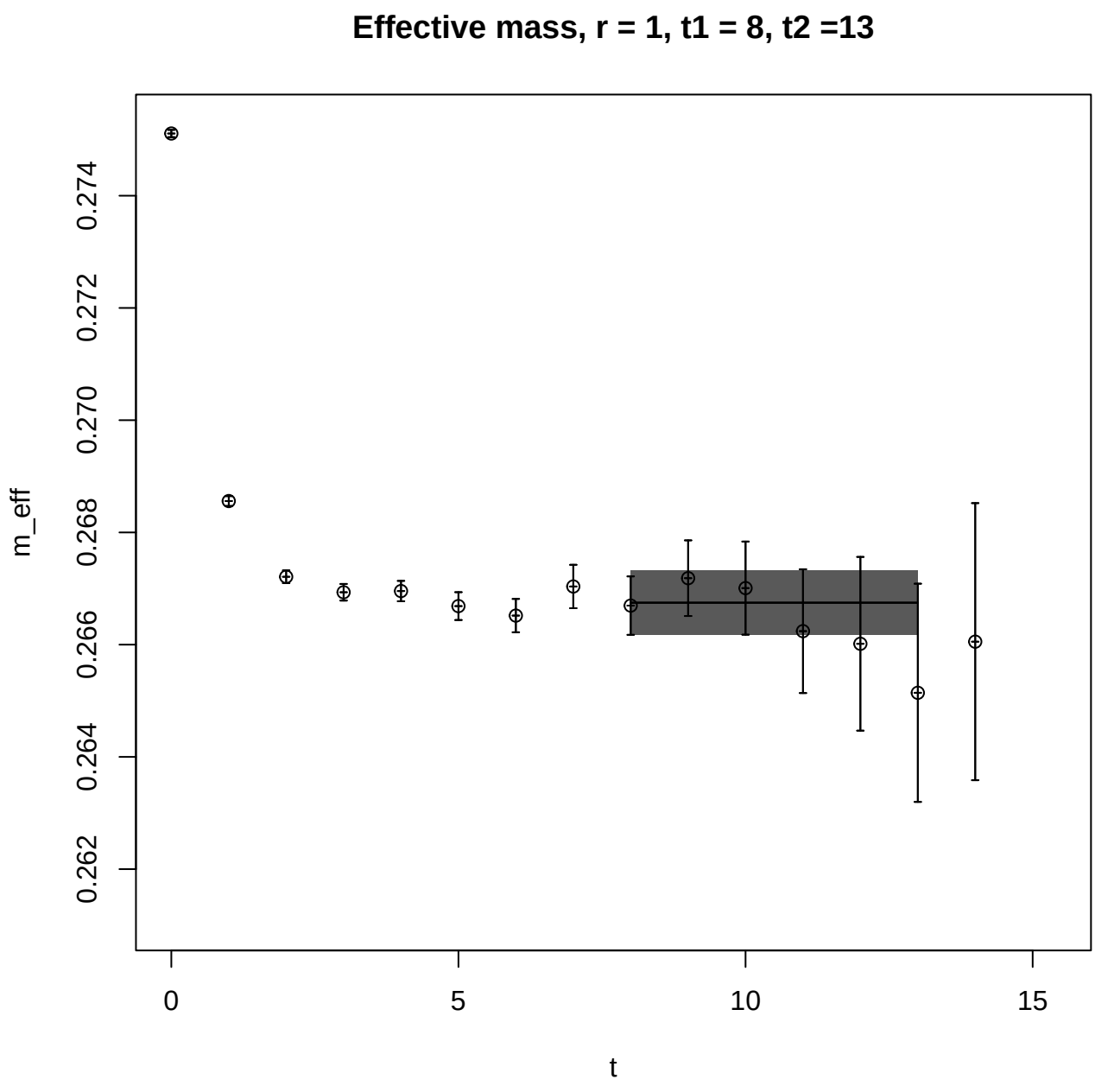
t


Effective mass, r = 1, t1 = 8, t2 =13
m_eff
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0.264
0.266
0.268
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0.272
0.274
0
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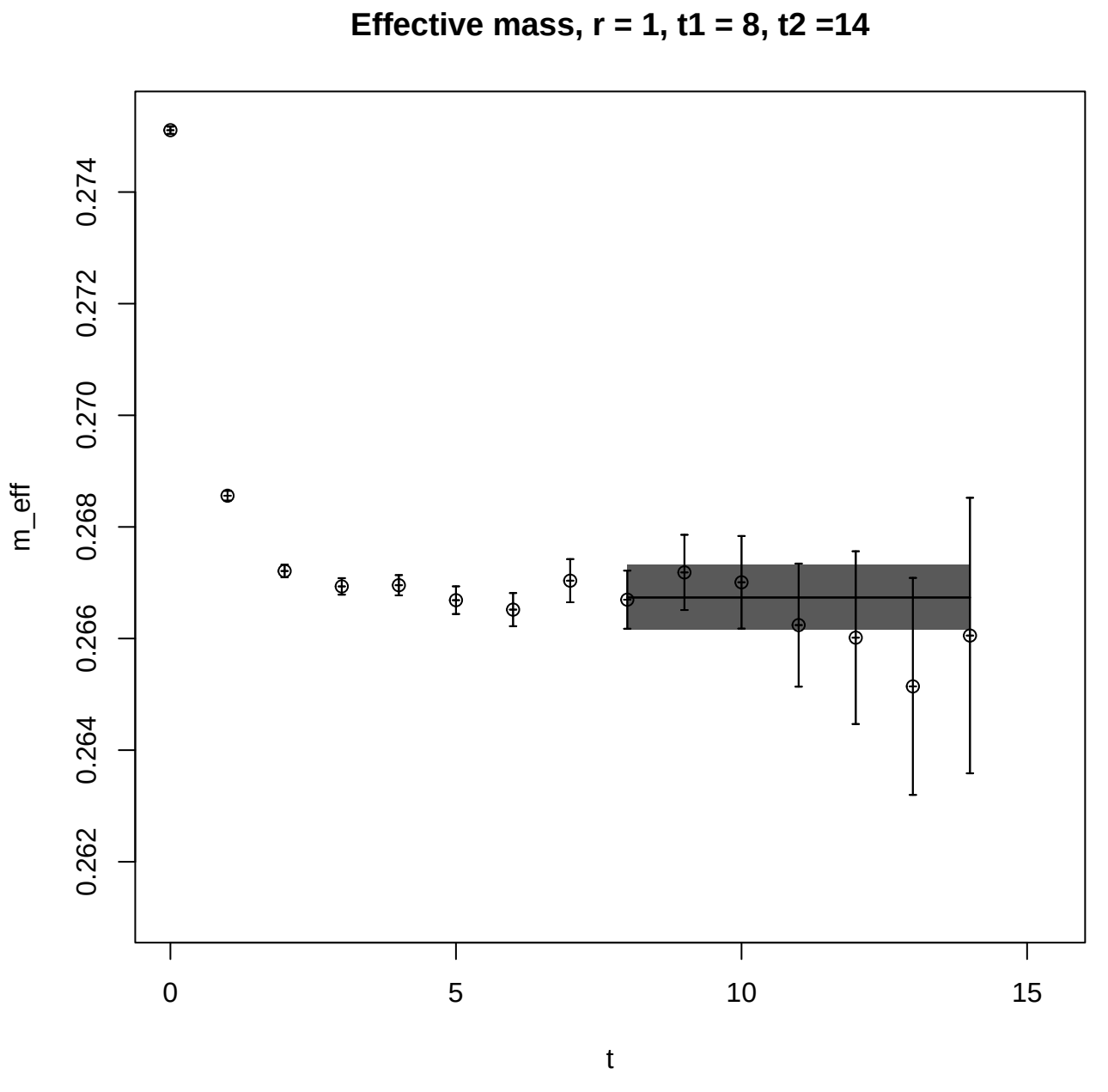
15
t


Effective mass, r = 1, t1 = 8, t2 =14
m_eff
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0.274
0
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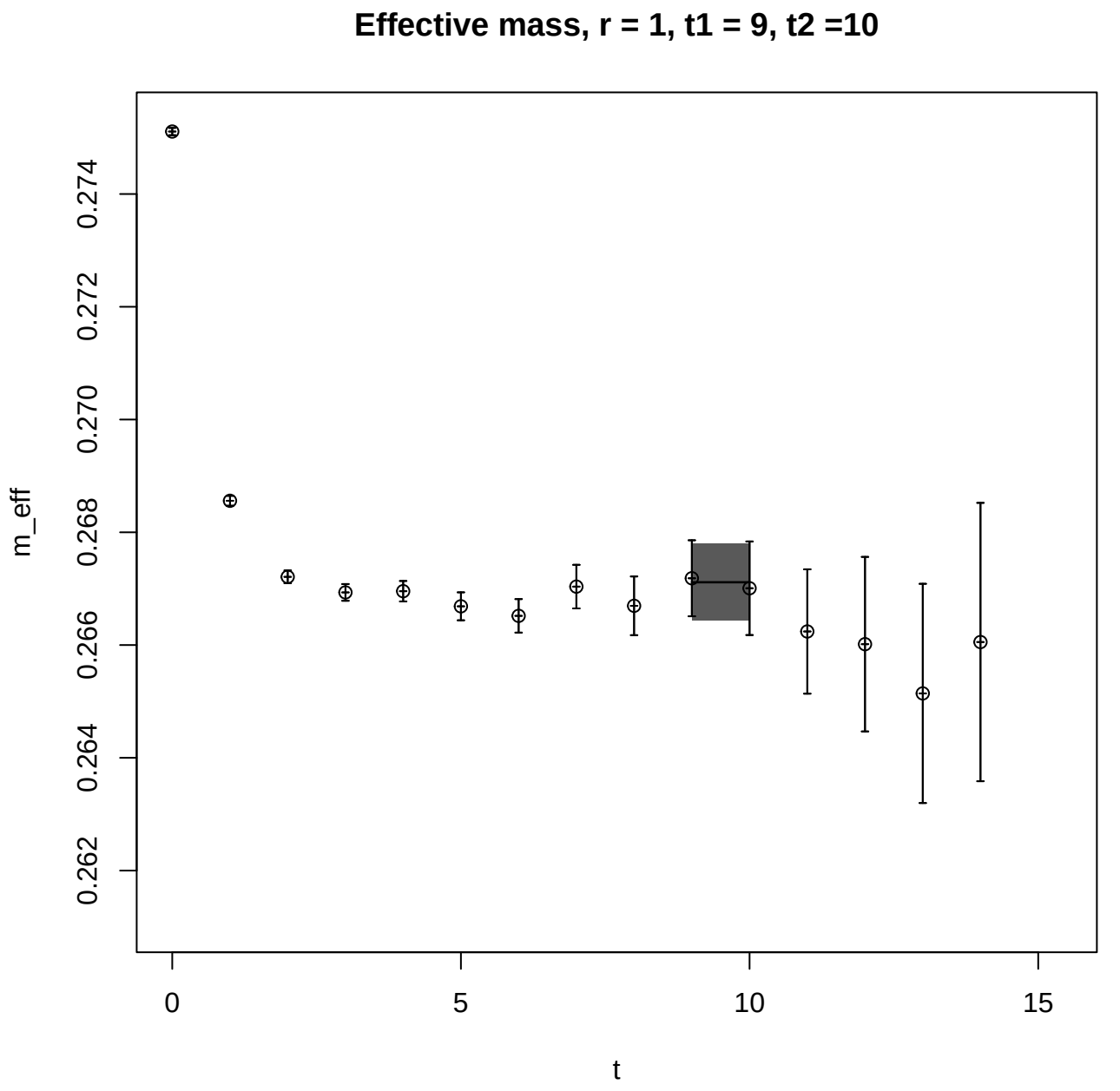
10
15
t


Effective mass, r = 1, t1 = 9, t2 =10
m_eff
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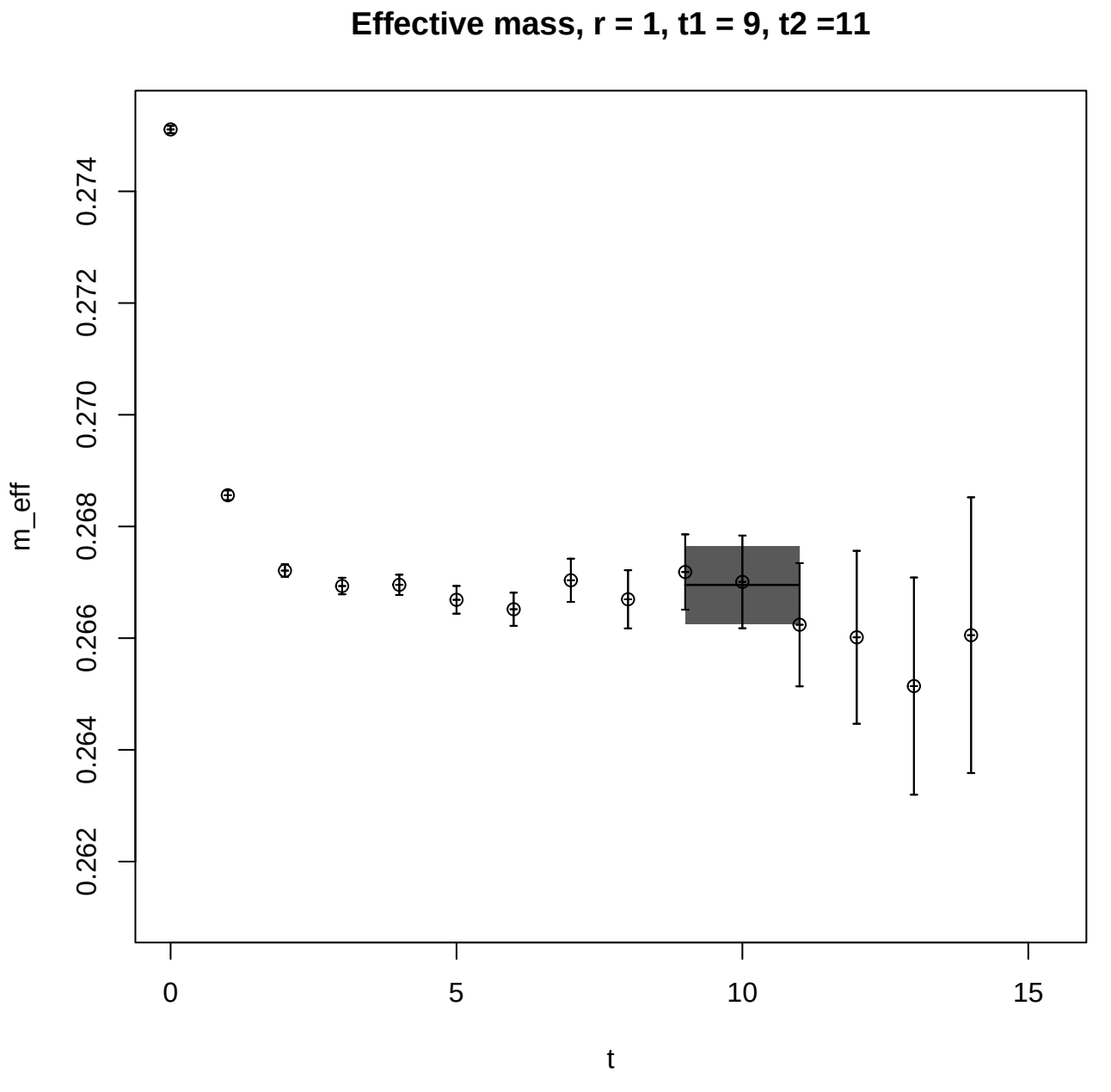
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =11
m_eff
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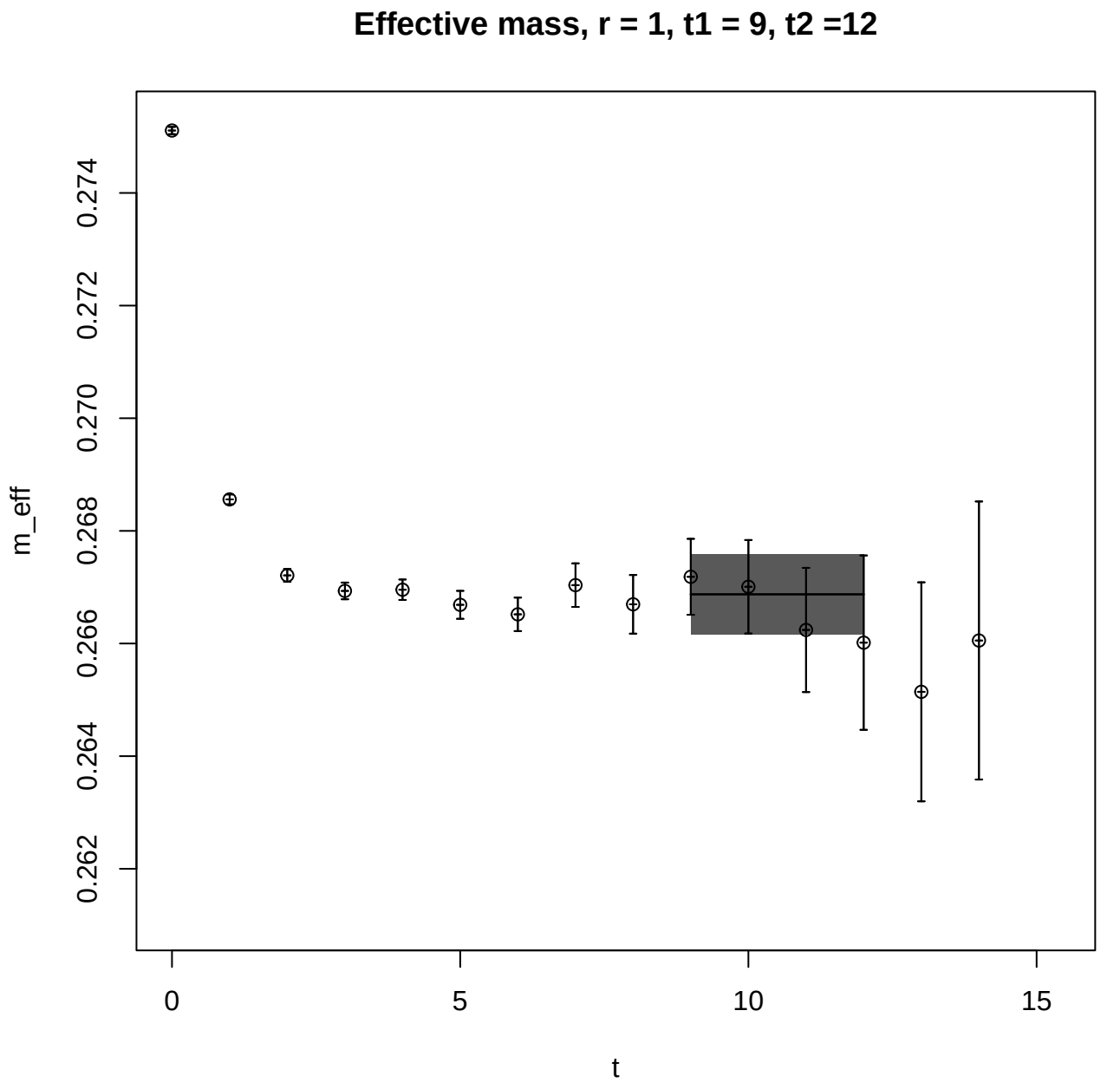
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =12
m_eff
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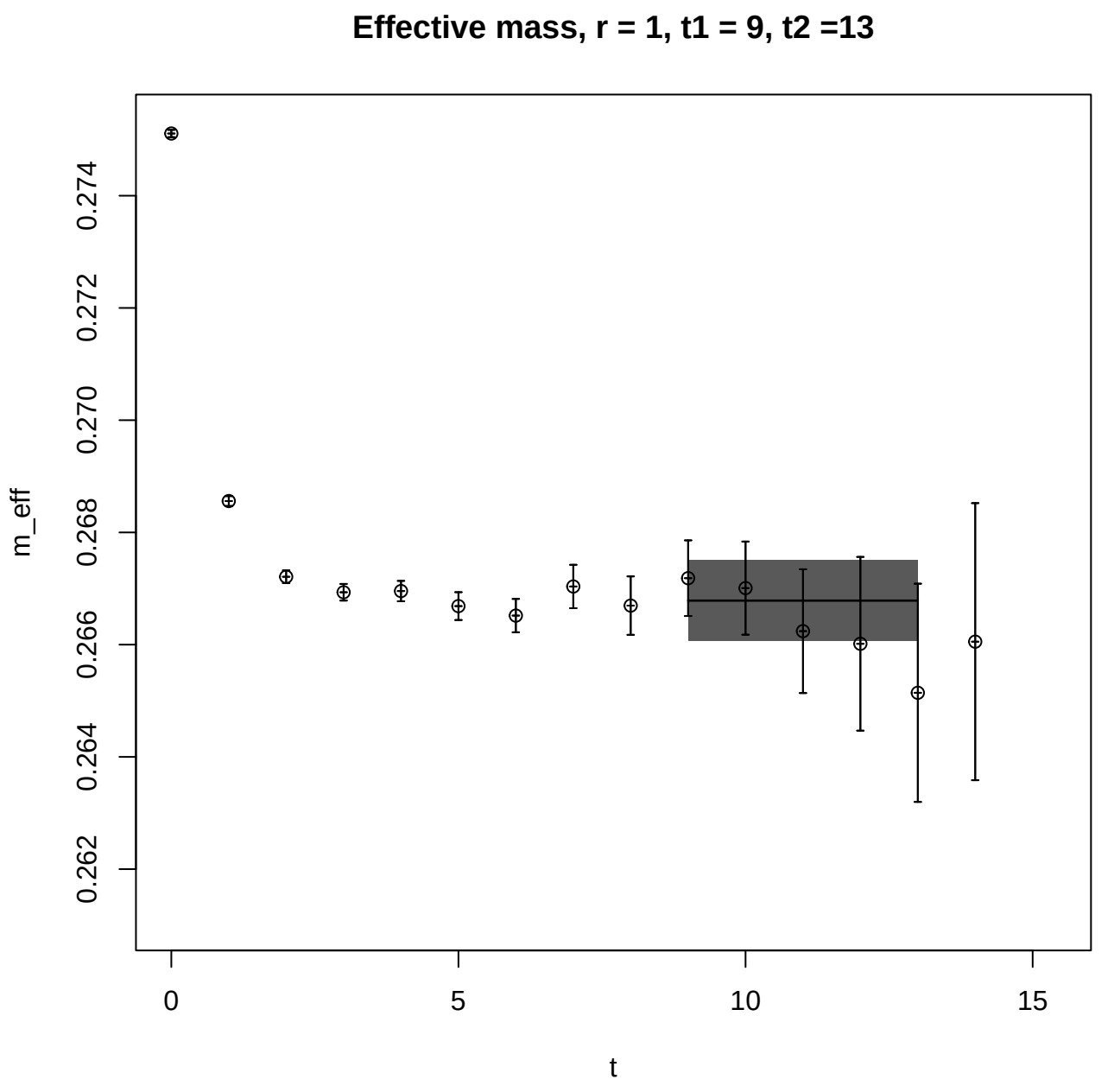
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =13
m_eff
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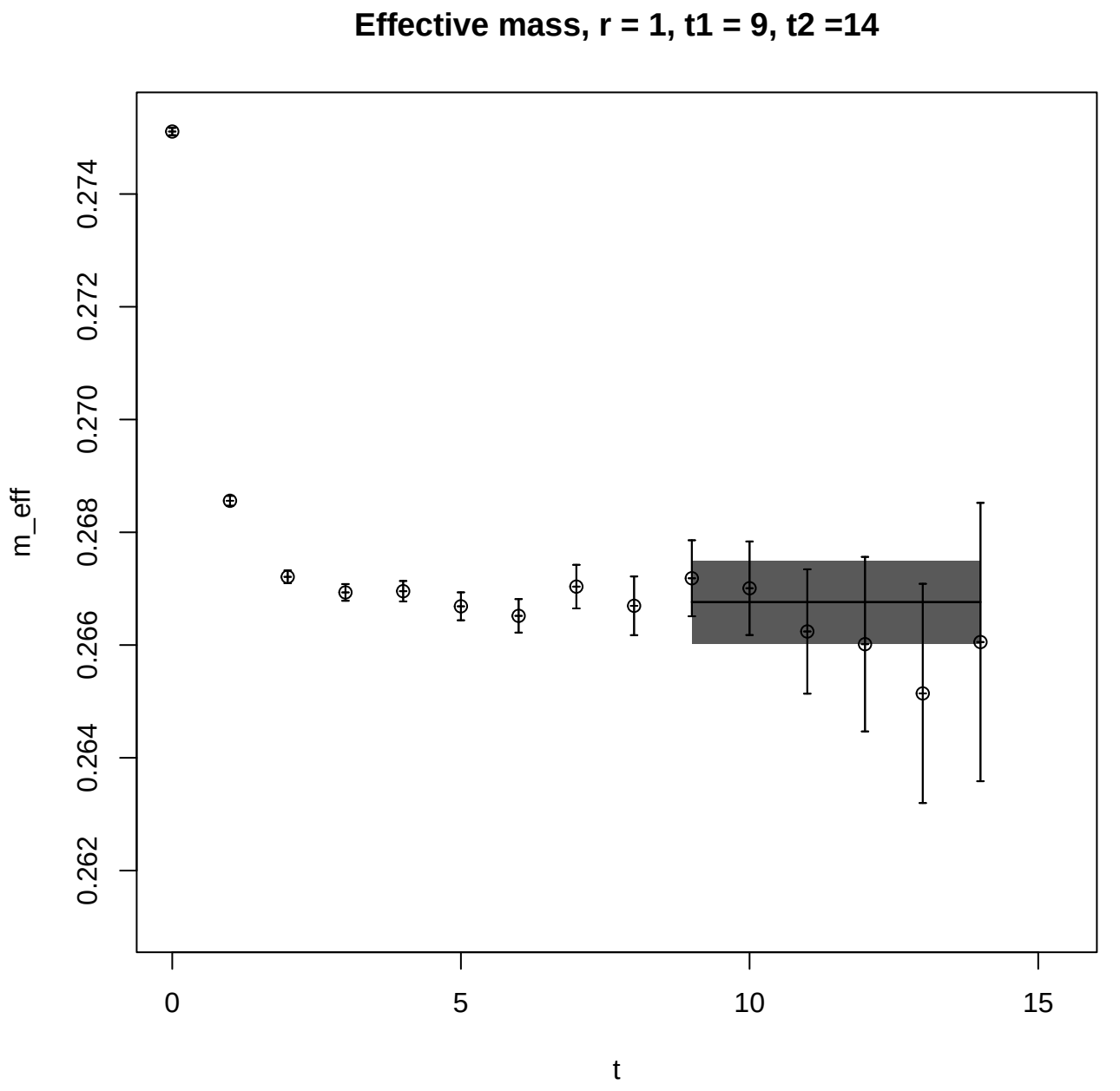
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =14
m_eff
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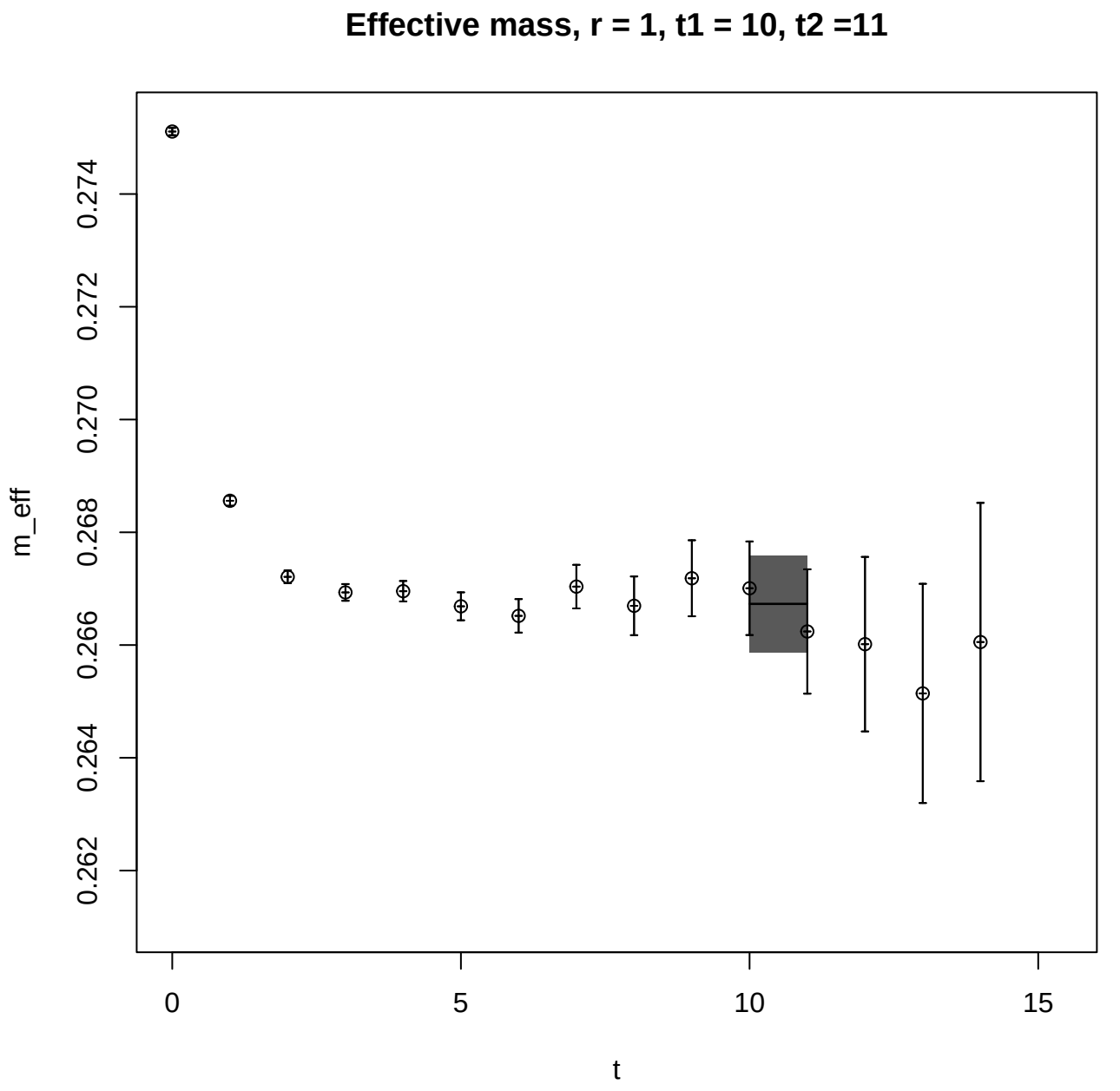
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 10, t2 =11
m_eff
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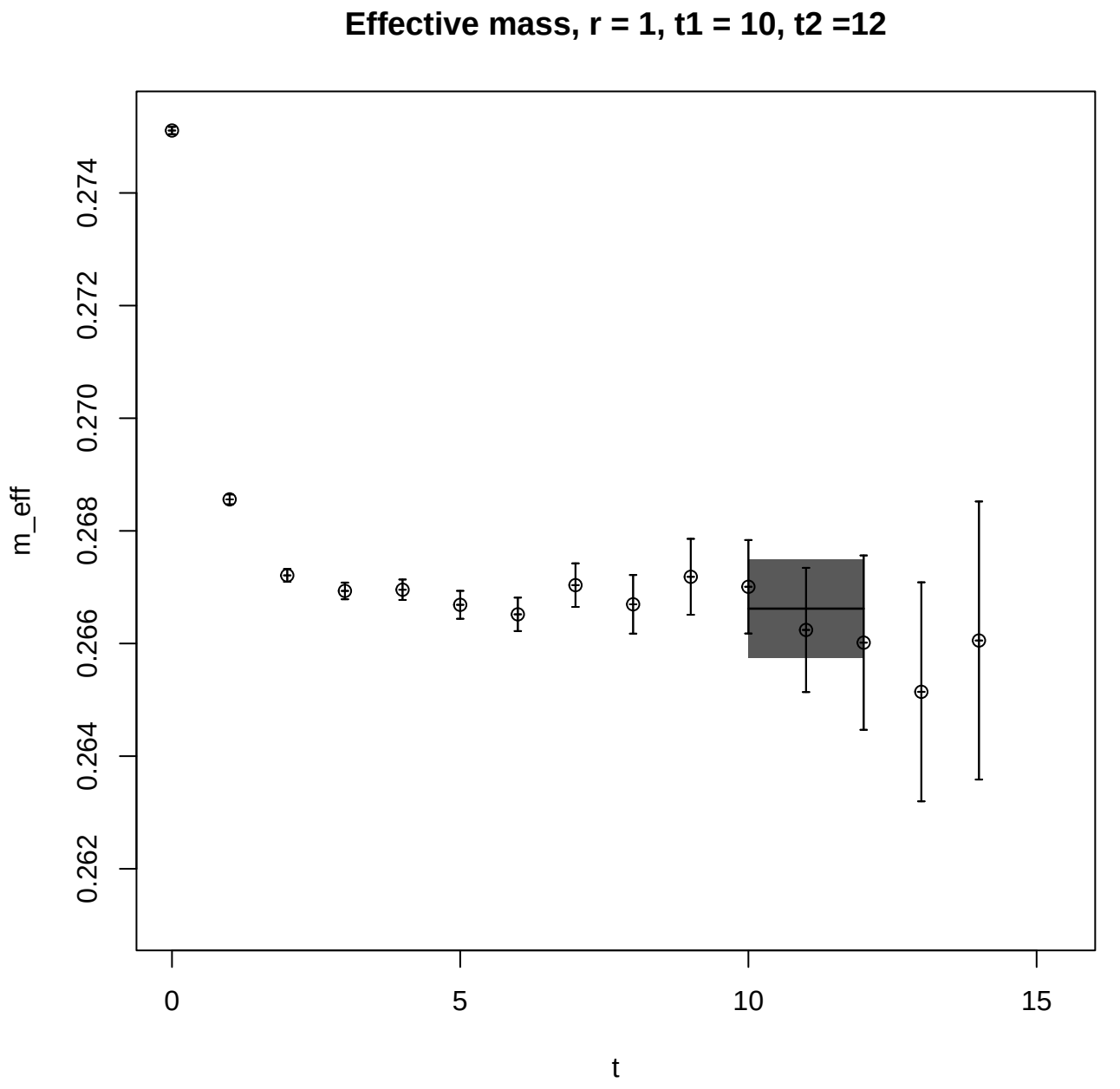
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 10, t2 =12
m_eff
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0.264
0.266
0.268
0.270
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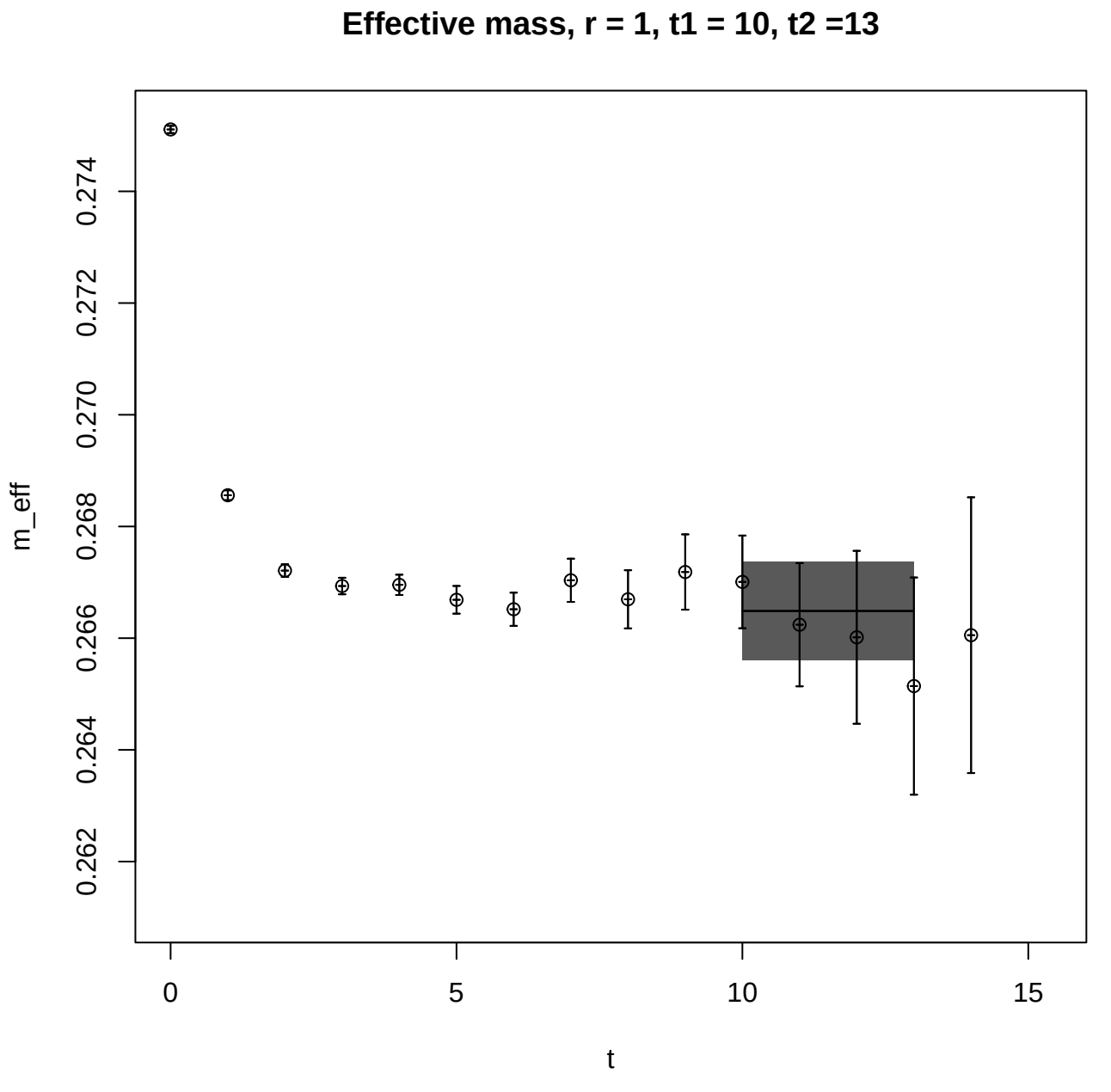
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 10, t2 =13
m_eff
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0.264
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0.268
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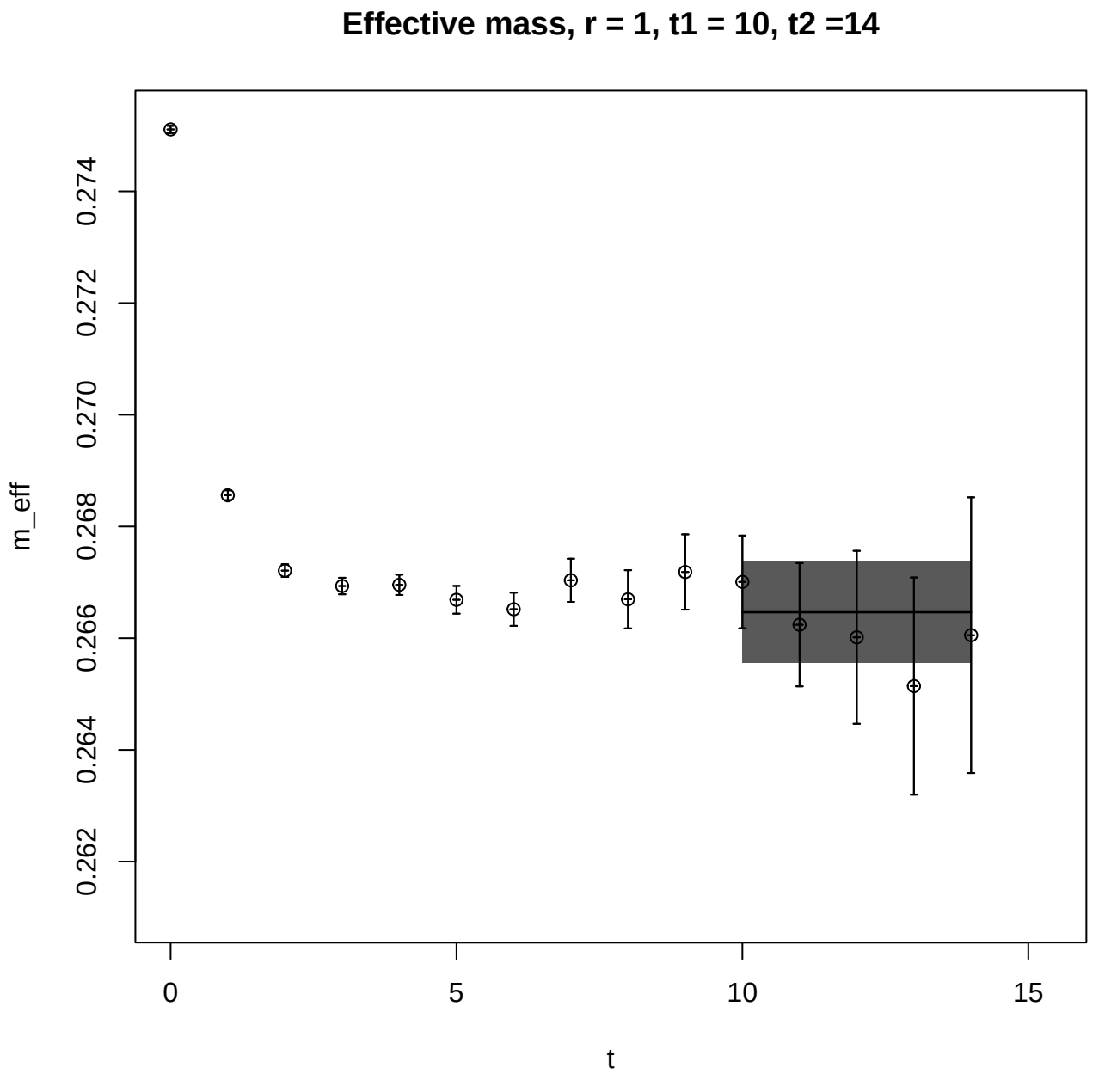
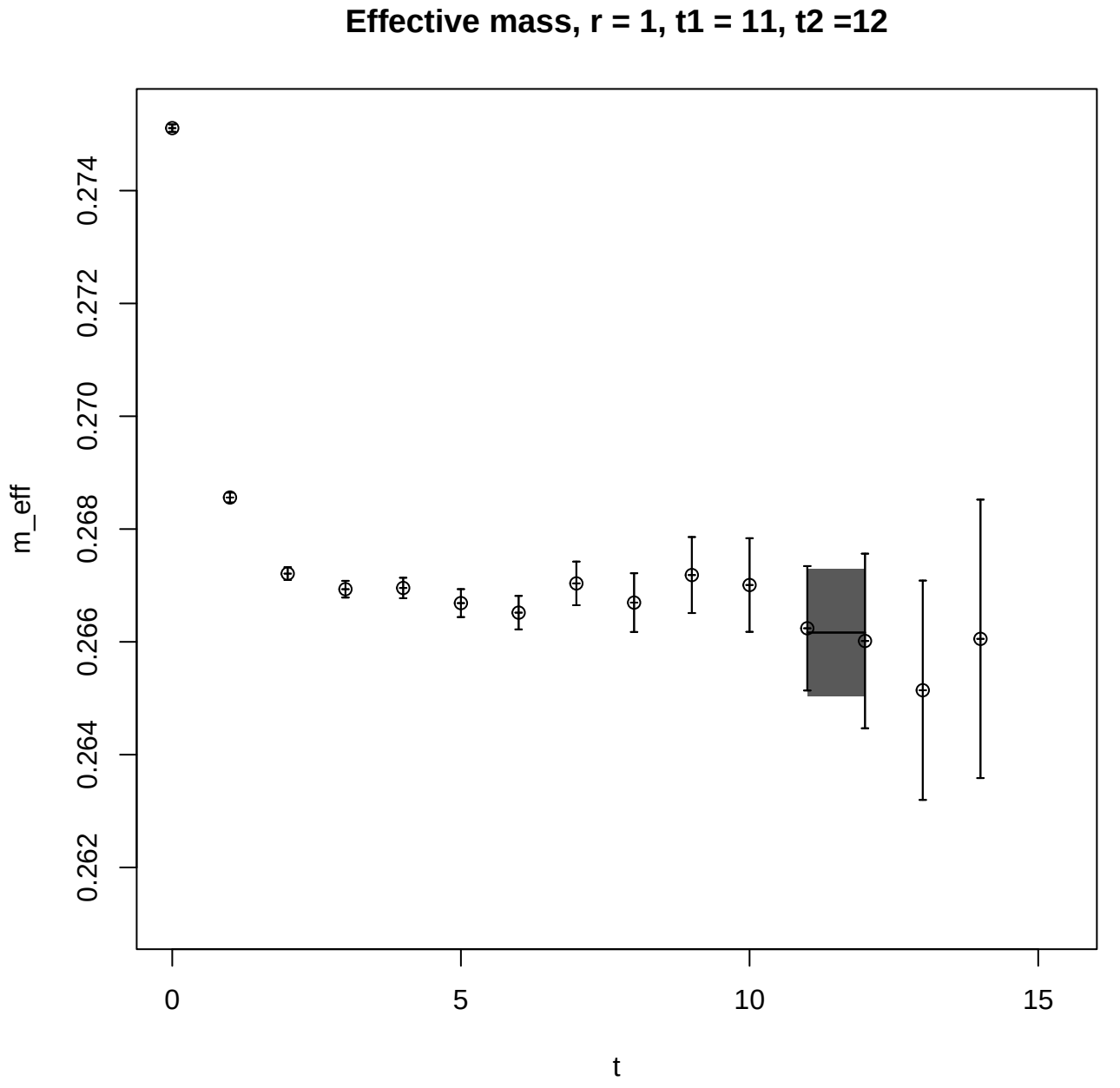
15
t


Effective mass, r = 1, t1 = 10, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 11, t2 =12
m_eff
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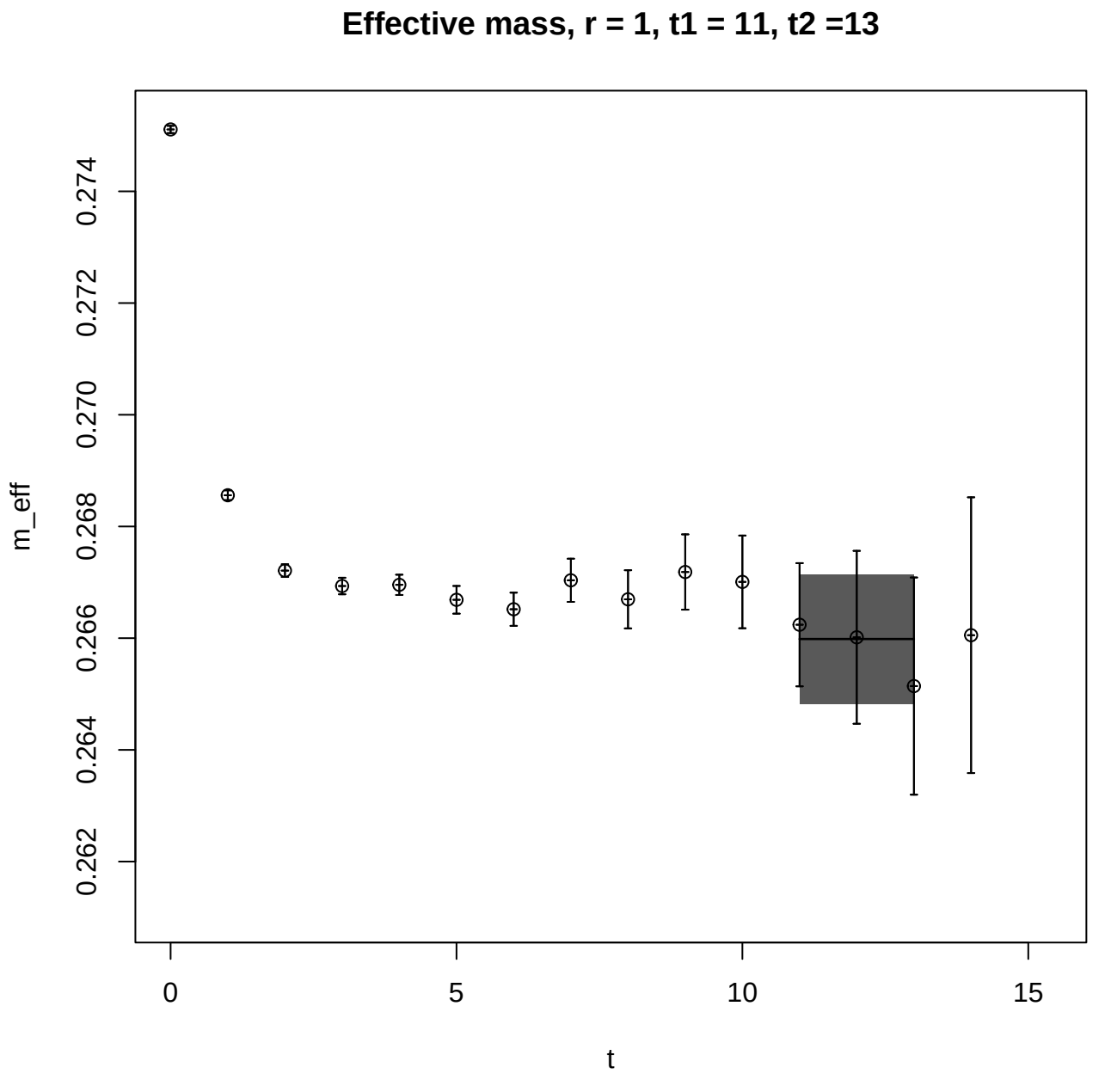
5
10
15
t


Effective mass, r = 1, t1 = 11, t2 =13
m_eff
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0.264
0.266
0.268
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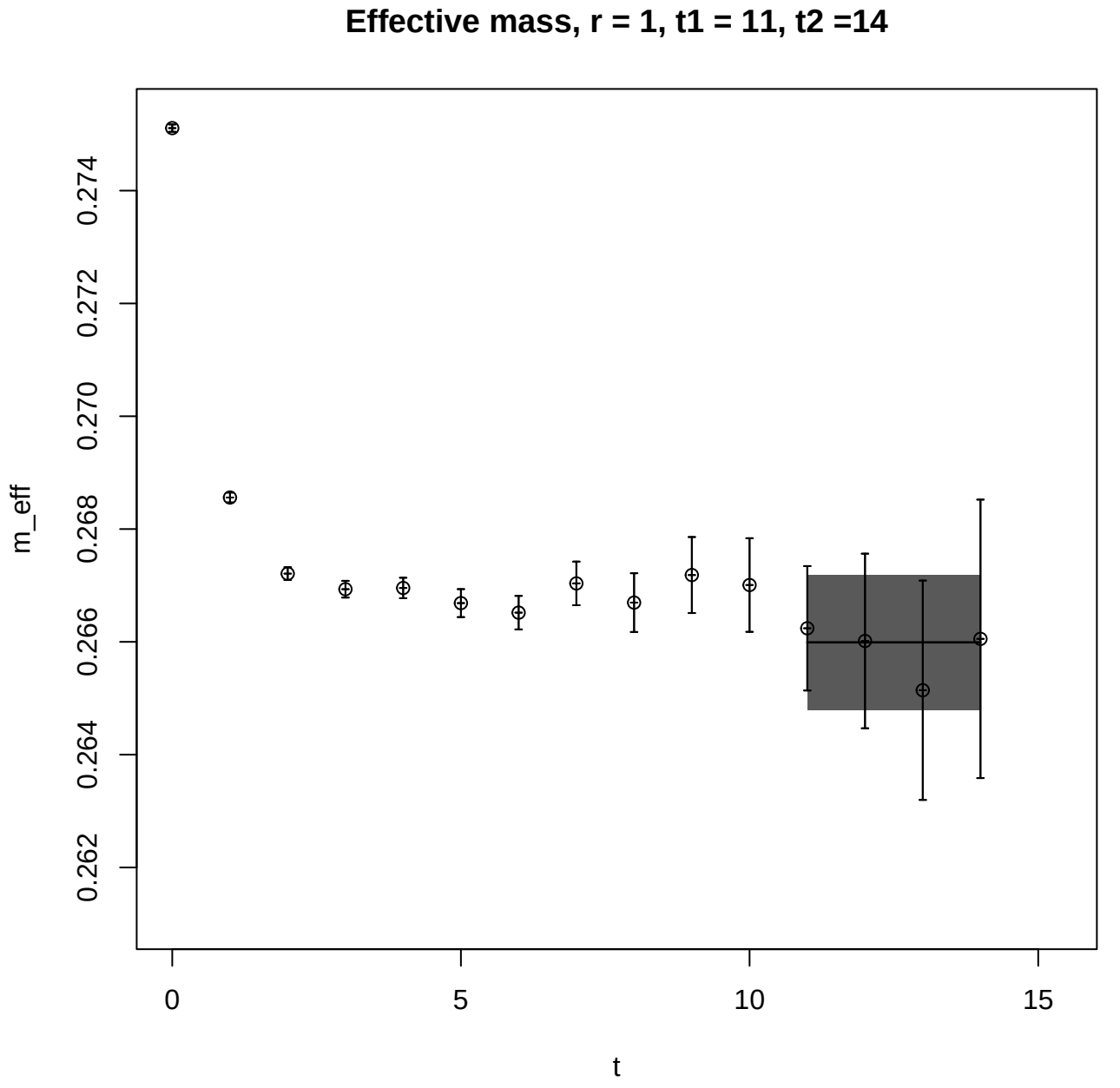
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 11, t2 =14
m_eff
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0.264
0.266
0.268
0.270
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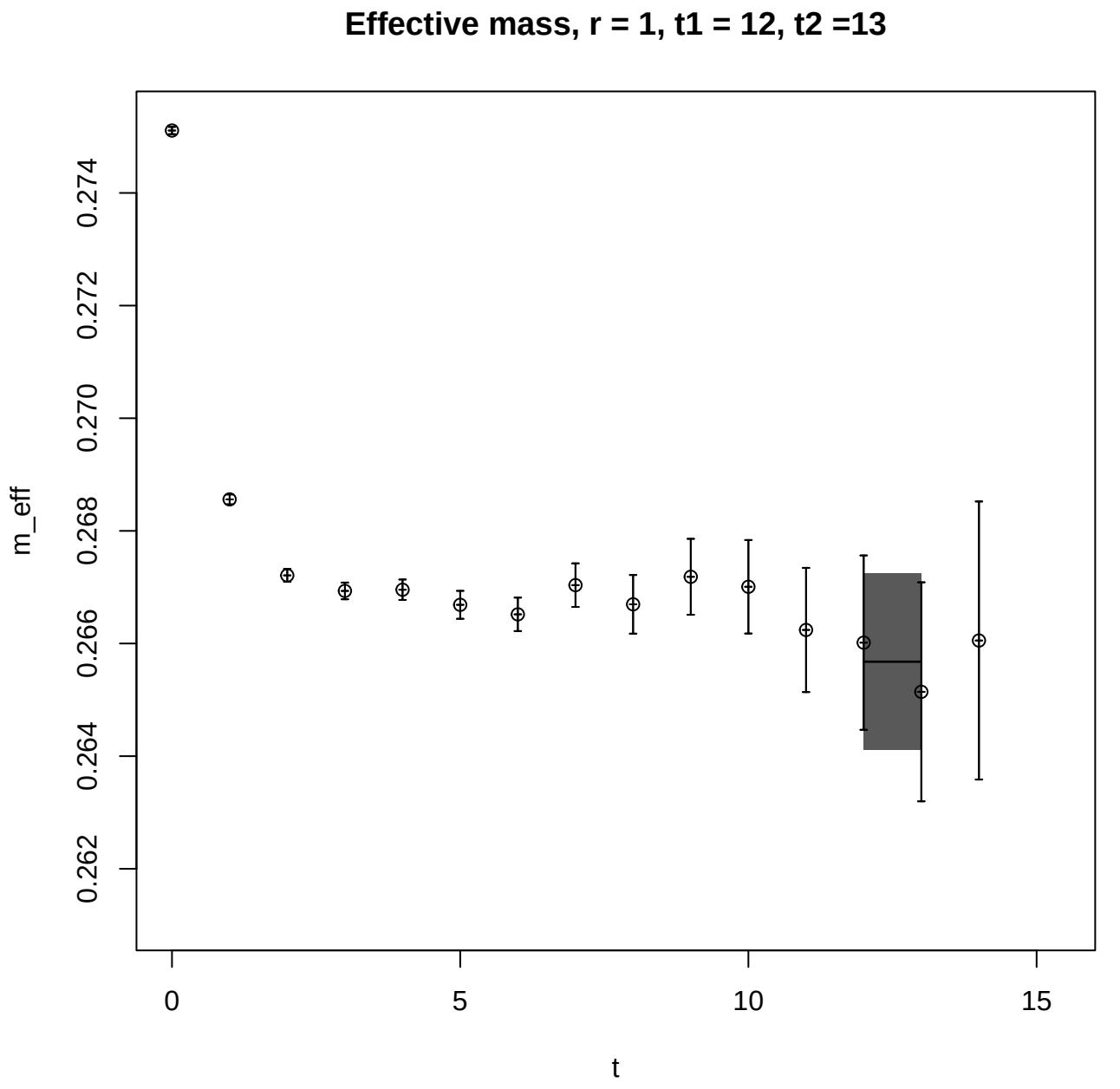
0.274
0
5
10
15
t


Effective mass, r = 1, t1 = 12, t2 =13
m_eff
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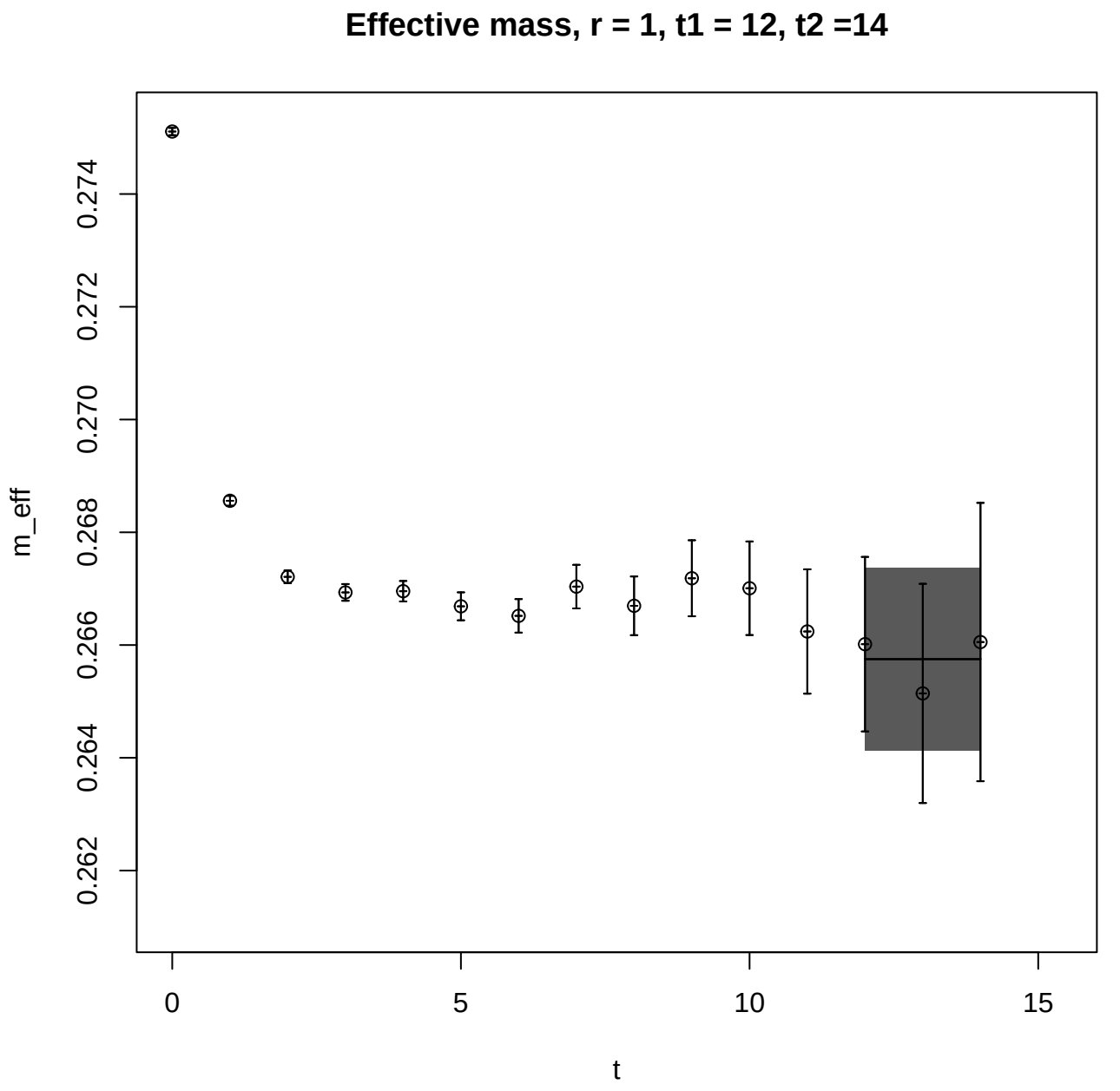
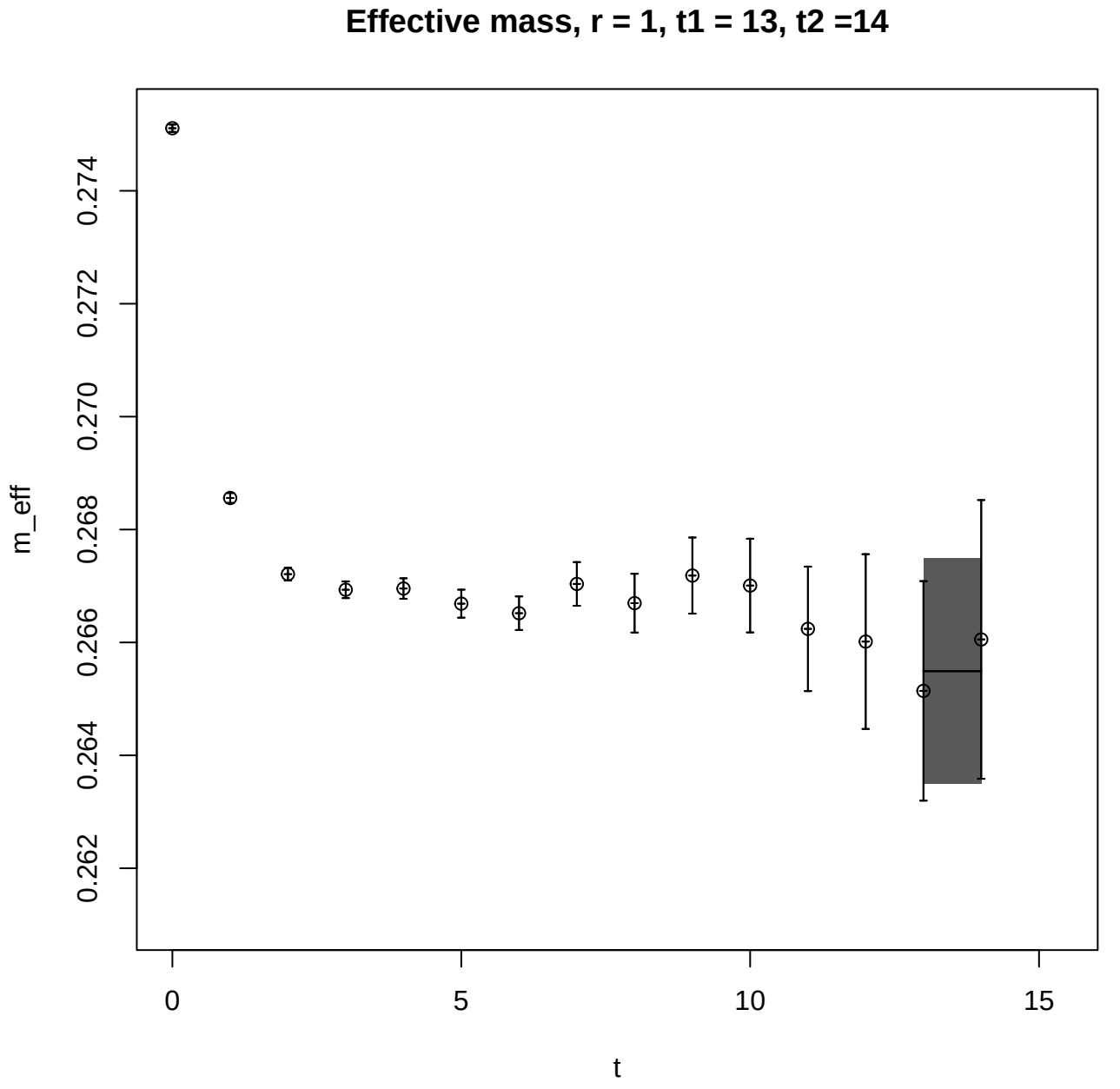
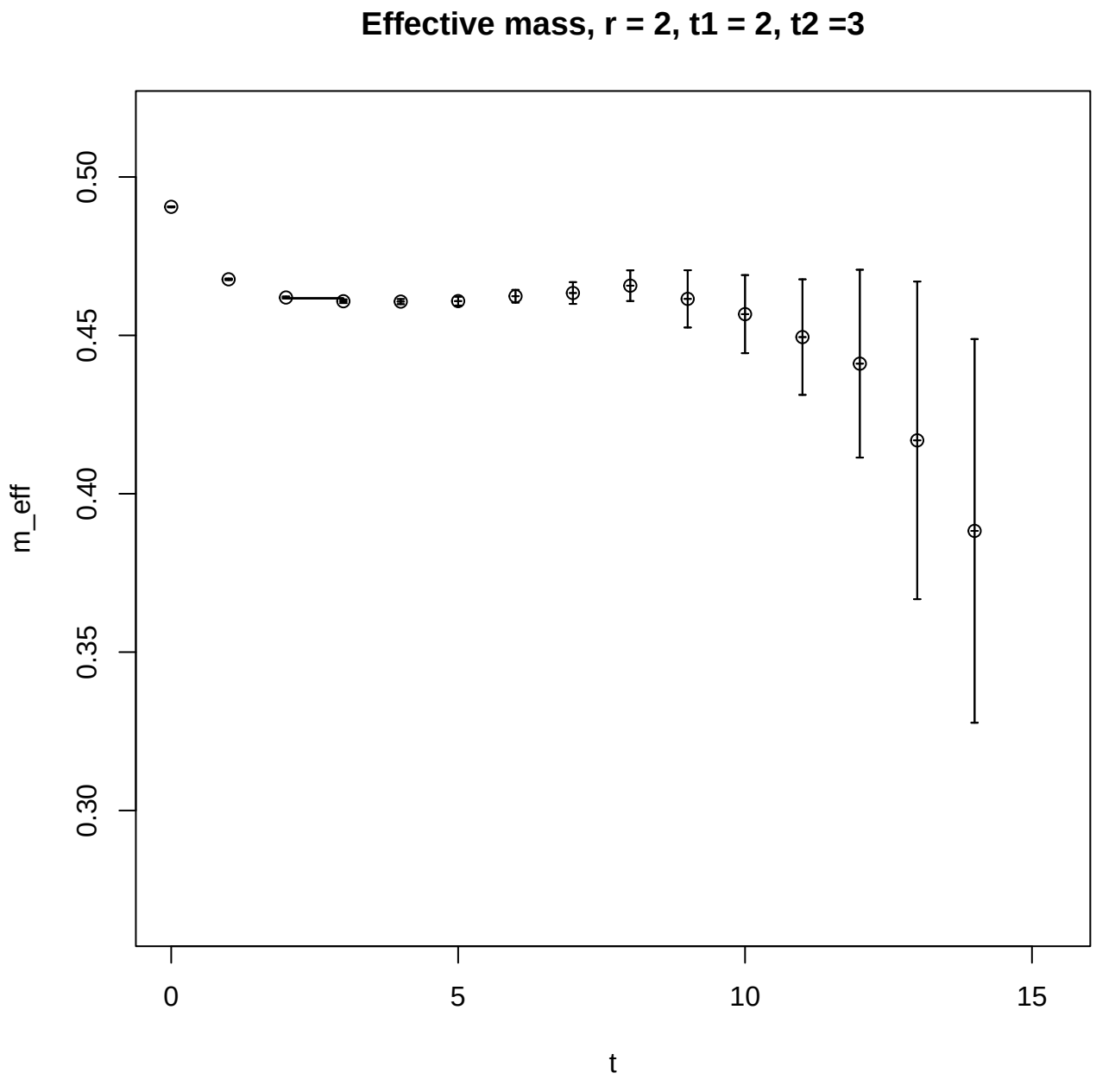
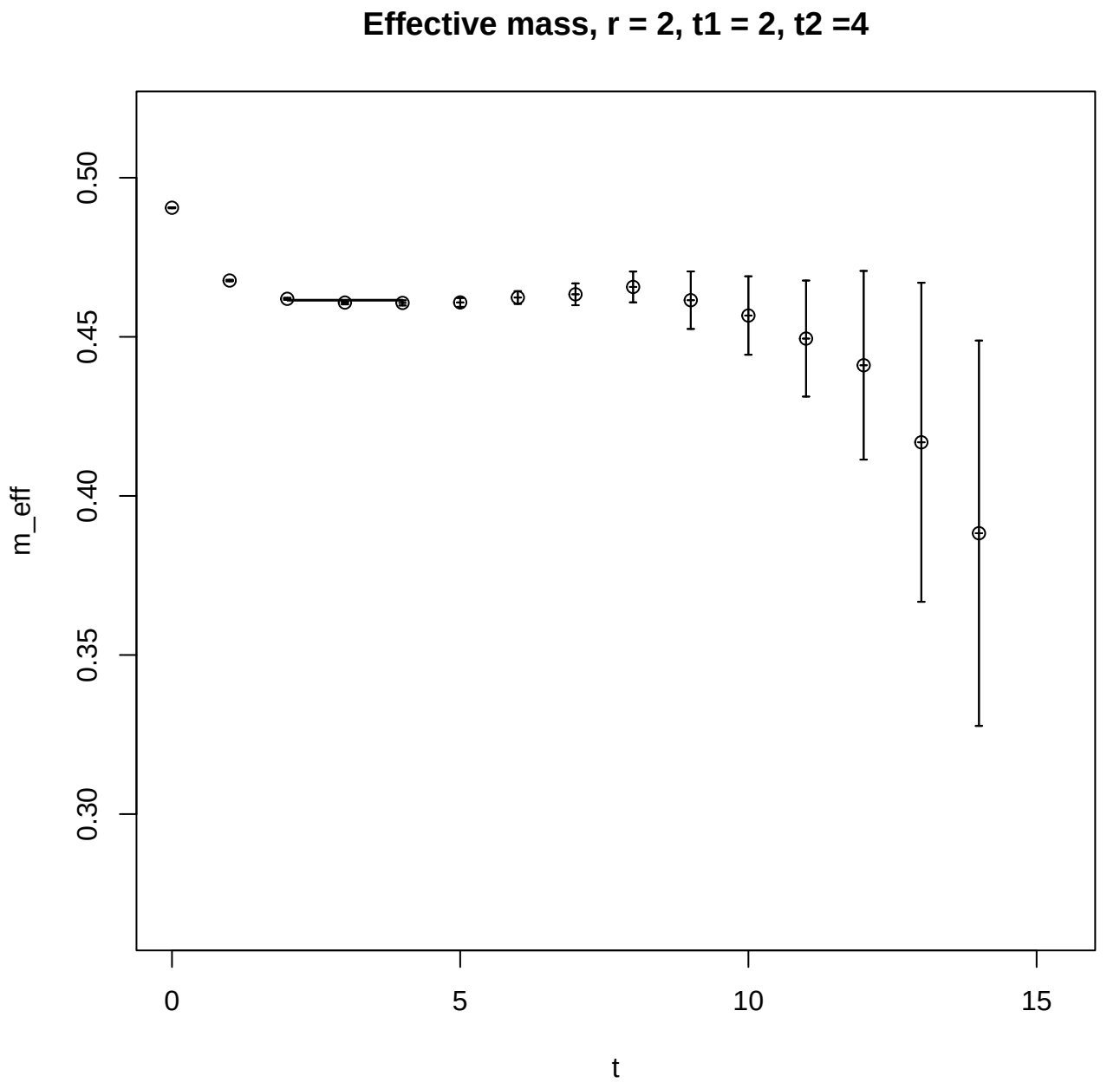
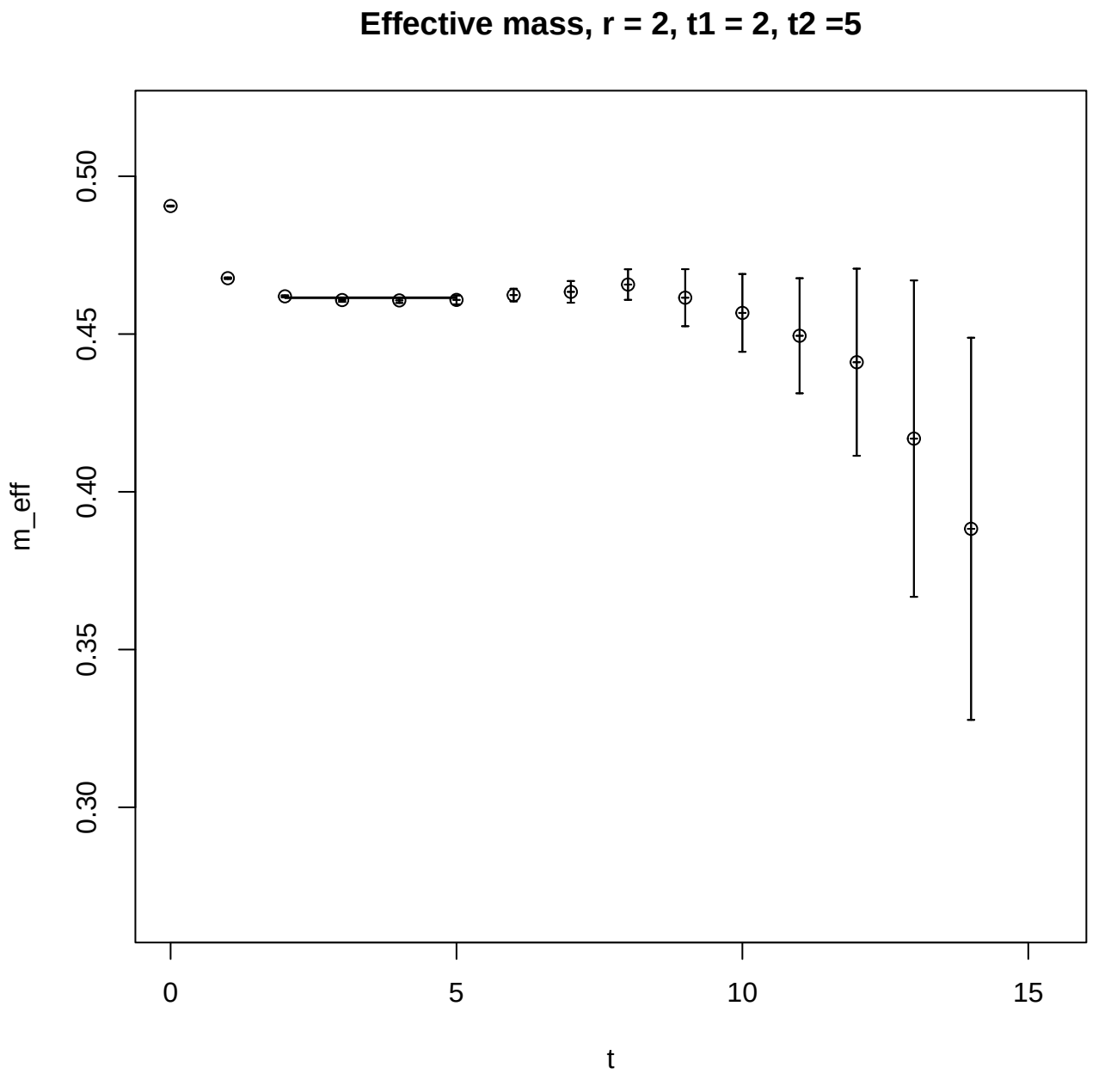
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

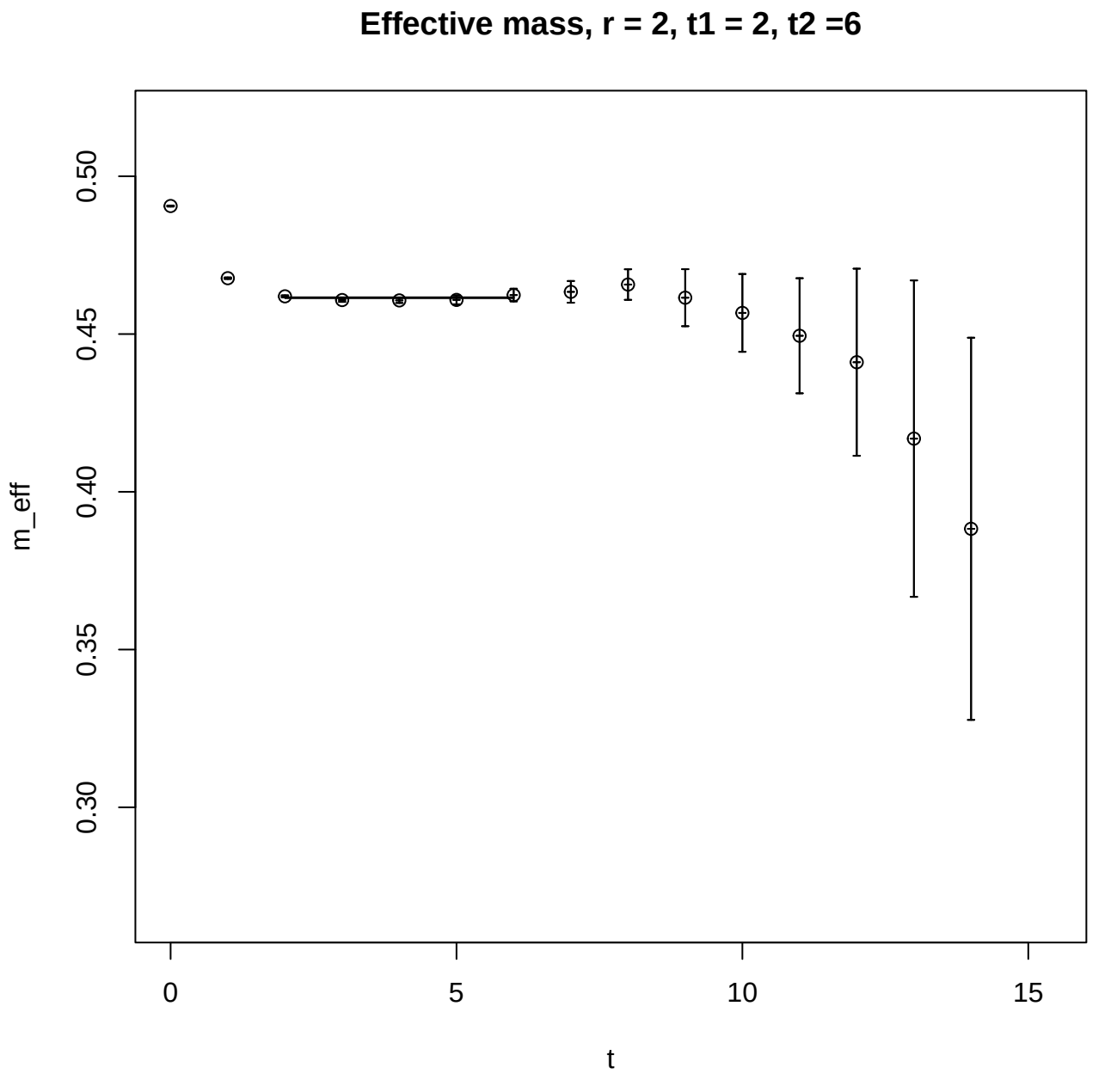

Effective mass, r = 1, t1 = 12, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

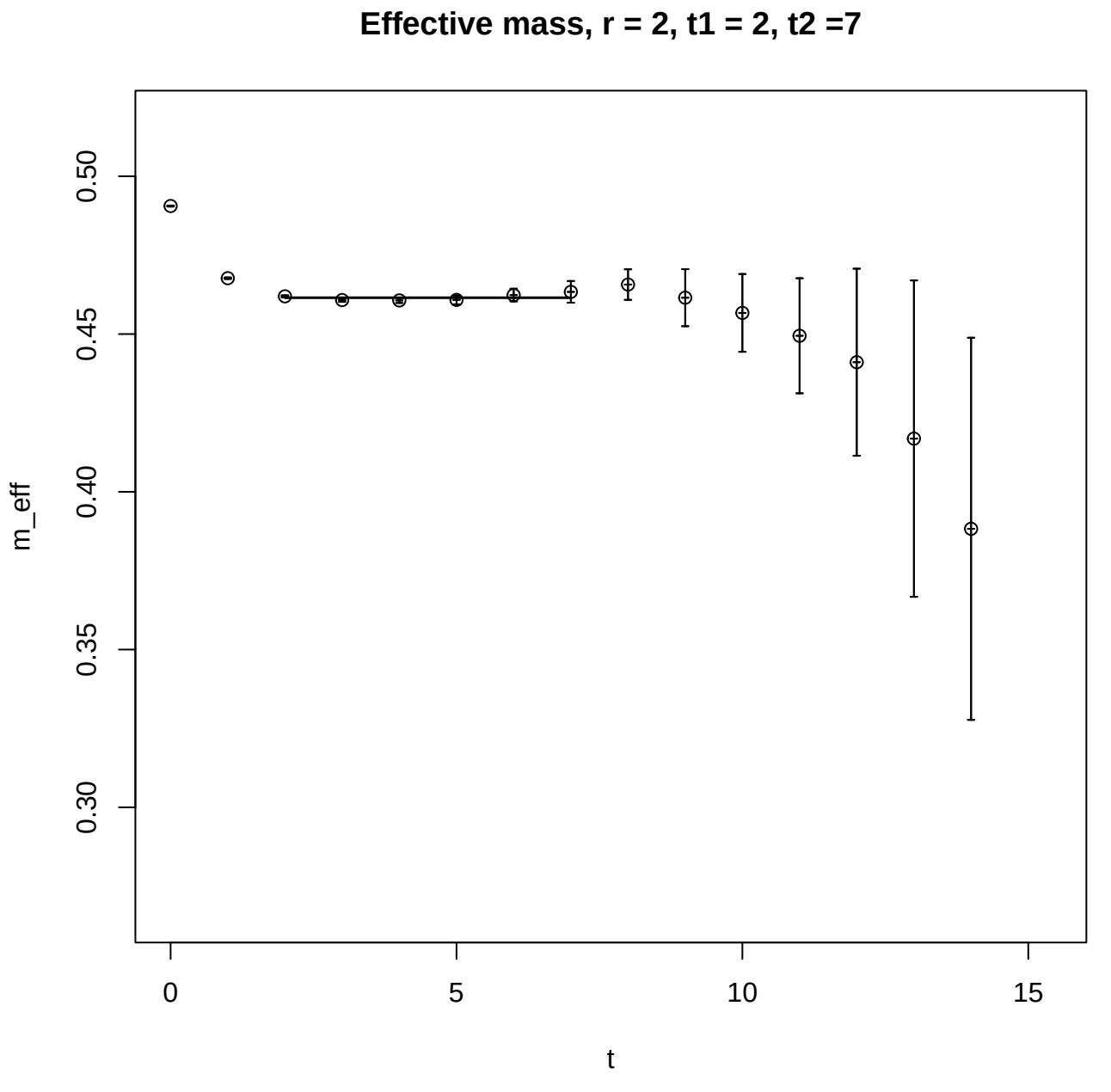

Effective mass, r = 1, t1 = 13, t2 =14
m_eff
0.262
0.264
0.266
0.268
0.270
0.272
0.274
0
5
10
15
t

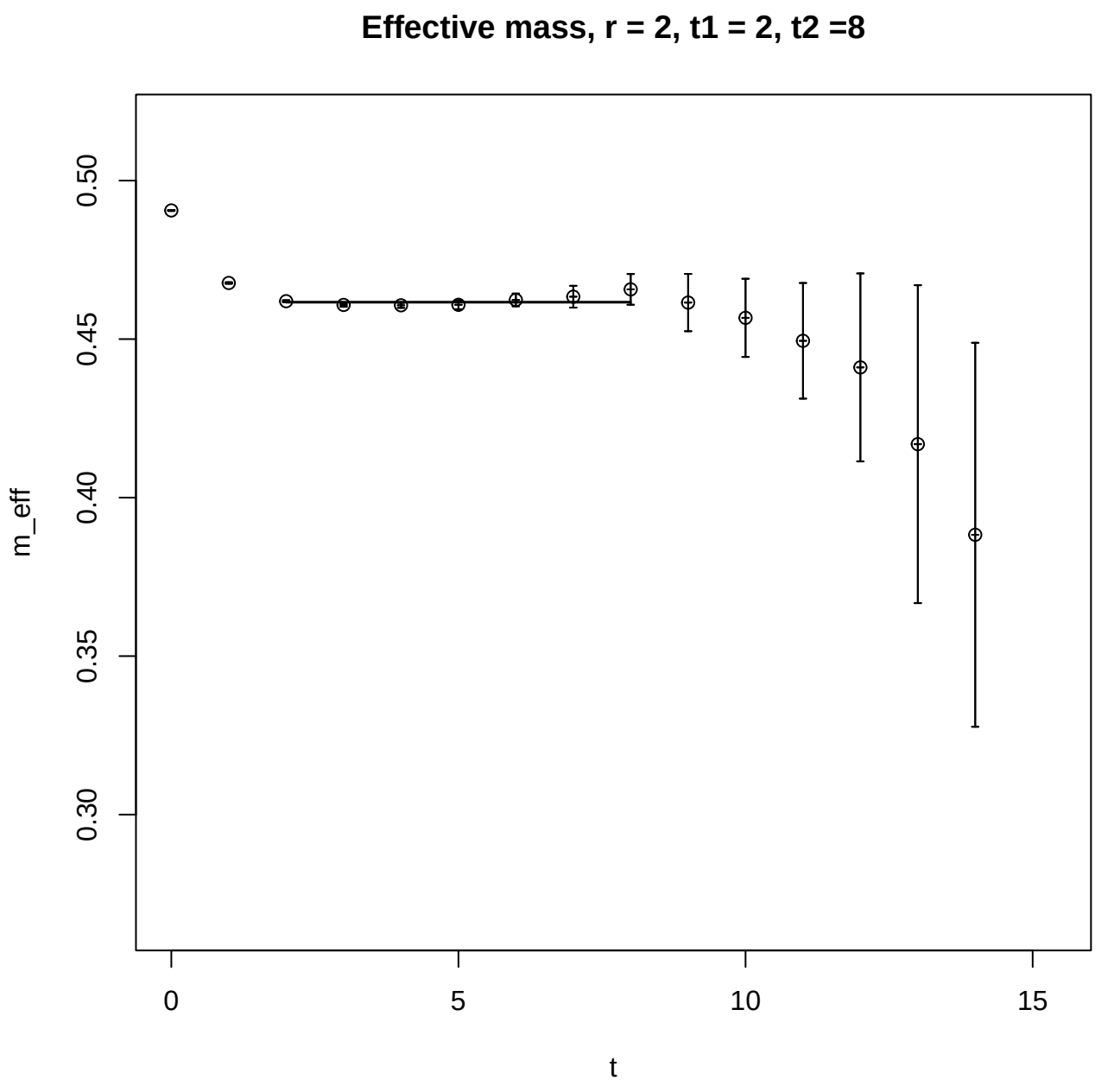

Effective mass, r = 2, t1 = 2, t2 =3
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

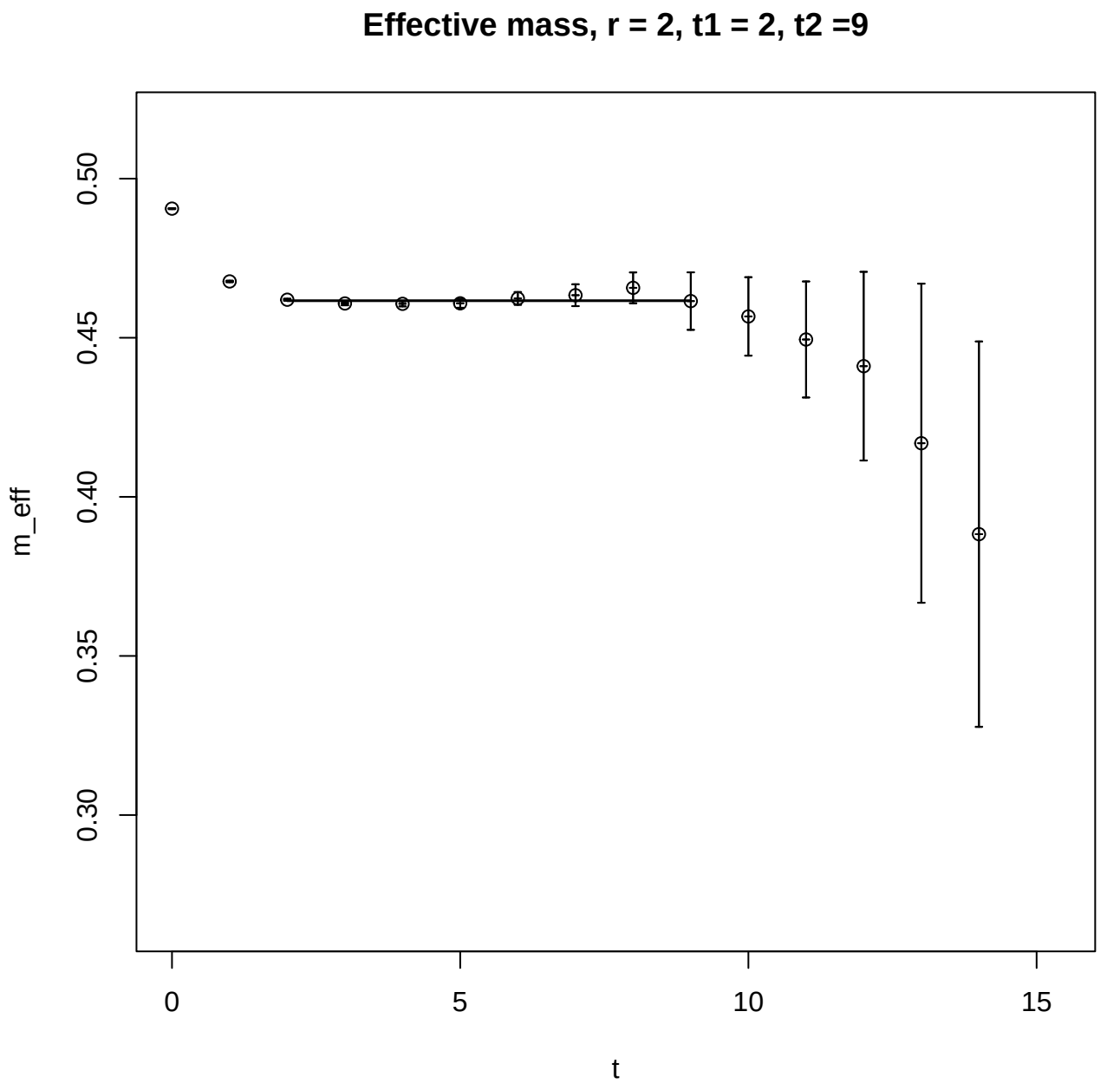

Effective mass, r = 2, t1 = 2, t2 =4
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

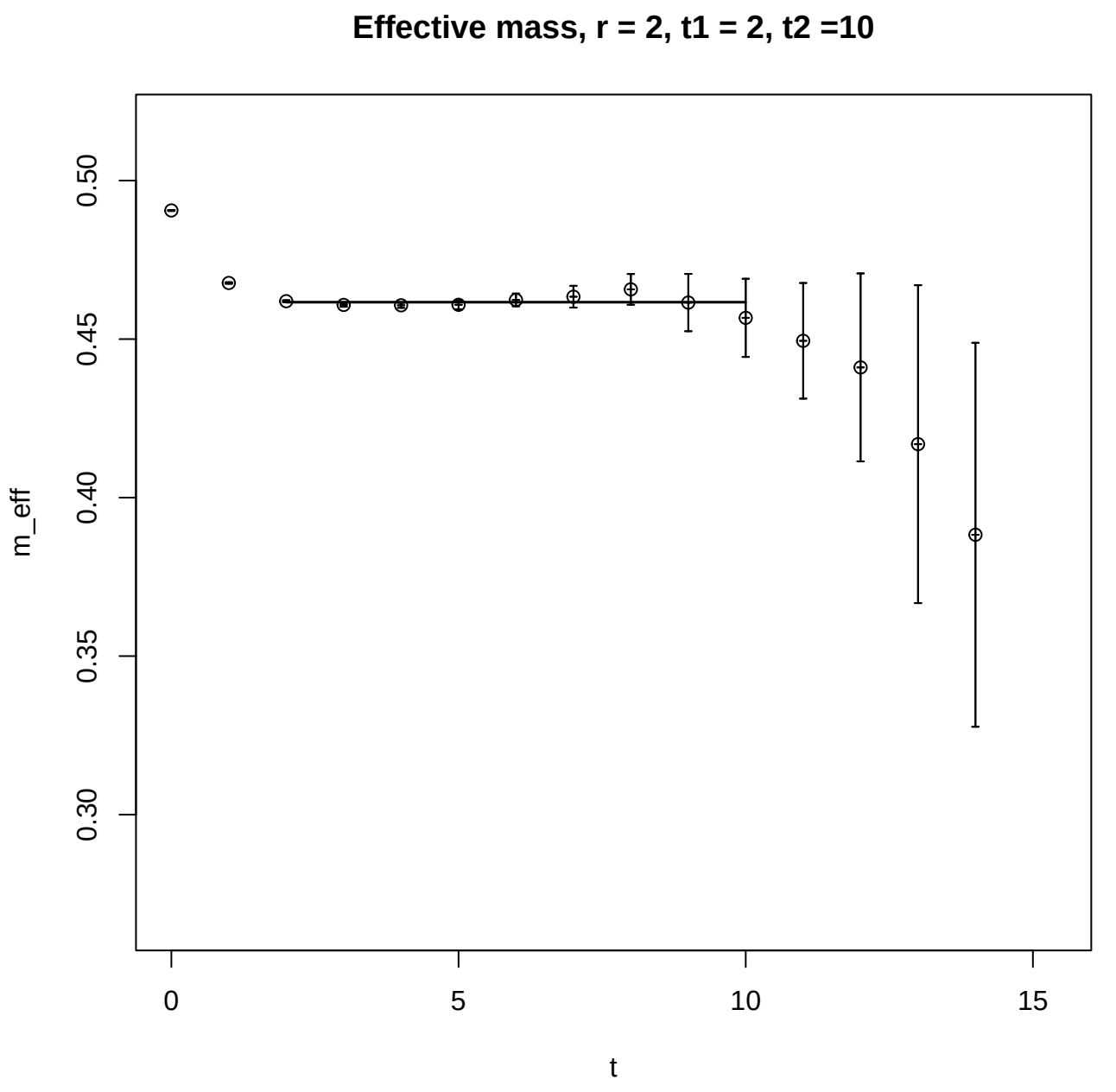

Effective mass, r = 2, t1 = 2, t2 =5
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

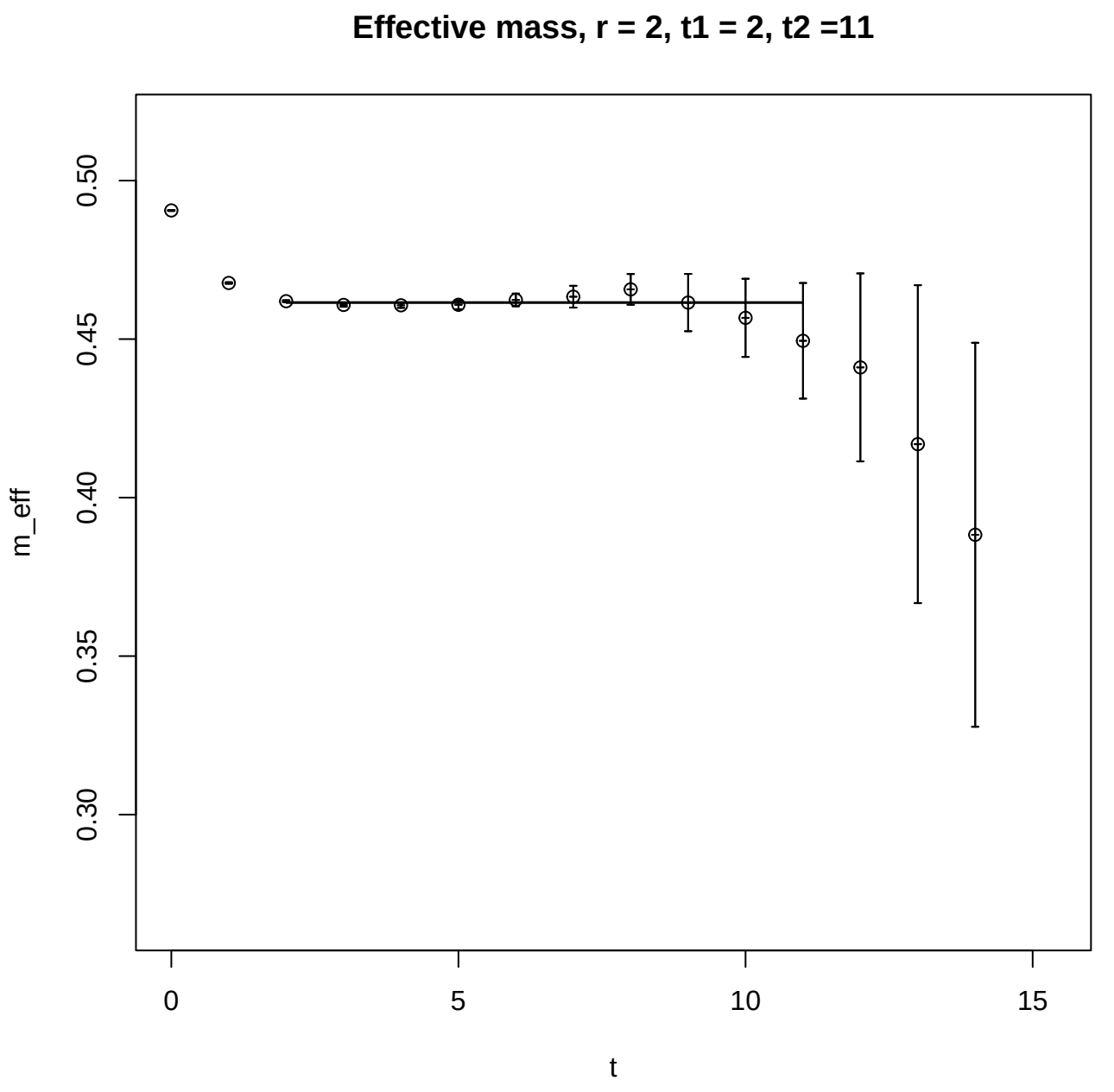

Effective mass, r = 2, t1 = 2, t2 =6
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

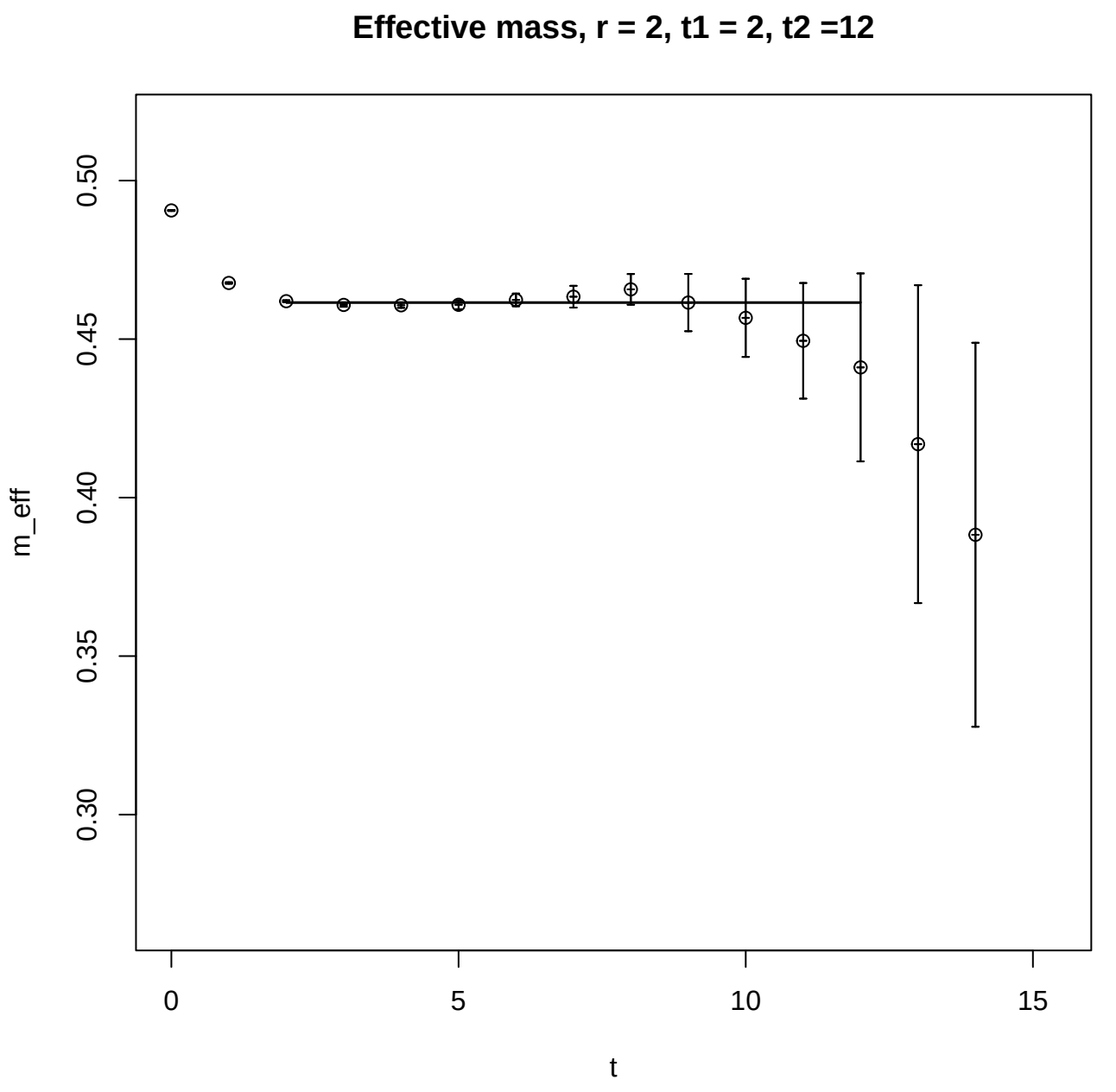

Effective mass, r = 2, t1 = 2, t2 =7
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

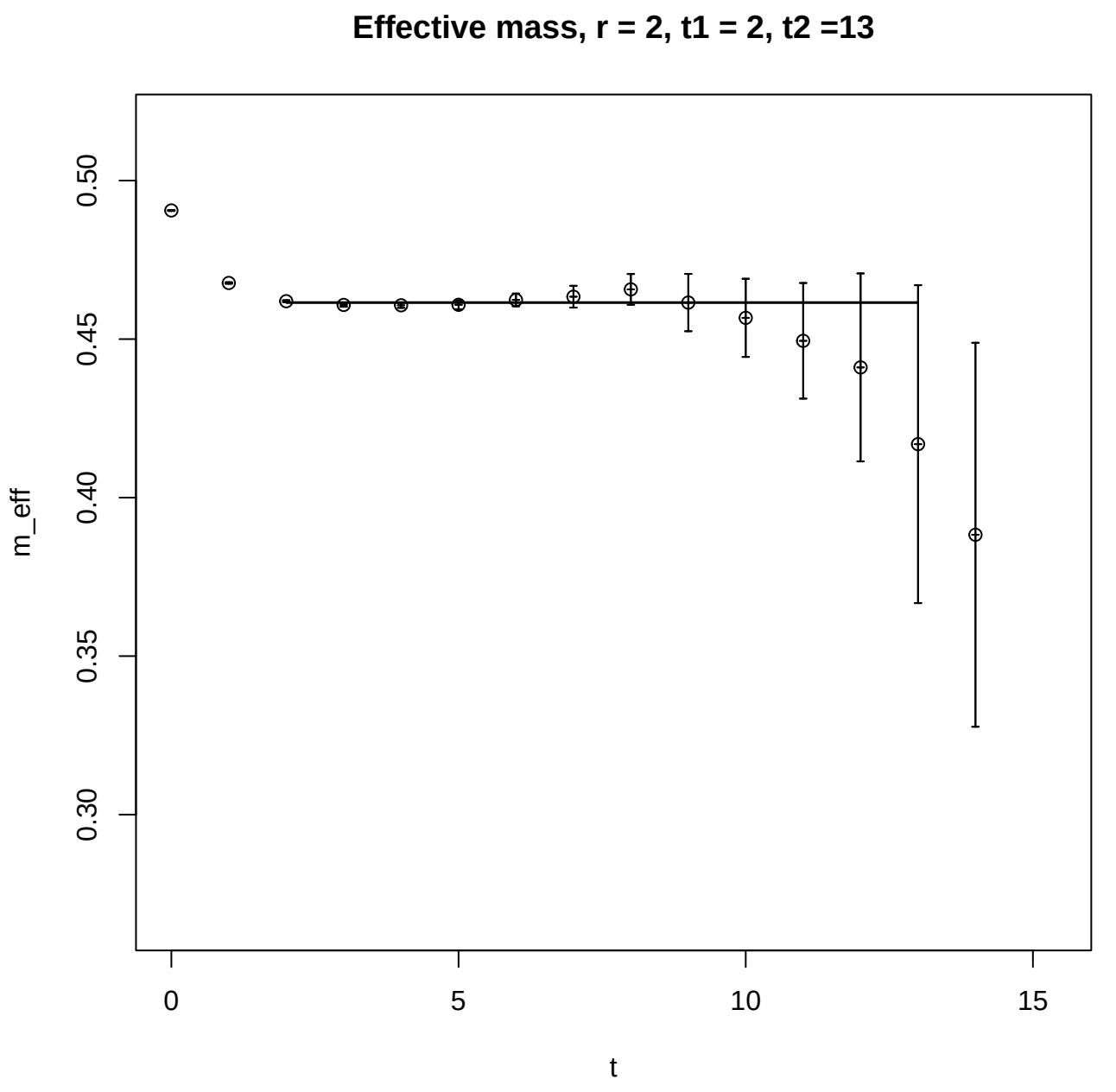

Effective mass, r = 2, t1 = 2, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

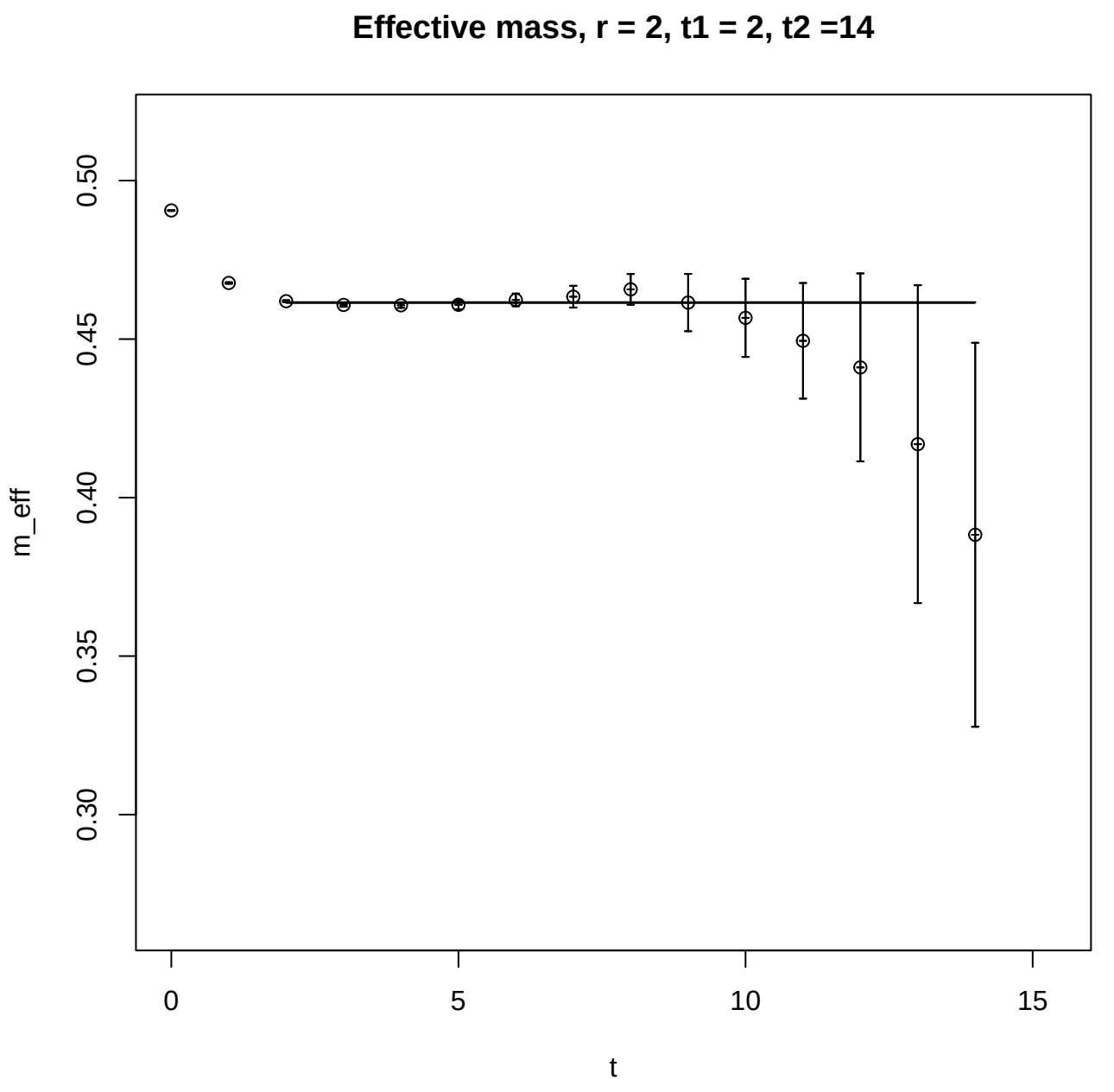

Effective mass, r = 2, t1 = 2, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

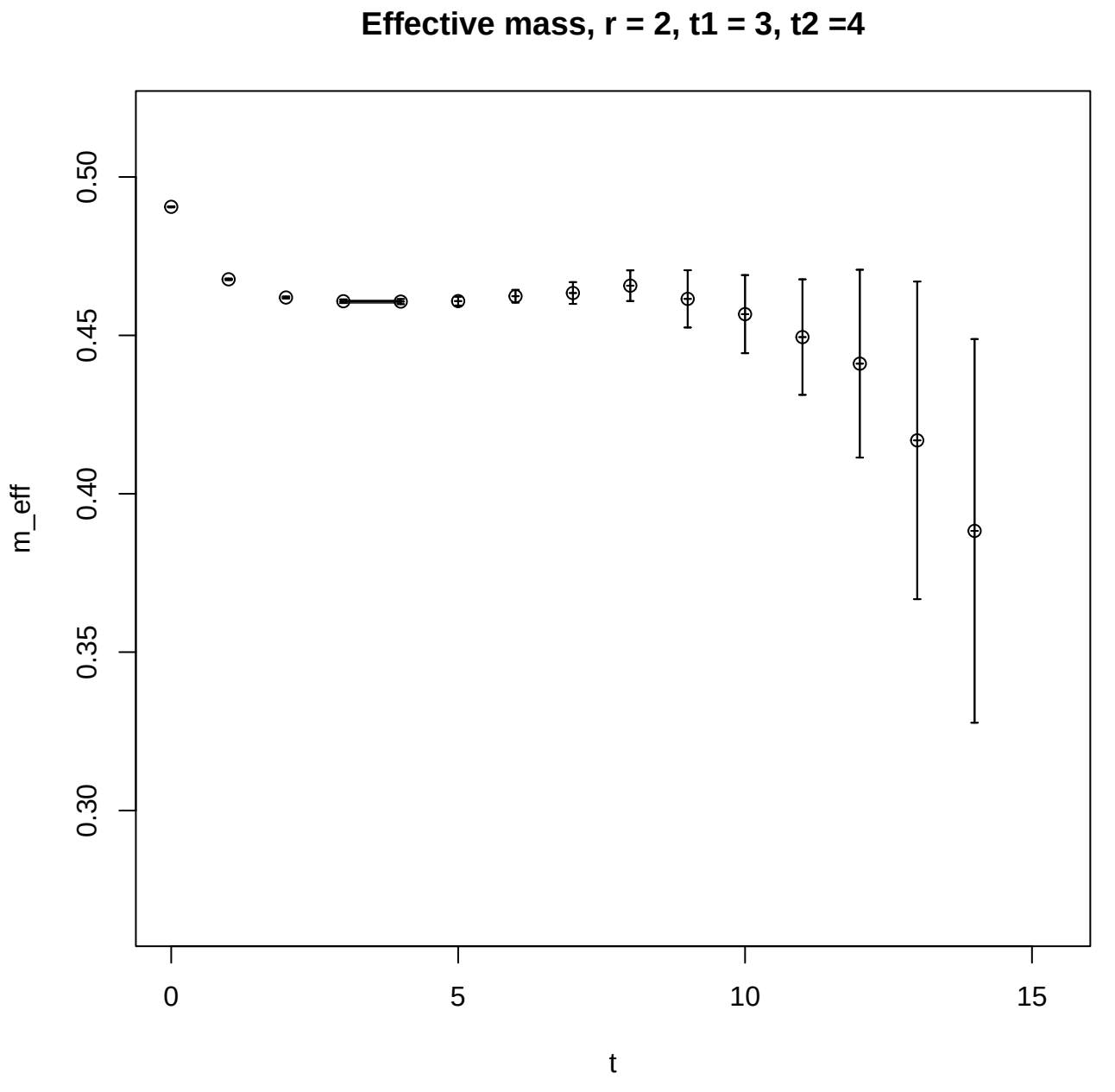

Effective mass, r = 2, t1 = 2, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

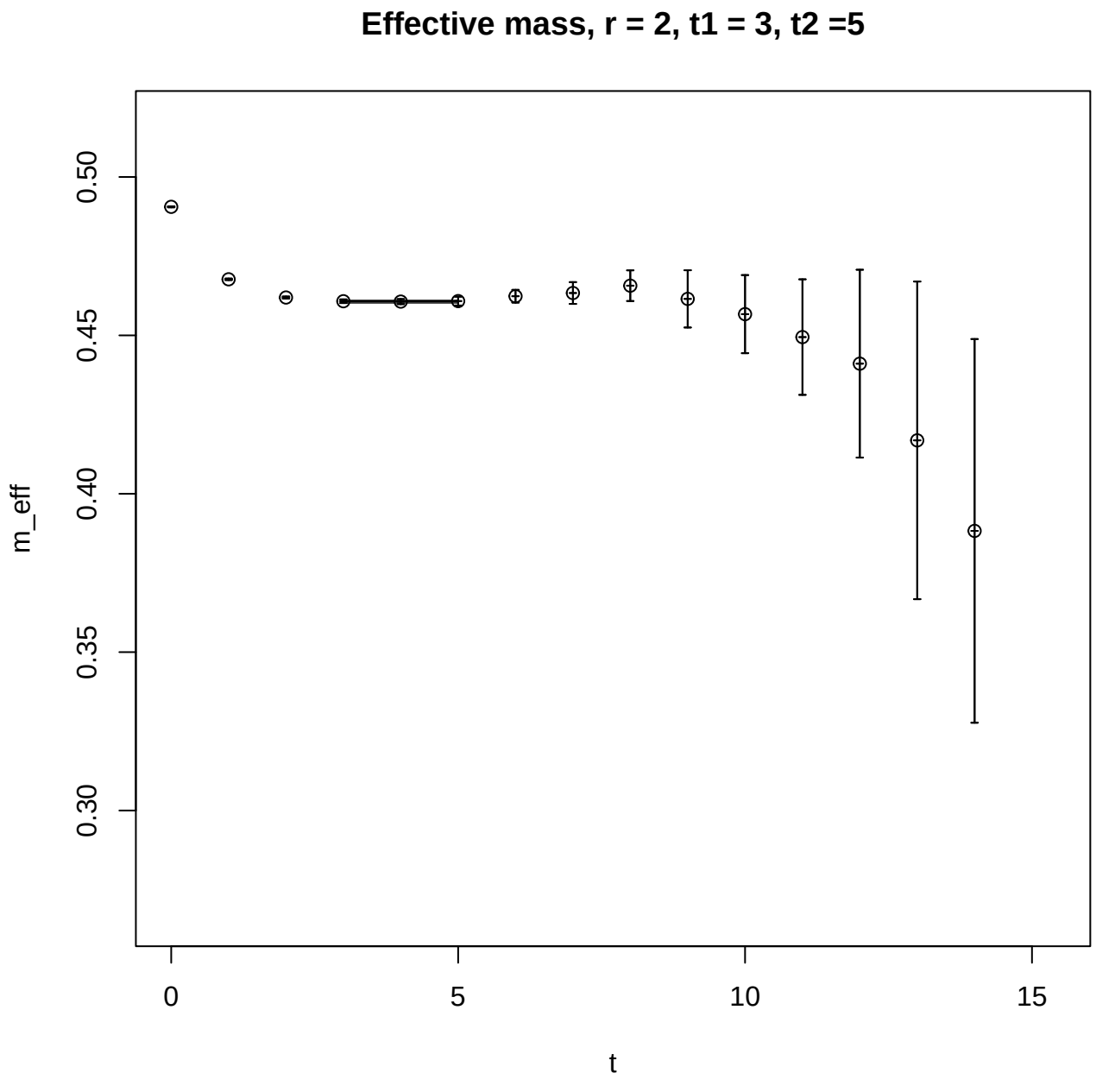

Effective mass, r = 2, t1 = 2, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

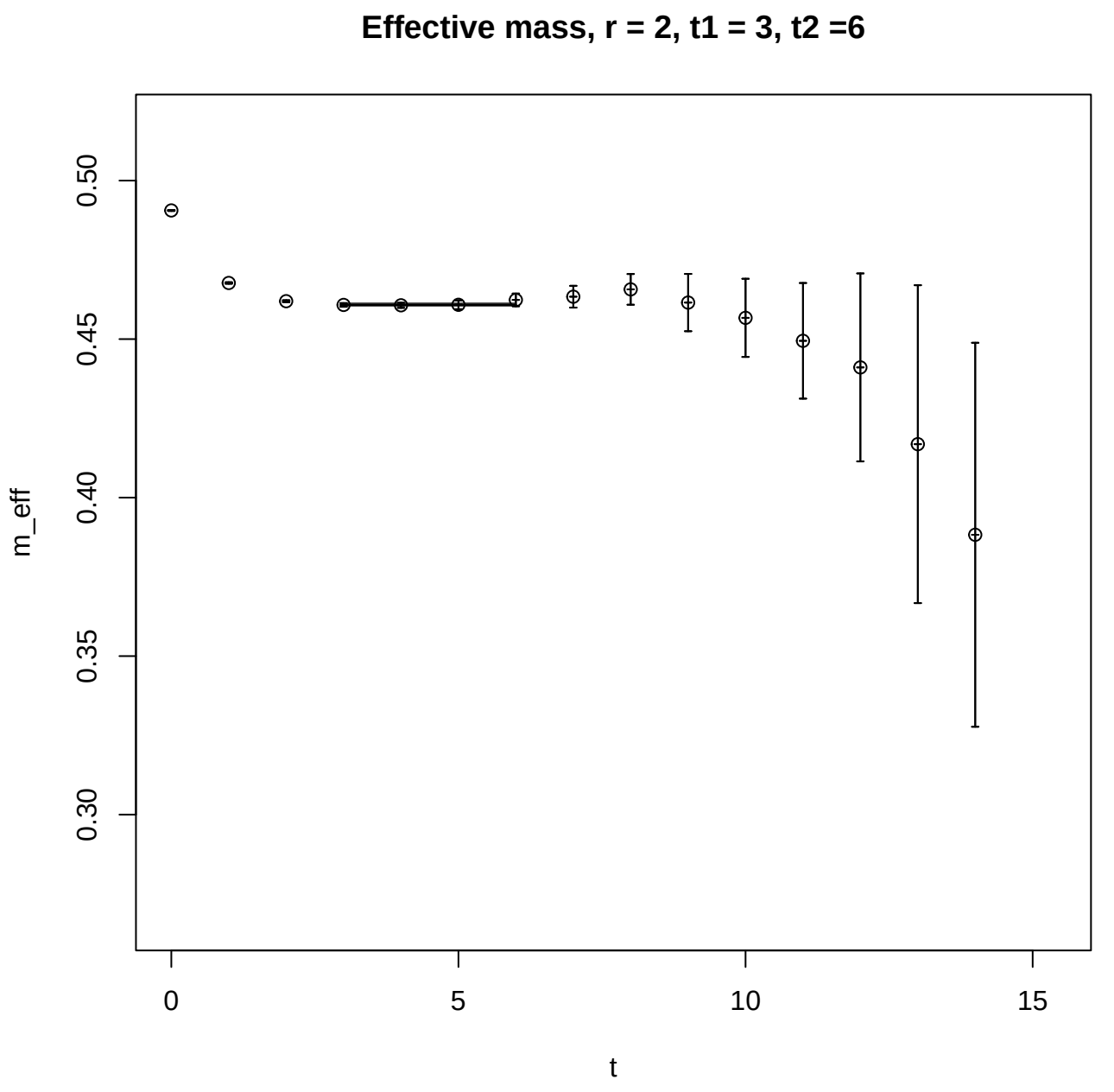

Effective mass, r = 2, t1 = 2, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

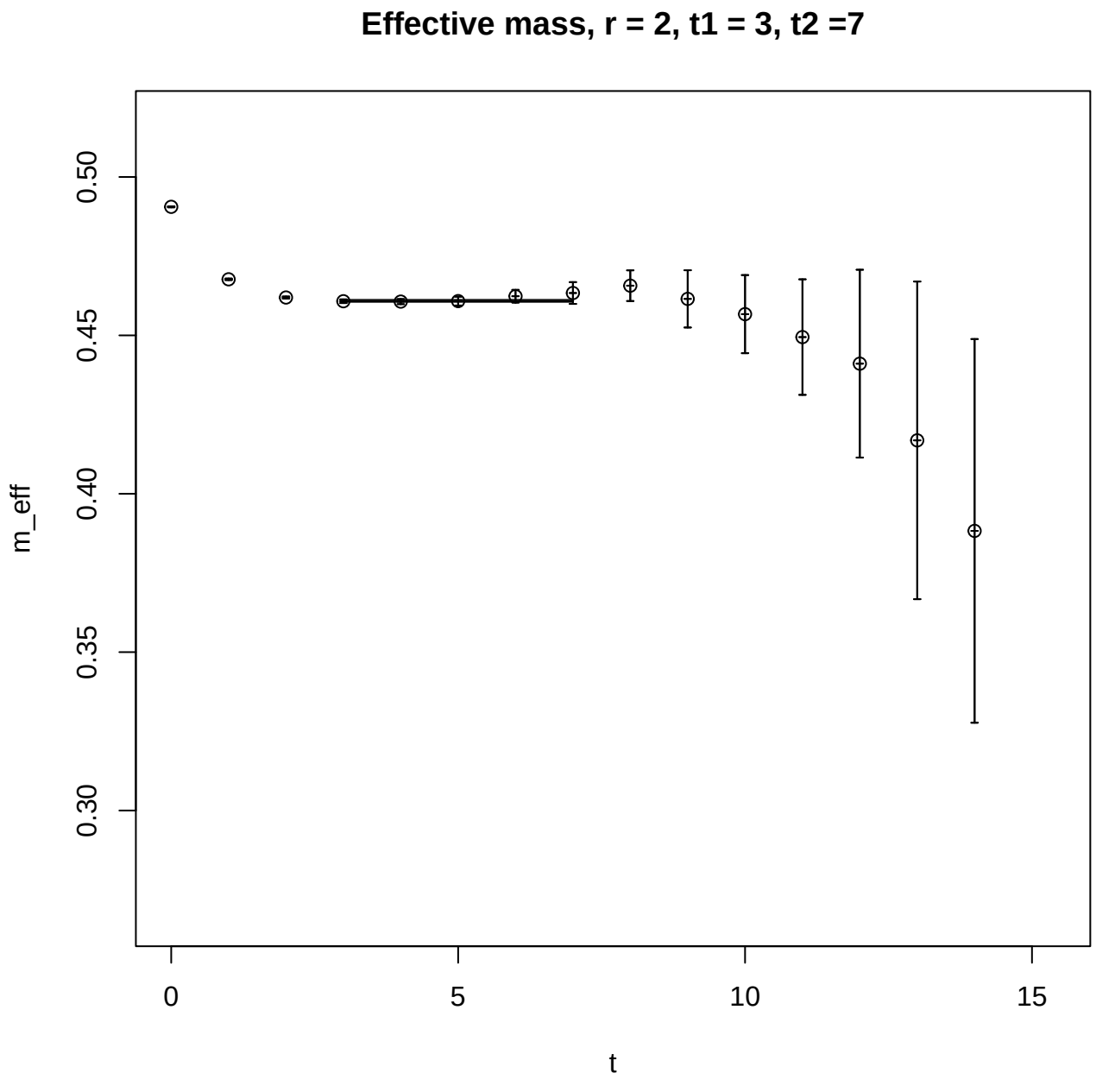

Effective mass, r = 2, t1 = 2, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

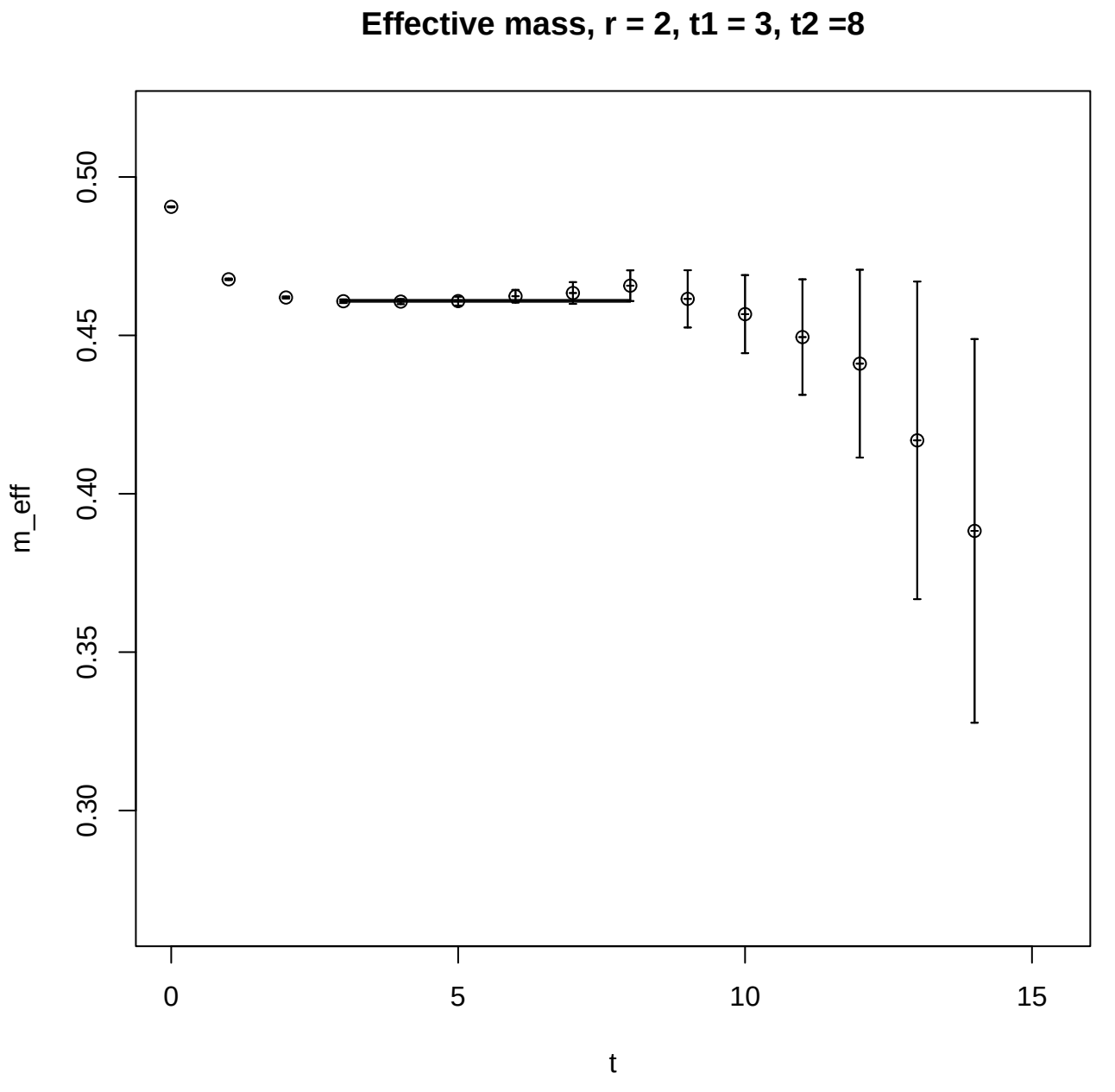

Effective mass, r = 2, t1 = 2, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

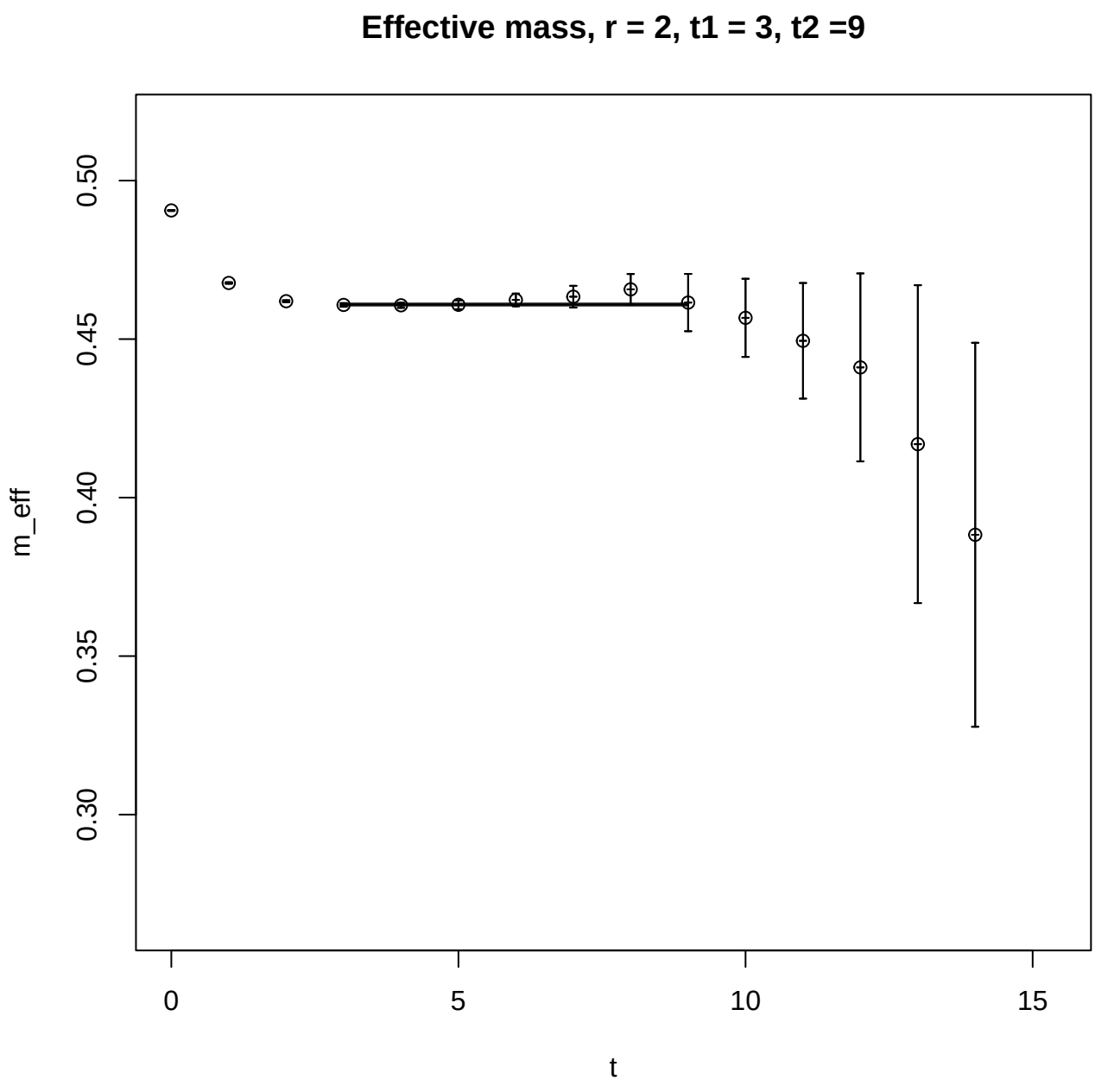

Effective mass, r = 2, t1 = 3, t2 =4
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

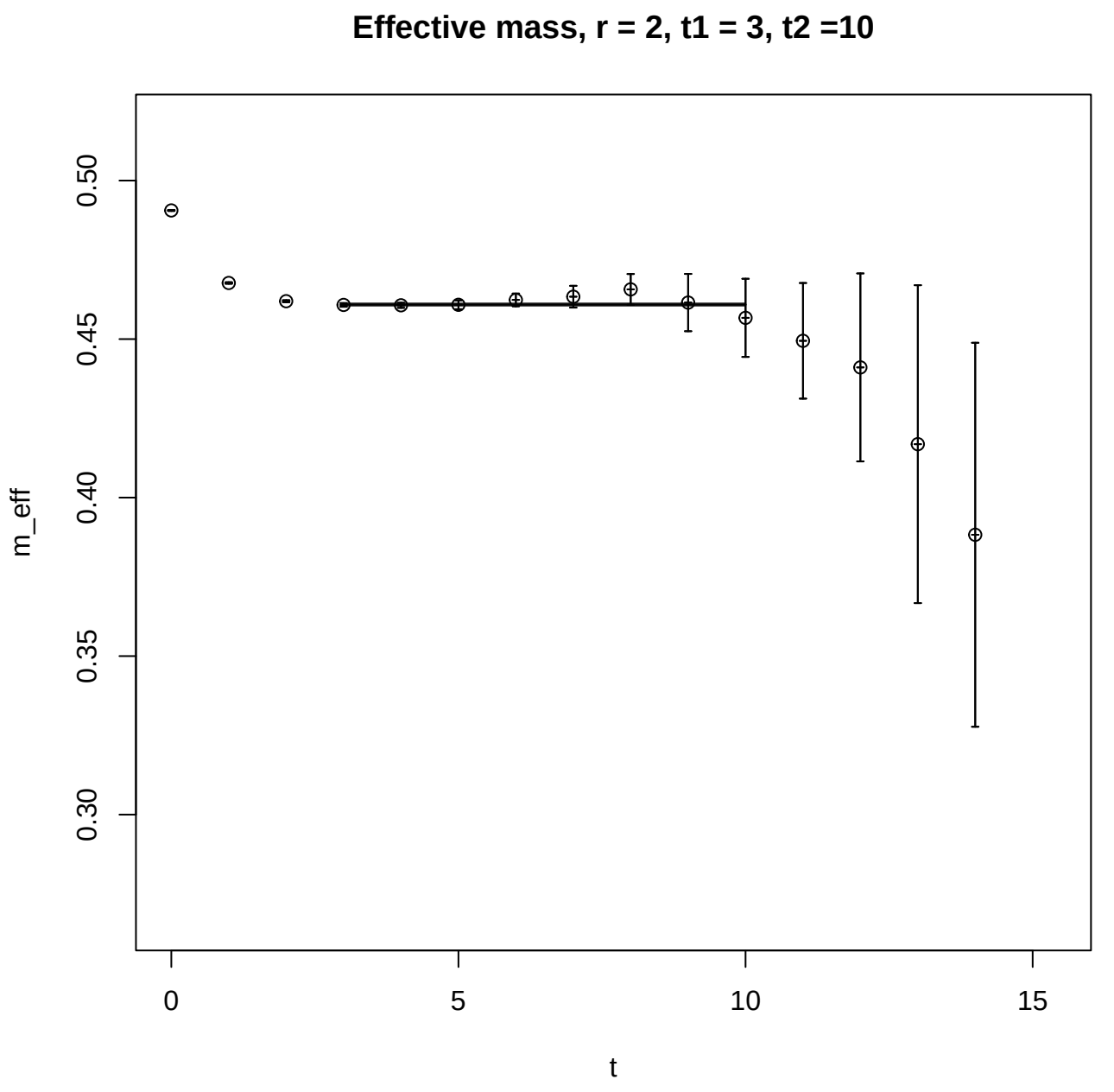

Effective mass, r = 2, t1 = 3, t2 =5
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

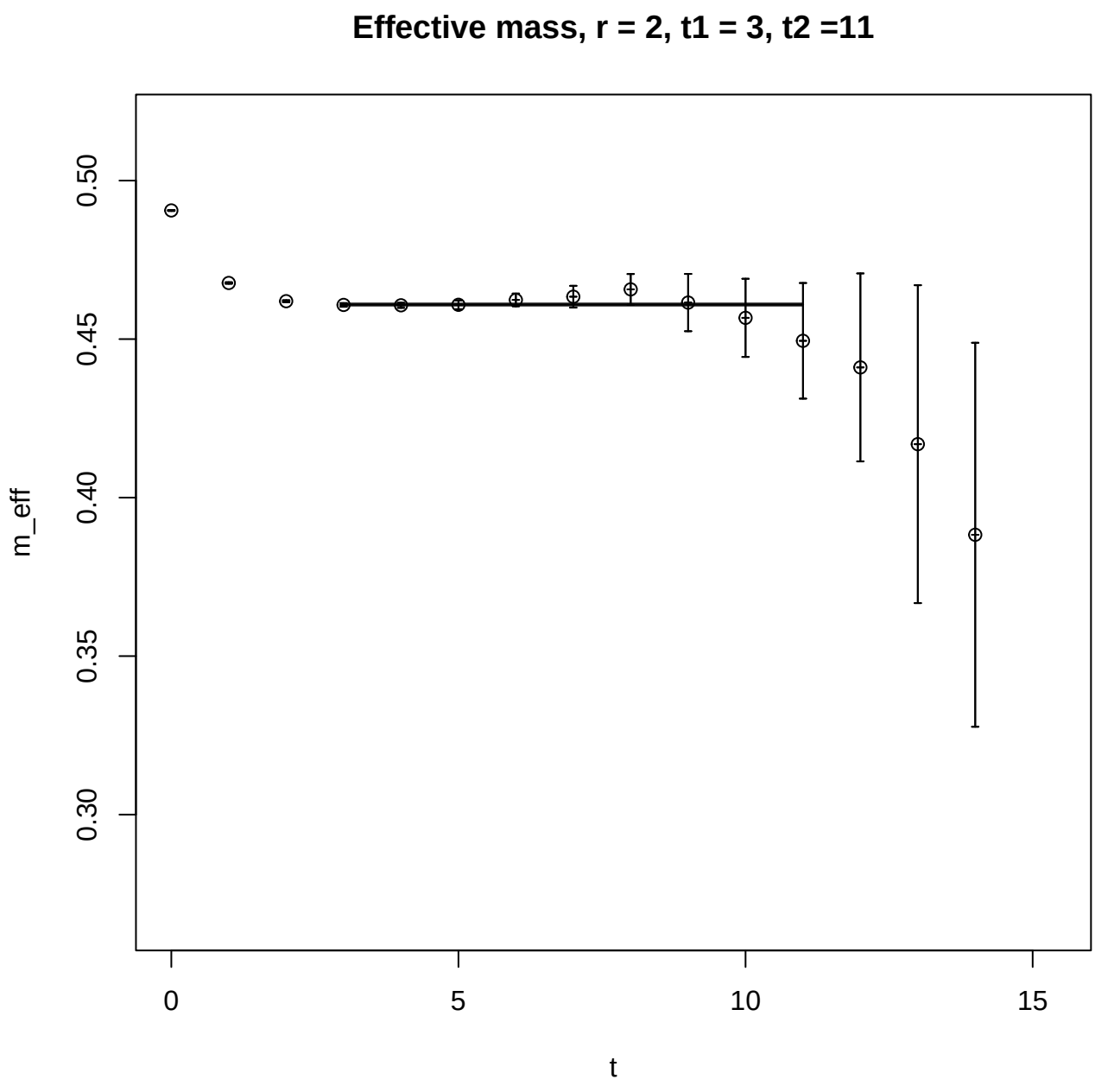

Effective mass, r = 2, t1 = 3, t2 =6
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

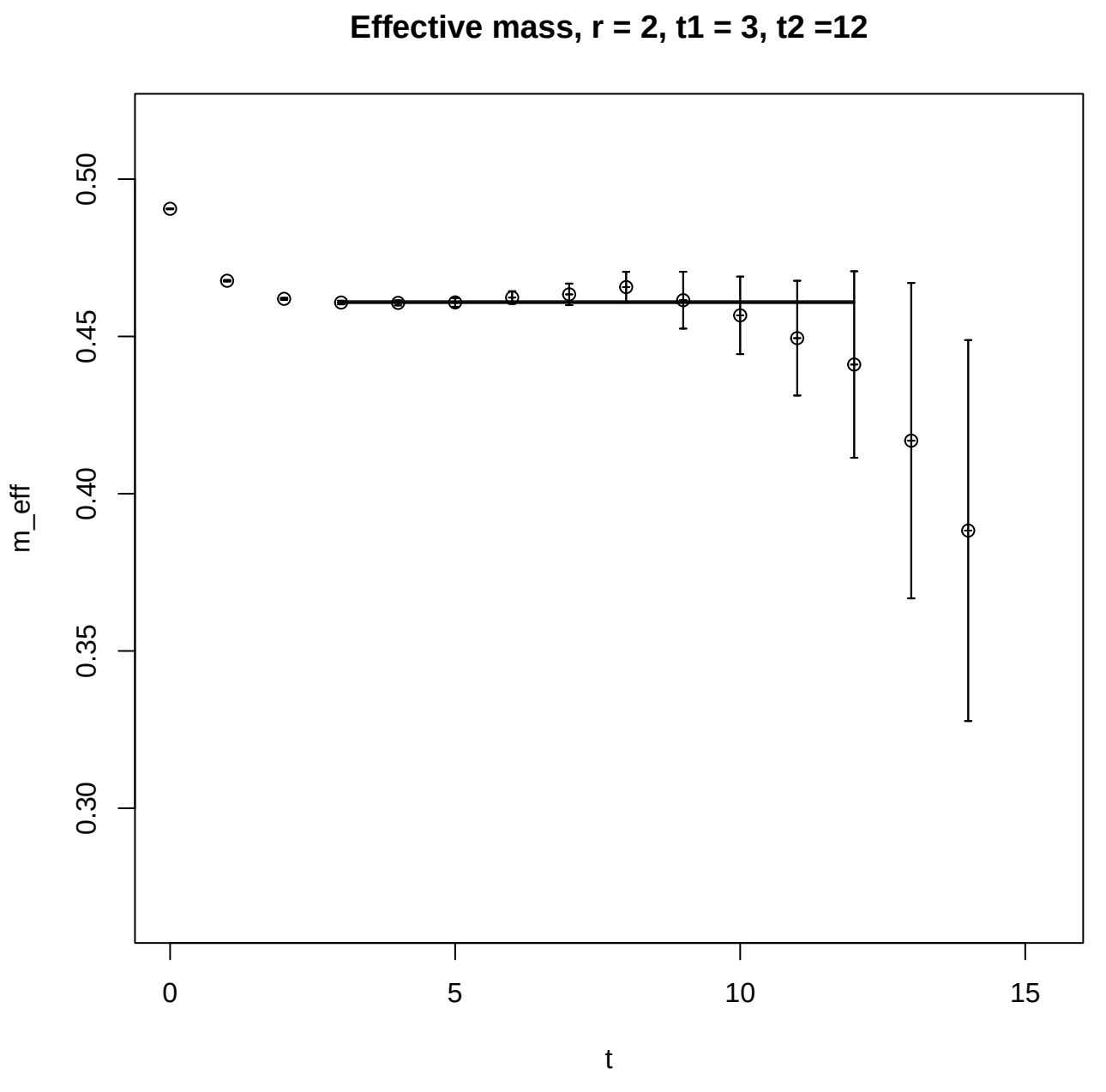

Effective mass, r = 2, t1 = 3, t2 =7
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

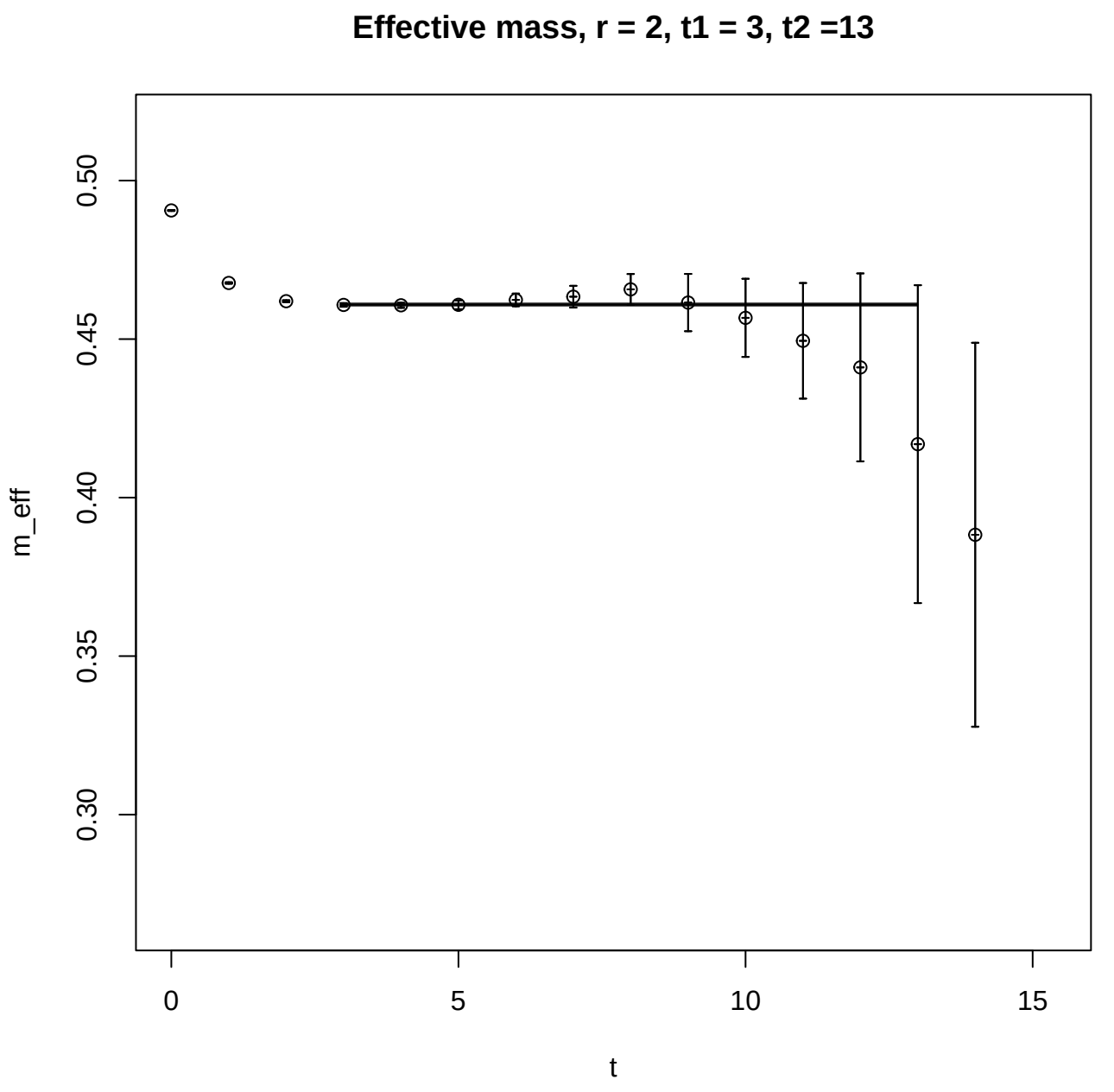

Effective mass, r = 2, t1 = 3, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

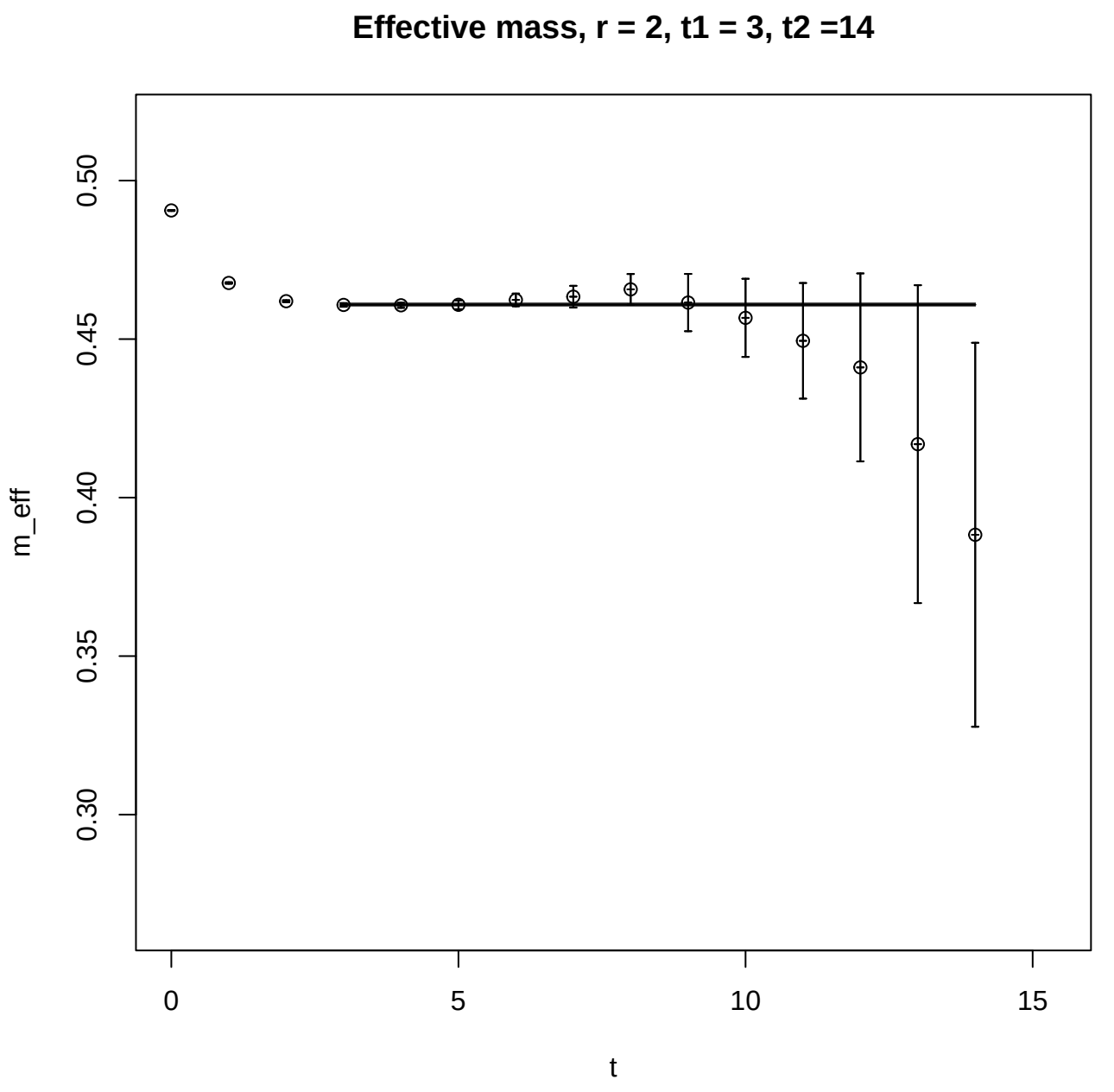

Effective mass, r = 2, t1 = 3, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

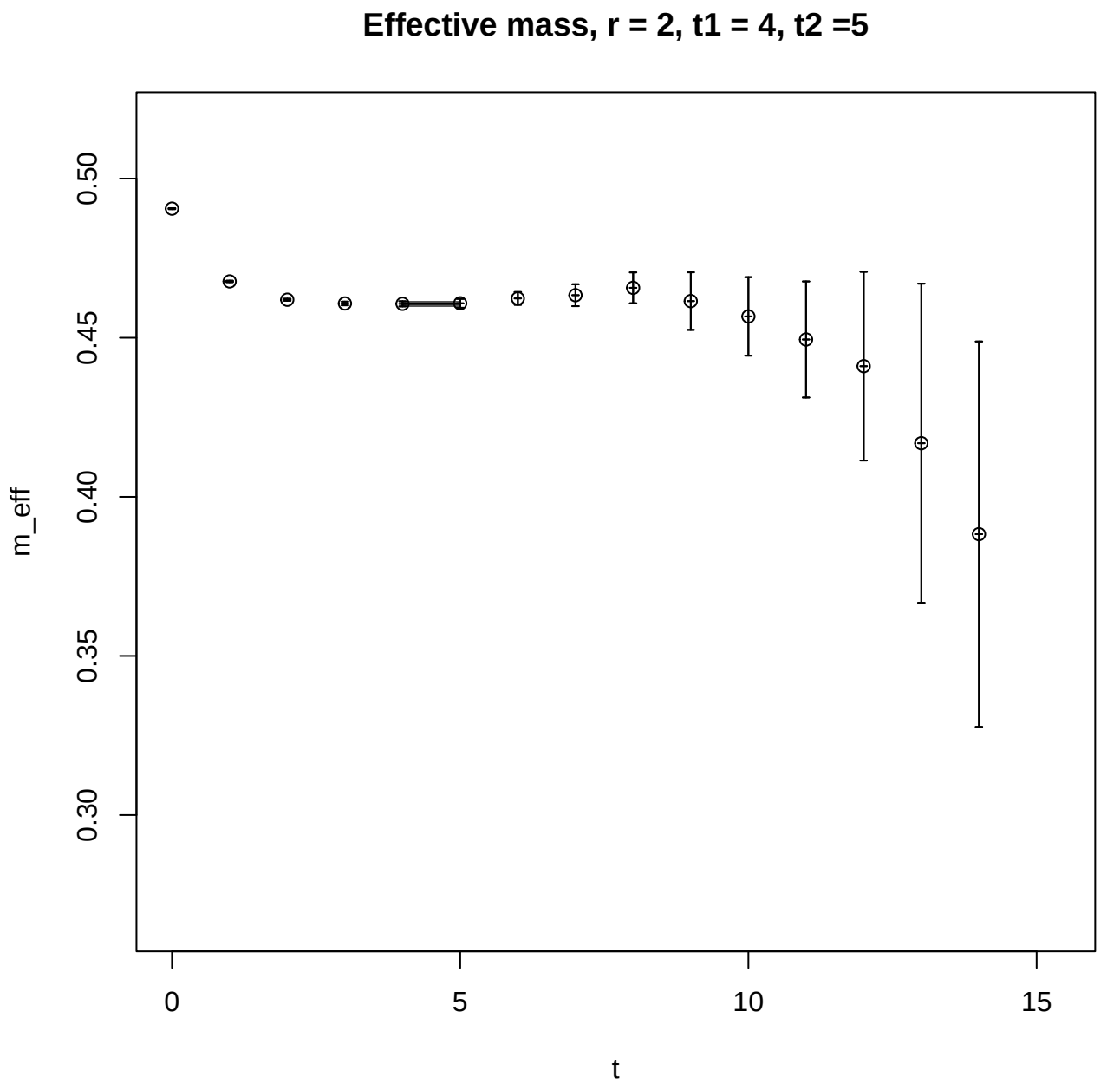

Effective mass, r = 2, t1 = 3, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

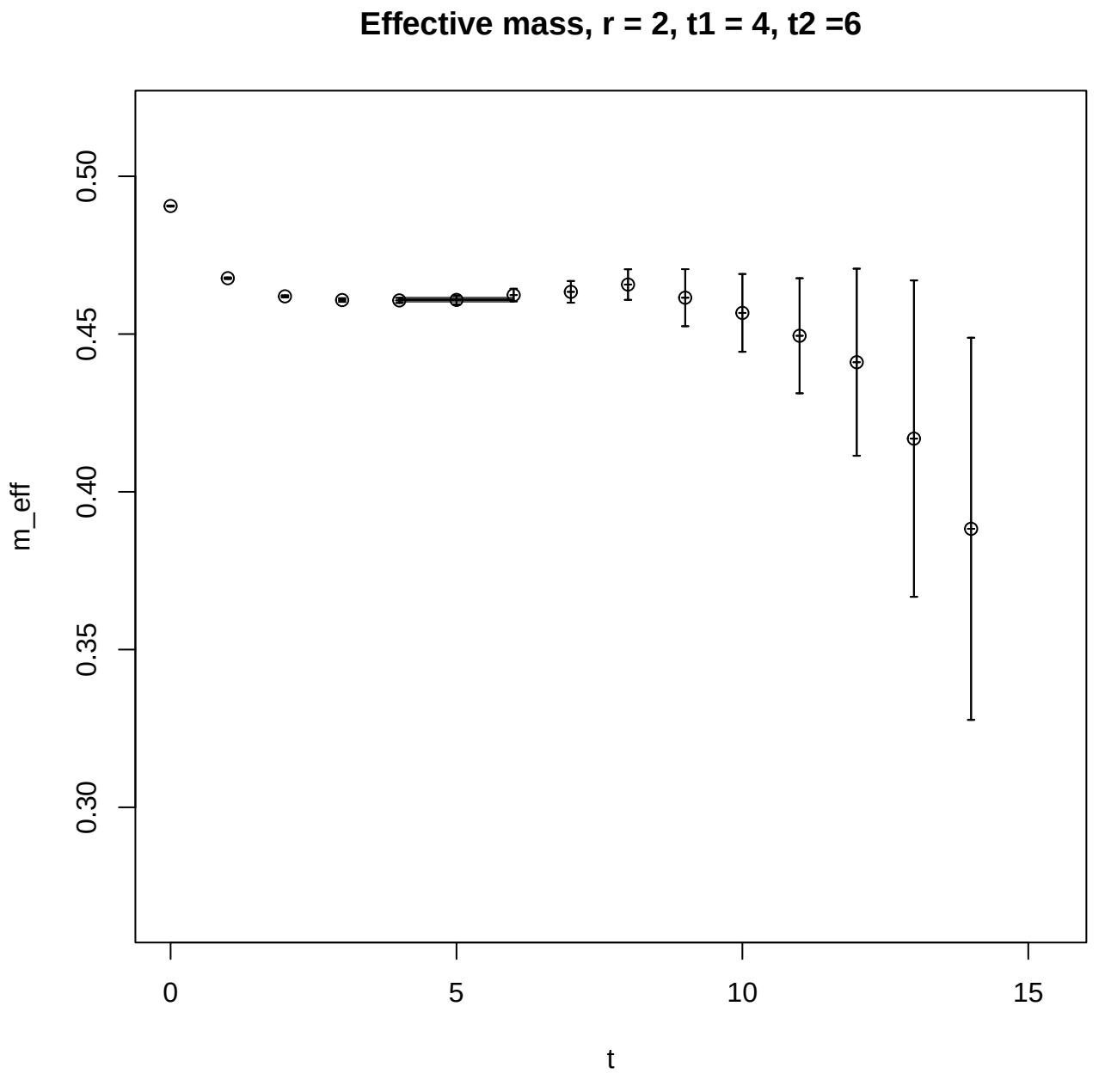

Effective mass, r = 2, t1 = 3, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

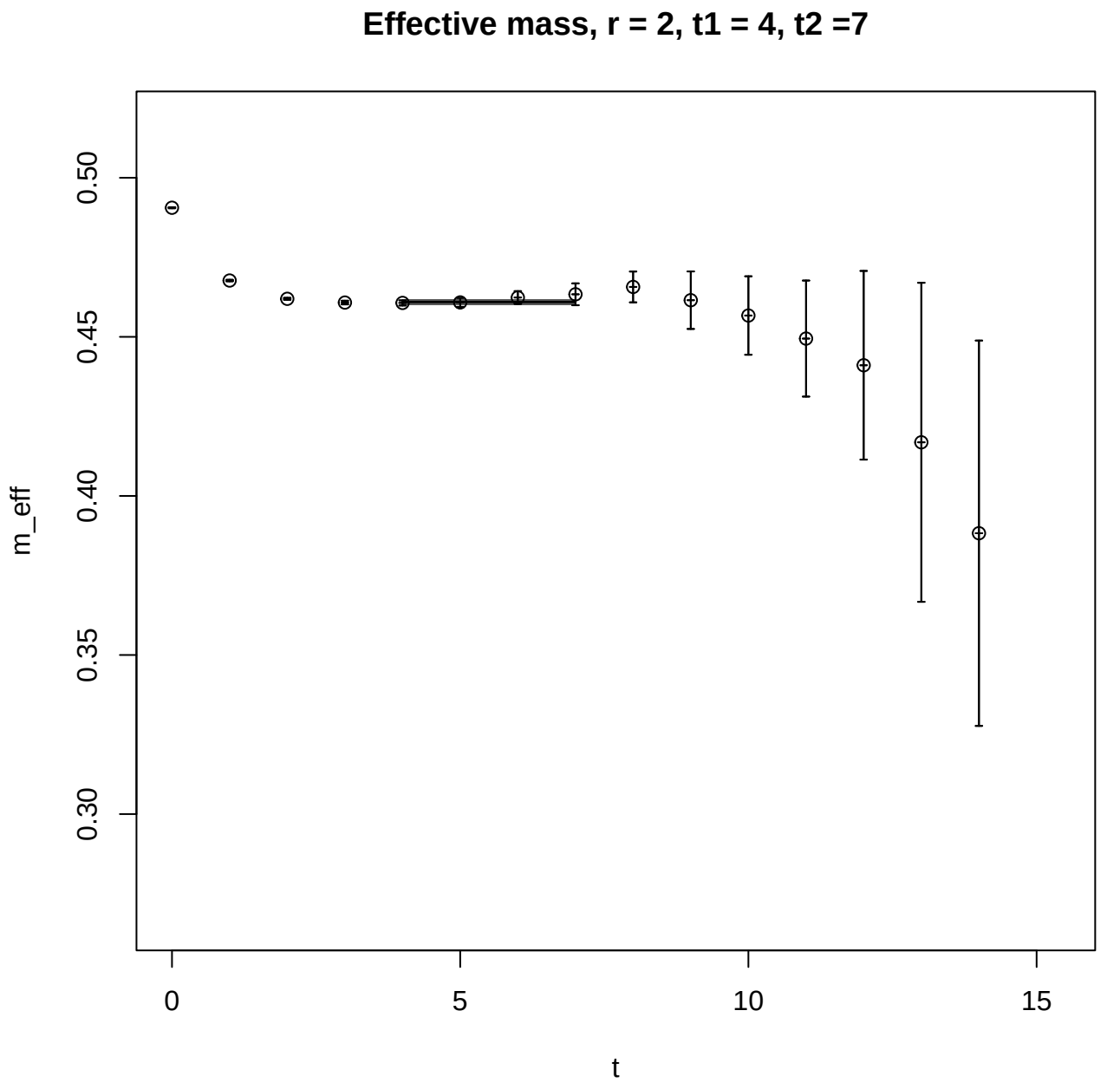

Effective mass, r = 2, t1 = 3, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

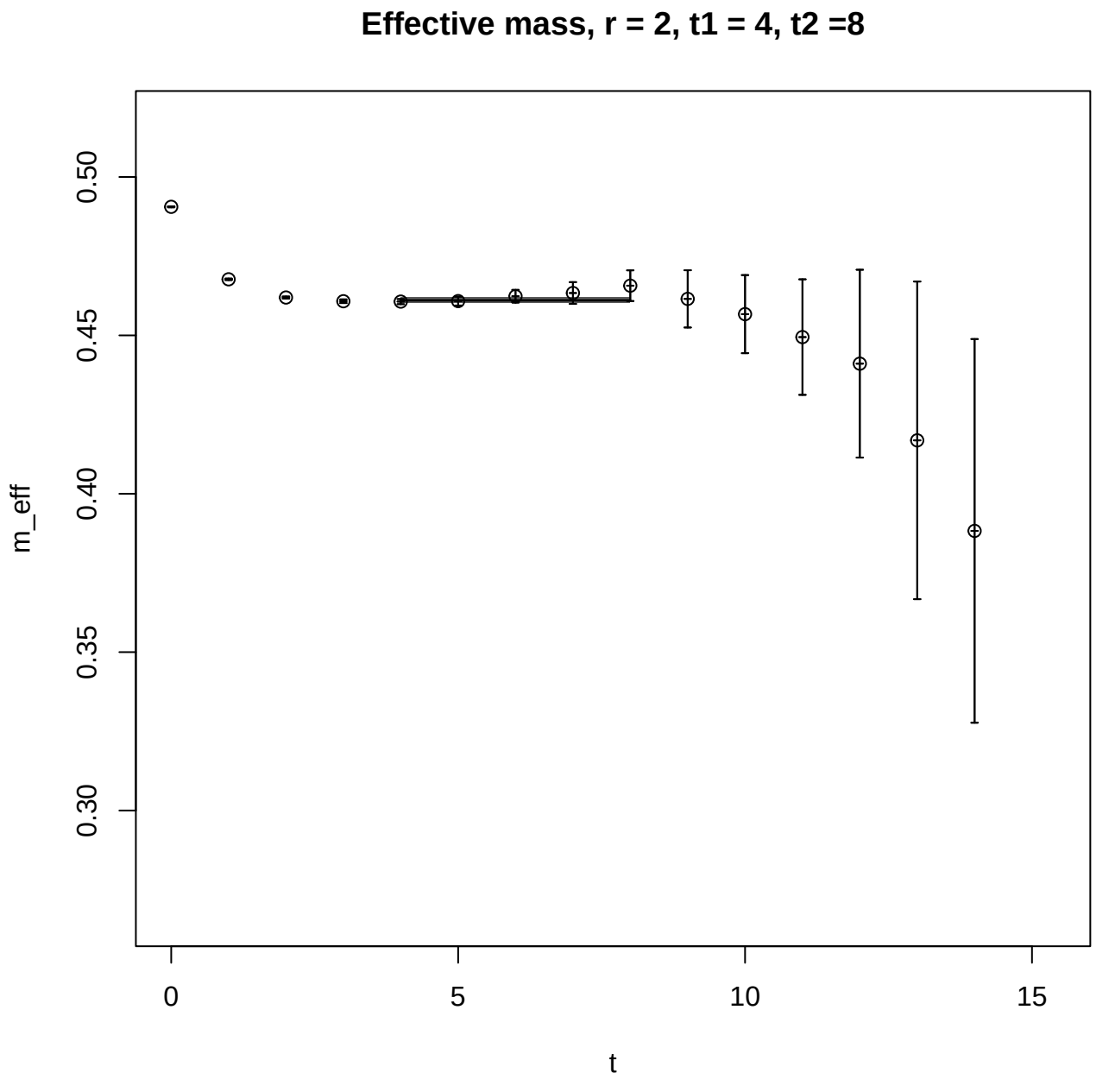

Effective mass, r = 2, t1 = 3, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

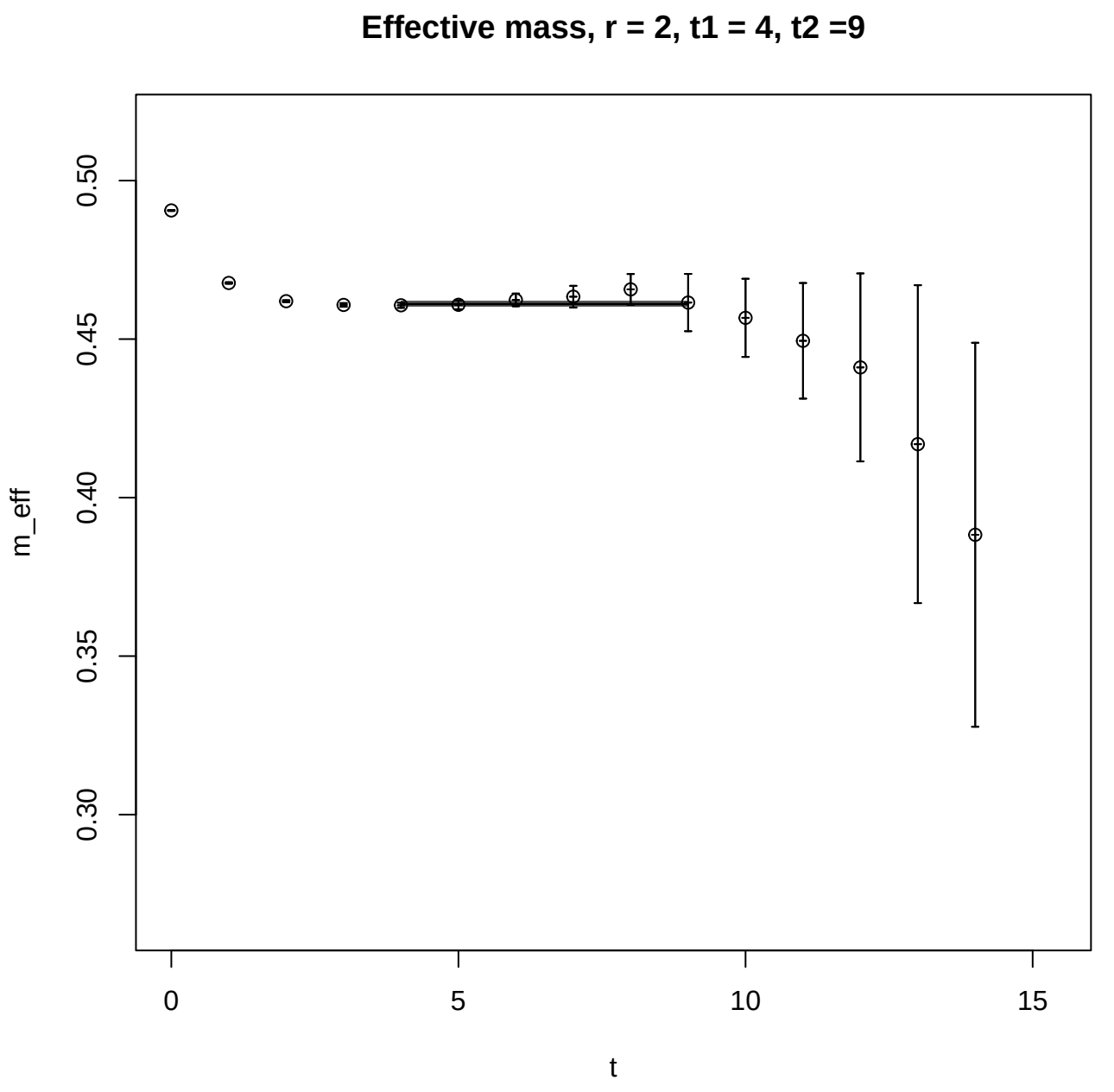

Effective mass, r = 2, t1 = 3, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

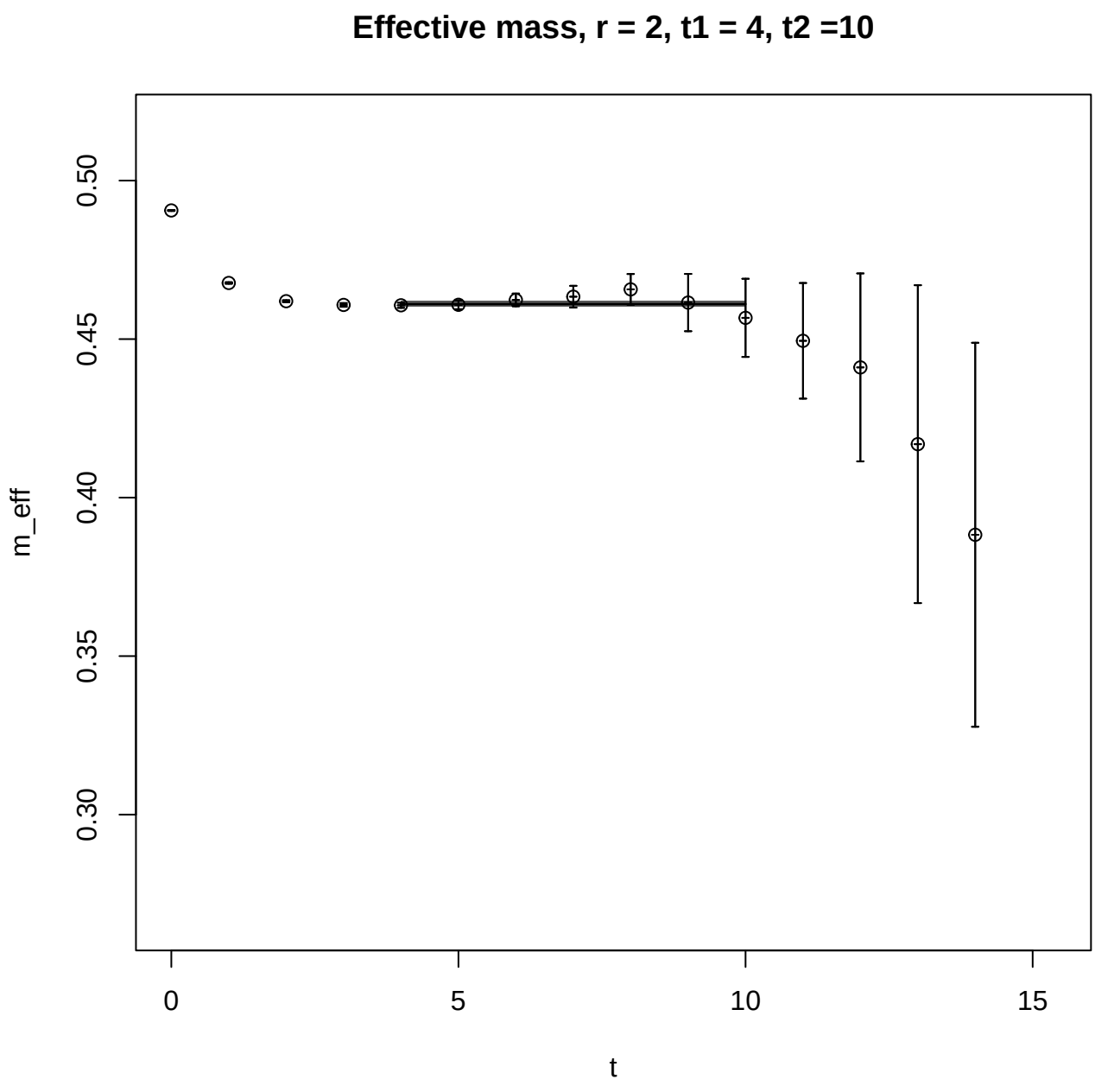

Effective mass, r = 2, t1 = 4, t2 =5
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

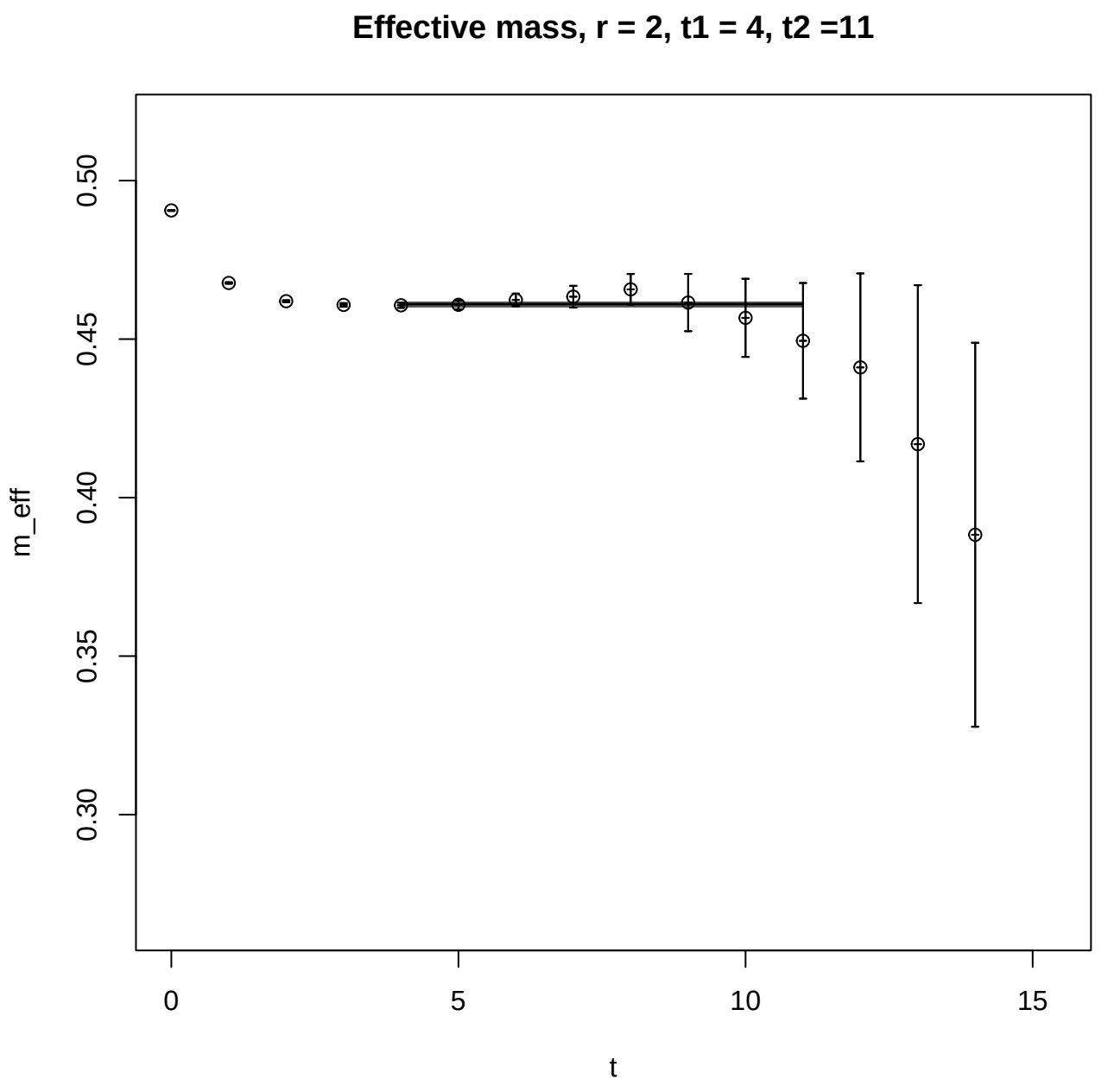

Effective mass, r = 2, t1 = 4, t2 =6
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

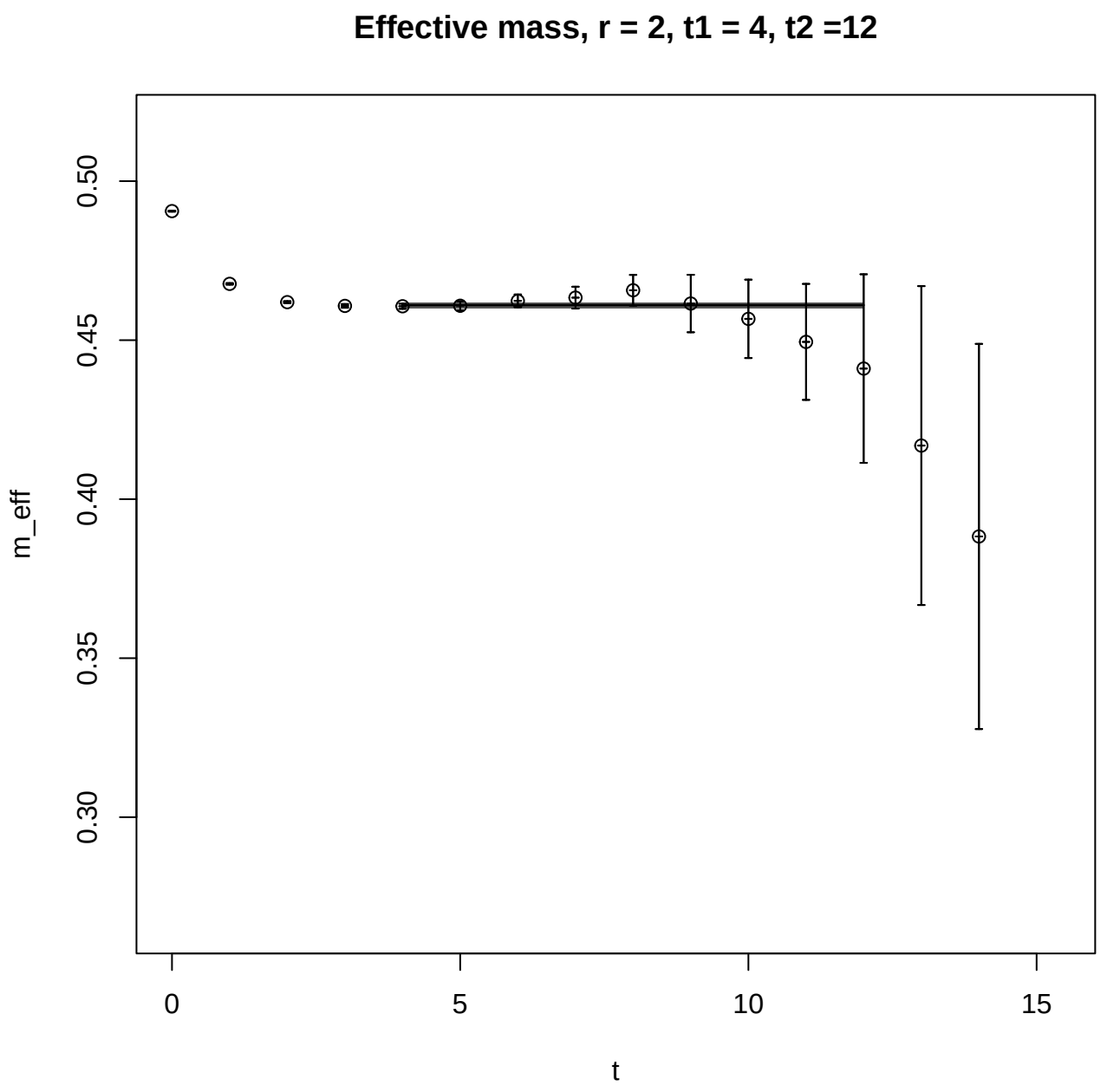

Effective mass, r = 2, t1 = 4, t2 =7
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

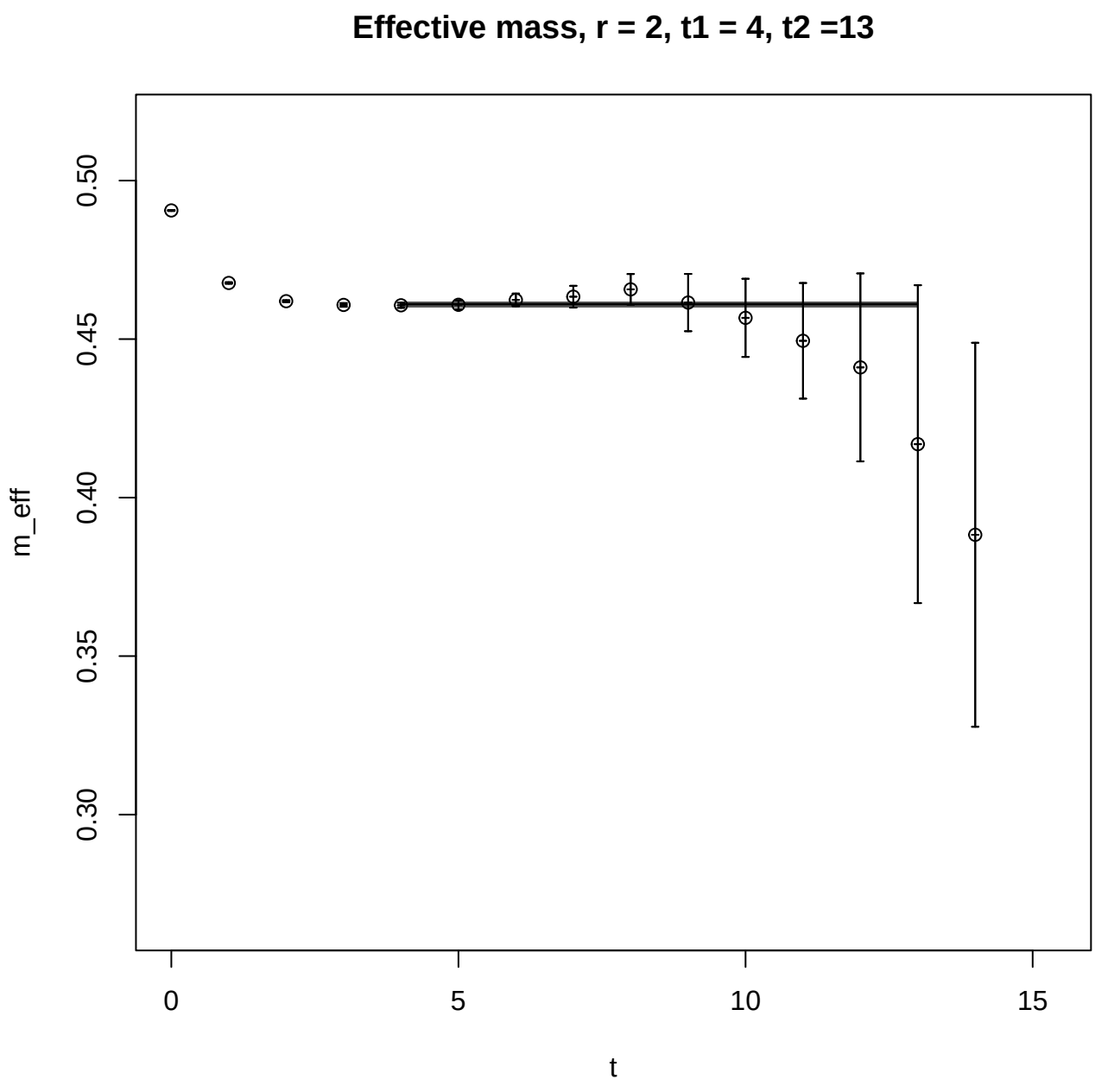

Effective mass, r = 2, t1 = 4, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

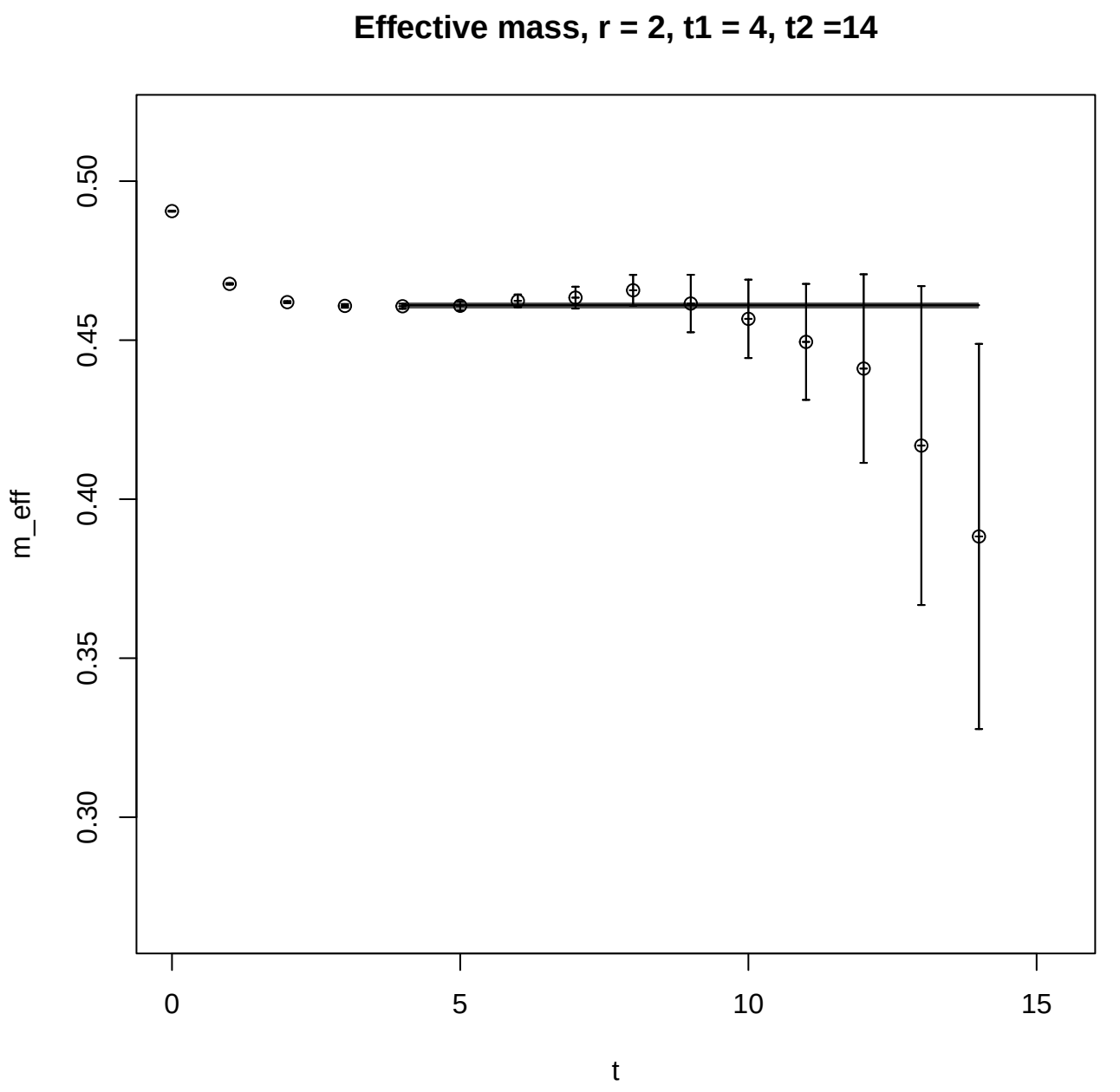

Effective mass, r = 2, t1 = 4, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

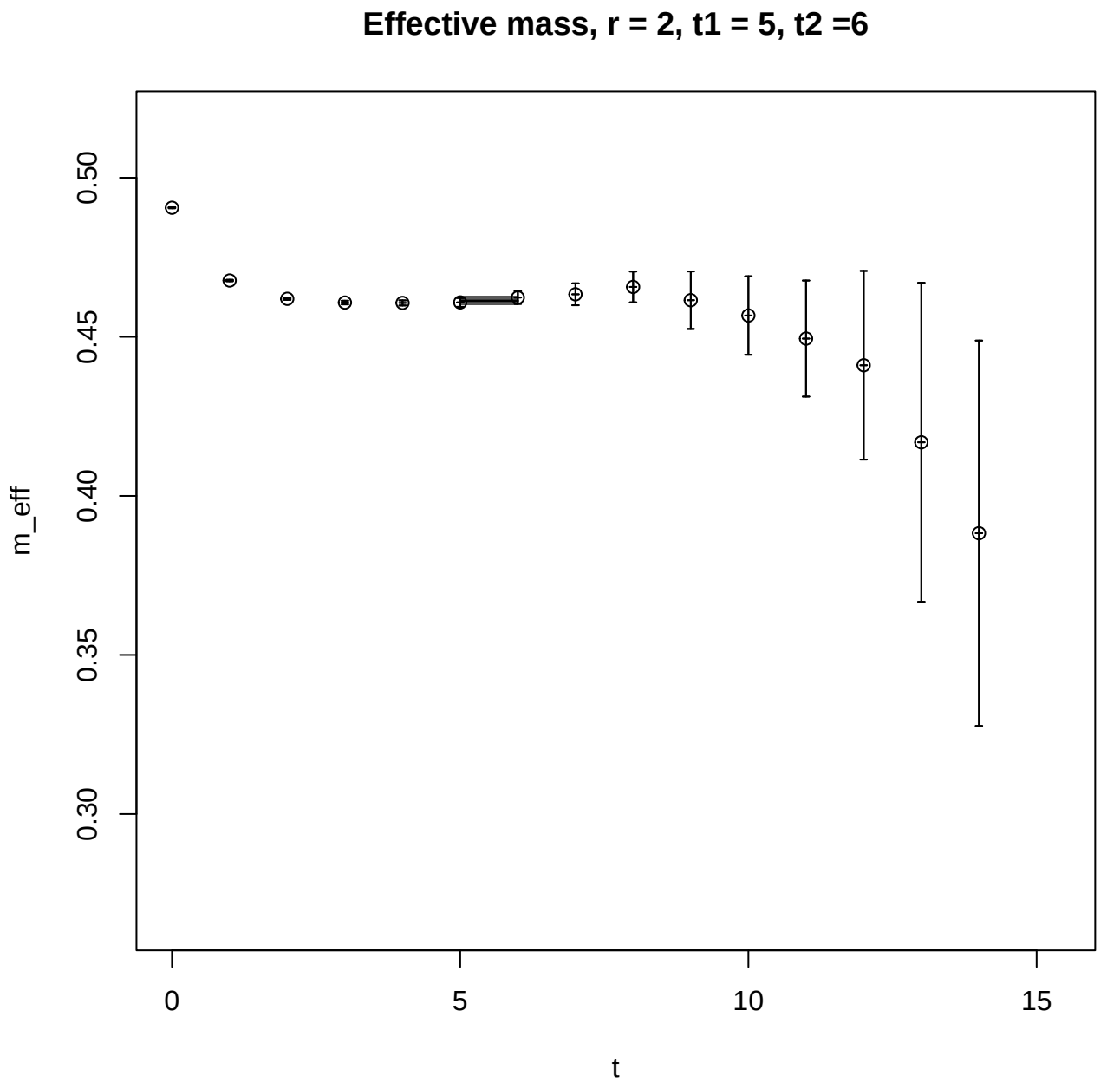

Effective mass, r = 2, t1 = 4, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

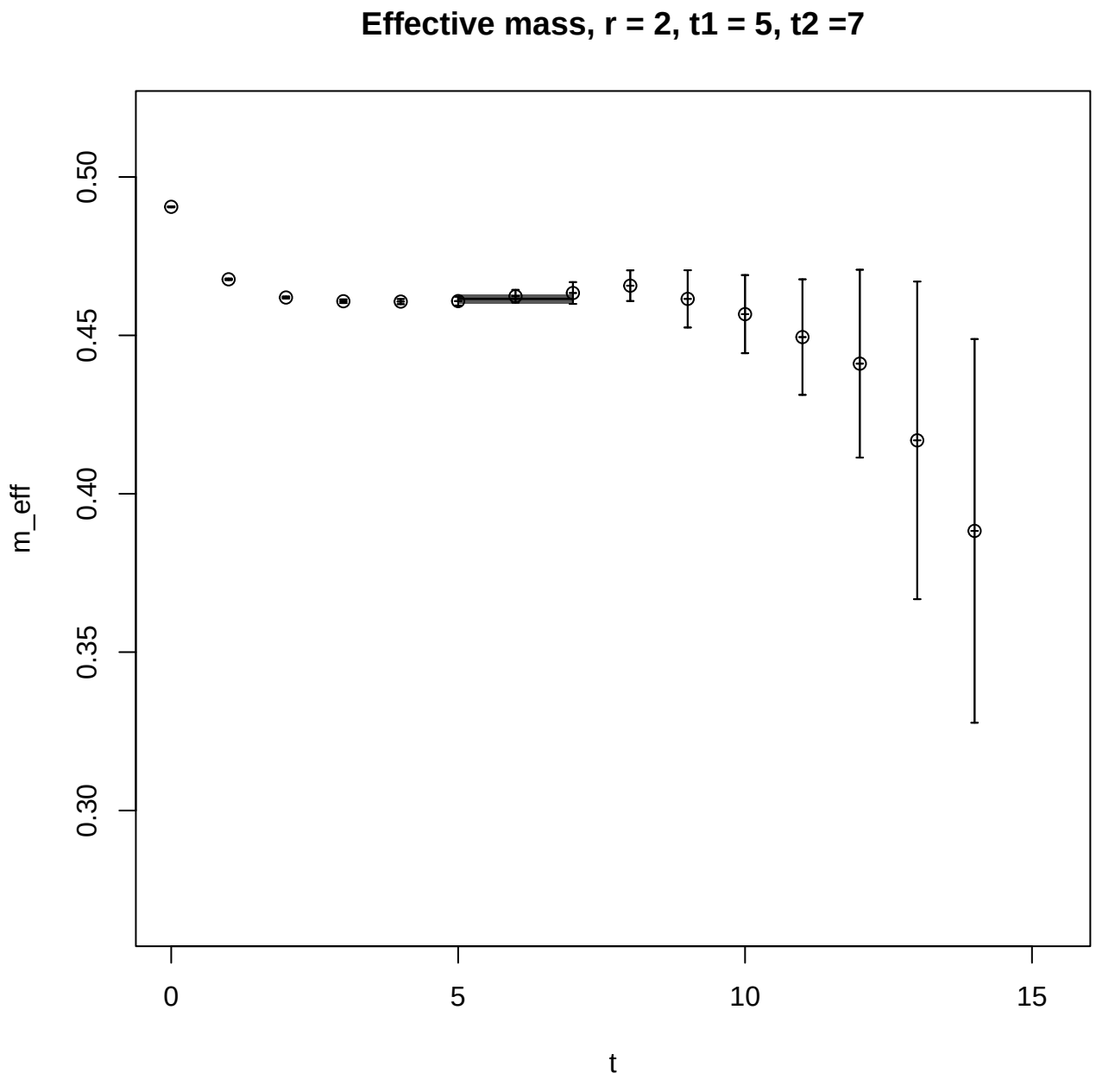

Effective mass, r = 2, t1 = 4, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

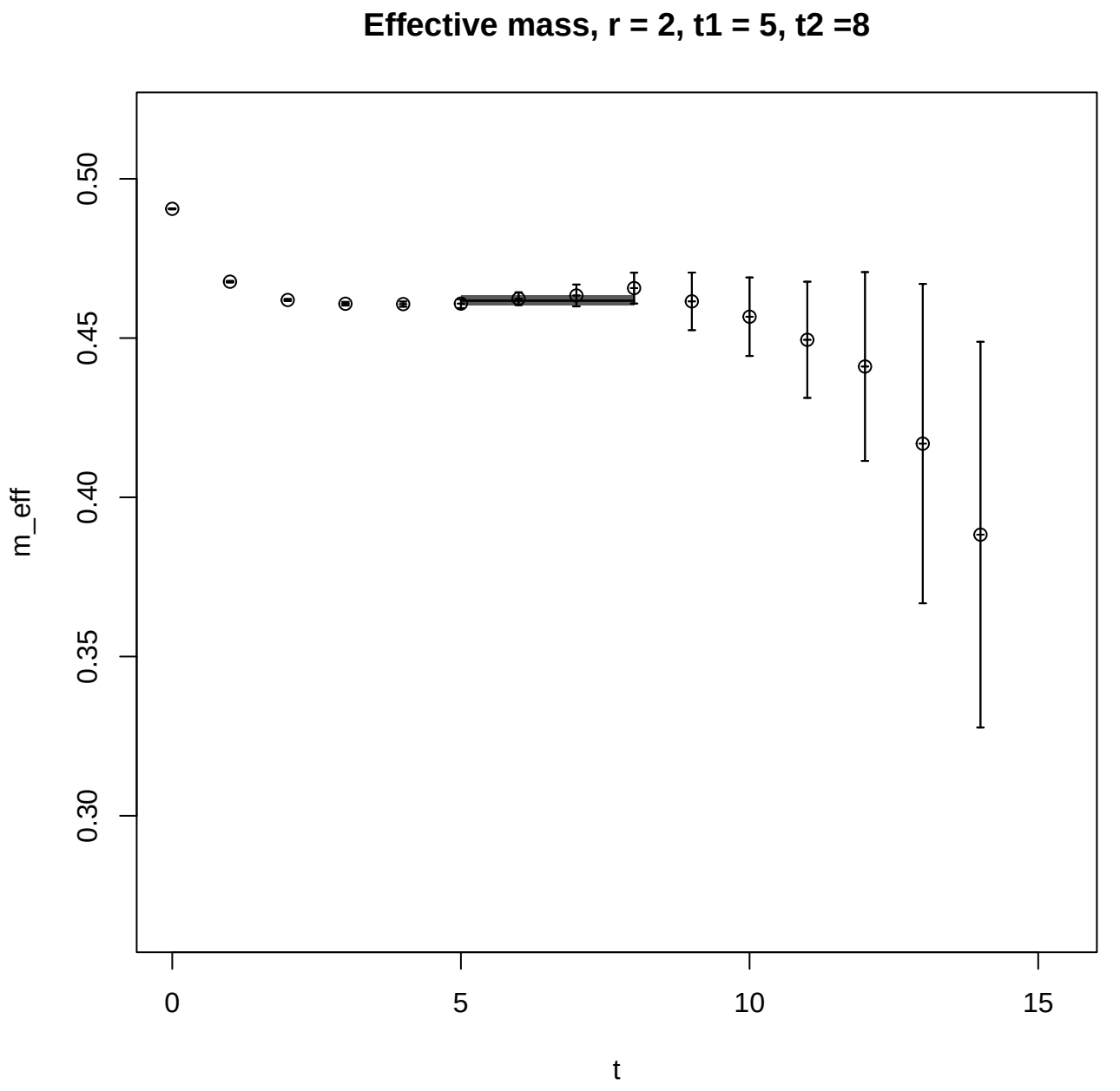

Effective mass, r = 2, t1 = 4, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

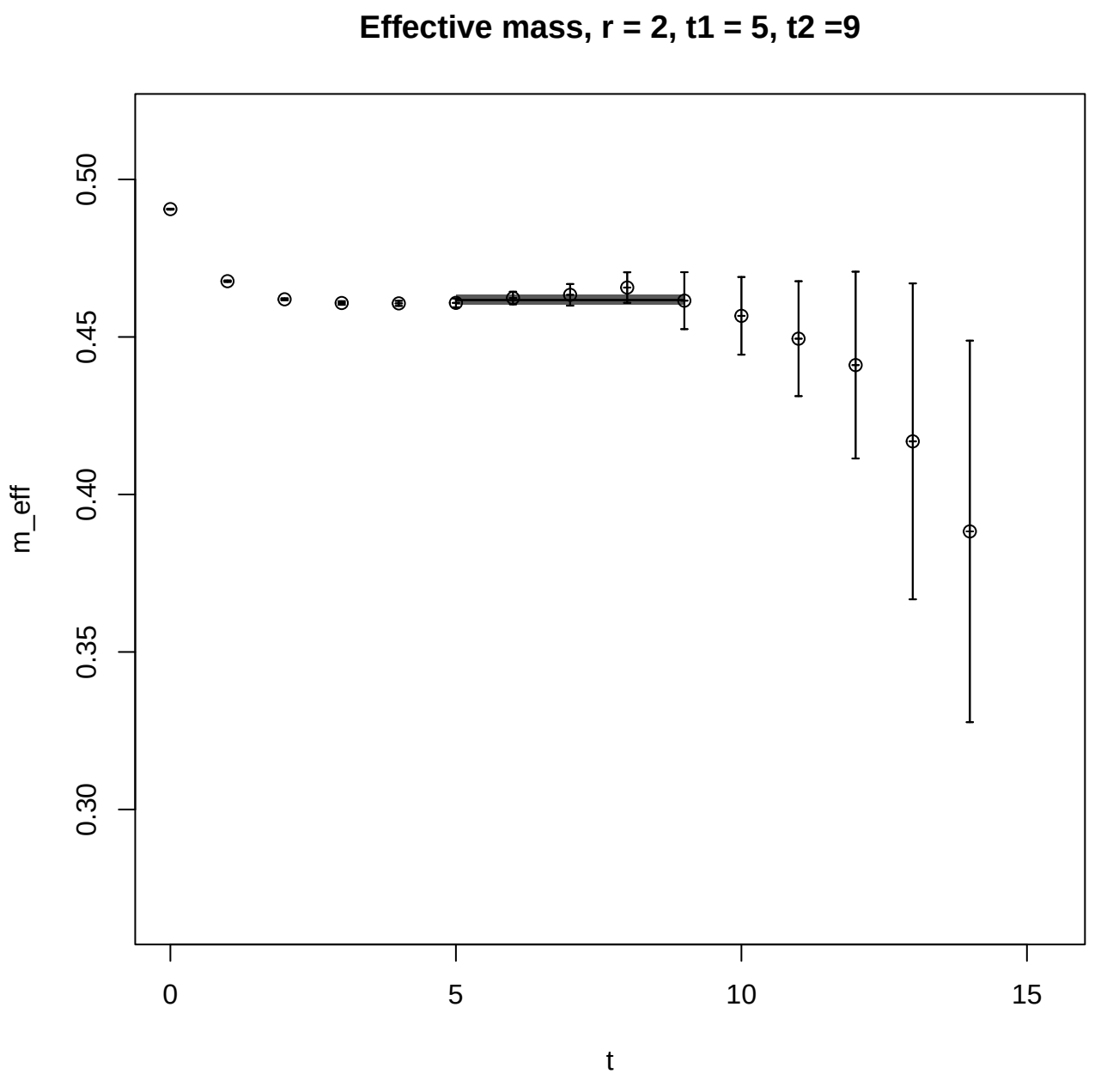

Effective mass, r = 2, t1 = 4, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

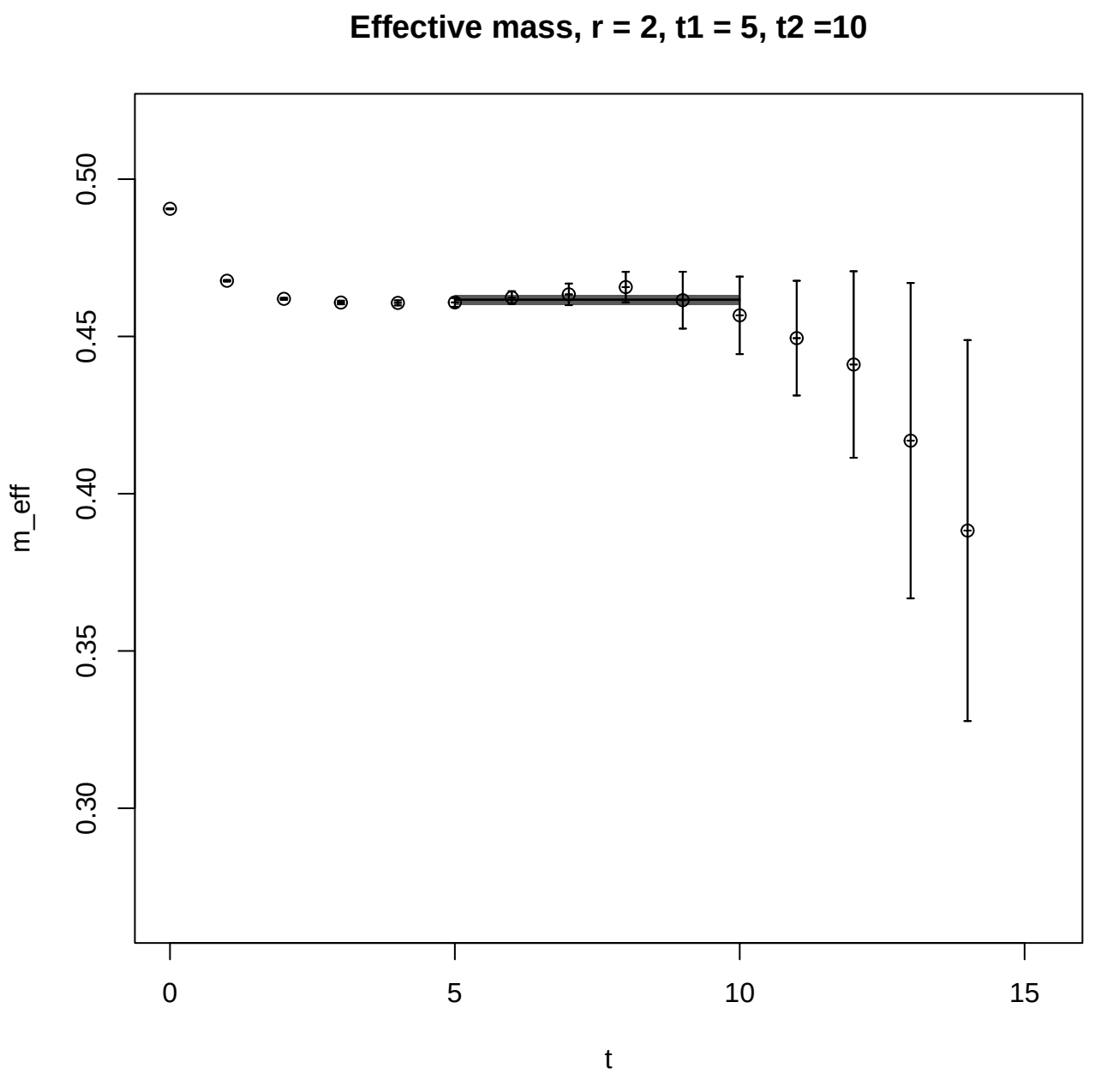

Effective mass, r = 2, t1 = 4, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

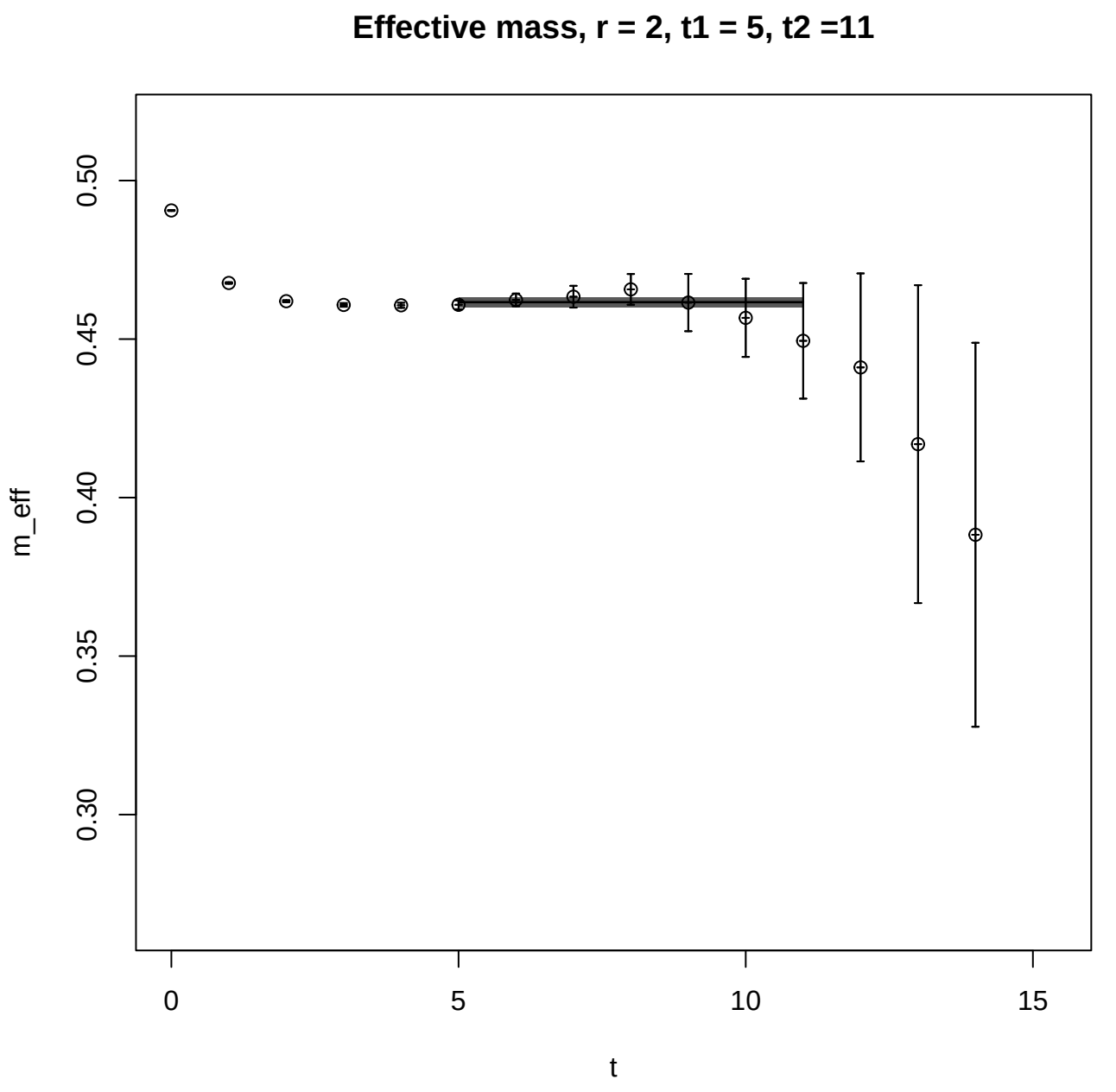

Effective mass, r = 2, t1 = 5, t2 =6
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

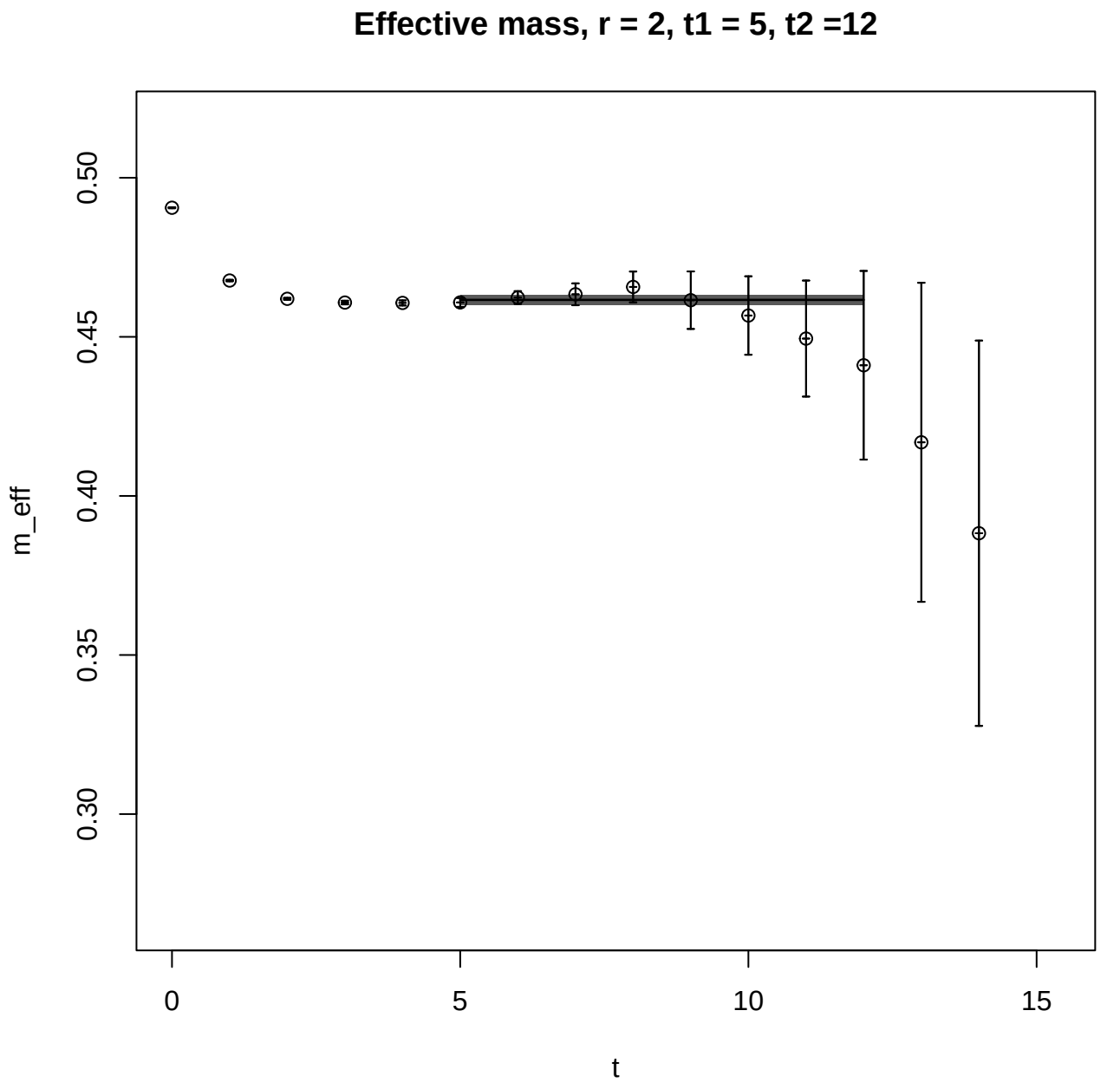

Effective mass, r = 2, t1 = 5, t2 =7
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

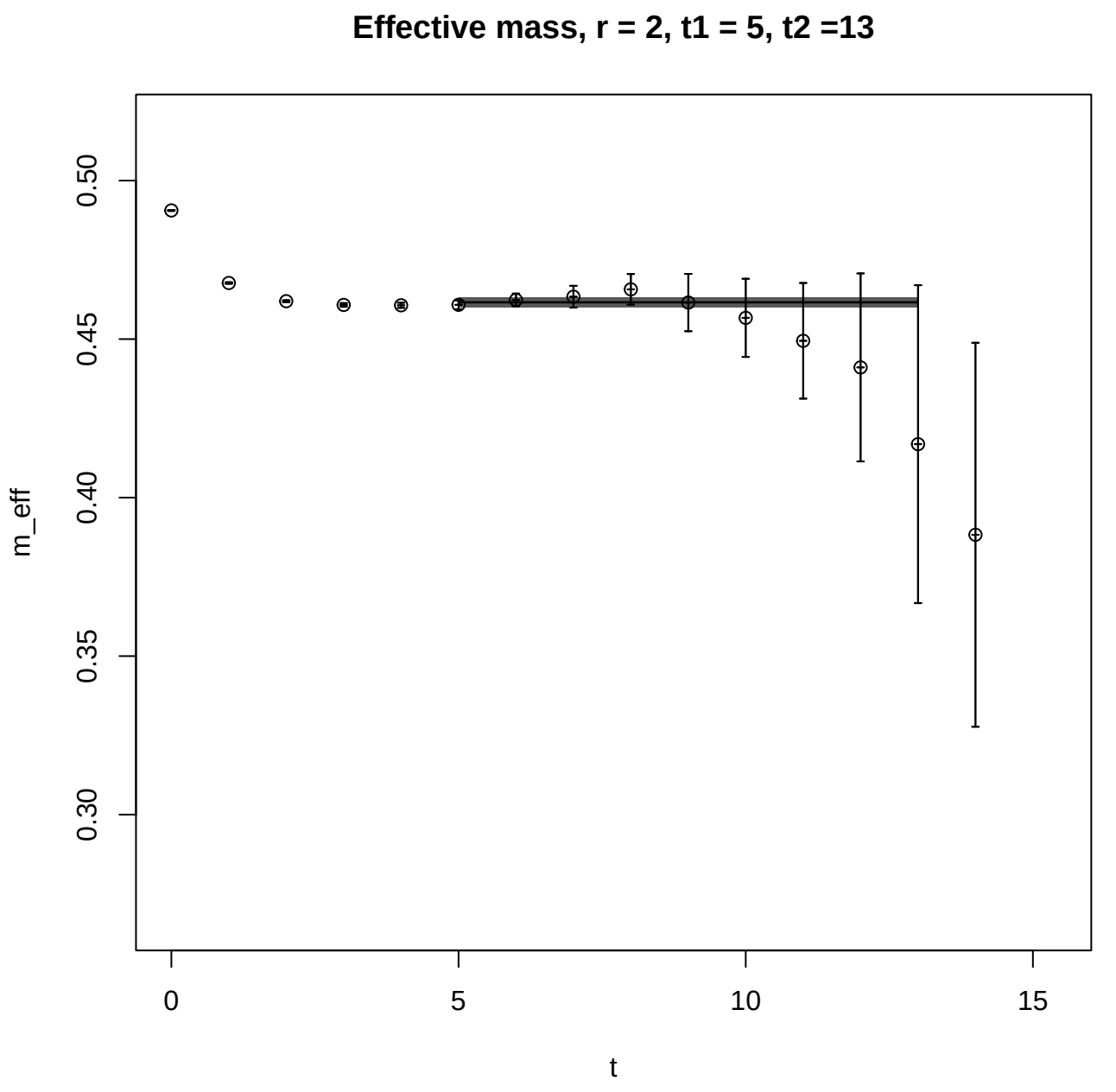

Effective mass, r = 2, t1 = 5, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

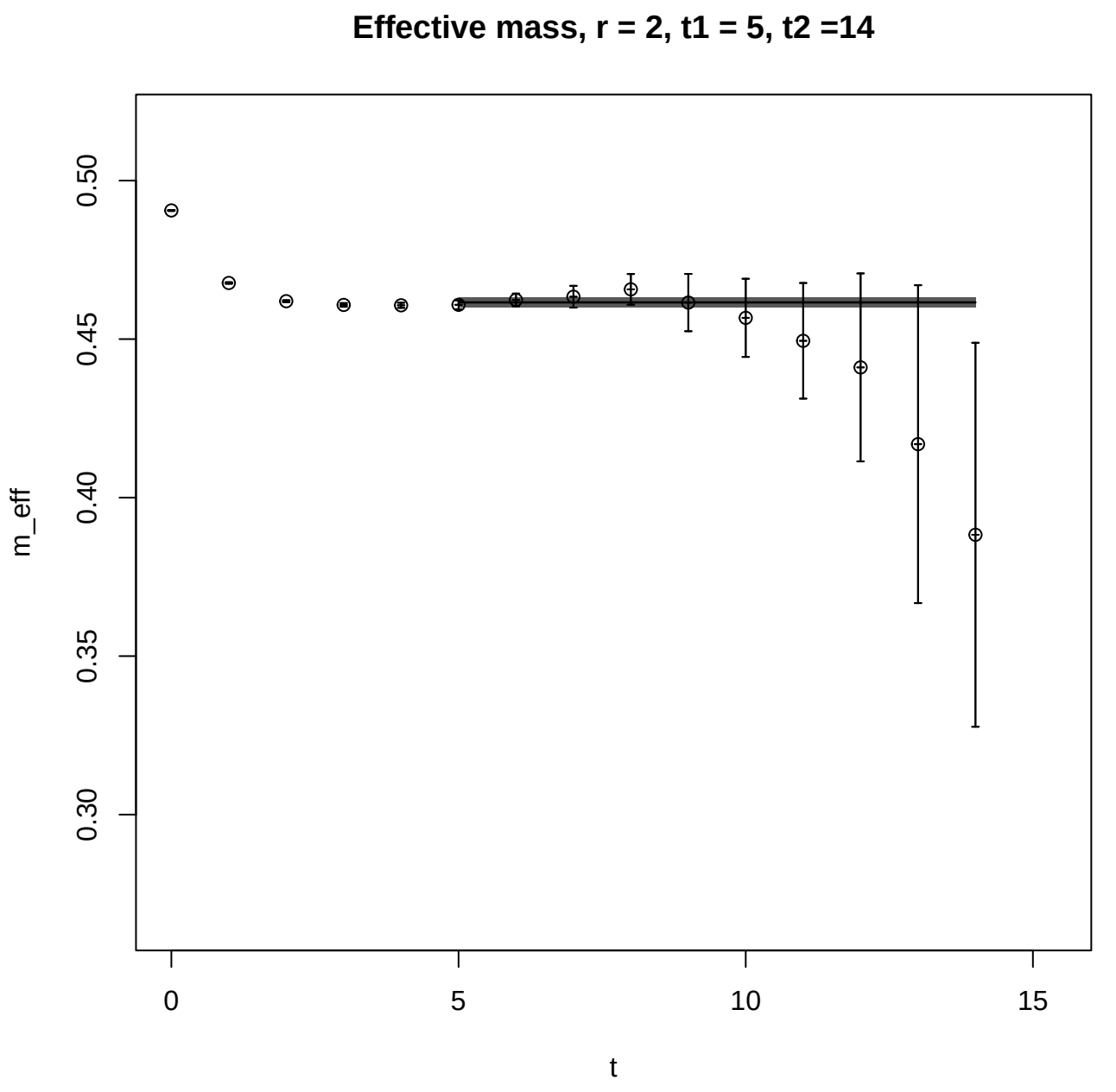

Effective mass, r = 2, t1 = 5, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

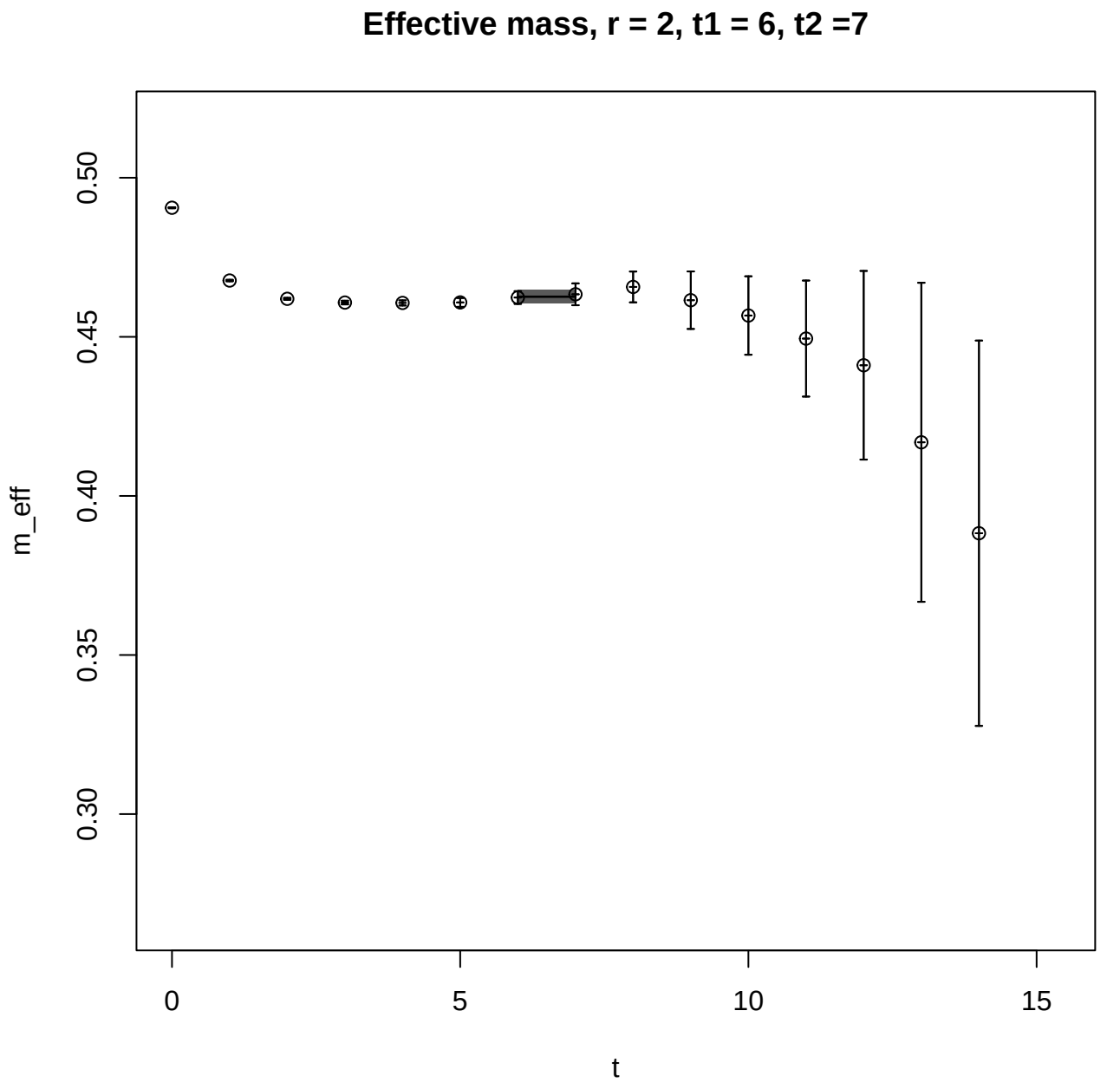

Effective mass, r = 2, t1 = 5, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

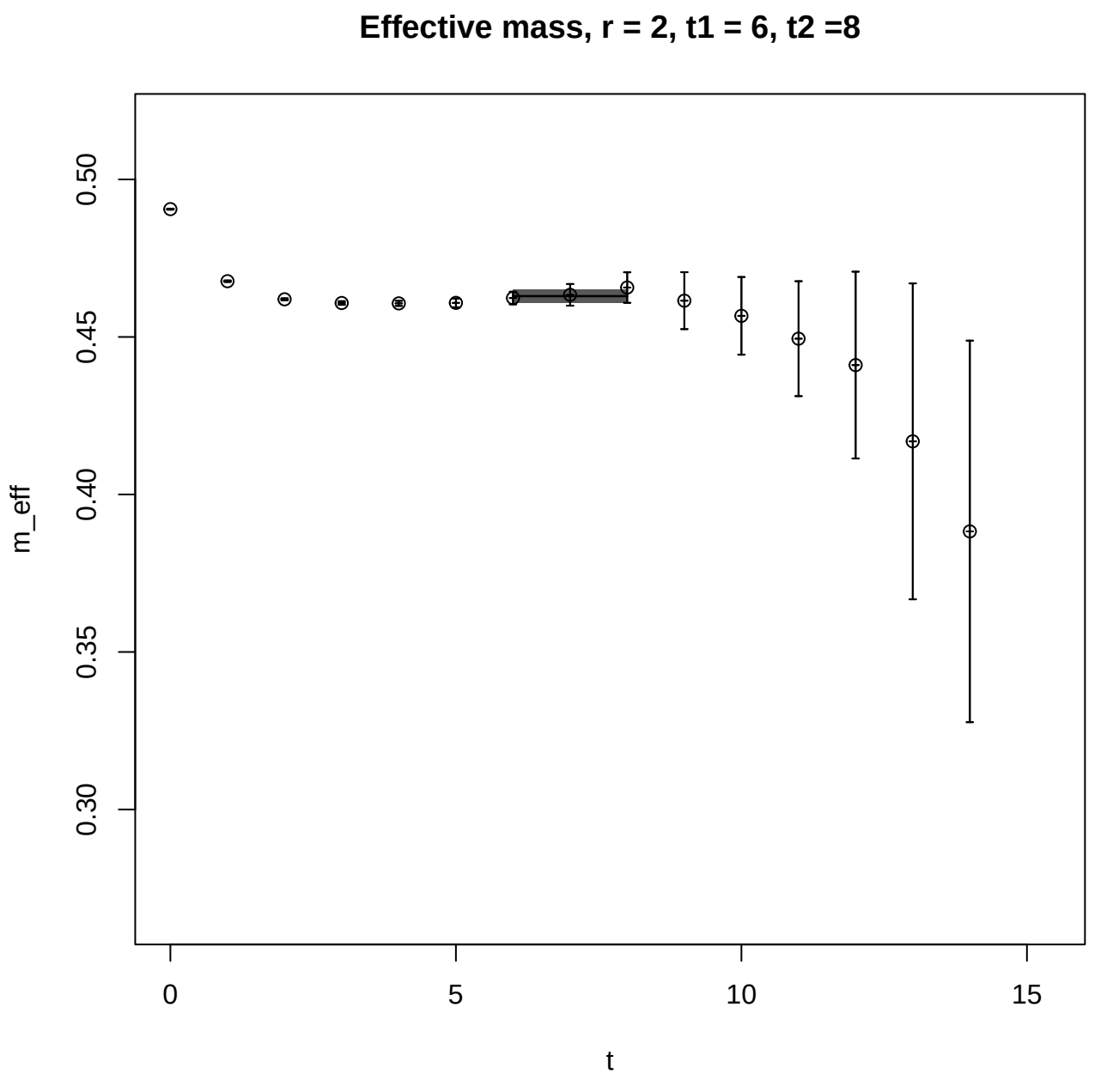

Effective mass, r = 2, t1 = 5, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

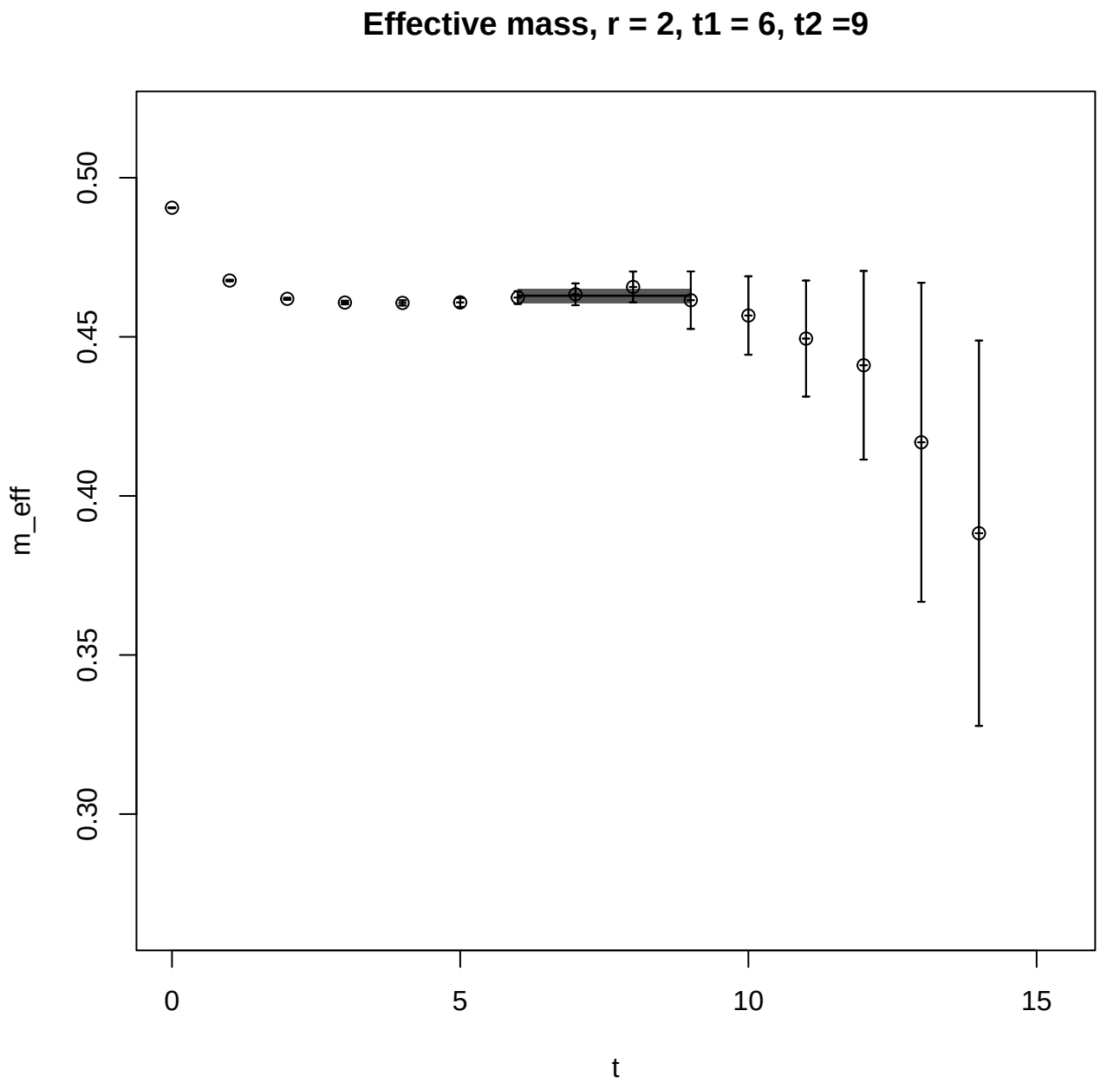

Effective mass, r = 2, t1 = 5, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

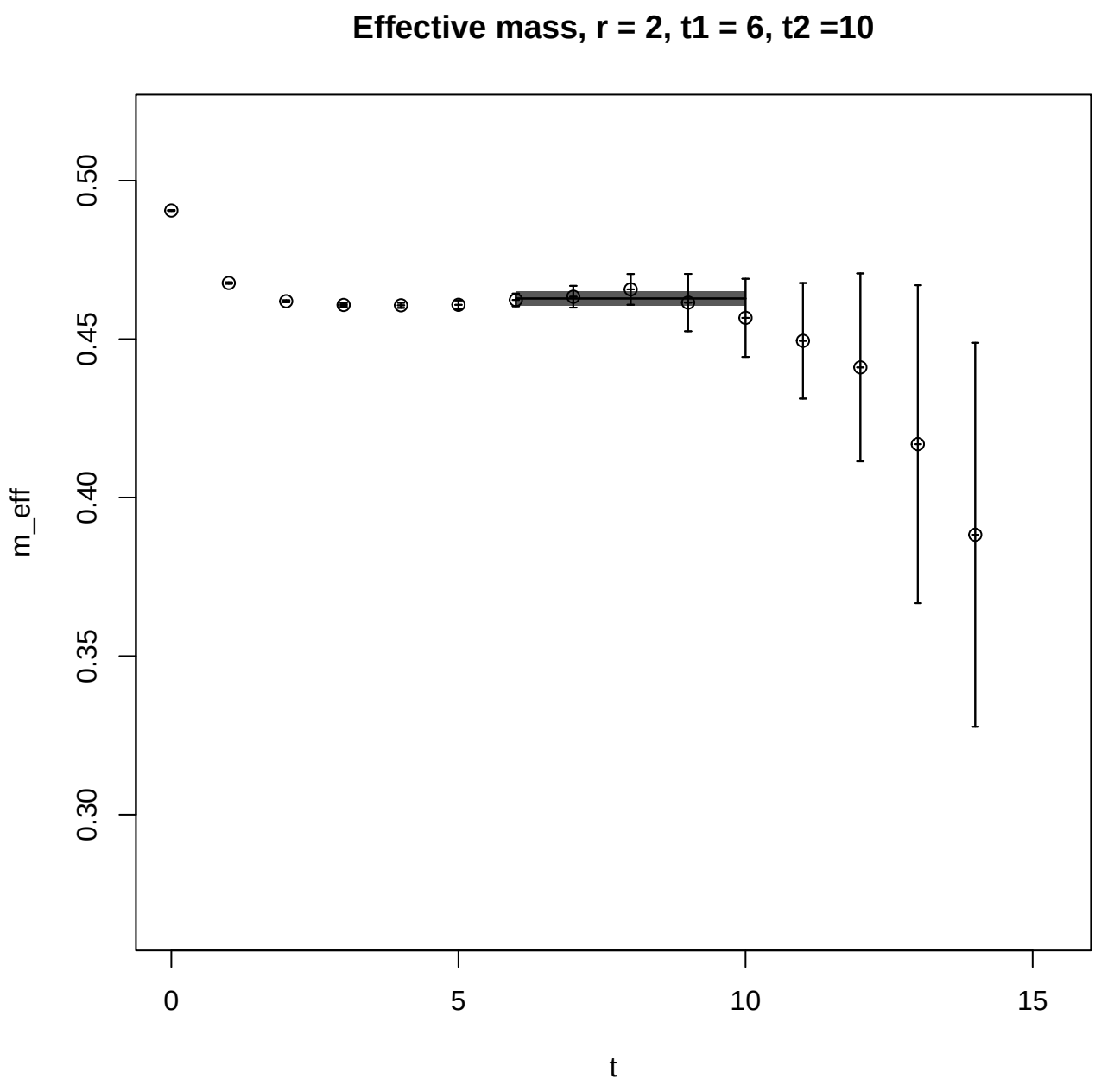

Effective mass, r = 2, t1 = 5, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

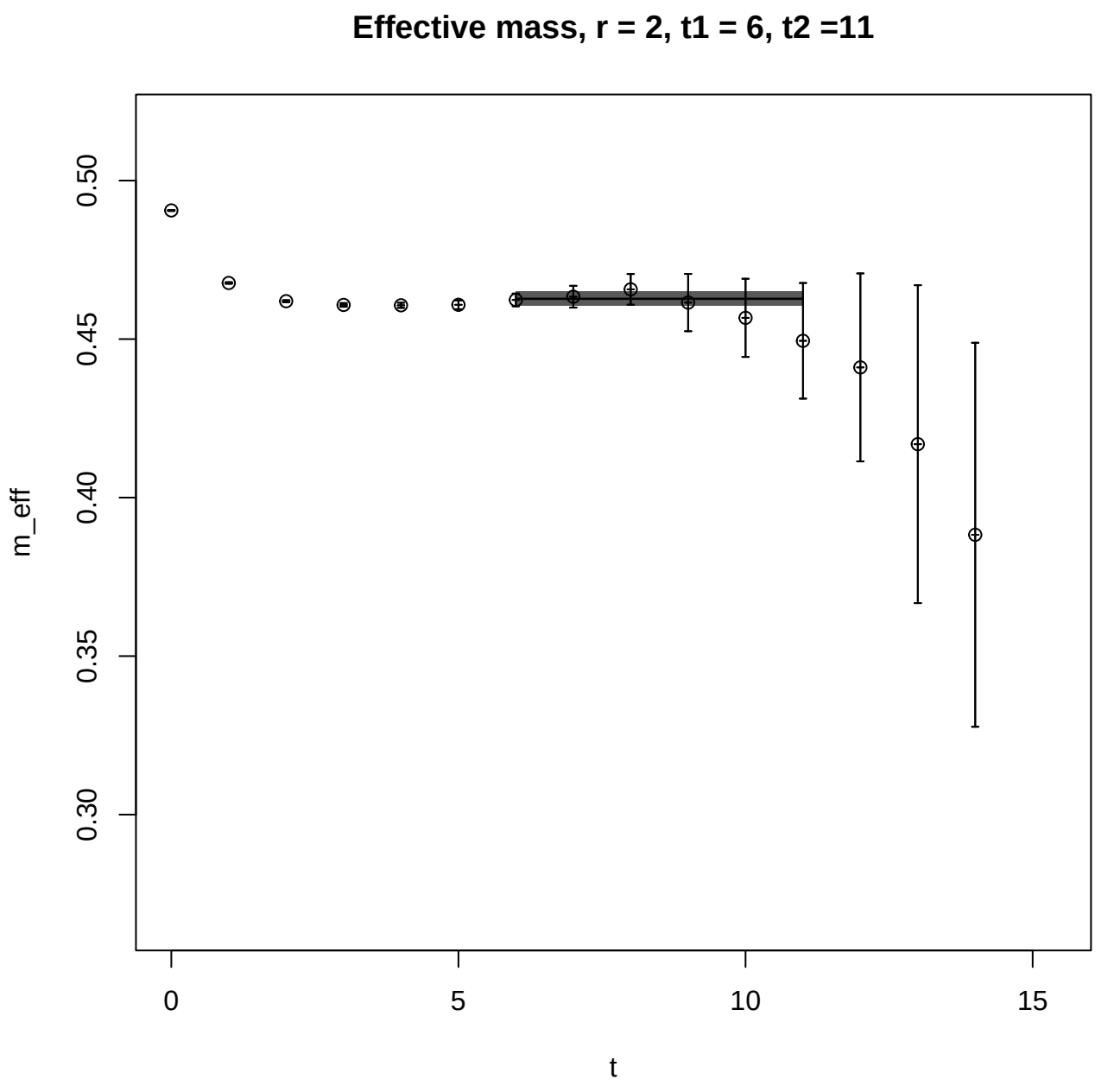

Effective mass, r = 2, t1 = 5, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

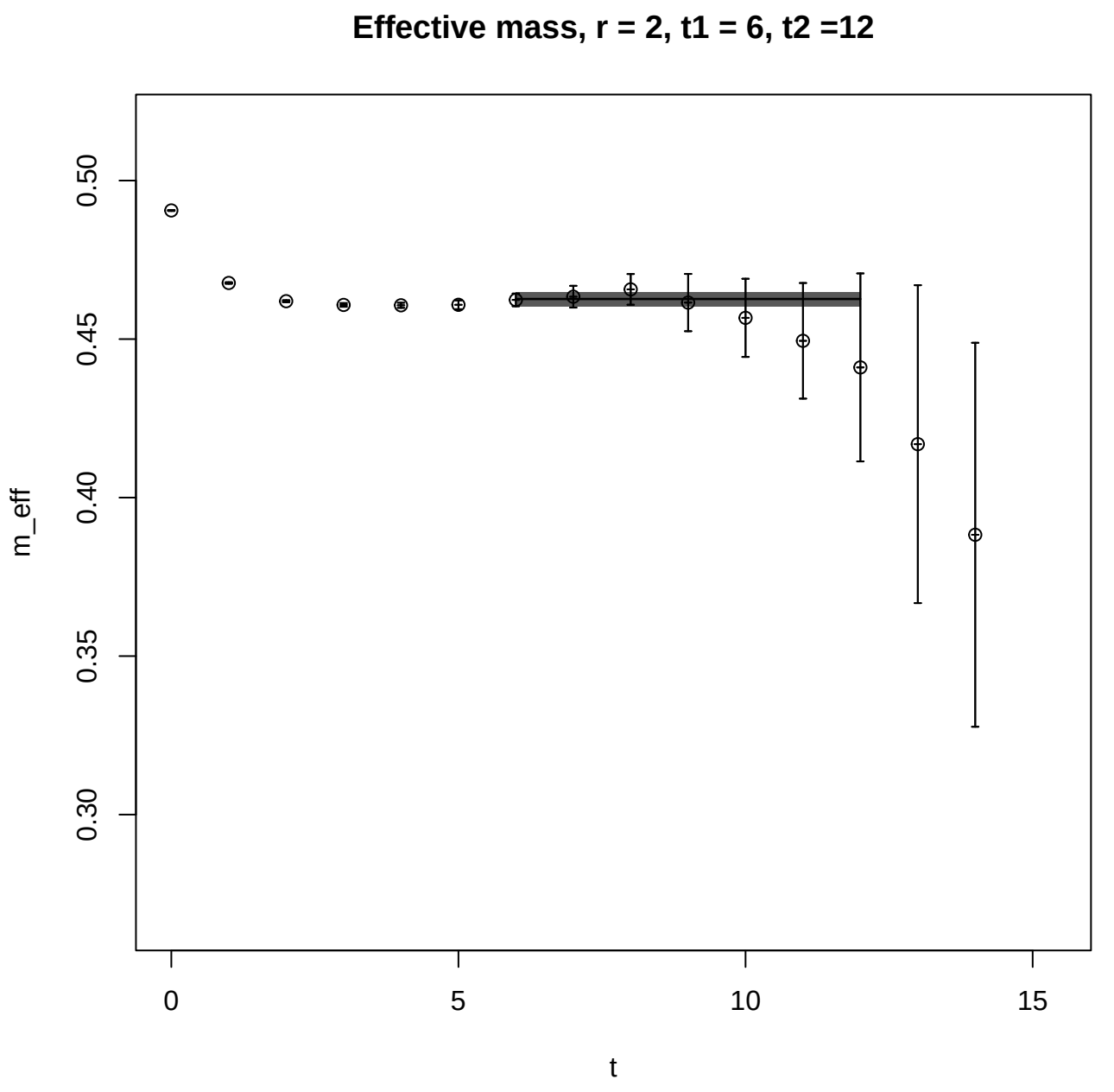

Effective mass, r = 2, t1 = 6, t2 =7
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

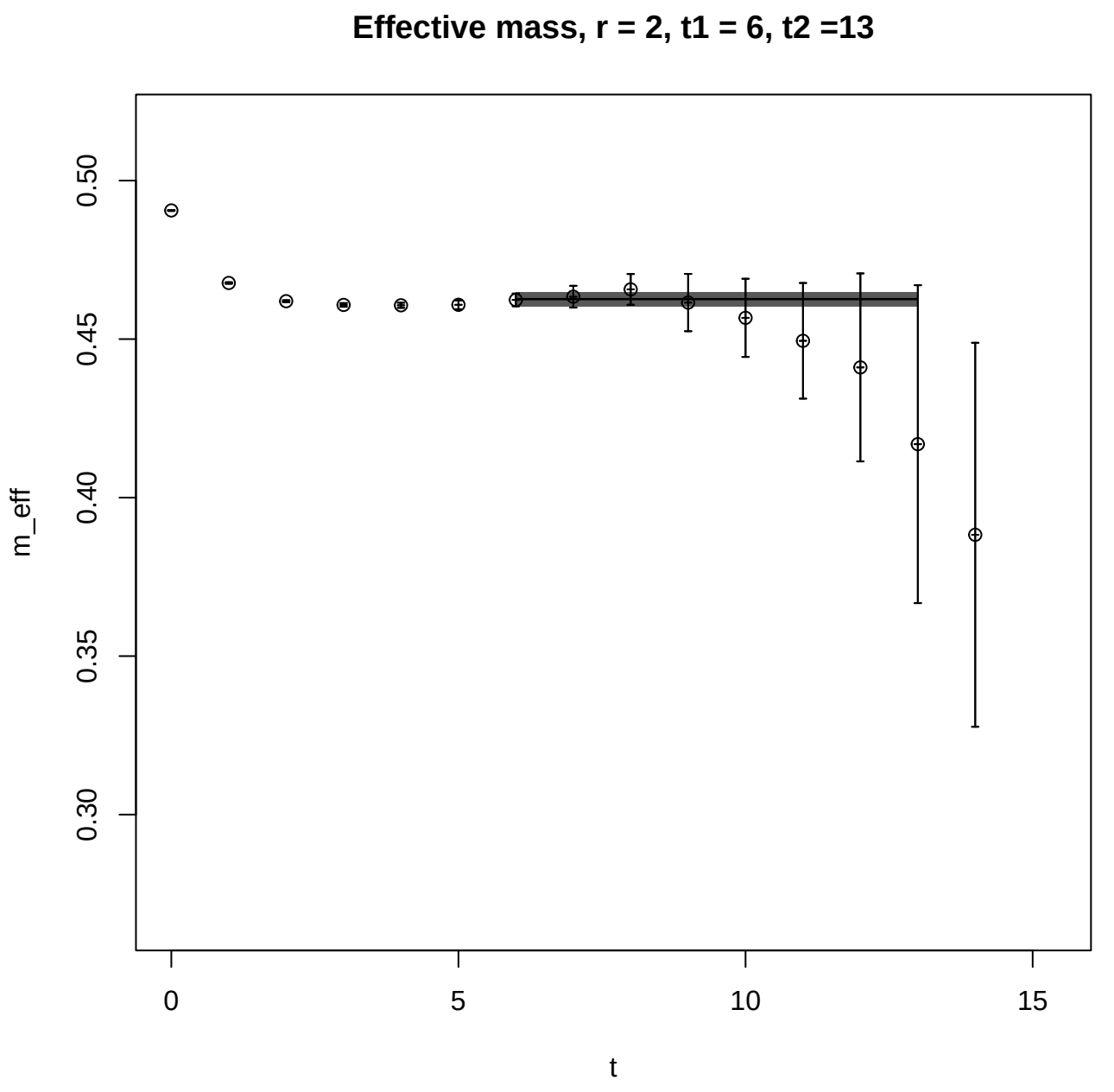

Effective mass, r = 2, t1 = 6, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

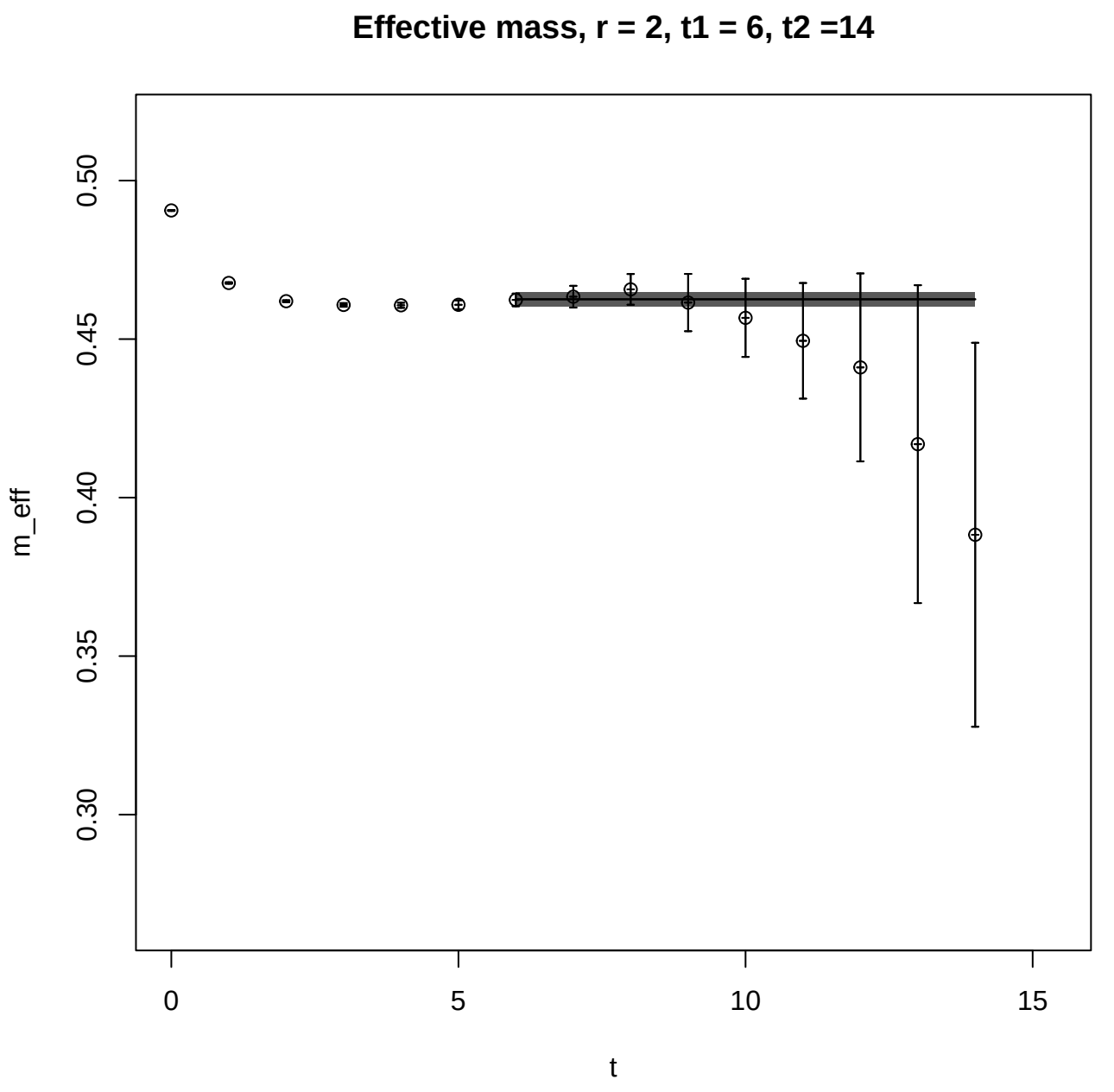

Effective mass, r = 2, t1 = 6, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

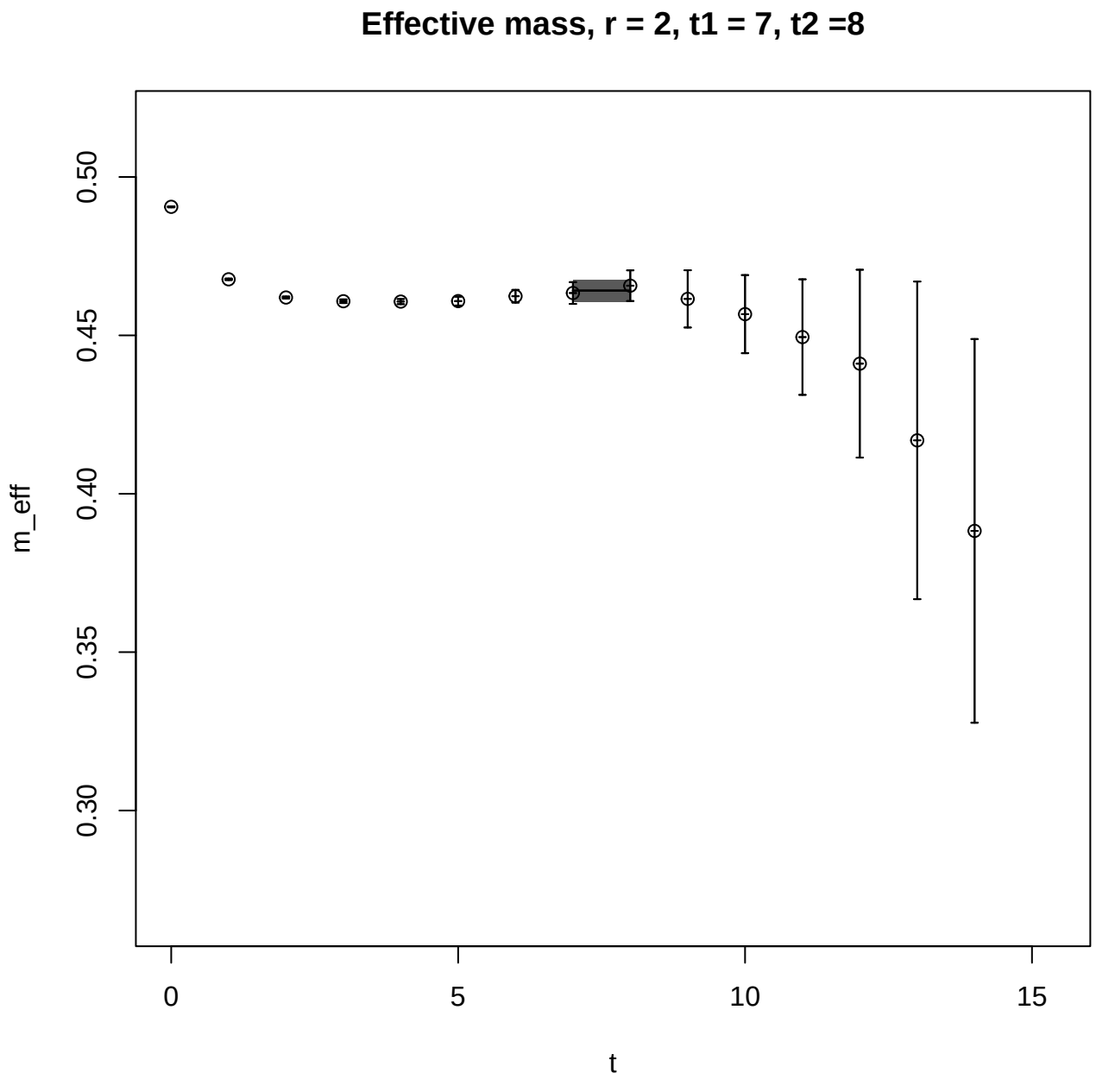

Effective mass, r = 2, t1 = 6, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

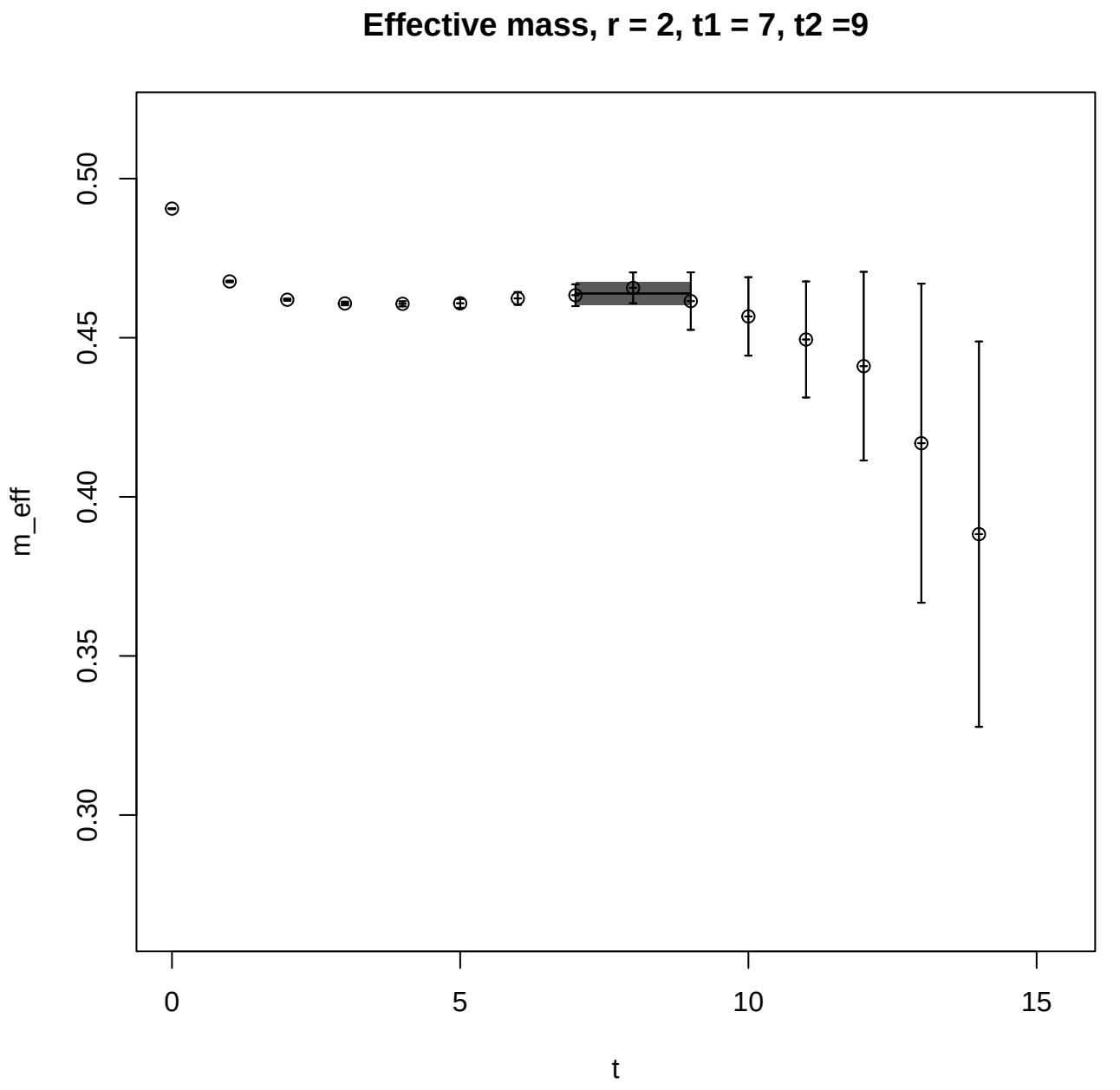

Effective mass, r = 2, t1 = 6, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

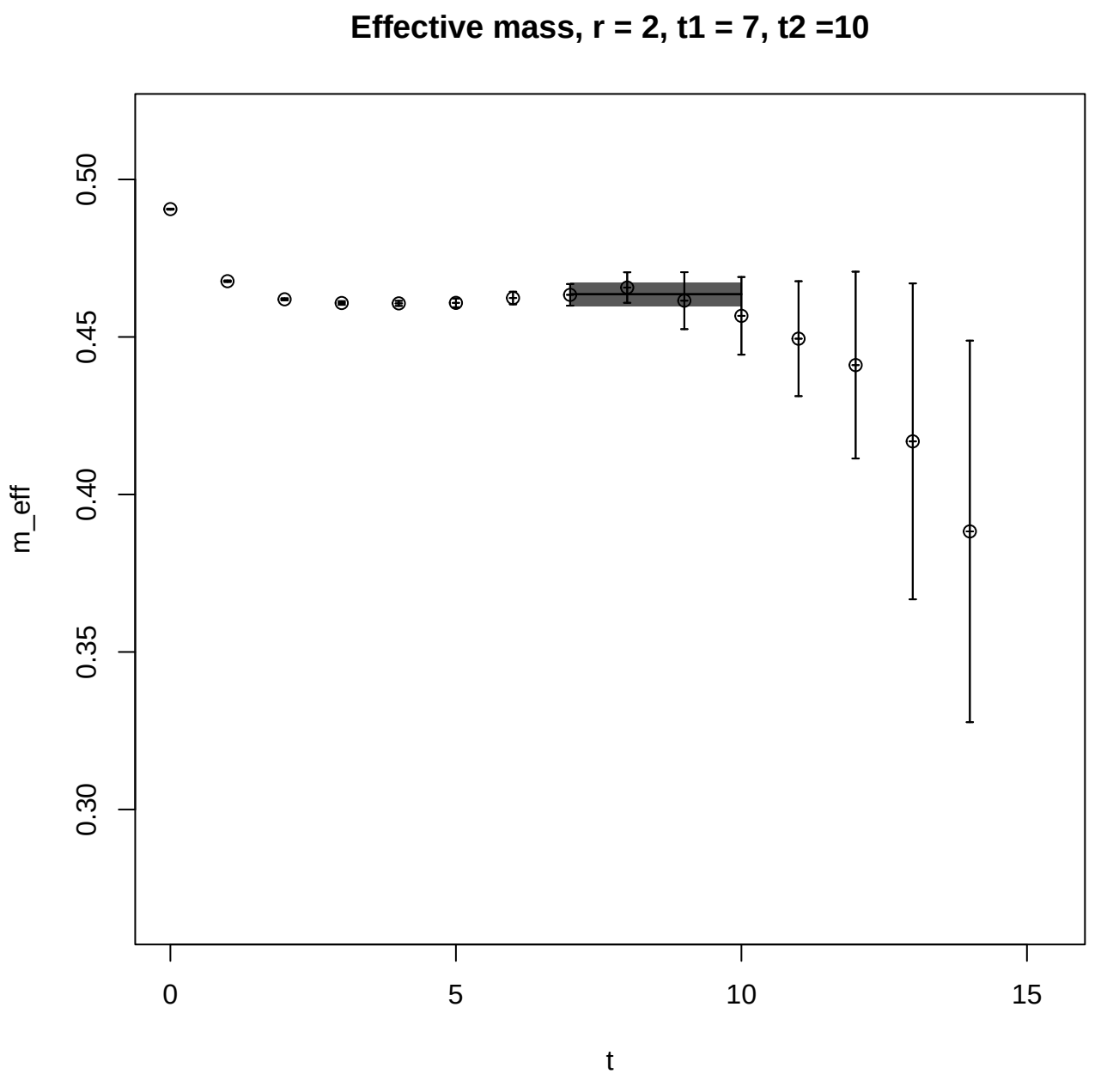

Effective mass, r = 2, t1 = 6, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

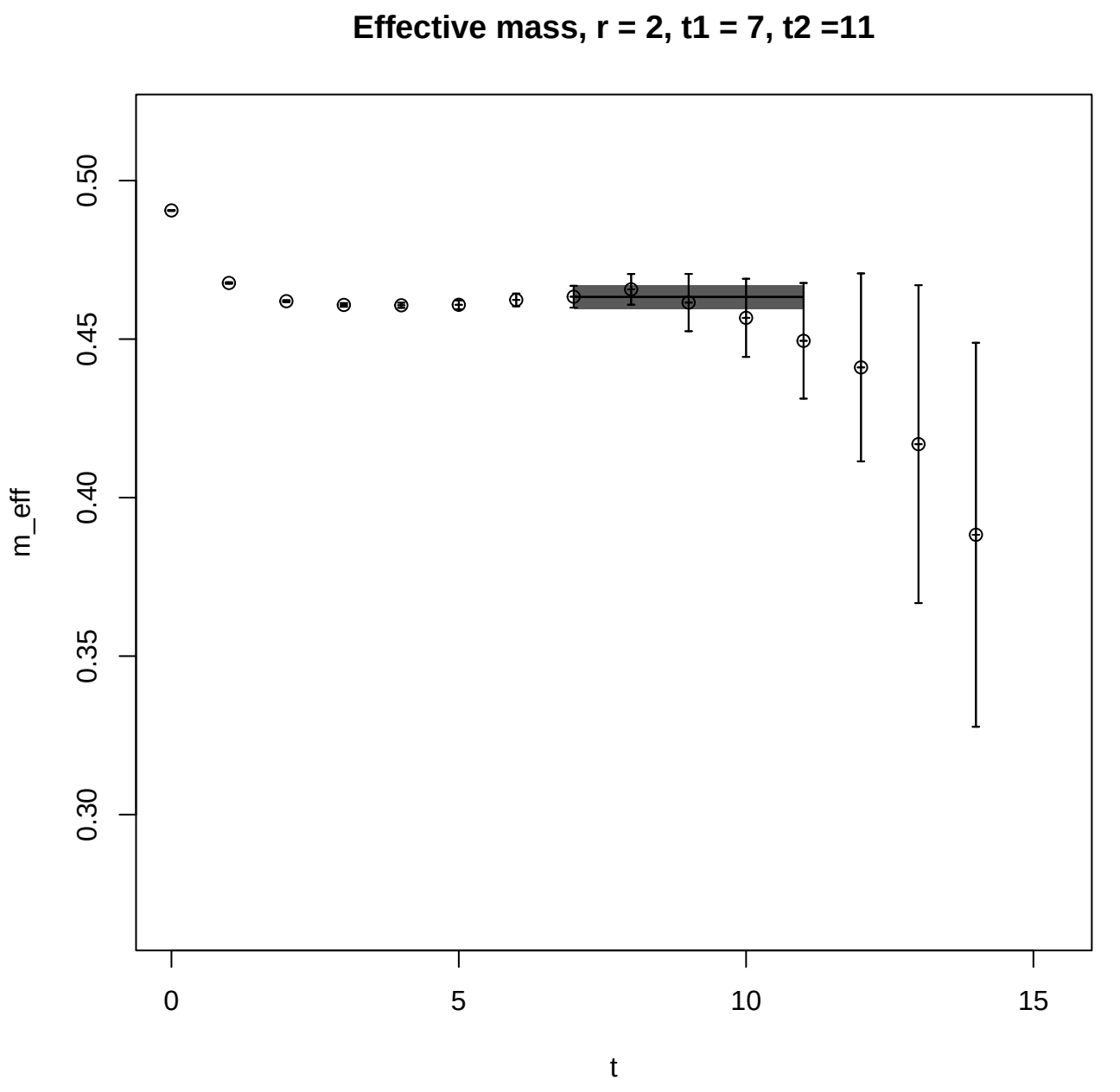

Effective mass, r = 2, t1 = 6, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

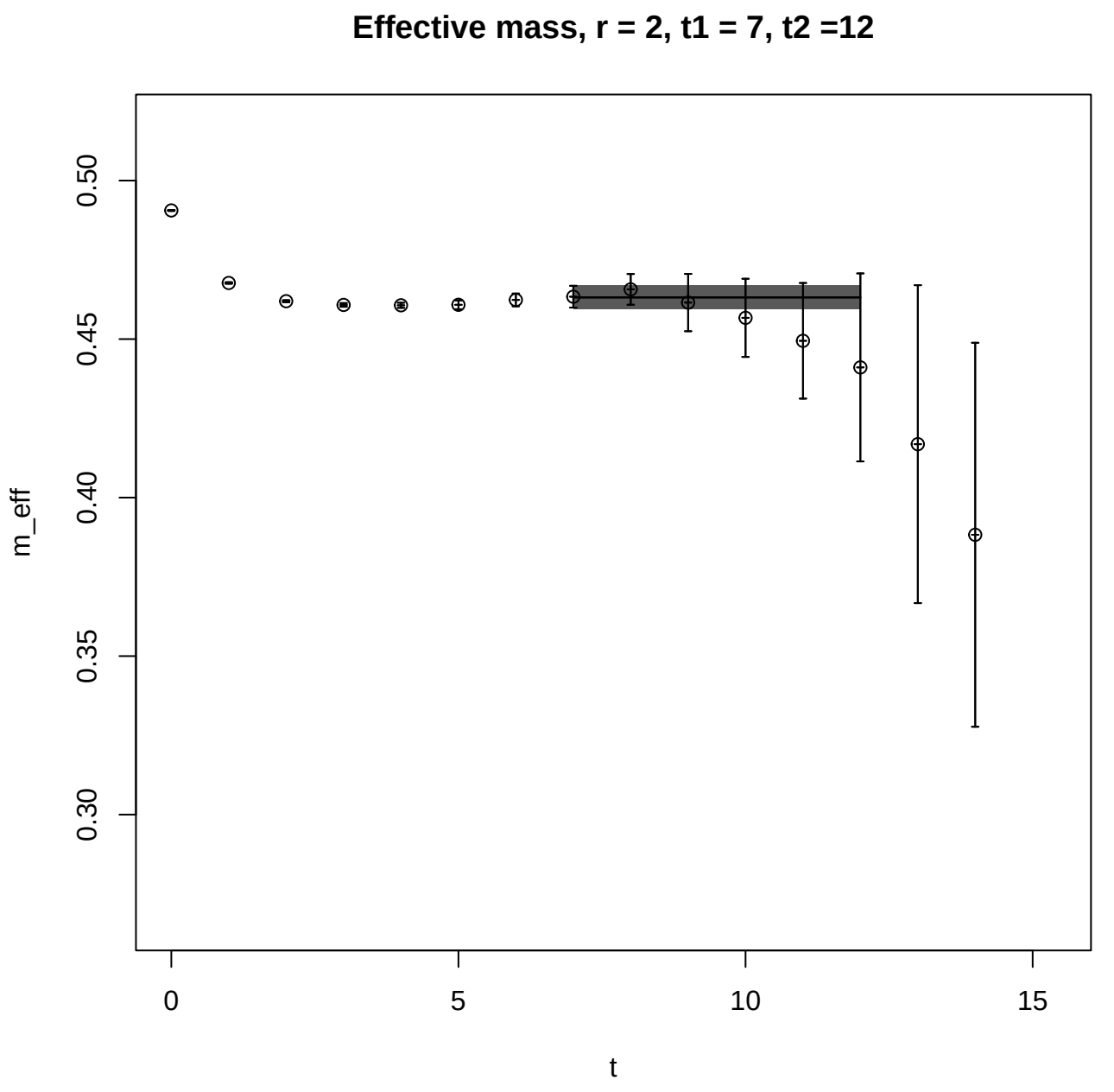

Effective mass, r = 2, t1 = 6, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

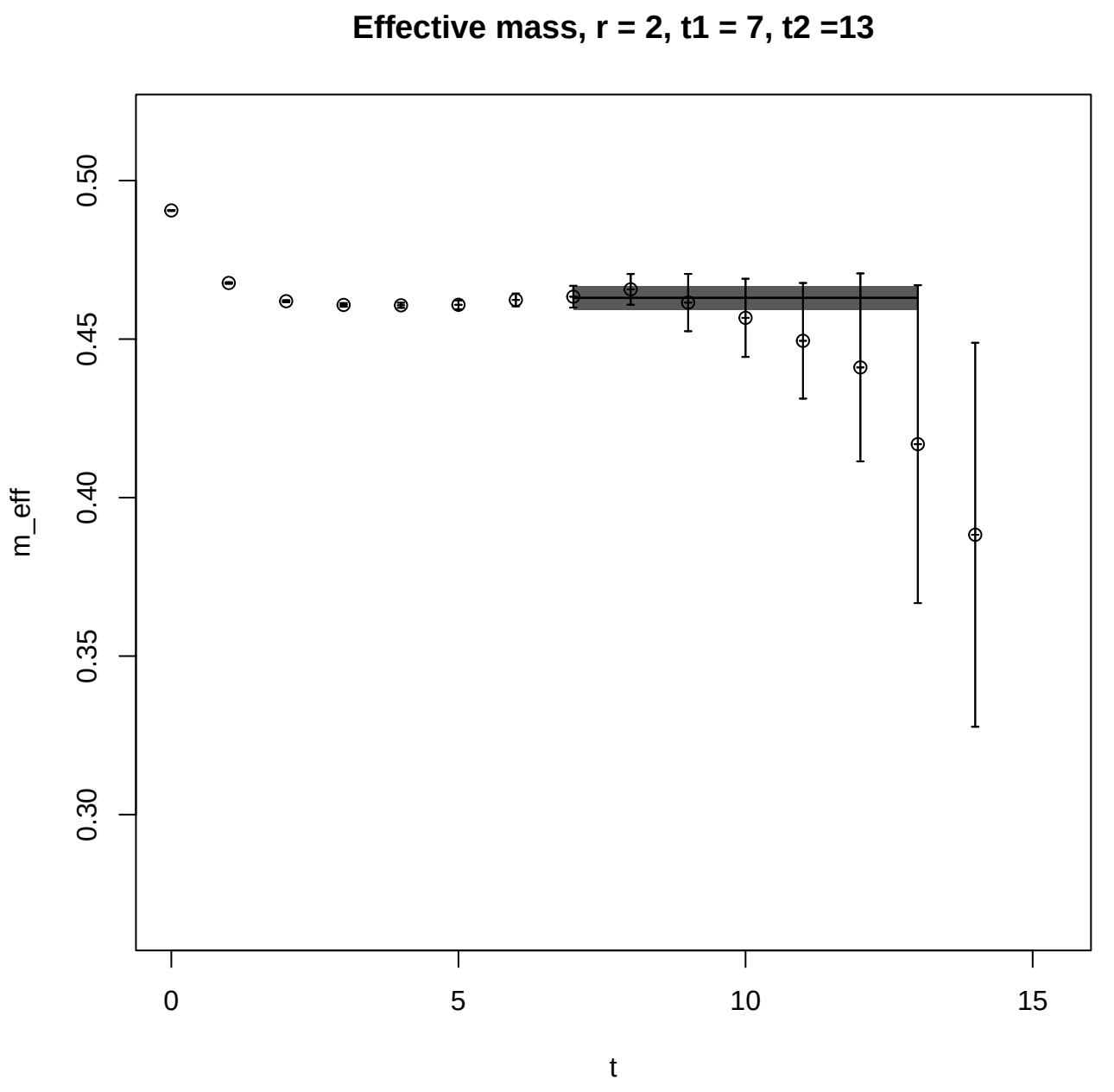

Effective mass, r = 2, t1 = 7, t2 =8
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

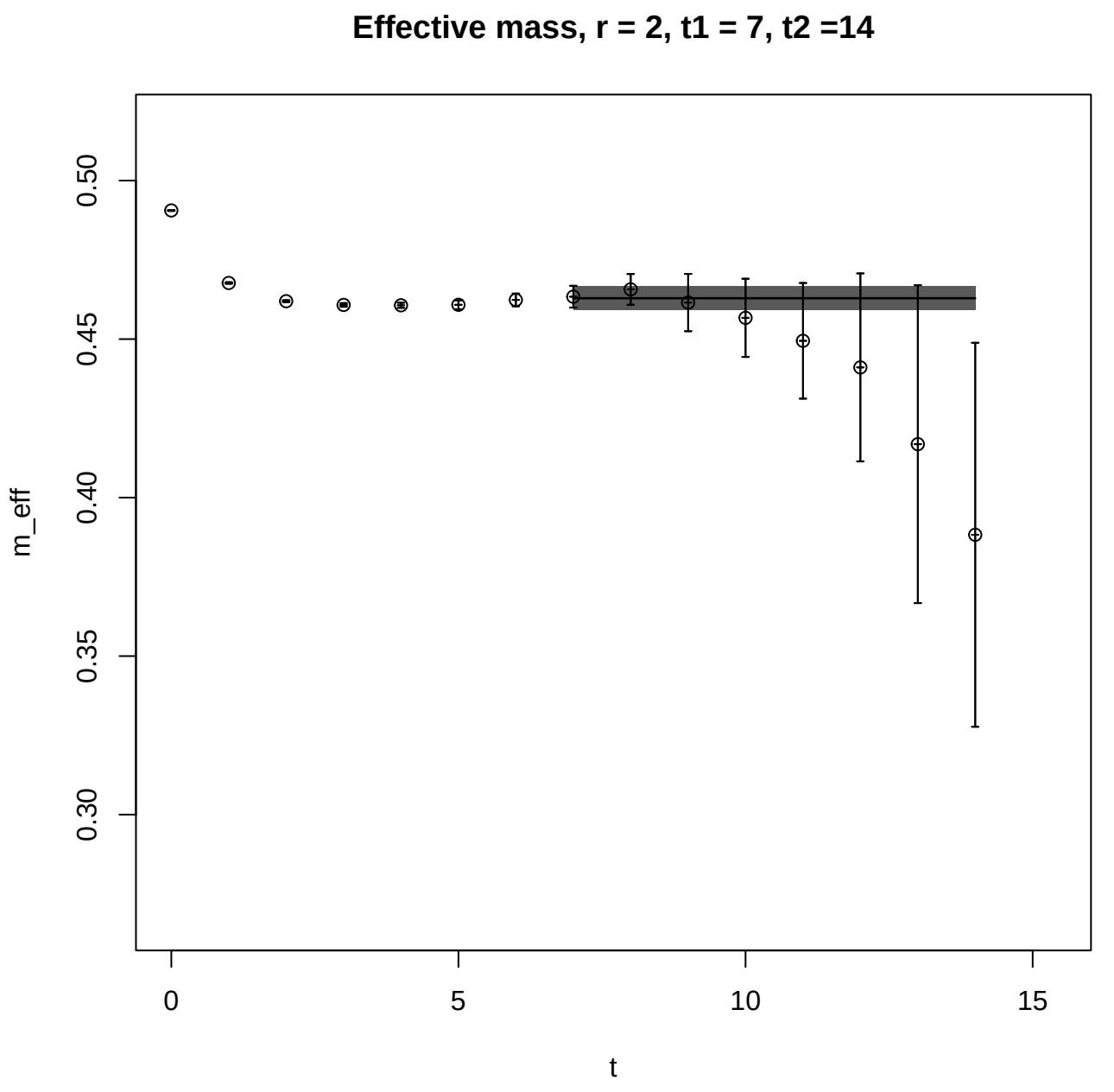

Effective mass, r = 2, t1 = 7, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

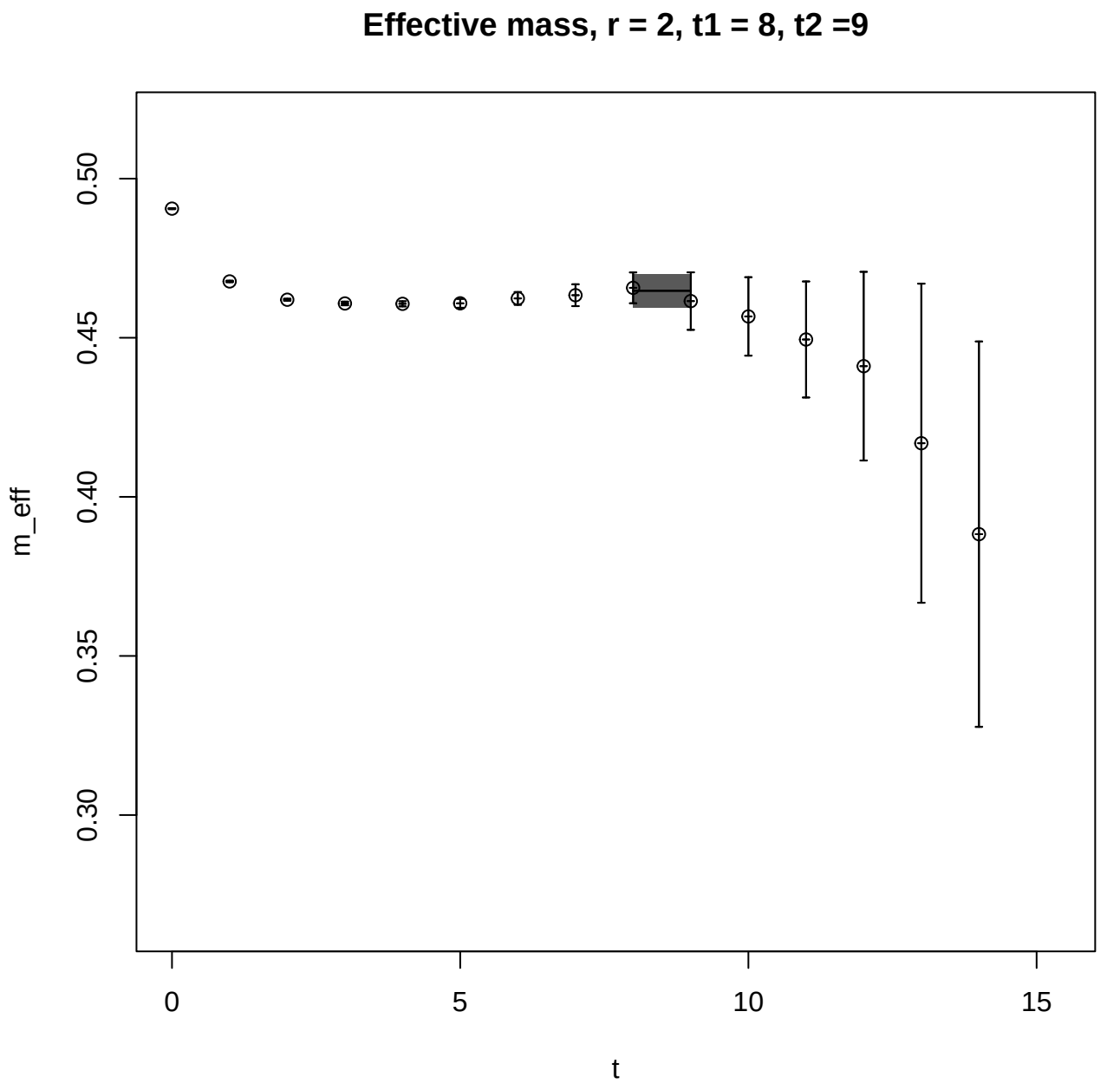

Effective mass, r = 2, t1 = 7, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

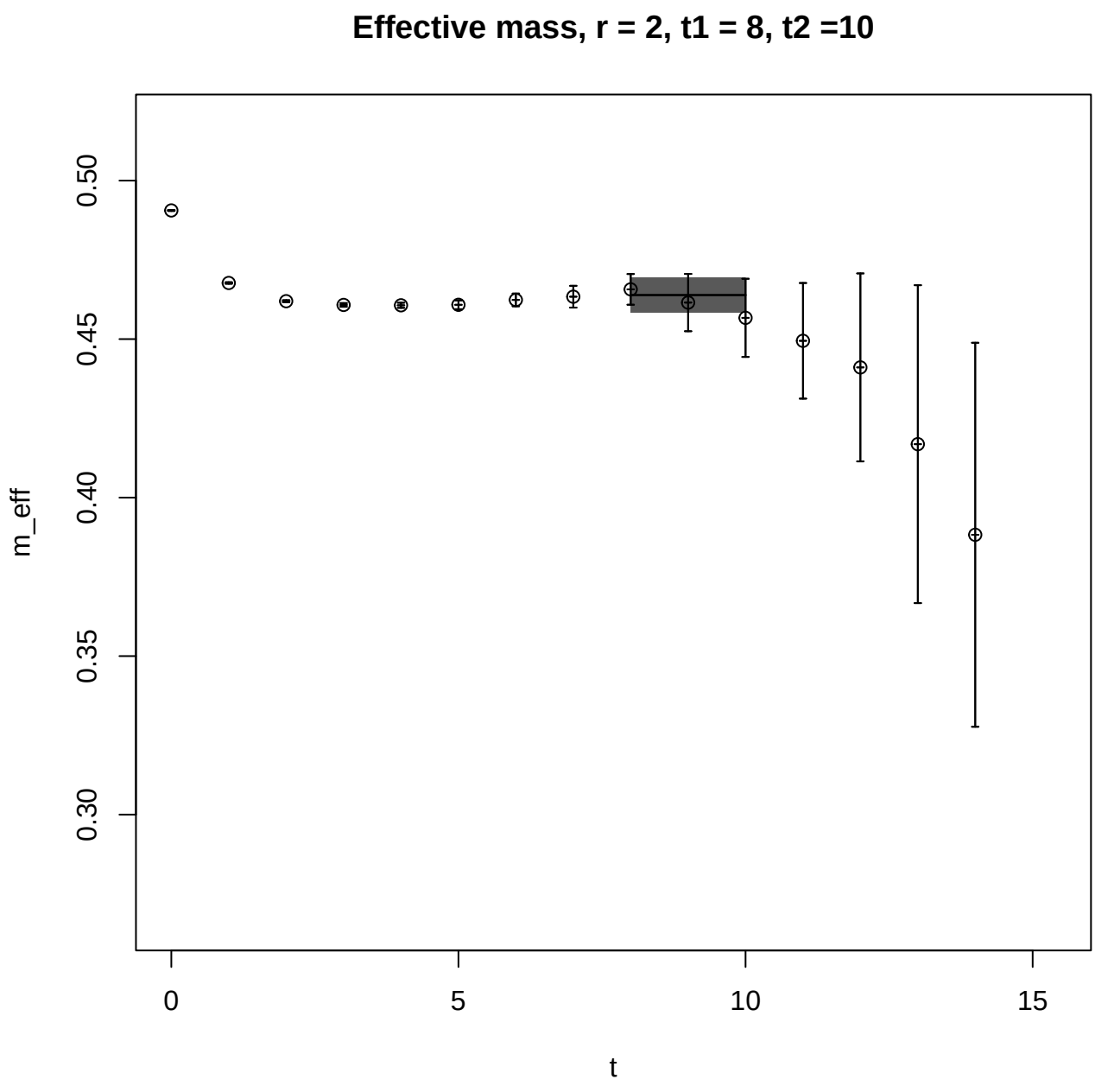

Effective mass, r = 2, t1 = 7, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

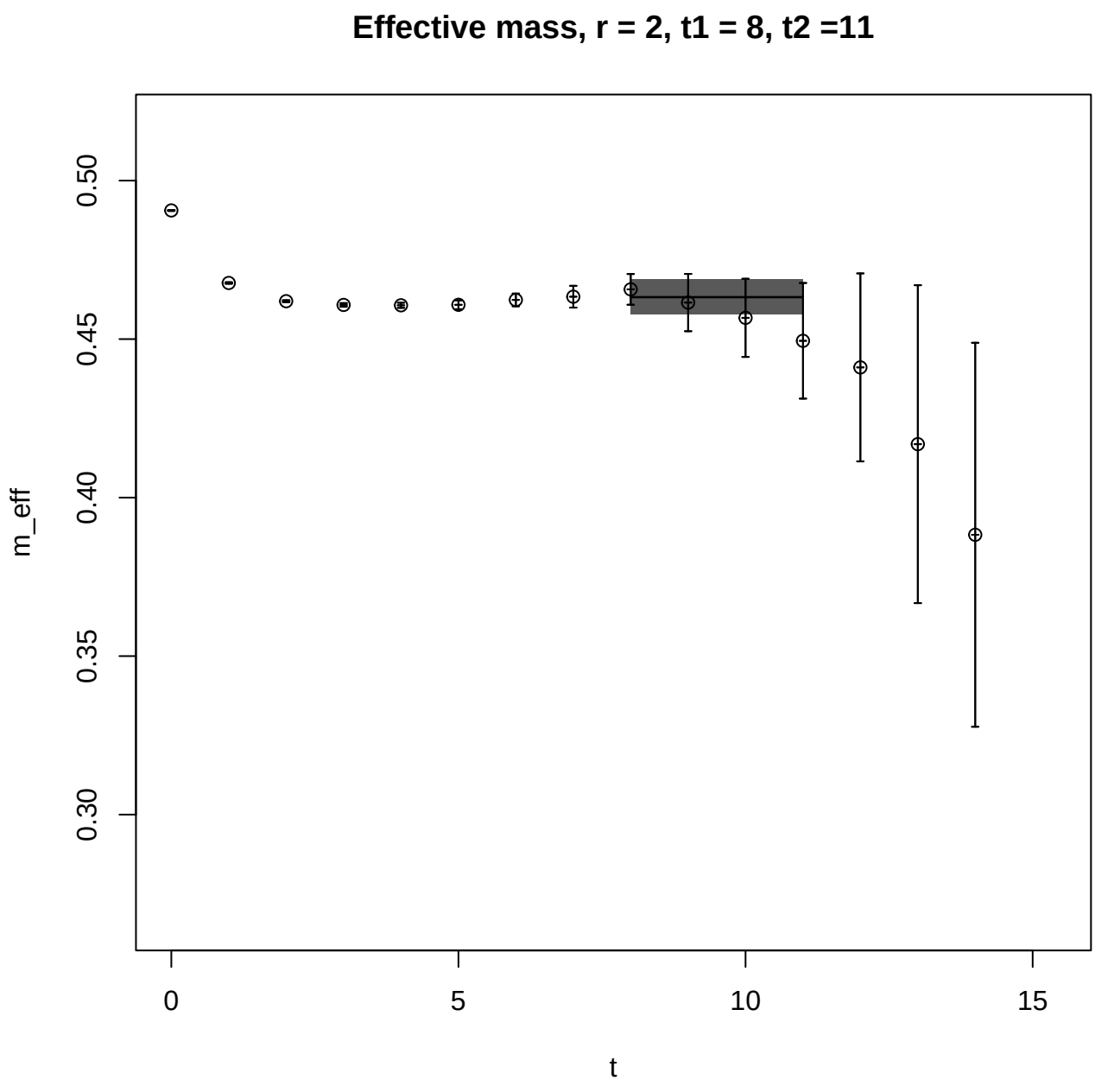

Effective mass, r = 2, t1 = 7, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

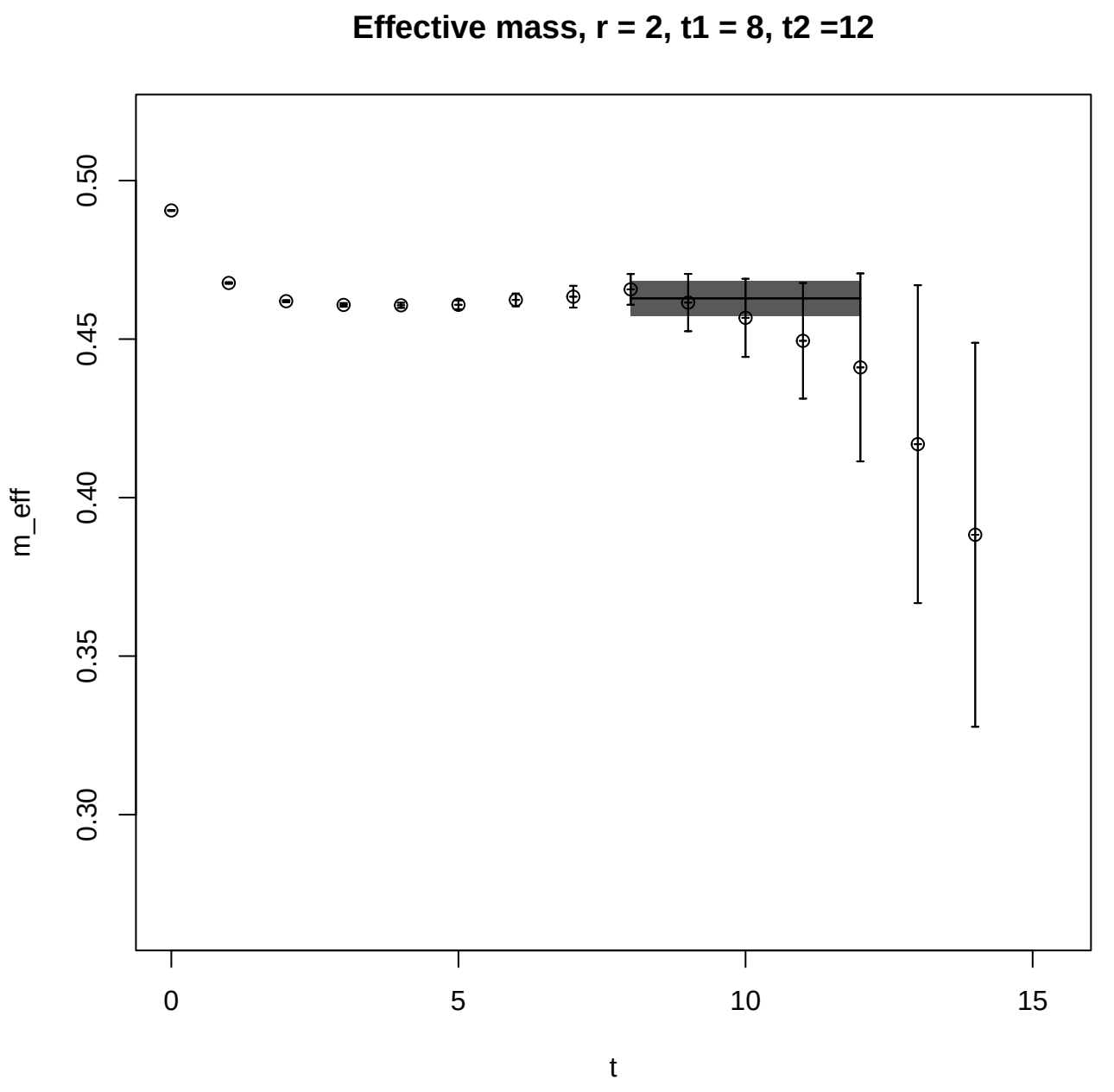

Effective mass, r = 2, t1 = 7, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

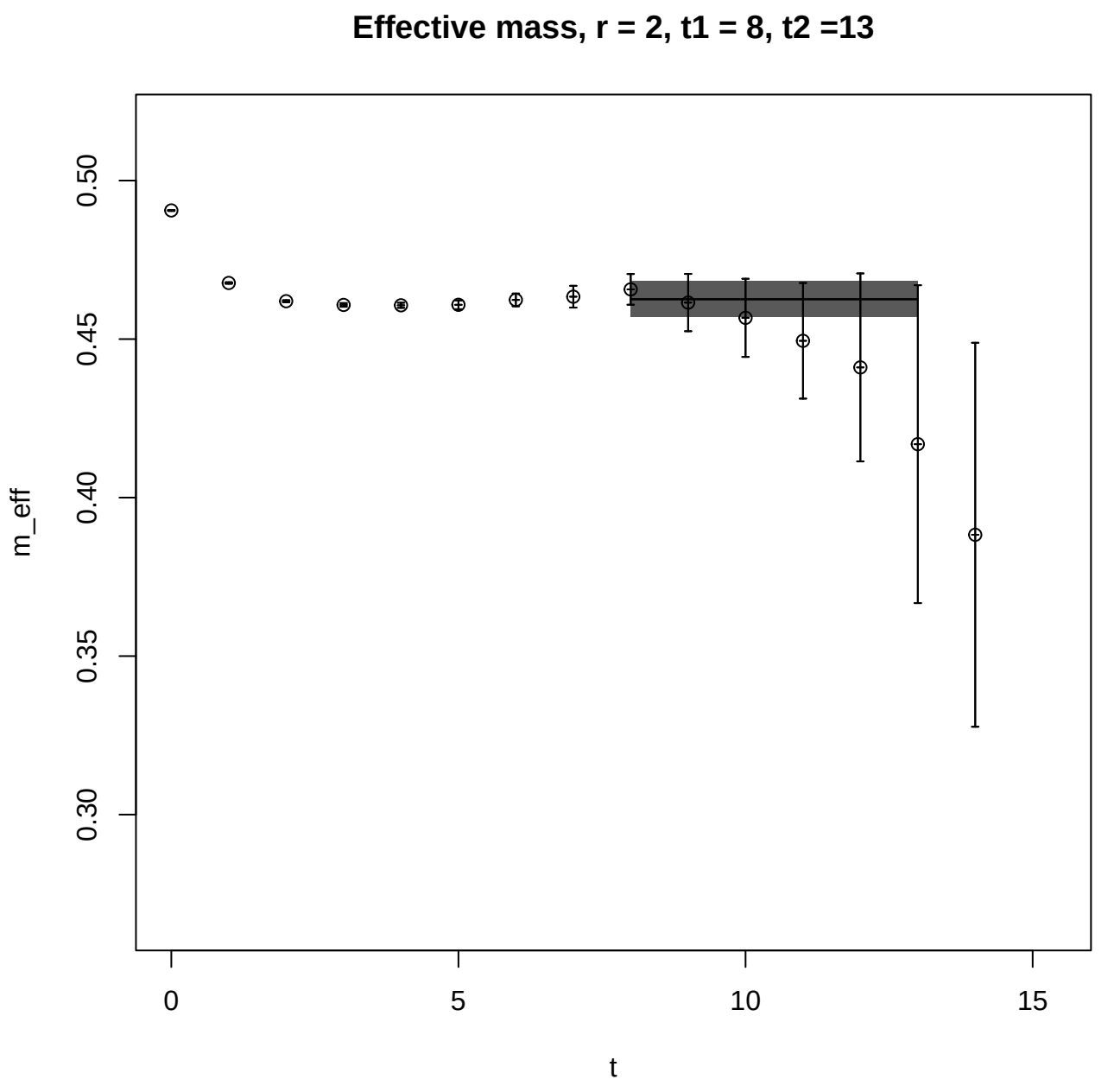

Effective mass, r = 2, t1 = 7, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

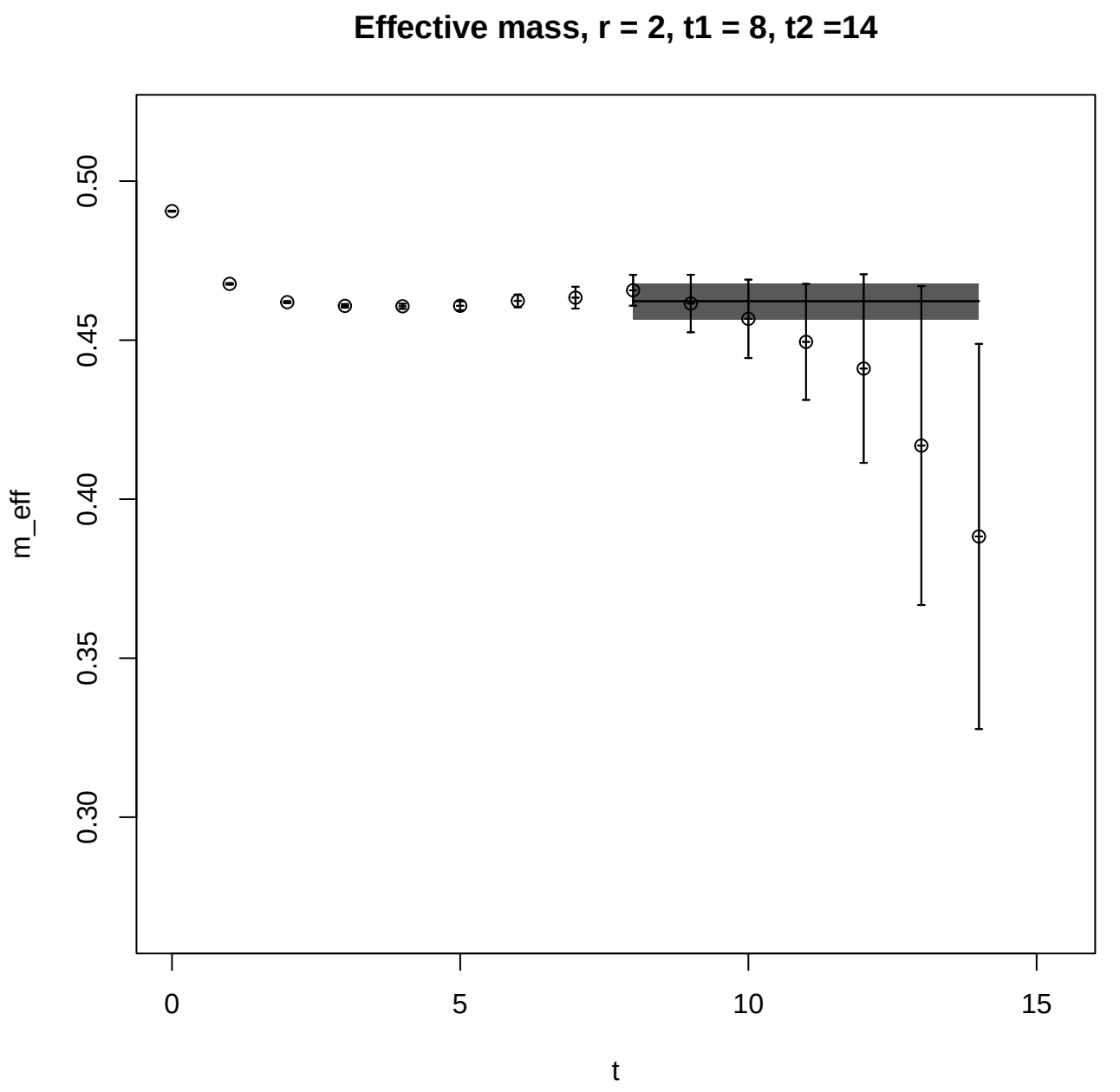

Effective mass, r = 2, t1 = 8, t2 =9
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =10
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =14
m_eff
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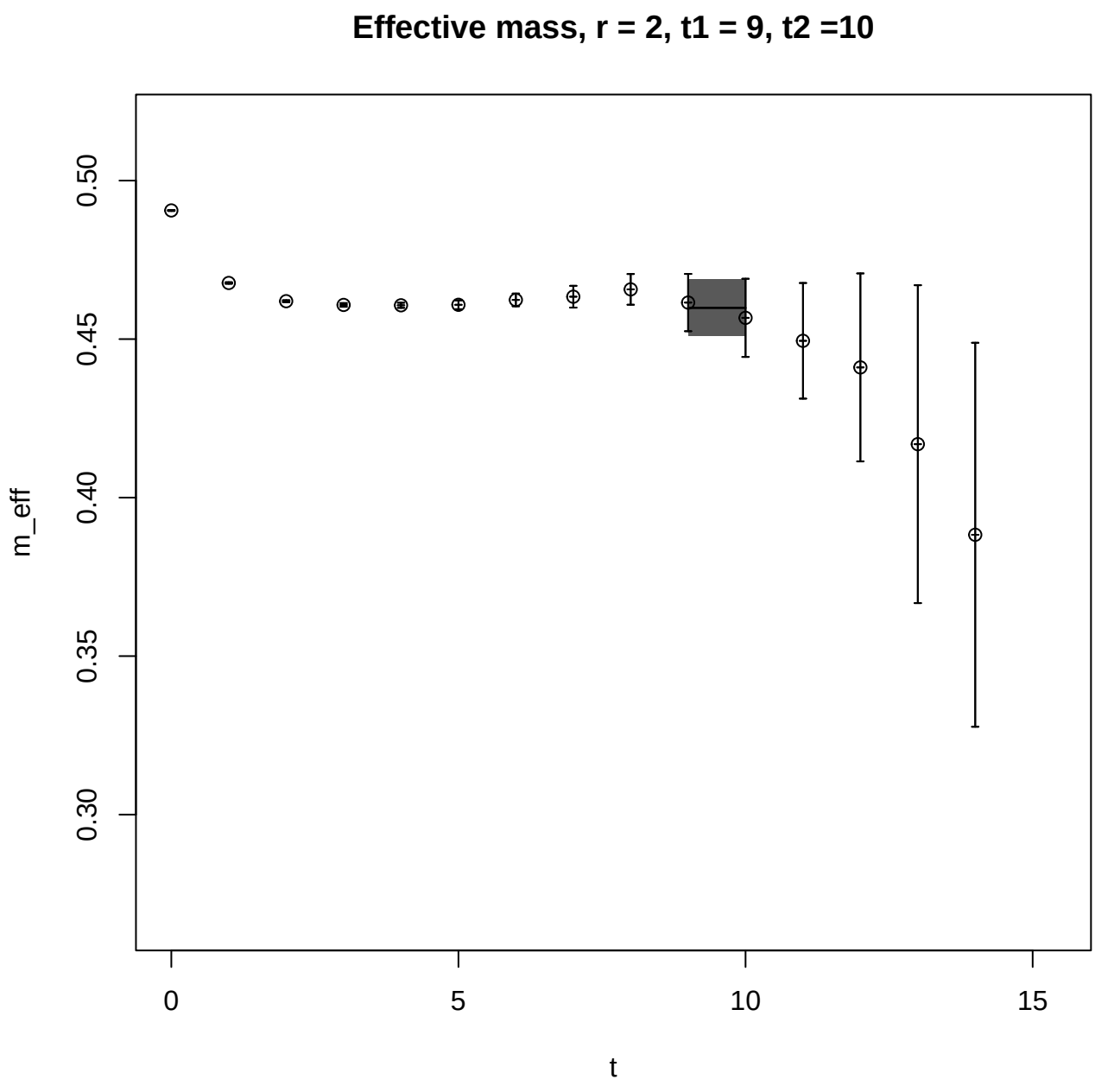
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 9, t2 =10
m_eff
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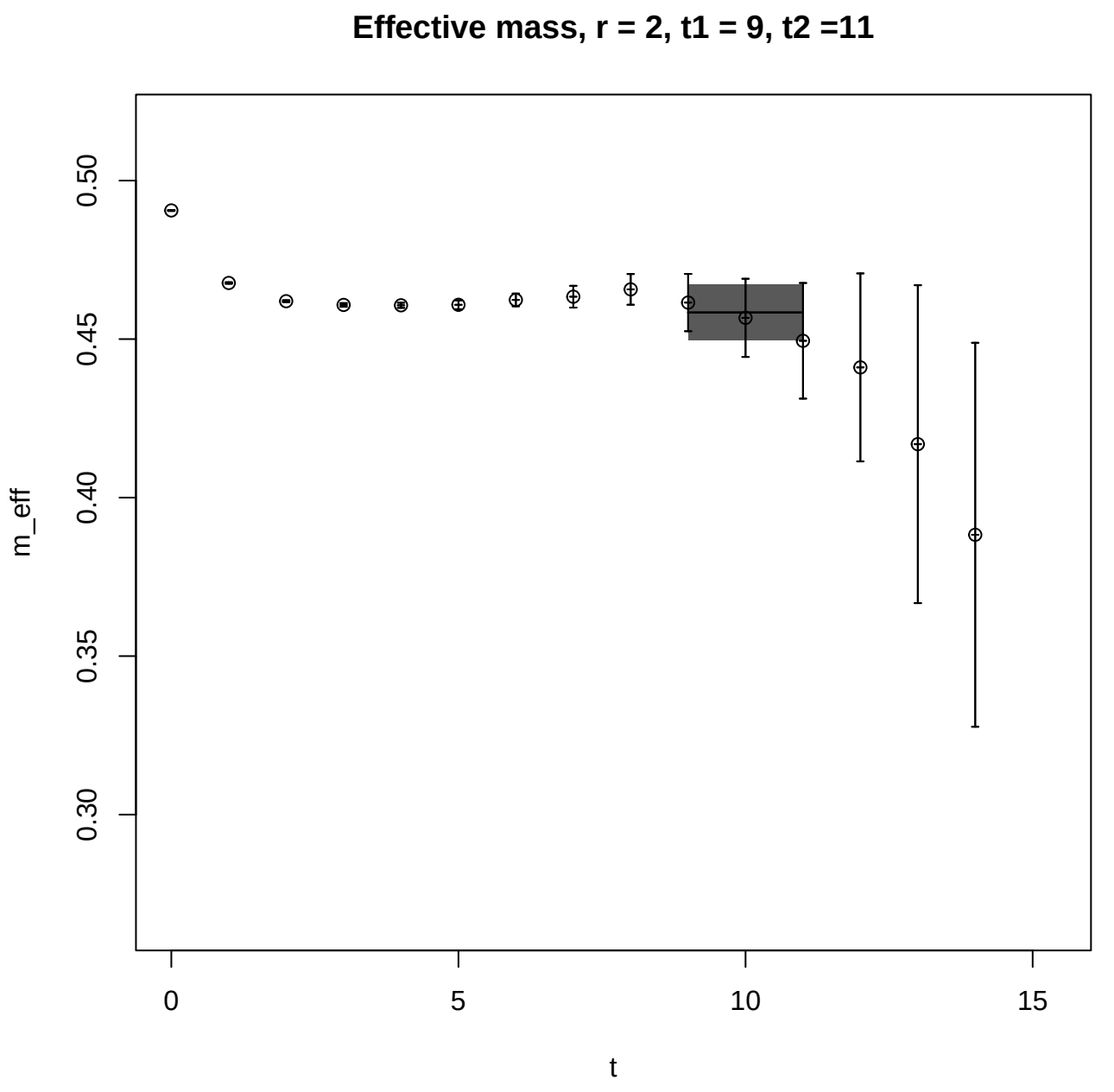
0.35
0.40
0.45
0.50
0
5
10
15
t


Effective mass, r = 2, t1 = 9, t2 =11
m_eff
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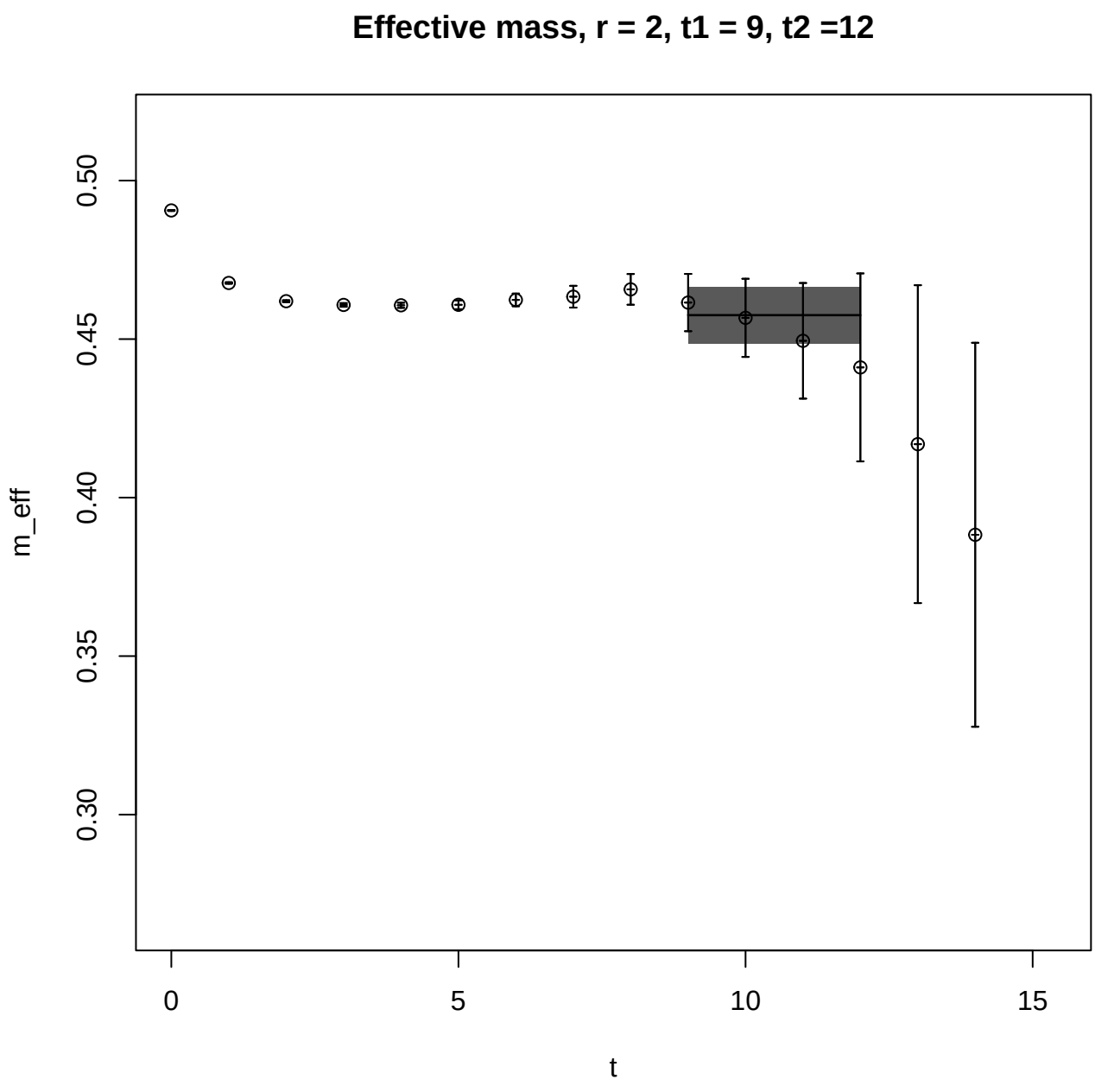
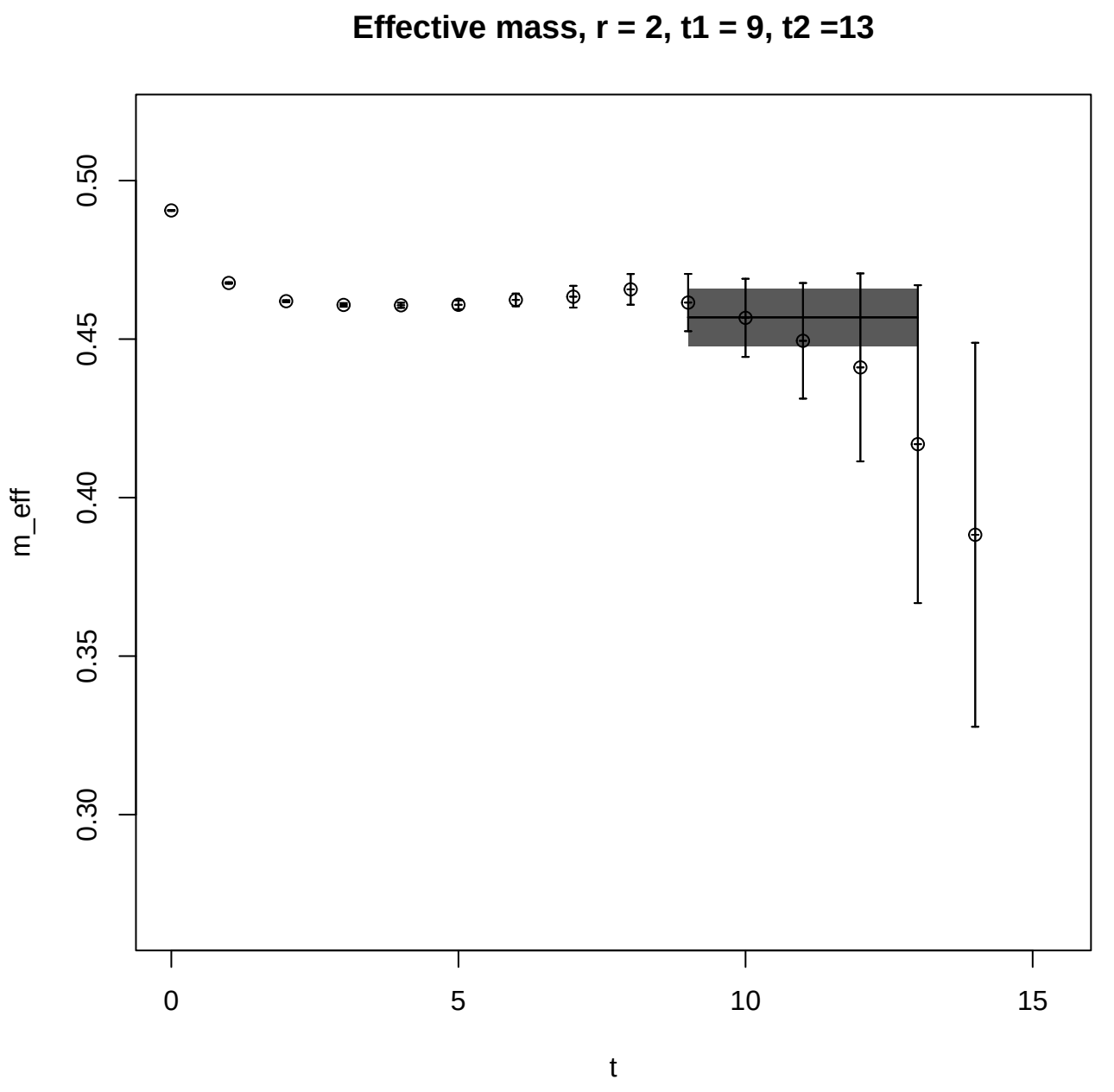
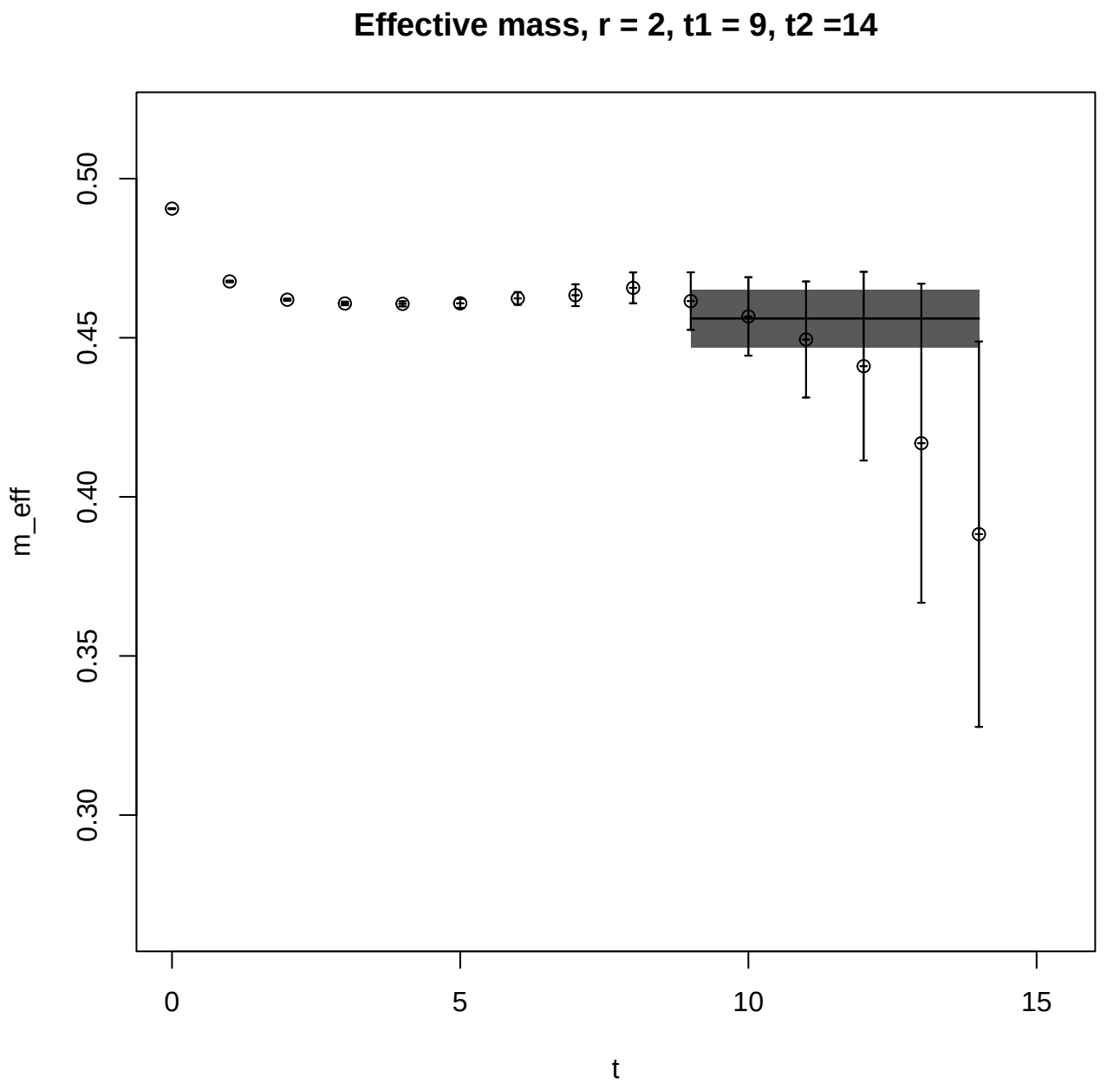
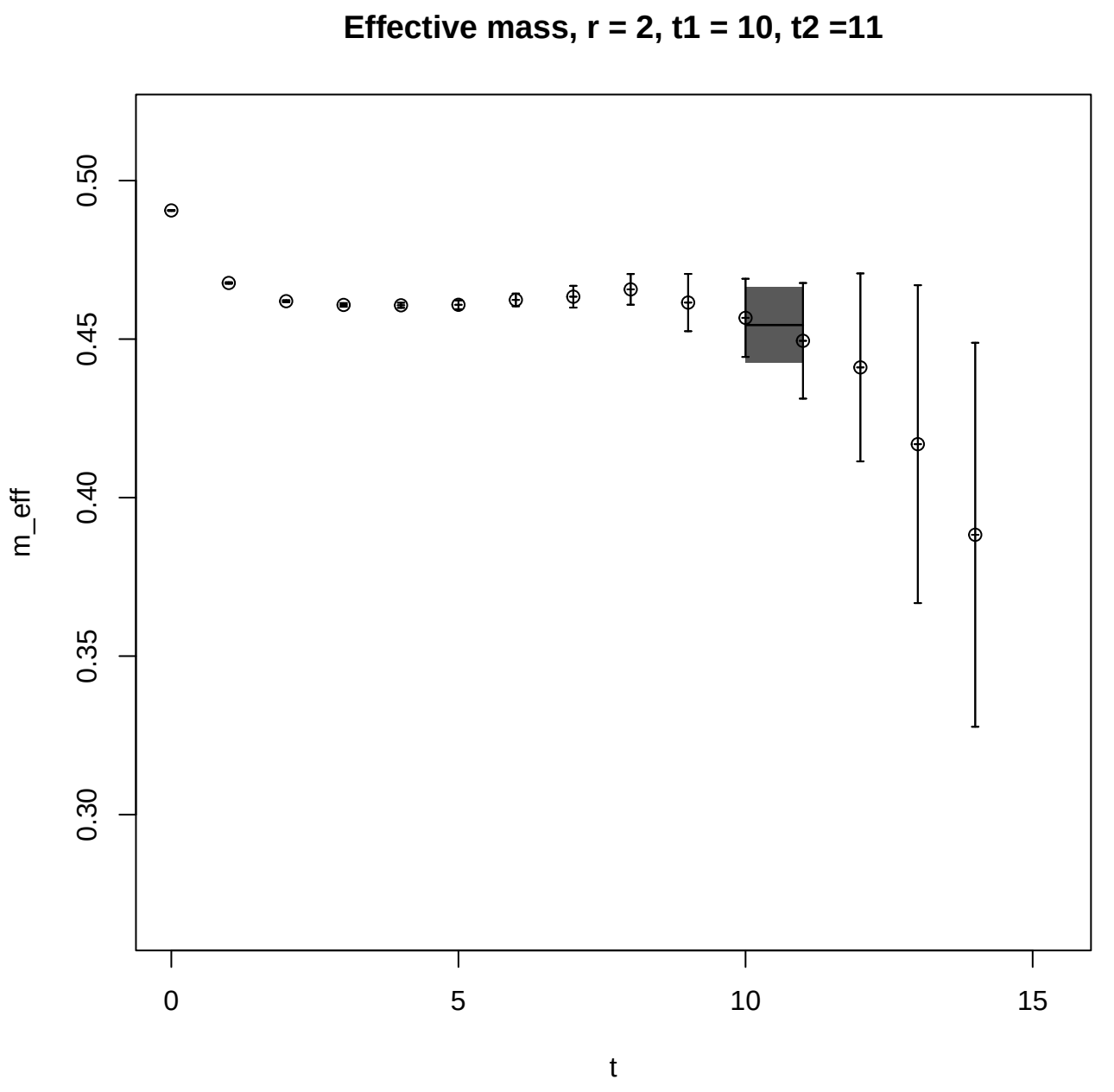
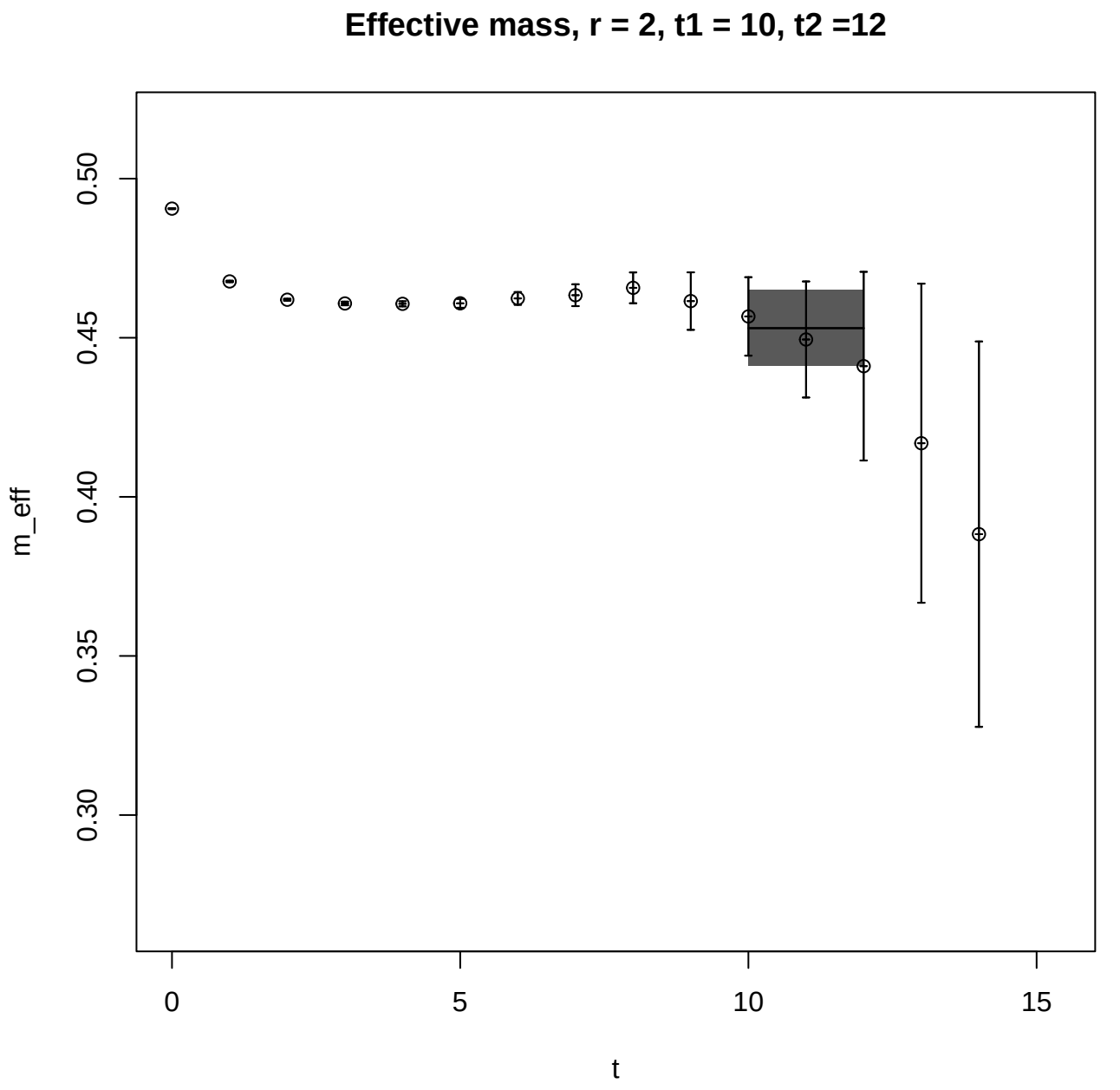
0.35
0.40
0.45
0.50
0
5
10
15
t

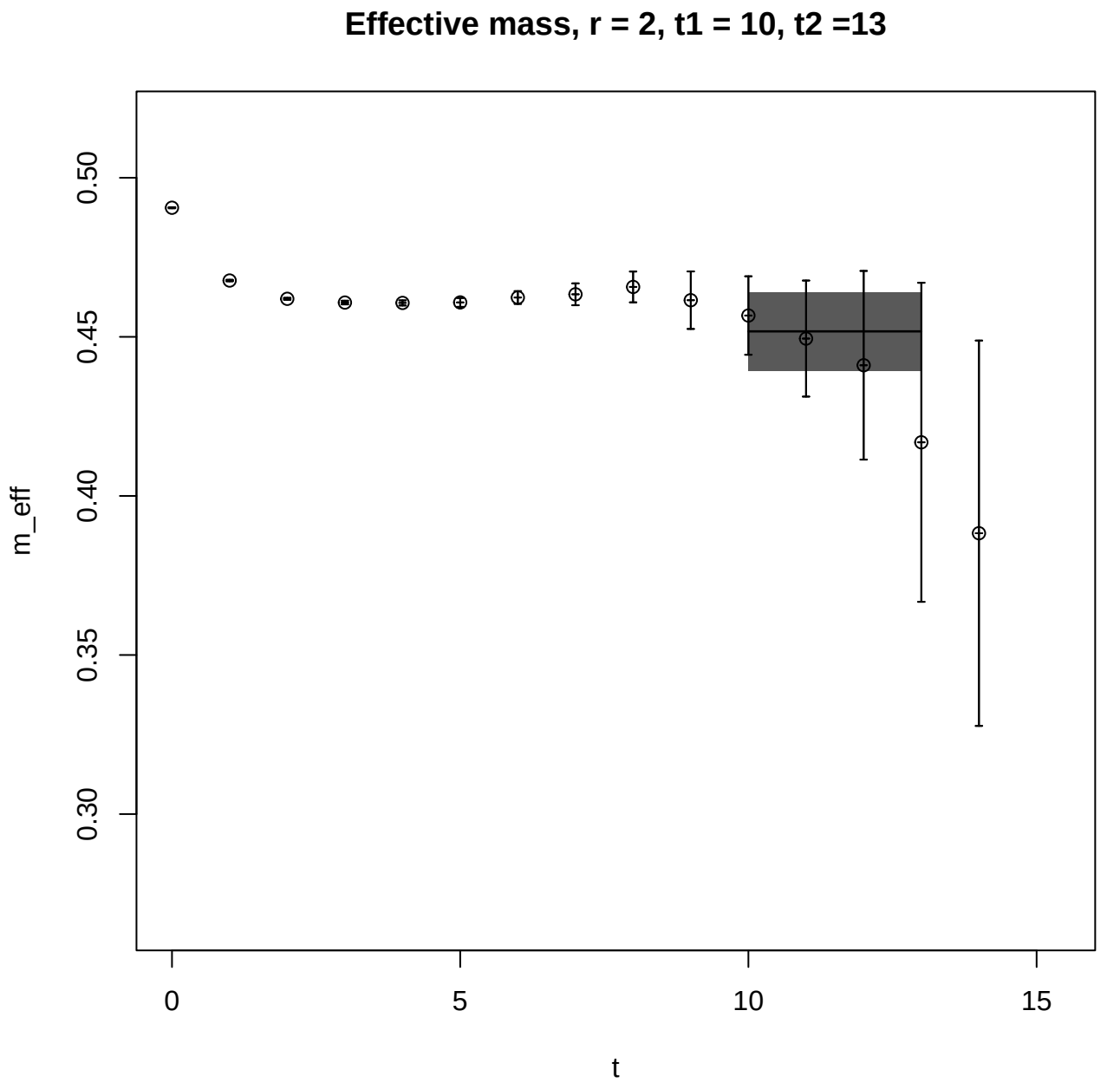

Effective mass, r = 2, t1 = 9, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

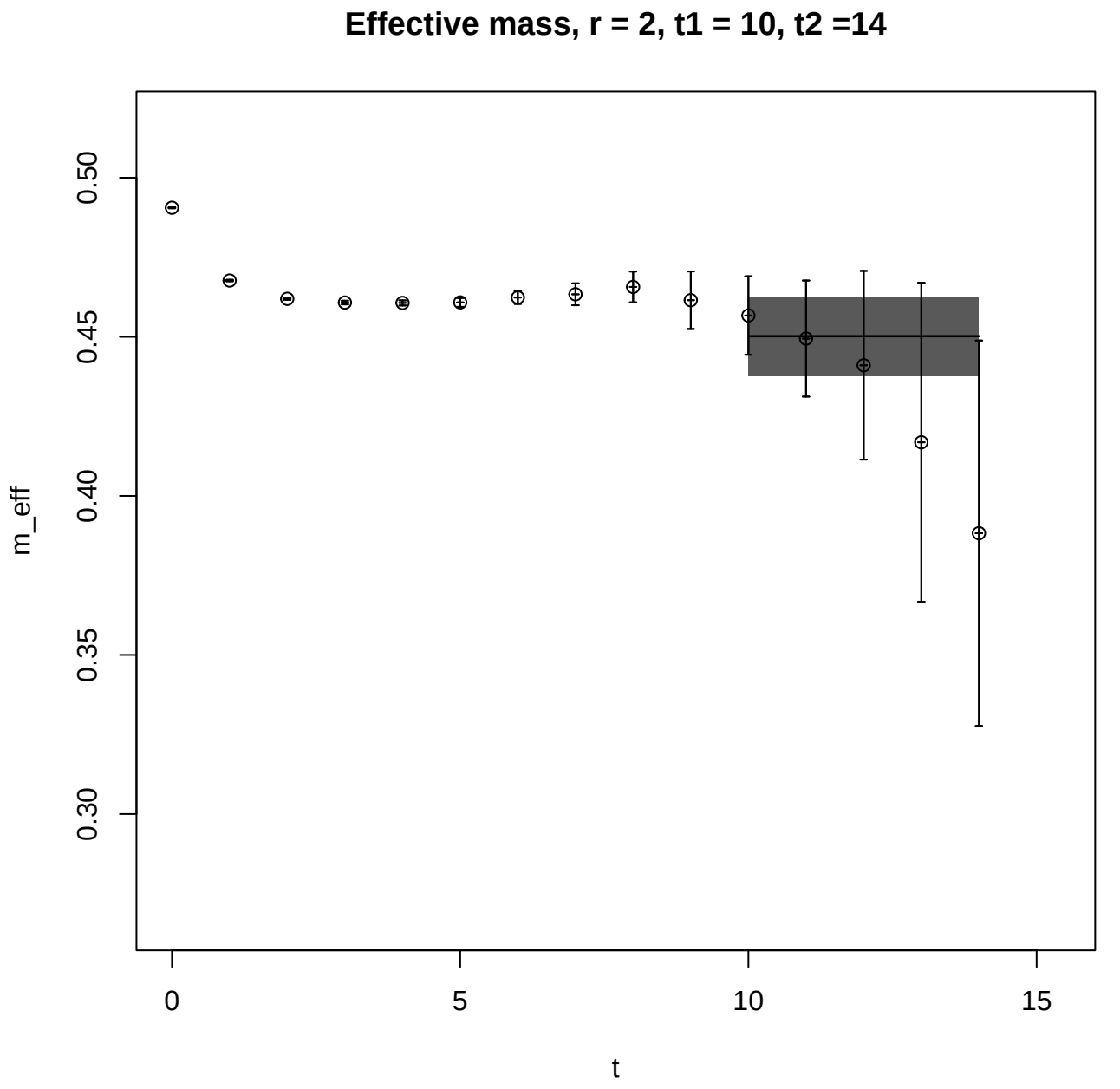

Effective mass, r = 2, t1 = 9, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

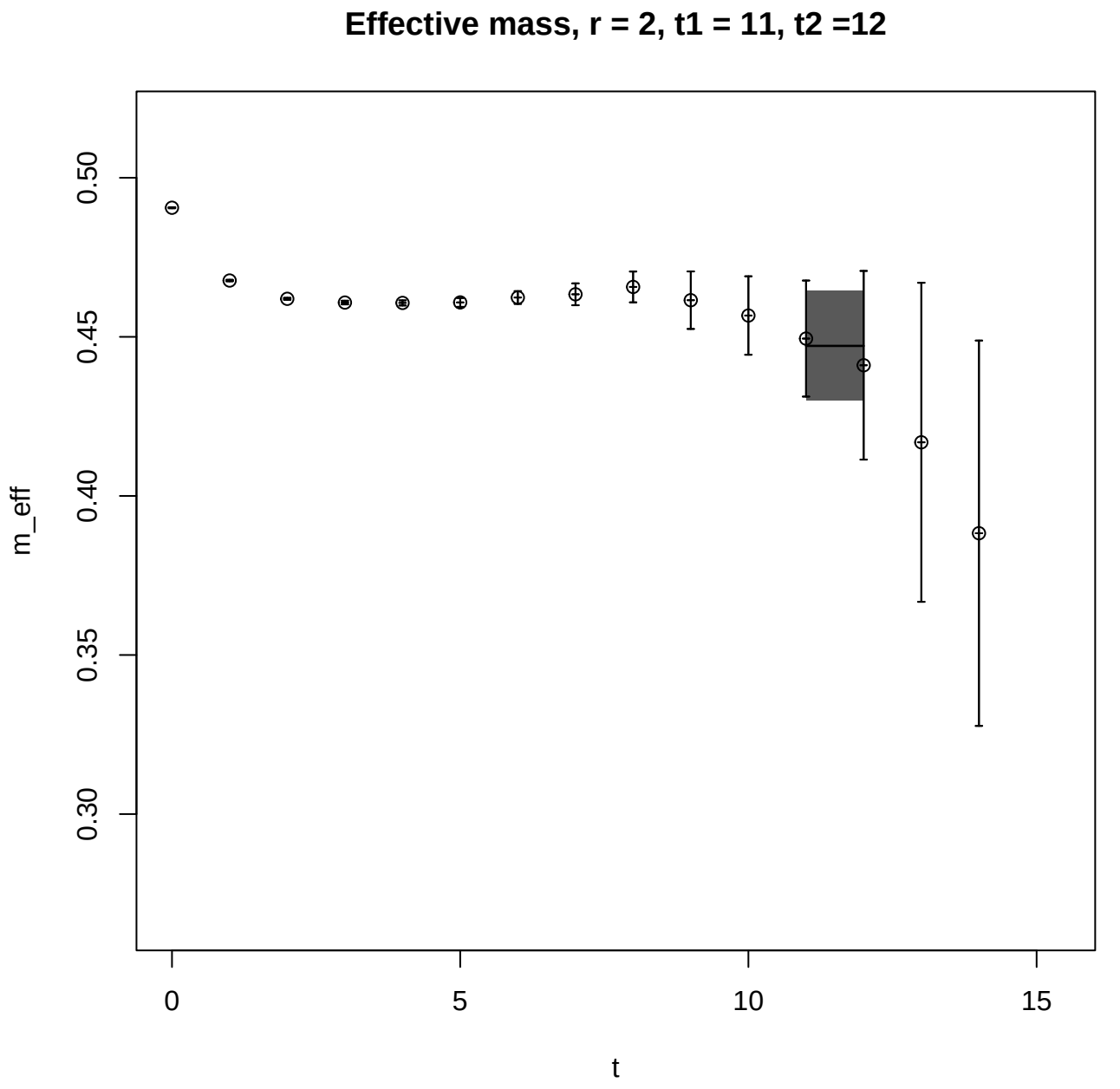

Effective mass, r = 2, t1 = 9, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

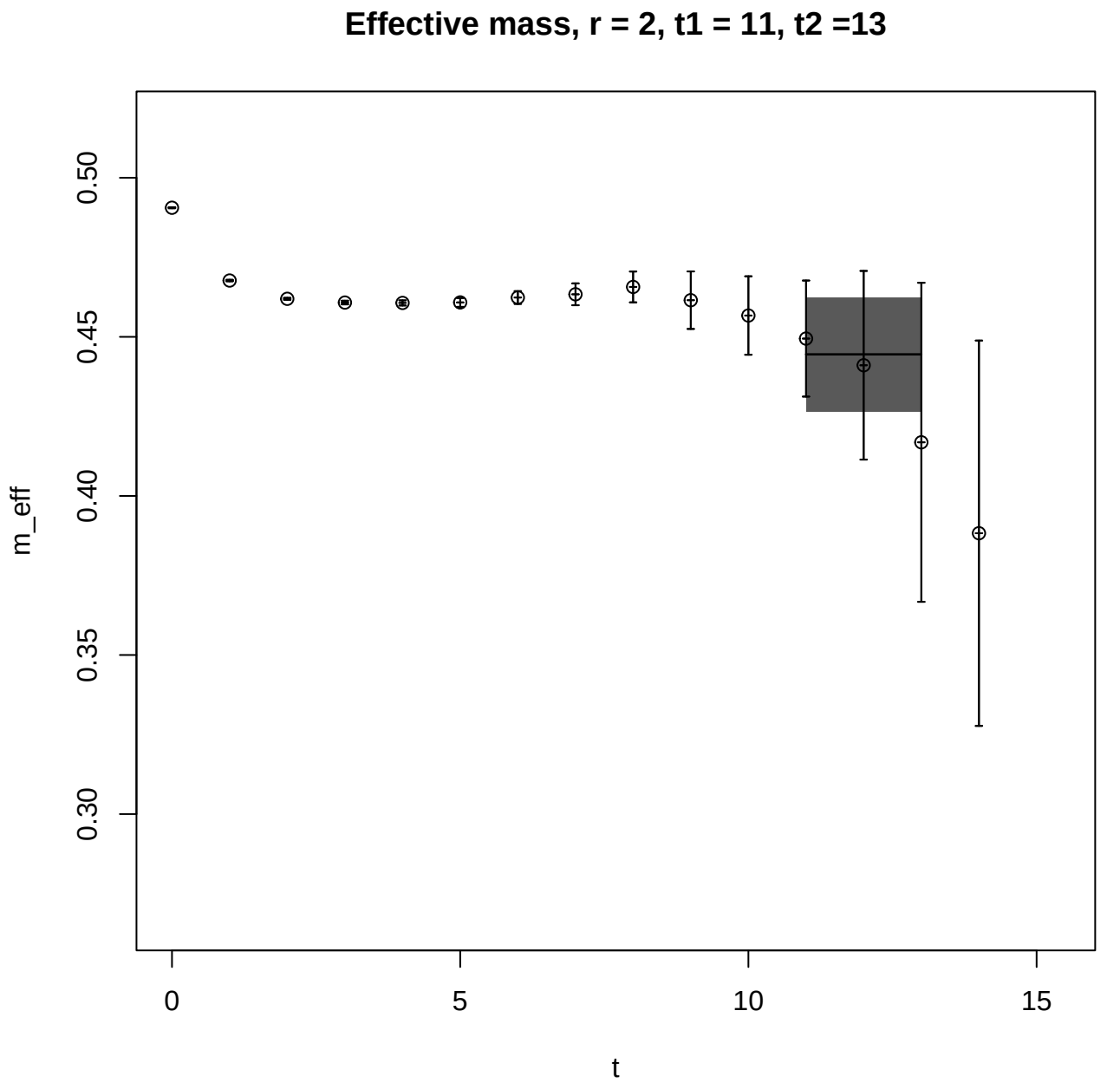

Effective mass, r = 2, t1 = 10, t2 =11
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

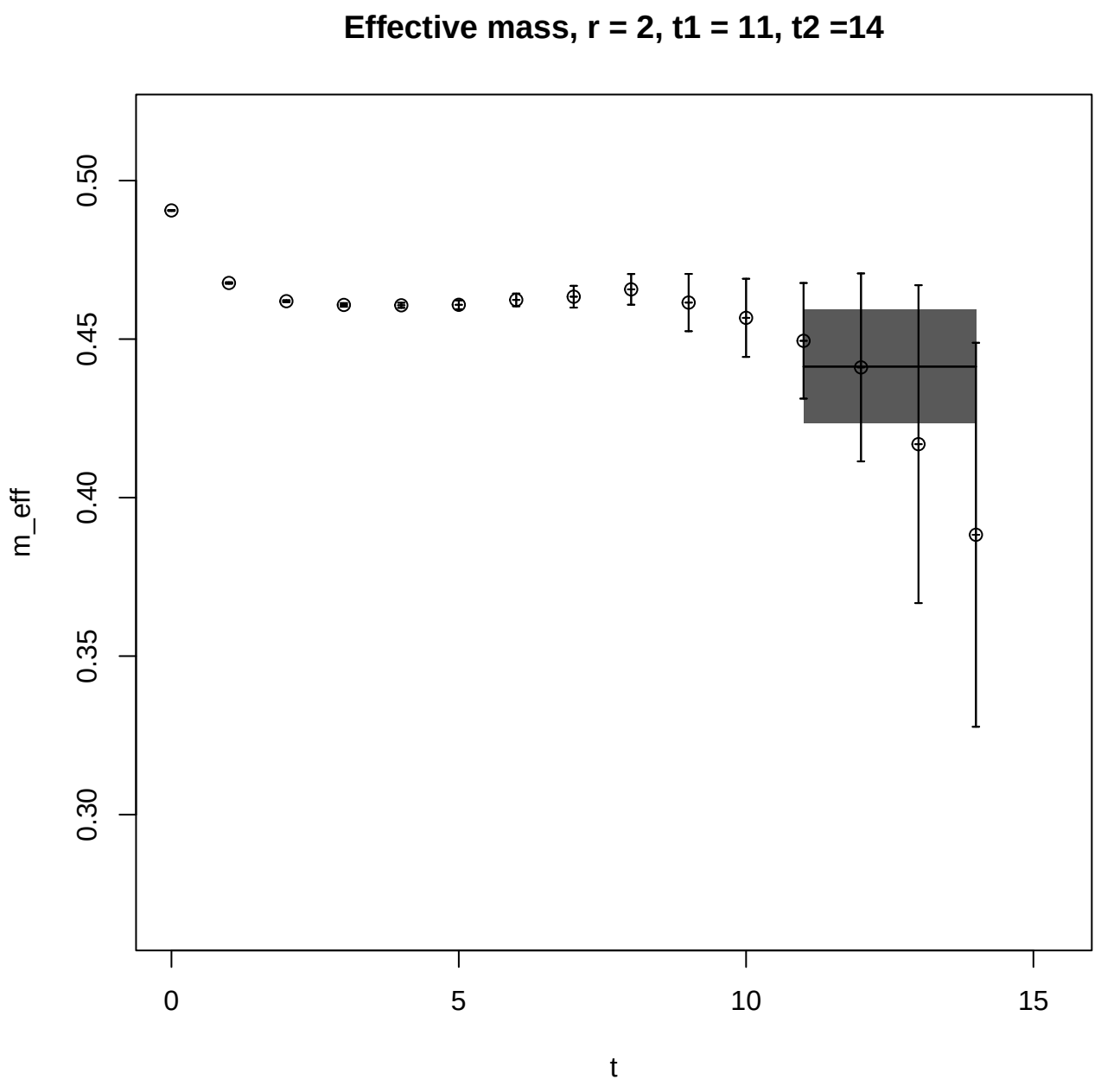

Effective mass, r = 2, t1 = 10, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

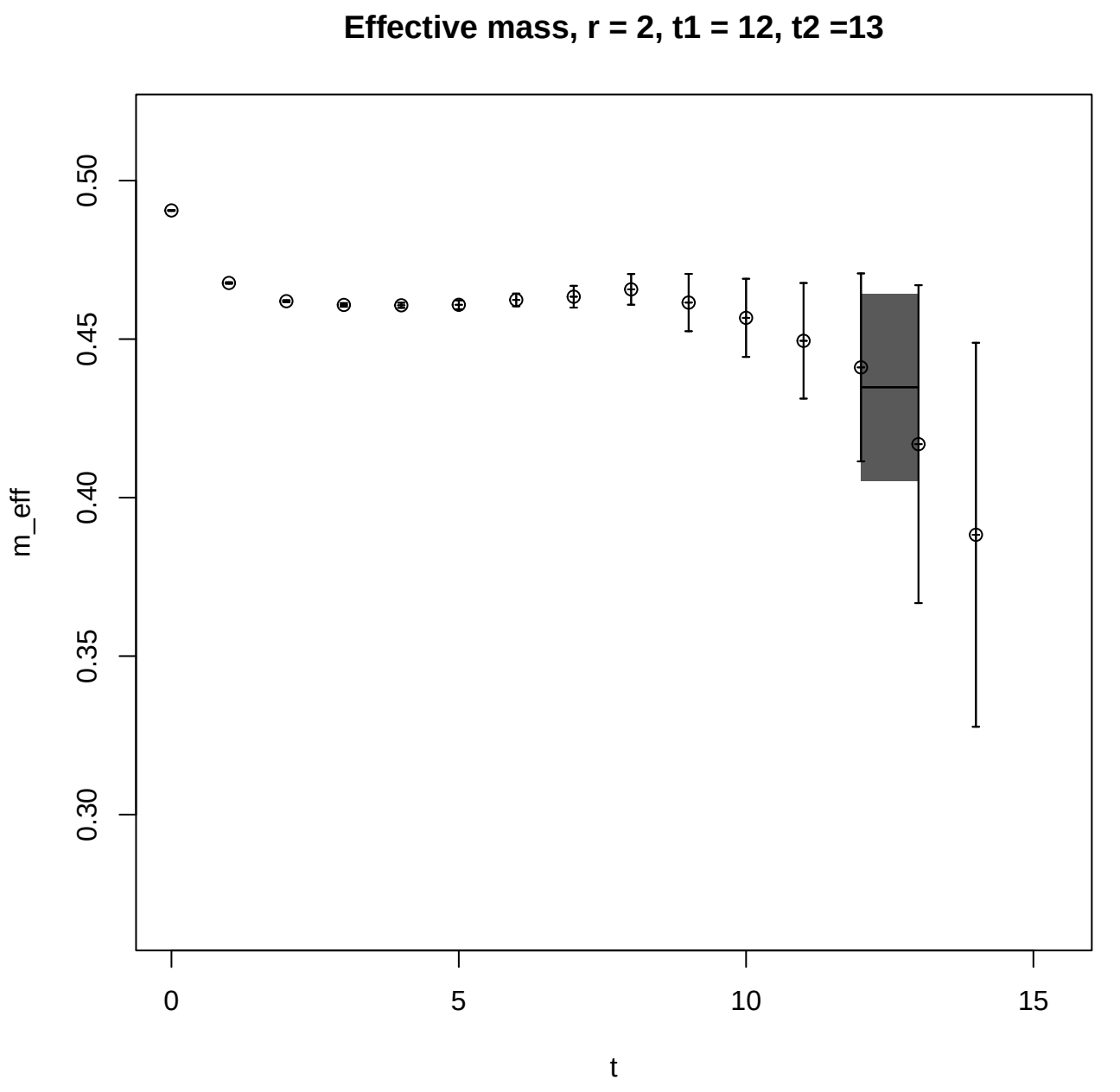

Effective mass, r = 2, t1 = 10, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

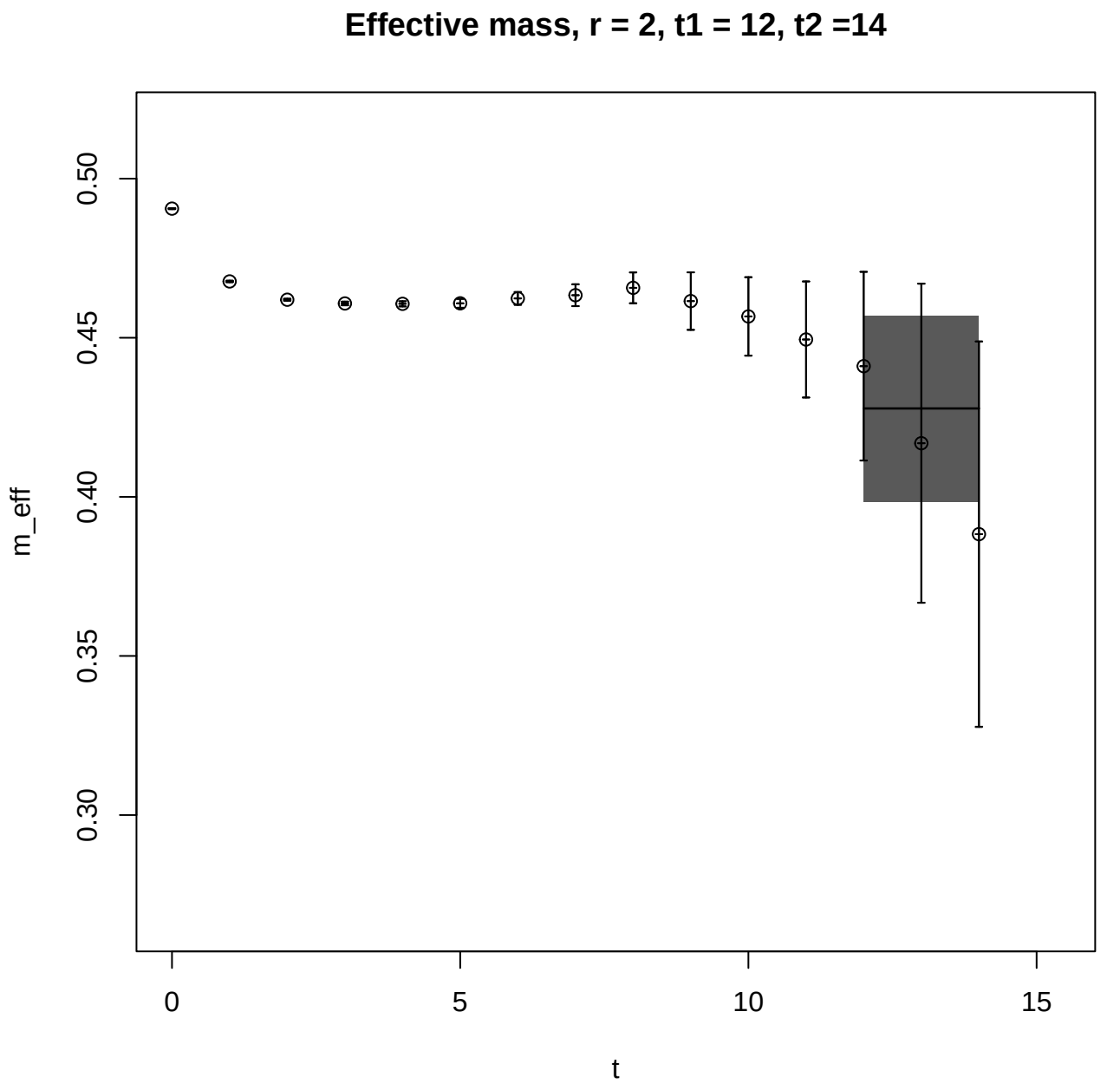

Effective mass, r = 2, t1 = 10, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

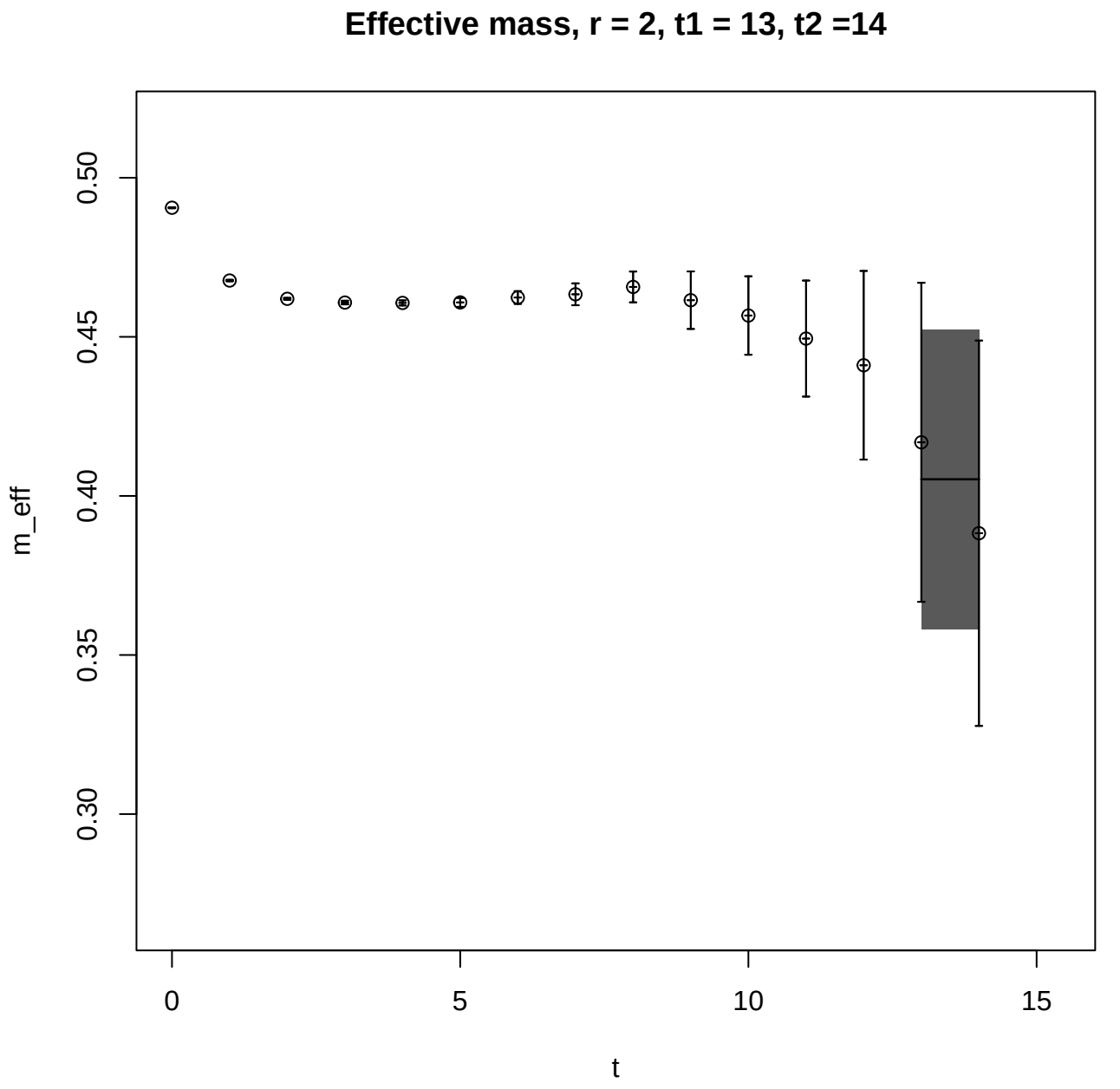

Effective mass, r = 2, t1 = 11, t2 =12
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

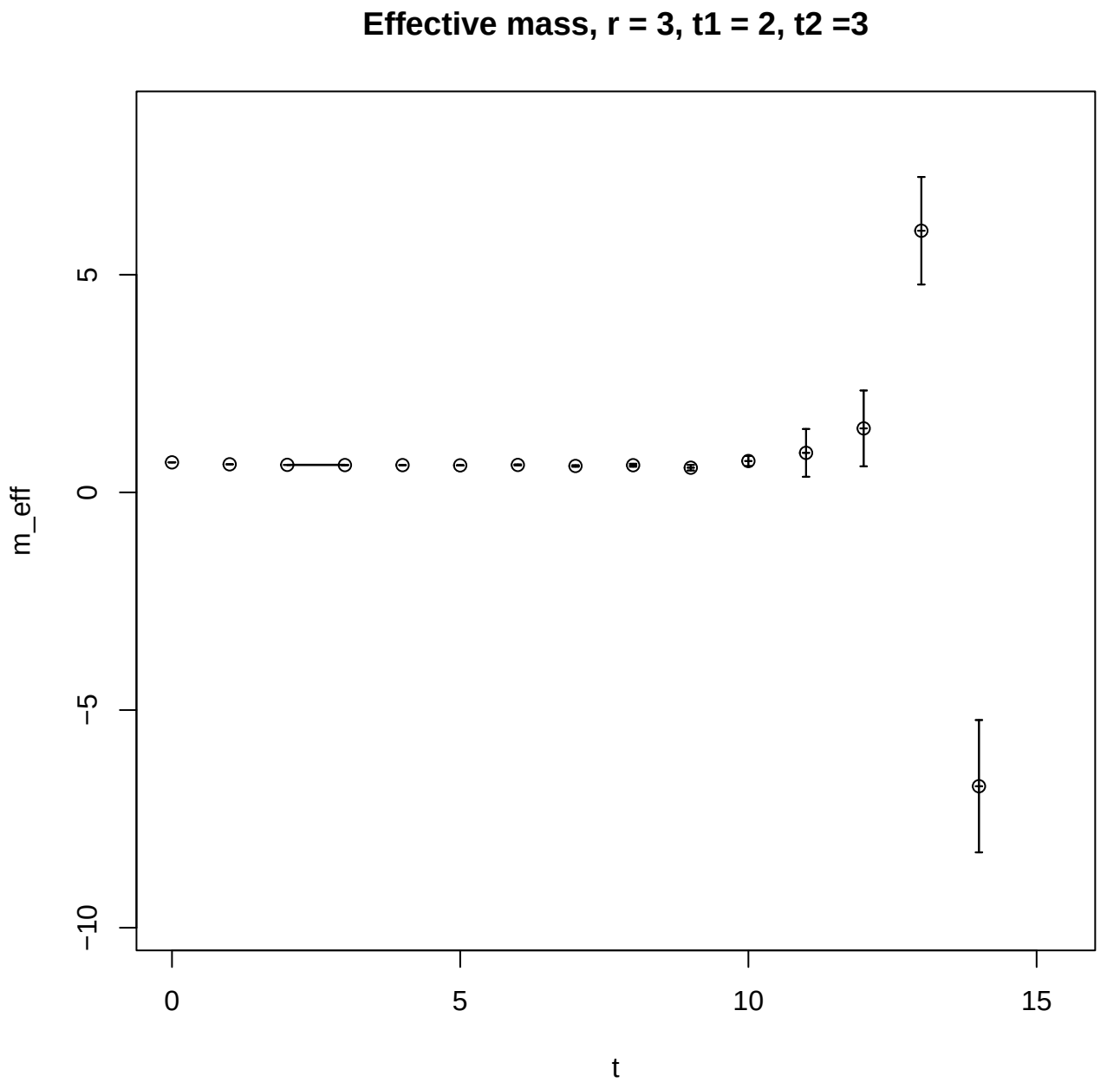

Effective mass, r = 2, t1 = 11, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

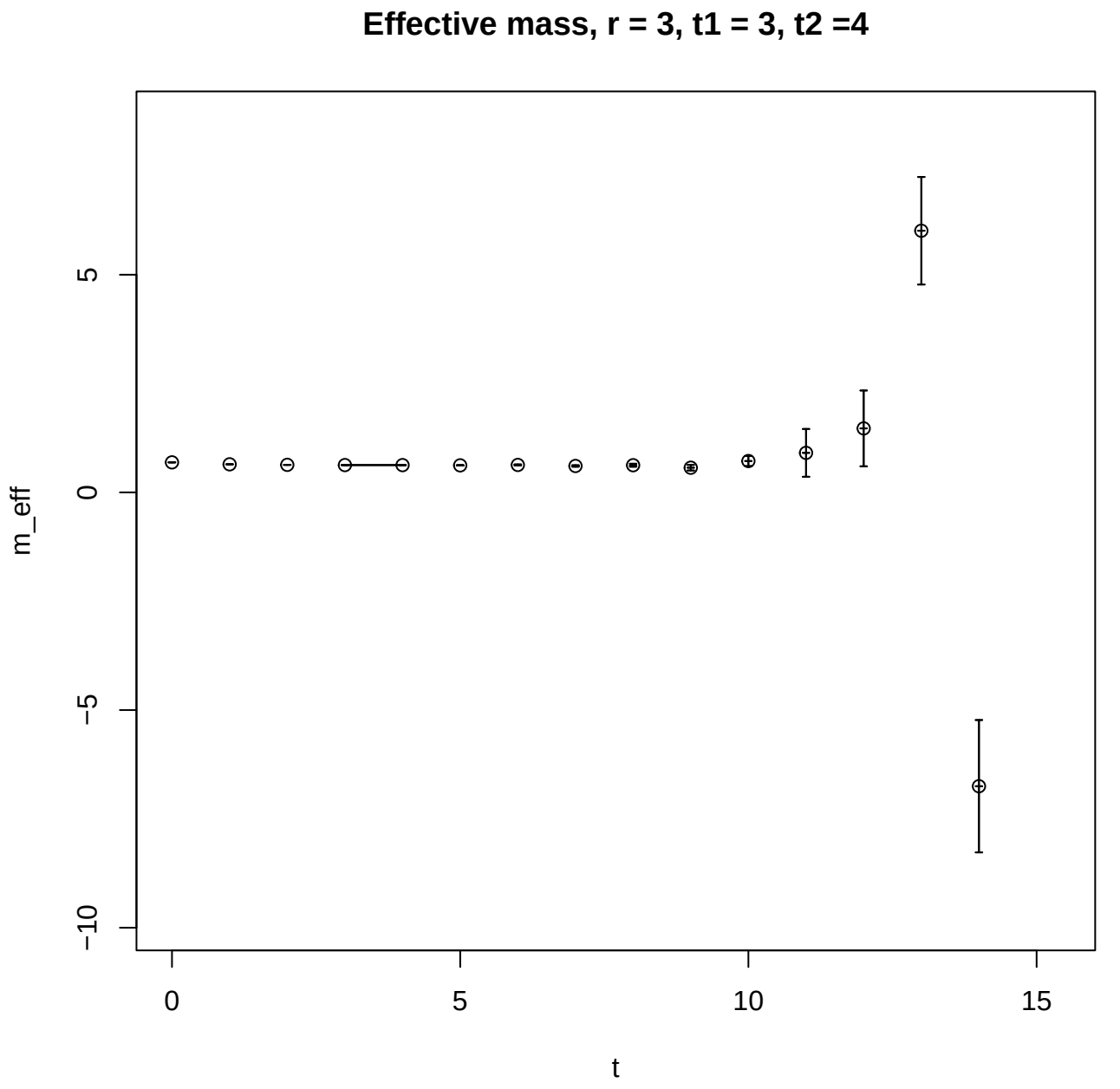

Effective mass, r = 2, t1 = 11, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

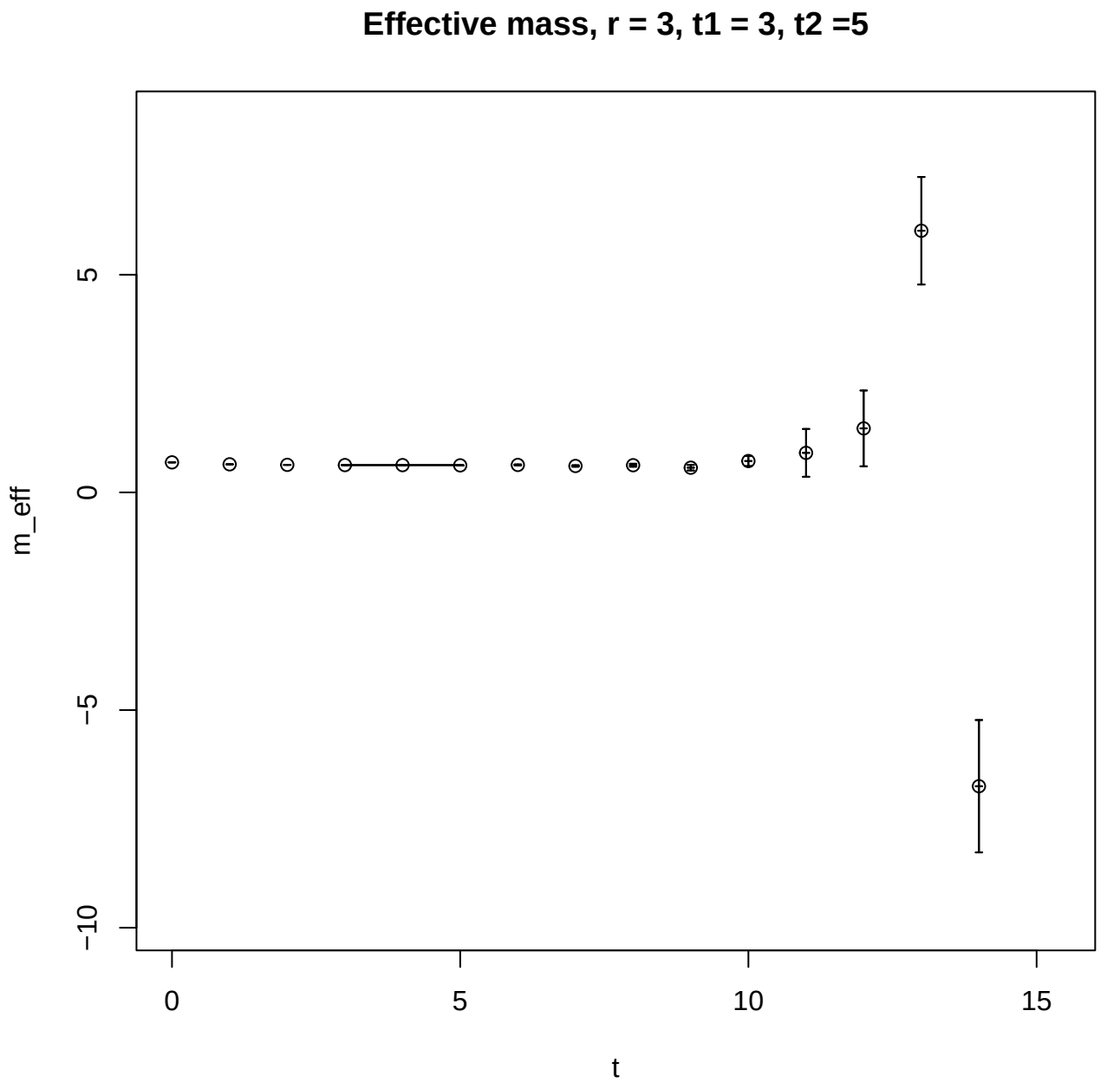

Effective mass, r = 2, t1 = 12, t2 =13
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

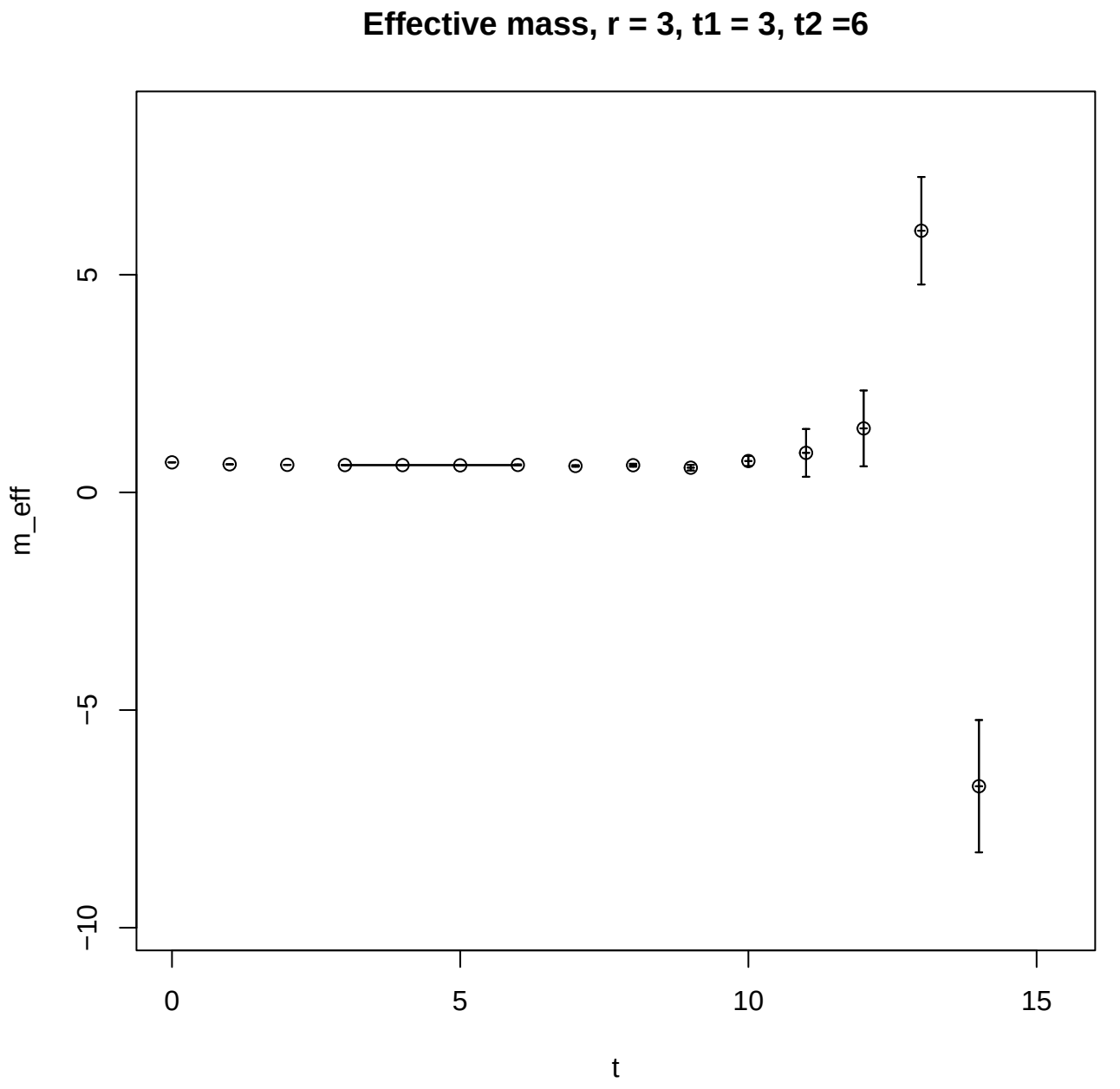

Effective mass, r = 2, t1 = 12, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

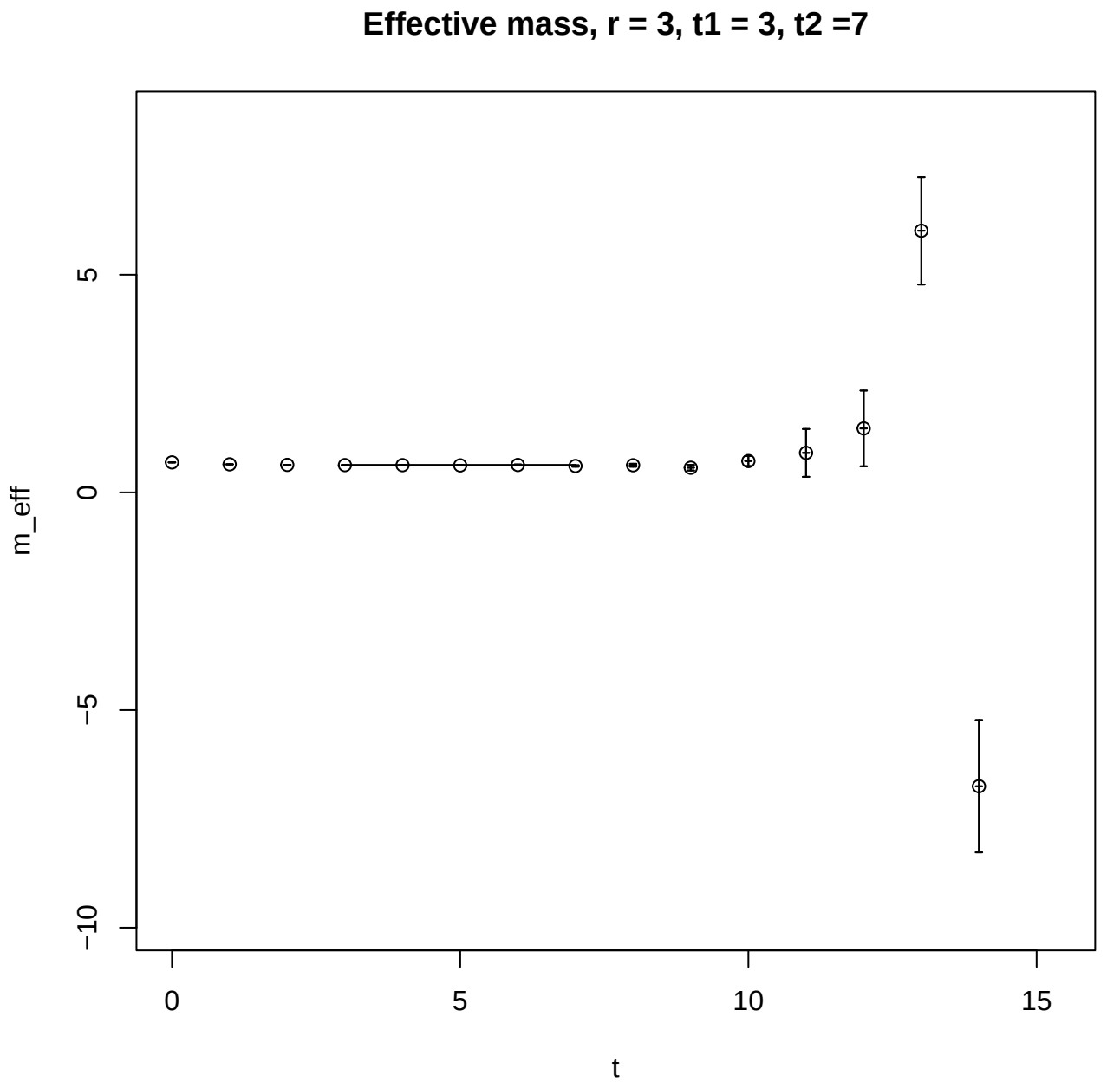

Effective mass, r = 2, t1 = 13, t2 =14
m_eff
0.30
0.35
0.40
0.45
0.50
0
5
10
15
t

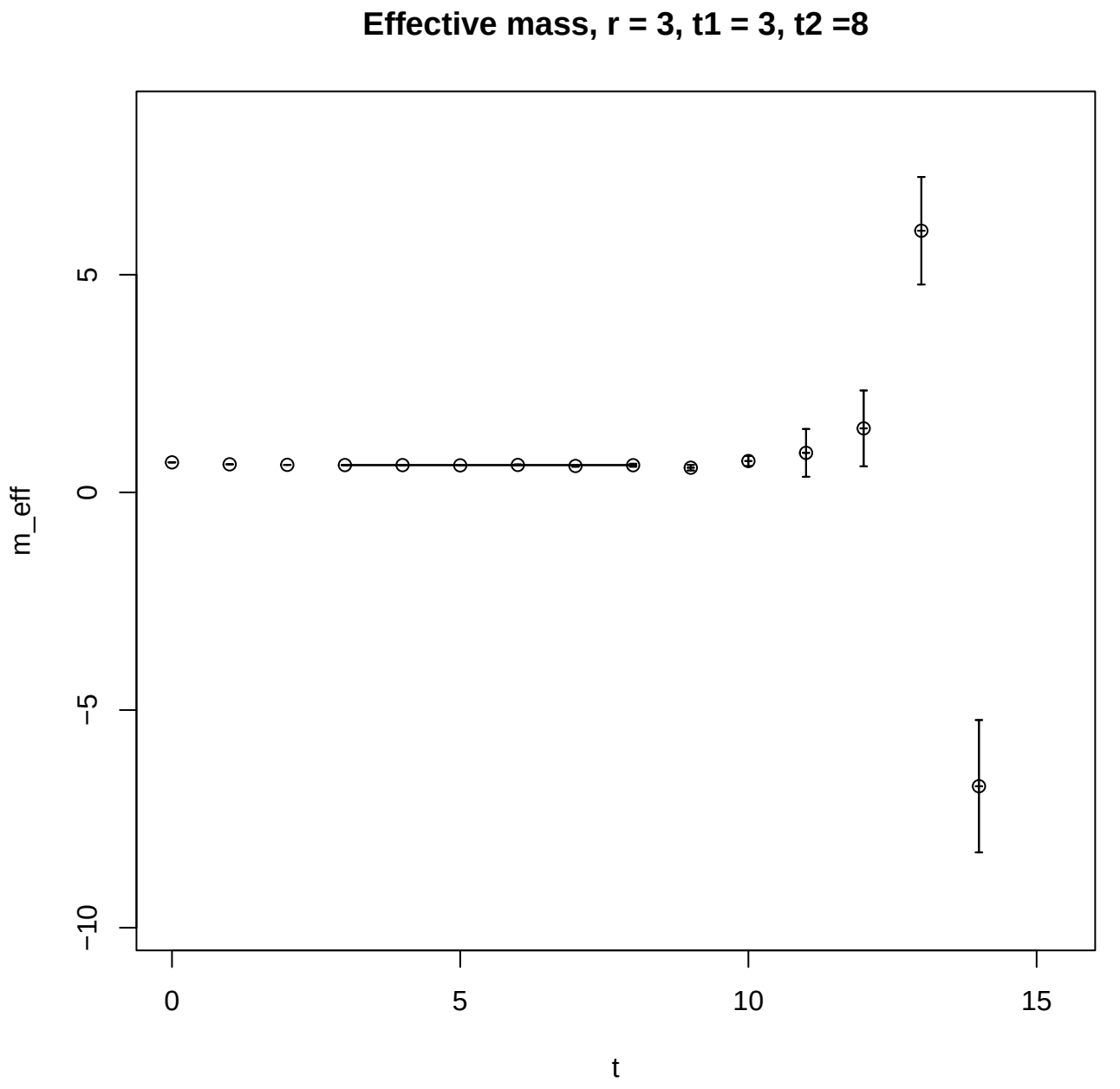

Effective mass, r = 3, t1 = 2, t2 =3
m_eff
5
0
−5
−10
0
5
10
15
t

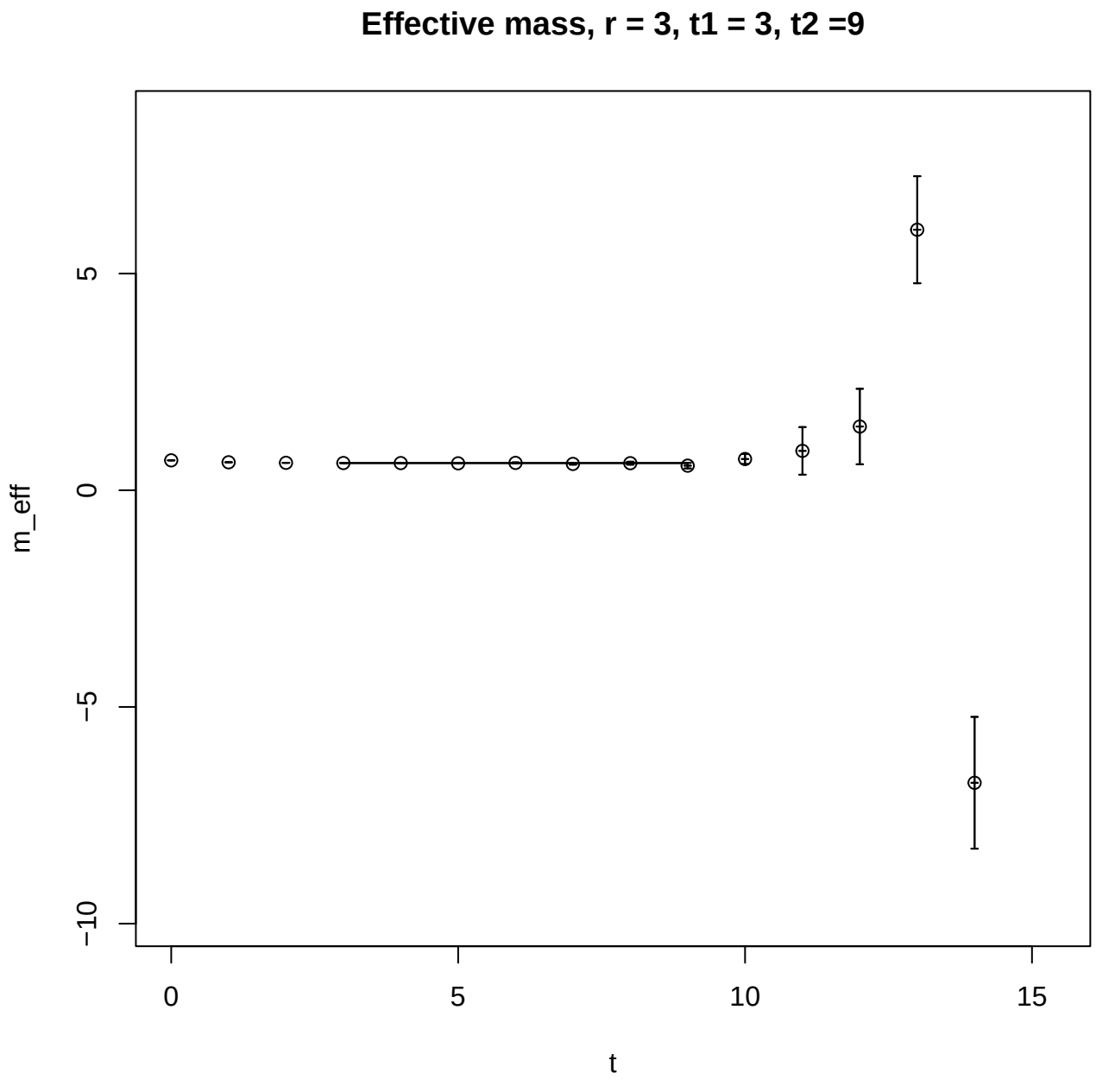

Effective mass, r = 3, t1 = 3, t2 =4
m_eff
5
0
−5
−10
0
5
10
15
t

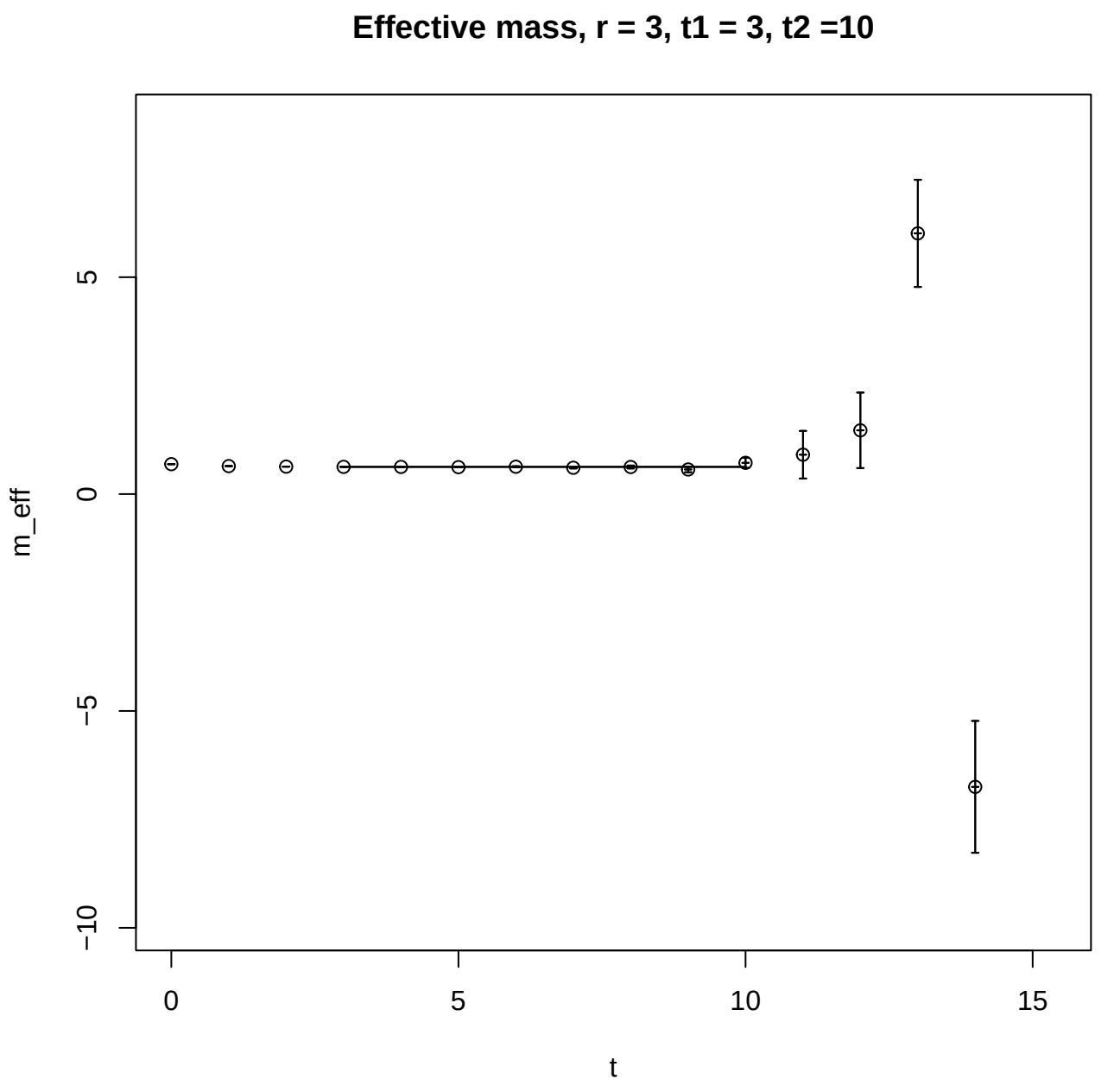

Effective mass, r = 3, t1 = 3, t2 =5
m_eff
t
5
0
−5
−10
0
5
10
15

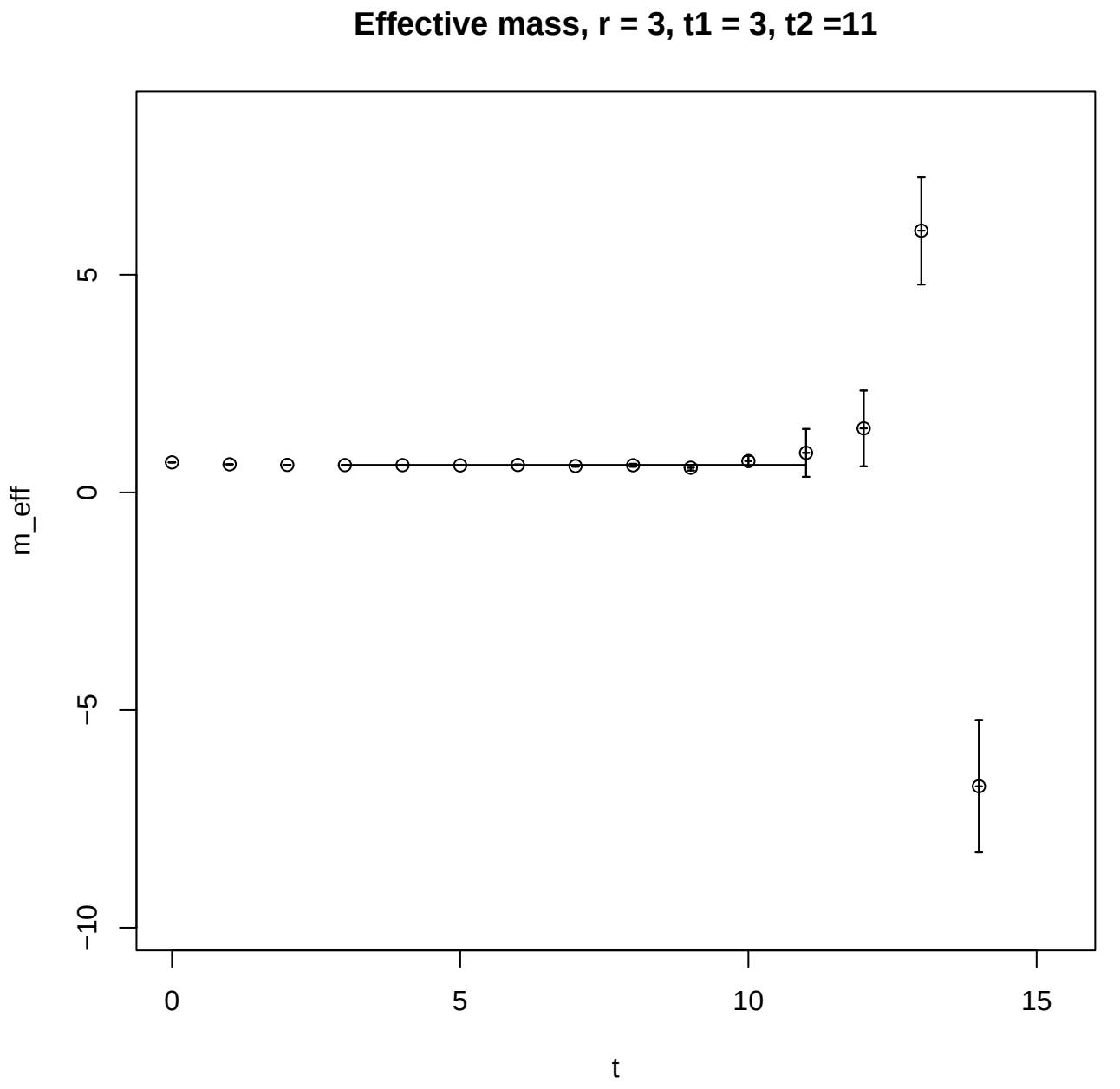

Effective mass, r = 3, t1 = 3, t2 =6
m_eff
t
5
0
−5
−10
0
5
10
15

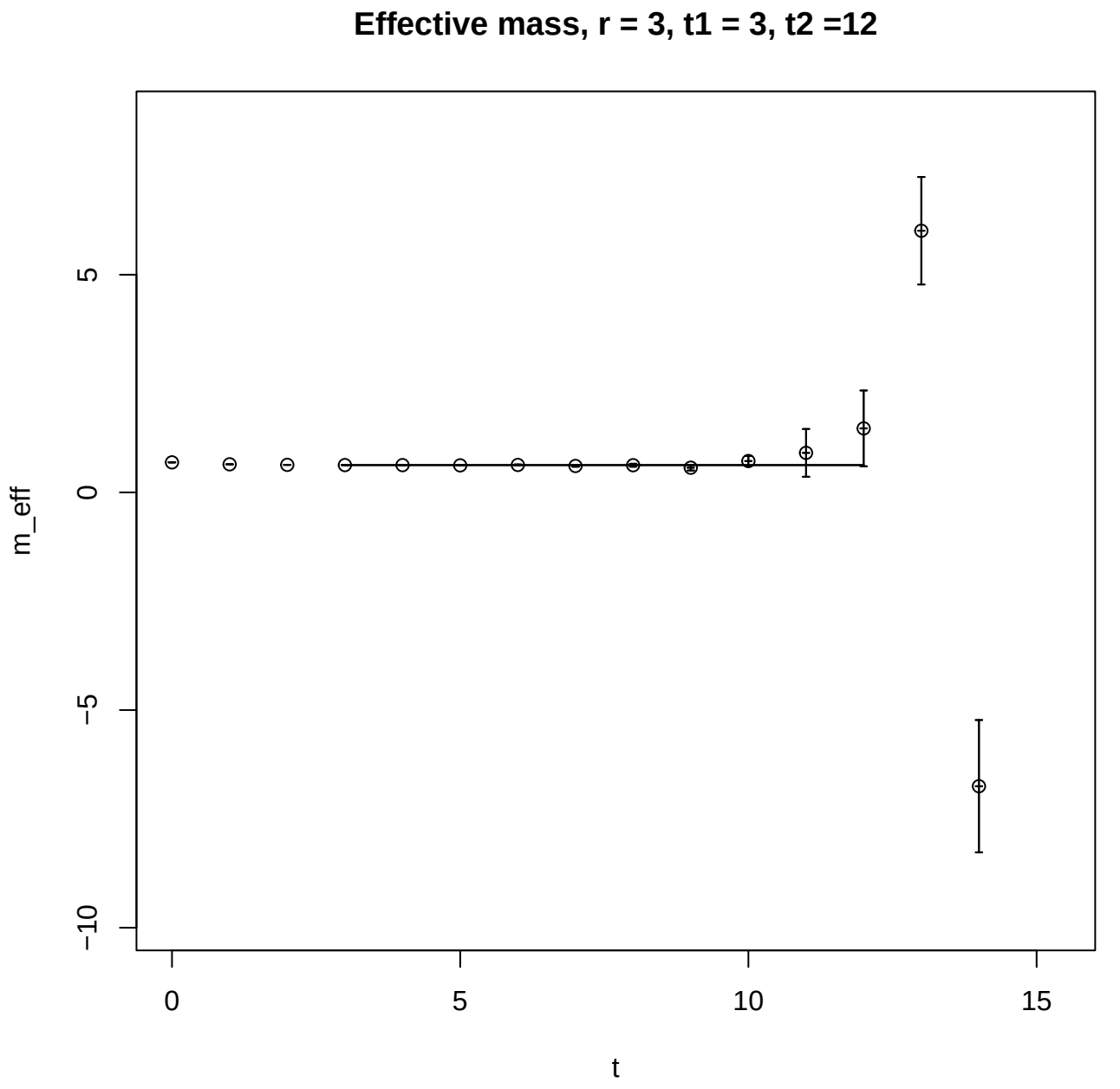

Effective mass, r = 3, t1 = 3, t2 =7
m_eff
t
5
0
−5
−10
0
5
10
15

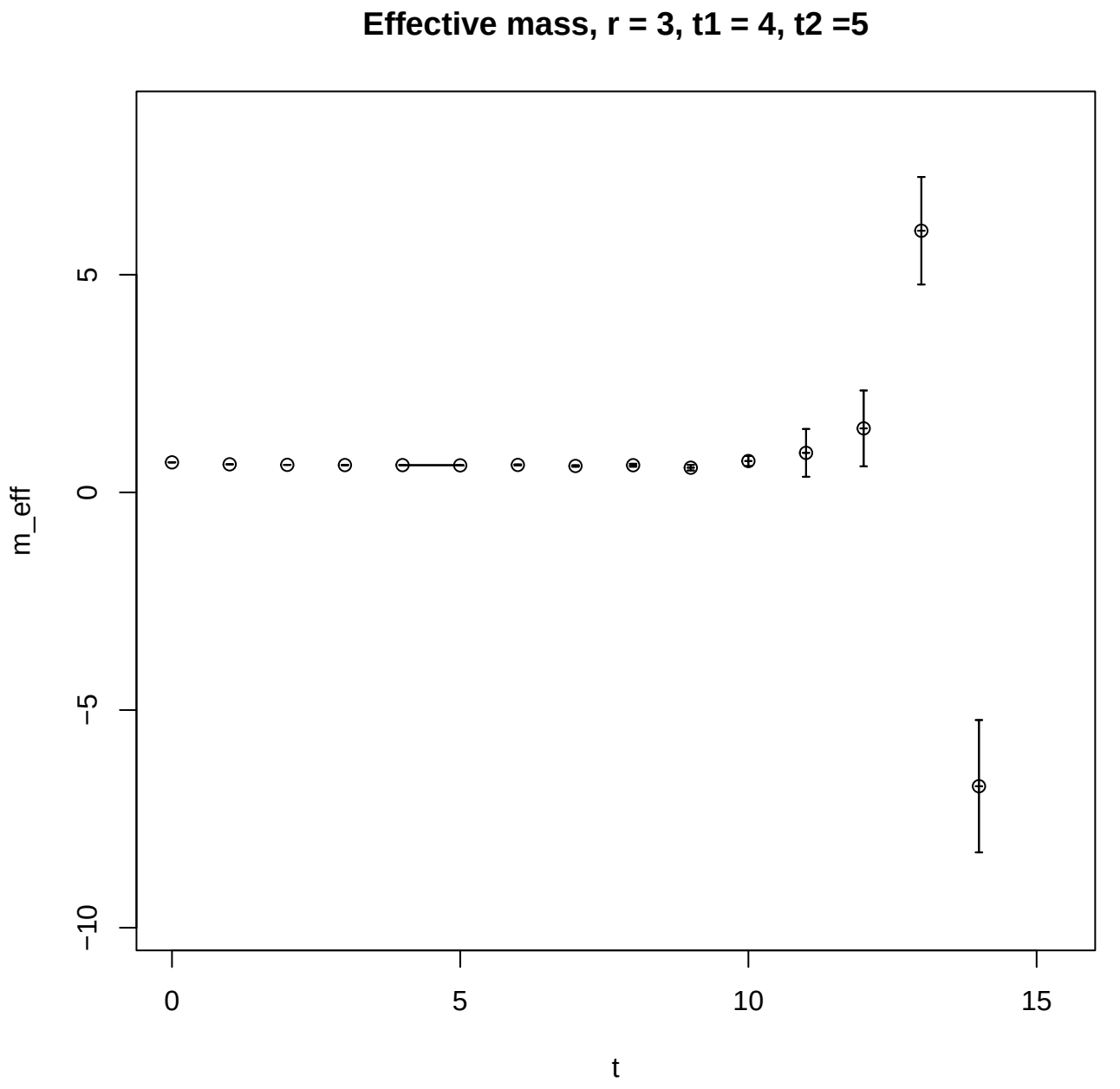

Effective mass, r = 3, t1 = 3, t2 =8
m_eff
t
5
0
−5
−10
0
5
10
15

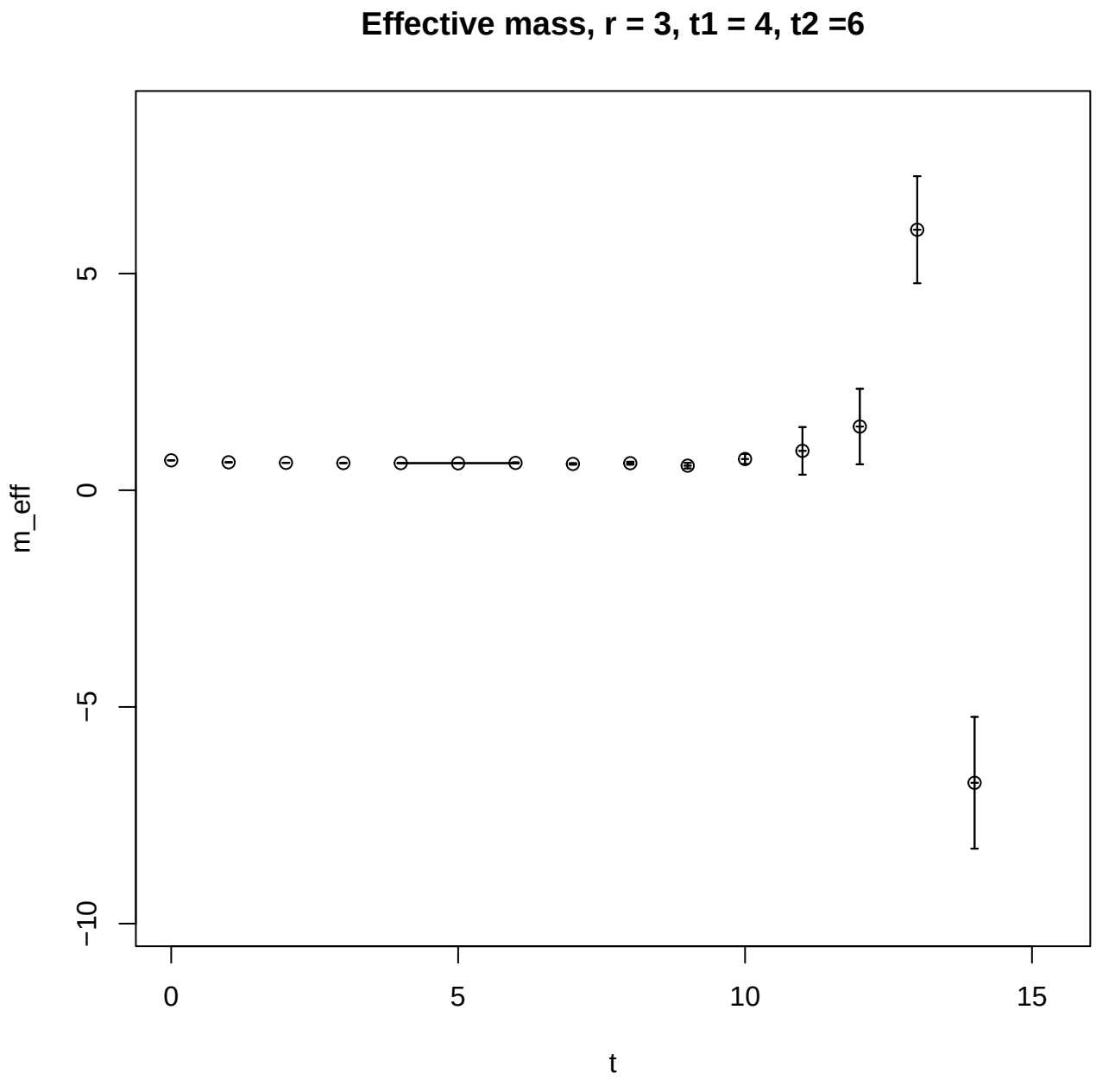

Effective mass, r = 3, t1 = 3, t2 =9
m_eff
t
5
0
−5
−10
0
5
10
15

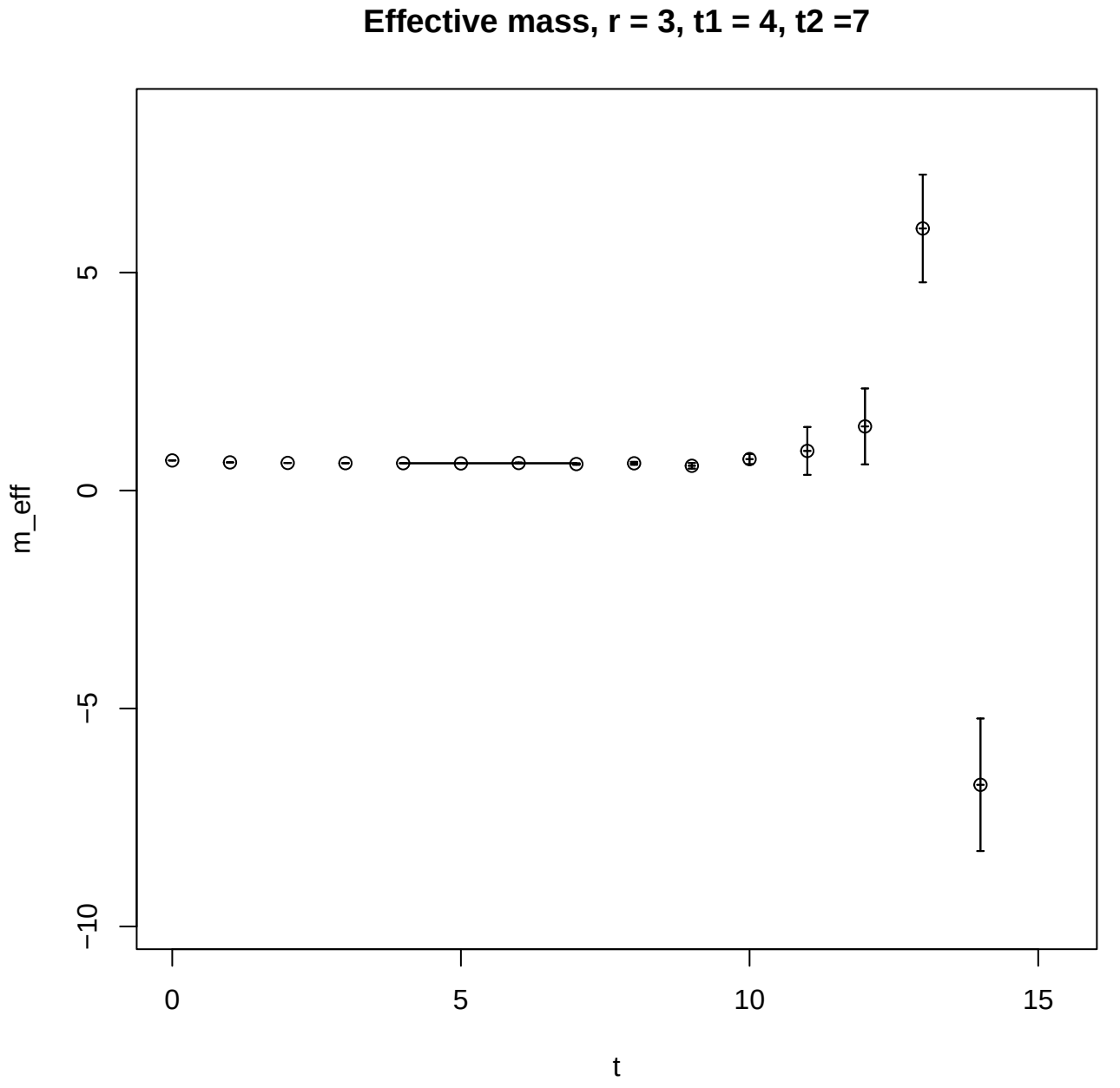

Effective mass, r = 3, t1 = 3, t2 =10
m_eff
t
5
0
−5
−10
0
5
10
15

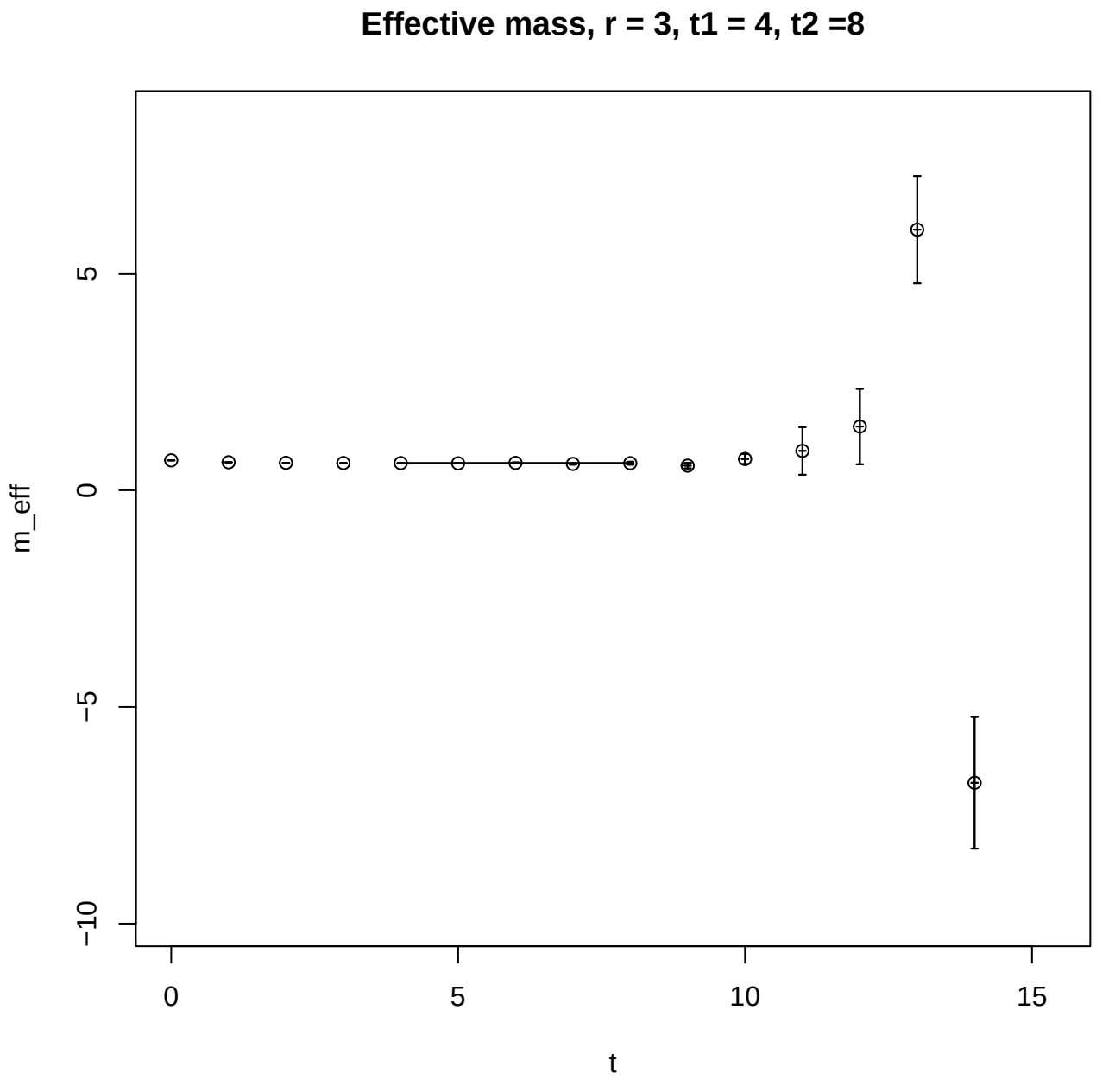

Effective mass, r = 3, t1 = 3, t2 =11
m_eff
t
5
0
−5
−10
0
5
10
15

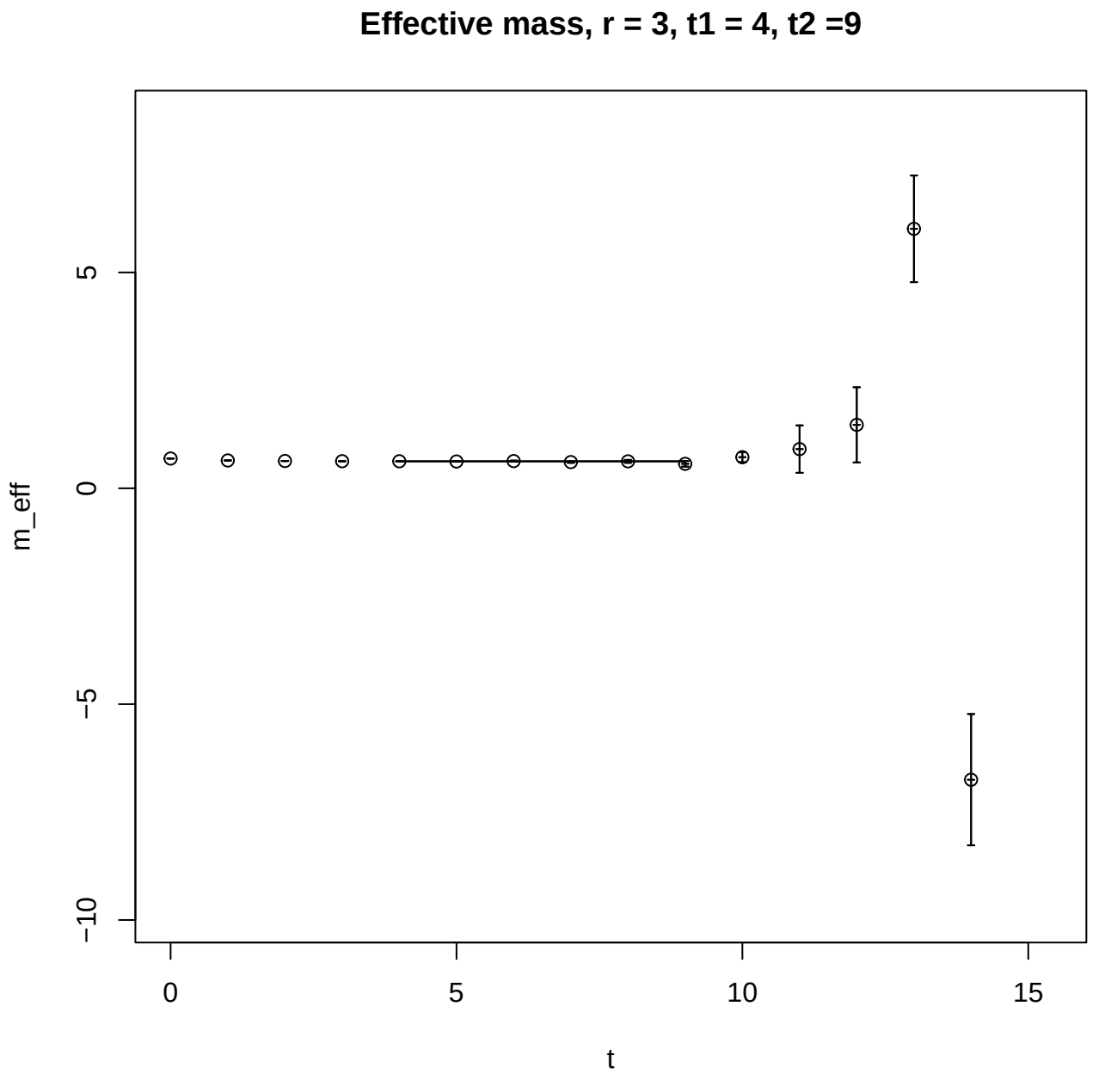

Effective mass, r = 3, t1 = 3, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

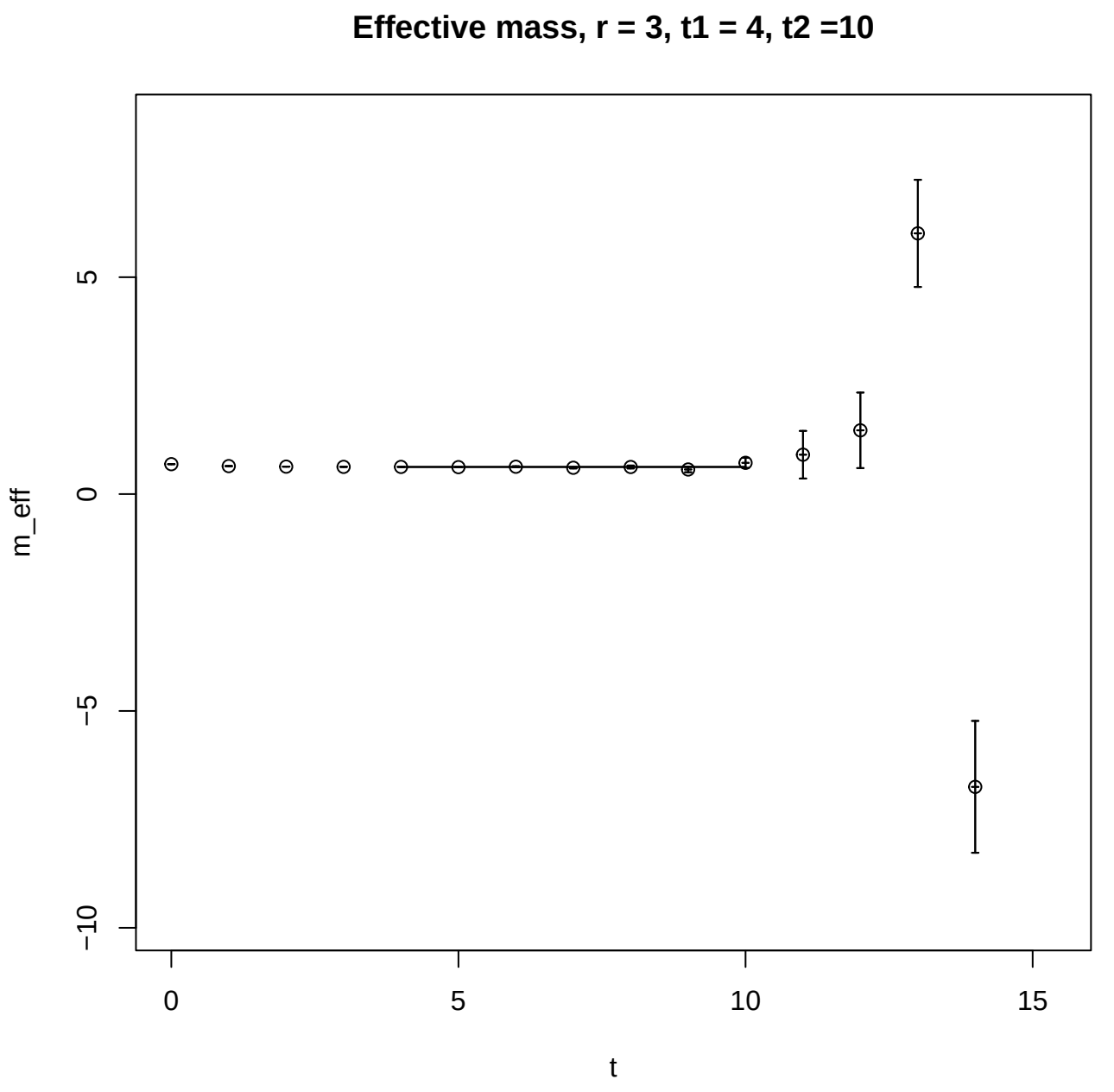

Effective mass, r = 3, t1 = 4, t2 =5
m_eff
5
0
−5
−10
0
5
10
15
t

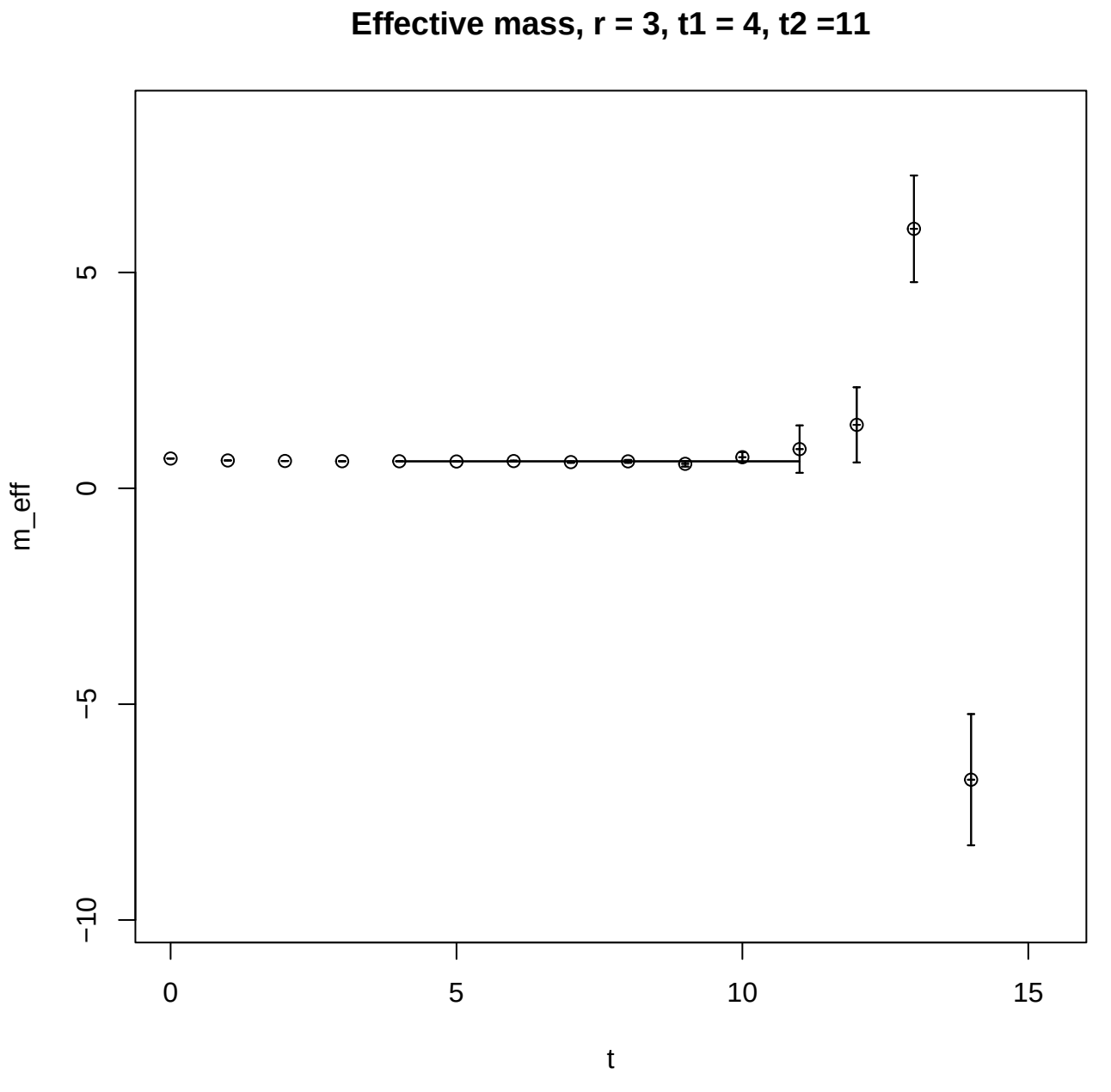

Effective mass, r = 3, t1 = 4, t2 =6
m_eff
5
0
−5
−10
0
5
10
15
t

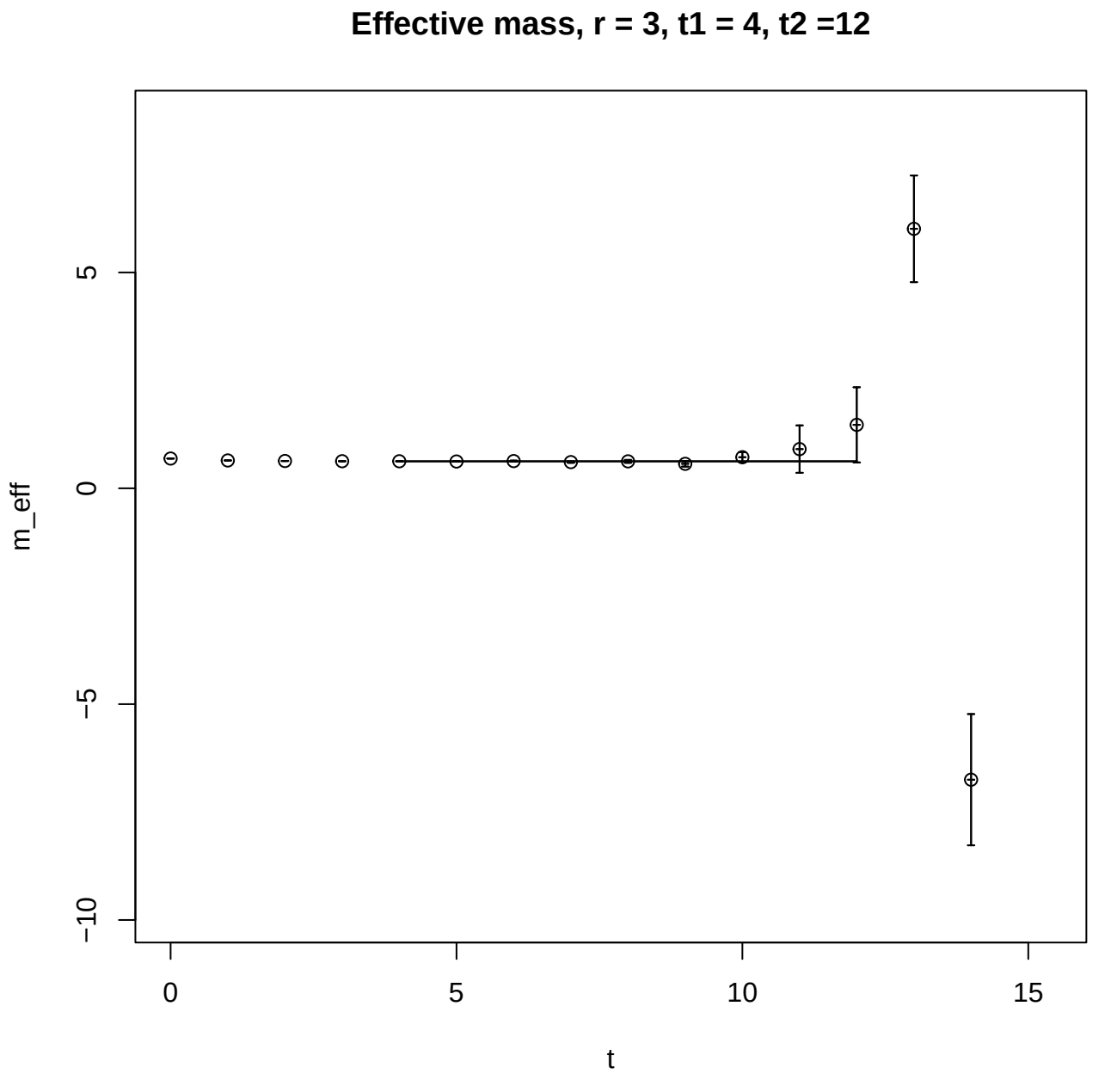

Effective mass, r = 3, t1 = 4, t2 =7
m_eff
t
5
0
−5
−10
0
5
10
15

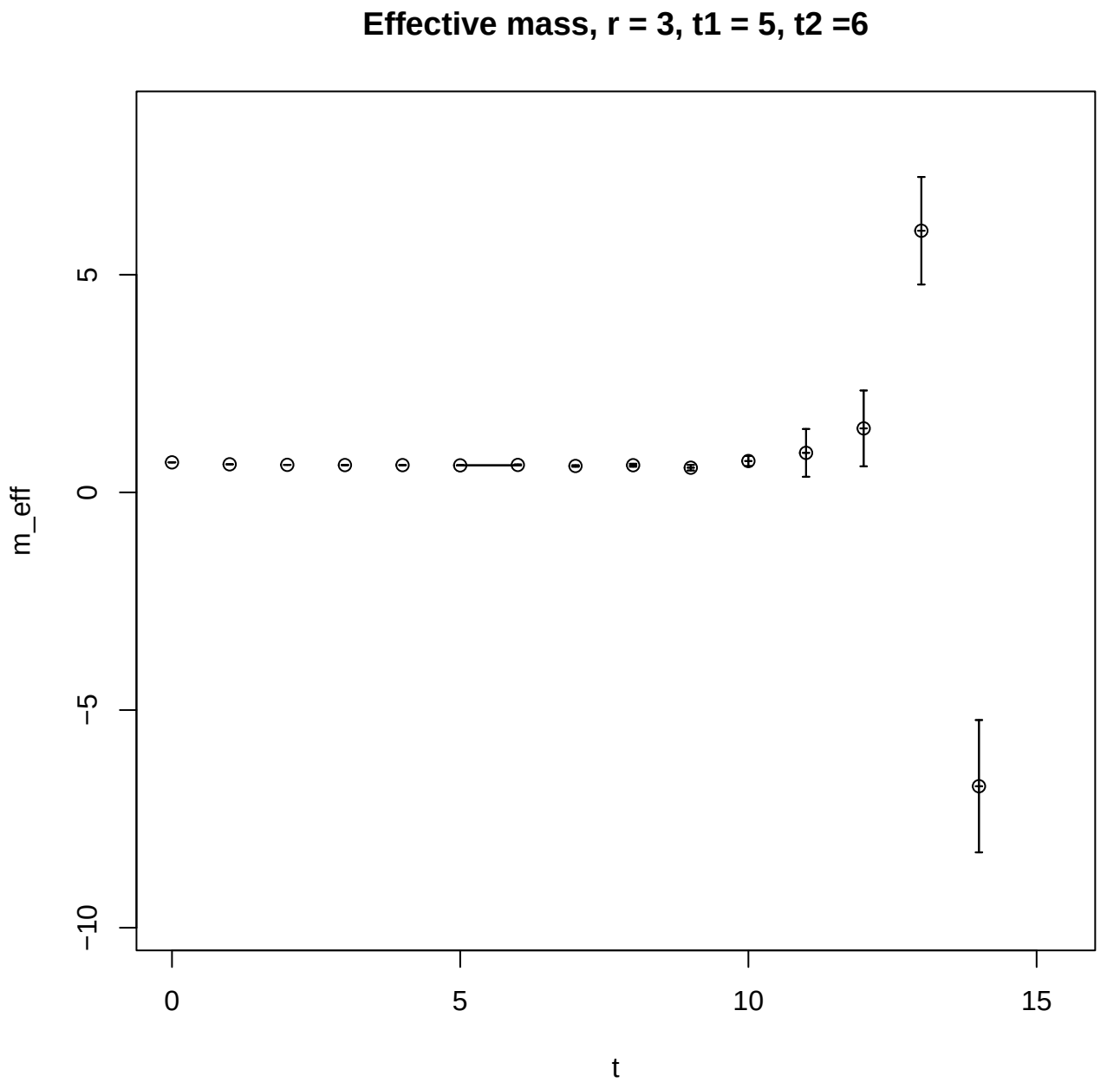

Effective mass, r = 3, t1 = 4, t2 =8
m_eff
t
5
0
−5
−10
0
5
10
15

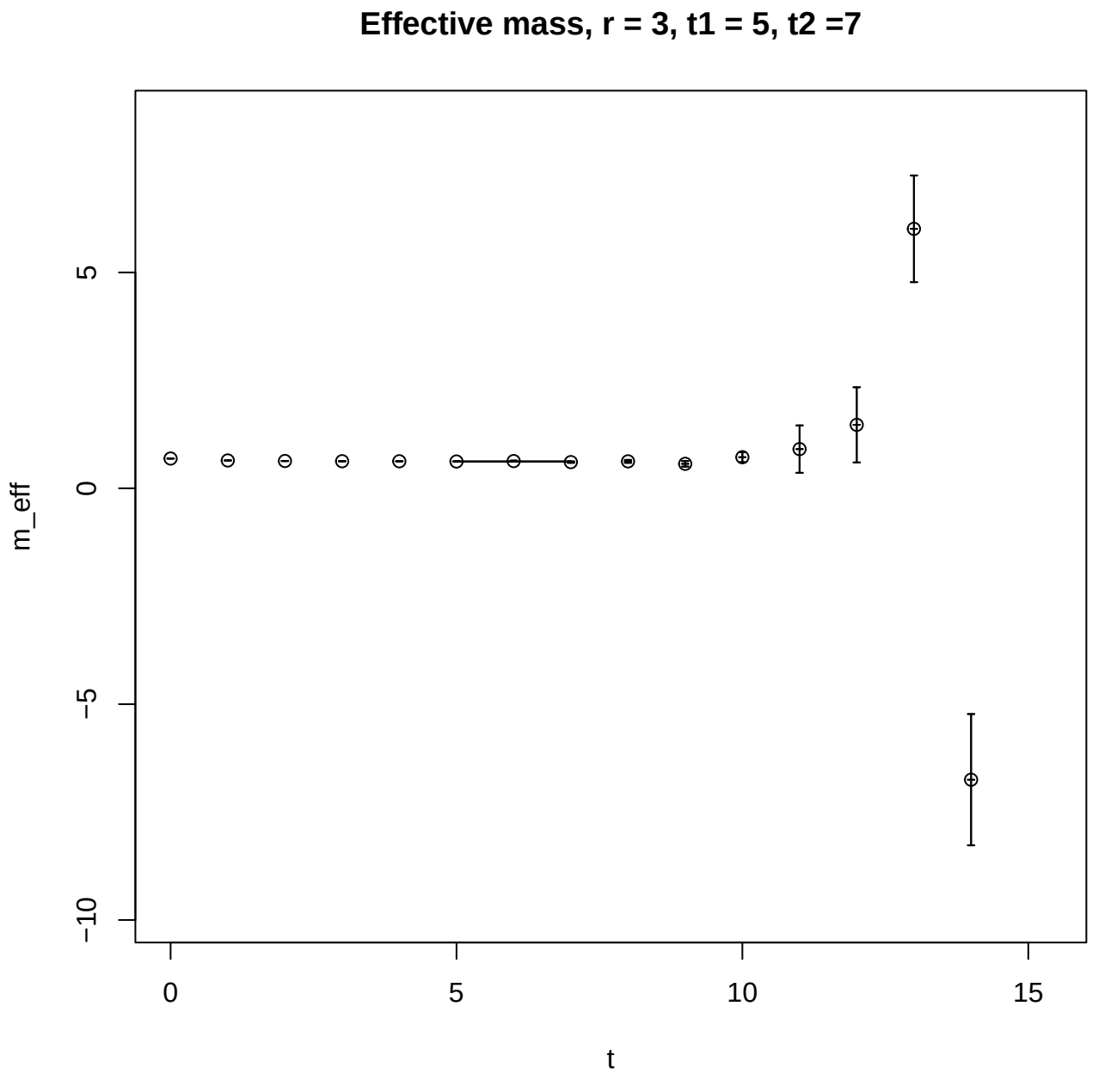

Effective mass, r = 3, t1 = 4, t2 =9
m_eff
t
5
0
−5
−10
0
5
10
15

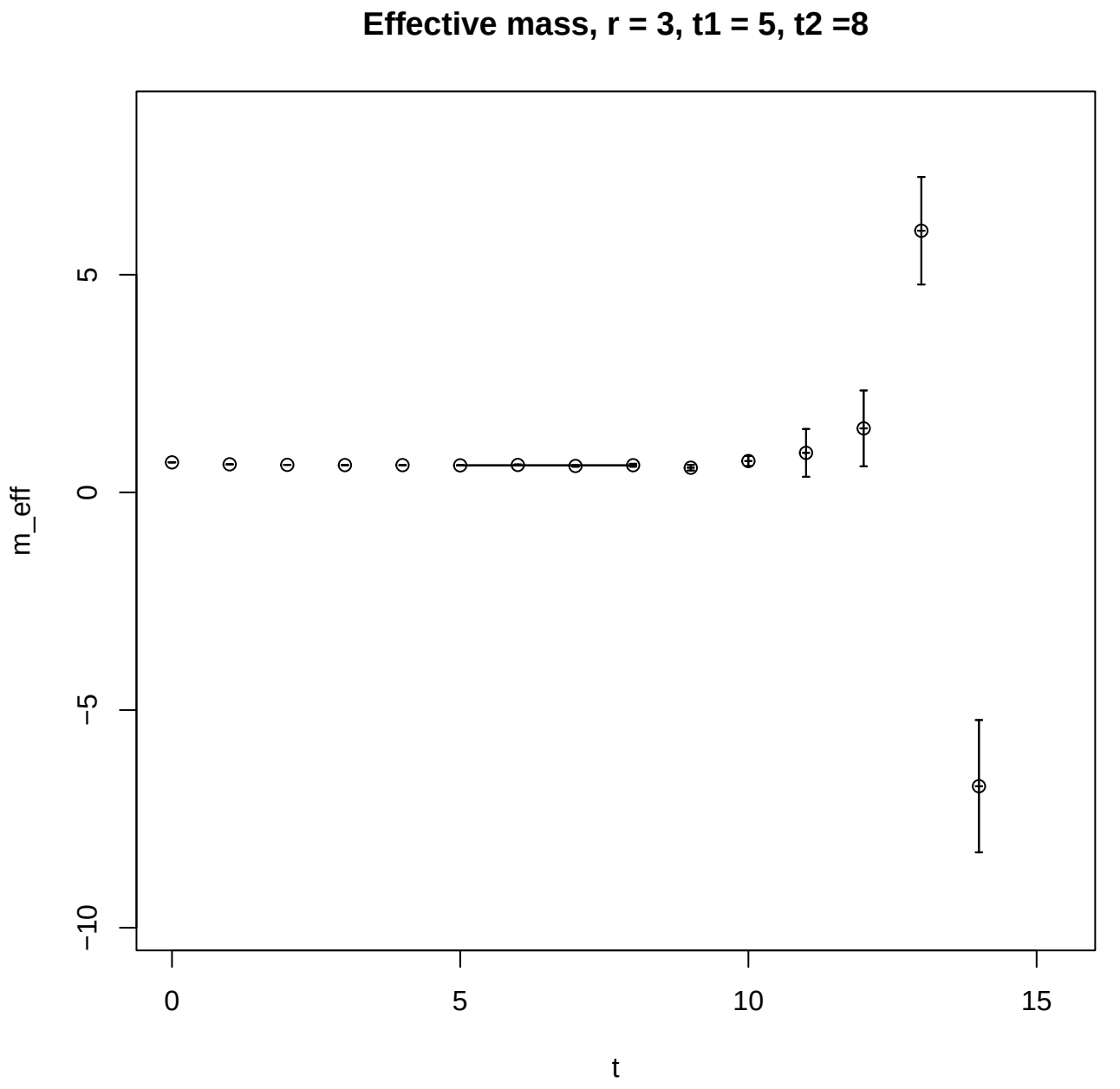

Effective mass, r = 3, t1 = 4, t2 =10
m_eff
t
5
0
−5
−10
0
5
10
15

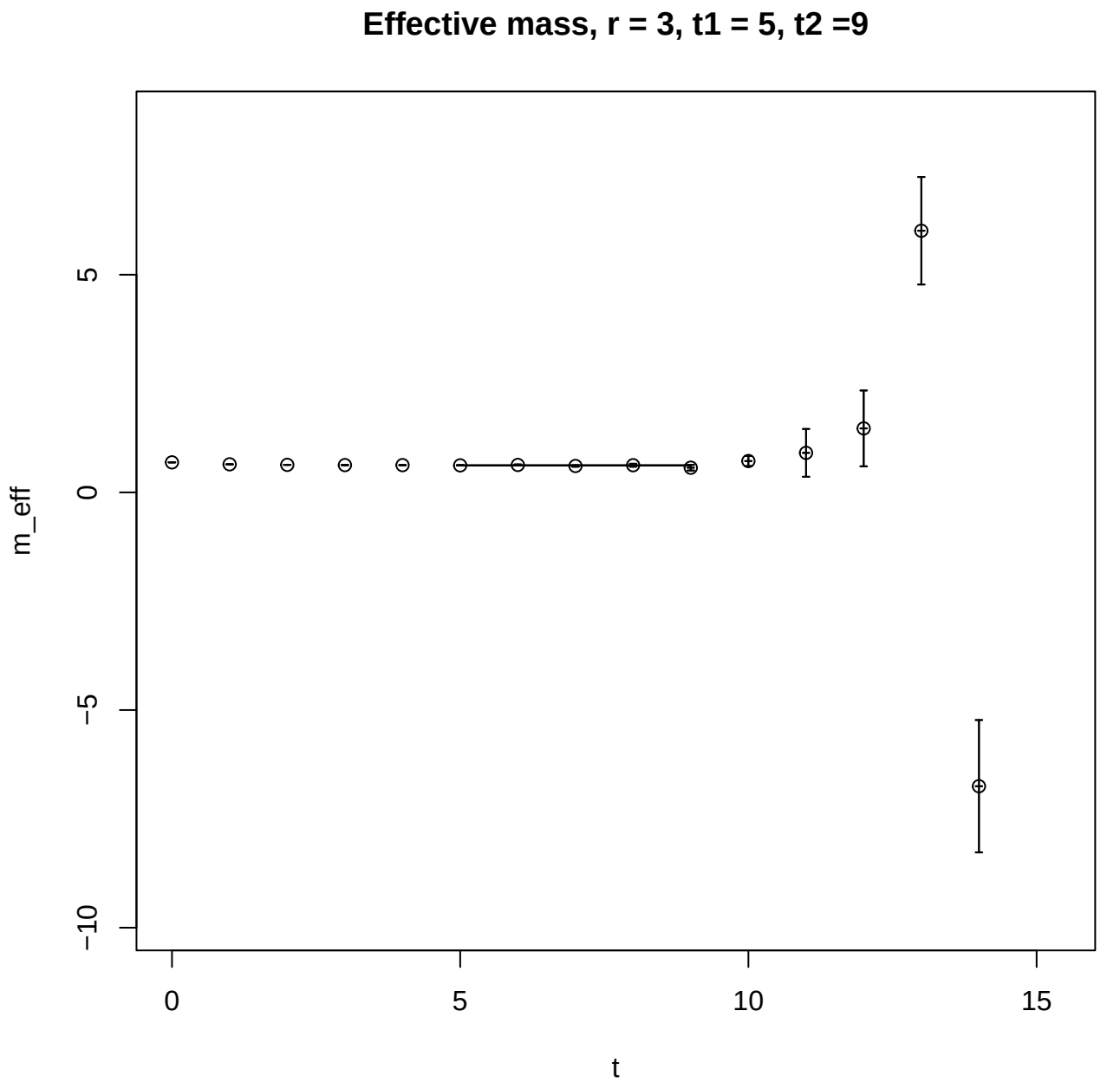

Effective mass, r = 3, t1 = 4, t2 =11
m_eff
t
5
0
−5
−10
0
5
10
15

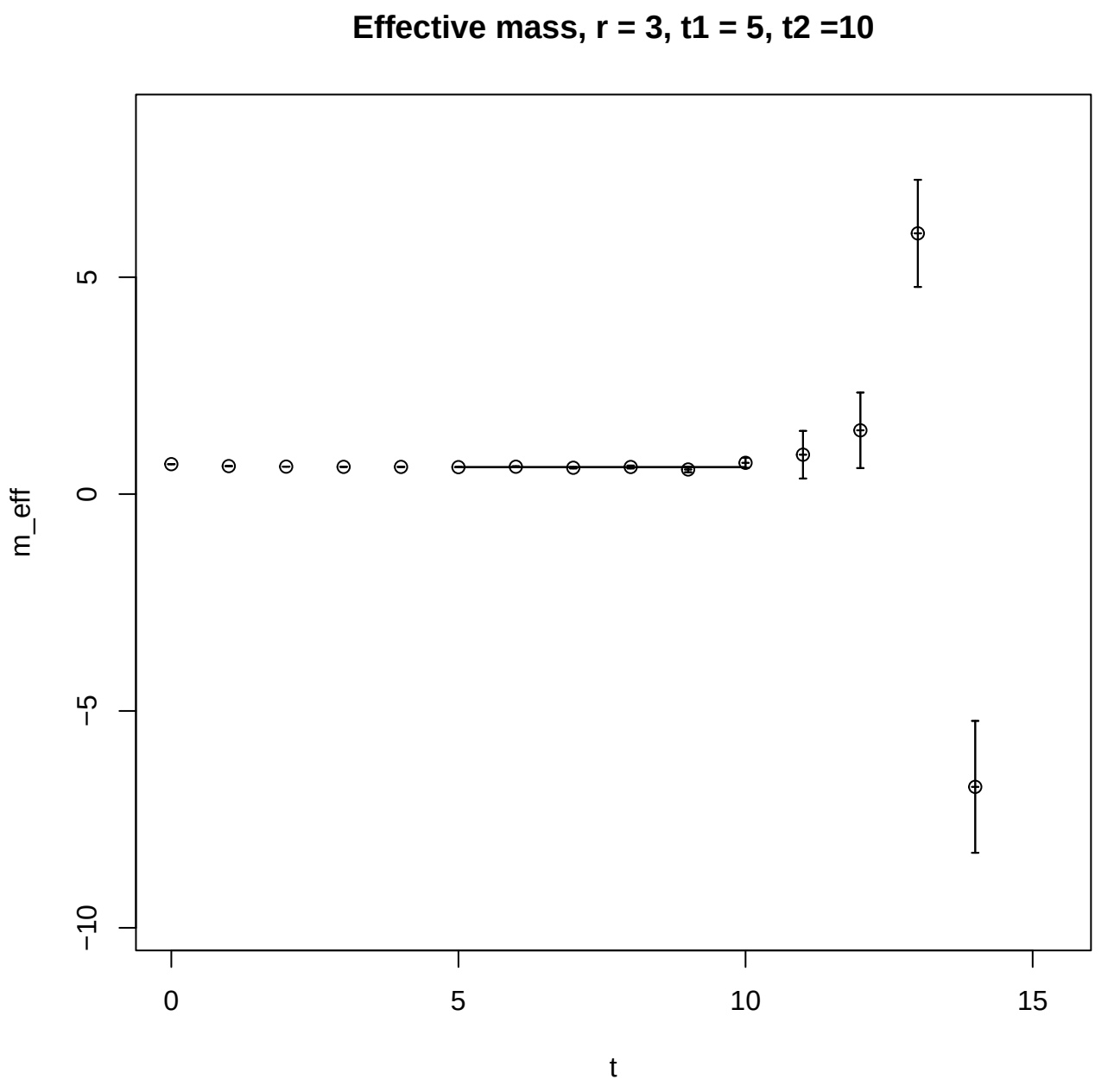

Effective mass, r = 3, t1 = 4, t2 =12
m_eff
5
0
−5
−10
0
5
10
15
t

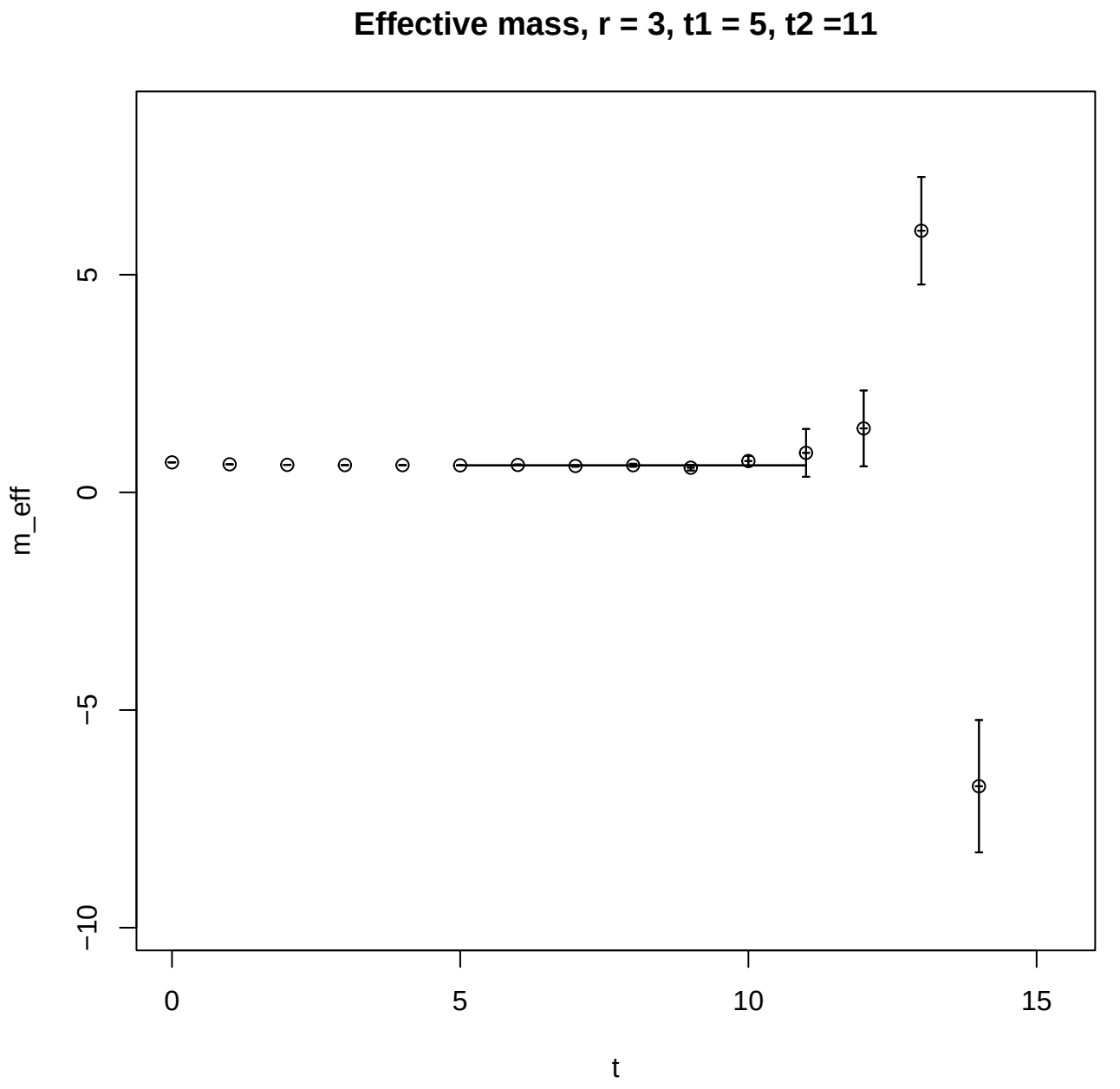

Effective mass, r = 3, t1 = 5, t2 =6
m_eff
5
0
−5
−10
0
5
10
15
t

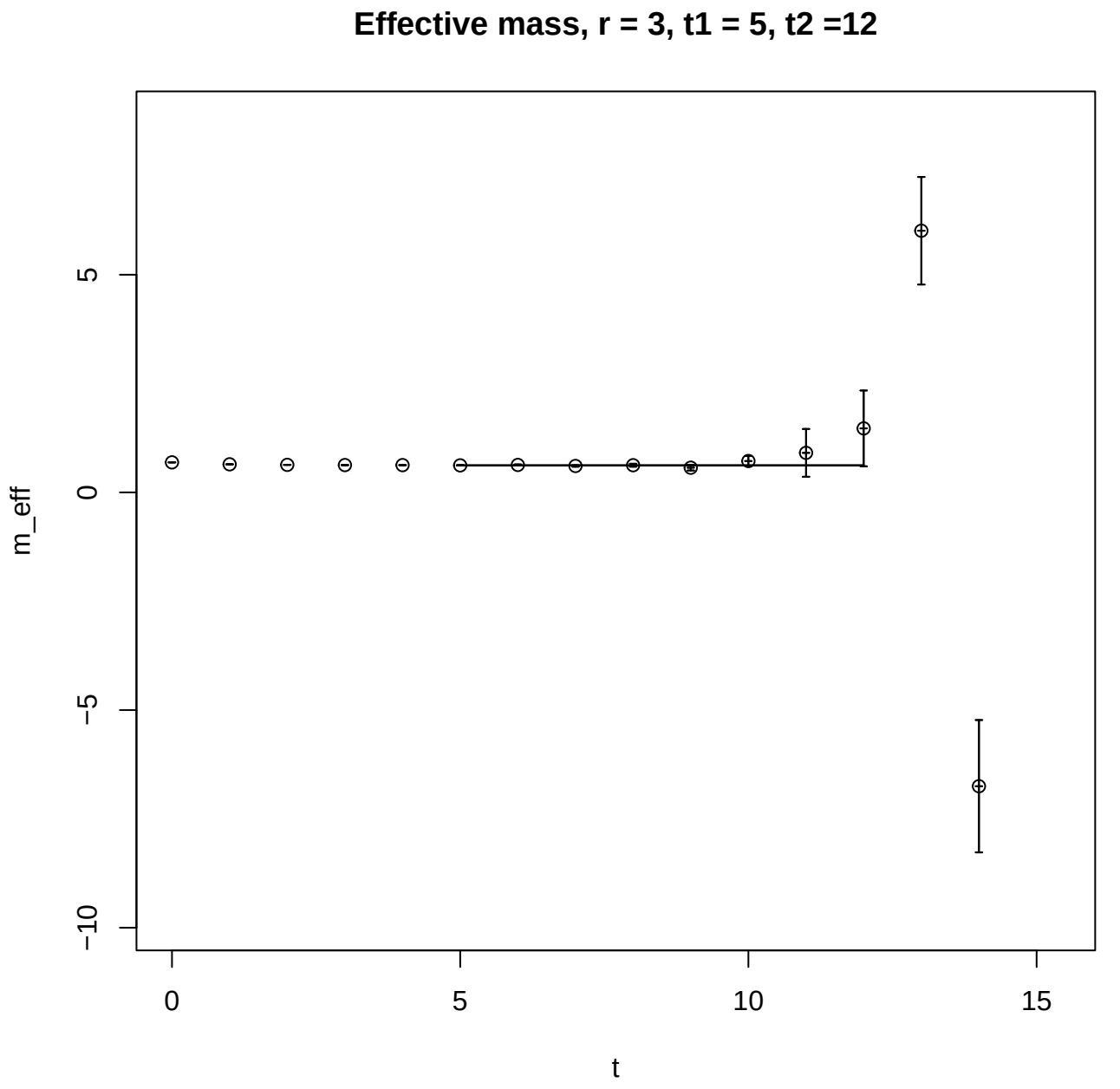

Effective mass, r = 3, t1 = 5, t2 =7
m_eff
5
0
−5
−10
0
5
10
15
t

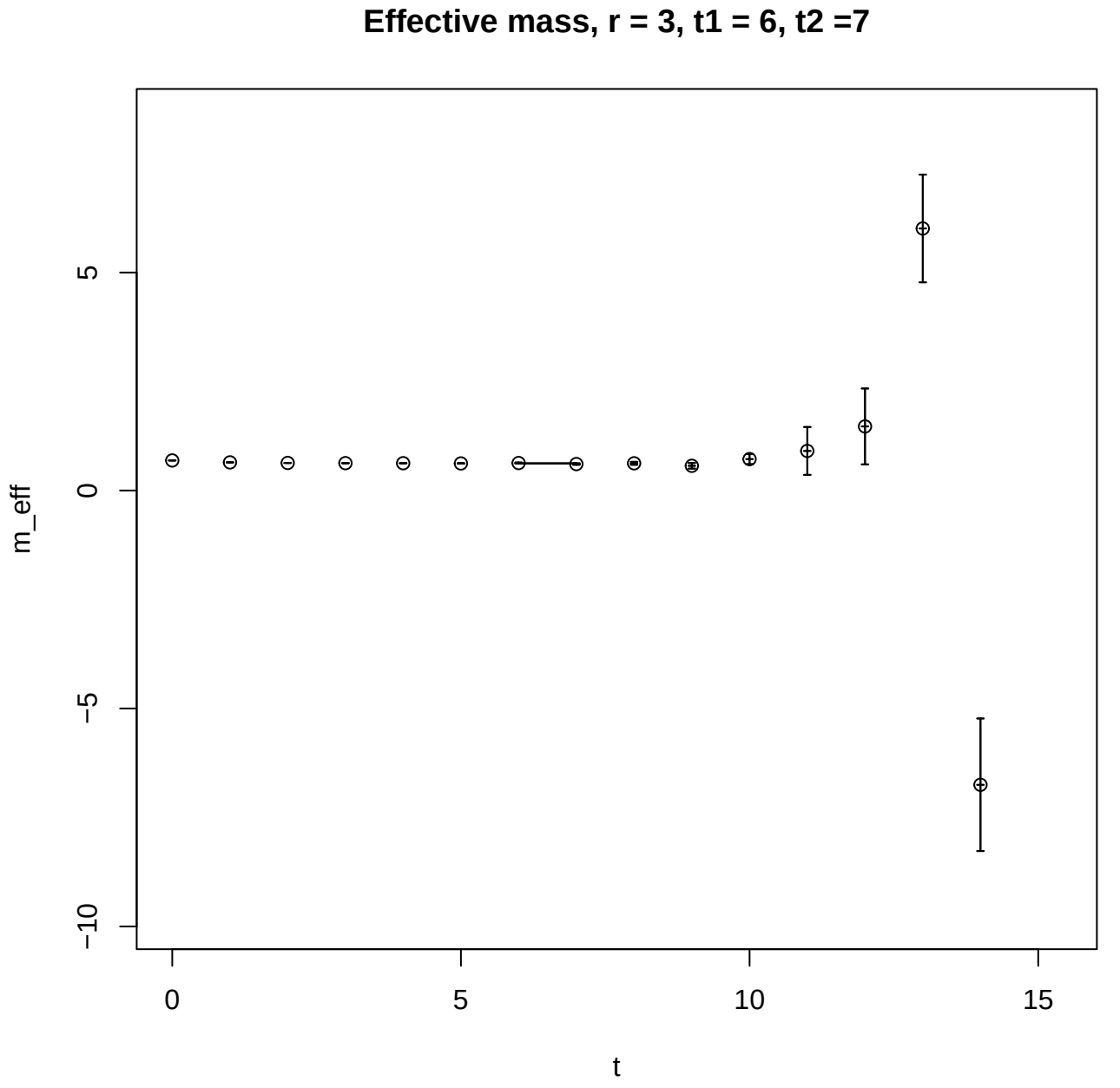

Effective mass, r = 3, t1 = 5, t2 =8
m_eff
t
5
0
−5
−10
0
5
10
15

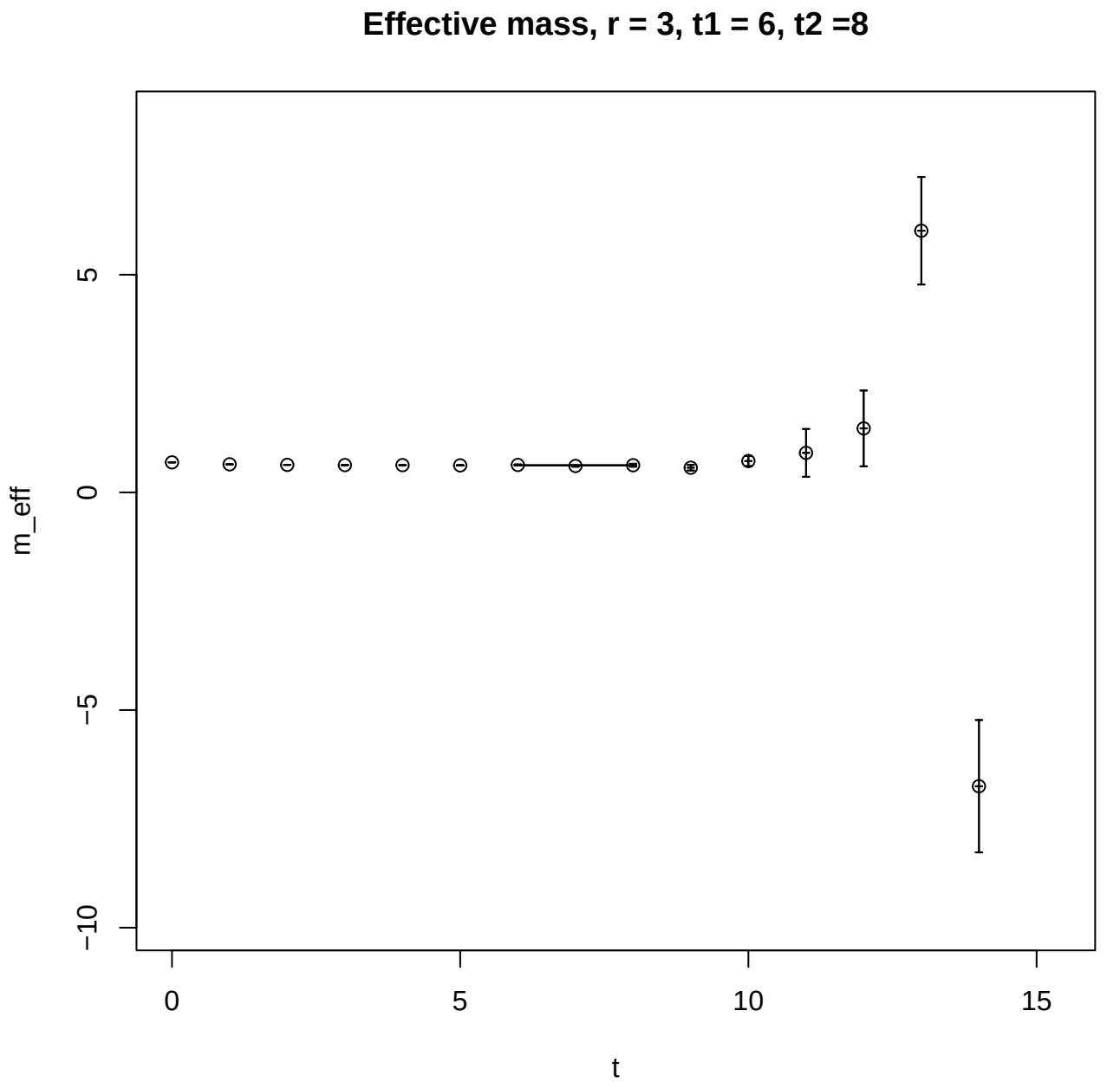

Effective mass, r = 3, t1 = 5, t2 =9
m_eff
5
0
−5
−10
0
5
10
15
t

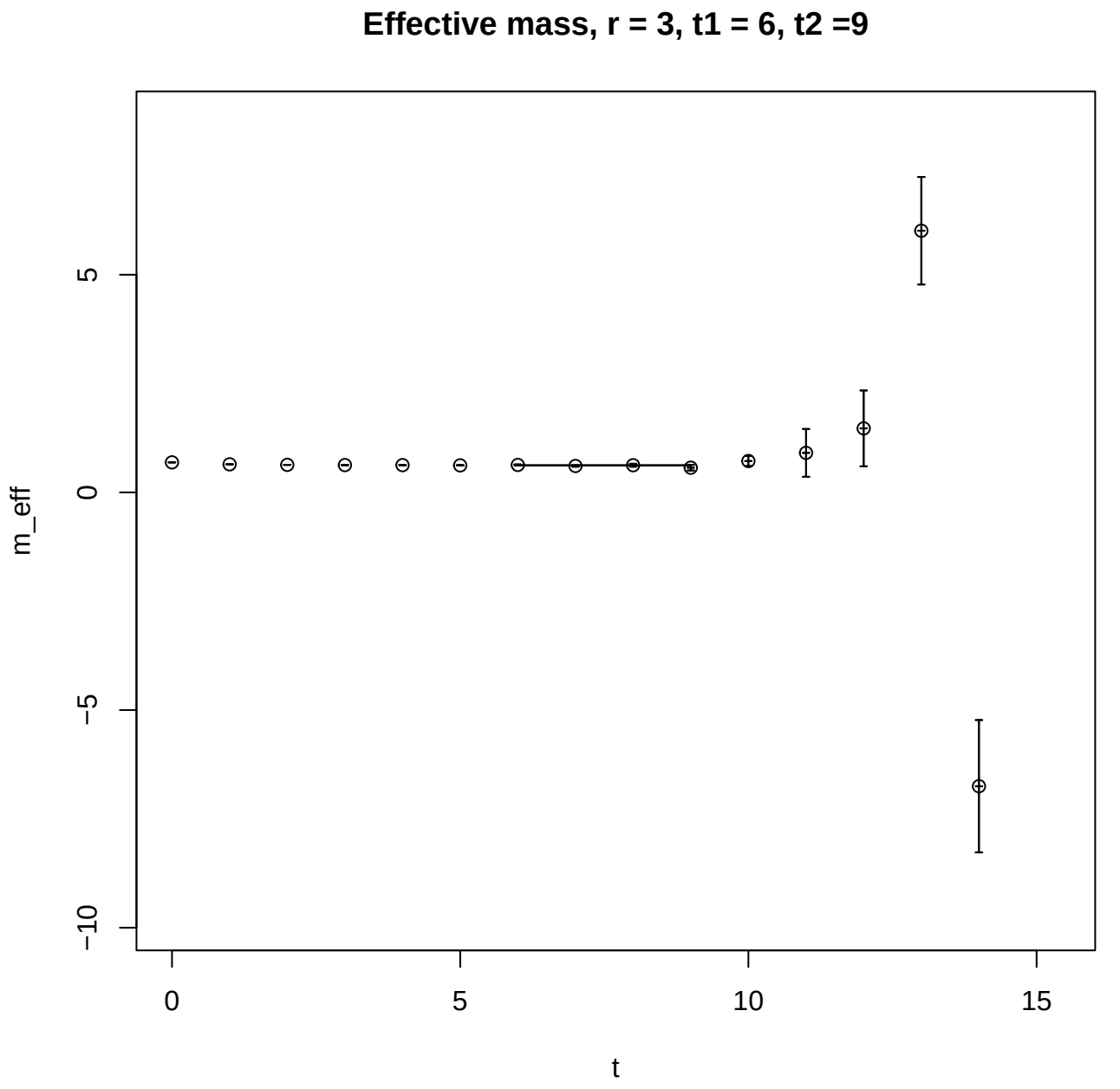

Effective mass, r = 3, t1 = 5, t2 =10
m_eff
5
0
−5
−10
0
5
10
15
t

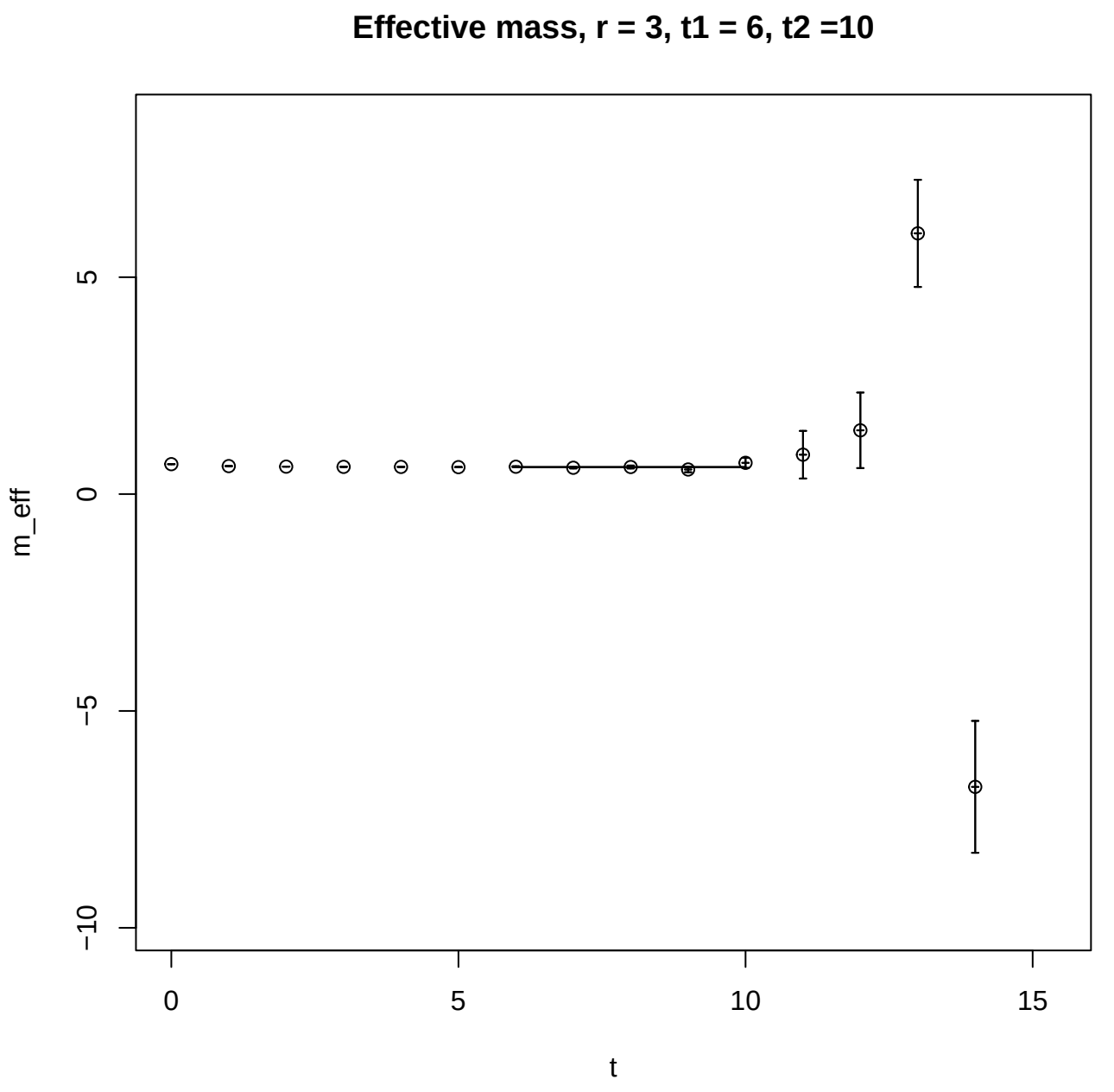

Effective mass, r = 3, t1 = 5, t2 =11
m_eff
5
0
−5
−10
0
5
10
15
t

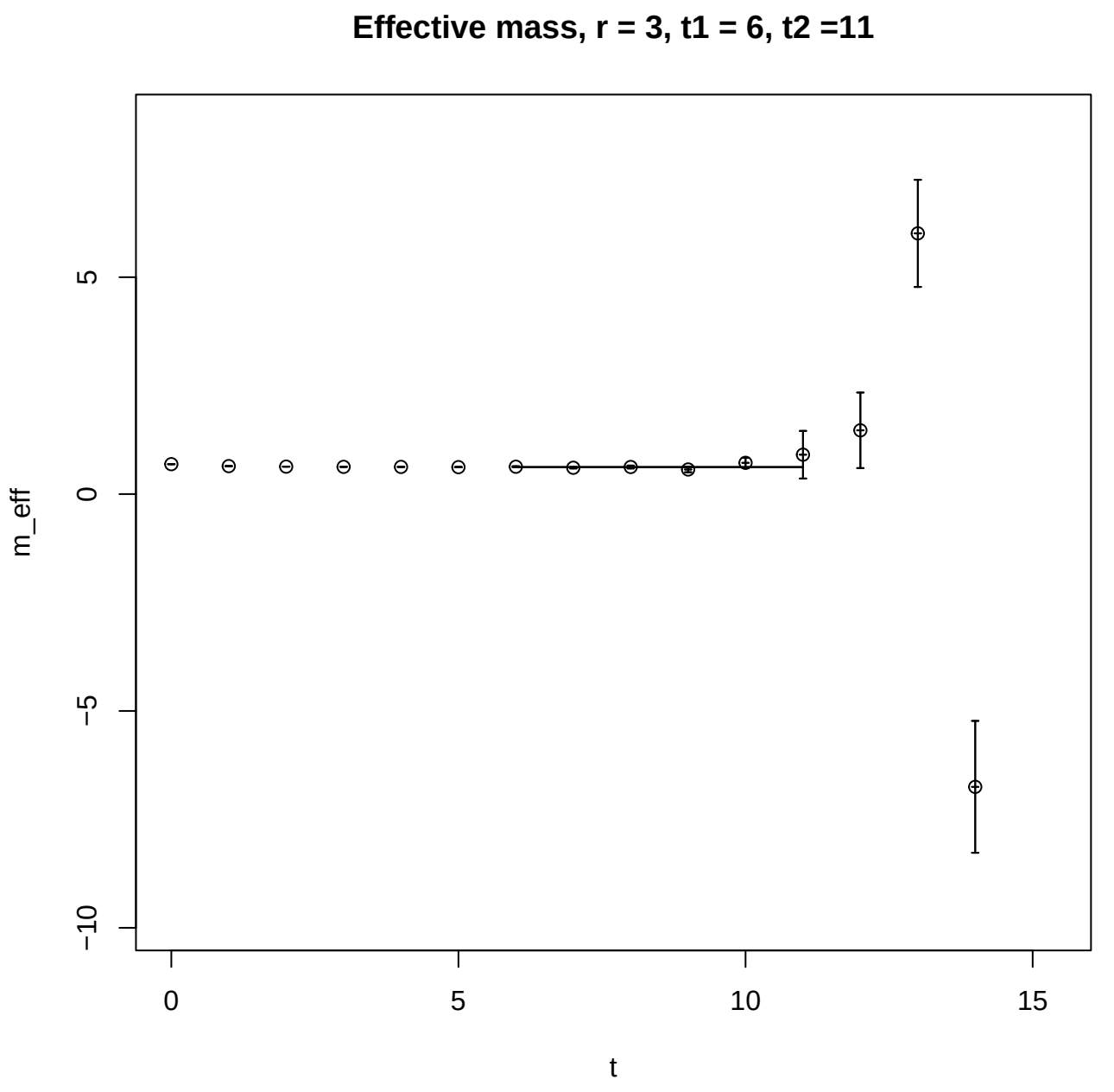

Effective mass, r = 3, t1 = 5, t2 =12
m_eff
5
0
−5
−10
0
5
10
15
t

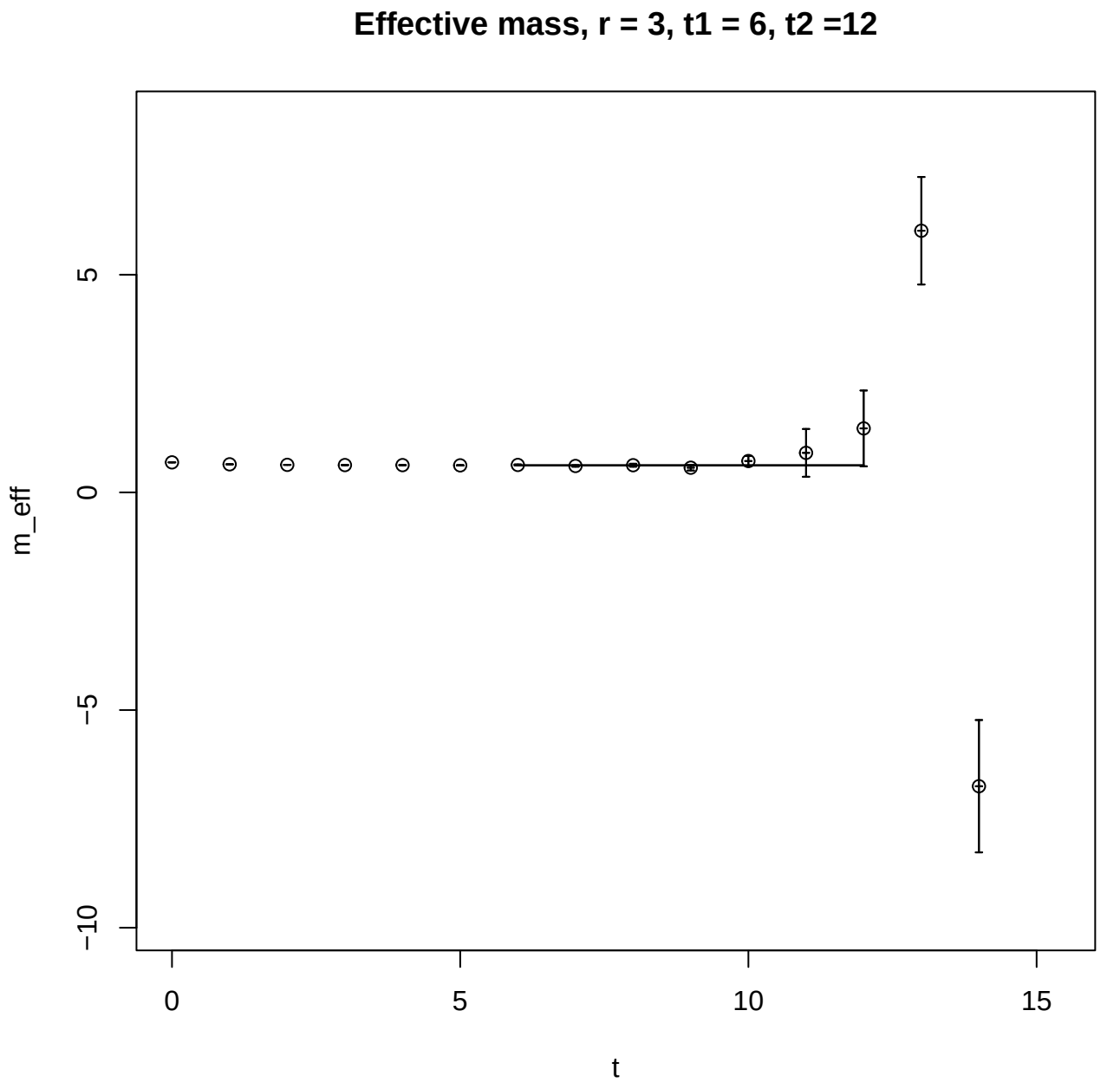

Effective mass, r = 3, t1 = 6, t2 =7
m_eff
5
0
−5
−10
0
5
10
15
t

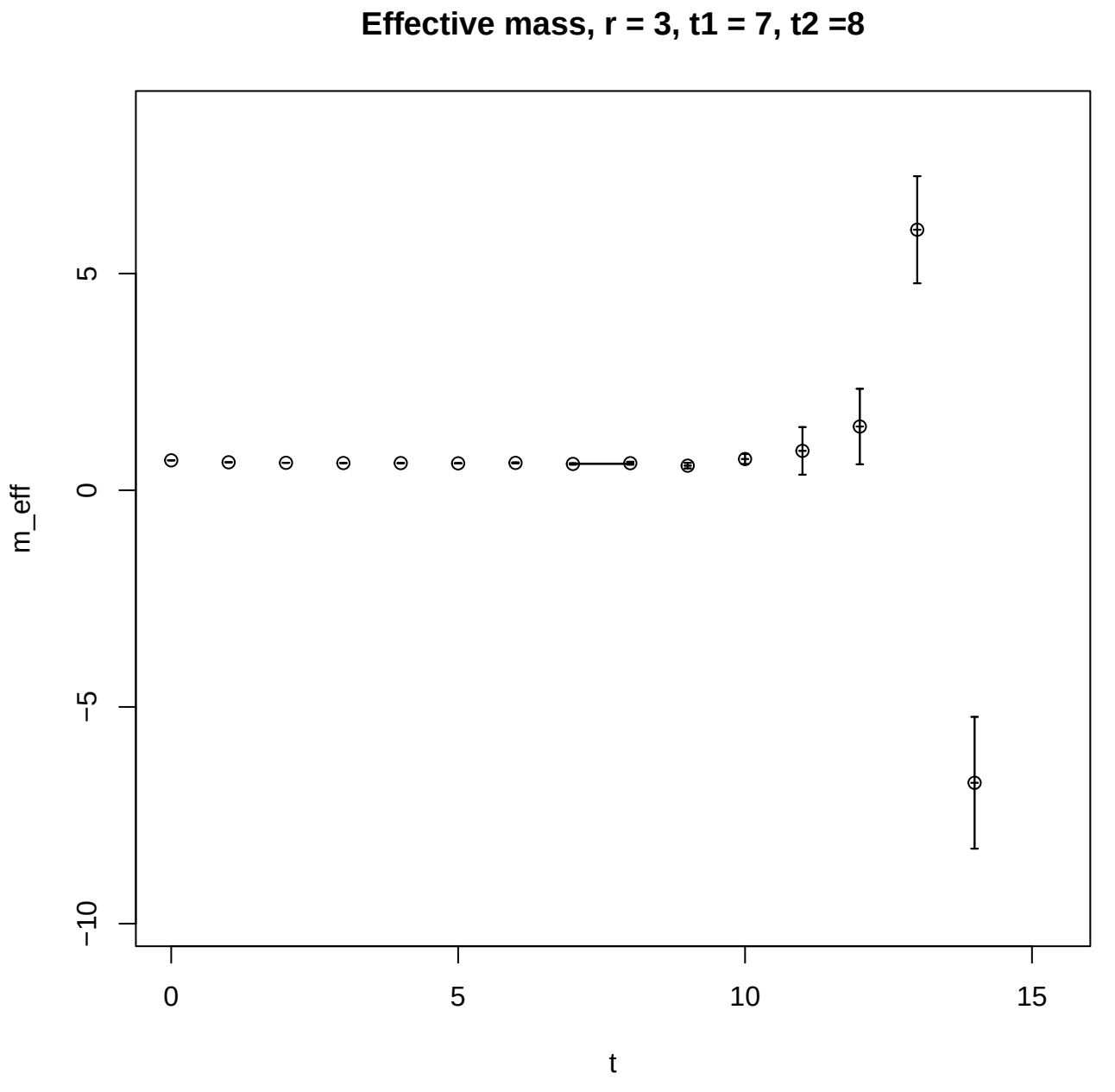

Effective mass, r = 3, t1 = 6, t2 =8
m_eff
t
5
0
−5
−10
0
5
10
15

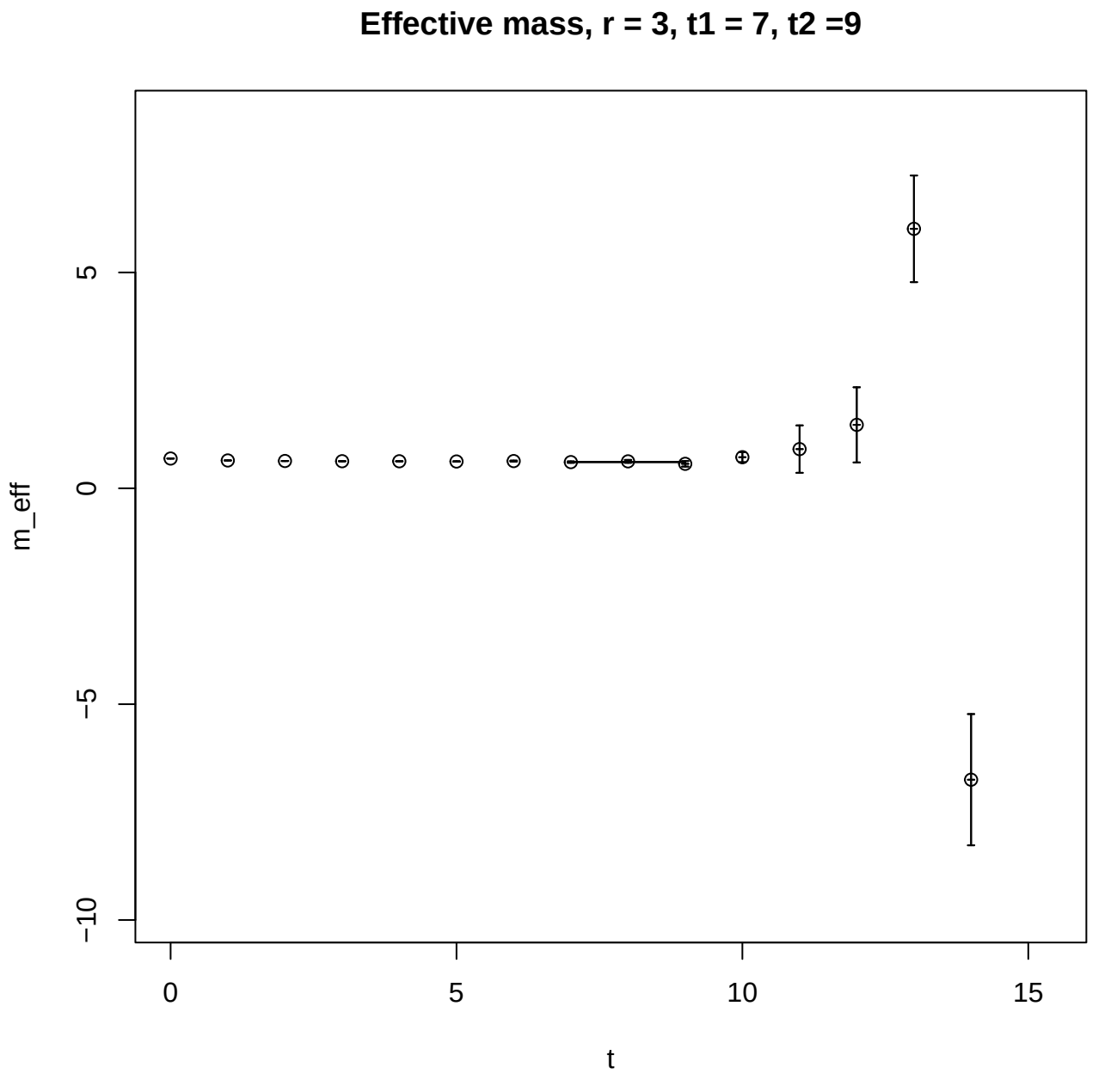

Effective mass, r = 3, t1 = 6, t2 =9
m_eff
t
5
0
−5
−10
0
5
10
15

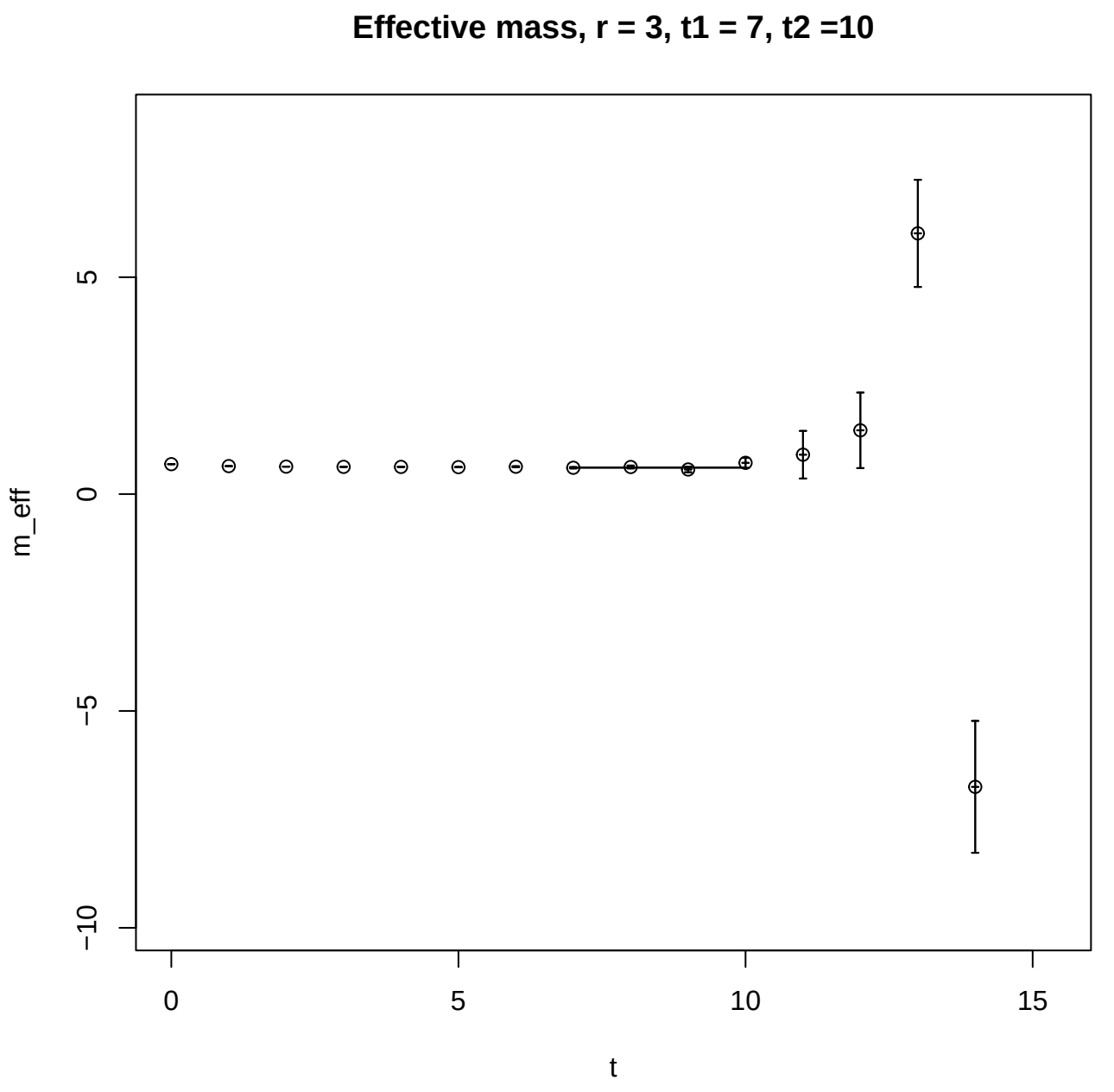

Effective mass, r = 3, t1 = 6, t2 =10
m_eff
5
0
−5
−10
0
5
10
15
t

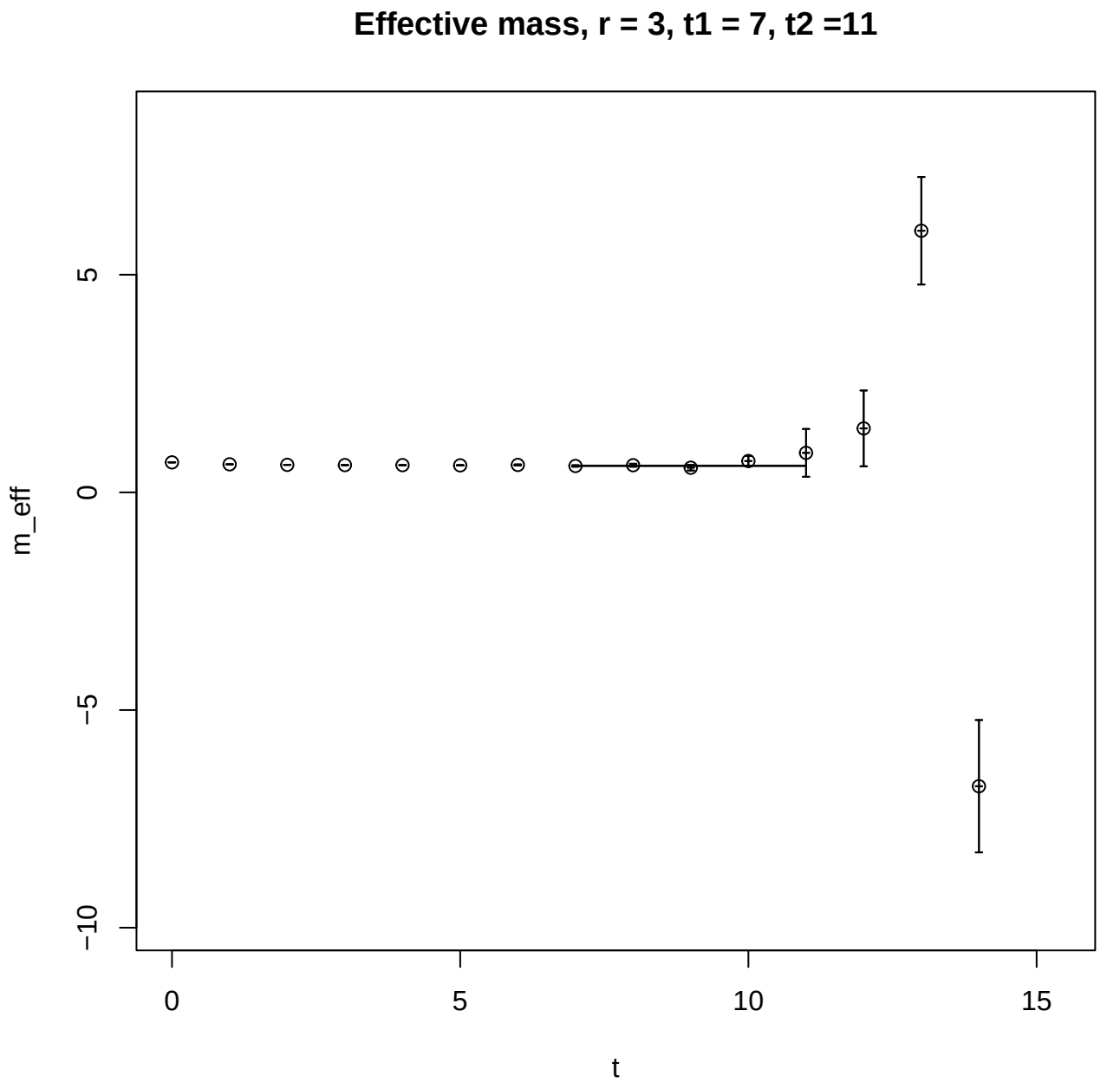

Effective mass, r = 3, t1 = 6, t2 =11
m_eff
5
0
−5
−10
0
5
10
15
t

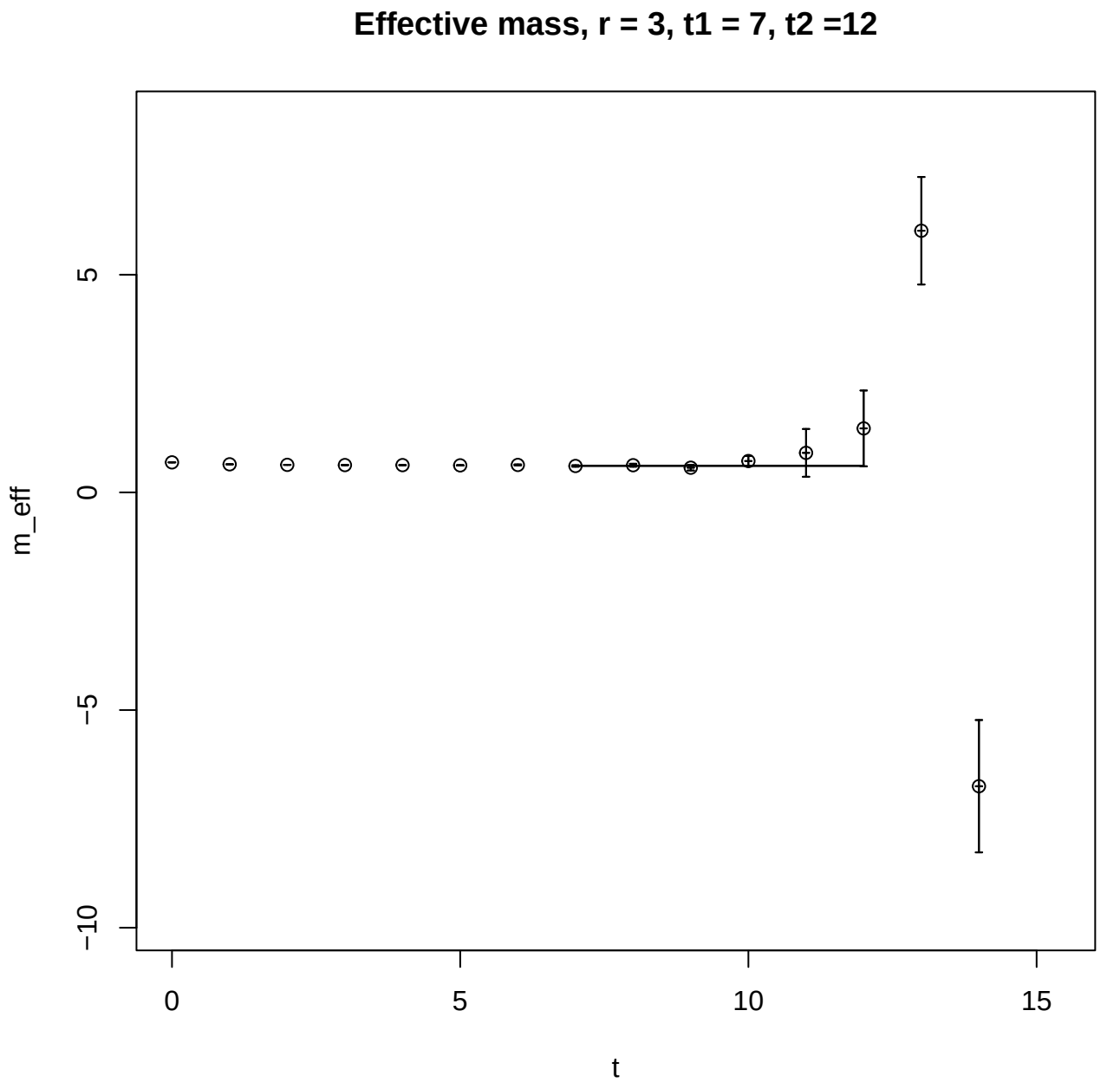

Effective mass, r = 3, t1 = 6, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

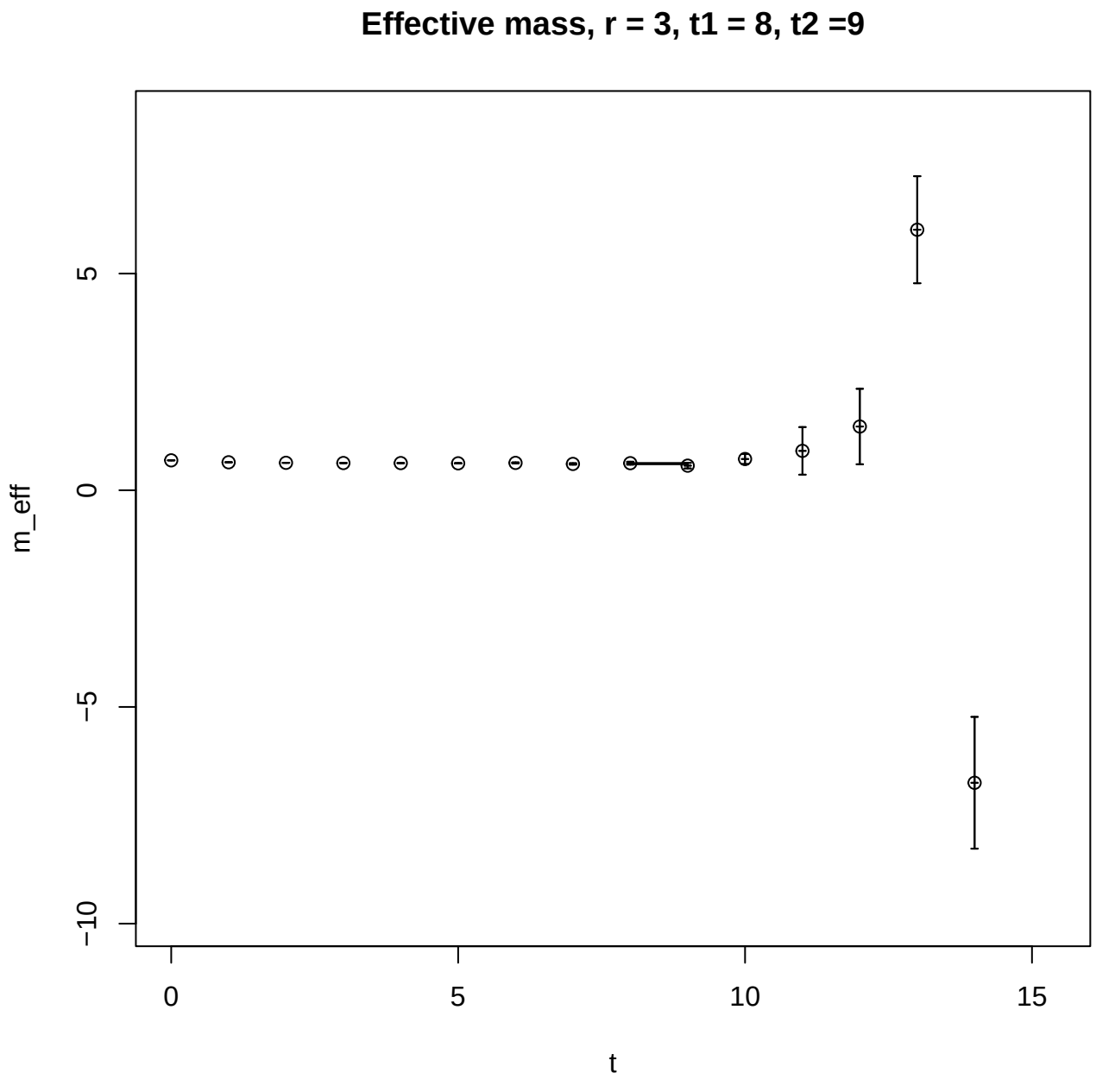

Effective mass, r = 3, t1 = 7, t2 =8
m_eff
5
0
−5
−10
0
5
10
15
t

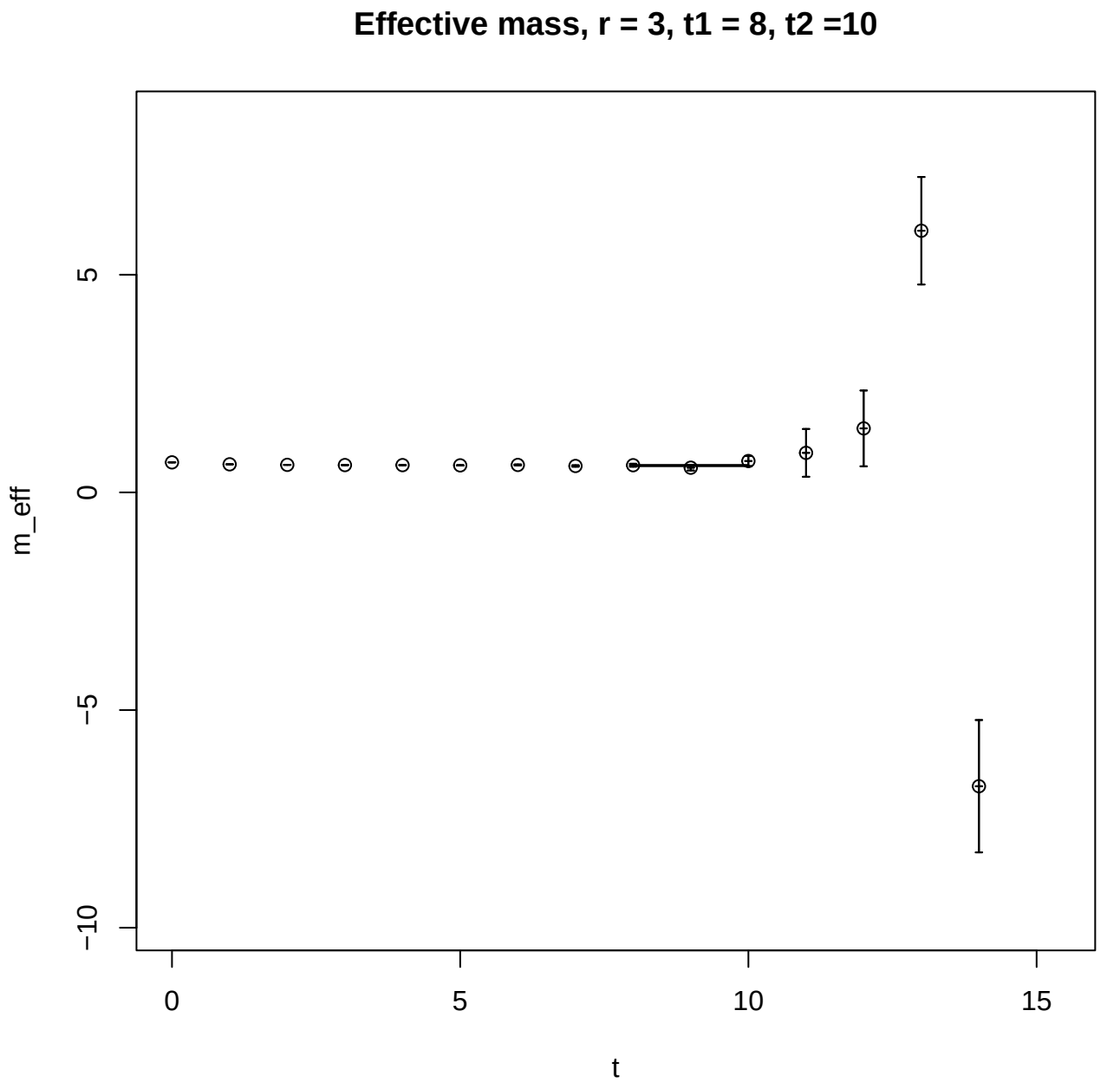

Effective mass, r = 3, t1 = 7, t2 =9
m_eff
t
5
0
−5
−10
0
5
10
15

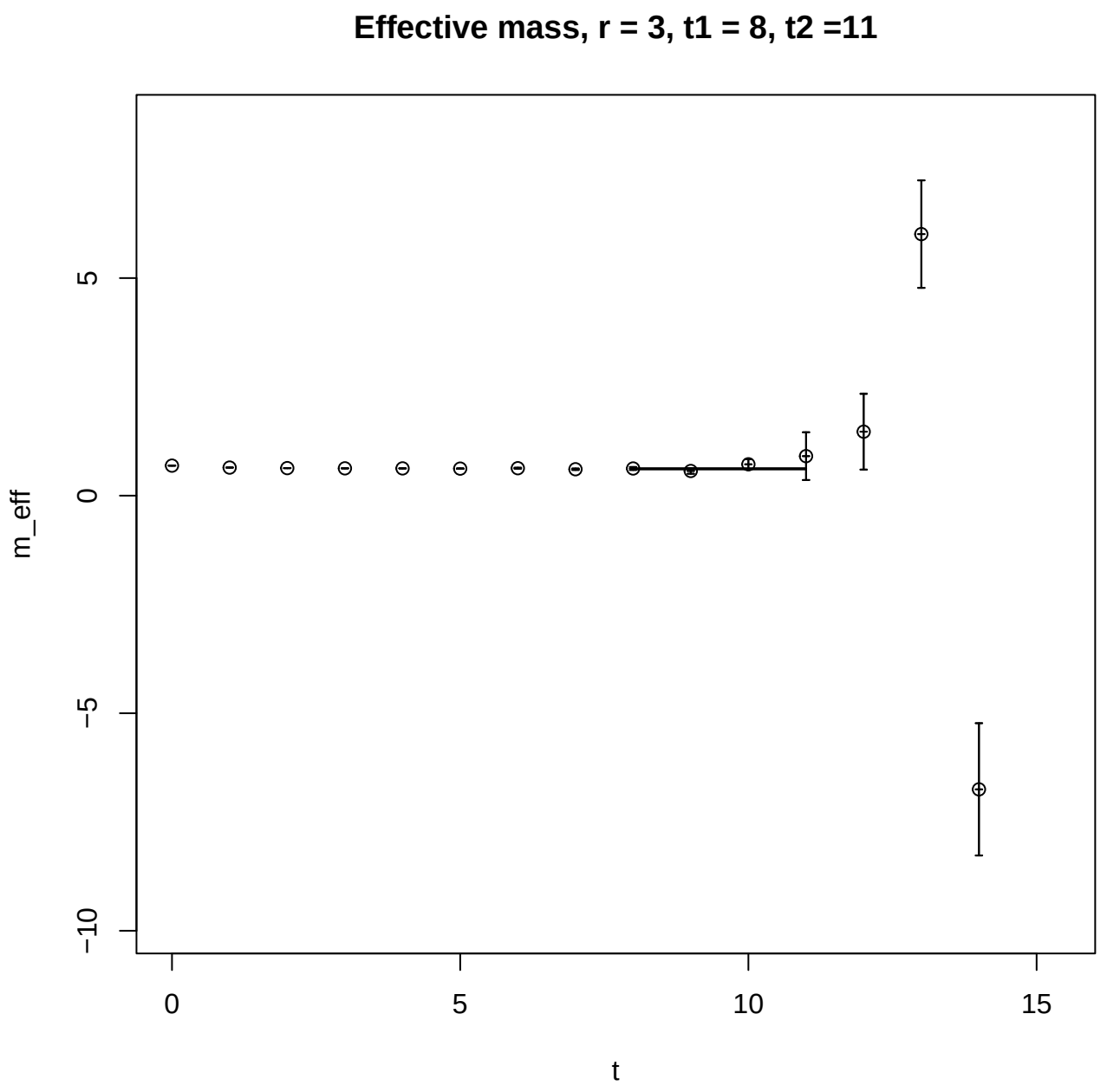

Effective mass, r = 3, t1 = 7, t2 =10
m_eff
t
5
0
−5
−10
0
5
10
15

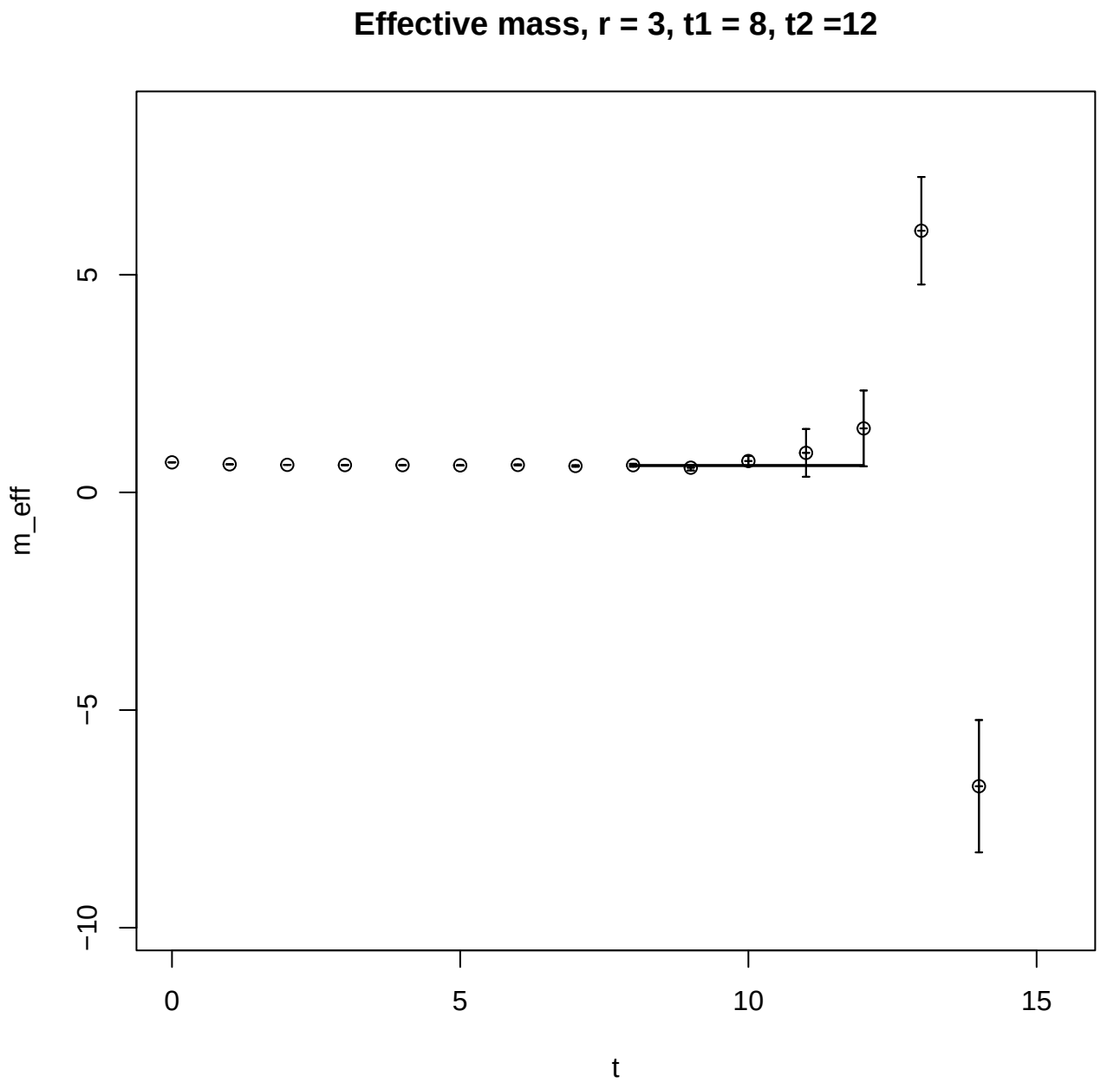

Effective mass, r = 3, t1 = 7, t2 =11
m_eff
5
0
−5
−10
0
5
10
15
t

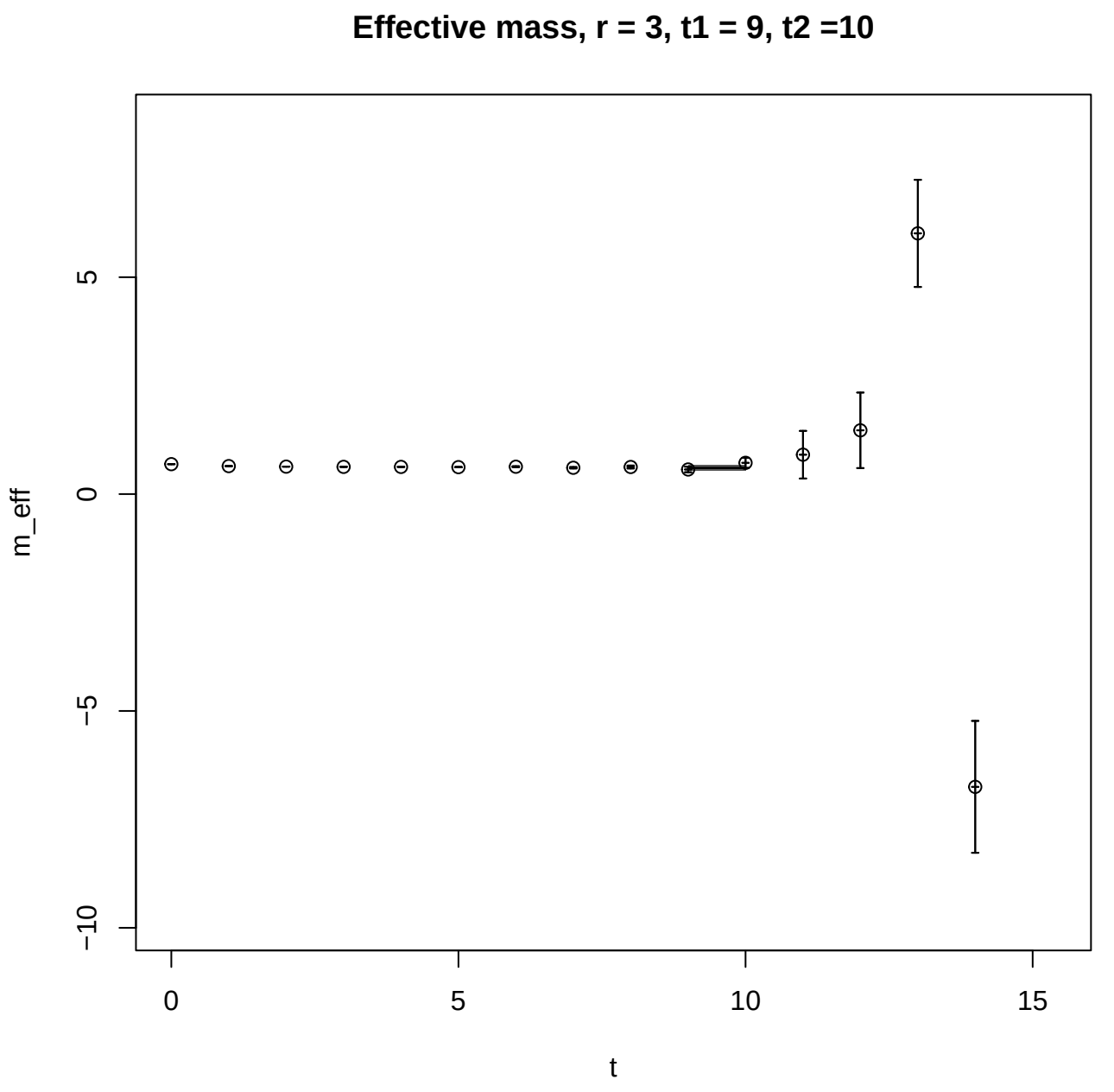

Effective mass, r = 3, t1 = 7, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

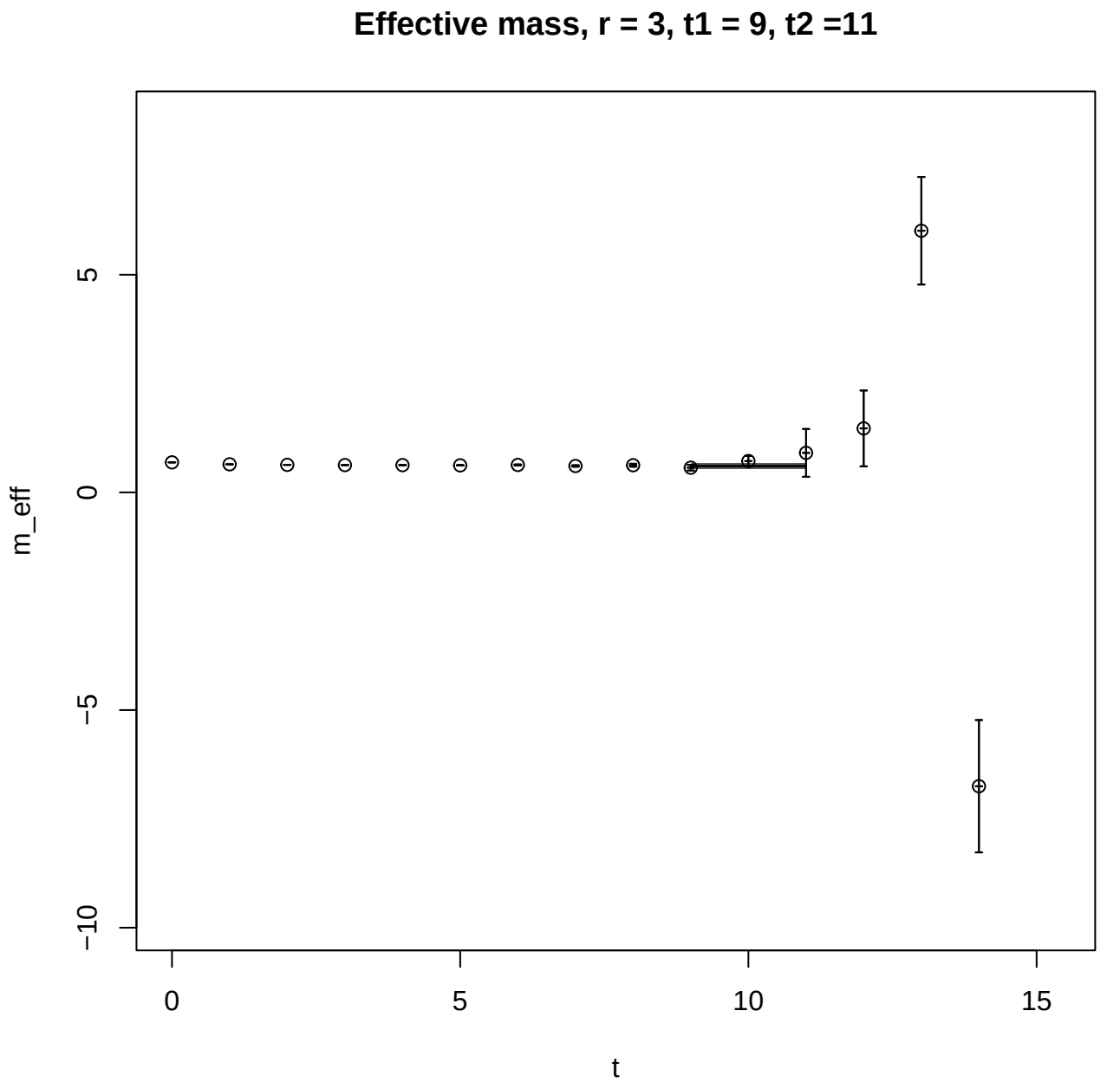

Effective mass, r = 3, t1 = 8, t2 =9
m_eff
5
0
−5
−10
0
5
10
15
t

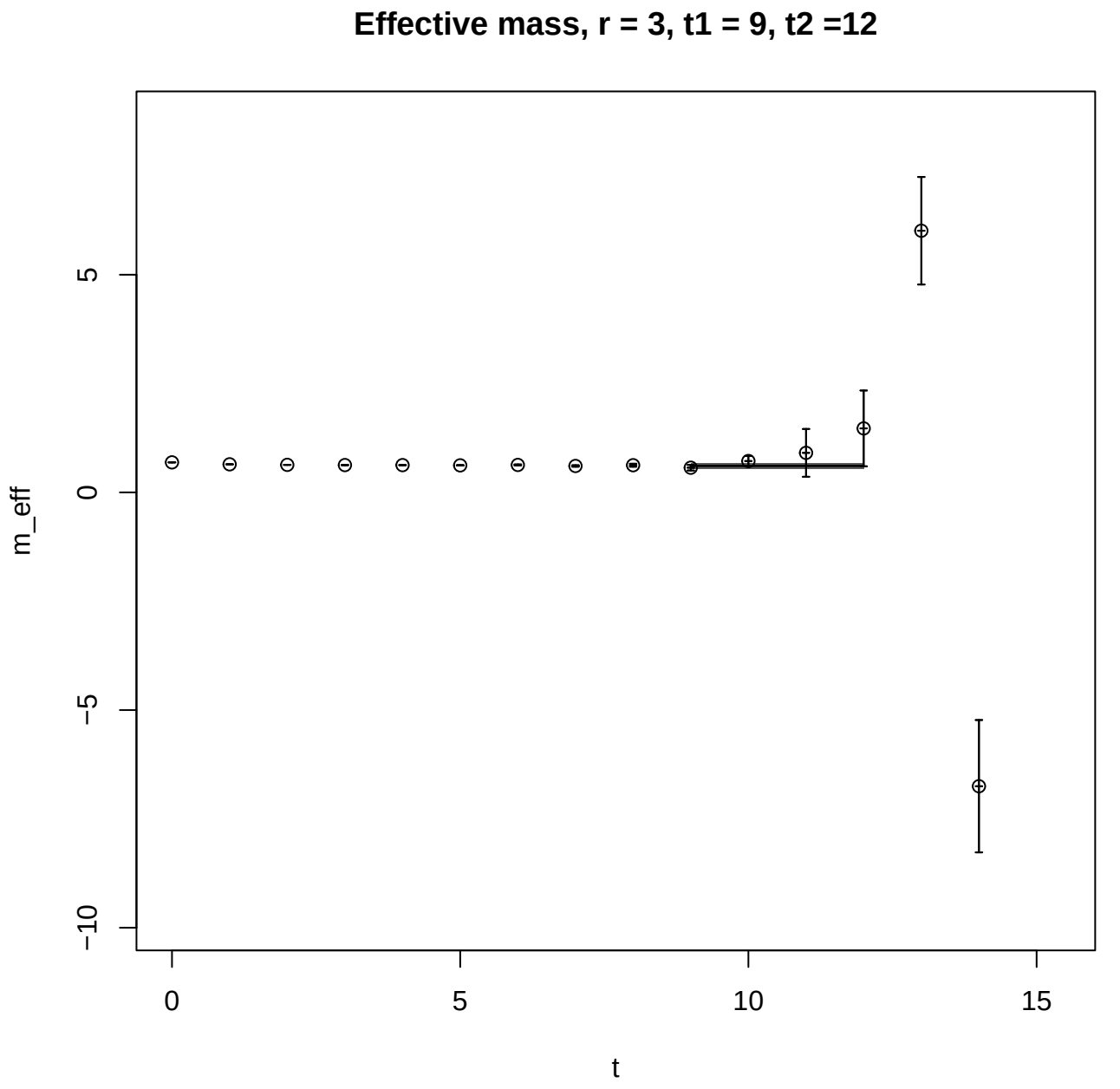

Effective mass, r = 3, t1 = 8, t2 =10
m_eff
t
5
0
−5
−10
0
5
10
15

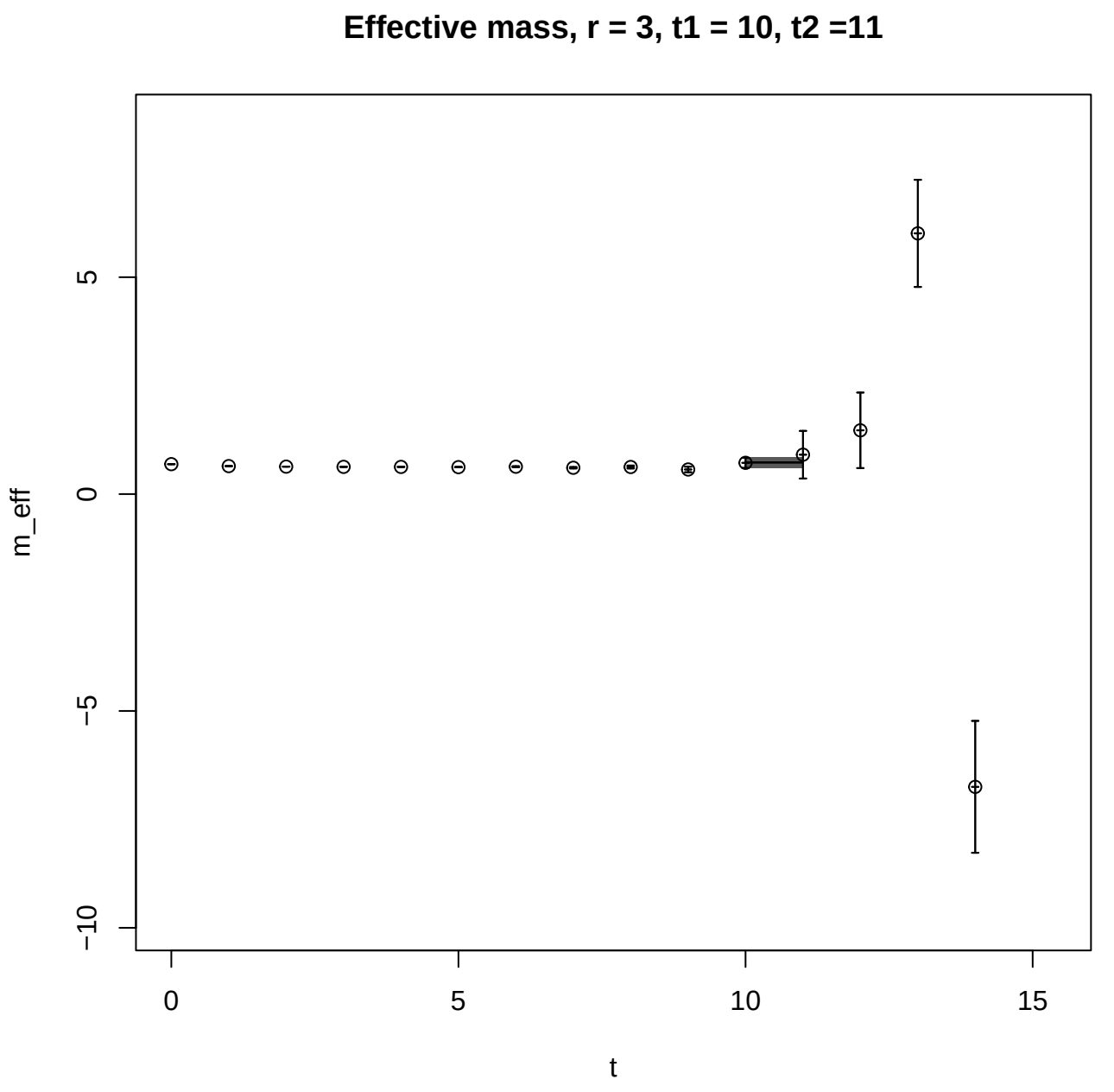

Effective mass, r = 3, t1 = 8, t2 =11
m_eff
t
5
0
−5
−10
0
5
10
15

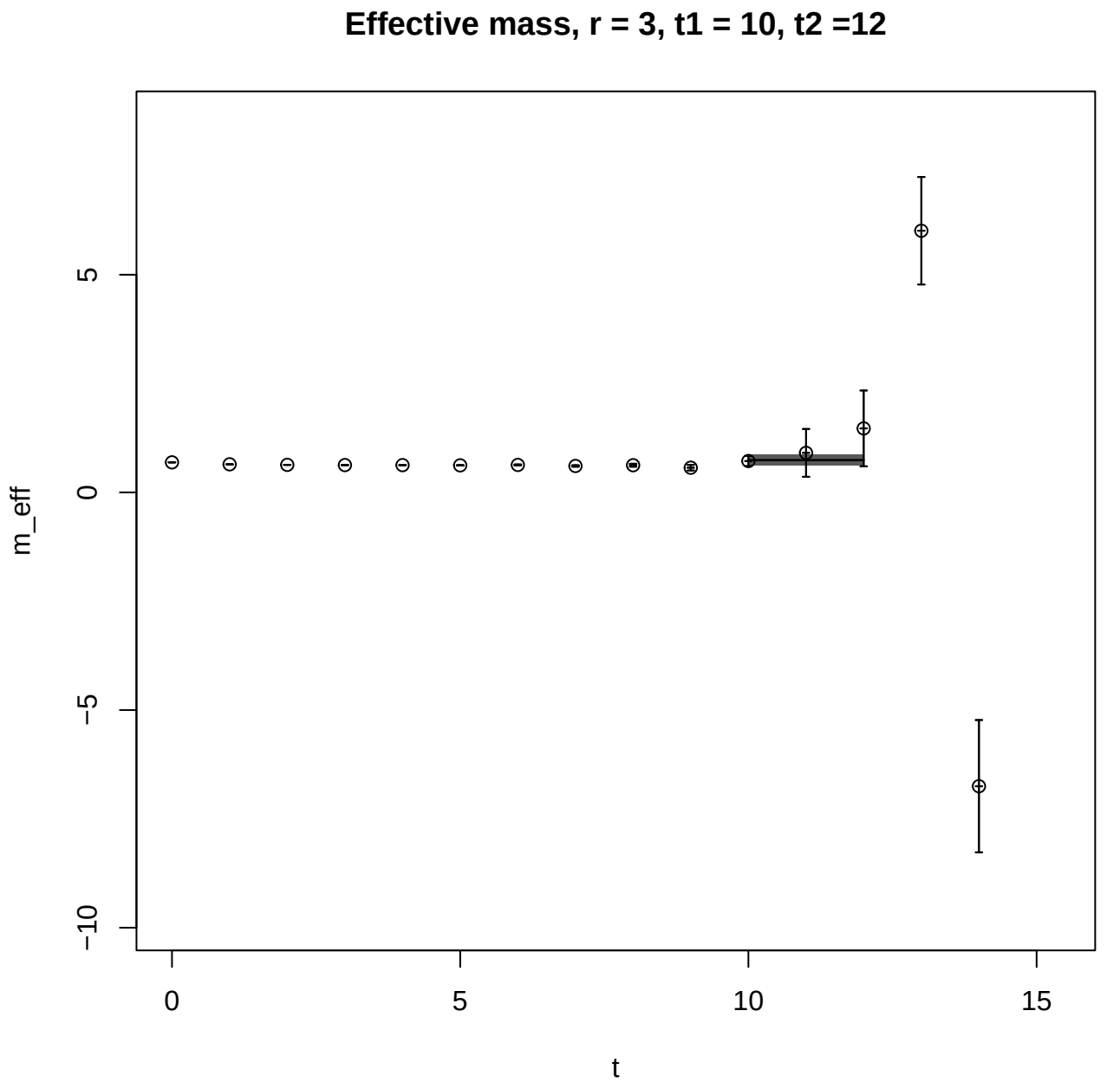

Effective mass, r = 3, t1 = 8, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

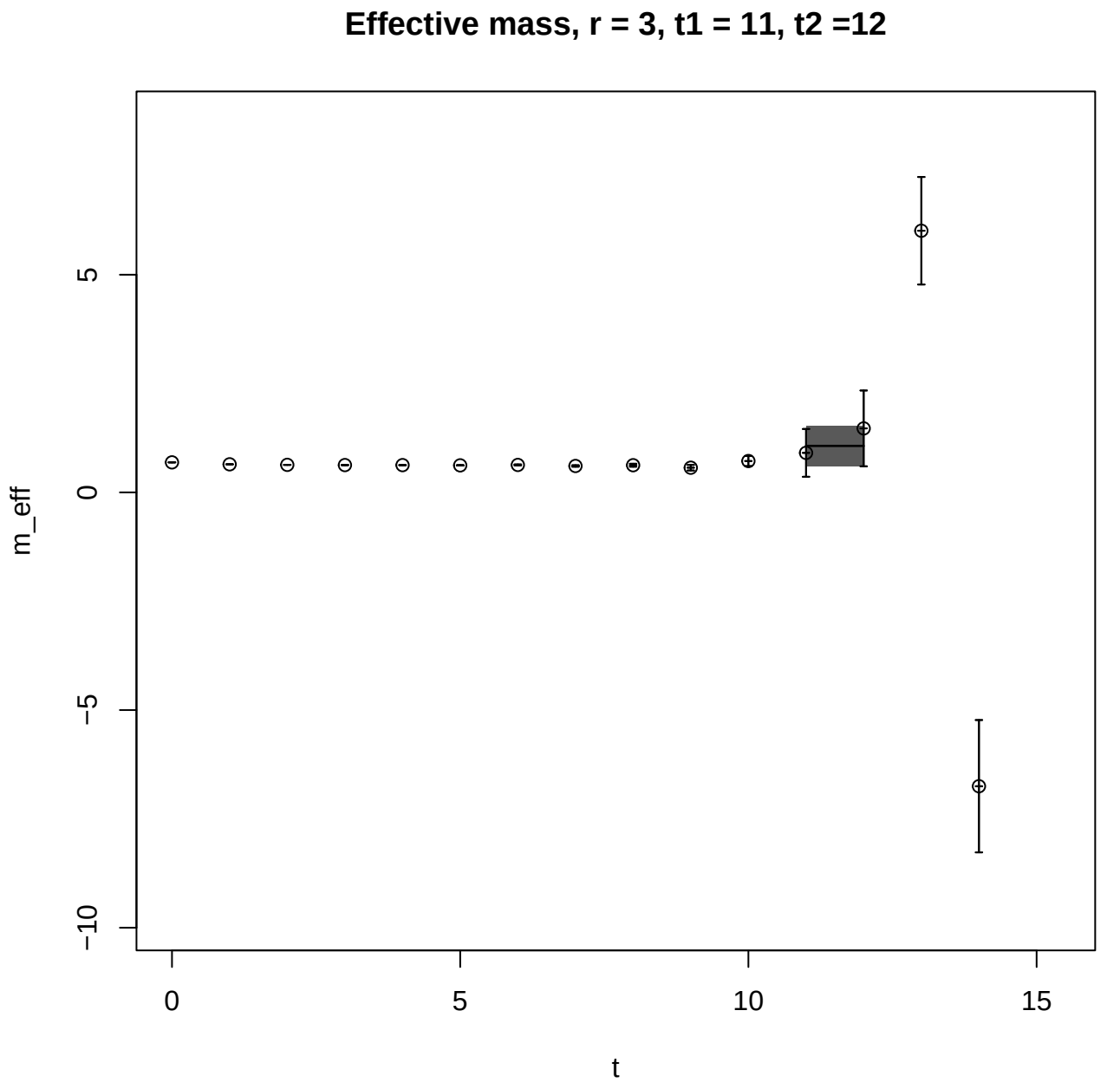

Effective mass, r = 3, t1 = 9, t2 =10
m_eff
5
0
−5
−10
0
5
10
15
t

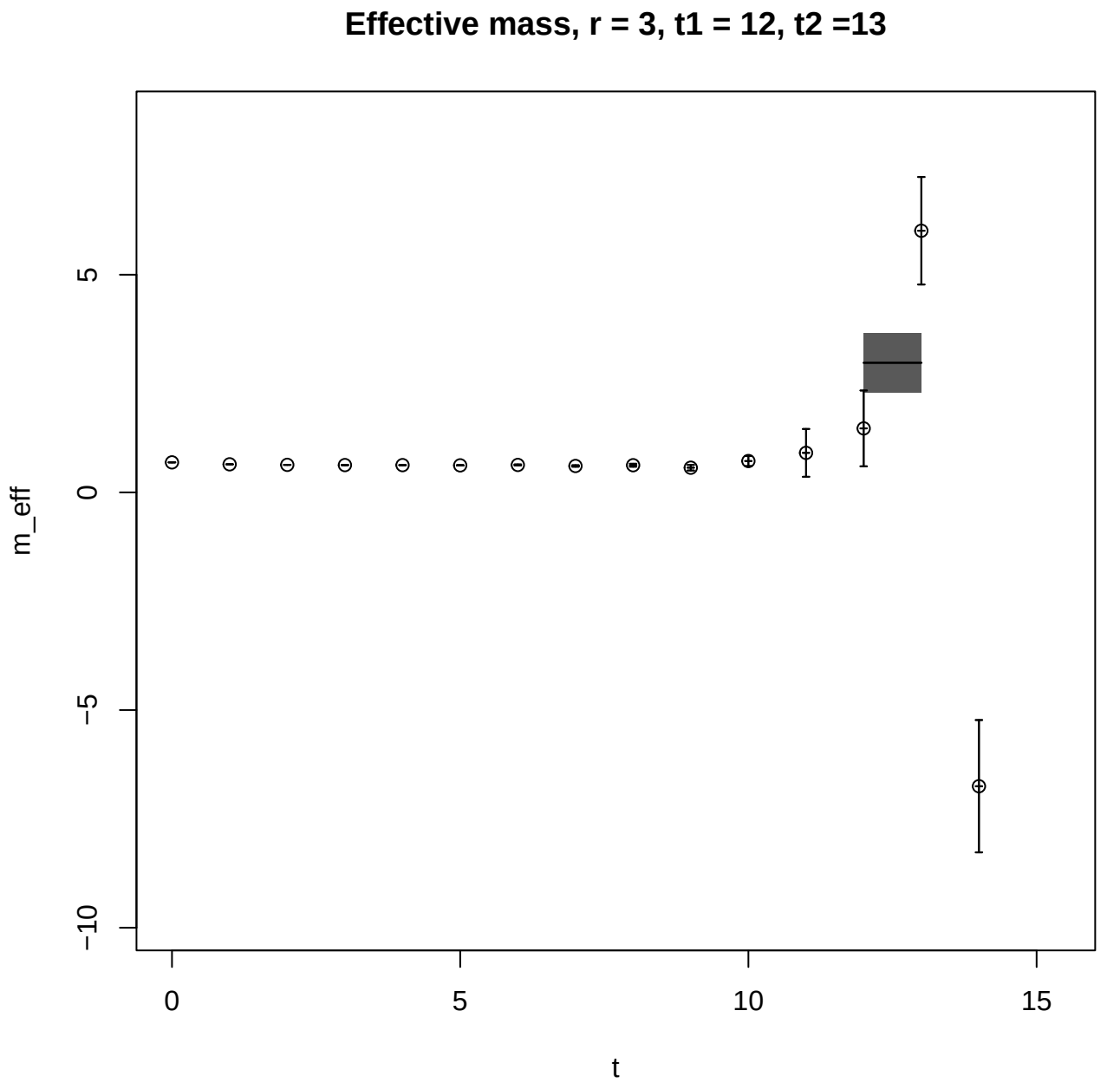

Effective mass, r = 3, t1 = 9, t2 =11
m_eff
5
0
−5
−10
0
5
10
15
t

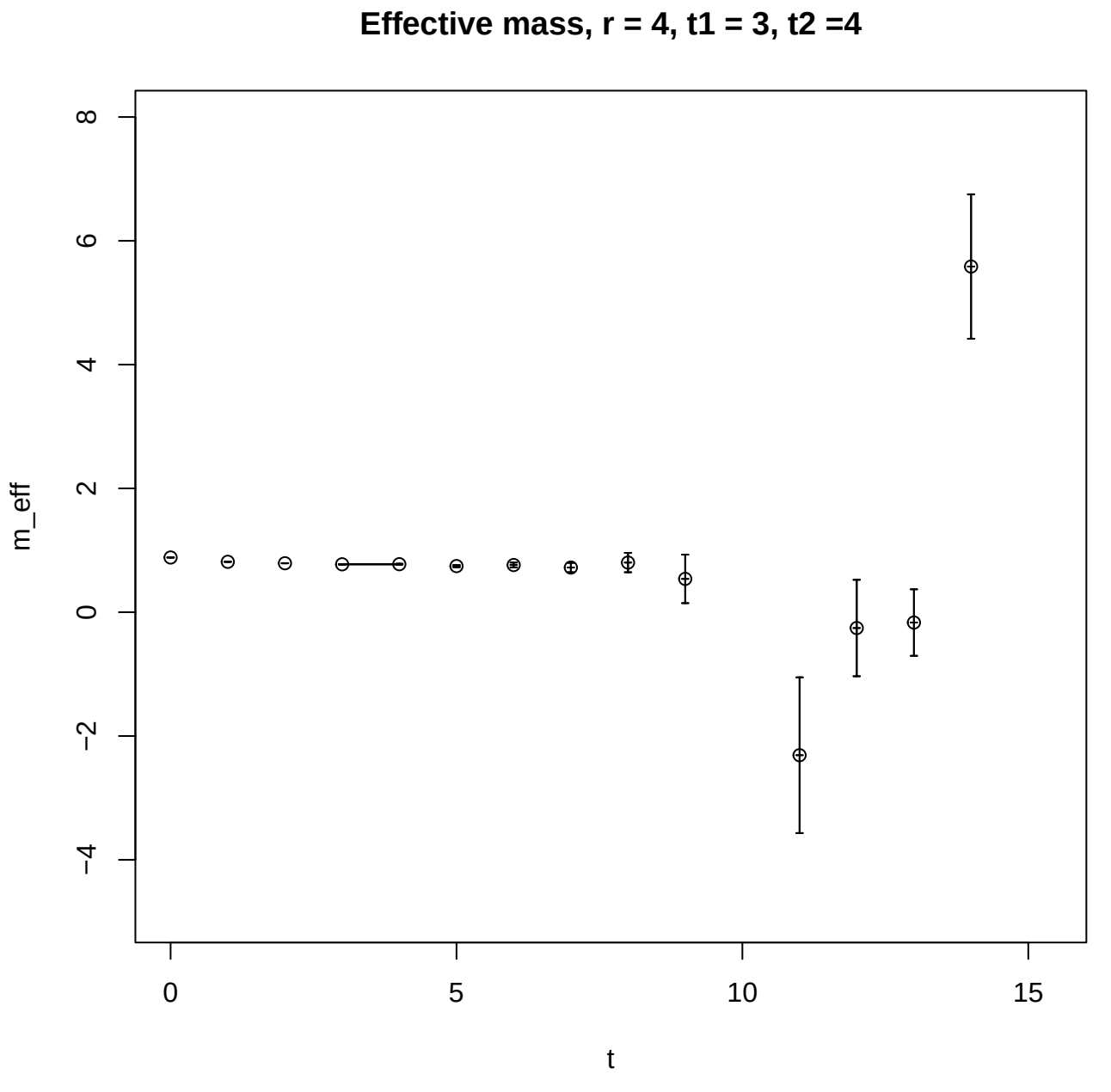

Effective mass, r = 3, t1 = 9, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

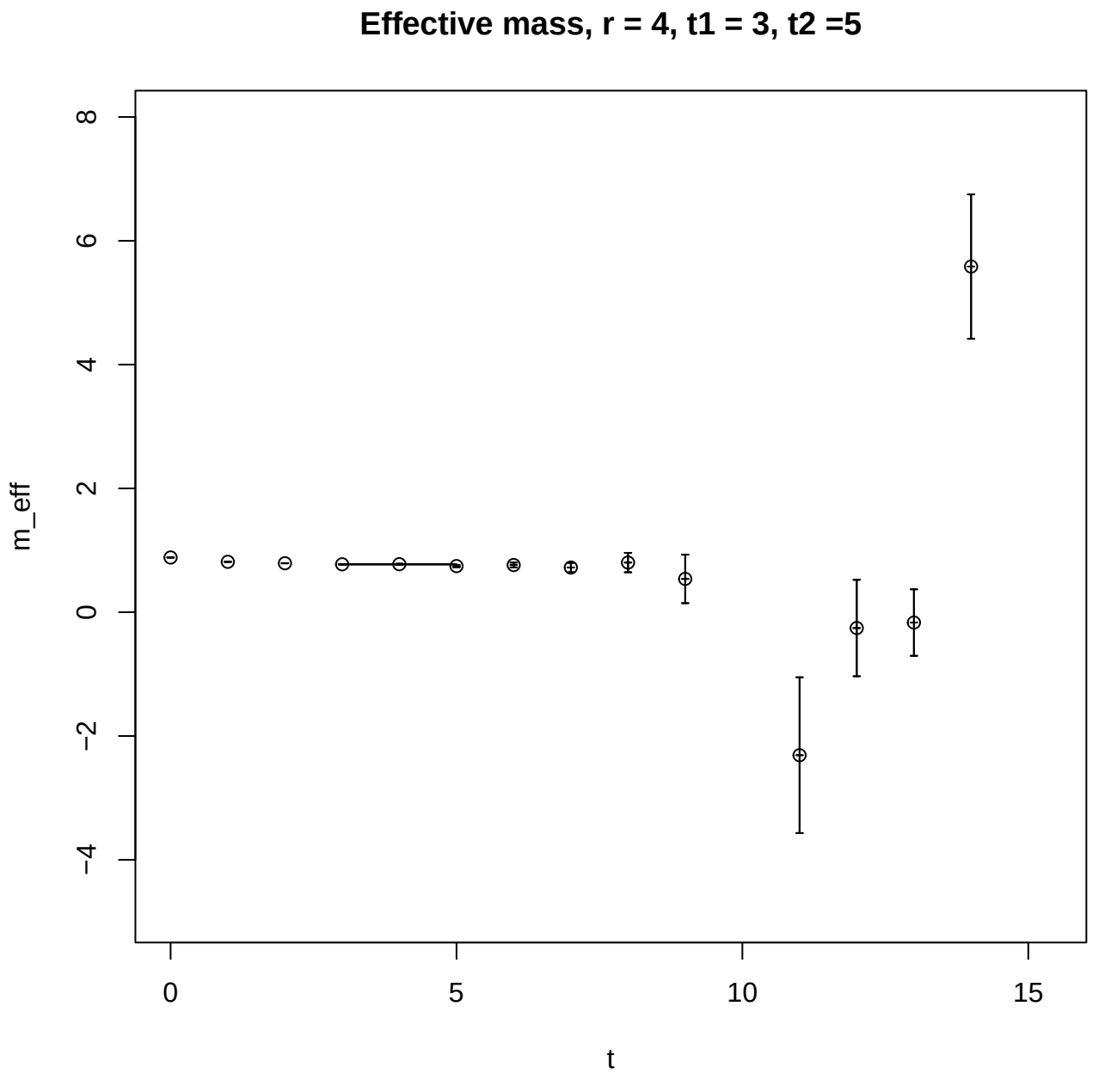

Effective mass, r = 3, t1 = 10, t2 =11
m_eff
t
5
0
−5
−10
0
5
10
15

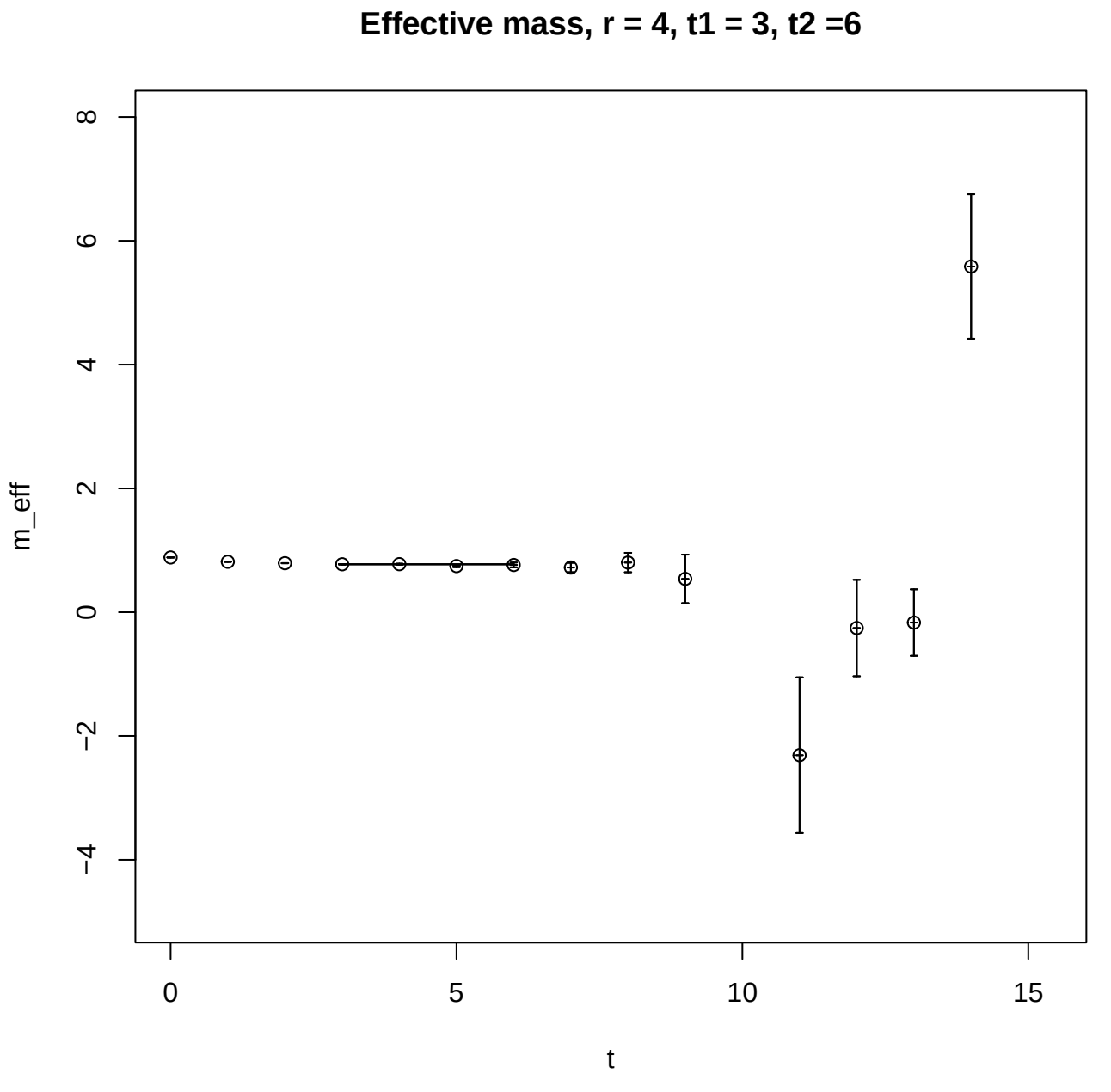

Effective mass, r = 3, t1 = 10, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

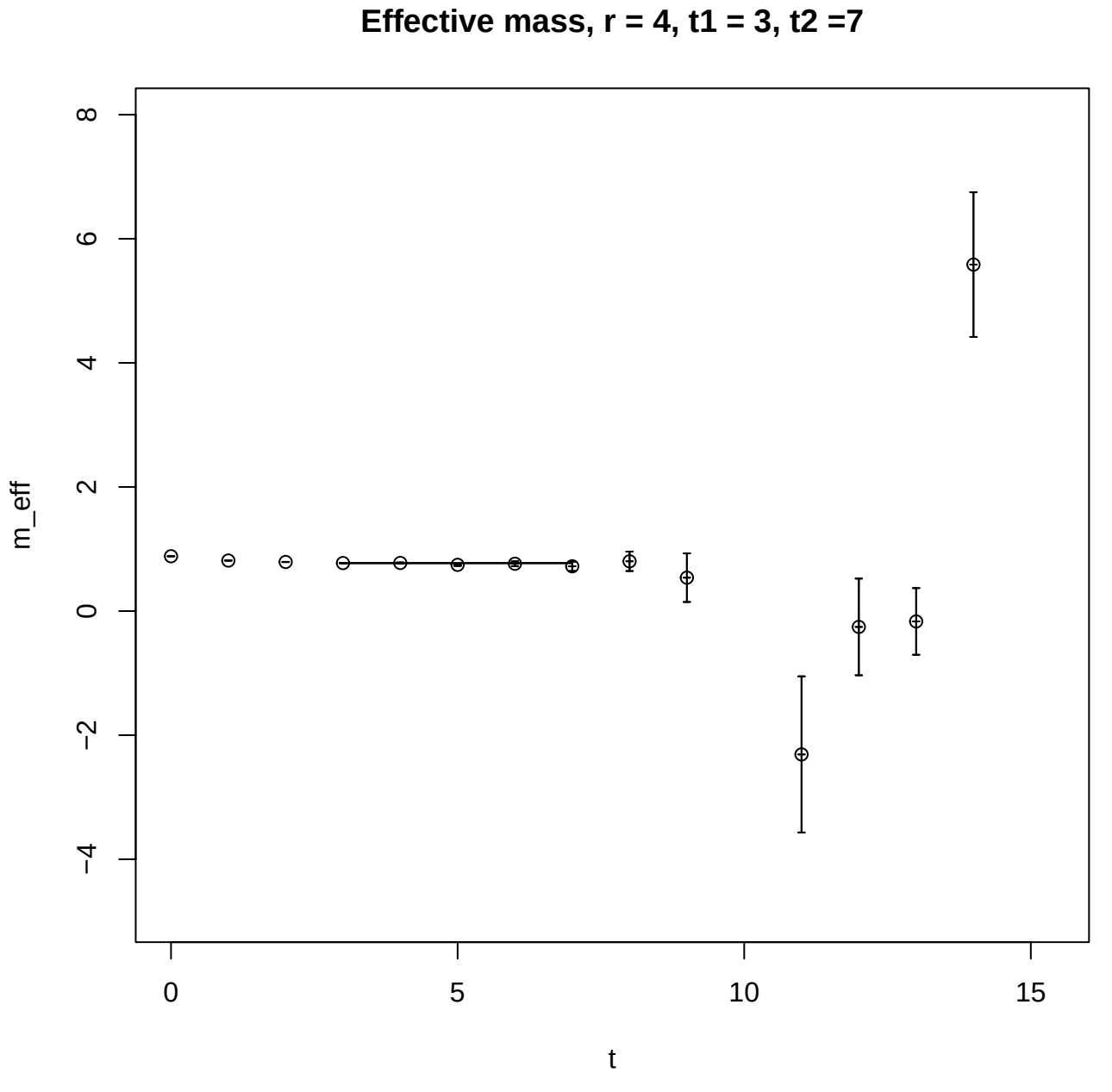

Effective mass, r = 3, t1 = 11, t2 =12
m_eff
t
5
0
−5
−10
0
5
10
15

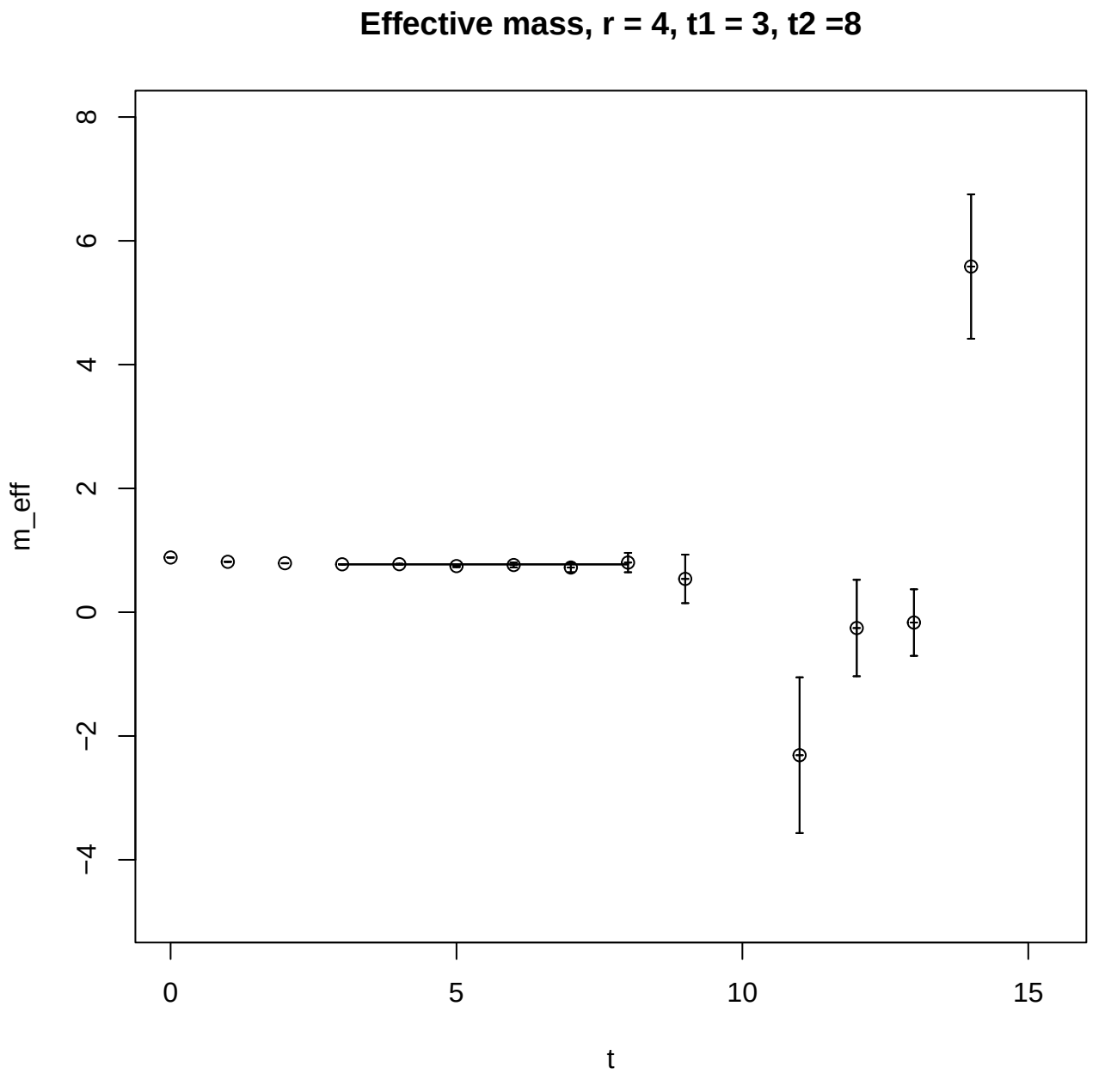

Effective mass, r = 3, t1 = 12, t2 =13
m_eff
t
5
0
−5
−10
0
5
10
15

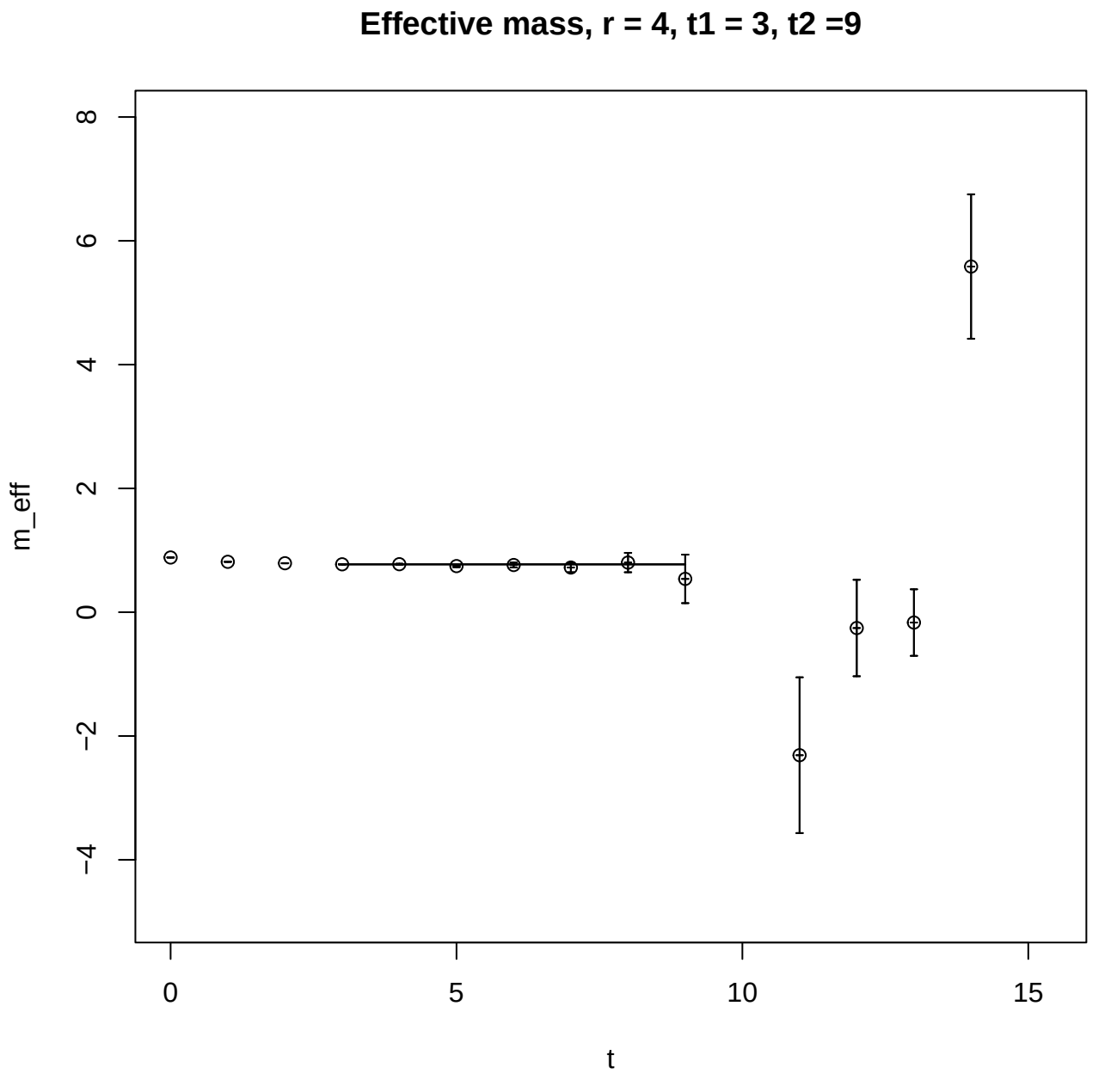

Effective mass, r = 4, t1 = 3, t2 =4
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

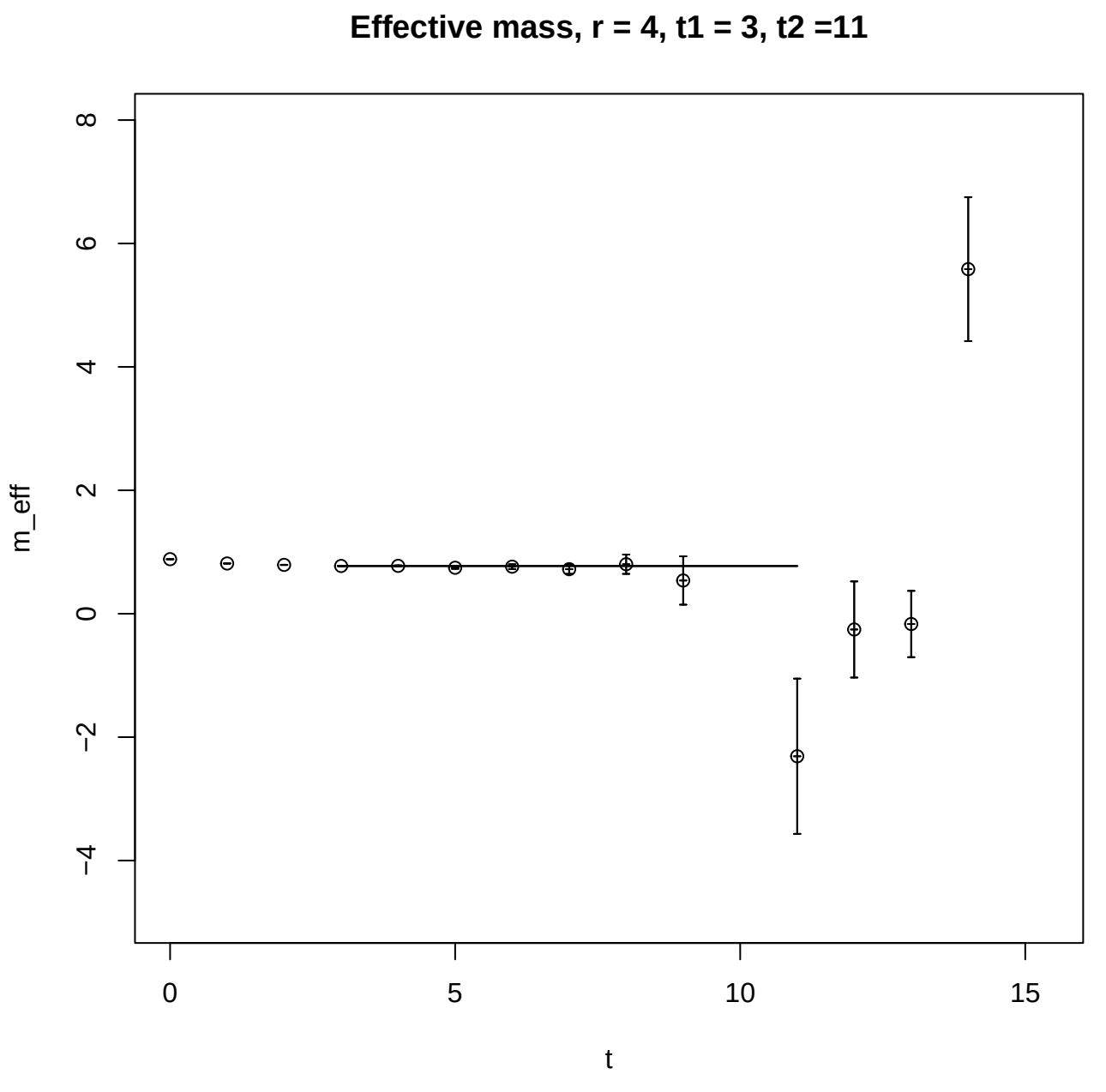

Effective mass, r = 4, t1 = 3, t2 =5
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

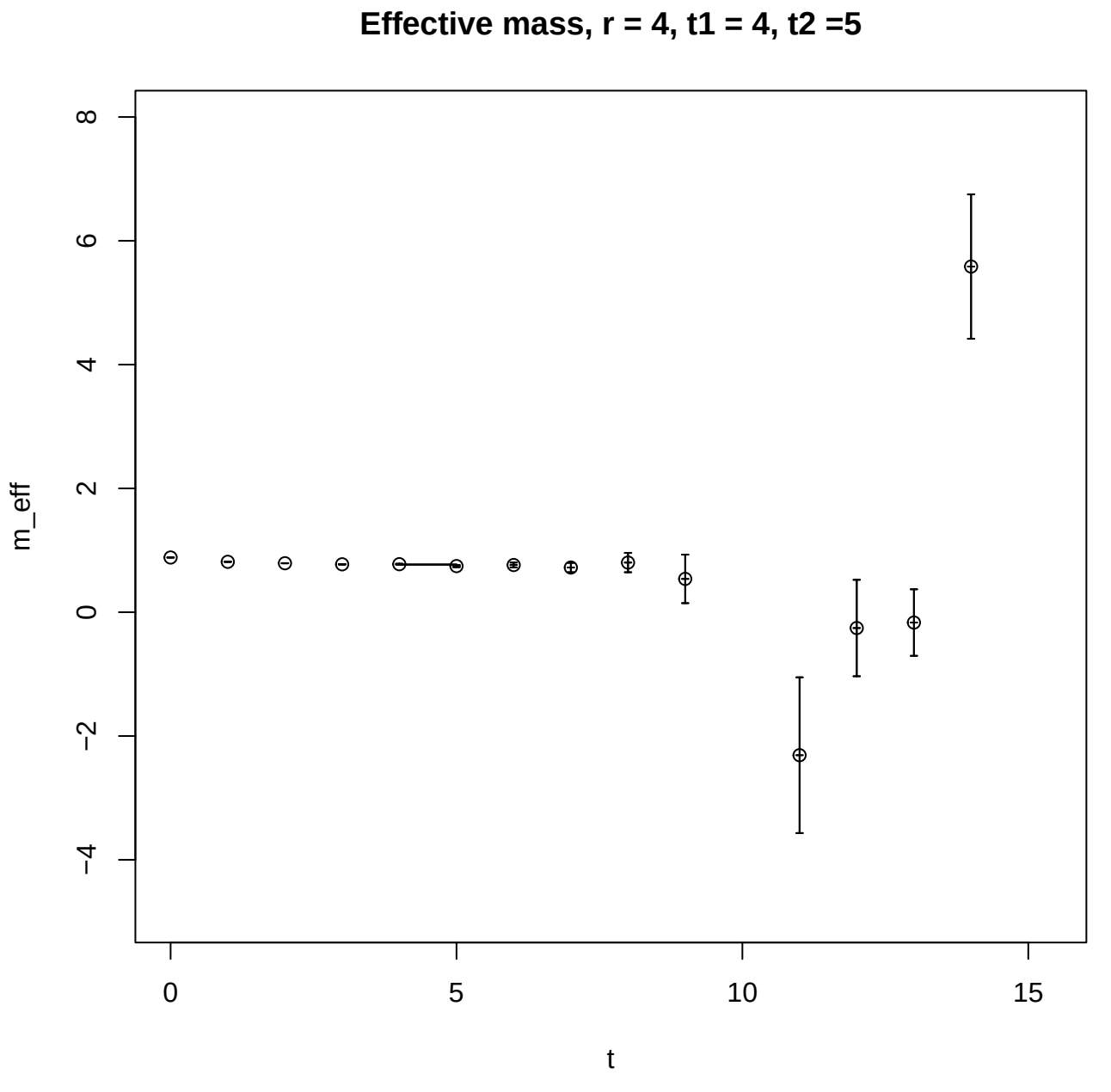

Effective mass, r = 4, t1 = 3, t2 =6
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

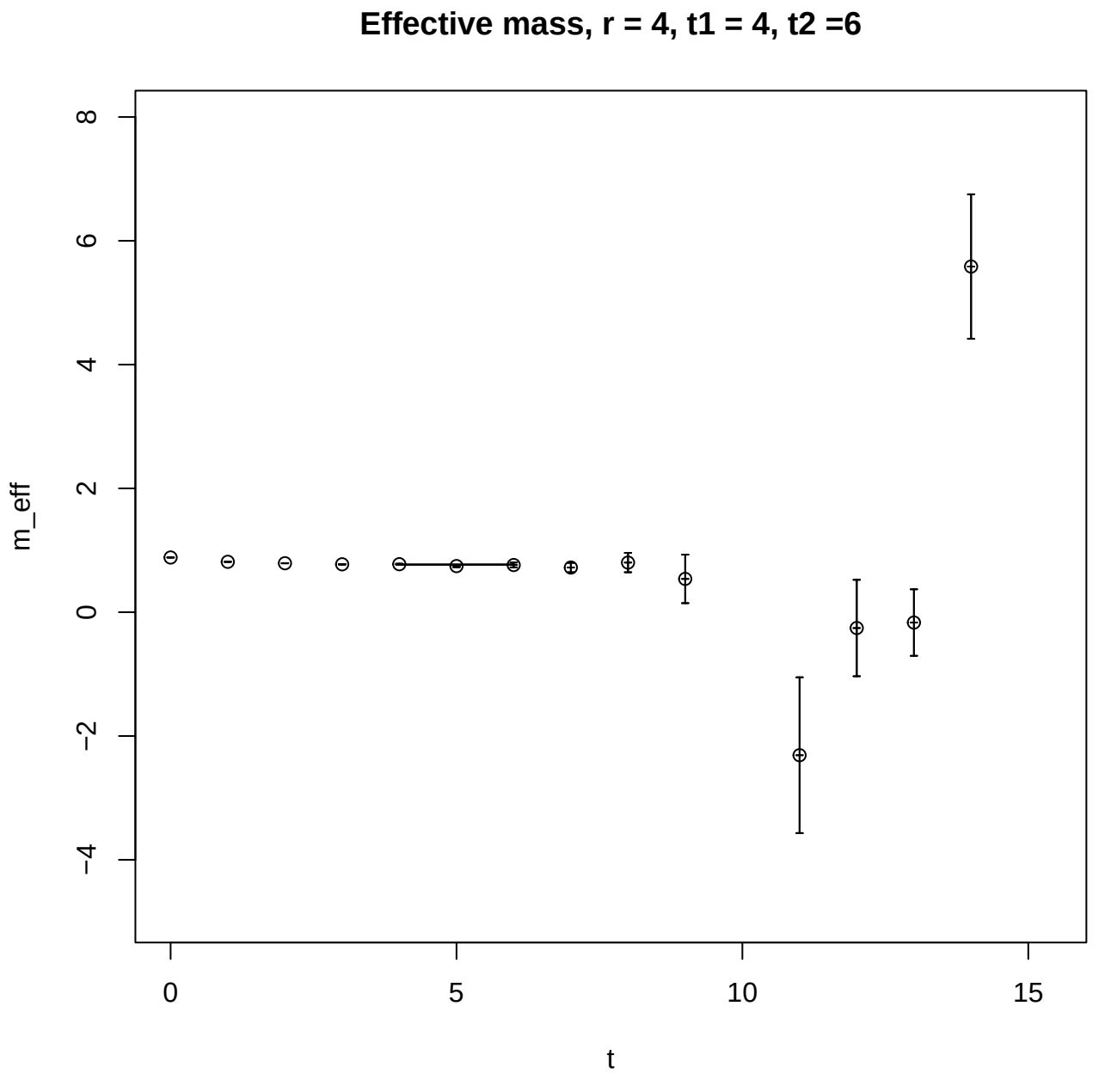

Effective mass, r = 4, t1 = 3, t2 =7
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

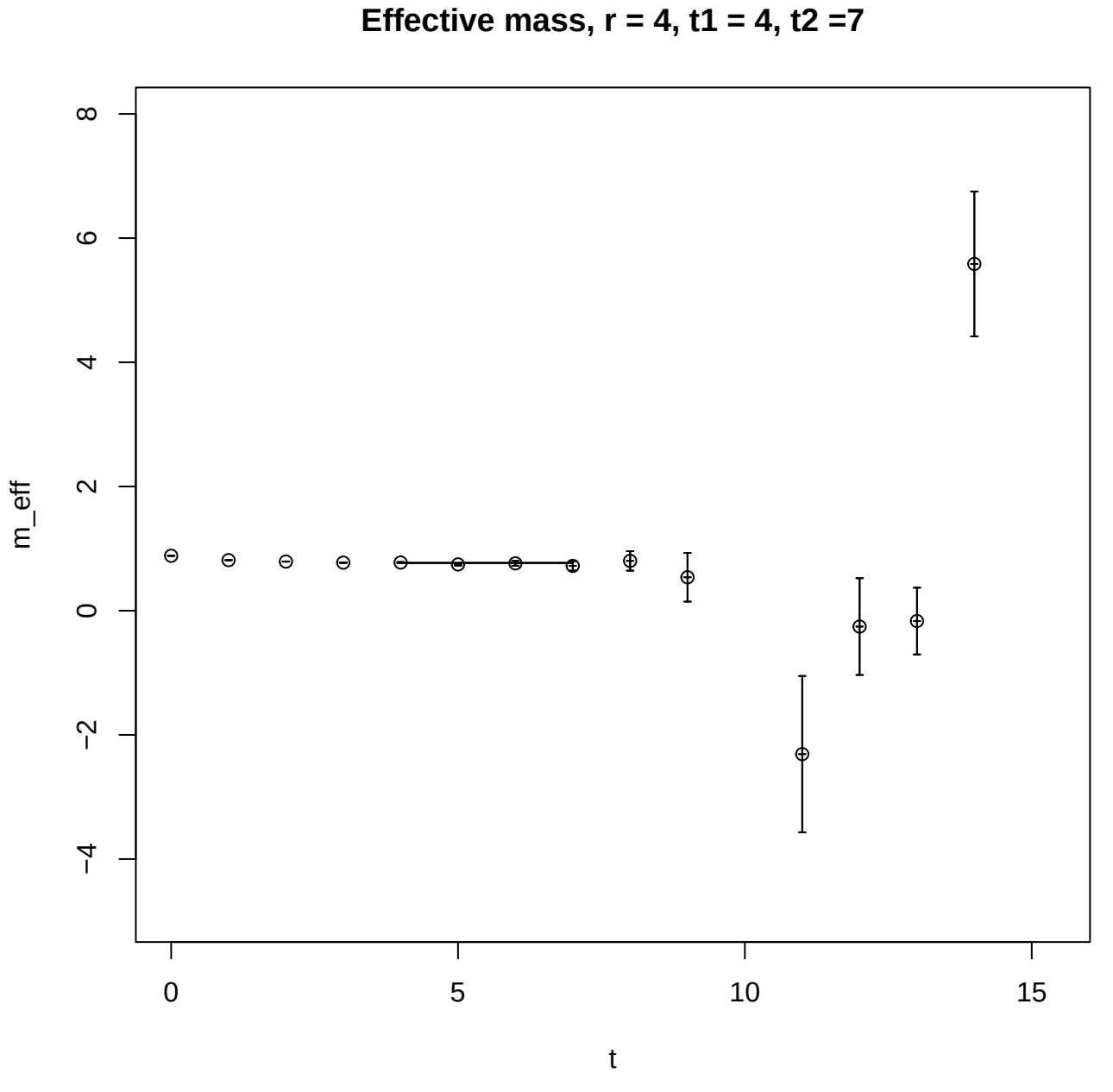

Effective mass, r = 4, t1 = 3, t2 =8
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

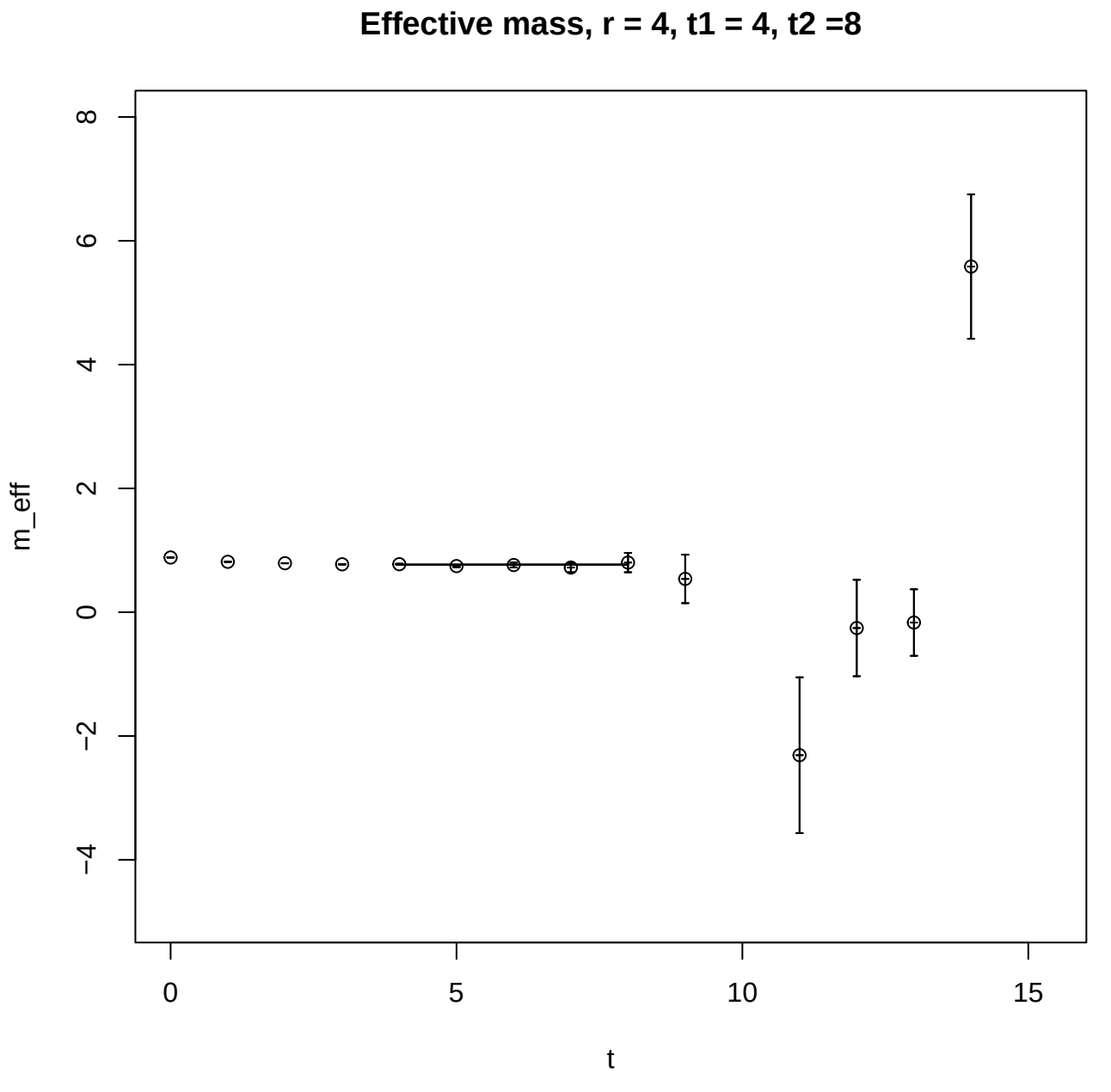

Effective mass, r = 4, t1 = 3, t2 =9
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

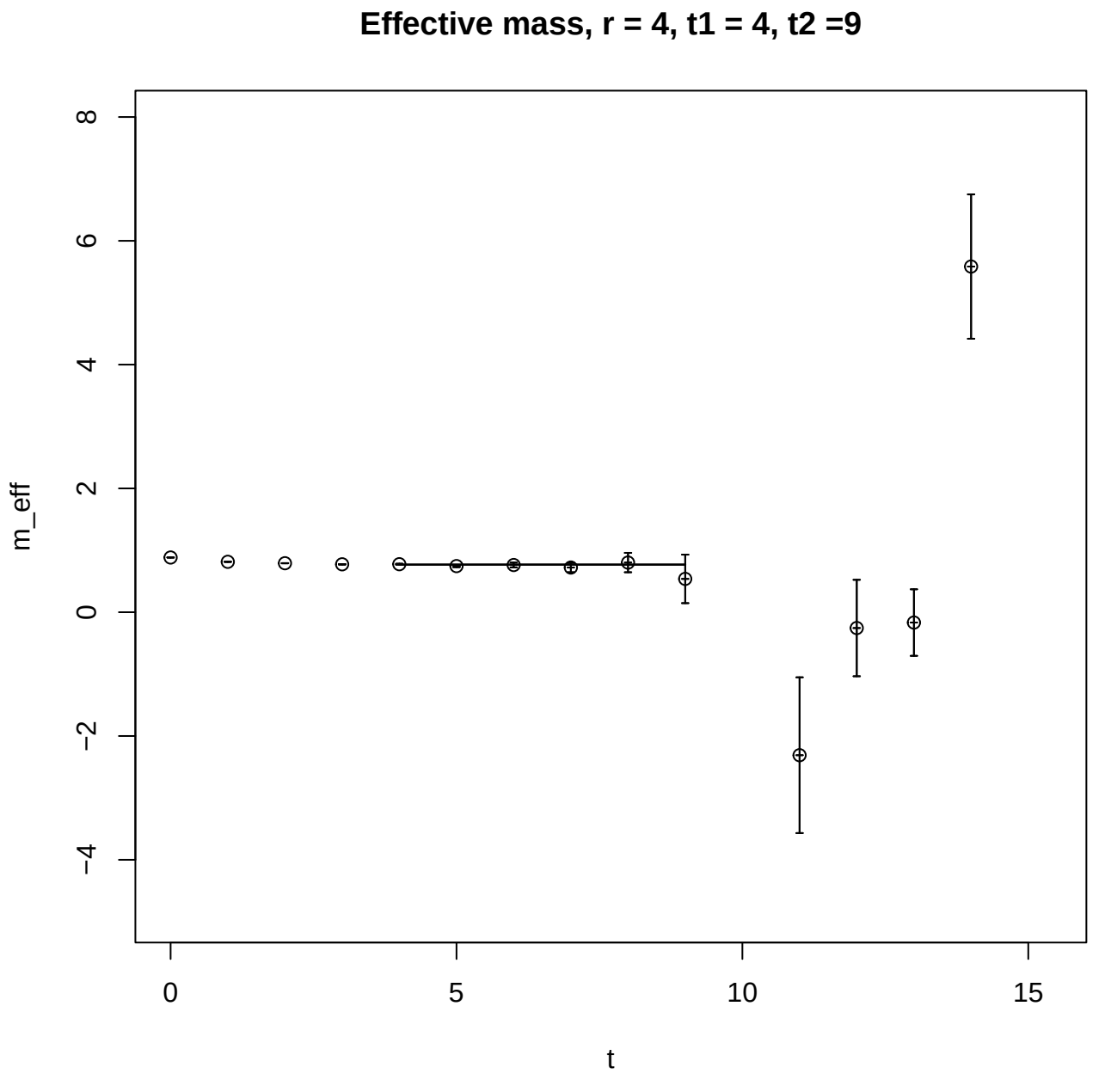

Effective mass, r = 4, t1 = 3, t2 =11
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

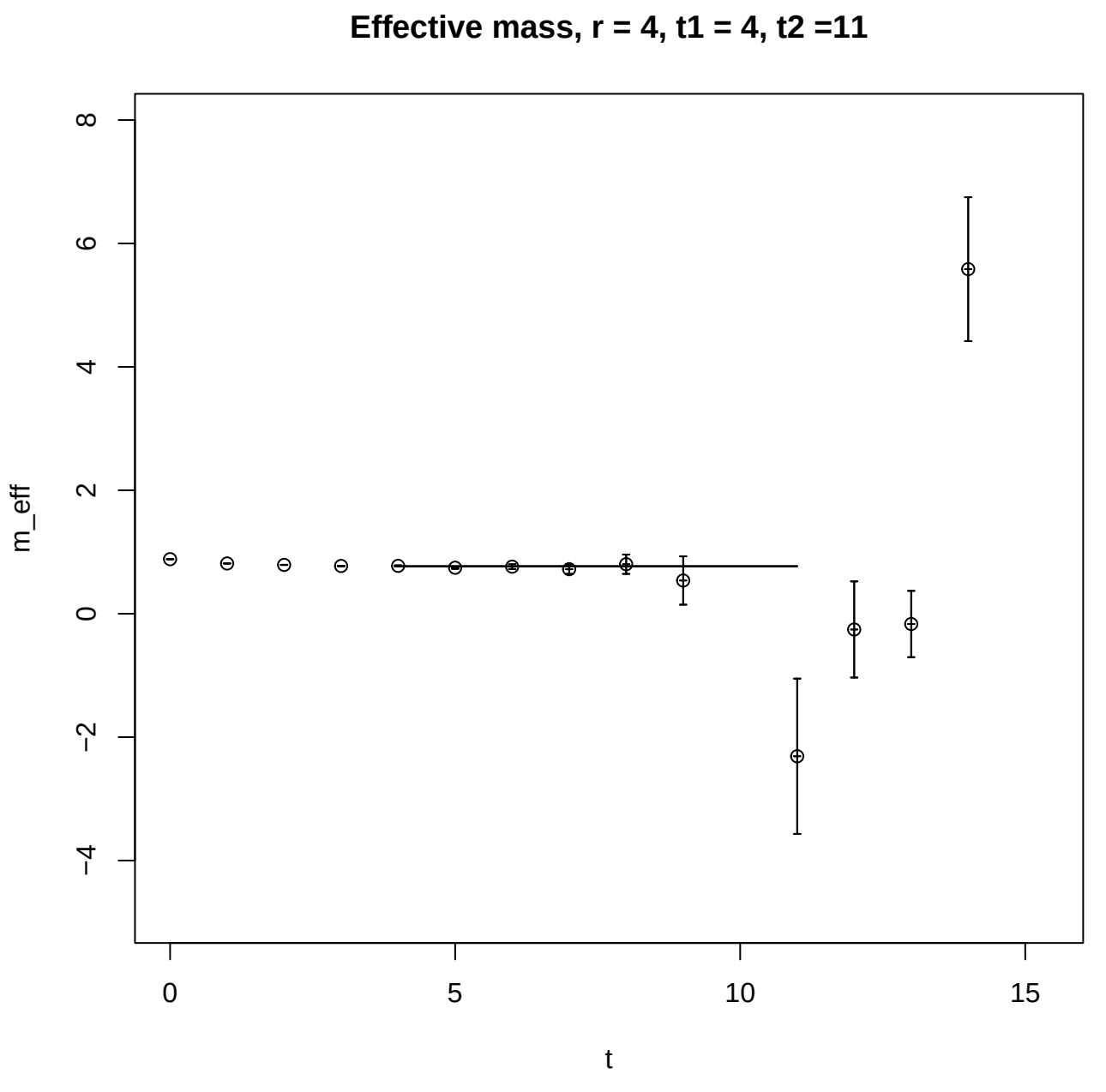

Effective mass, r = 4, t1 = 4, t2 =5
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

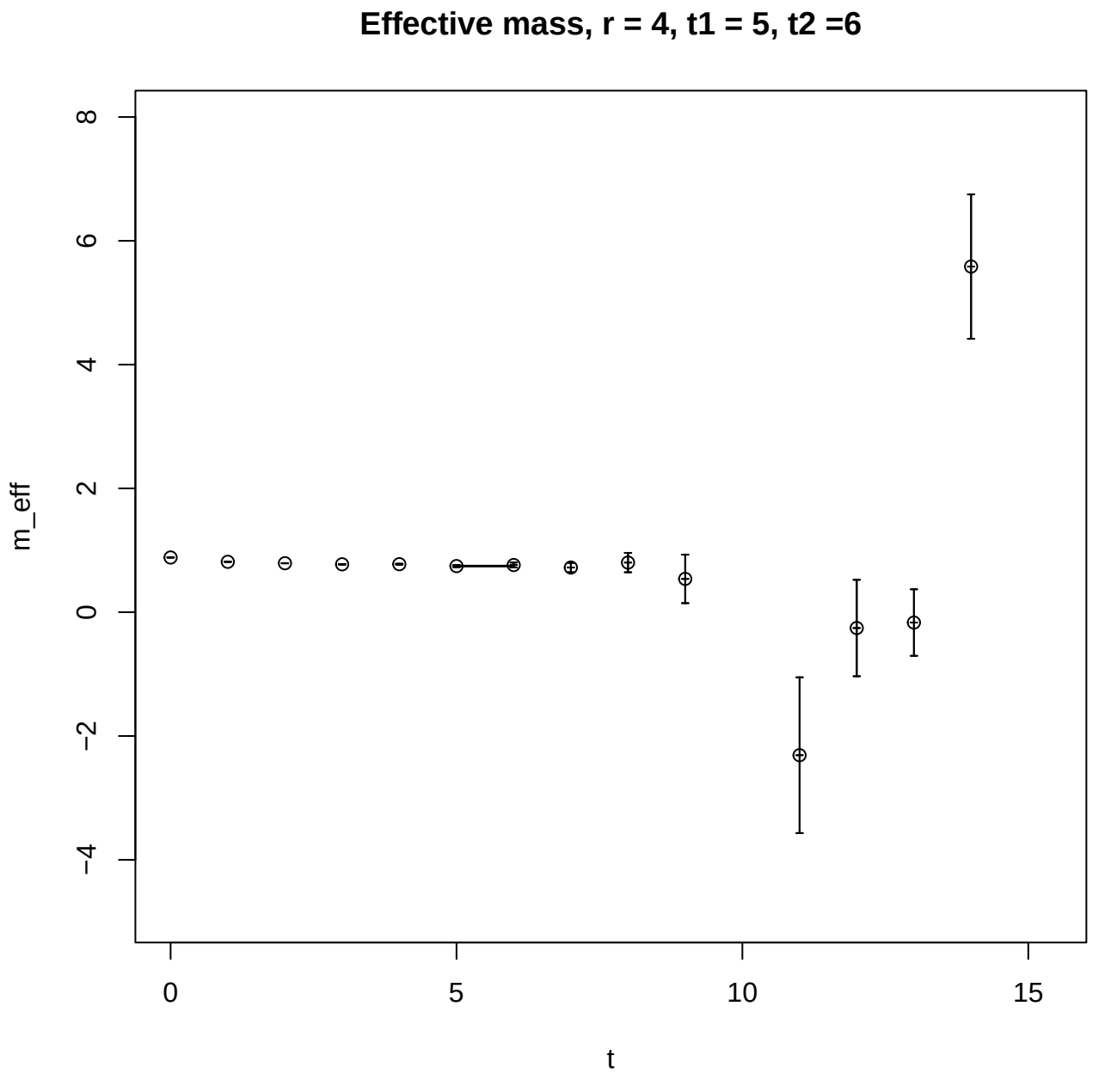

Effective mass, r = 4, t1 = 4, t2 =6
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

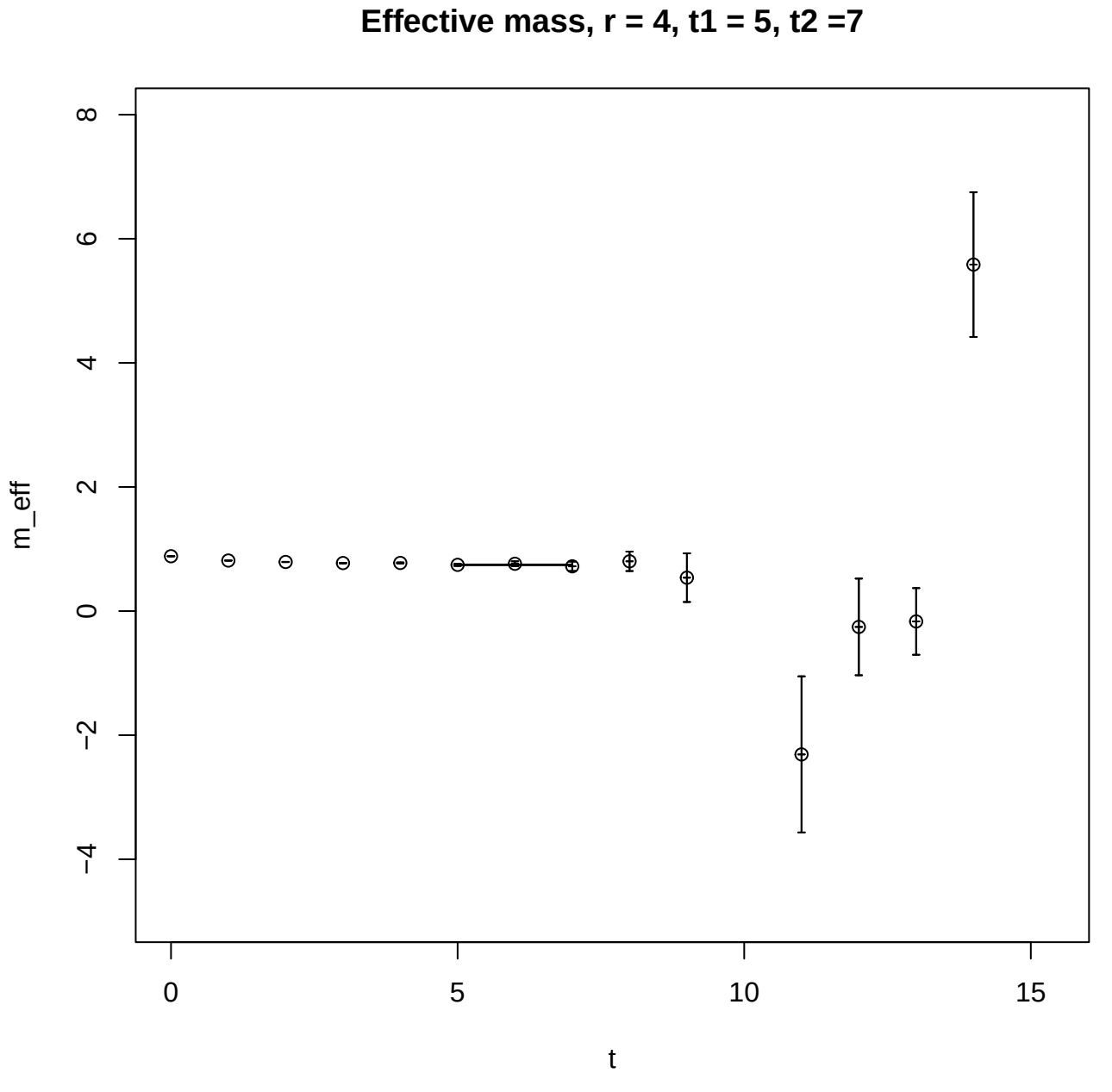

Effective mass, r = 4, t1 = 4, t2 =7
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

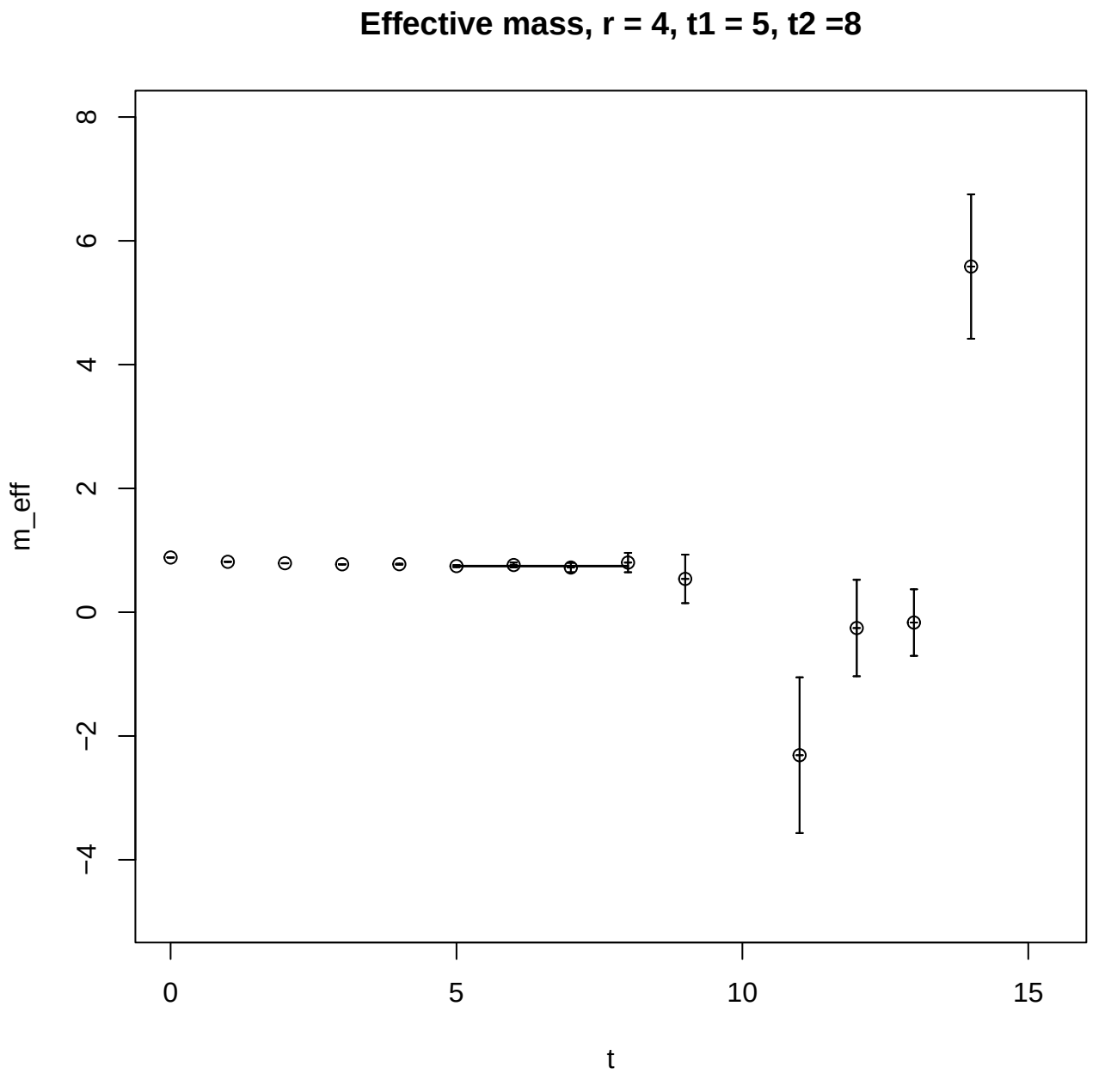

Effective mass, r = 4, t1 = 4, t2 =8
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

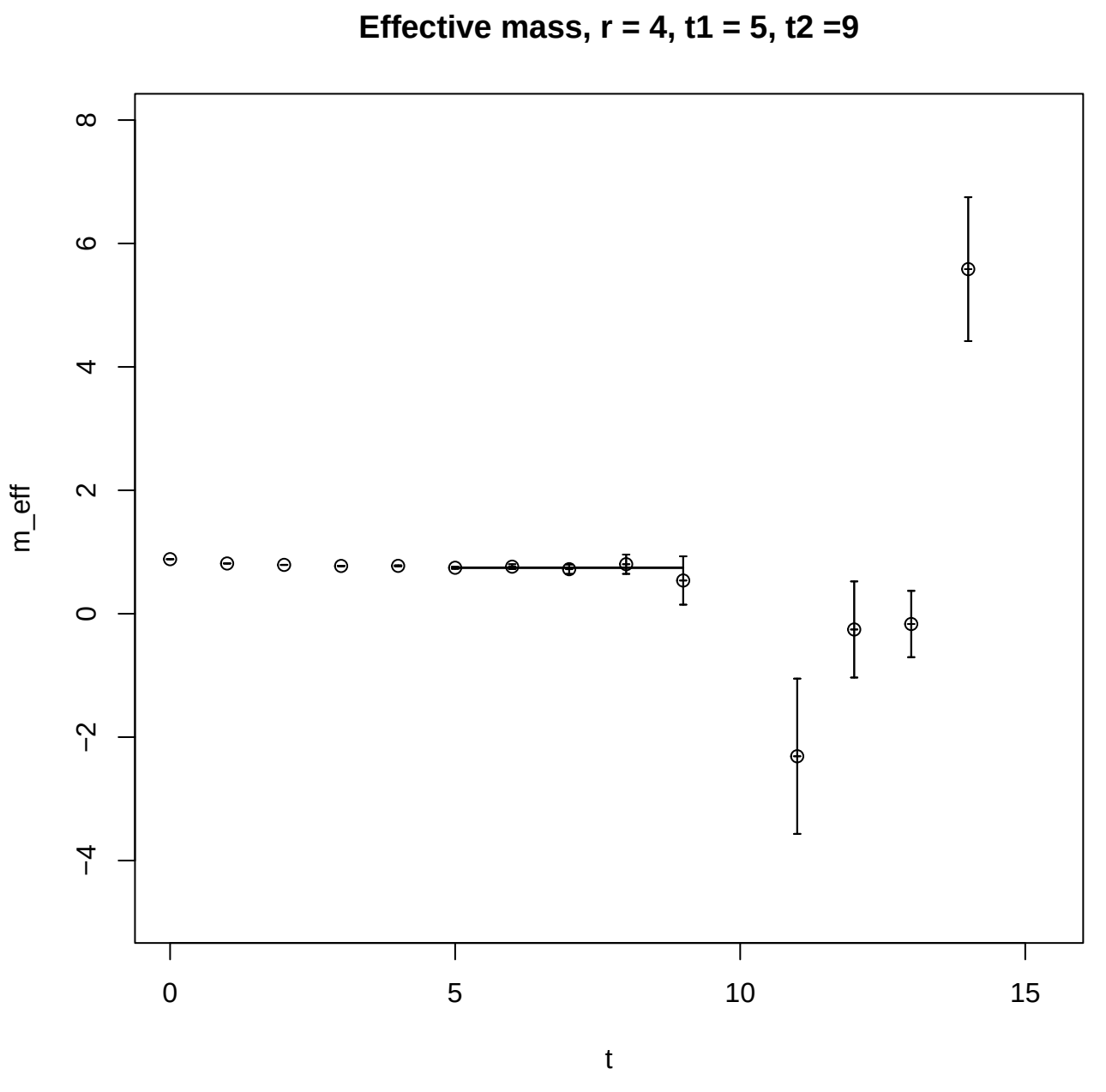

Effective mass, r = 4, t1 = 4, t2 =9
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

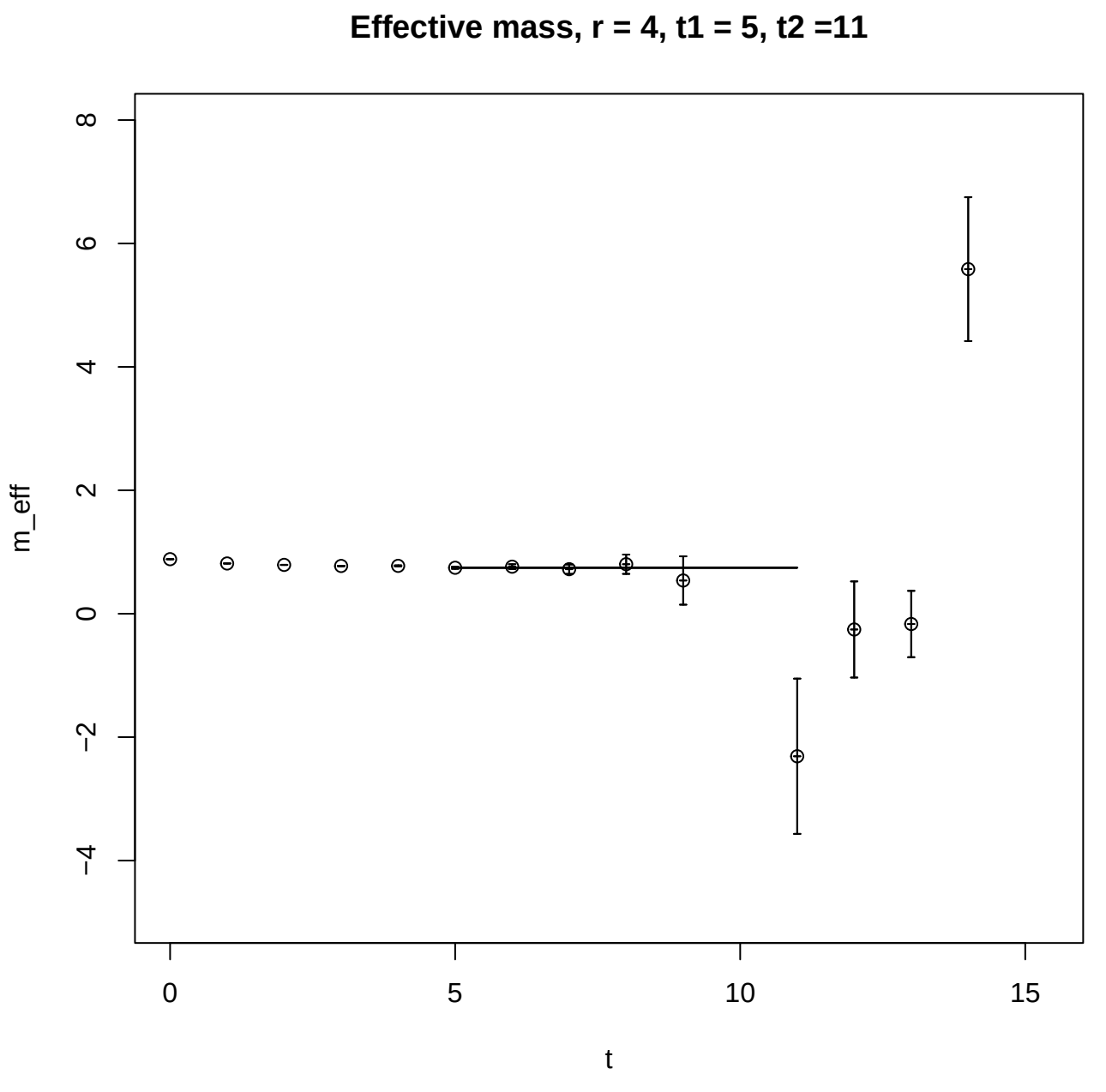

Effective mass, r = 4, t1 = 4, t2 =11
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

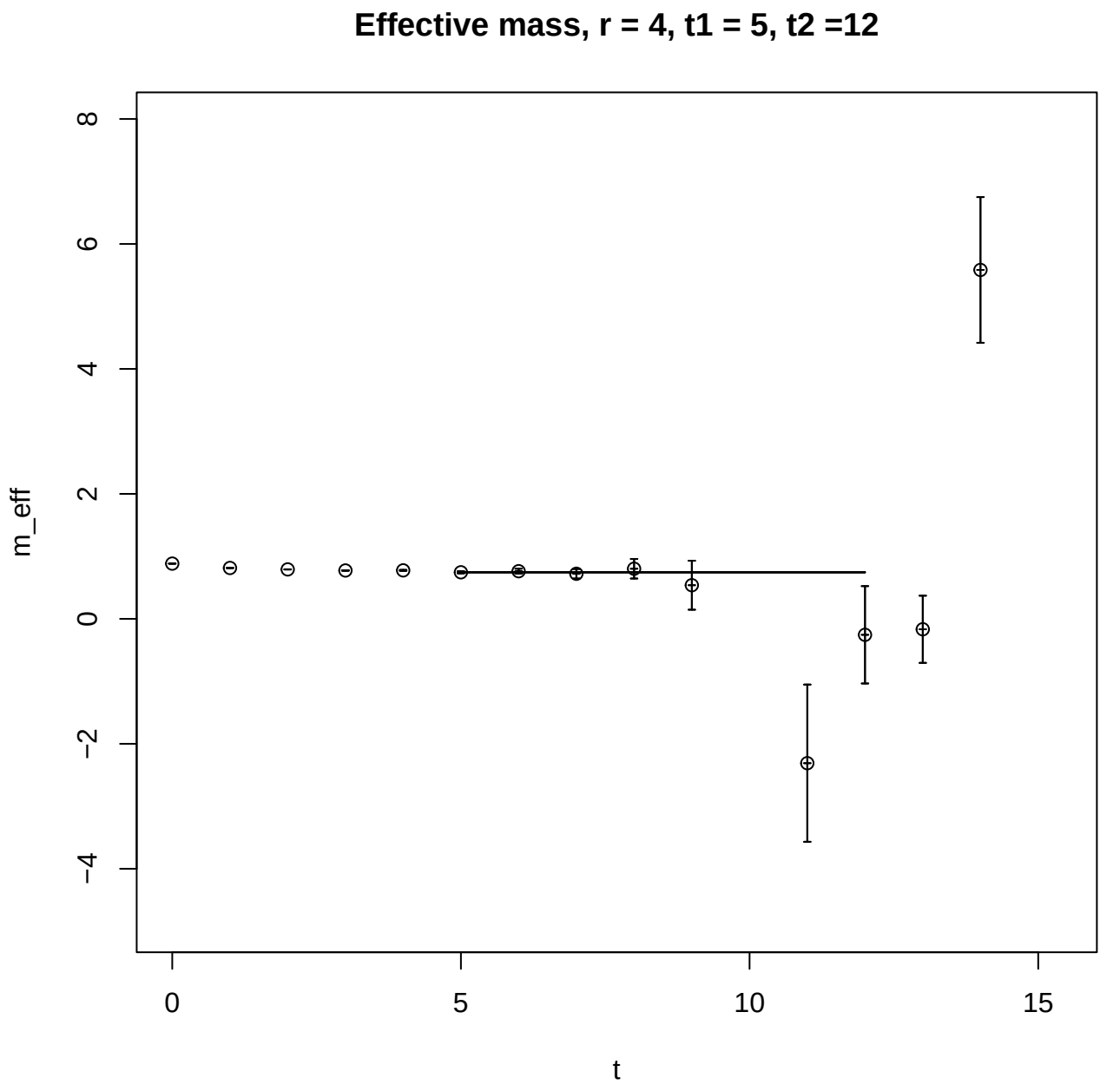

Effective mass, r = 4, t1 = 5, t2 =6
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

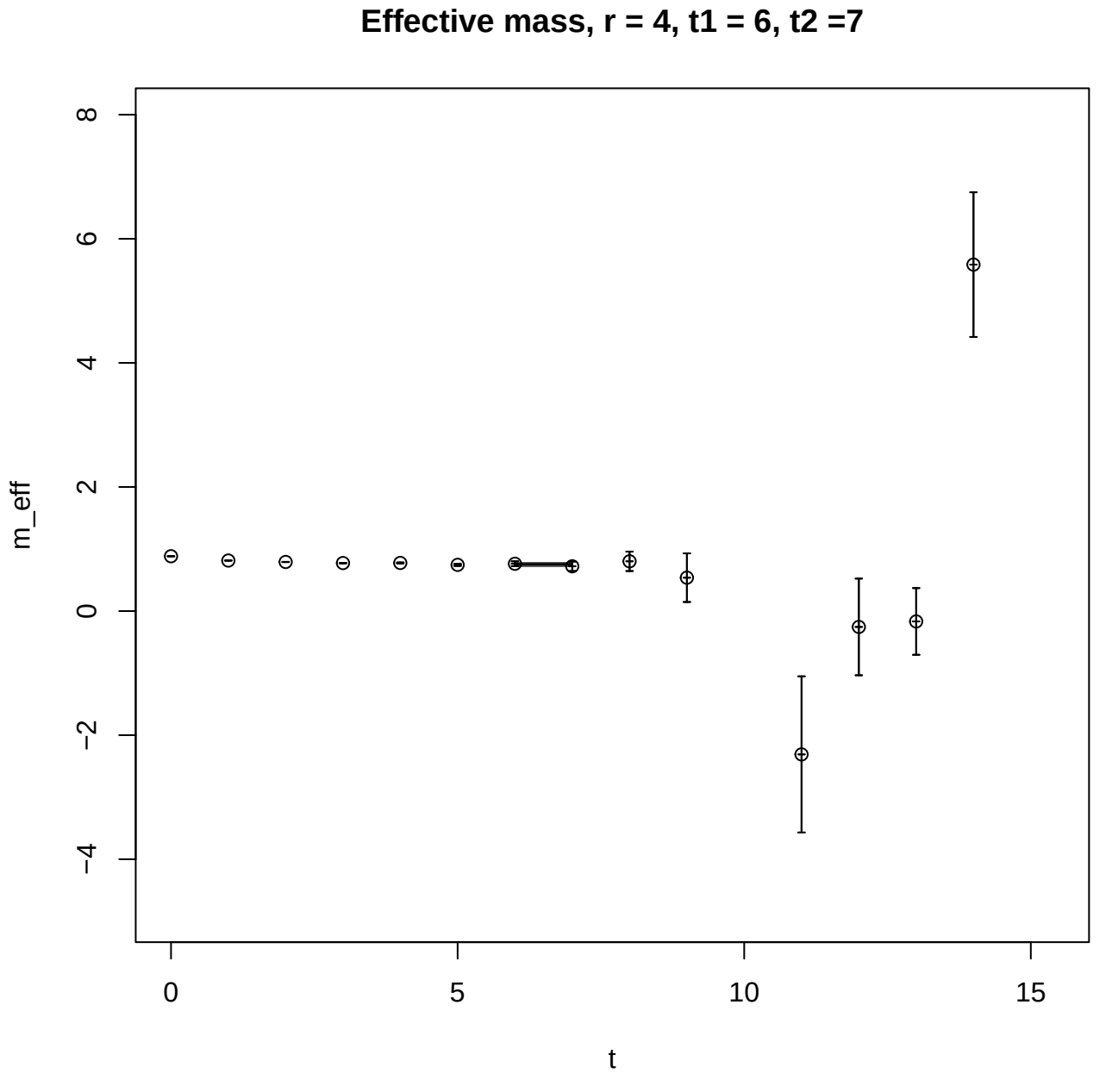

Effective mass, r = 4, t1 = 5, t2 =7
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

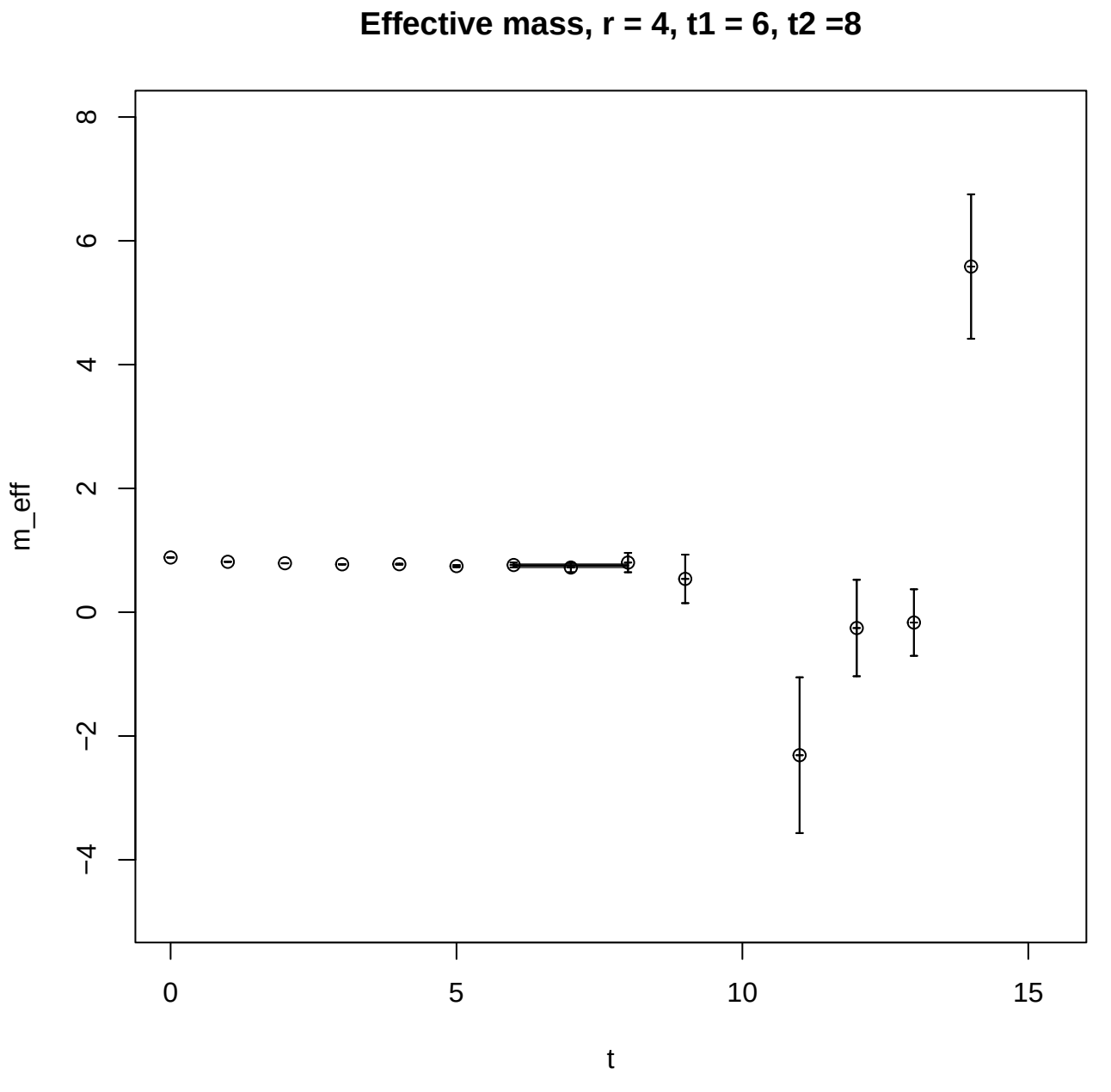

Effective mass, r = 4, t1 = 5, t2 =8
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

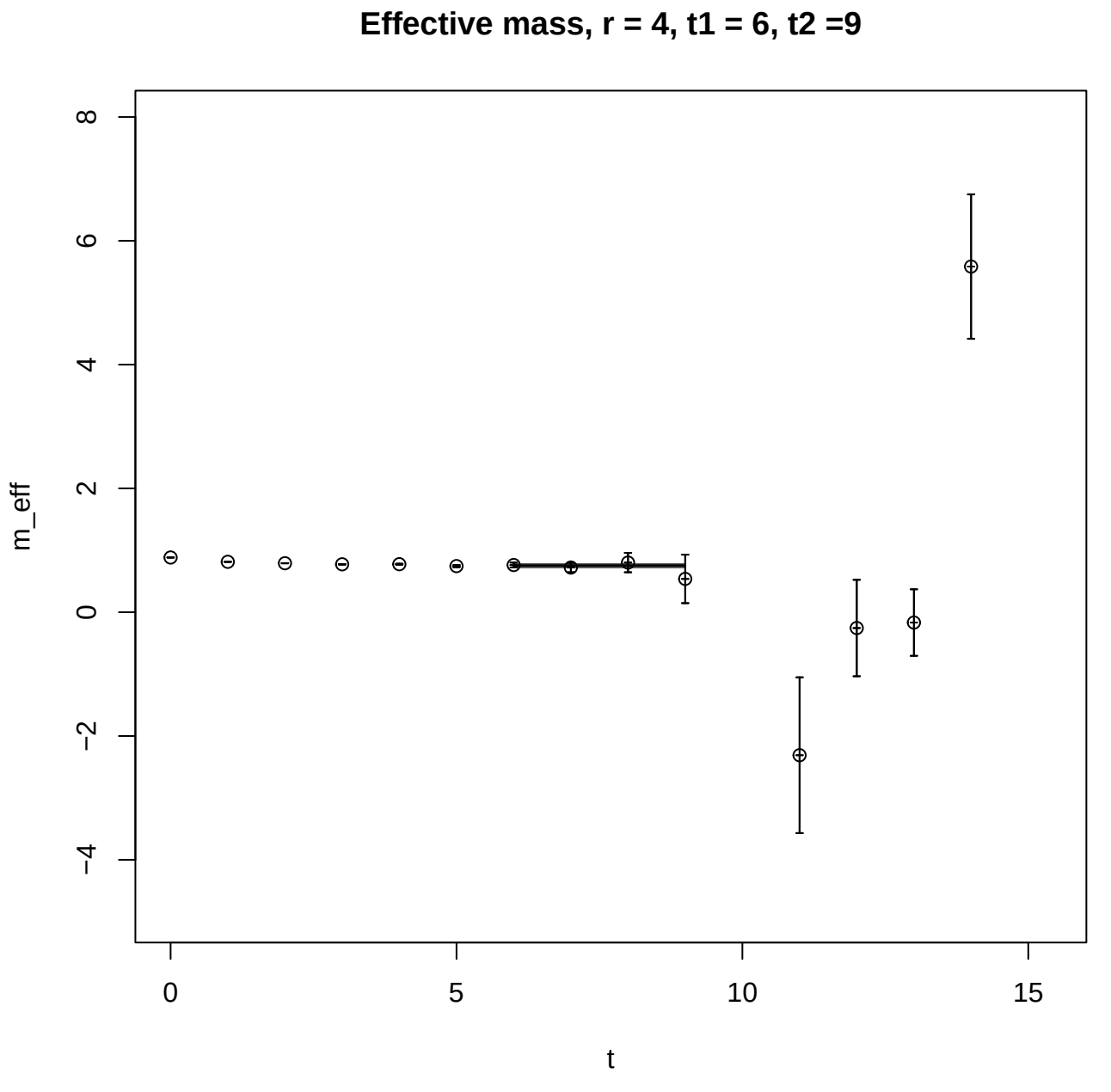

Effective mass, r = 4, t1 = 5, t2 =9
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

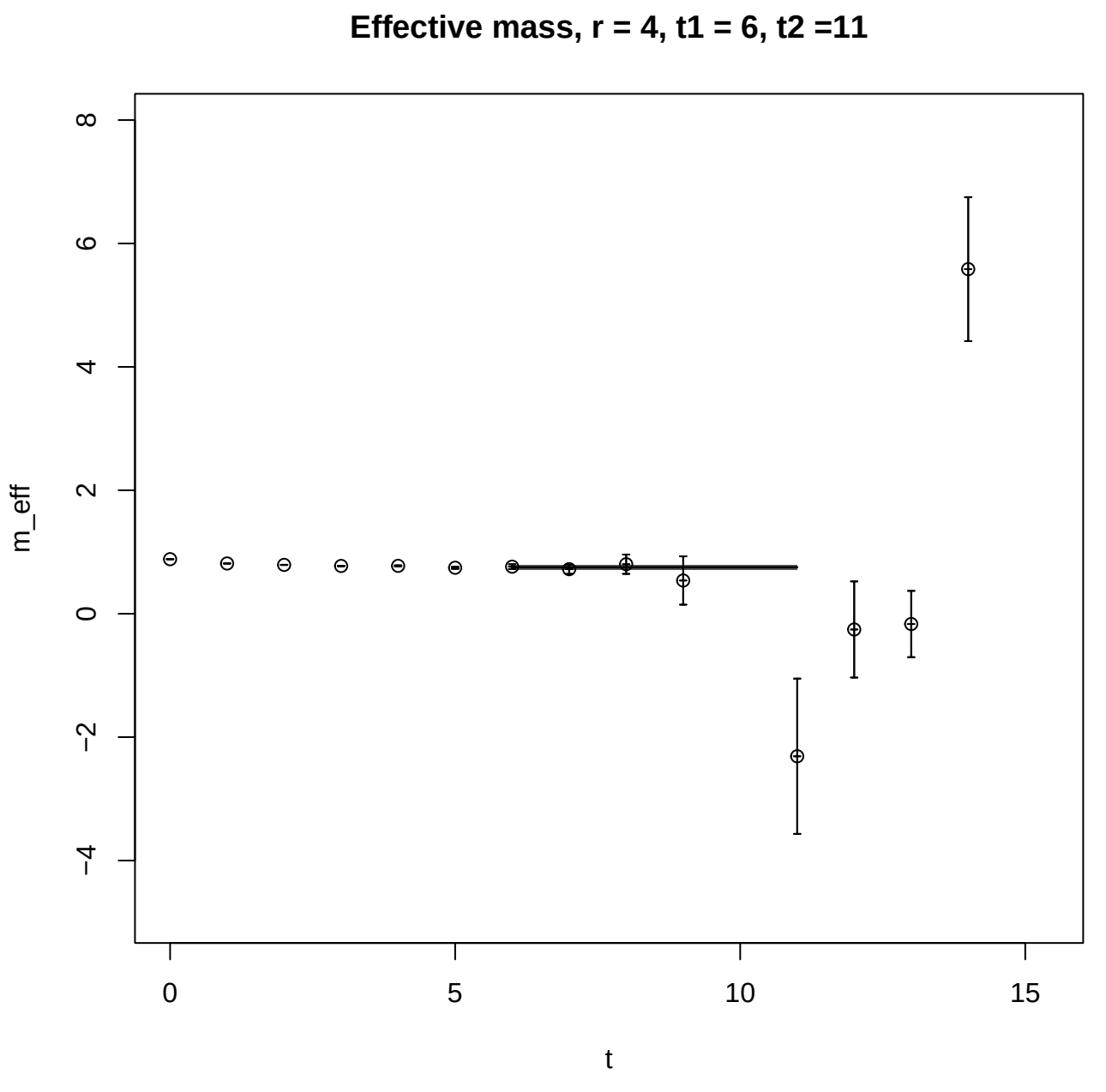

Effective mass, r = 4, t1 = 5, t2 =11
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

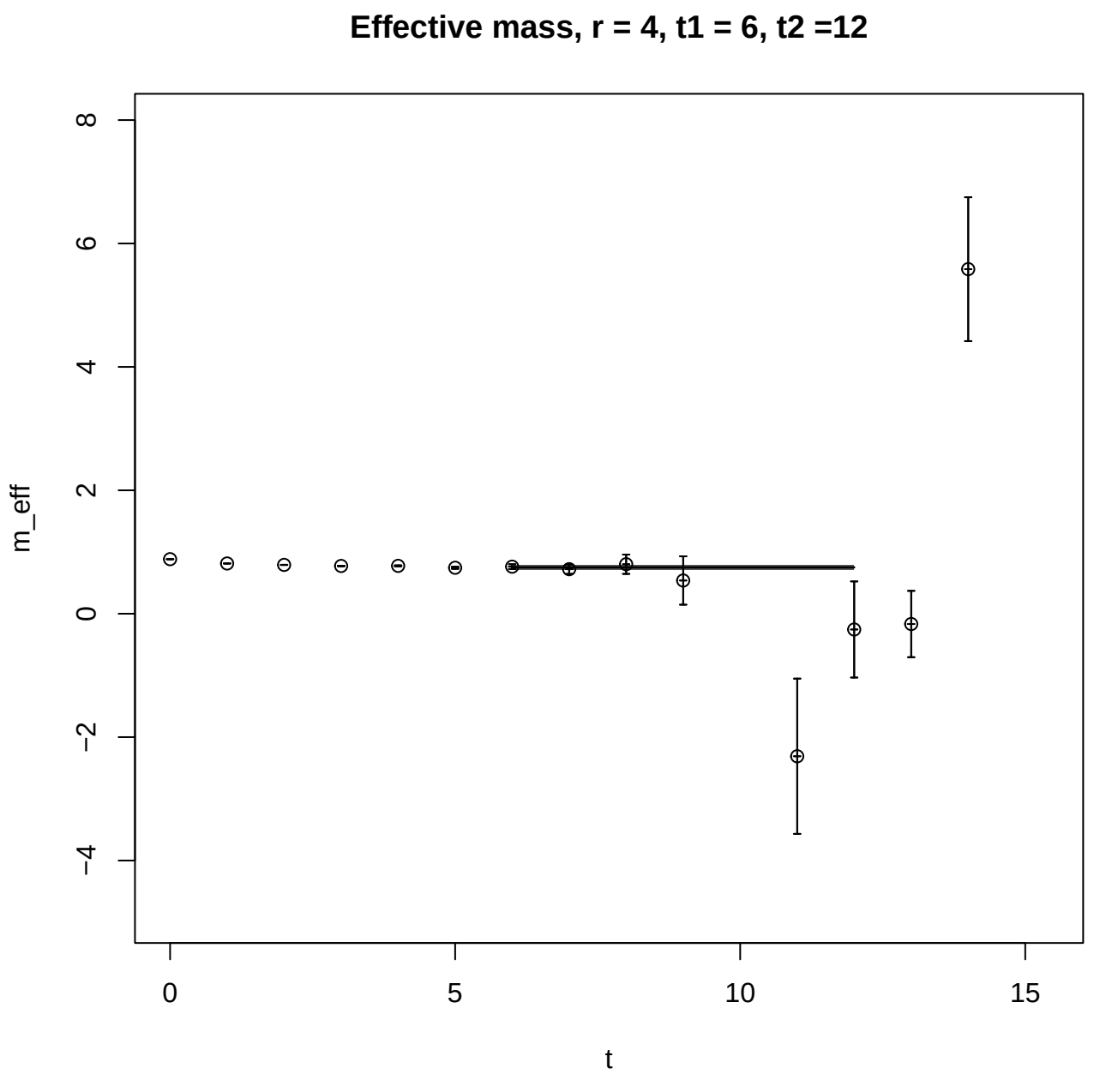

Effective mass, r = 4, t1 = 5, t2 =12
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

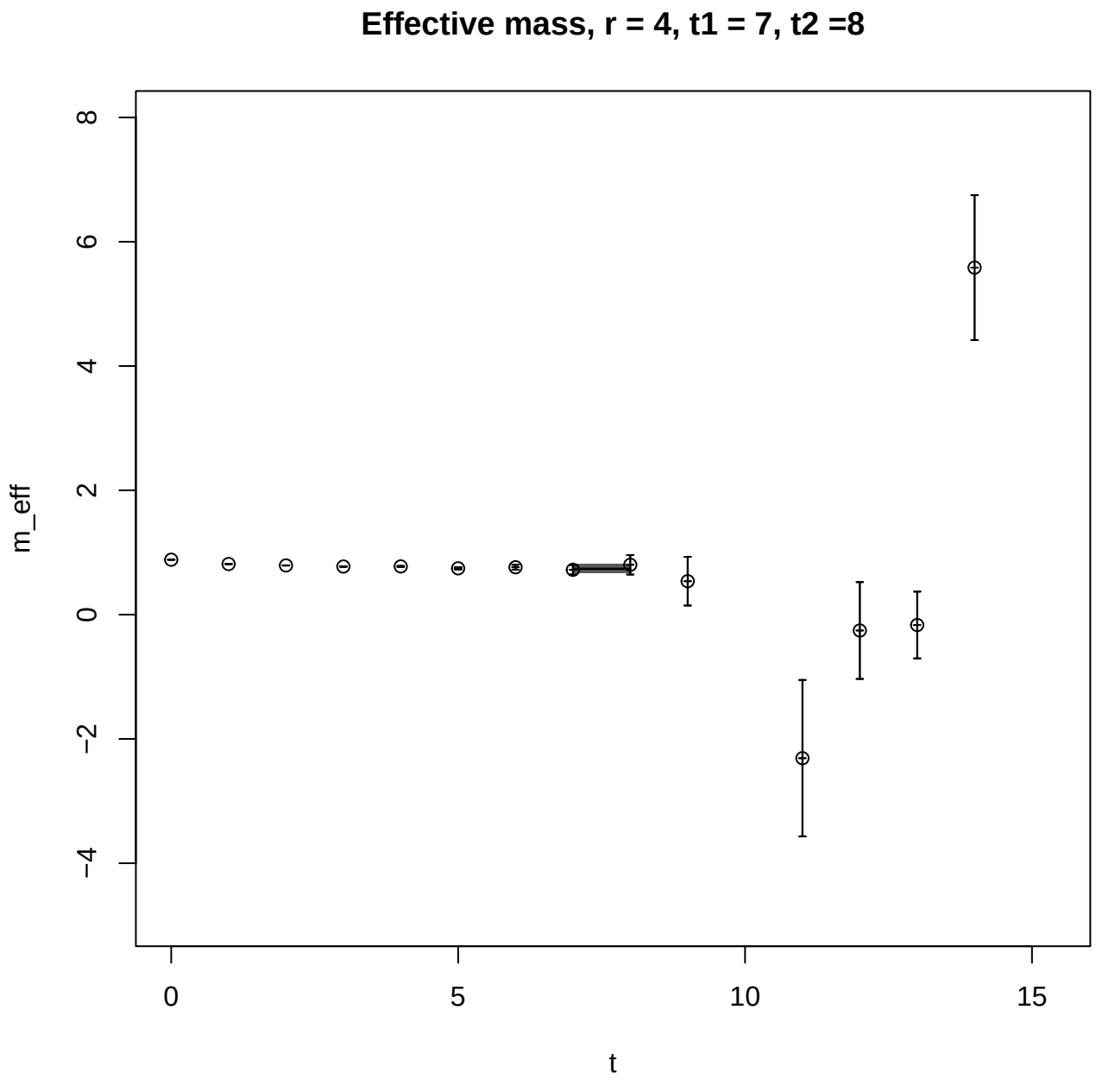

Effective mass, r = 4, t1 = 6, t2 =7
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

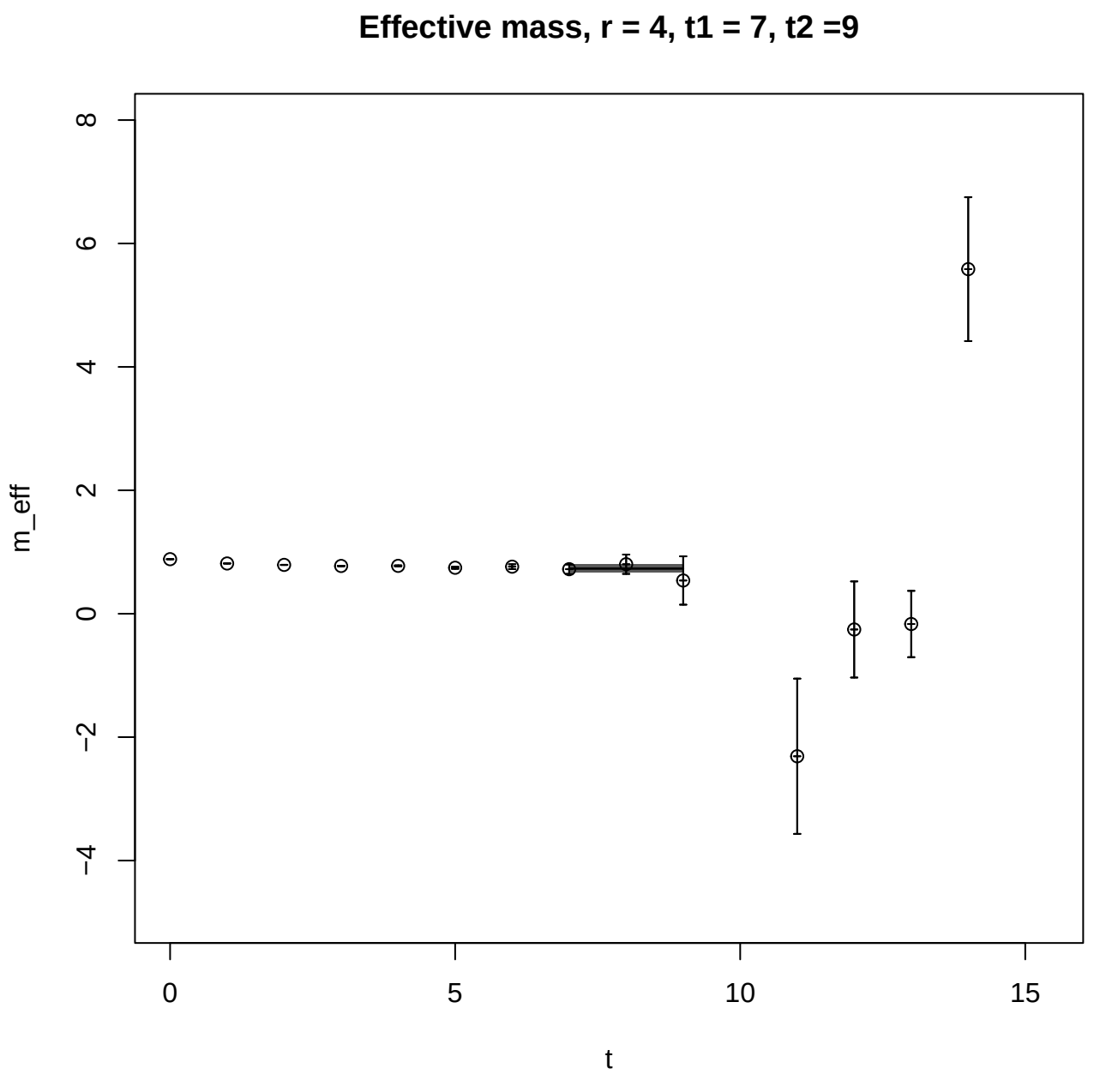

Effective mass, r = 4, t1 = 6, t2 =8
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

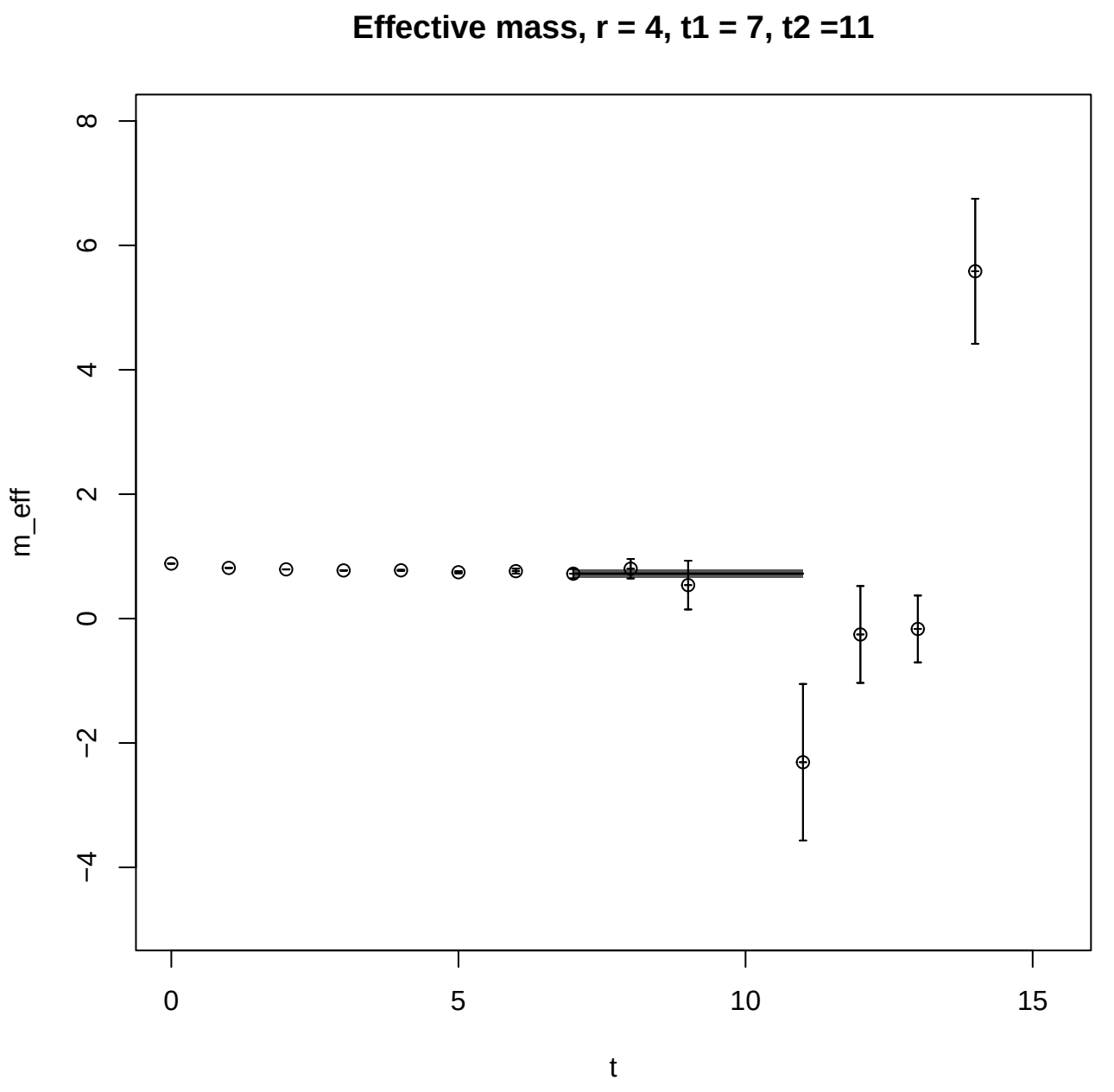

Effective mass, r = 4, t1 = 6, t2 =9
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

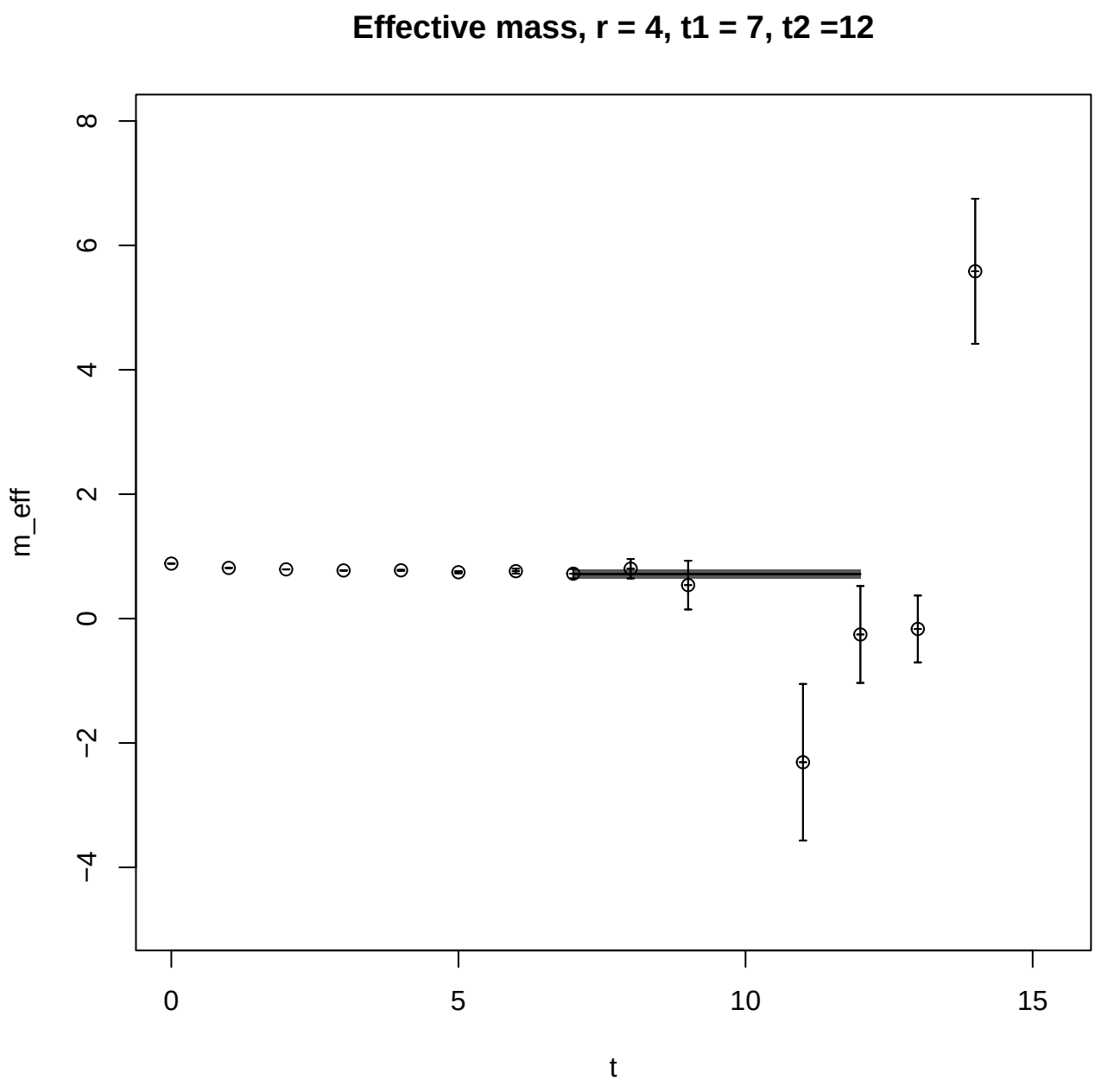

Effective mass, r = 4, t1 = 6, t2 =11
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

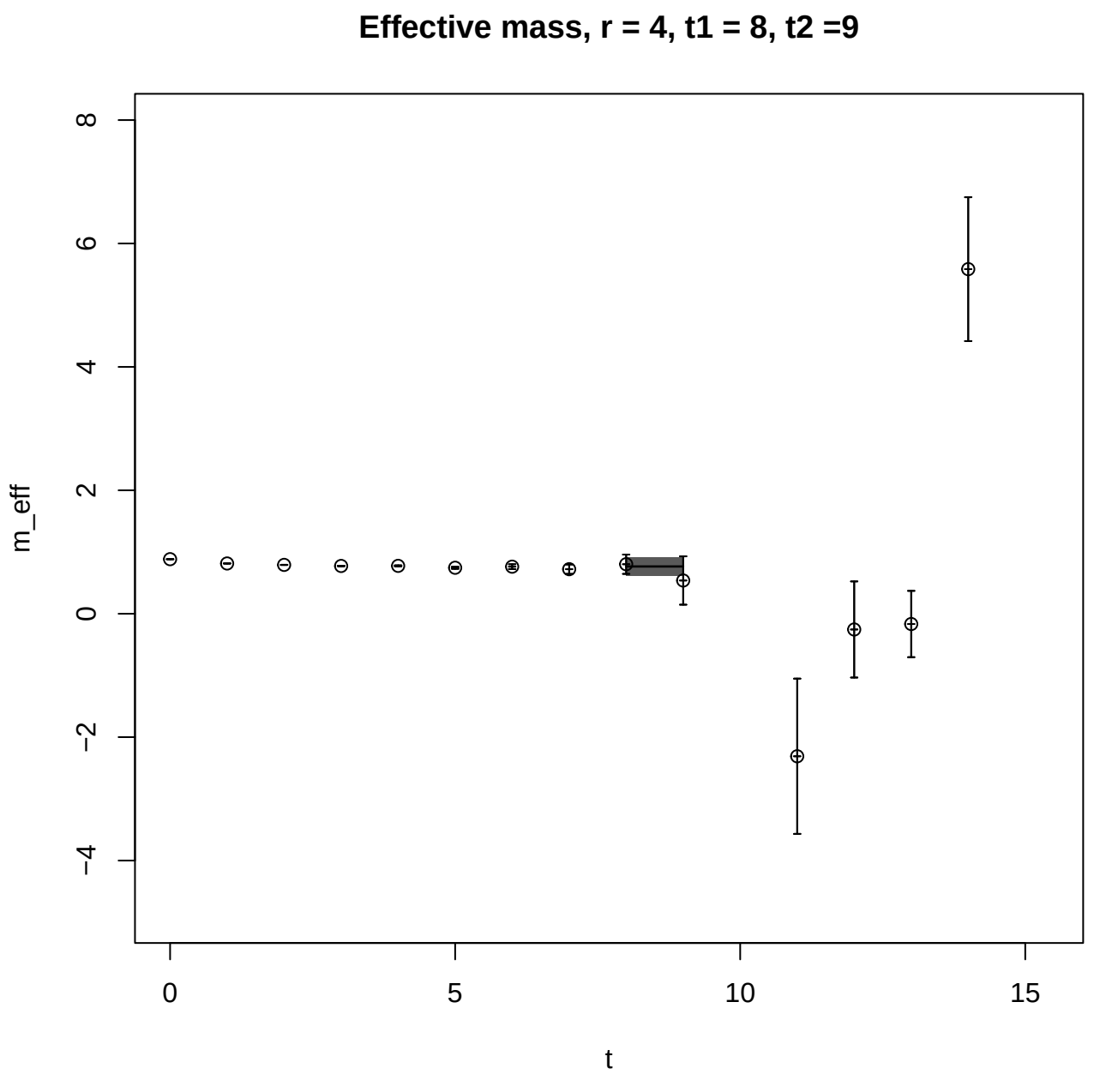

Effective mass, r = 4, t1 = 6, t2 =12
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

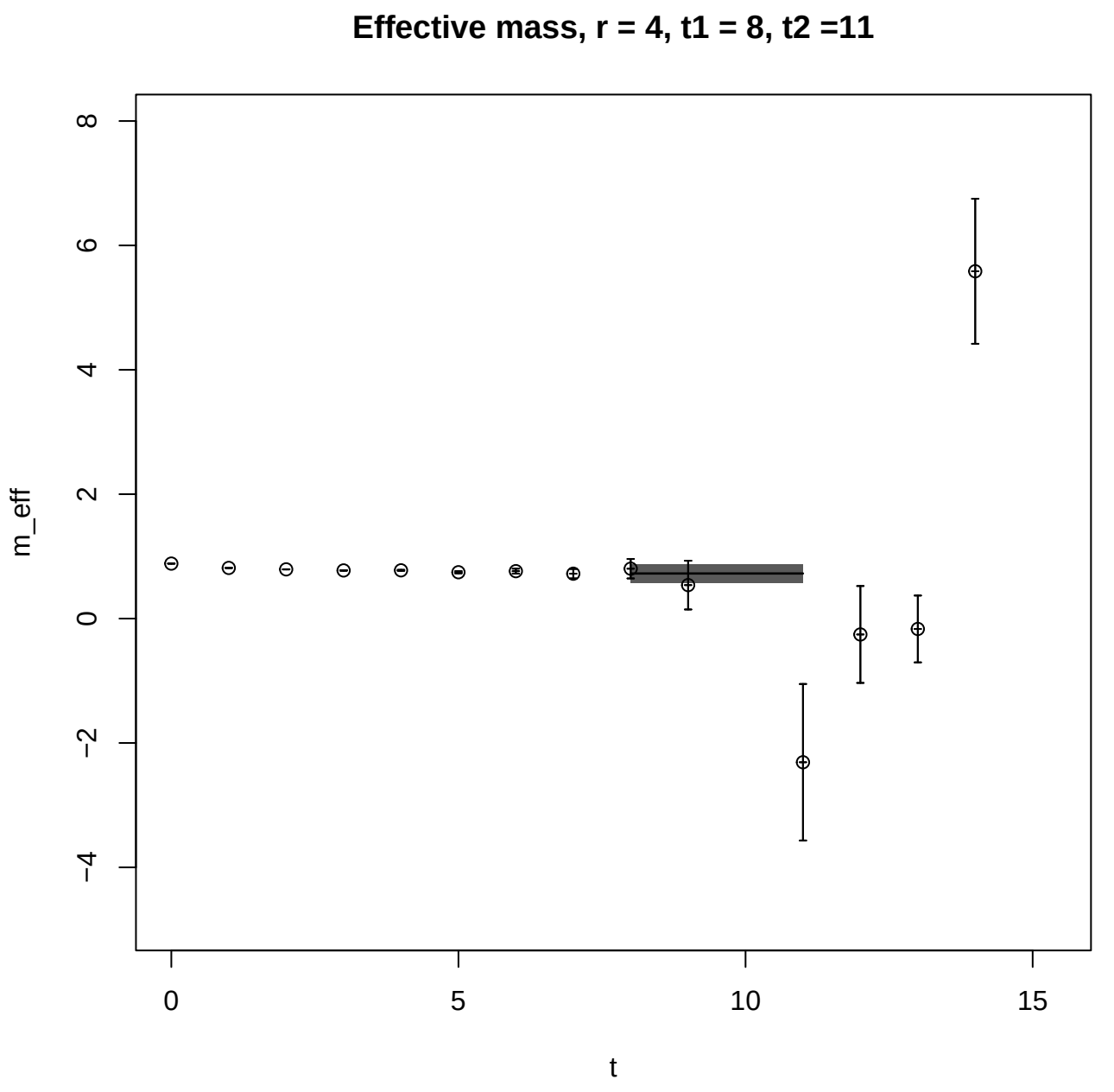

Effective mass, r = 4, t1 = 7, t2 =8
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

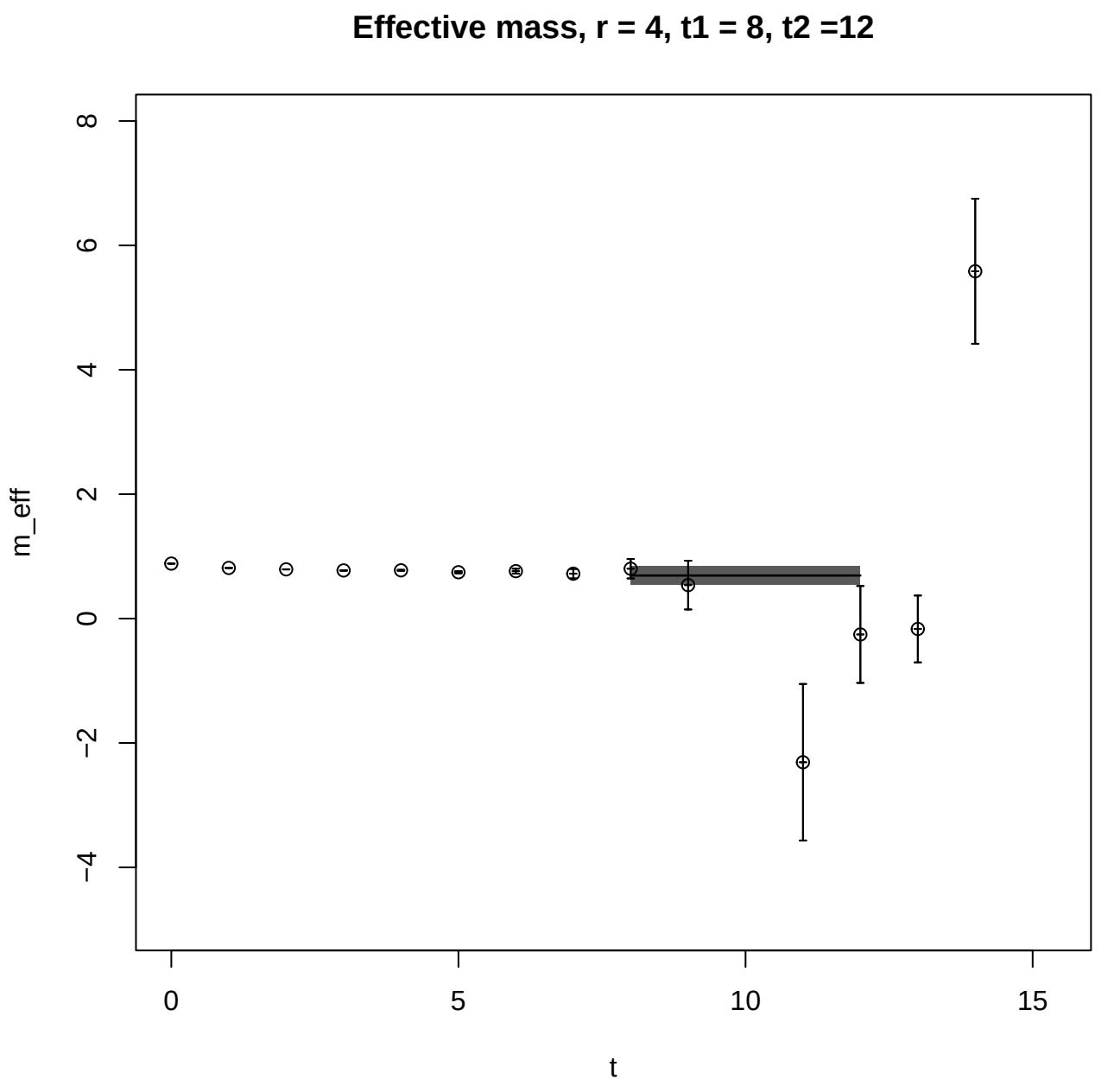

Effective mass, r = 4, t1 = 7, t2 =9
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

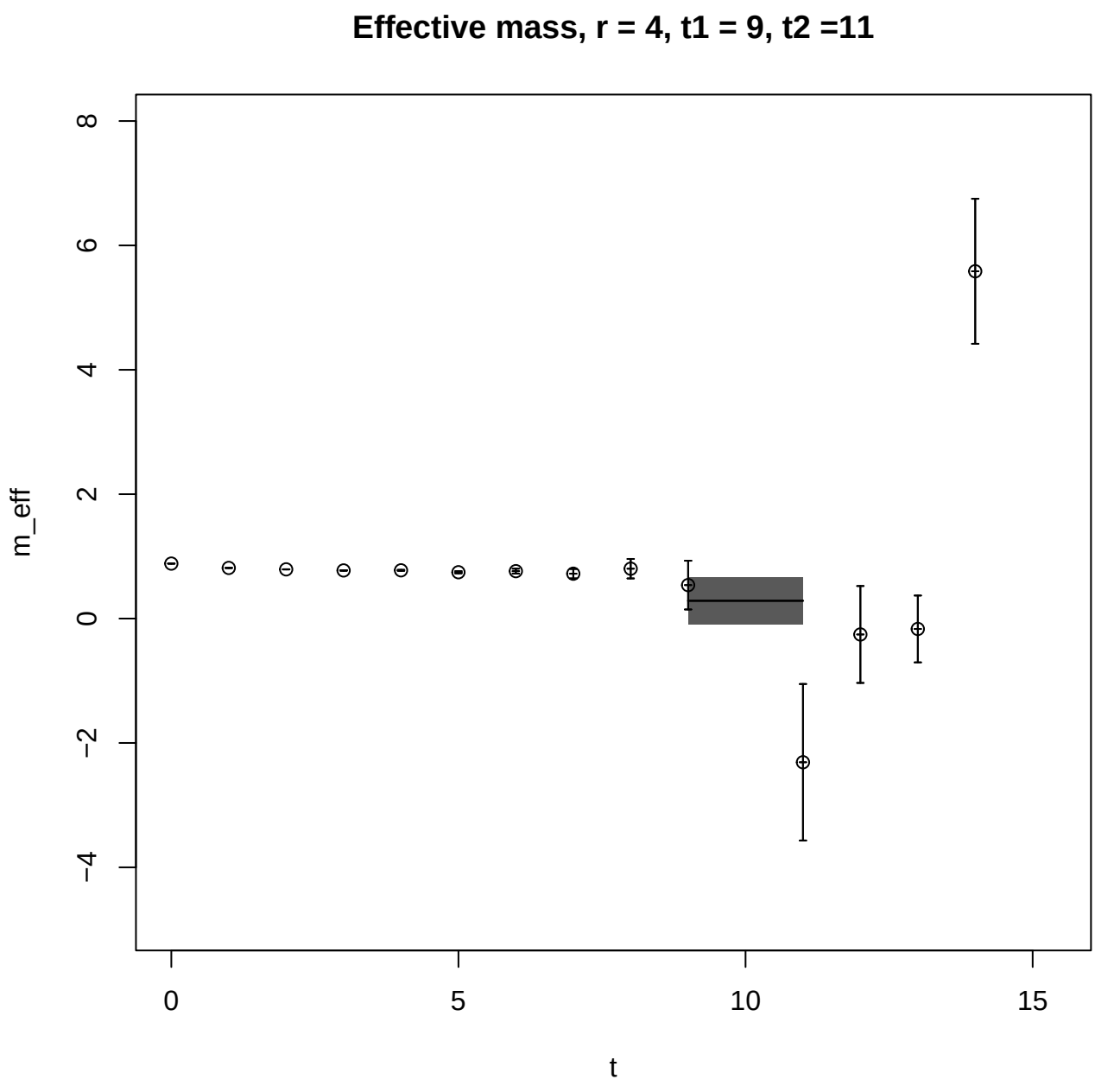

Effective mass, r = 4, t1 = 7, t2 =11
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

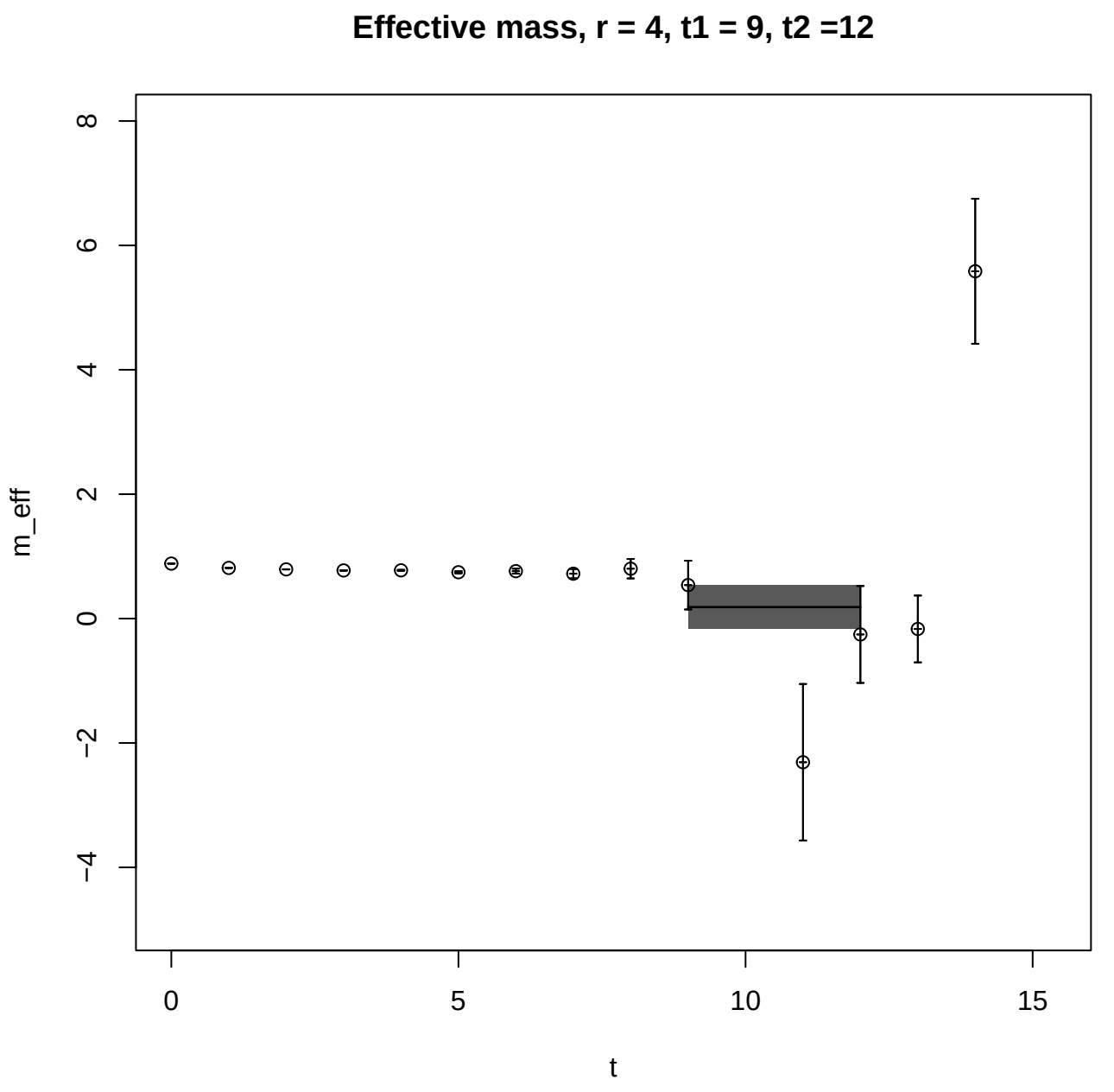

Effective mass, r = 4, t1 = 7, t2 =12
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

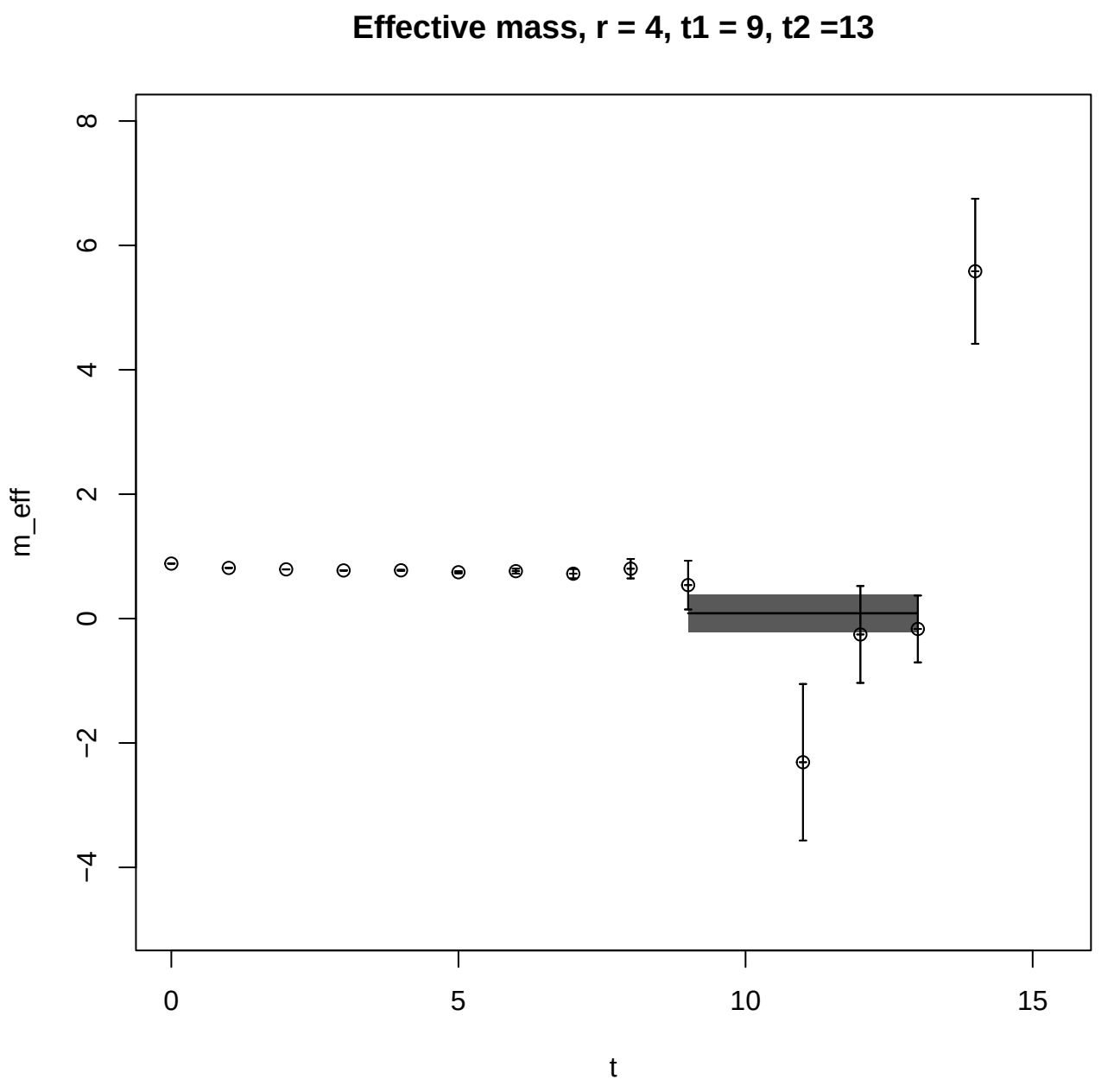

Effective mass, r = 4, t1 = 8, t2 =9
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

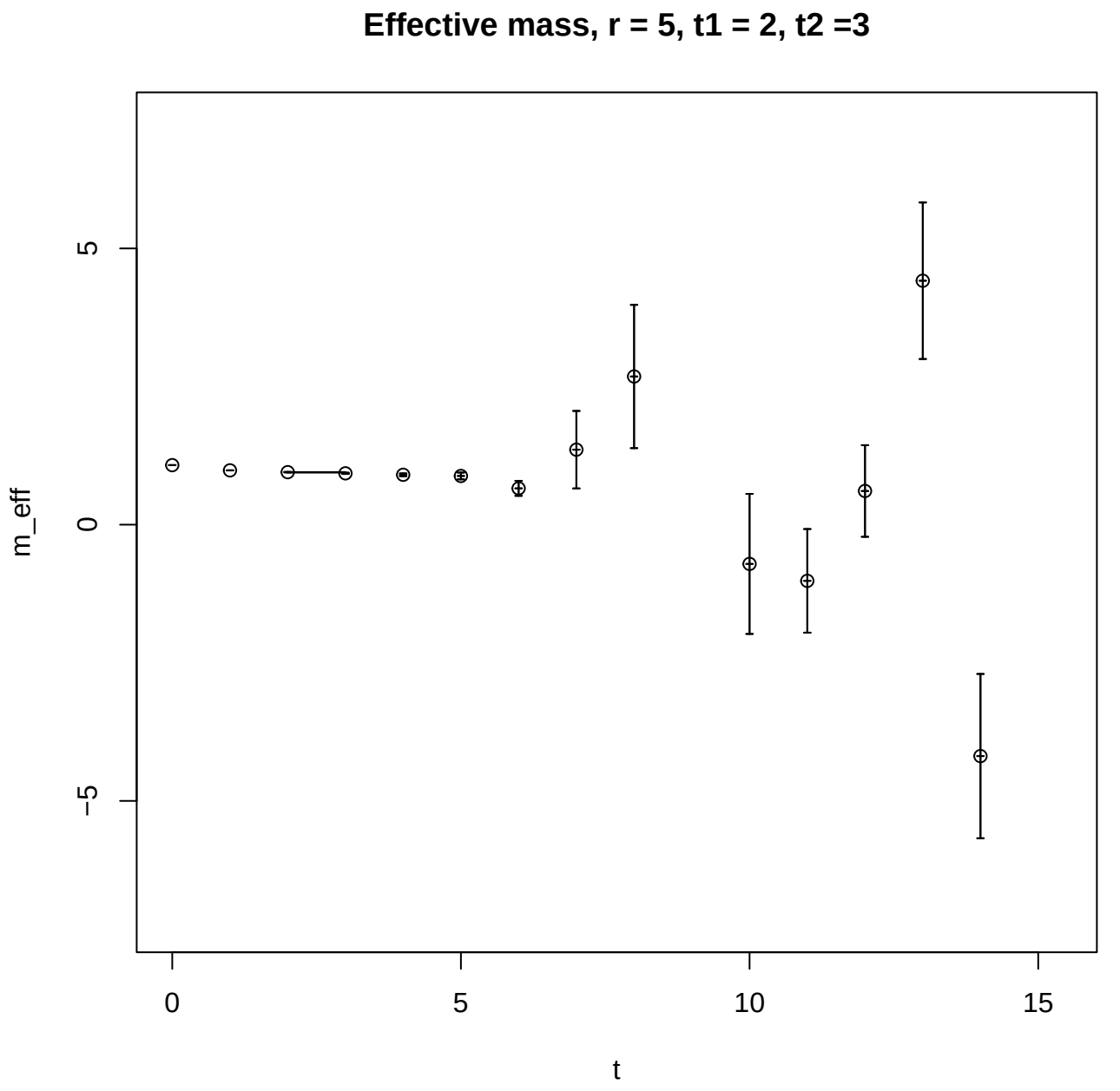

Effective mass, r = 4, t1 = 8, t2 =11
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

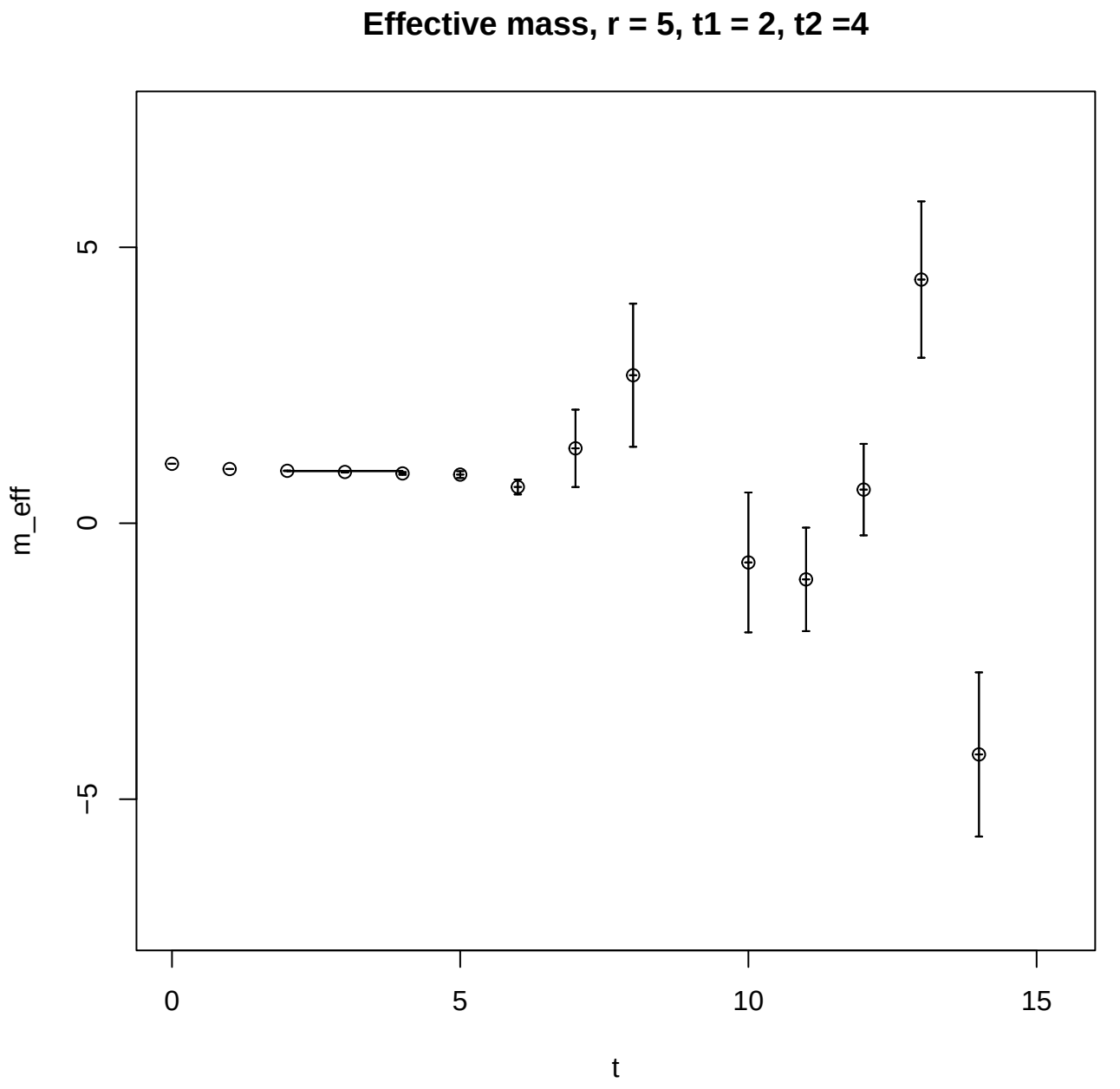

Effective mass, r = 4, t1 = 8, t2 =12
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

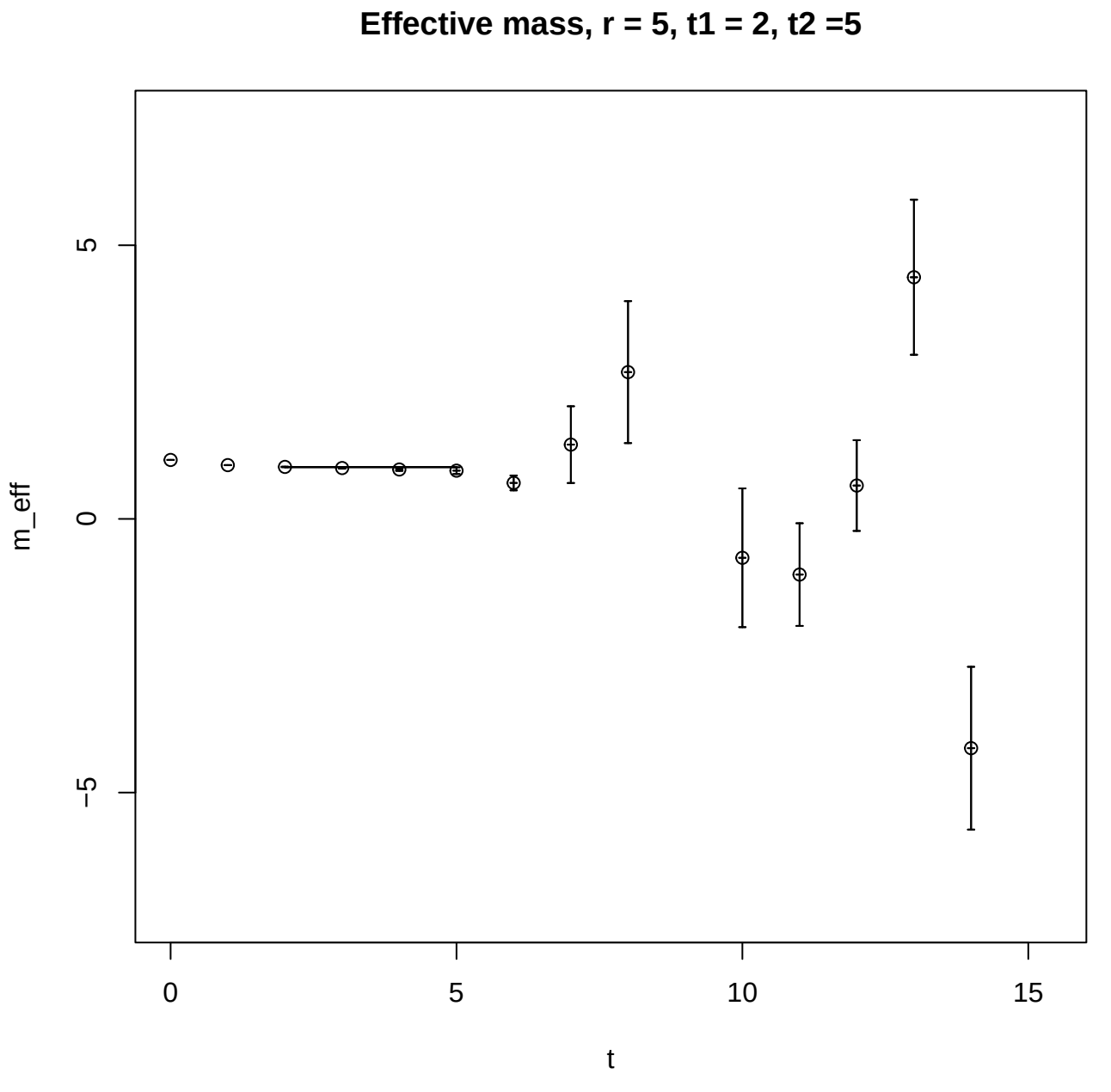

Effective mass, r = 4, t1 = 9, t2 =11
m_eff
8
6
4
2
0
−2
−4
0
5
10
15
t

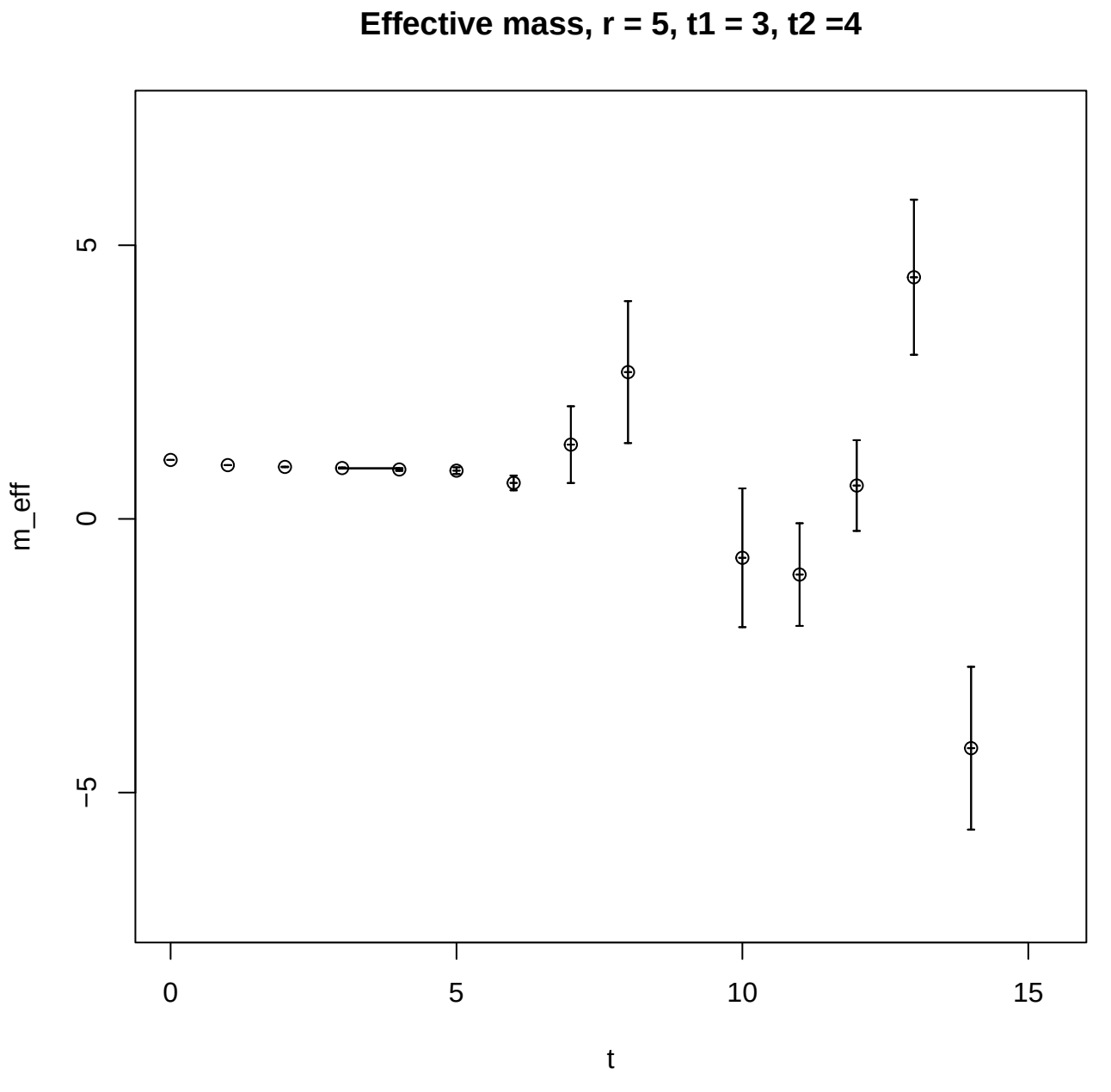

Effective mass, r = 4, t1 = 9, t2 =12
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

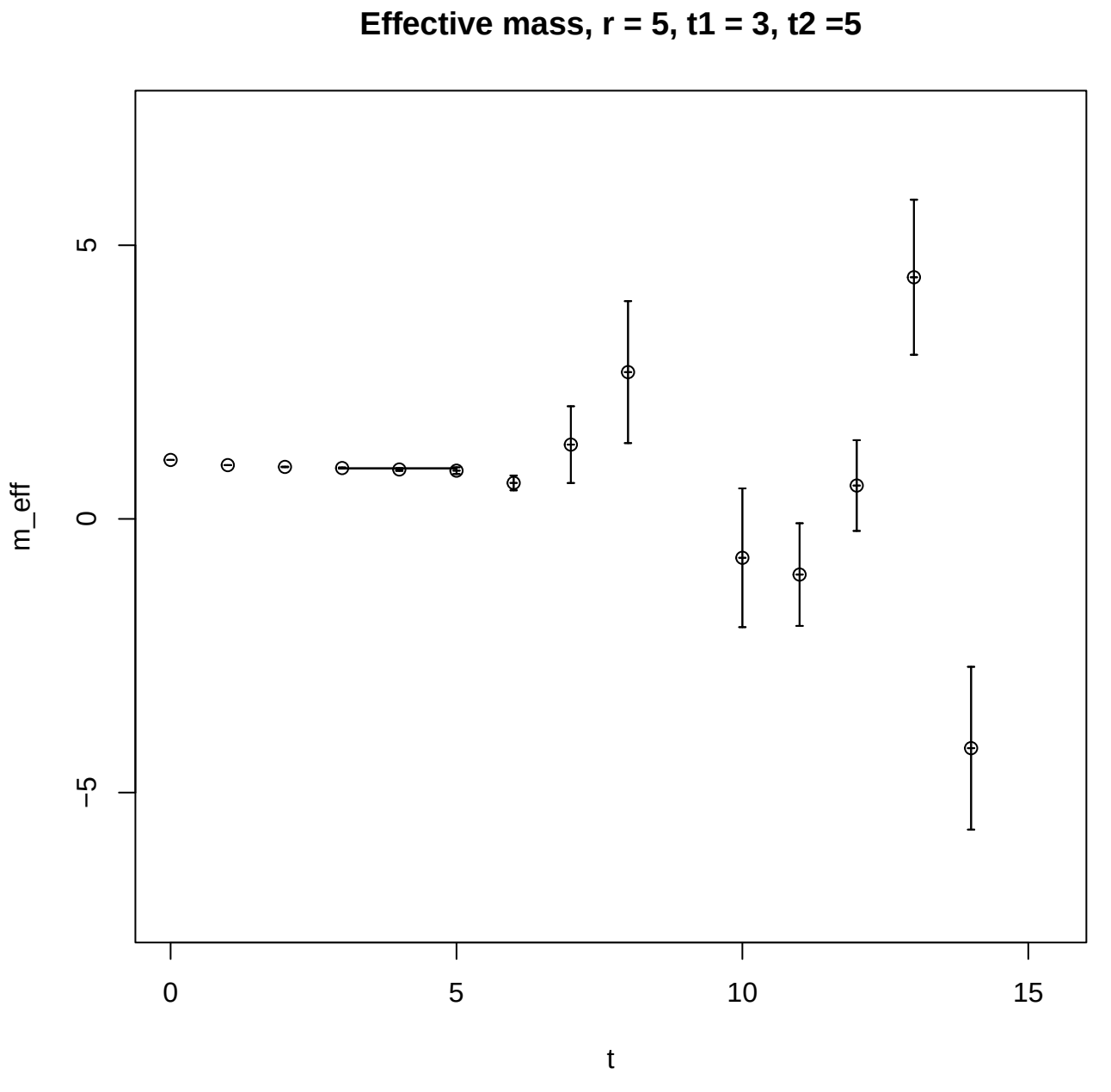

Effective mass, r = 4, t1 = 9, t2 =13
m_eff
t
8
6
4
2
0
−2
−4
0
5
10
15

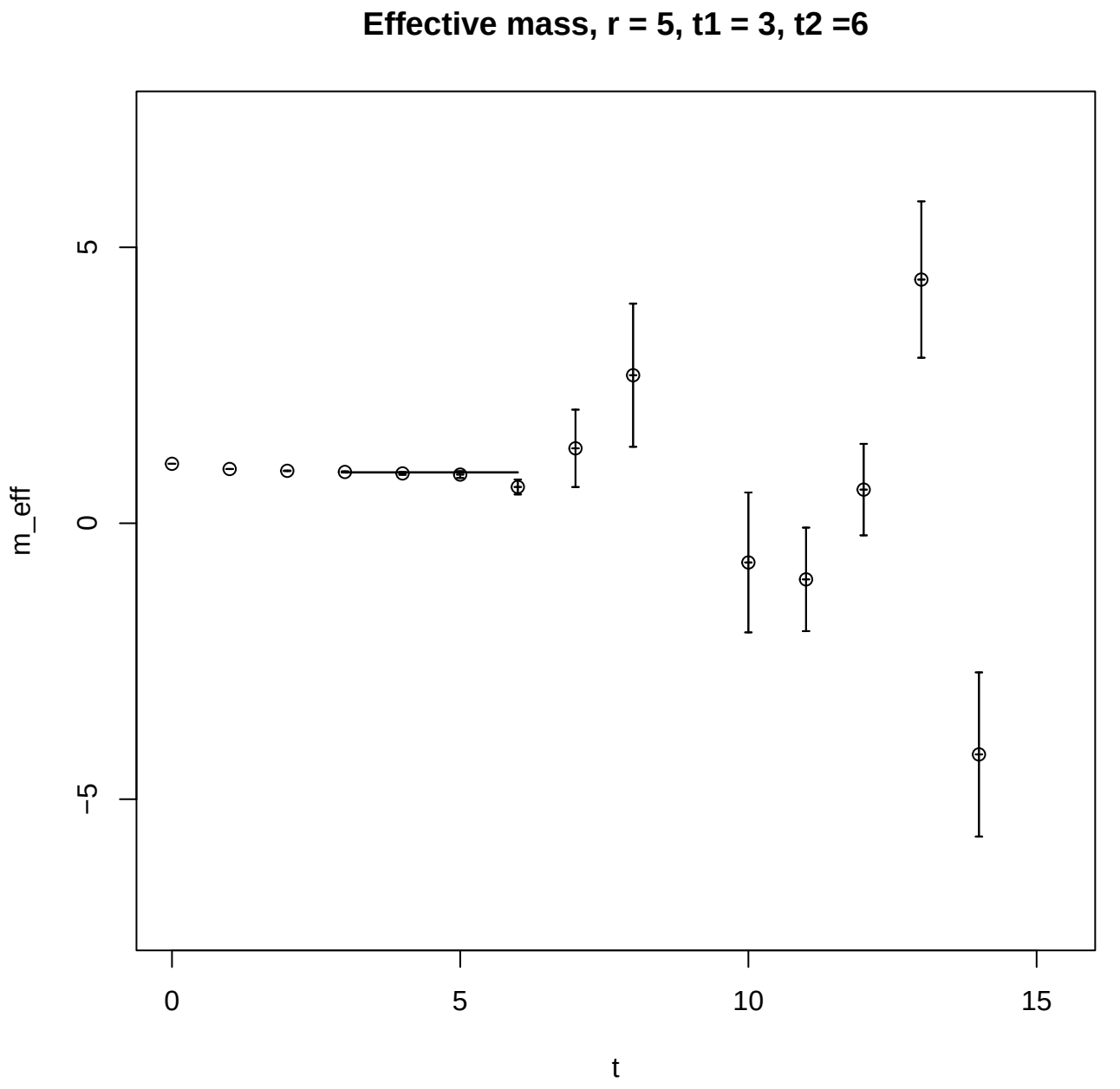

Effective mass, r = 5, t1 = 2, t2 =3
m_eff
5
0
−5
0
5
10
15
t

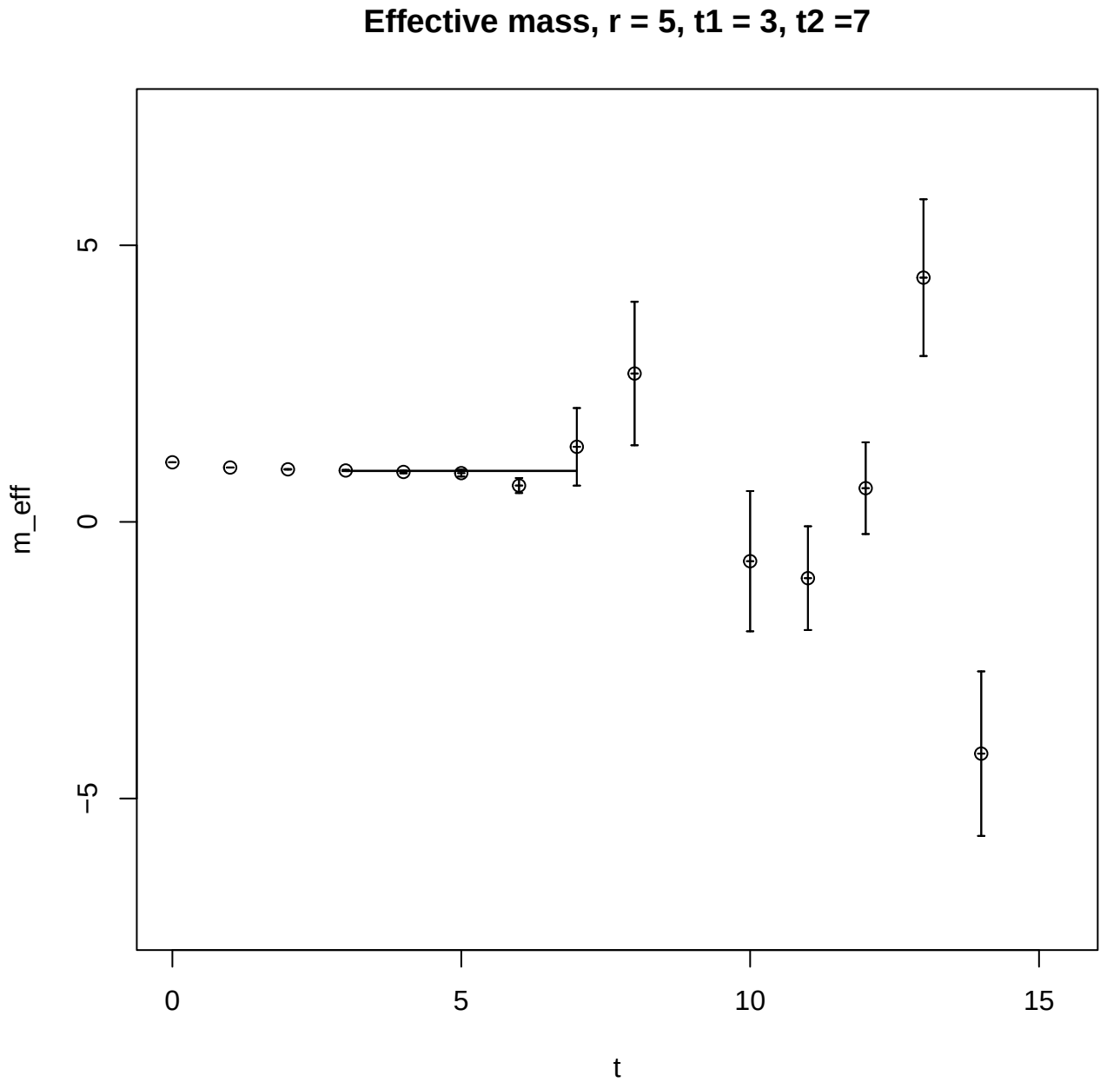

Effective mass, r = 5, t1 = 2, t2 =4
m_eff
5
0
−5
0
5
10
15
t

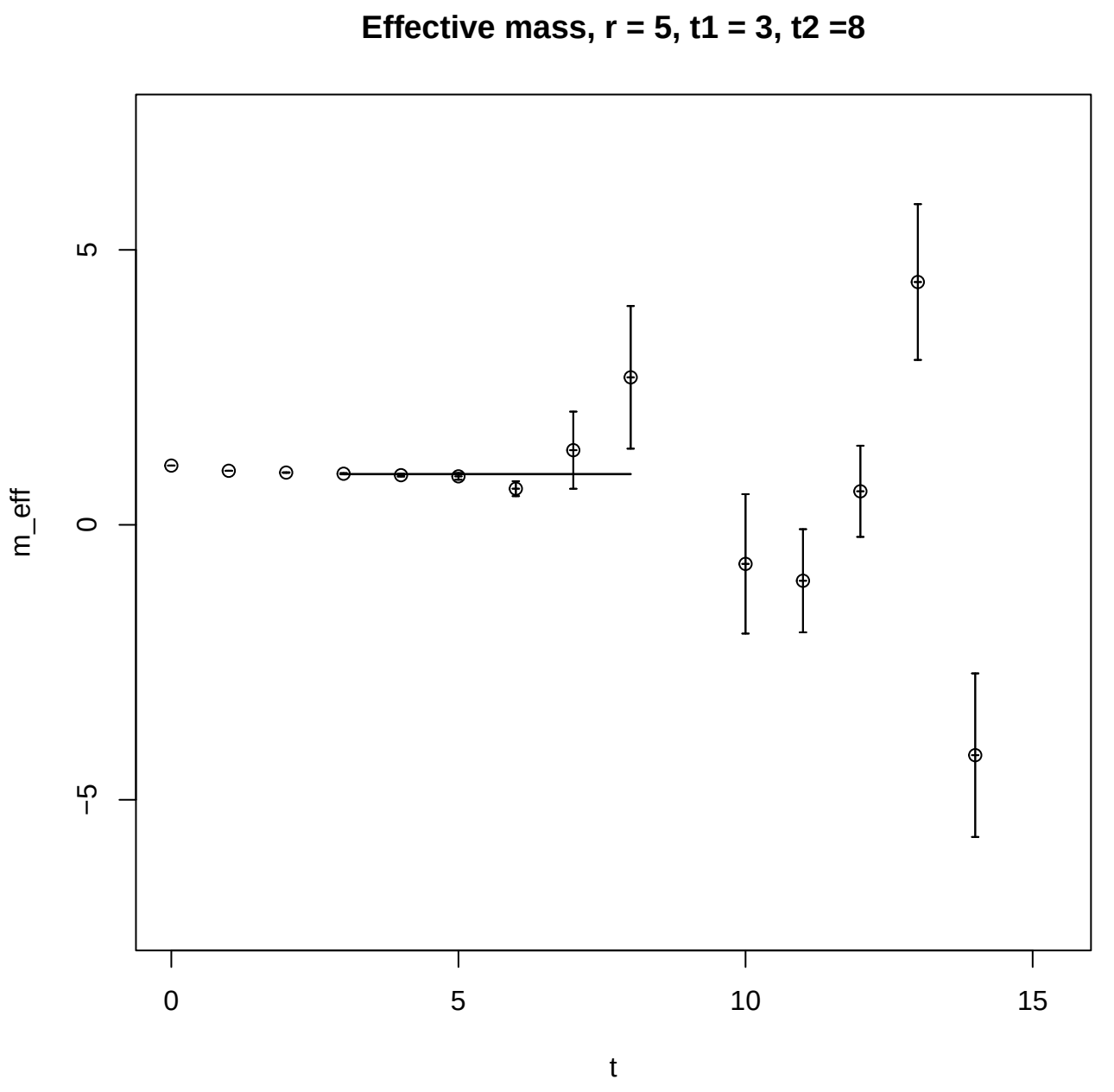

Effective mass, r = 5, t1 = 2, t2 =5
m_eff
5
0
−5
0
5
10
15
t

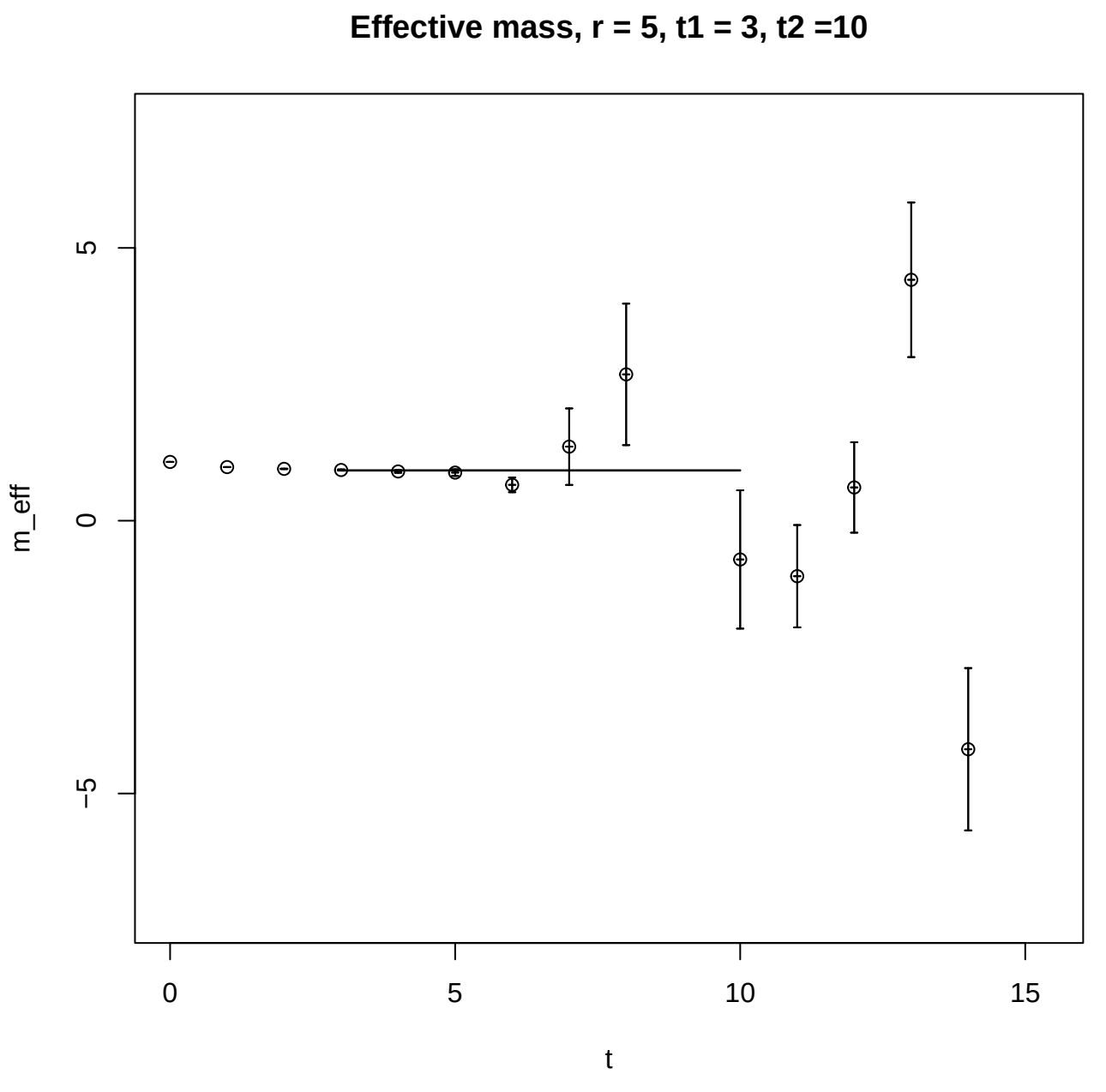

Effective mass, r = 5, t1 = 3, t2 =4
m_eff
5
0
−5
0
5
10
15
t

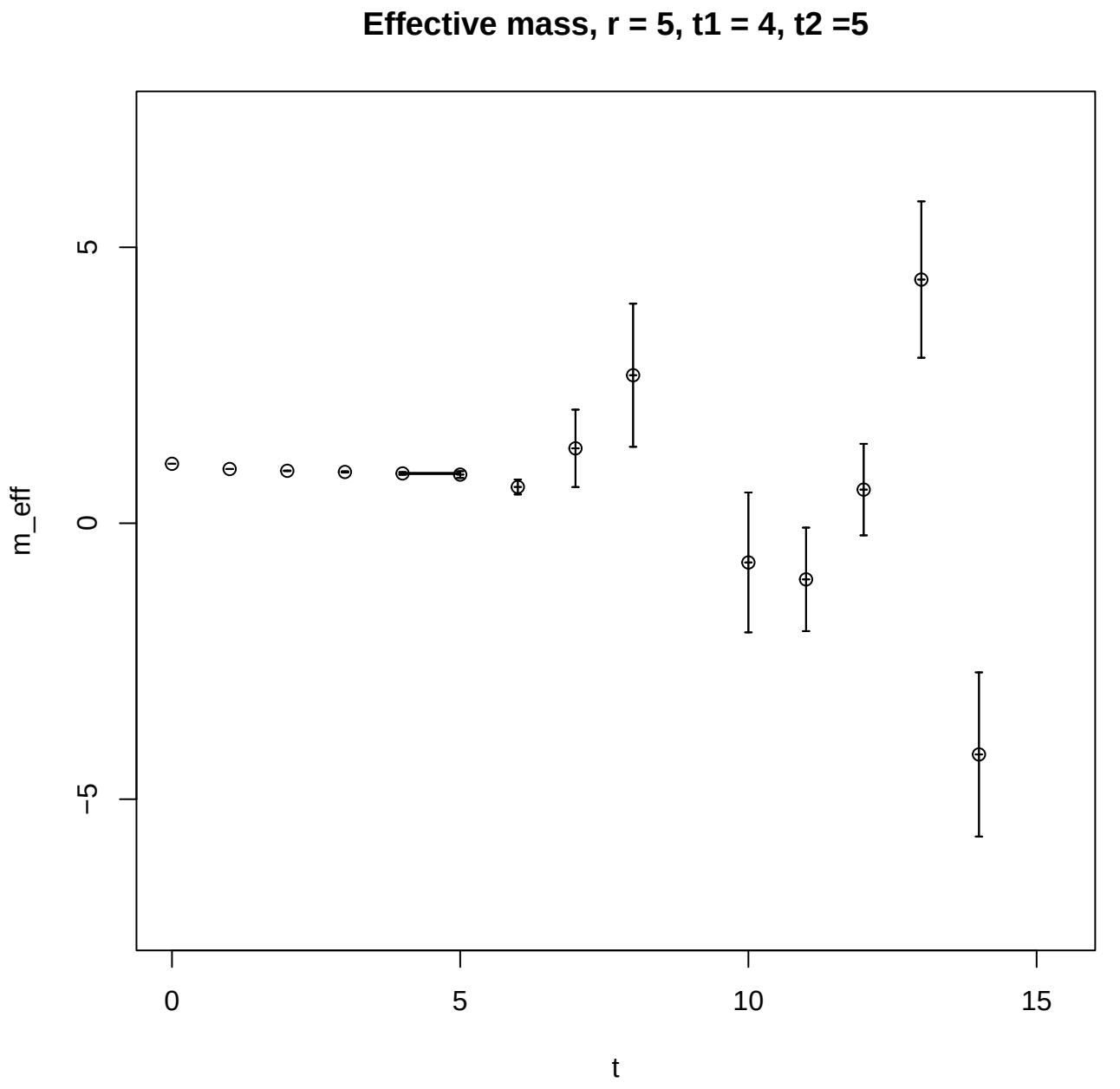

Effective mass, r = 5, t1 = 3, t2 =5
m_eff
5
0
−5
0
5
10
15
t

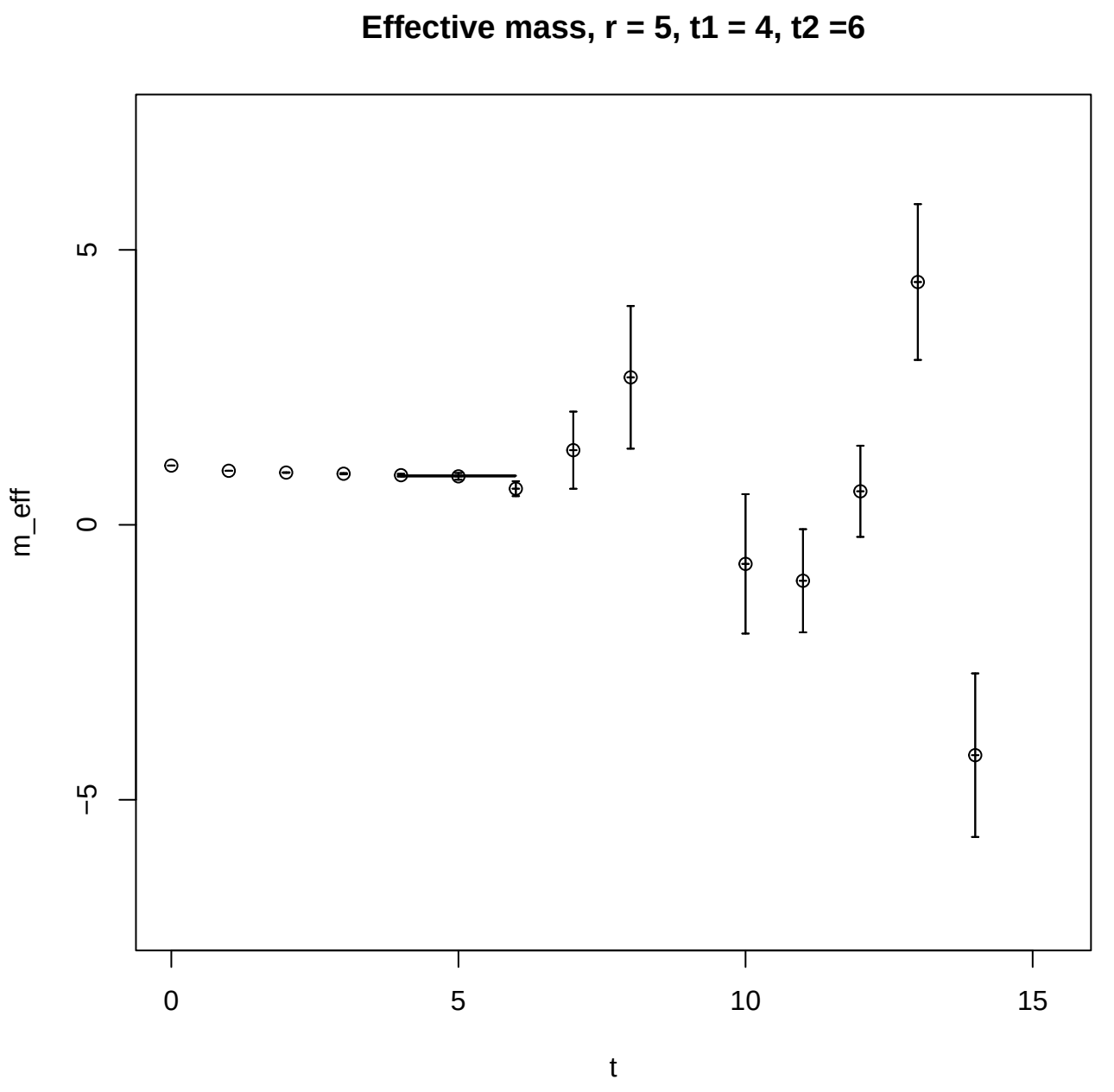

Effective mass, r = 5, t1 = 3, t2 =6
m_eff
5
0
−5
0
5
10
15
t

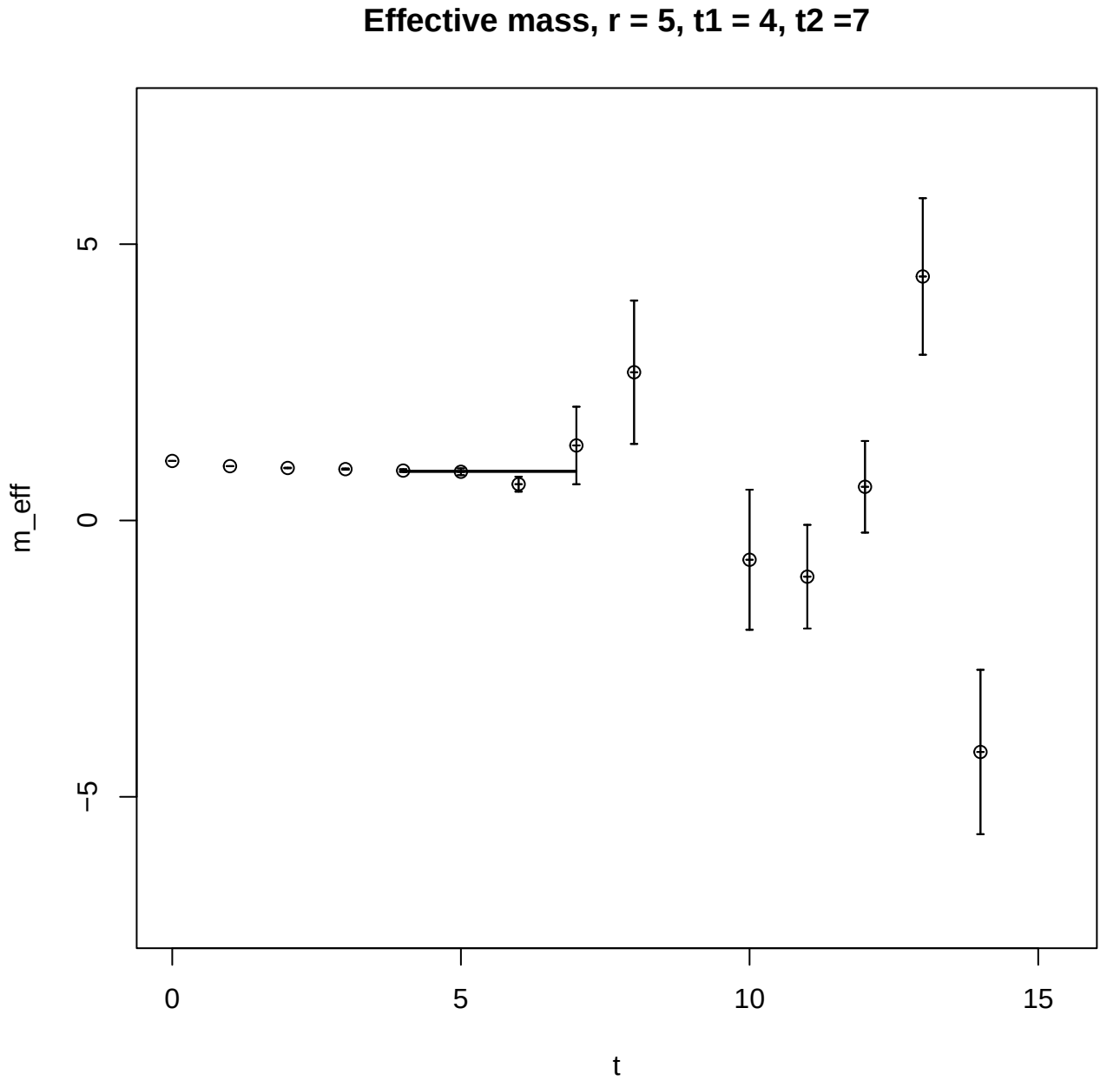

Effective mass, r = 5, t1 = 3, t2 =7
m_eff
5
0
−5
0
5
10
15
t

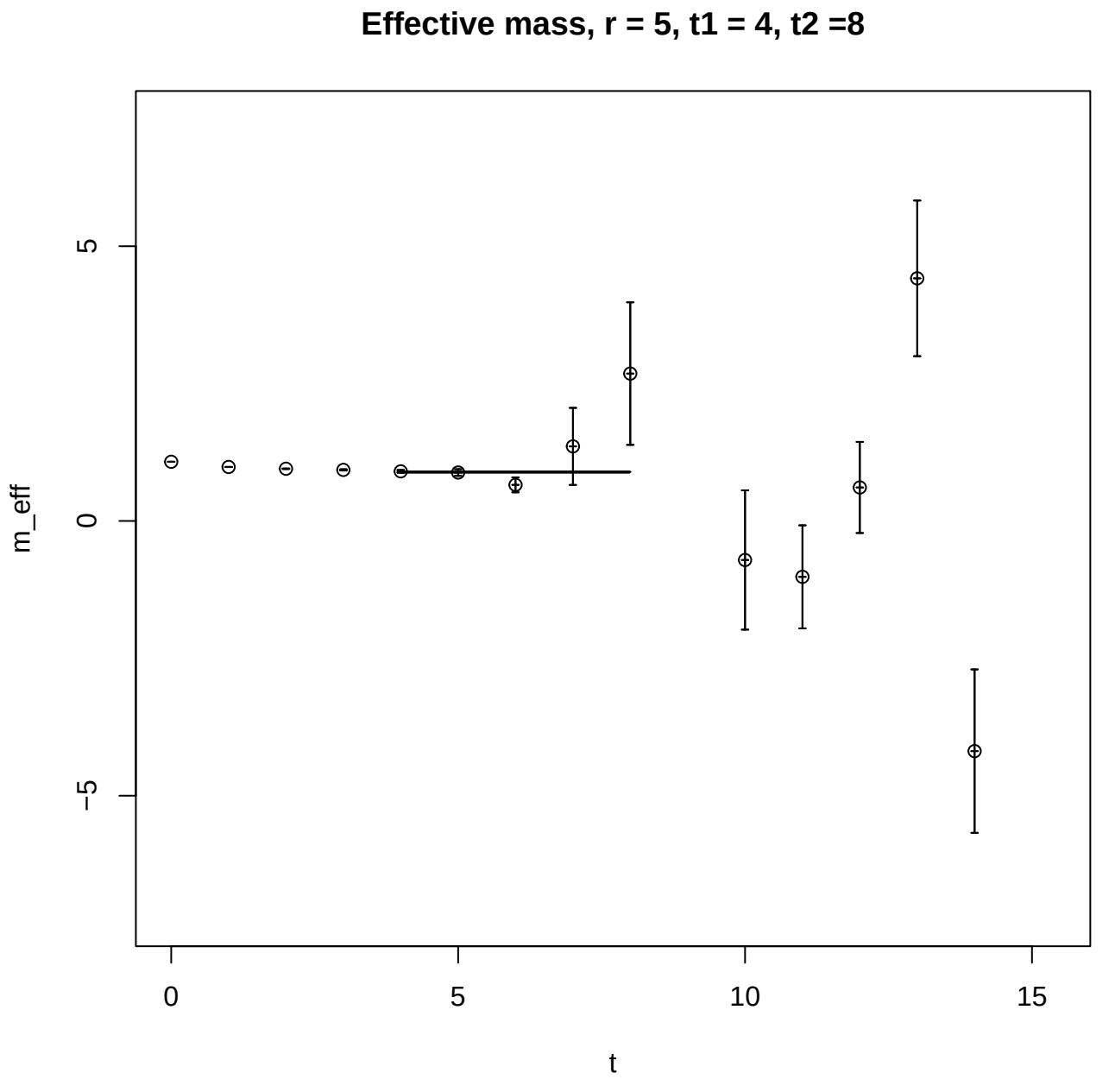

Effective mass, r = 5, t1 = 3, t2 =8
m_eff
5
0
−5
0
5
10
15
t

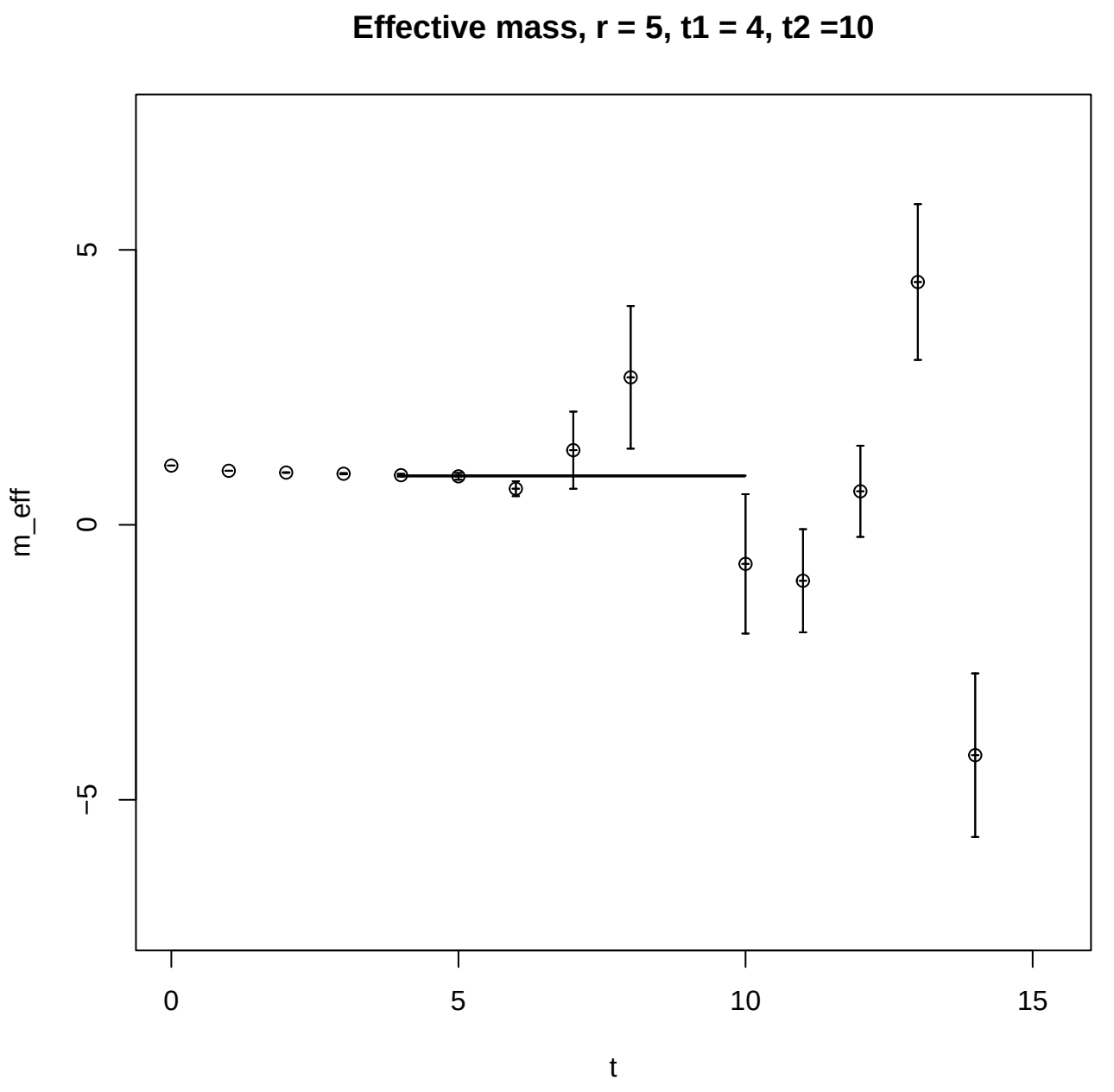

Effective mass, r = 5, t1 = 3, t2 =10
m_eff
5
0
−5
0
5
10
15
t

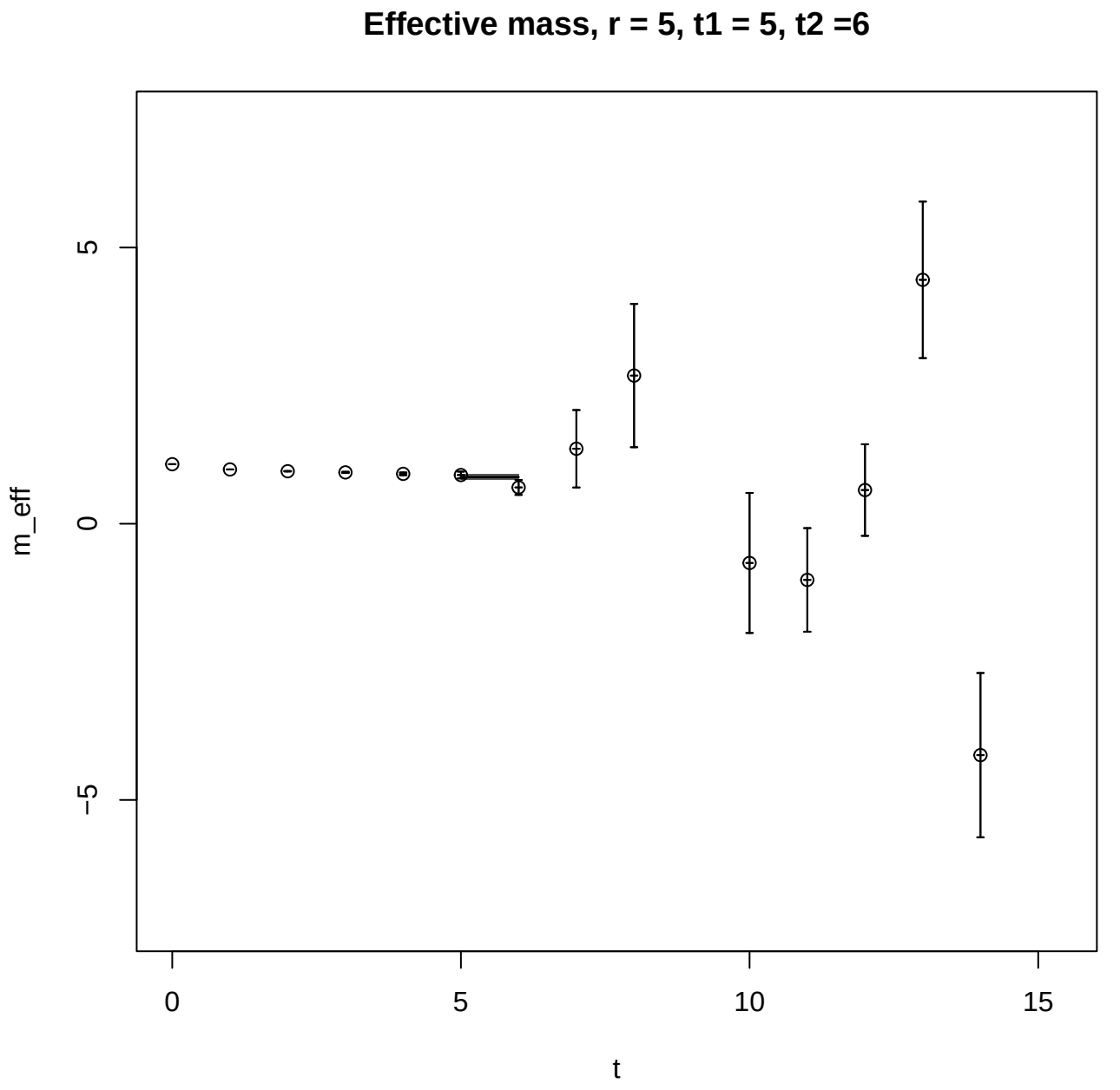

Effective mass, r = 5, t1 = 4, t2 =5
m_eff
5
0
−5
0
5
10
15
t

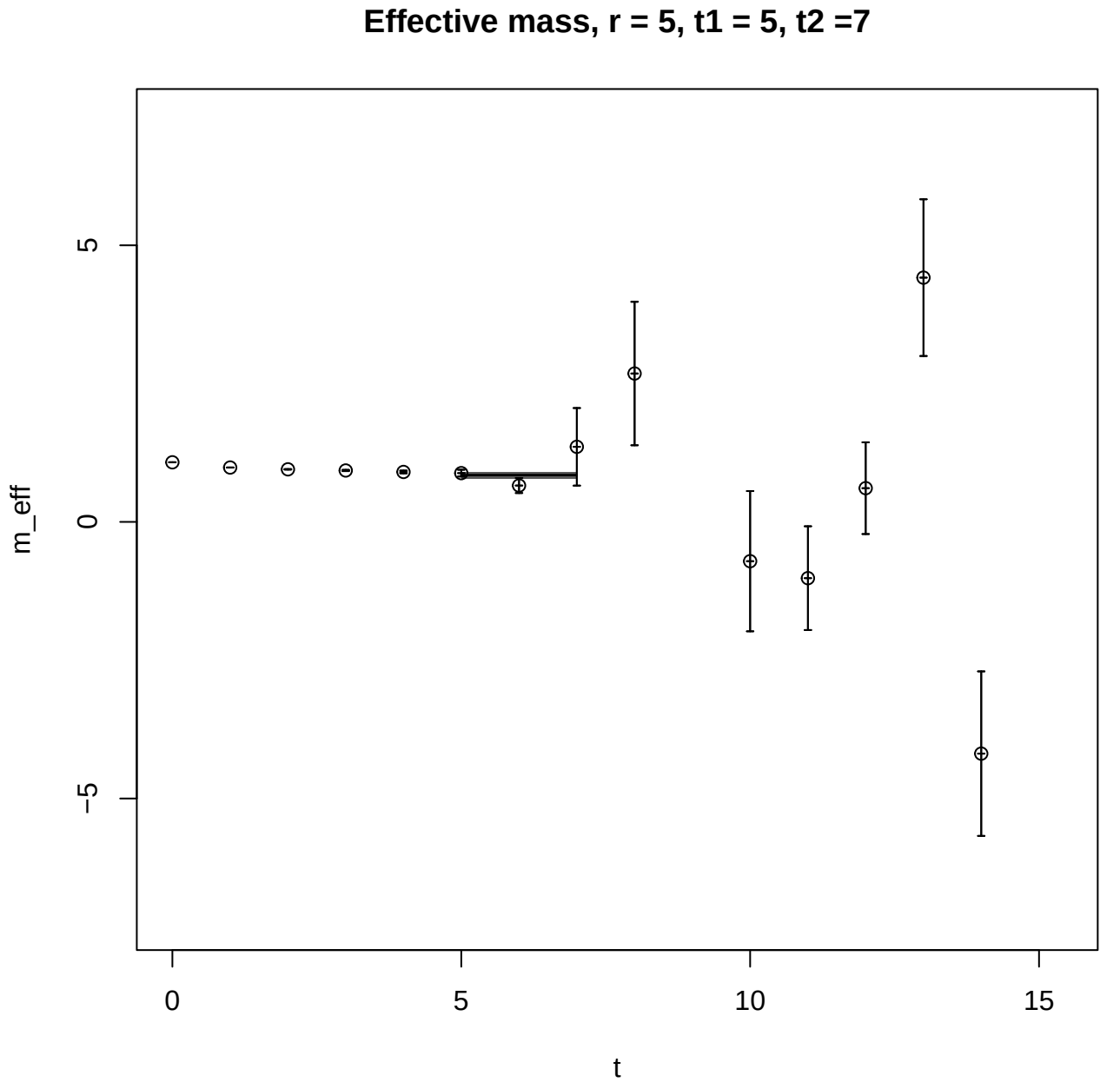

Effective mass, r = 5, t1 = 4, t2 =6
m_eff
5
0
−5
0
5
10
15
t

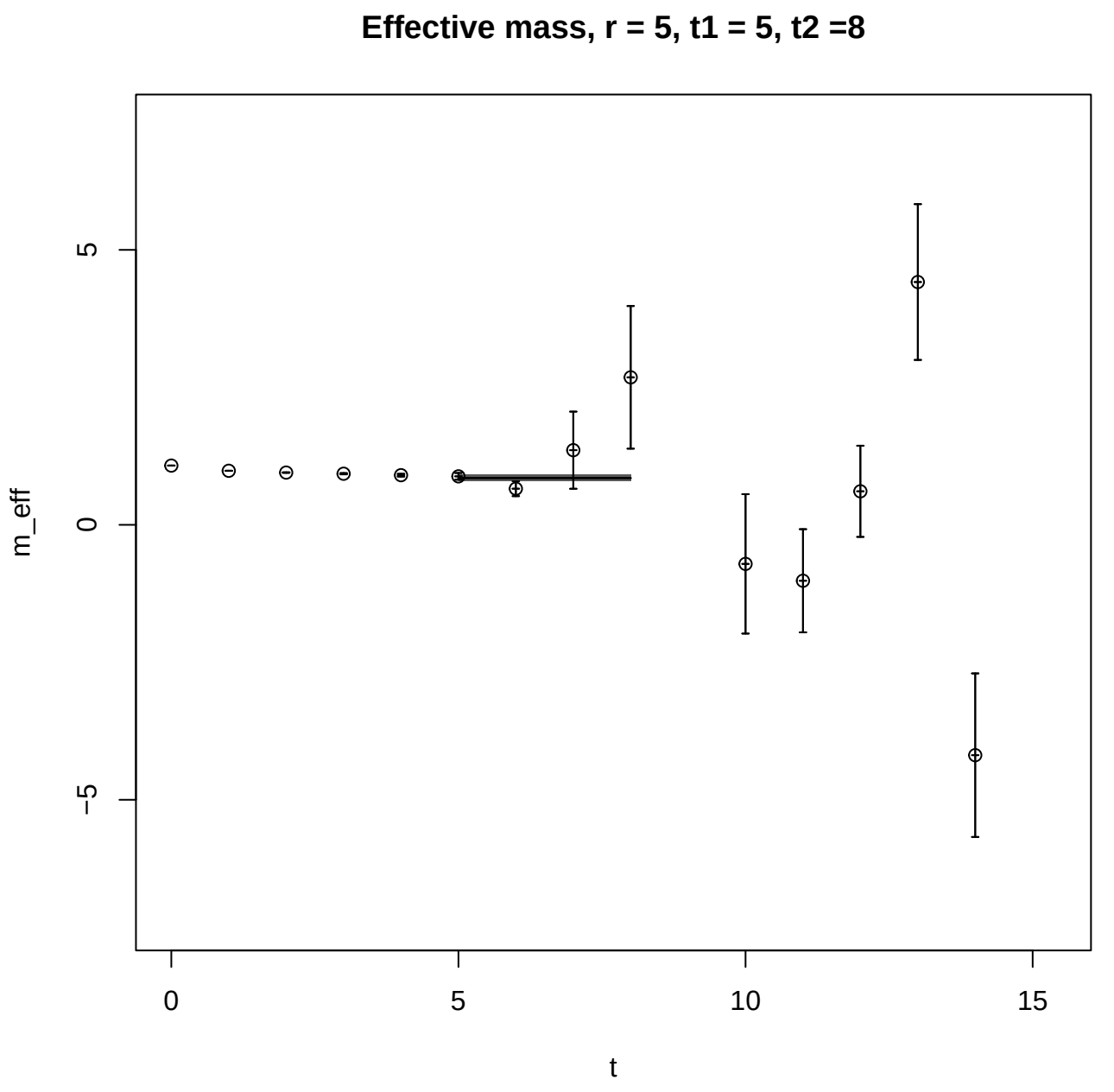

Effective mass, r = 5, t1 = 4, t2 =7
m_eff
5
0
−5
0
5
10
15
t

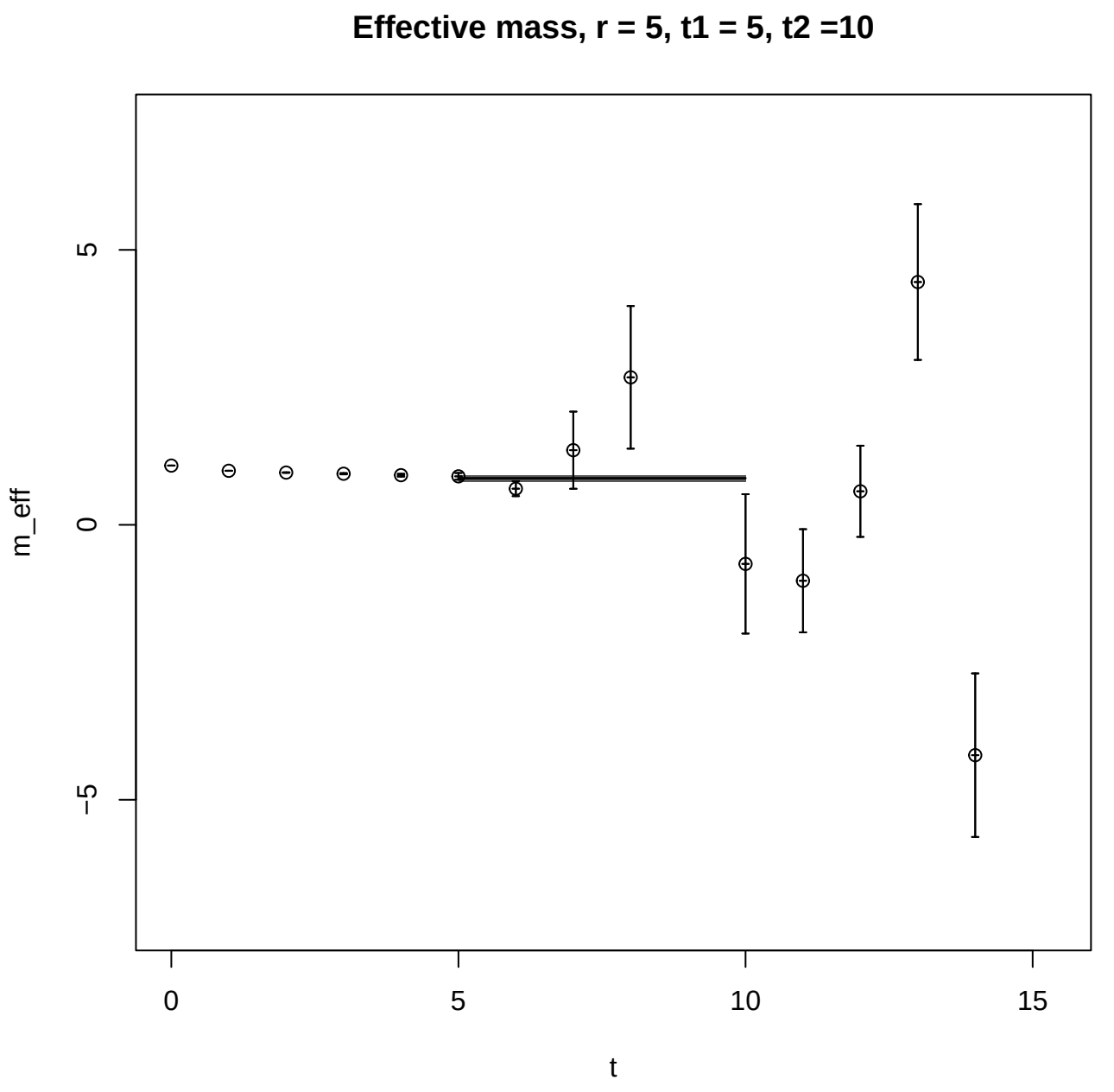

Effective mass, r = 5, t1 = 4, t2 =8
m_eff
5
0
−5
0
5
10
15
t

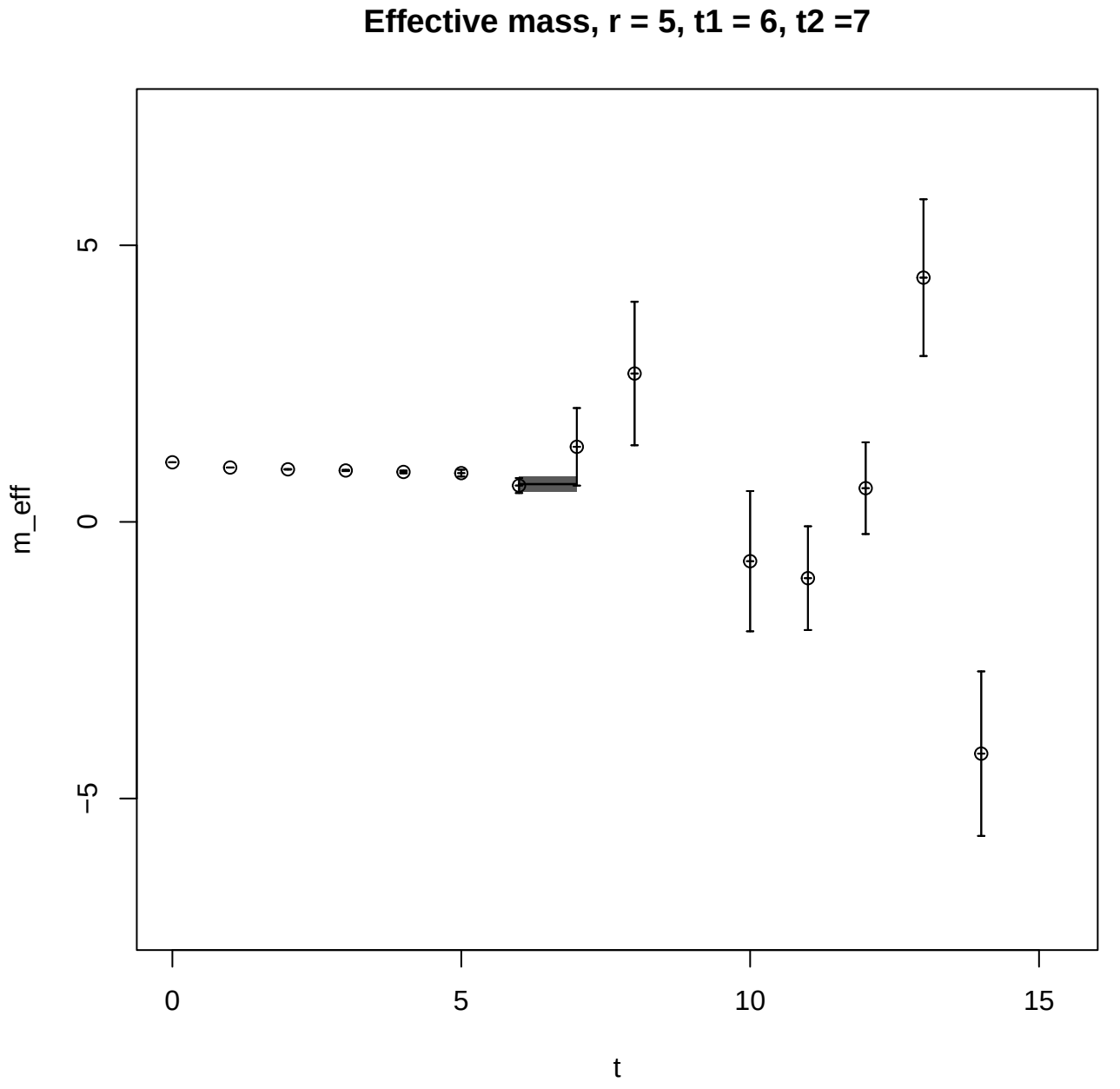

Effective mass, r = 5, t1 = 4, t2 =10
m_eff
5
0
−5
0
5
10
15
t

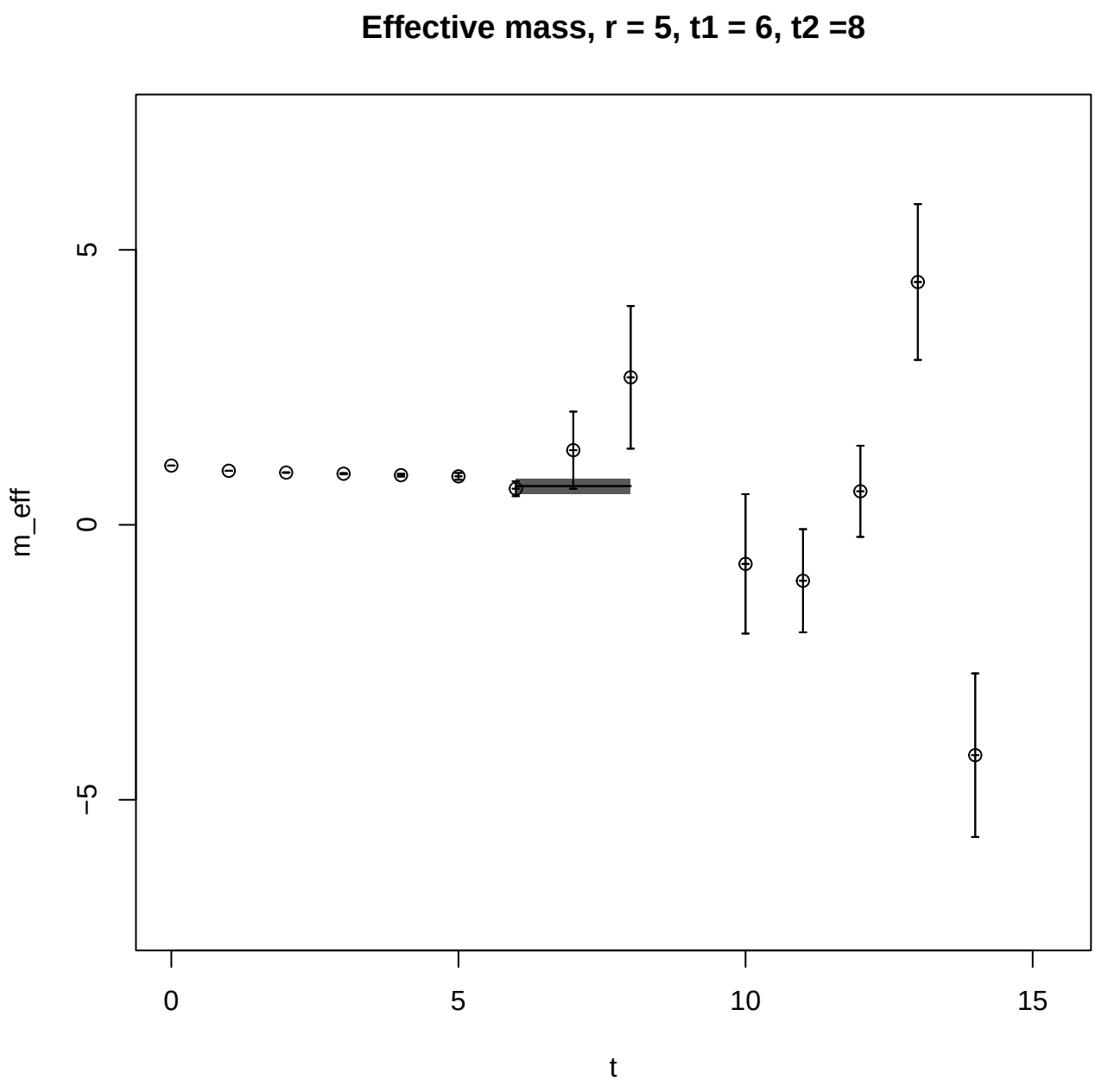

Effective mass, r = 5, t1 = 5, t2 =6
m_eff
5
0
−5
0
5
10
15
t

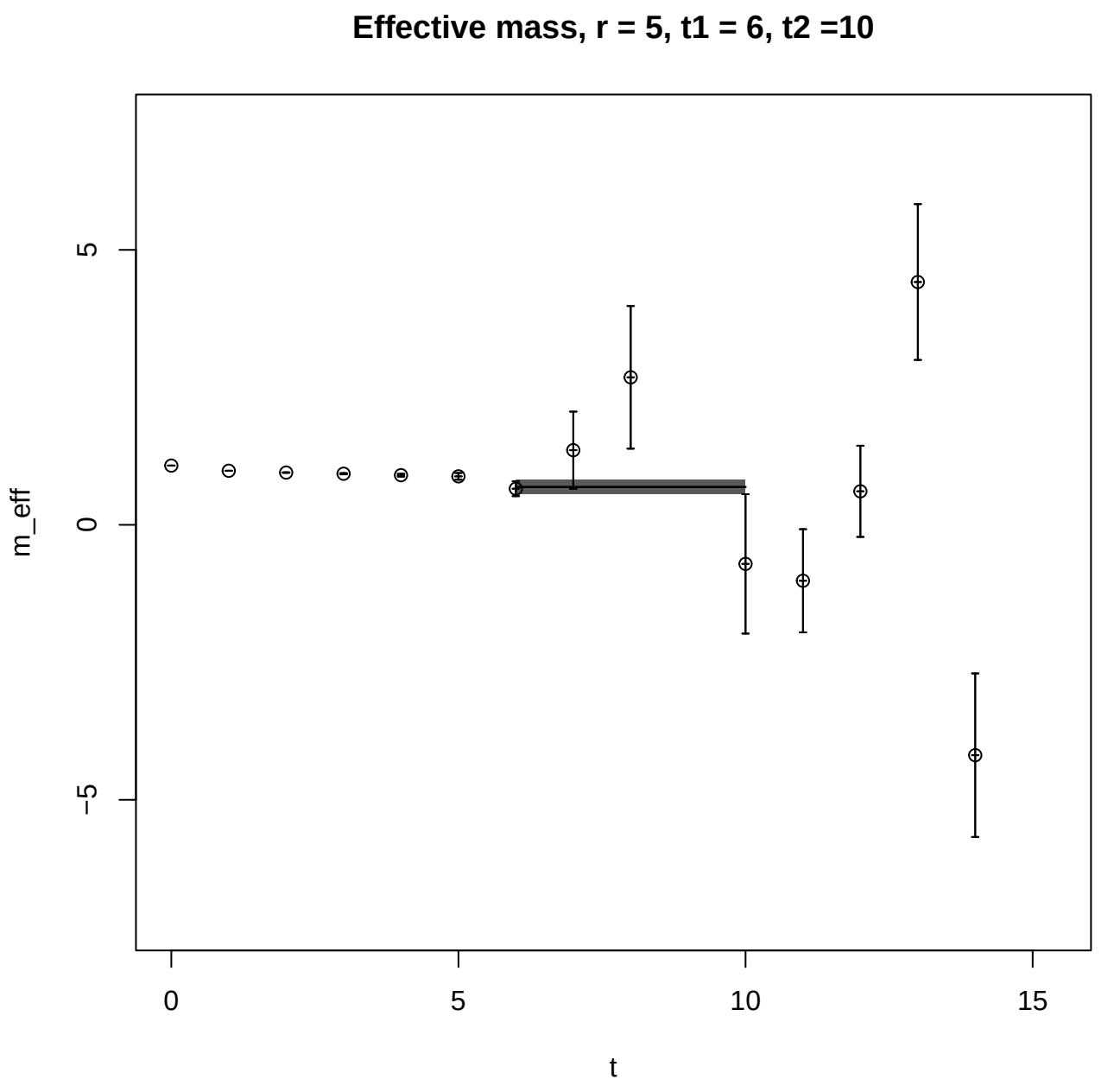

Effective mass, r = 5, t1 = 5, t2 =7
m_eff
5
0
−5
0
5
10
15
t

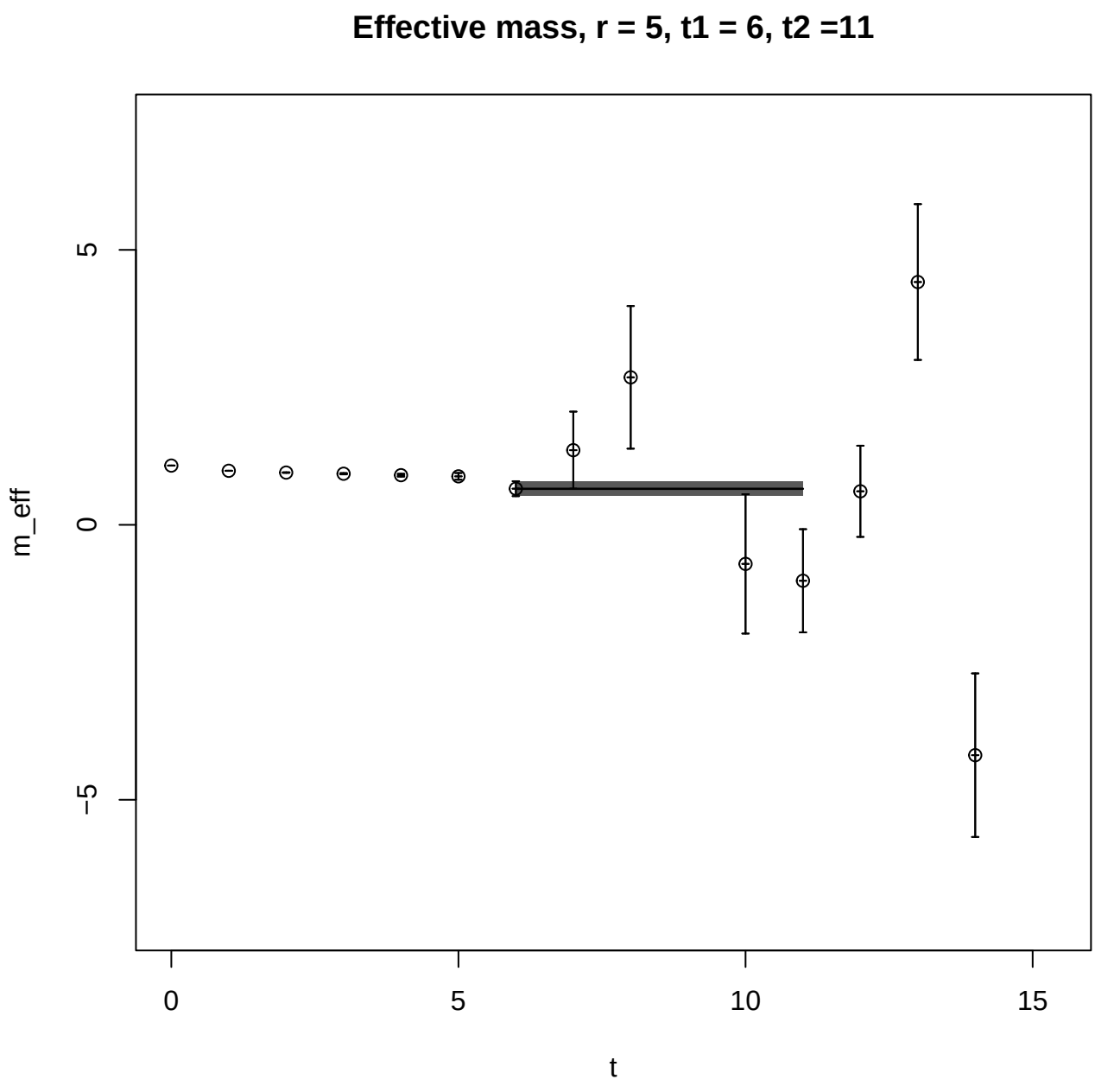

Effective mass, r = 5, t1 = 5, t2 =8
m_eff
5
0
−5
0
5
10
15
t

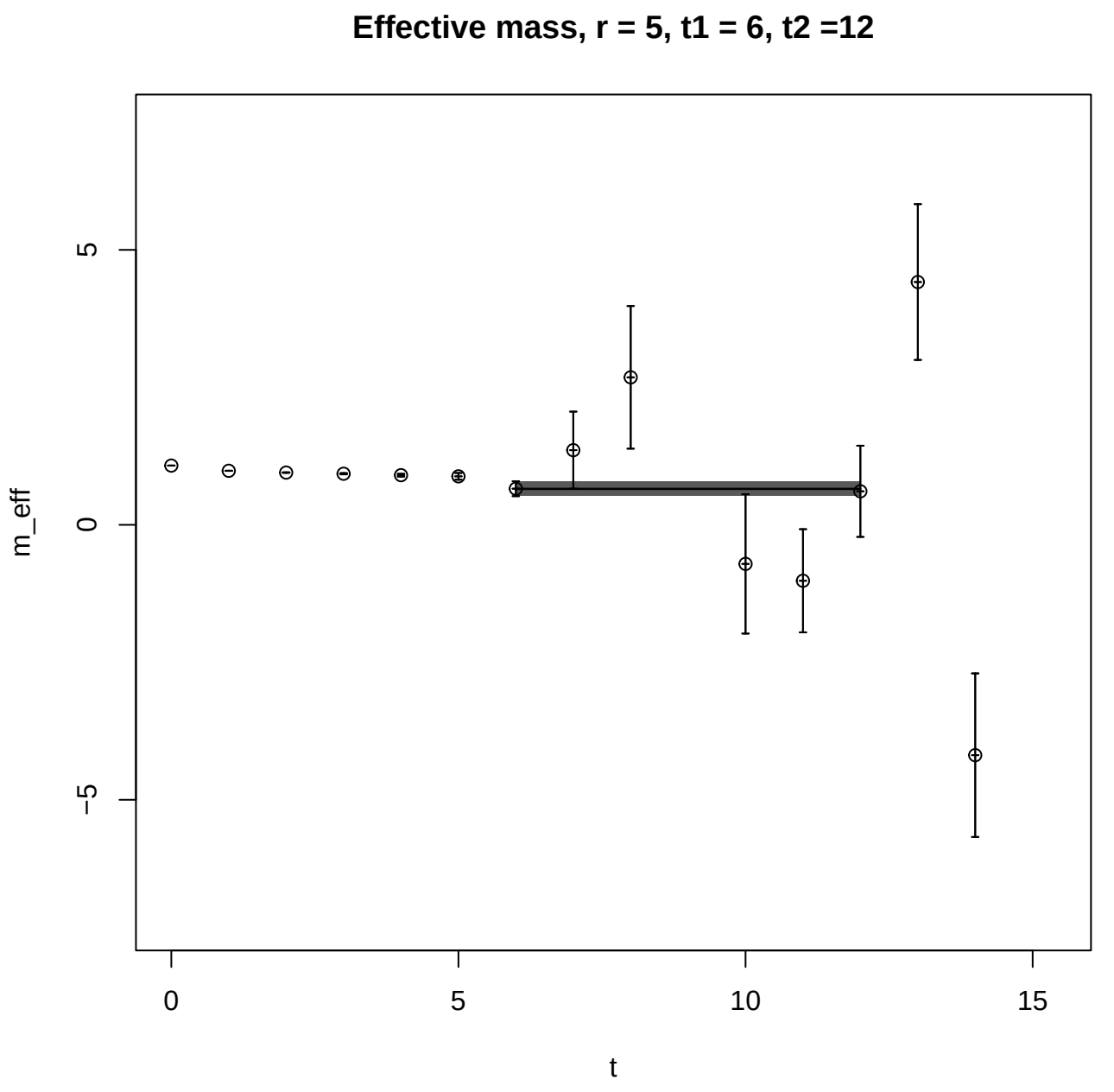

Effective mass, r = 5, t1 = 5, t2 =10
m_eff
5
0
−5
0
5
10
15
t

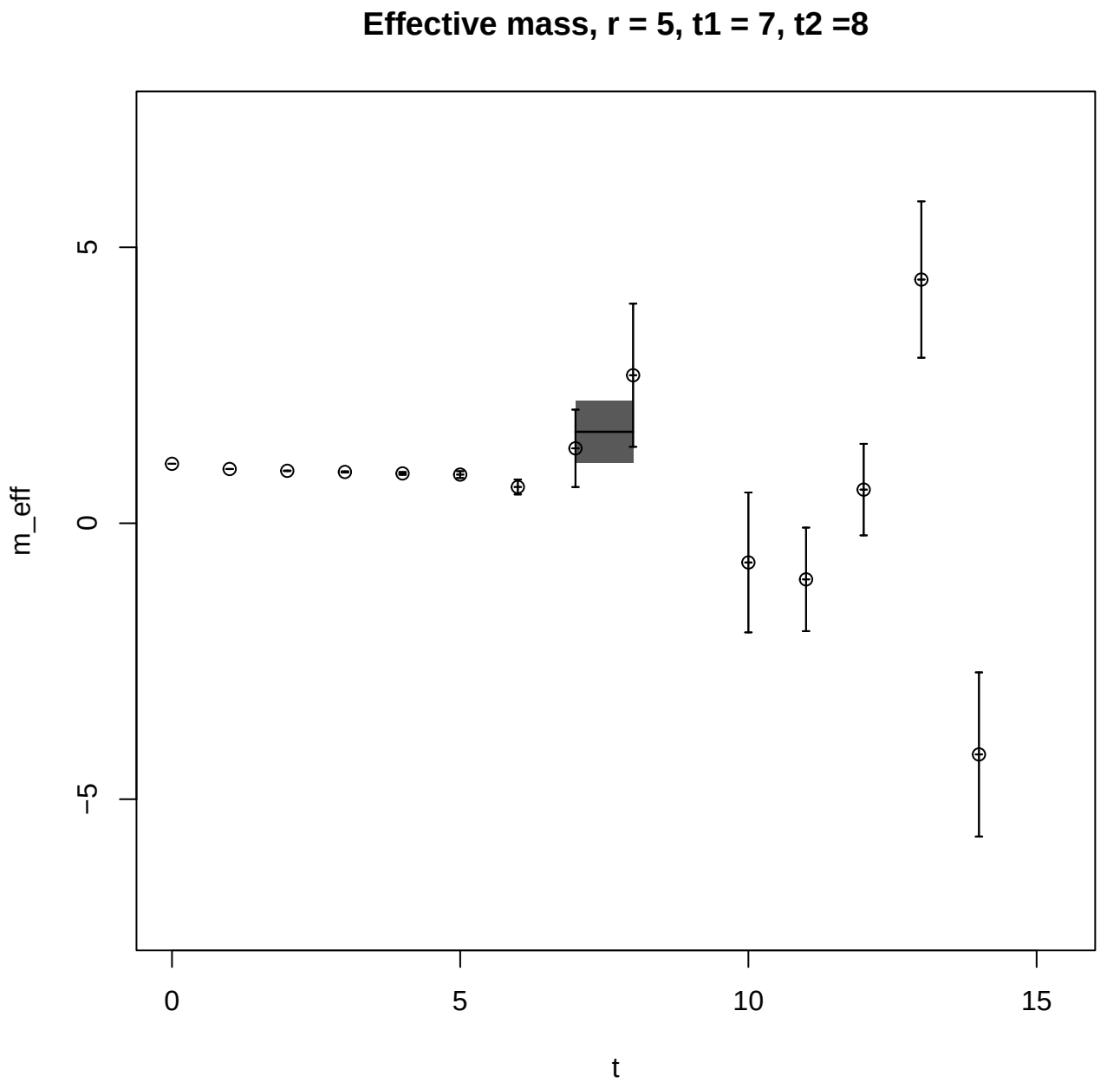

Effective mass, r = 5, t1 = 6, t2 =7
m_eff
5
0
−5
0
5
10
15
t

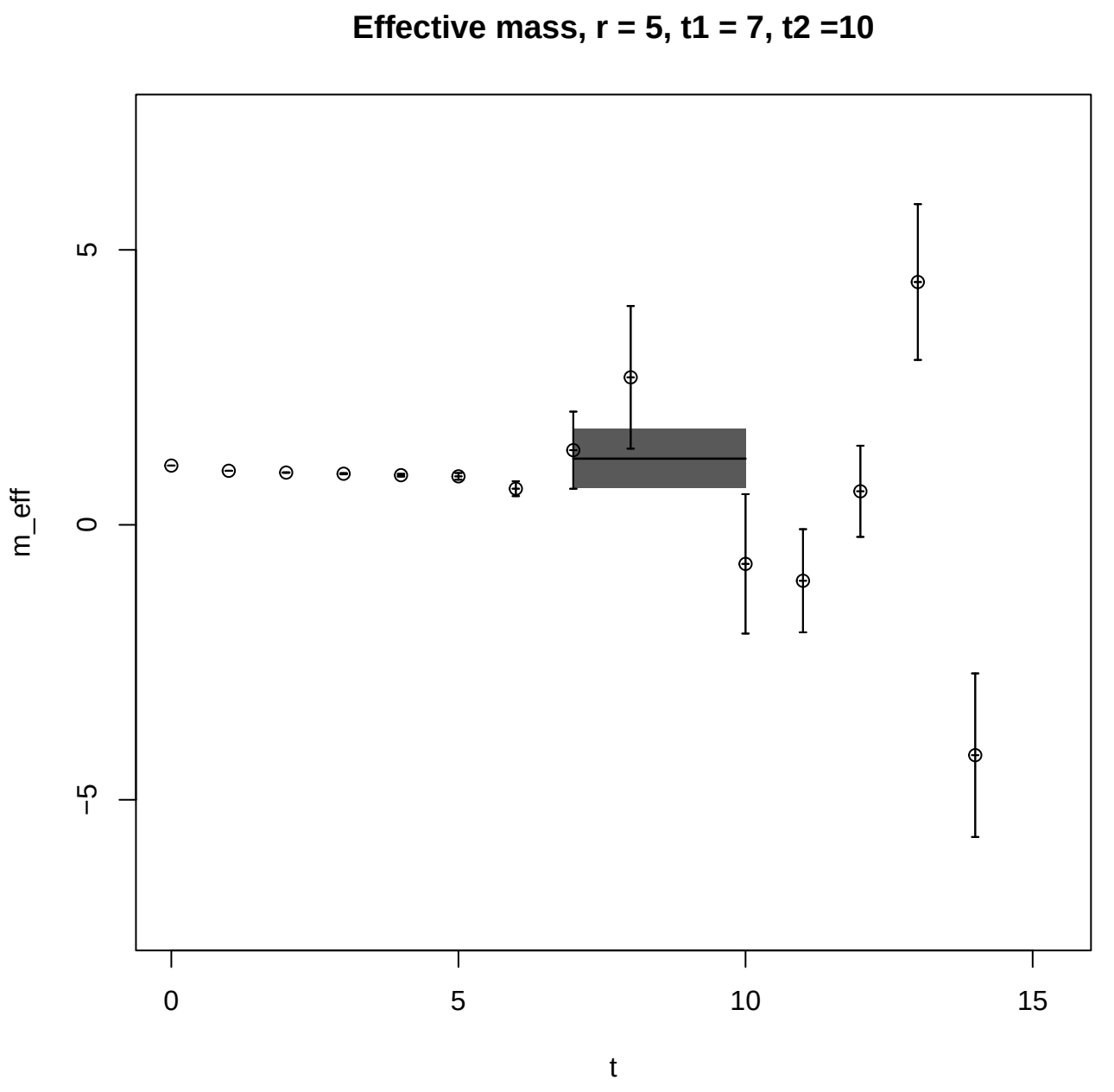

Effective mass, r = 5, t1 = 6, t2 =8
m_eff
5
0
−5
0
5
10
15
t

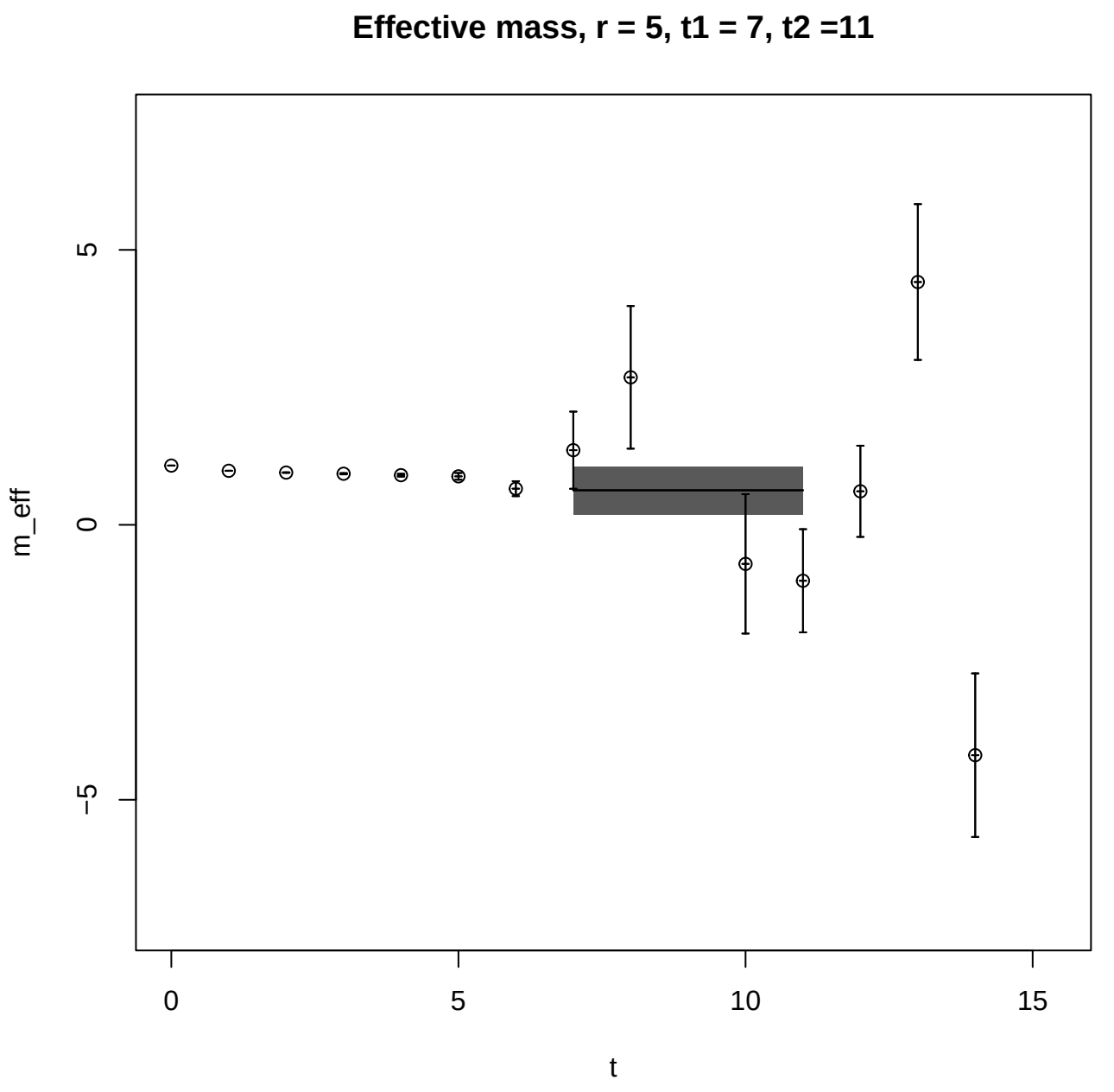

Effective mass, r = 5, t1 = 6, t2 =10
m_eff
5
0
−5
0
5
10
15
t

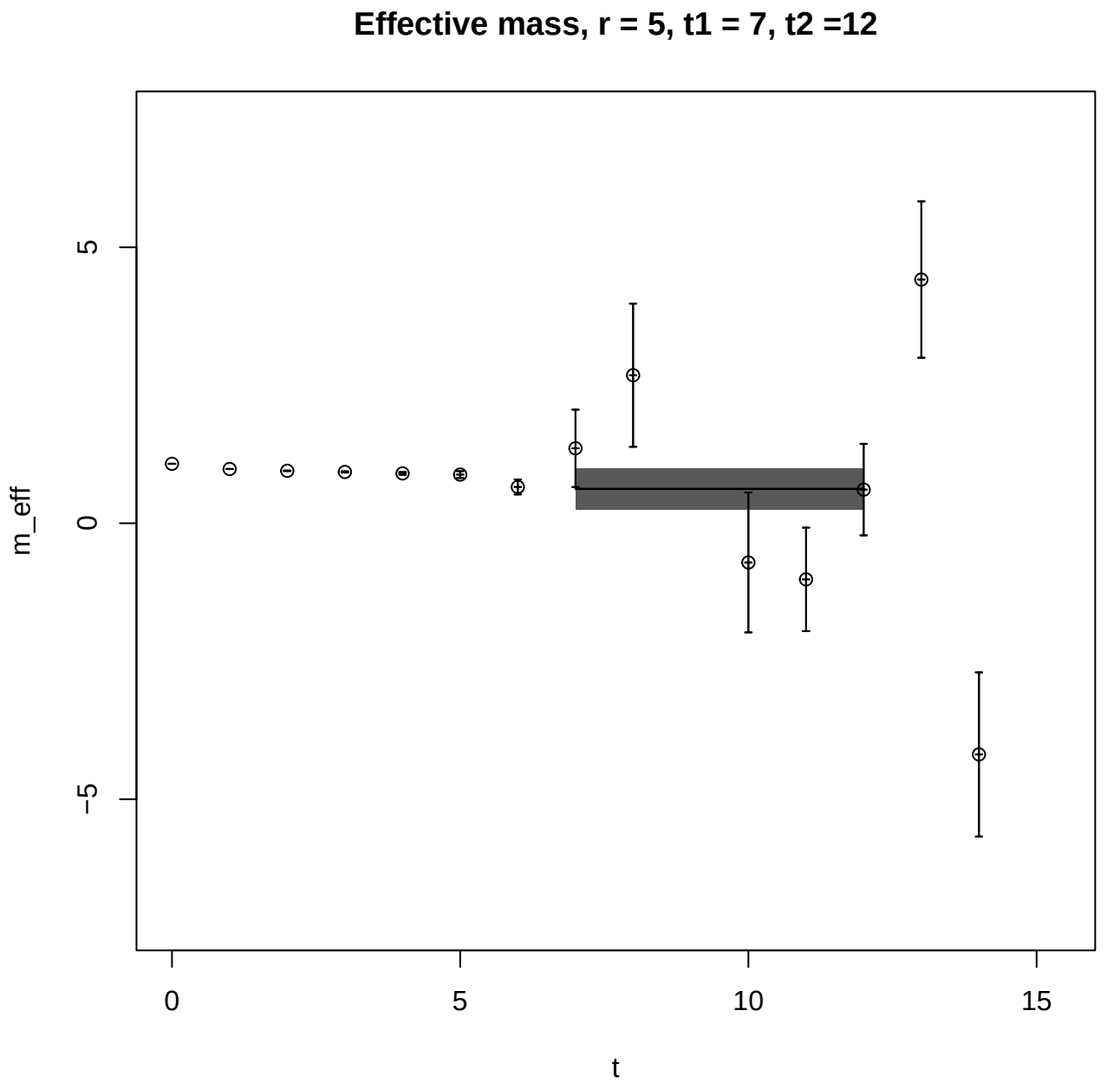

Effective mass, r = 5, t1 = 6, t2 =11
m_eff
5
0
−5
0
5
10
15
t

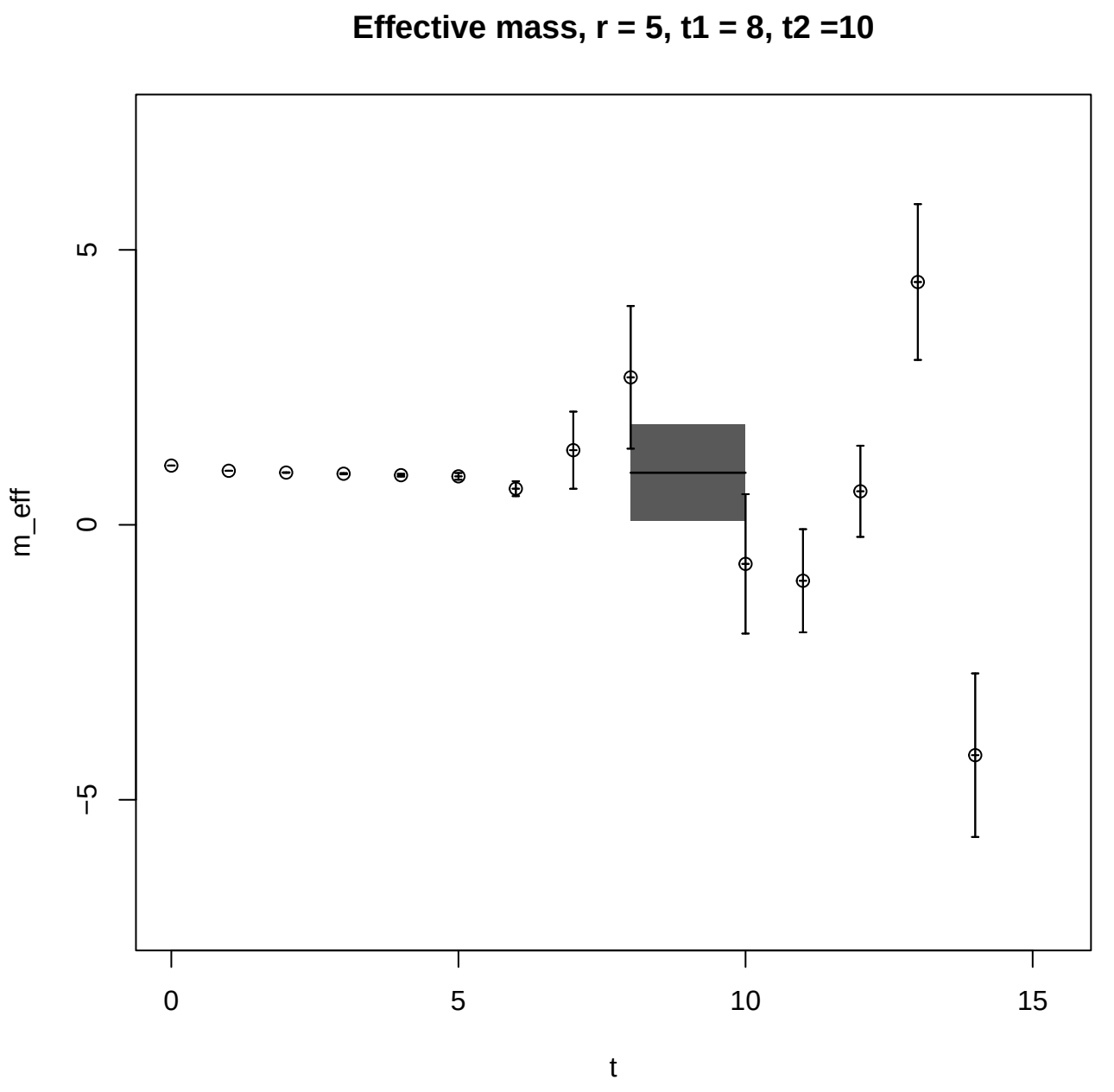

Effective mass, r = 5, t1 = 6, t2 =12
m_eff
5
0
−5
0
5
10
15
t

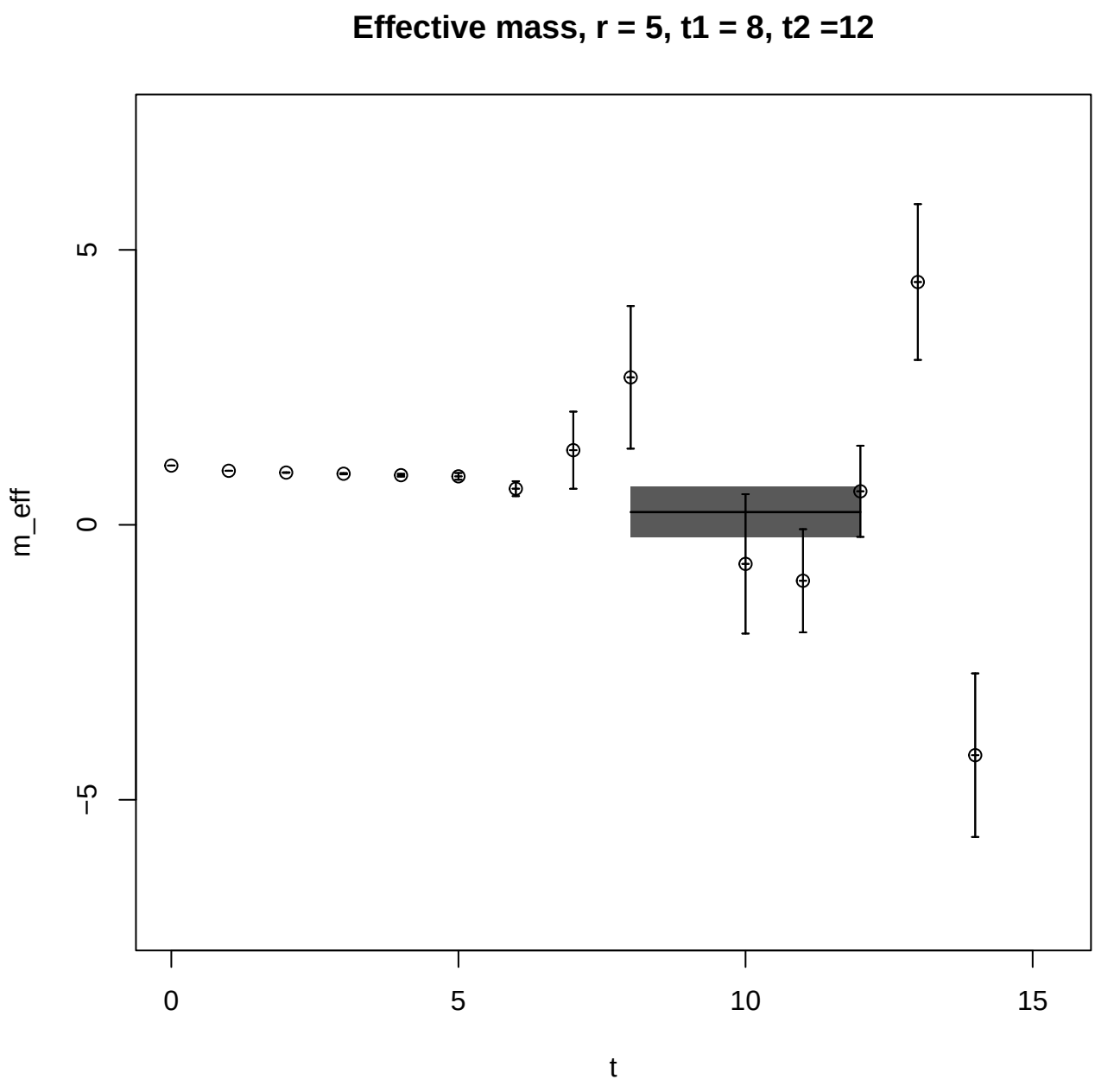

Effective mass, r = 5, t1 = 7, t2 =8
m_eff
5
0
−5
0
5
10
15
t

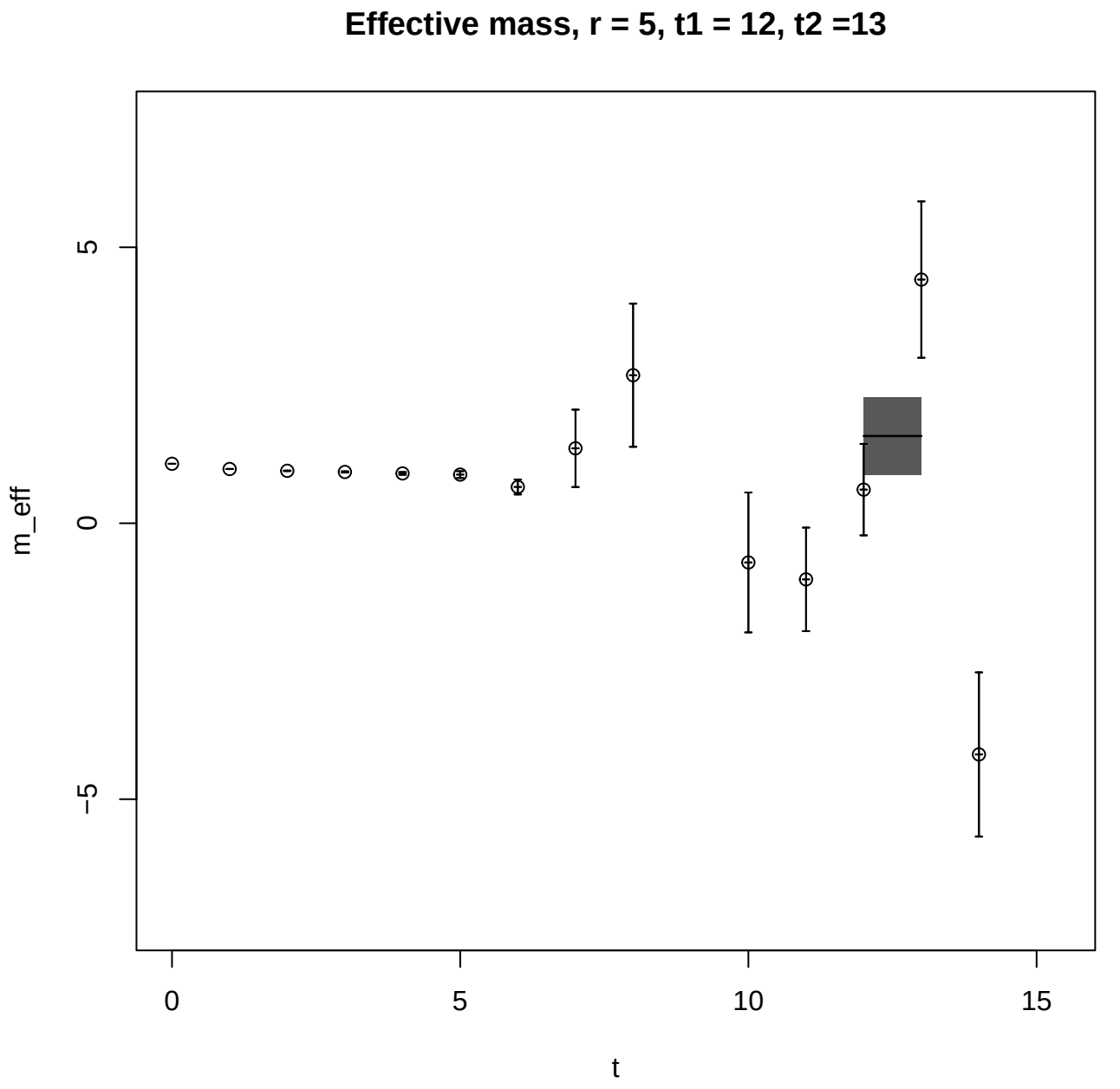

Effective mass, r = 5, t1 = 7, t2 =10
m_eff
5
0
−5
0
5
10
15
t

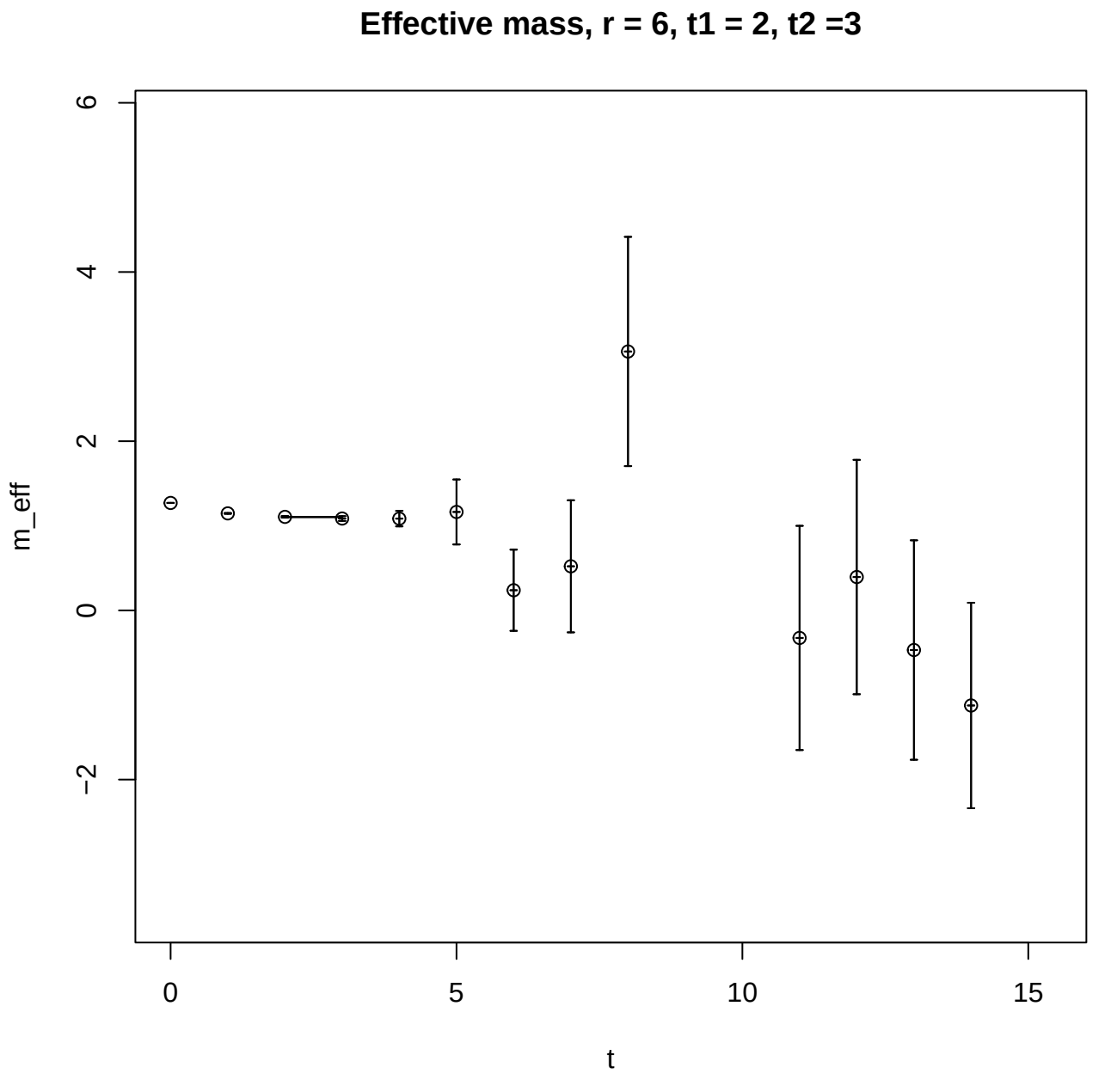

Effective mass, r = 5, t1 = 7, t2 =11
m_eff
5
0
−5
0
5
10
15
t

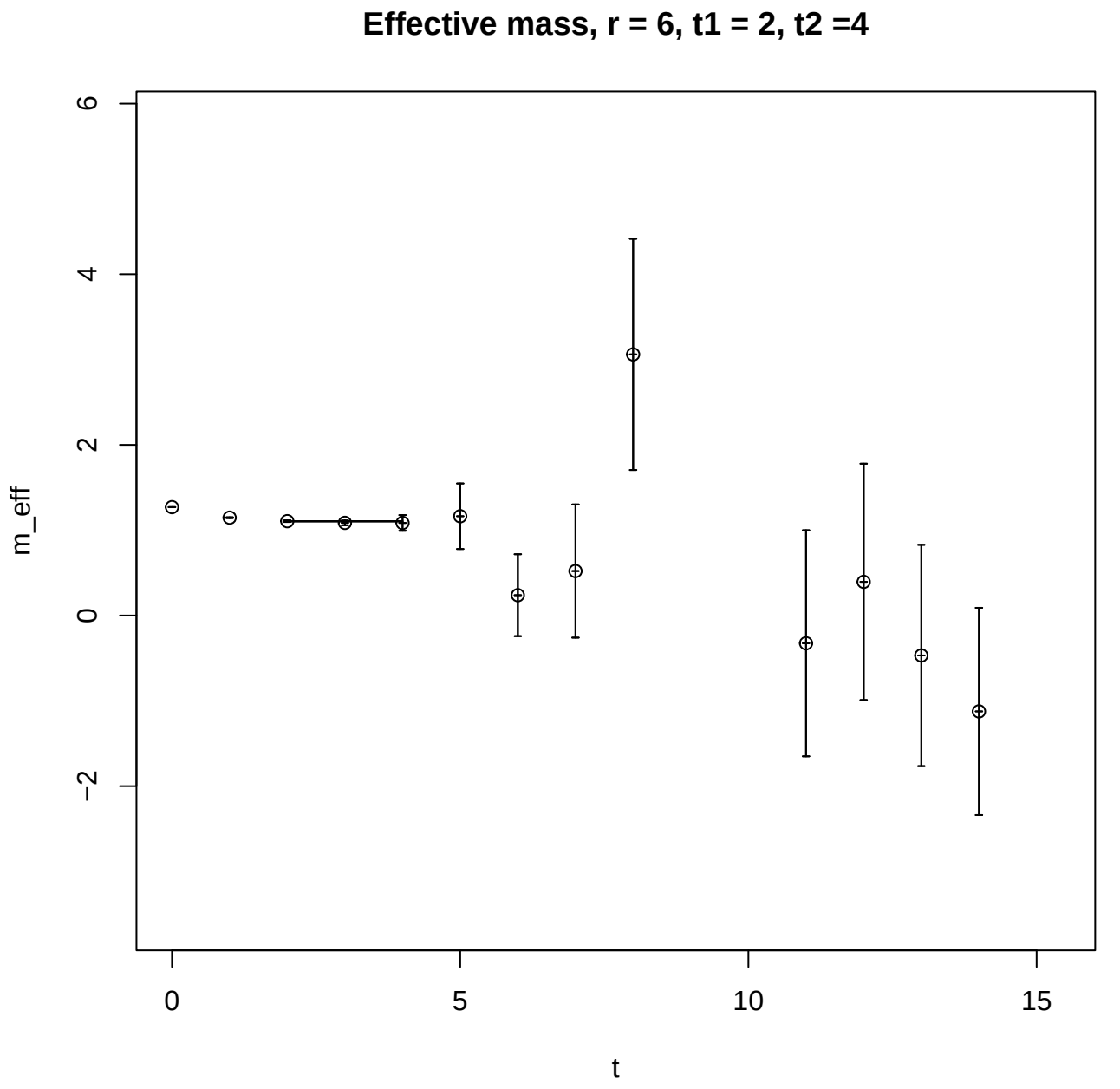

Effective mass, r = 5, t1 = 7, t2 =12
m_eff
5
0
−5
0
5
10
15
t

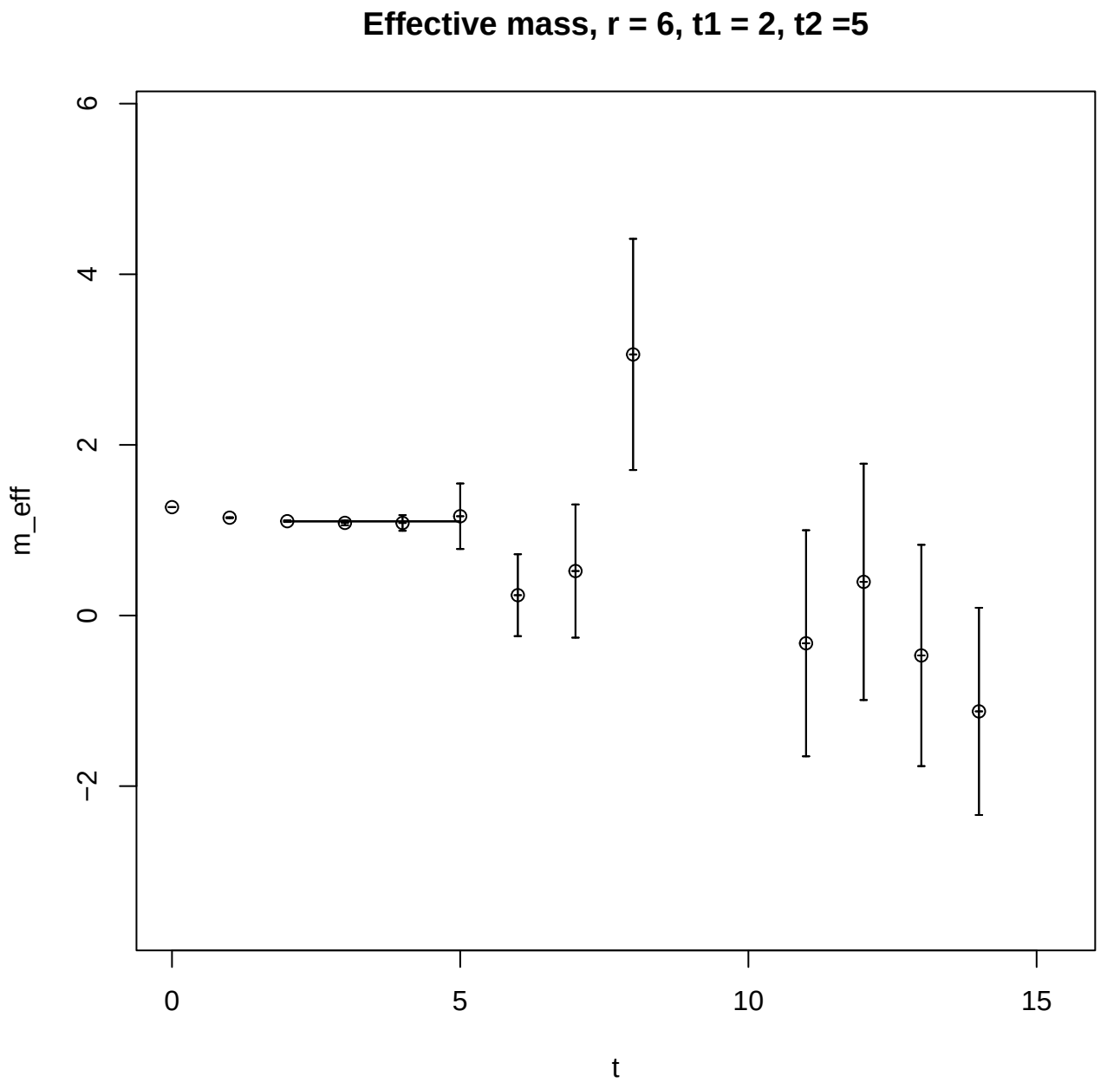

Effective mass, r = 5, t1 = 8, t2 =10
m_eff
5
0
−5
0
5
10
15
t

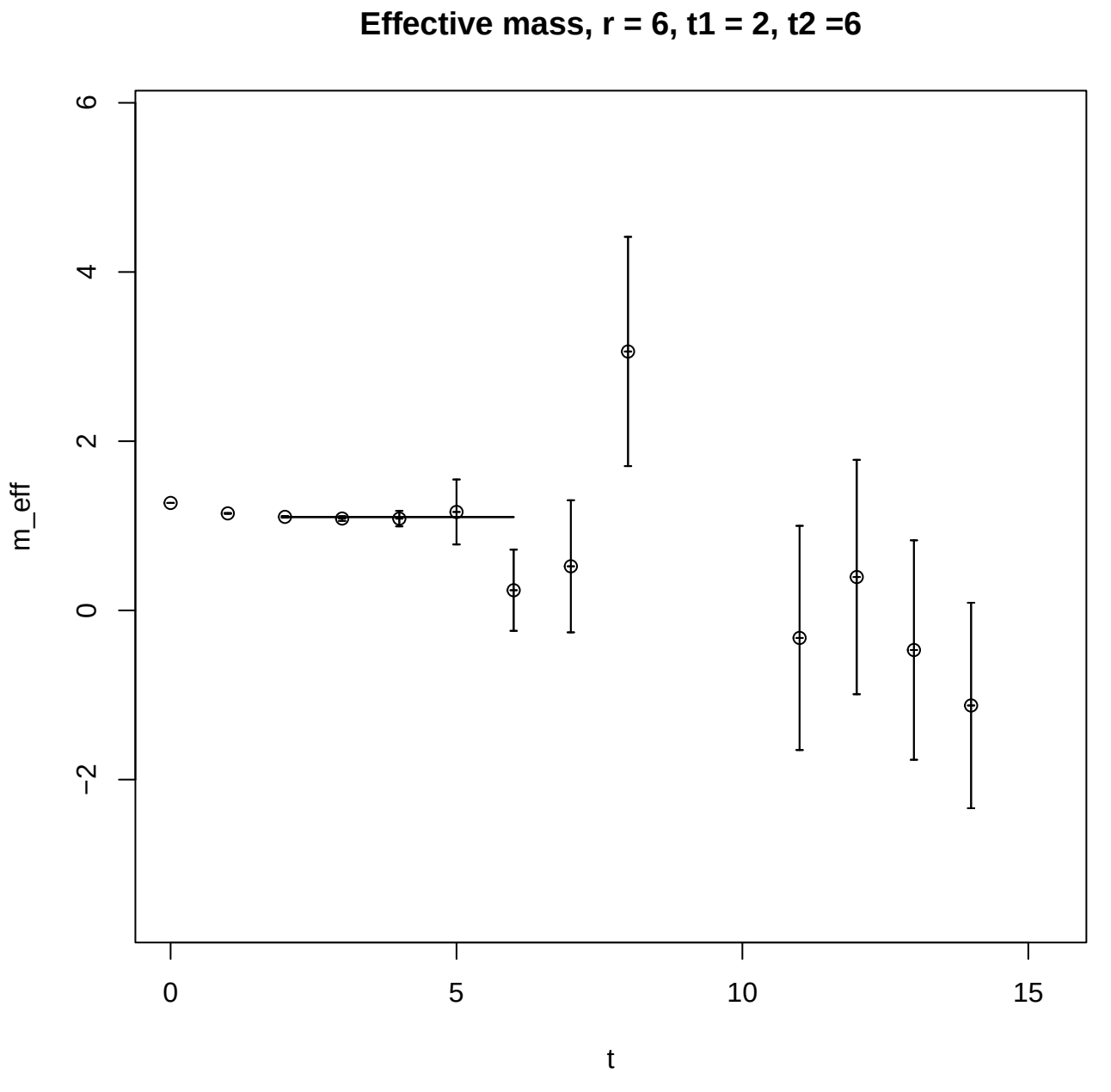

Effective mass, r = 5, t1 = 8, t2 =12
m_eff
5
0
−5
0
5
10
15
t

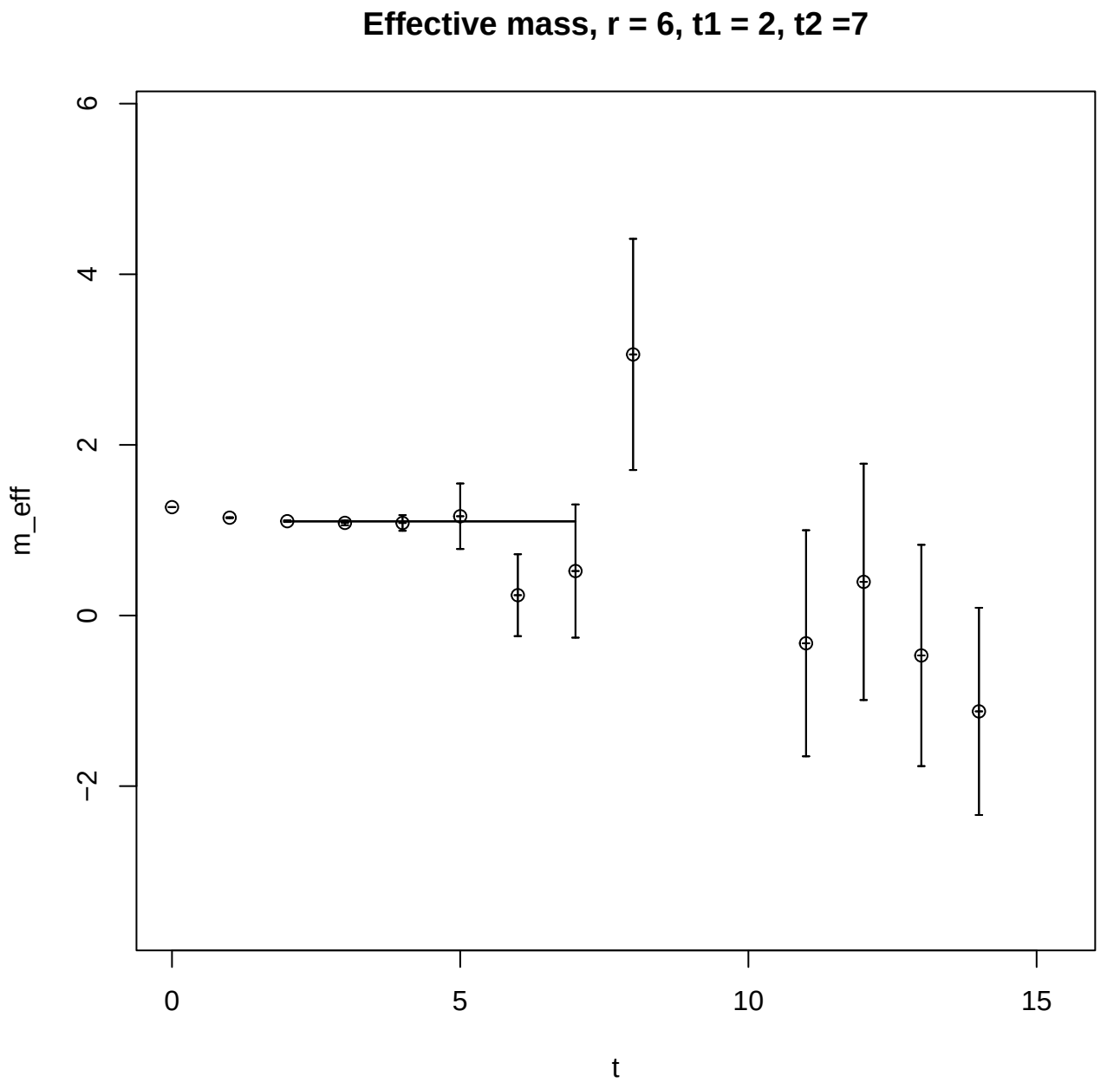

Effective mass, r = 5, t1 = 12, t2 =13
m_eff
5
0
−5
0
5
10
15
t

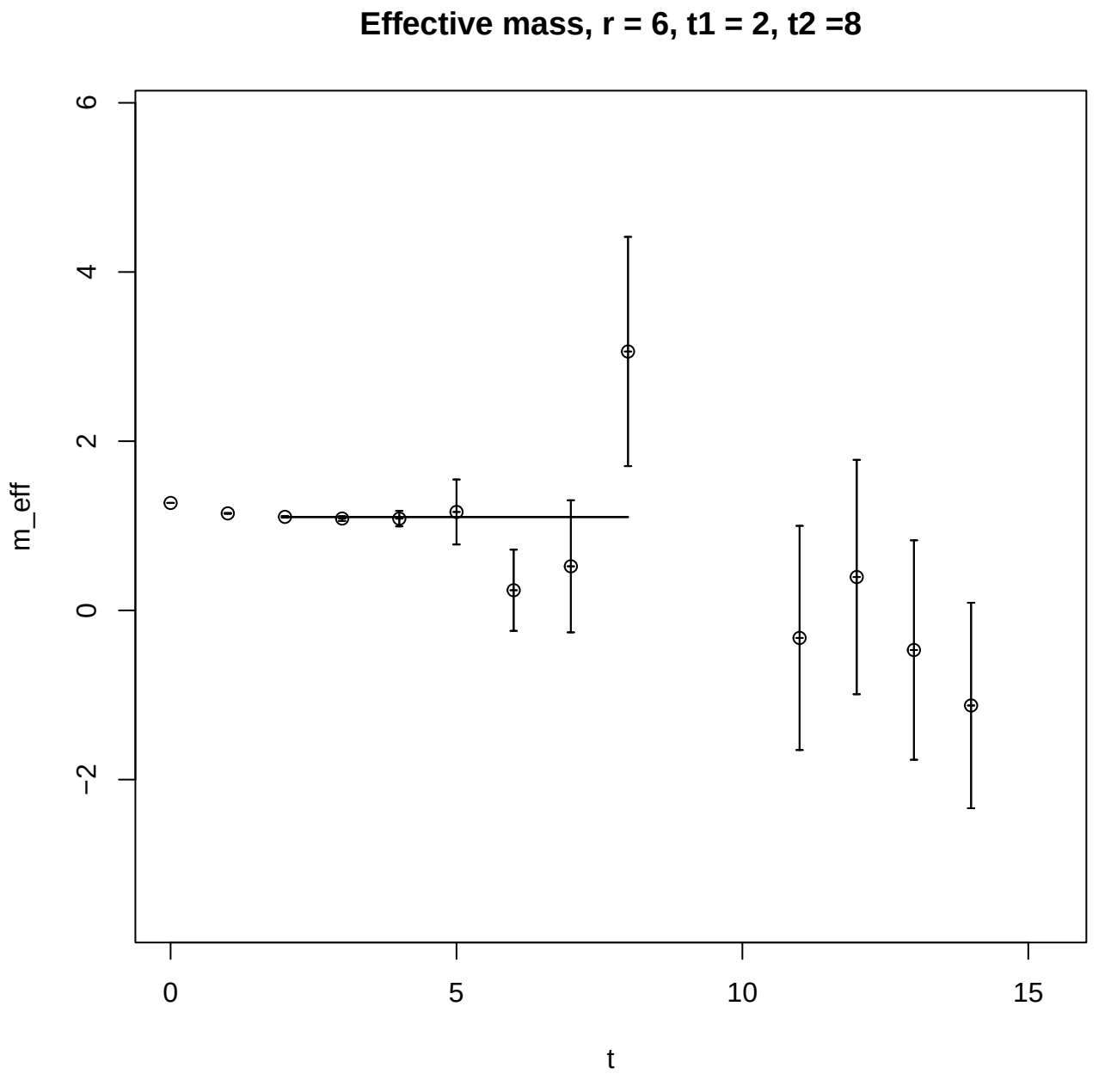

Effective mass, r = 6, t1 = 2, t2 =3
m_eff
6
4
2
0
−2
0
5
10
15
t

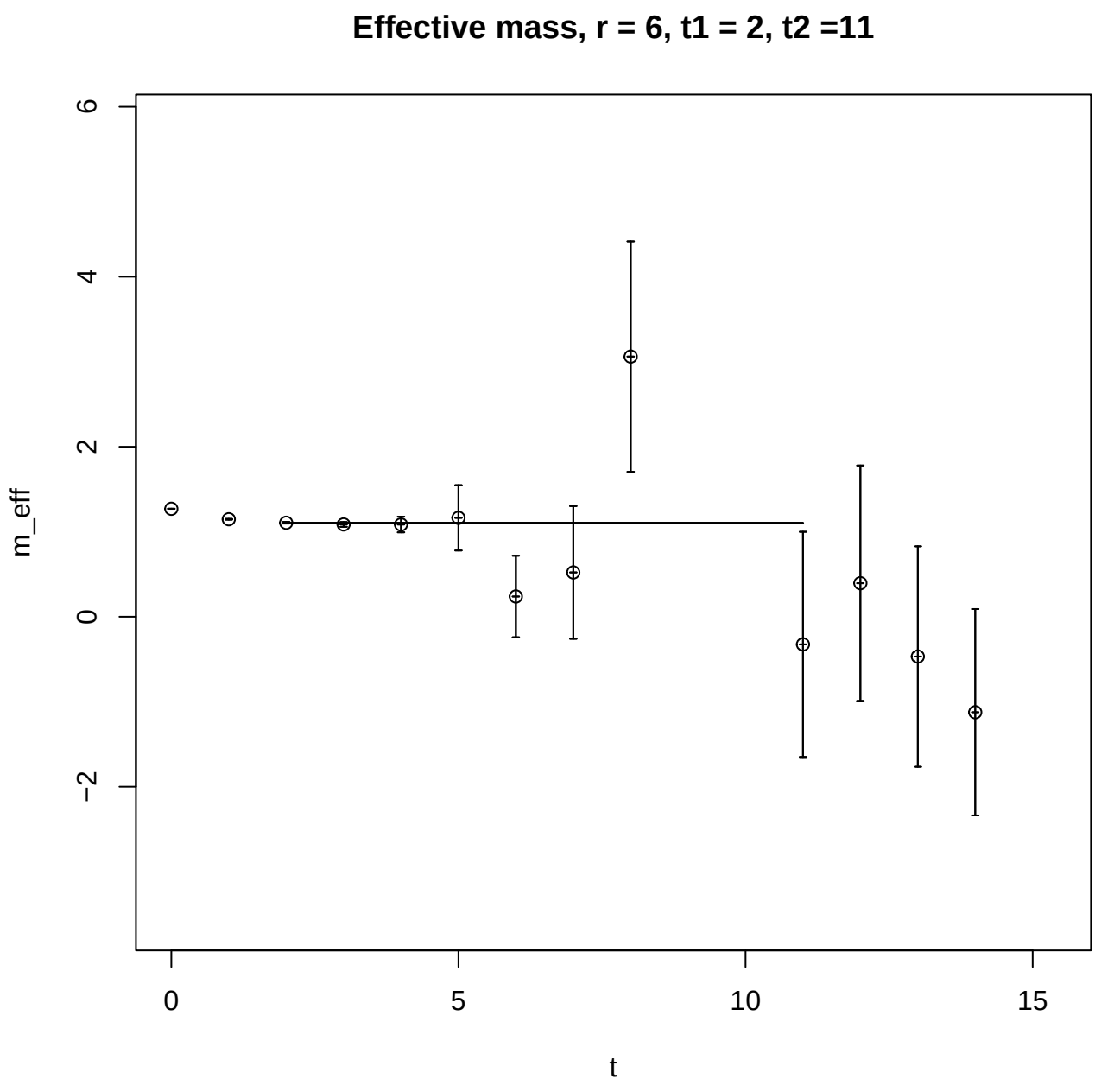

Effective mass, r = 6, t1 = 2, t2 =4
m_eff
6
4
2
0
−2
0
5
10
15
t

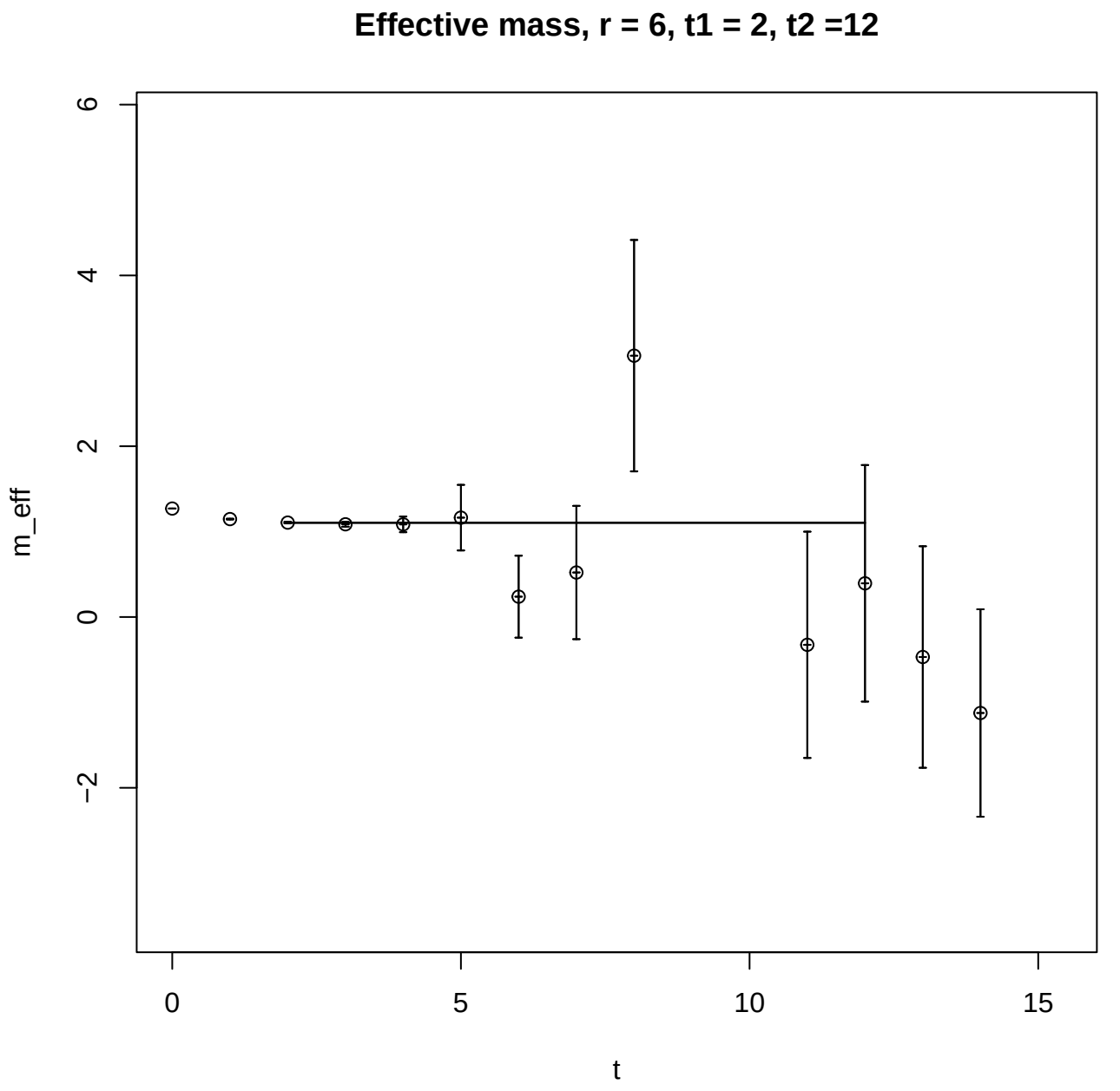

Effective mass, r = 6, t1 = 2, t2 =5
m_eff
6
4
2
0
−2
0
5
10
15
t

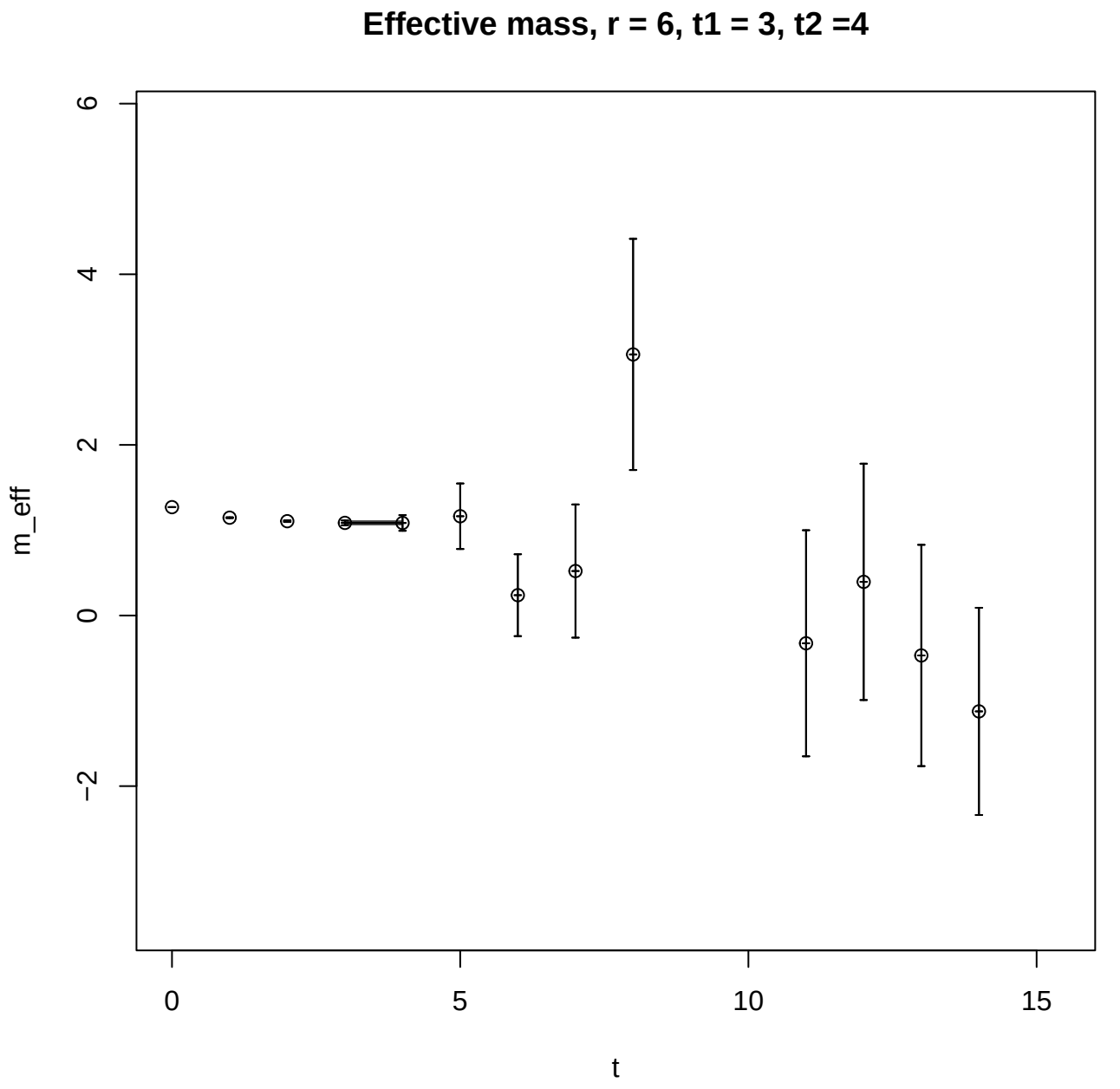

Effective mass, r = 6, t1 = 2, t2 =6
m_eff
6
4
2
0
−2
0
5
10
15
t

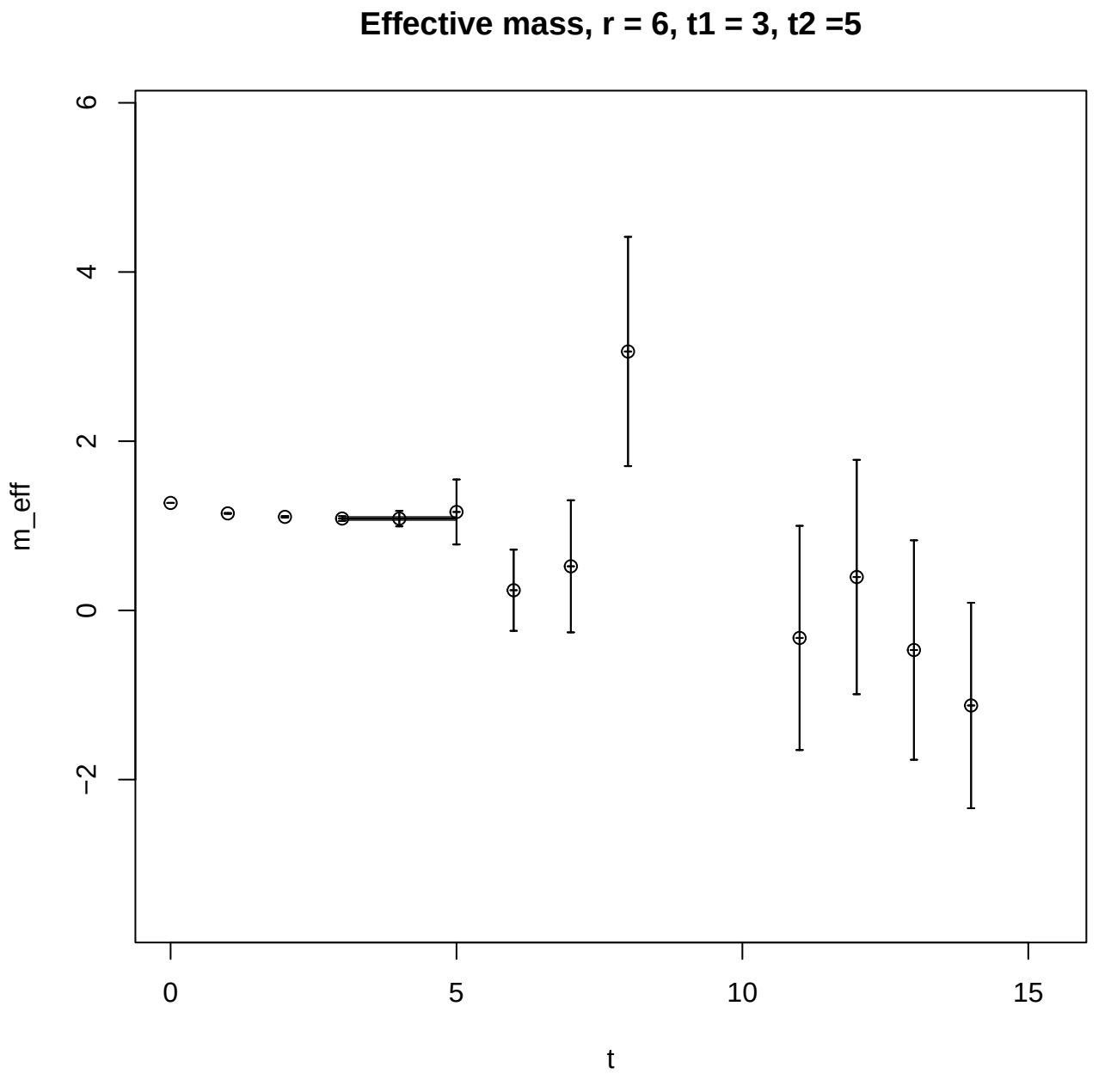

Effective mass, r = 6, t1 = 2, t2 =7
m_eff
6
4
2
0
−2
0
5
10
15
t

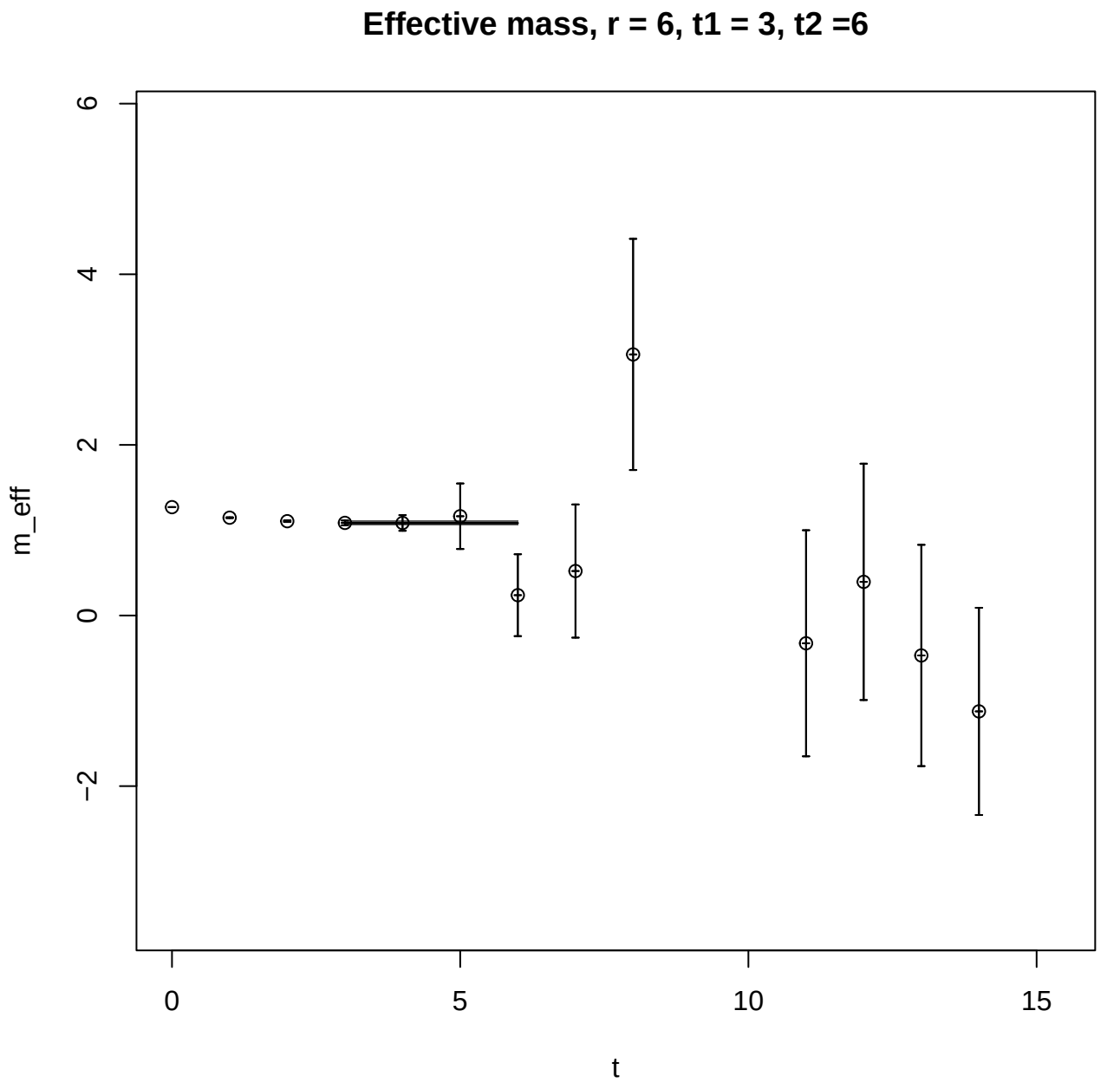

Effective mass, r = 6, t1 = 2, t2 =8
m_eff
6
4
2
0
−2
0
5
10
15
t

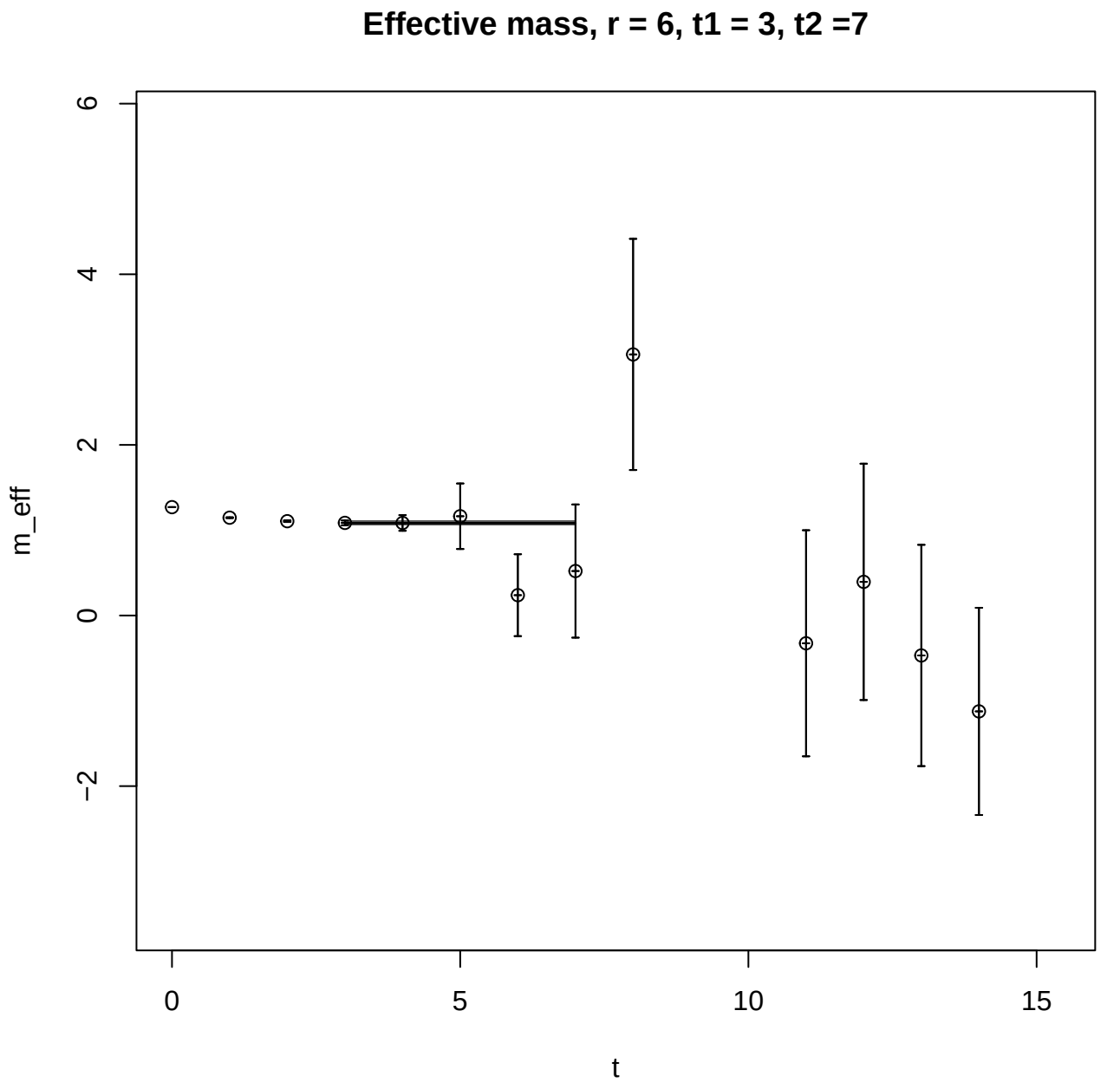

Effective mass, r = 6, t1 = 2, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

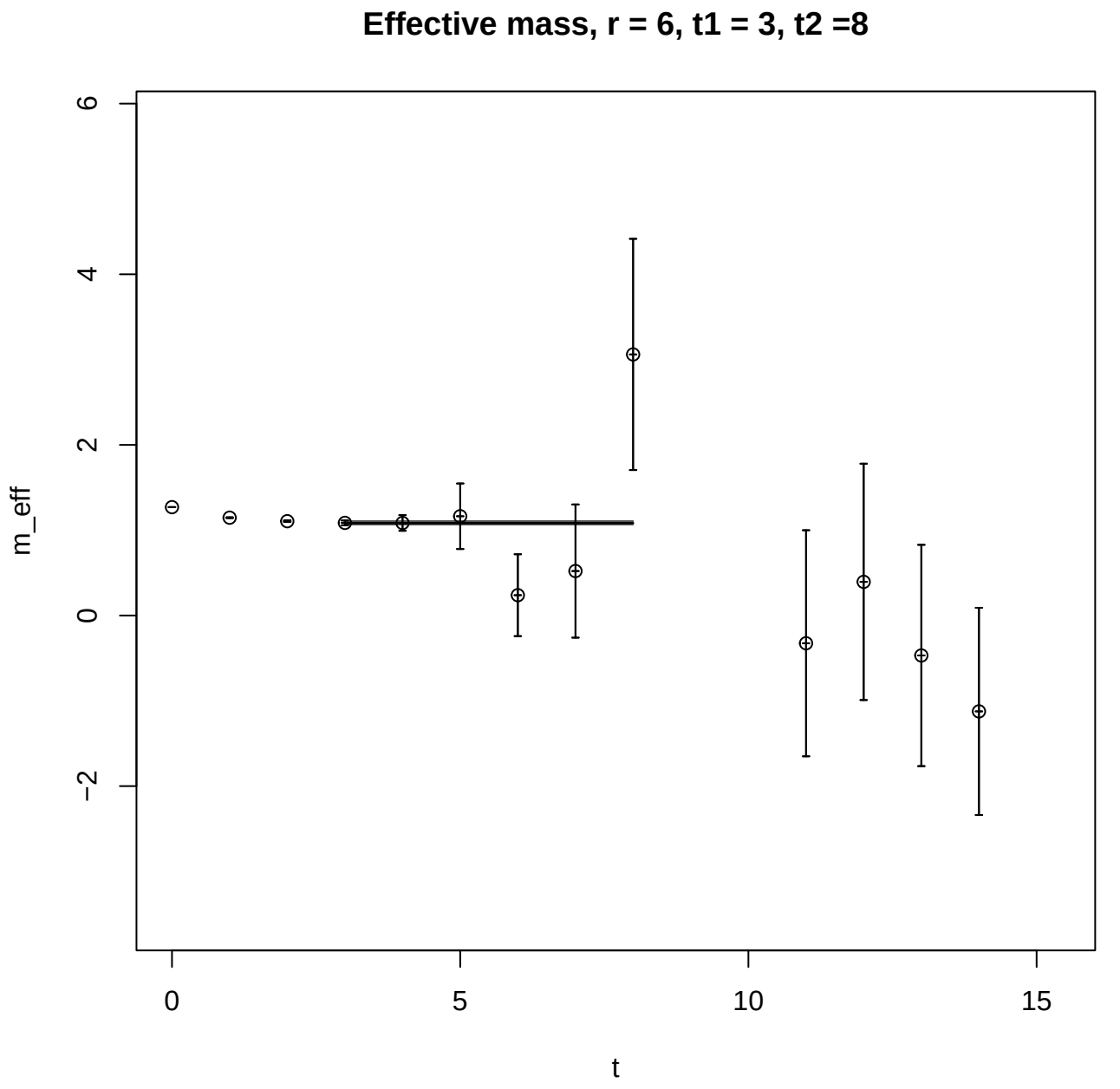

Effective mass, r = 6, t1 = 2, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

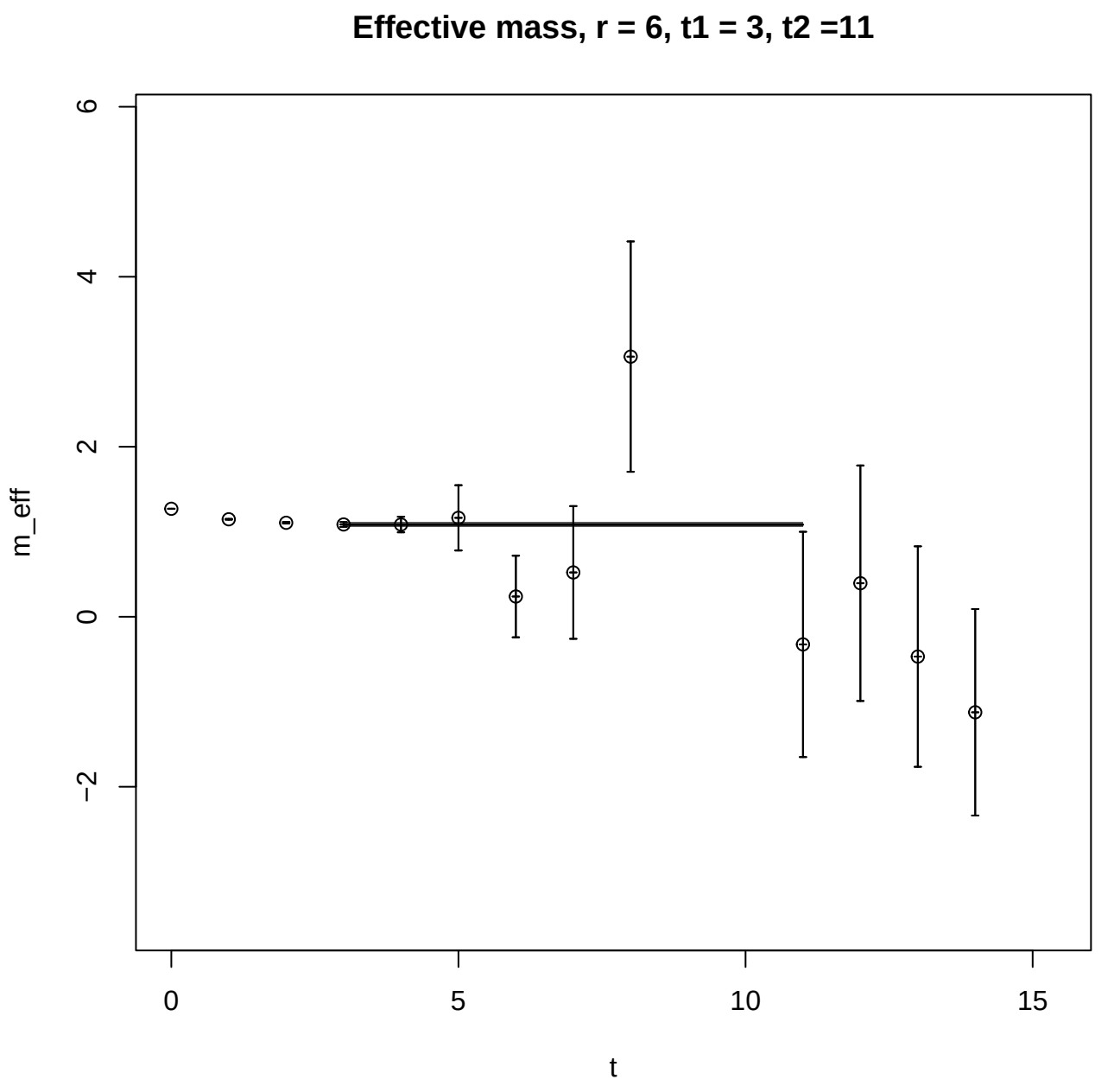

Effective mass, r = 6, t1 = 3, t2 =4
m_eff
6
4
2
0
−2
0
5
10
15
t

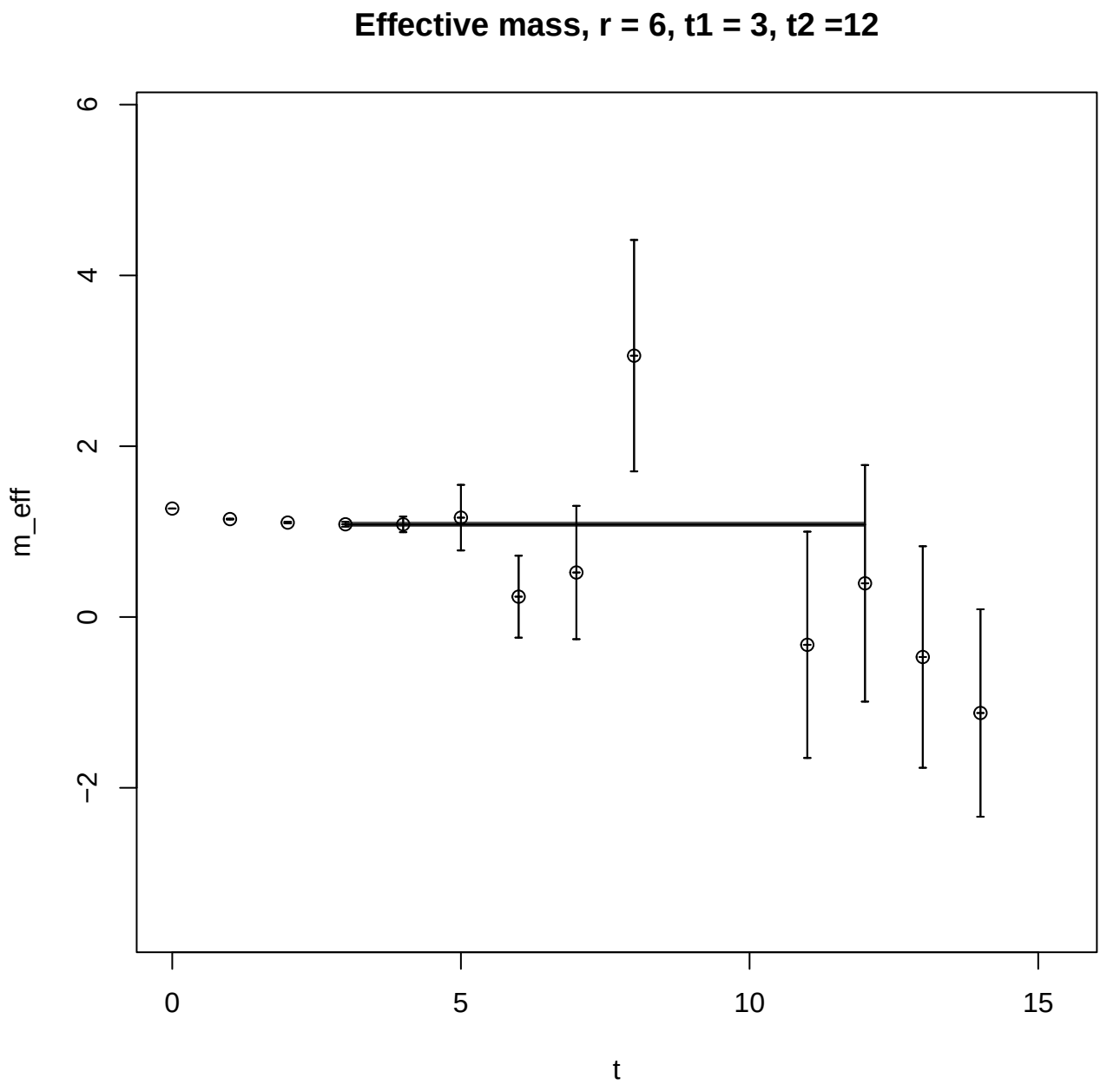

Effective mass, r = 6, t1 = 3, t2 =5
m_eff
t
6
4
2
0
−2
0
5
10
15

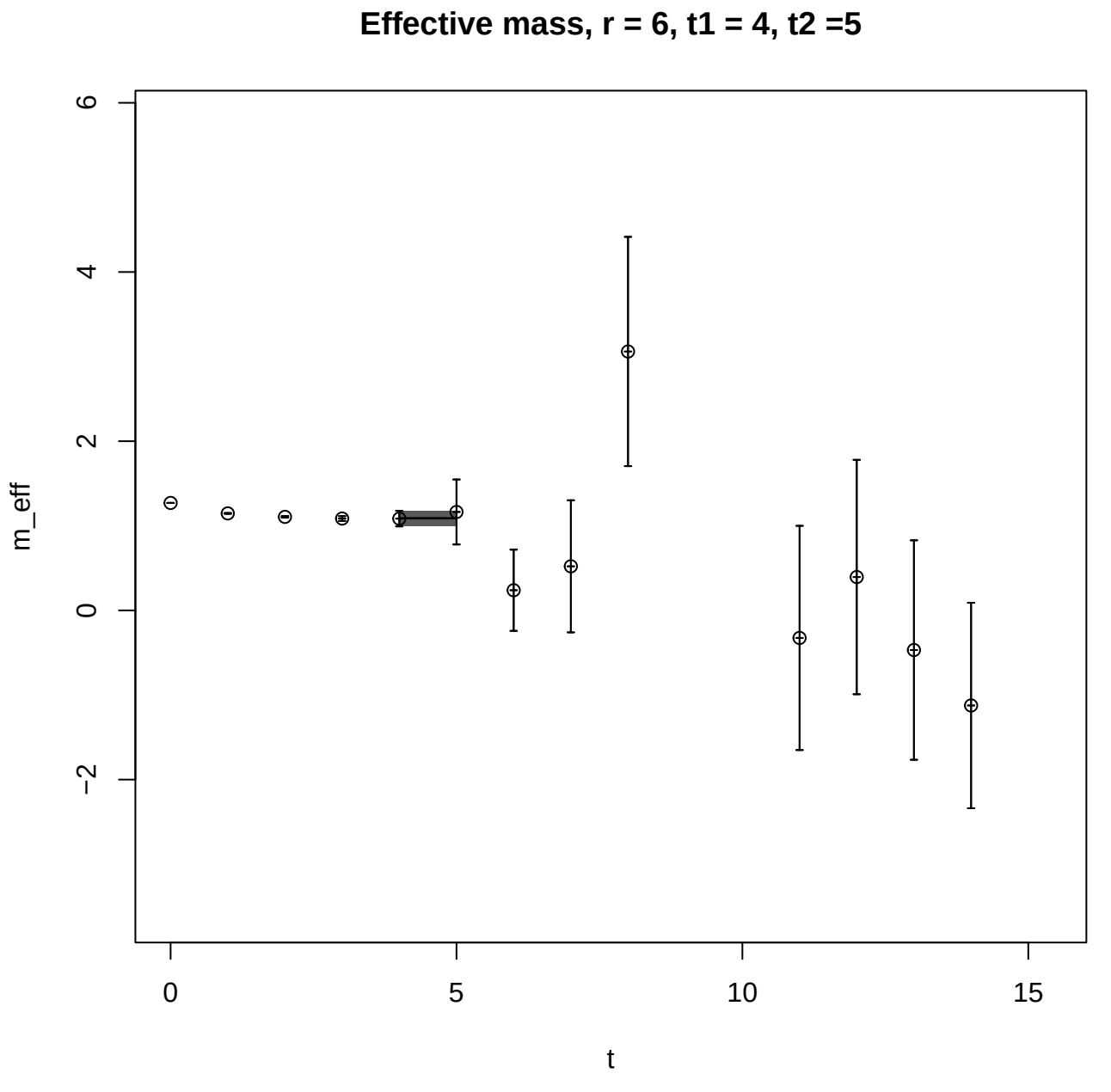

Effective mass, r = 6, t1 = 3, t2 =6
m_eff
6
4
2
0
−2
0
5
10
15
t

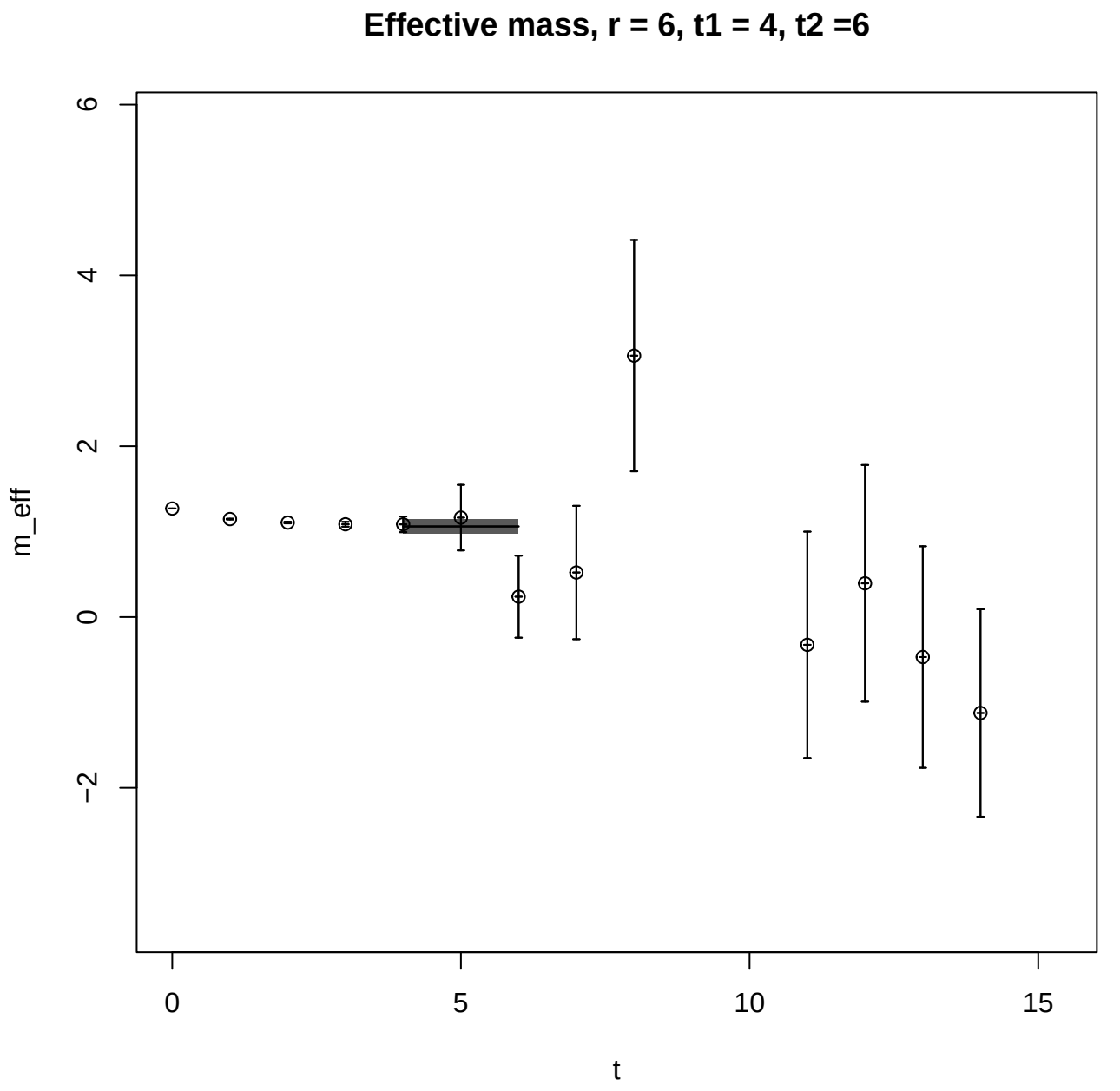

Effective mass, r = 6, t1 = 3, t2 =7
m_eff
6
4
2
0
−2
0
5
10
15
t

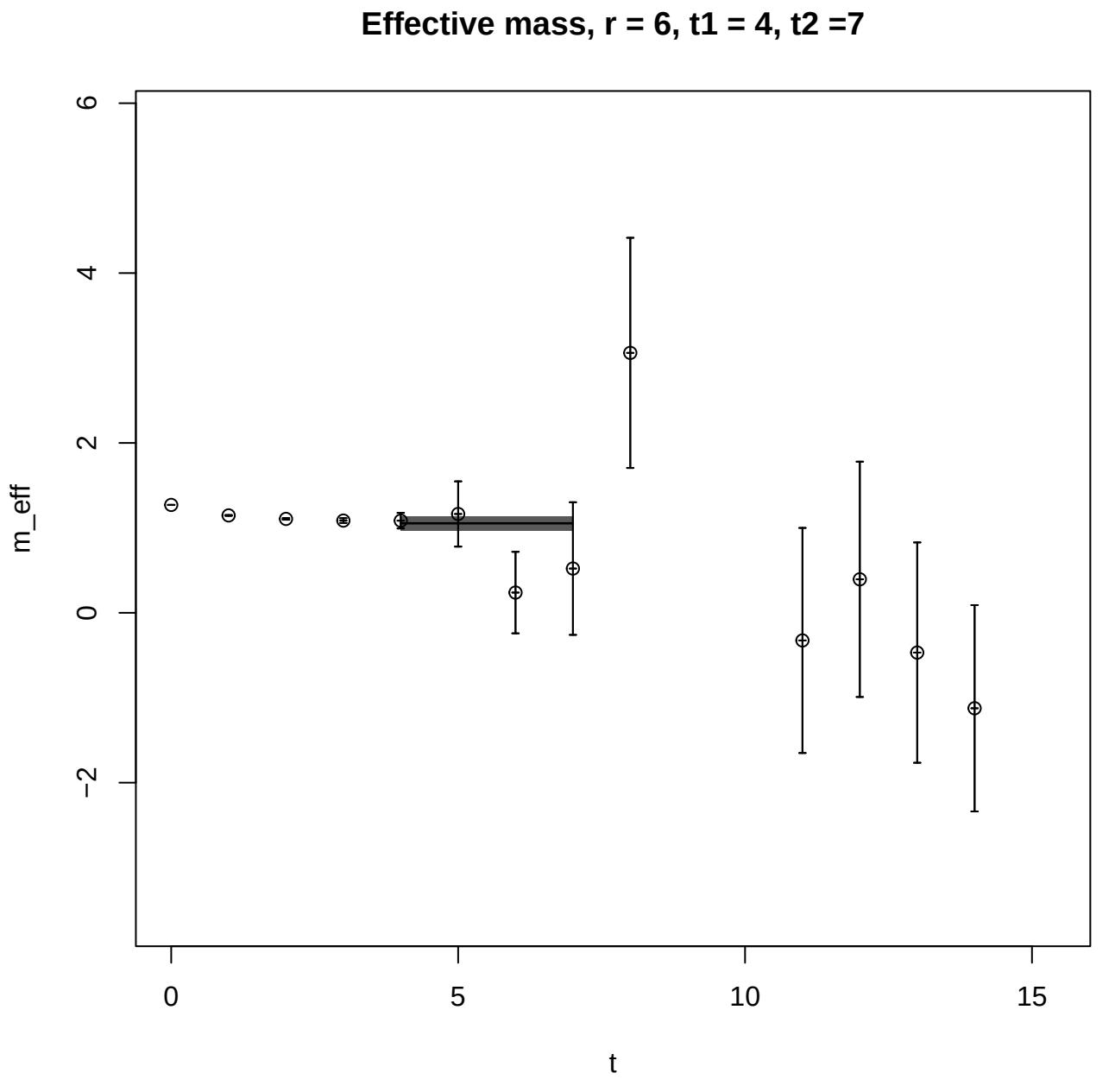

Effective mass, r = 6, t1 = 3, t2 =8
m_eff
t
6
4
2
0
−2
0
5
10
15

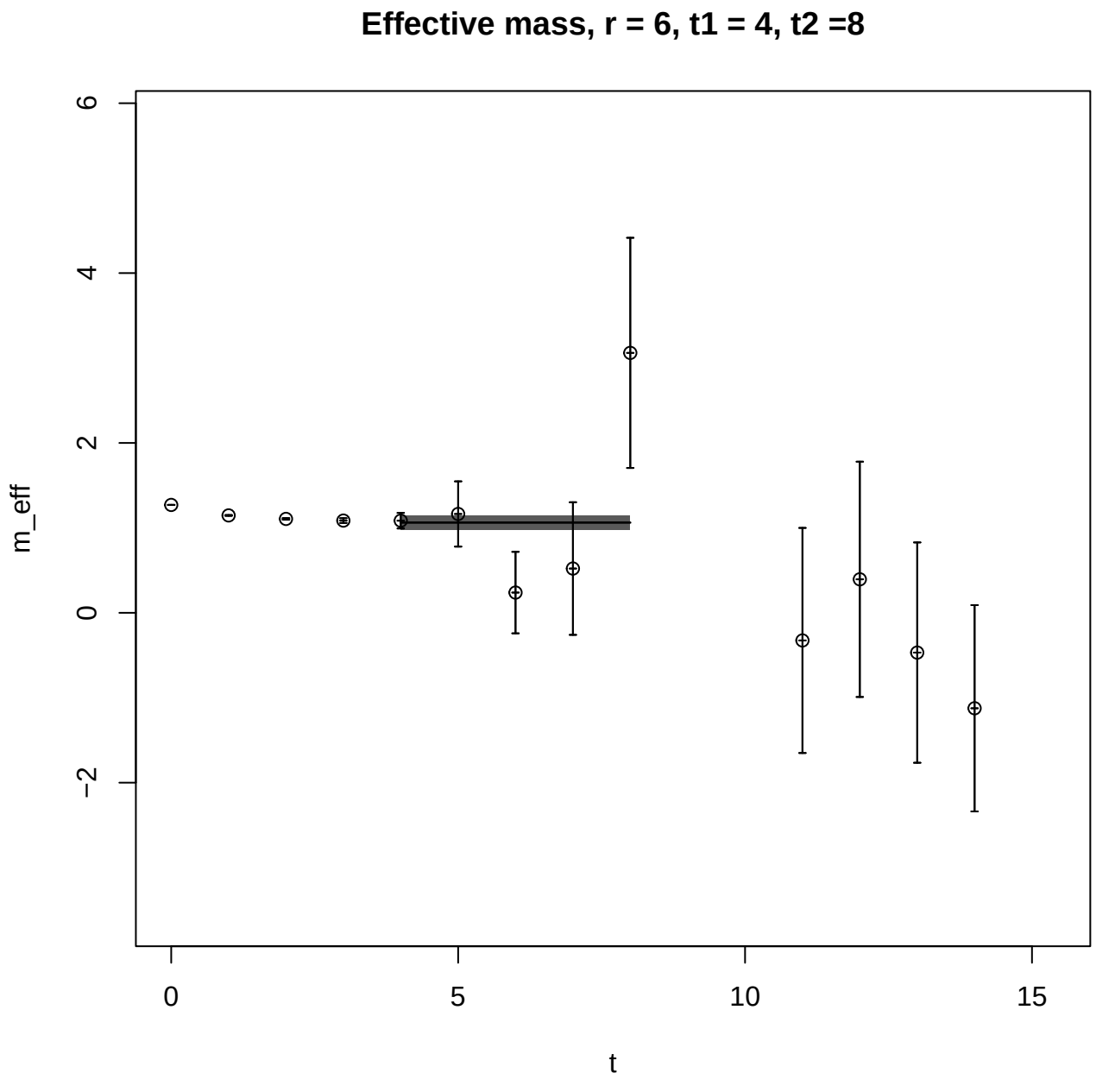

Effective mass, r = 6, t1 = 3, t2 =11
m_eff
6
4
2
0
−2
0
5
10
15
t

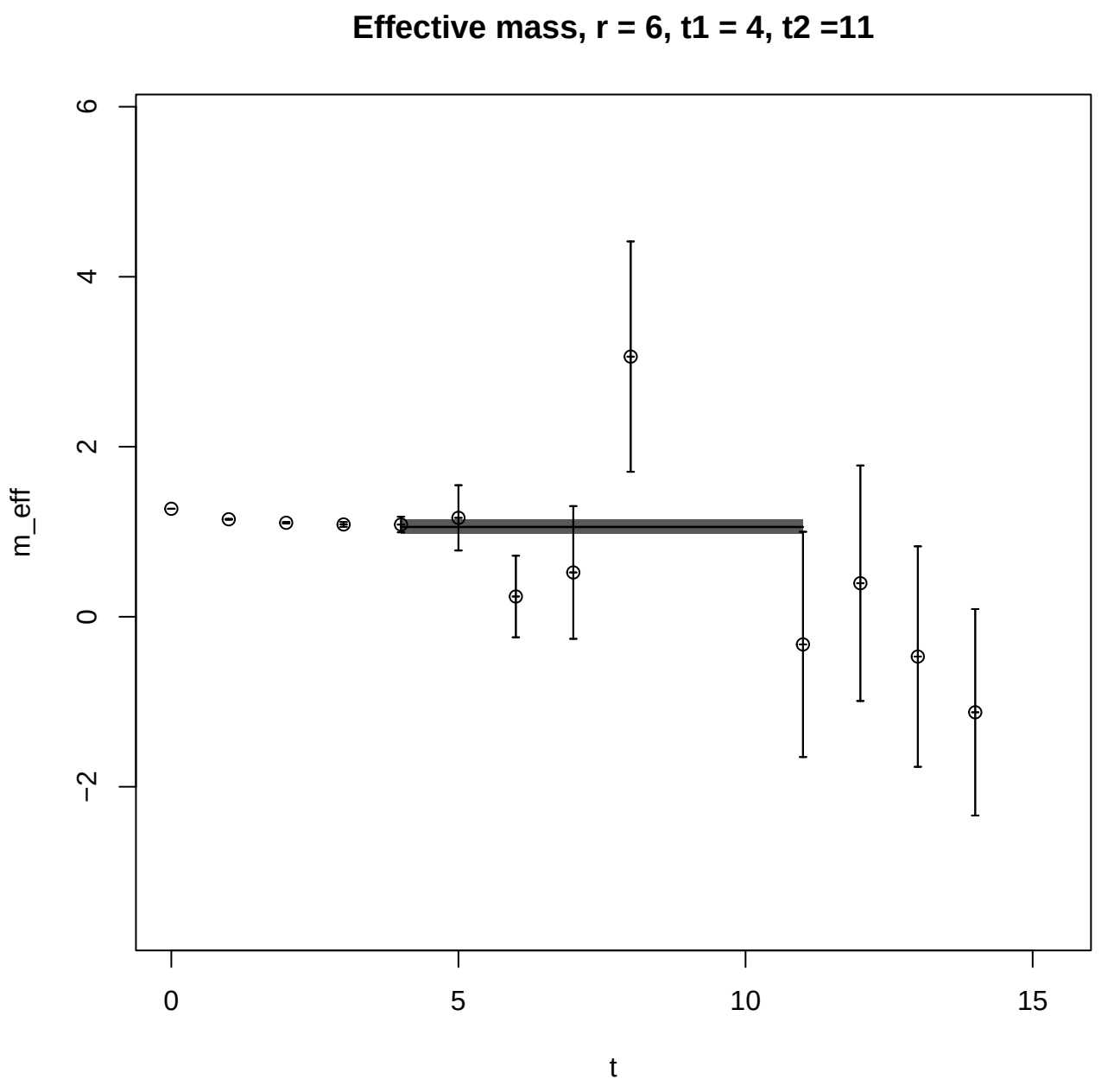

Effective mass, r = 6, t1 = 3, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

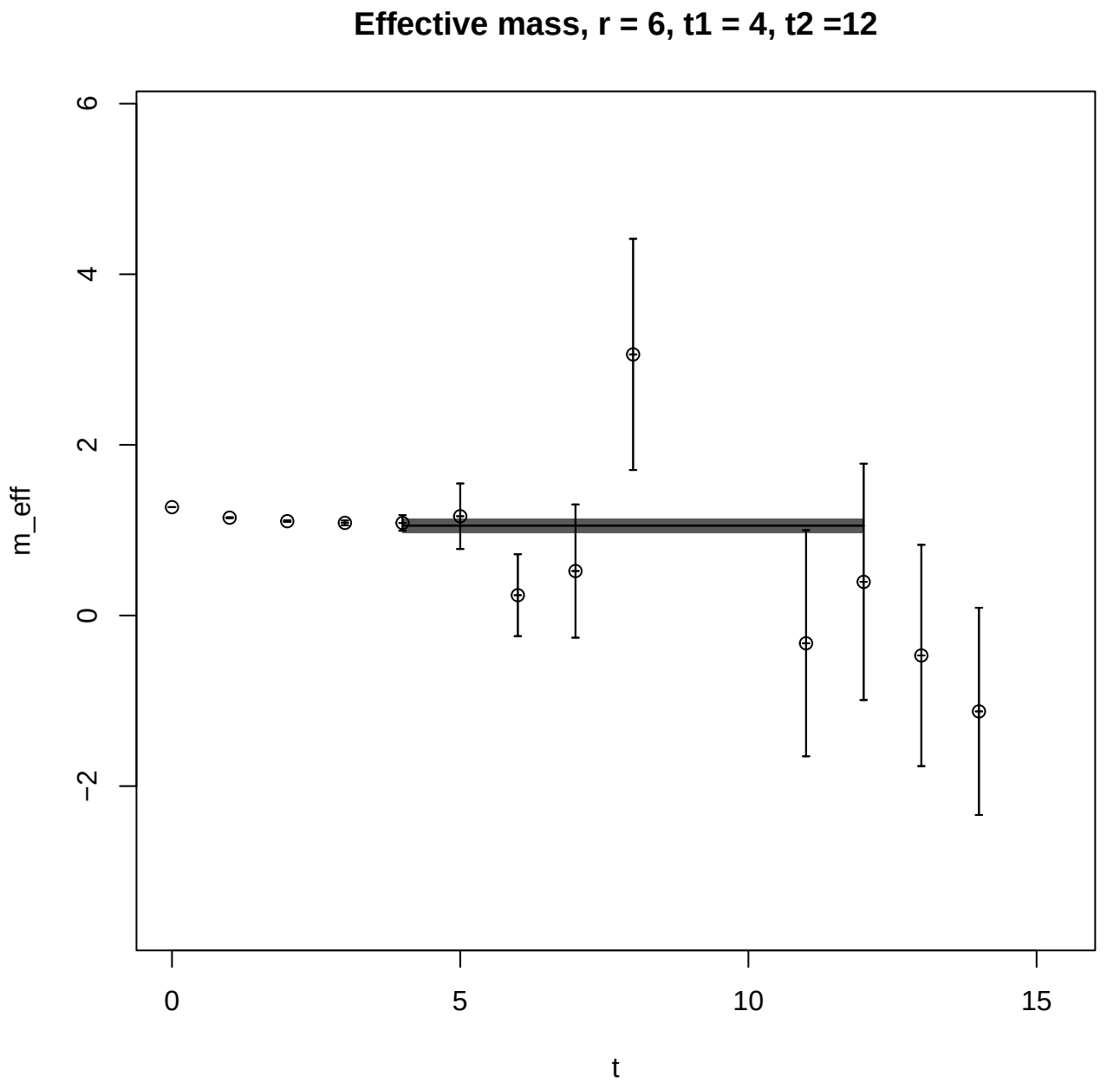

Effective mass, r = 6, t1 = 4, t2 =5
m_eff
6
4
2
0
−2
0
5
10
15
t

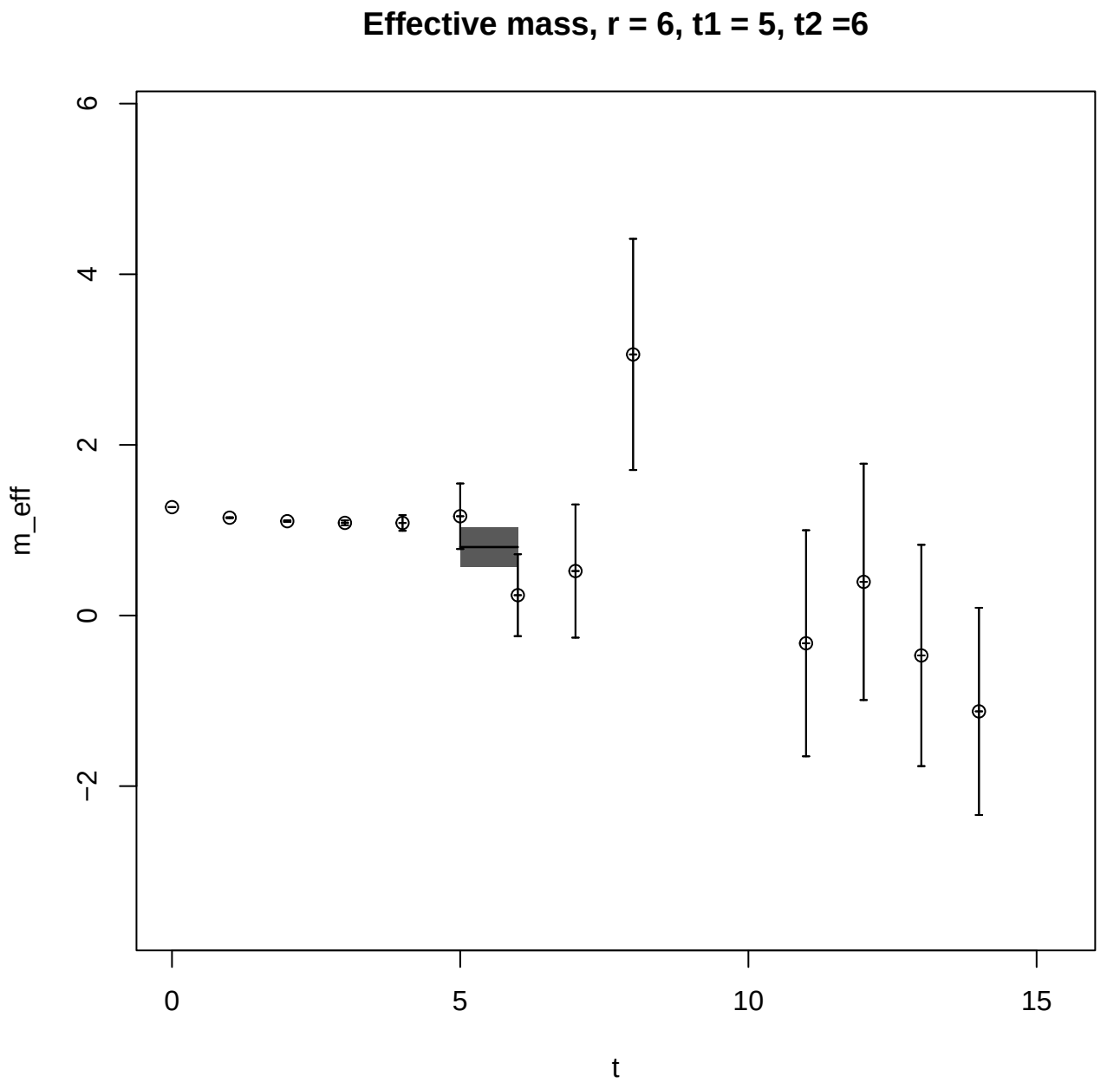

Effective mass, r = 6, t1 = 4, t2 =6
m_eff
t
6
4
2
0
−2
0
5
10
15

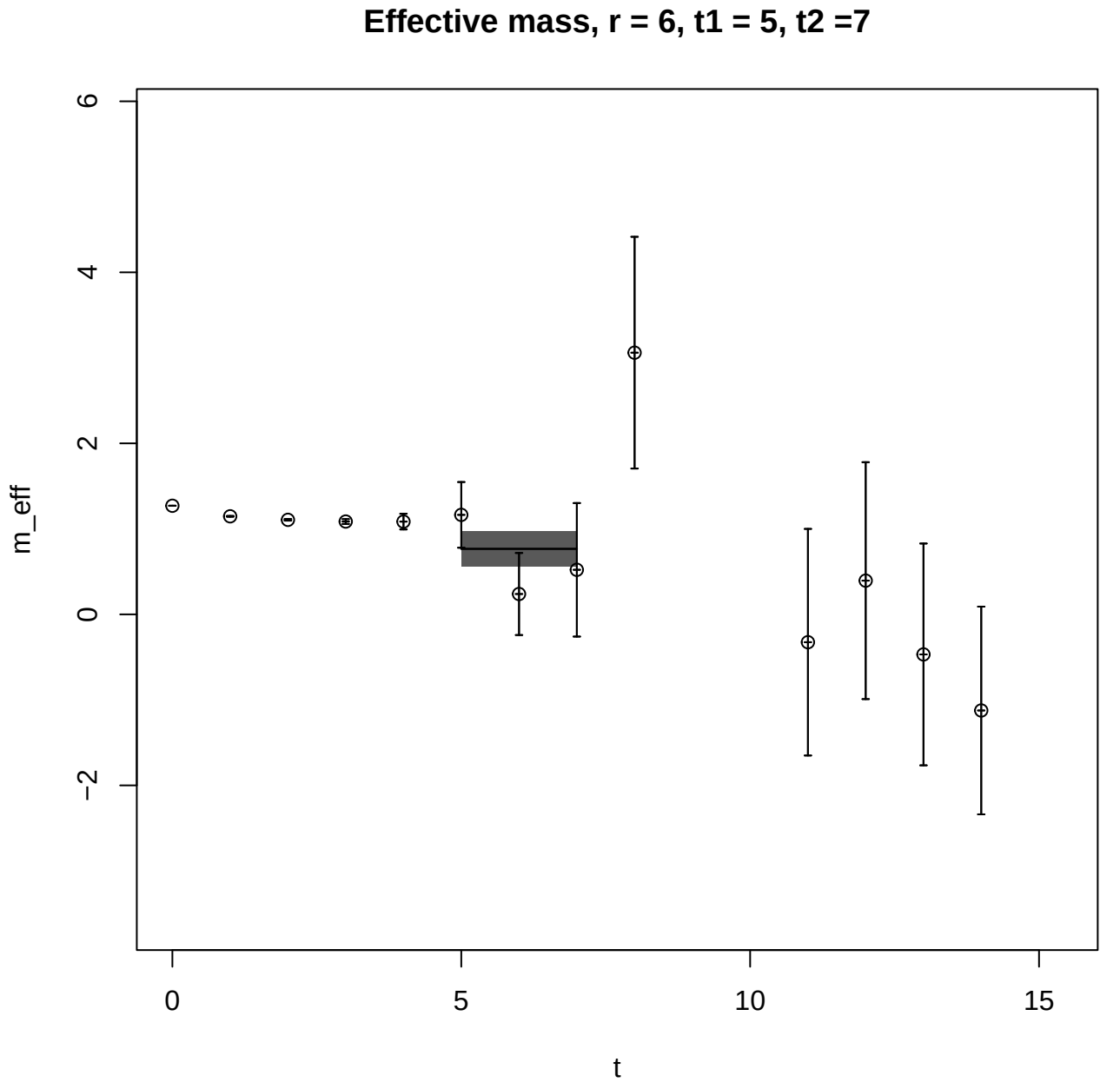

Effective mass, r = 6, t1 = 4, t2 =7
m_eff
t
6
4
2
0
−2
0
5
10
15

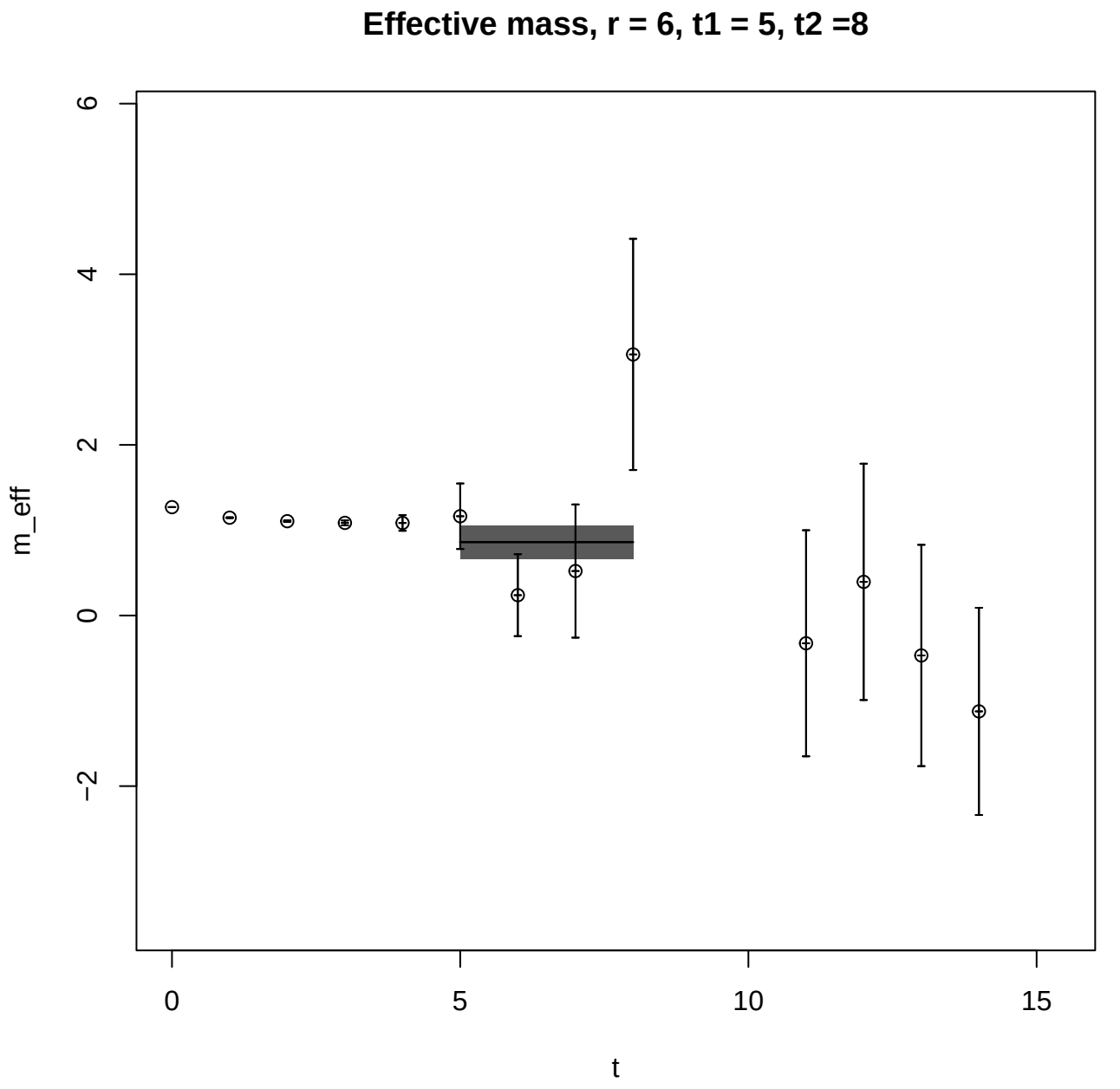

Effective mass, r = 6, t1 = 4, t2 =8
m_eff
t
6
4
2
0
−2
0
5
10
15

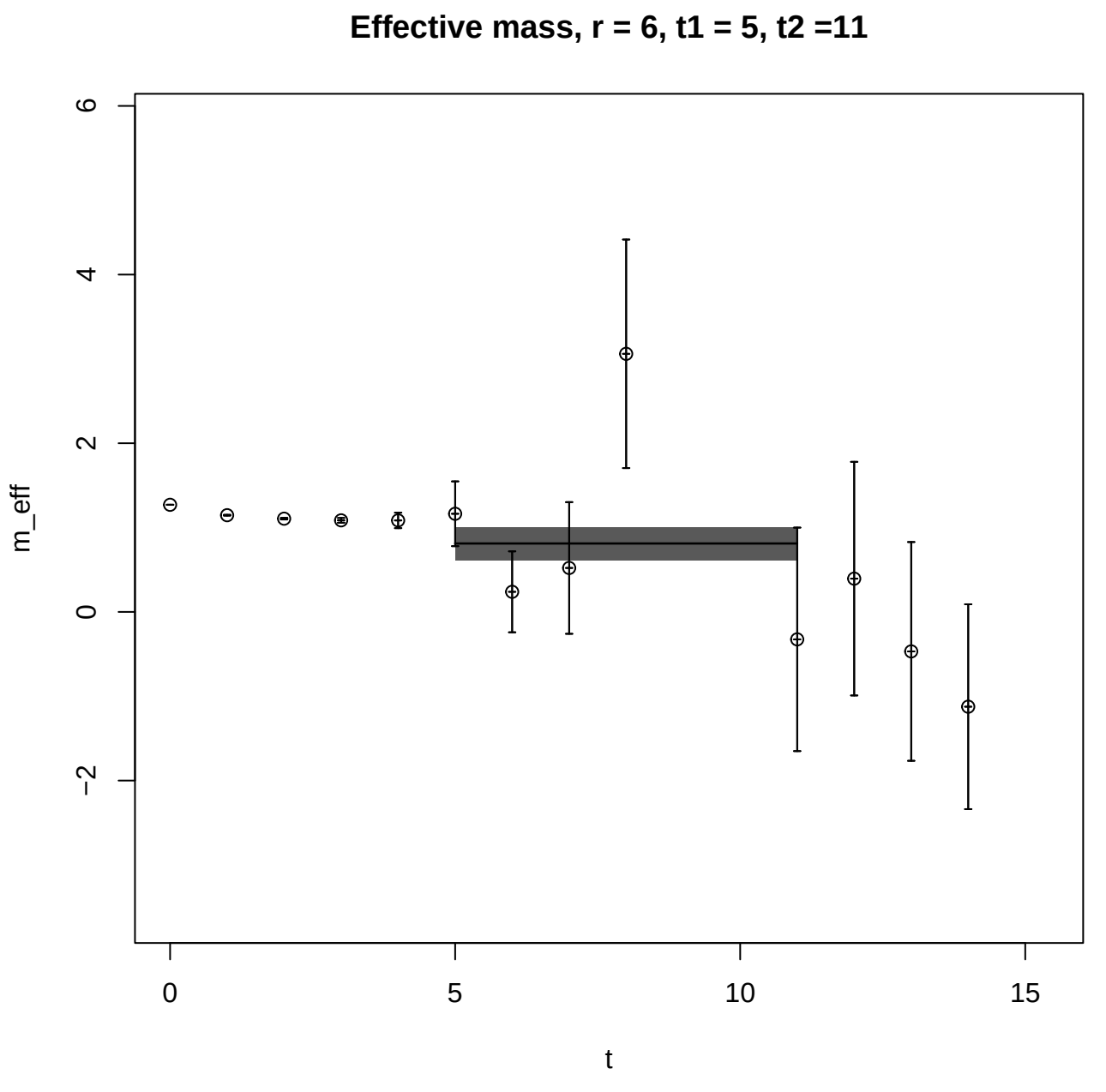

Effective mass, r = 6, t1 = 4, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

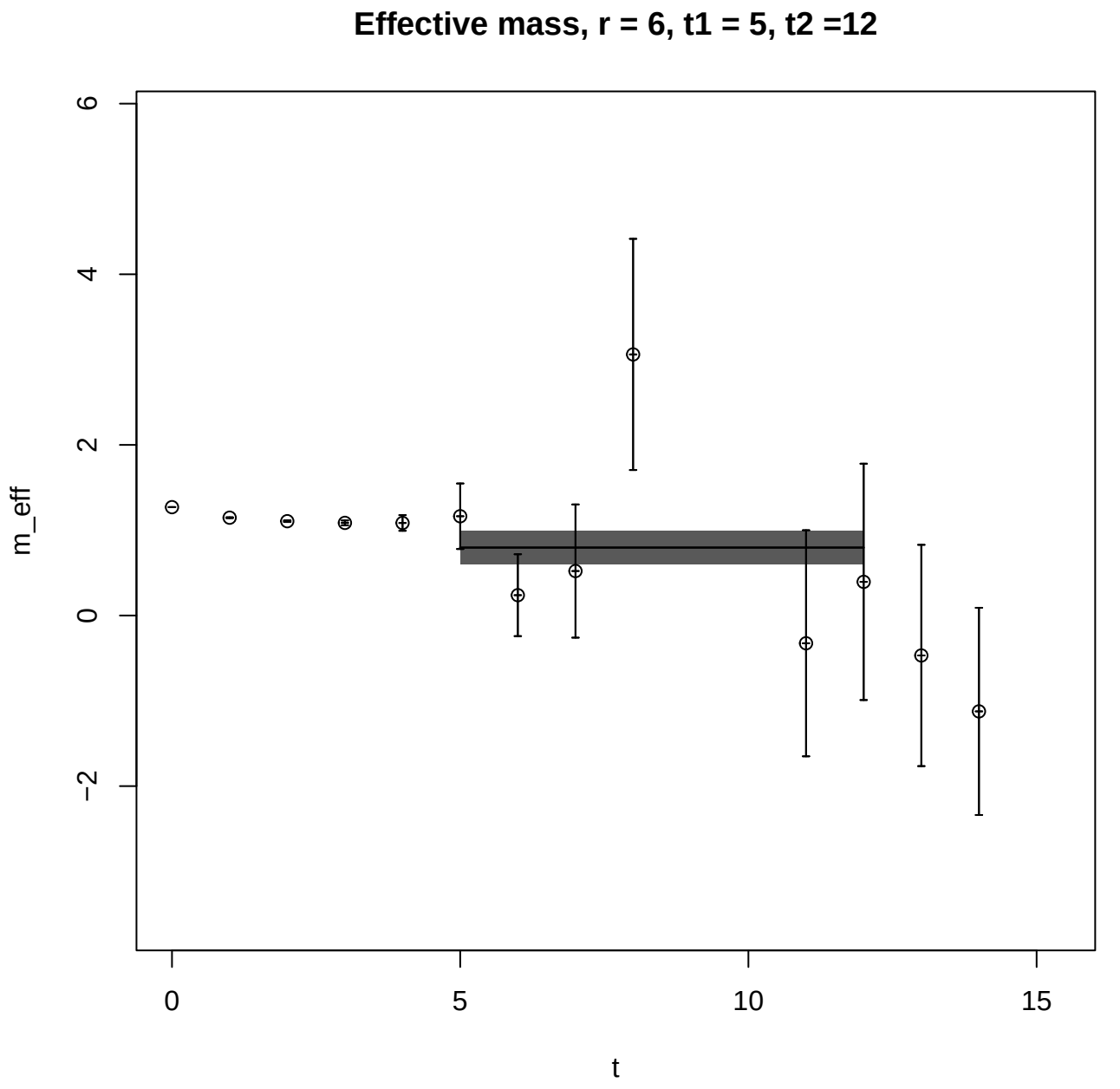

Effective mass, r = 6, t1 = 4, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

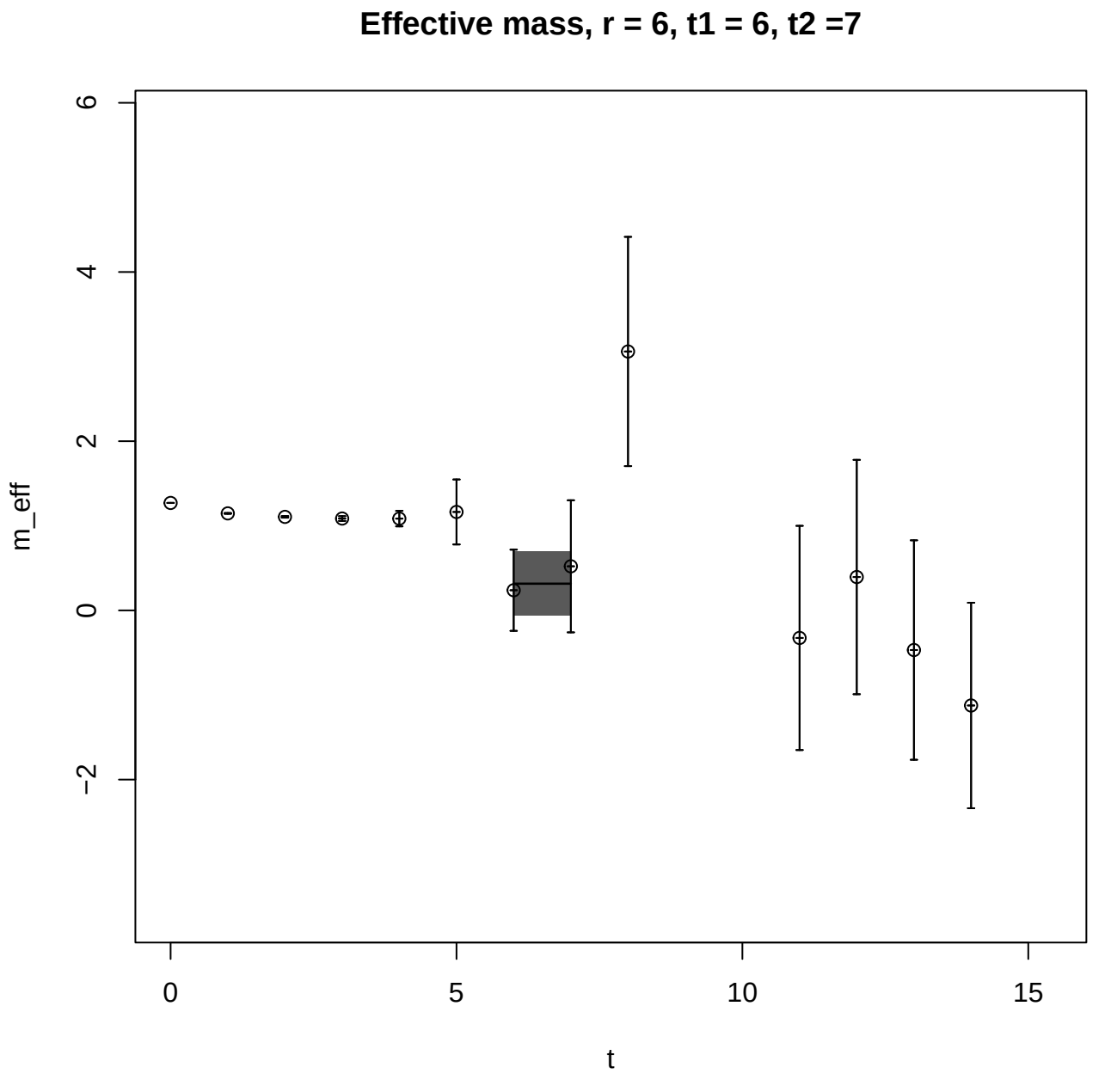

Effective mass, r = 6, t1 = 5, t2 =6
m_eff
6
4
2
0
−2
0
5
10
15
t

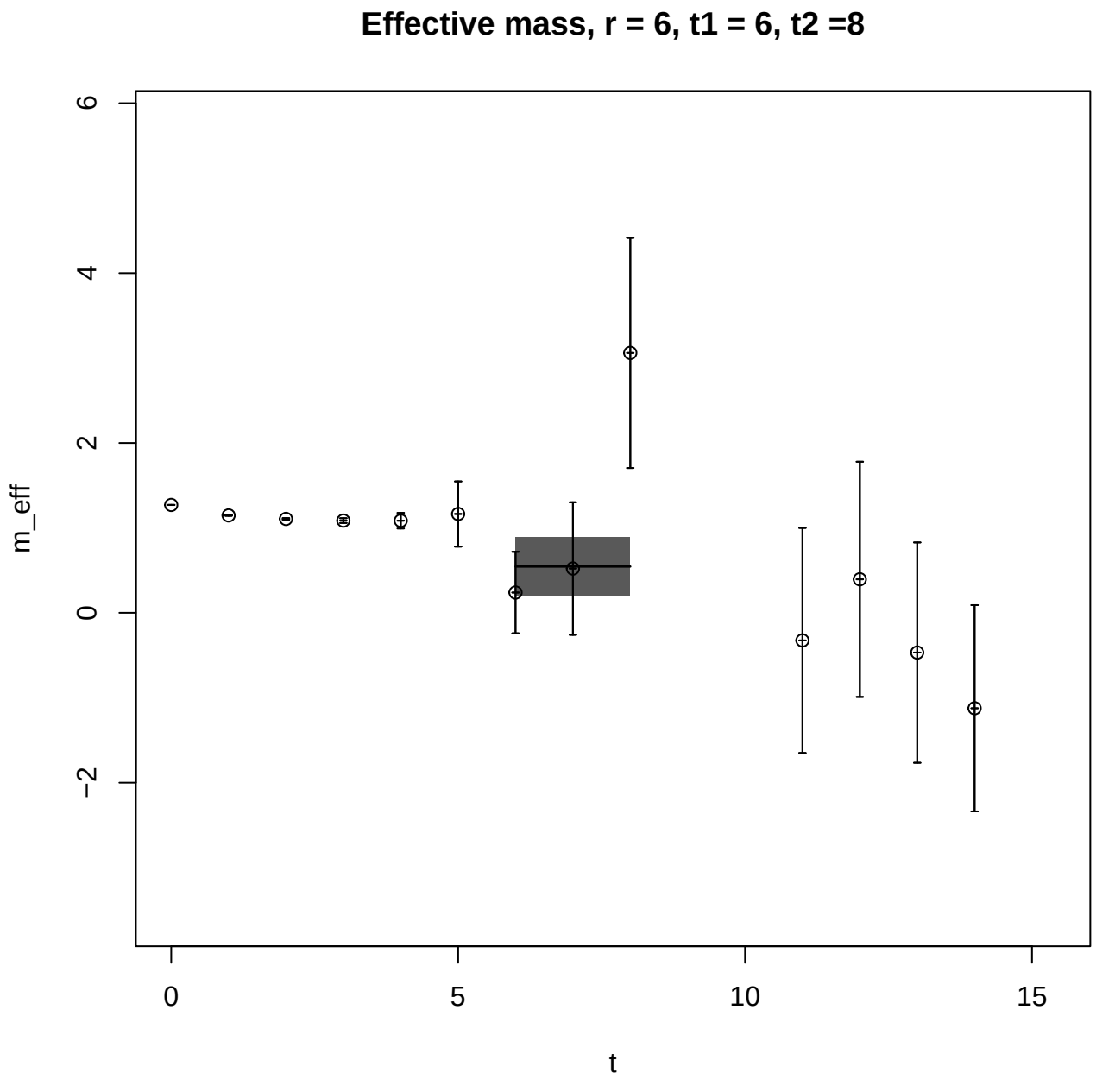

Effective mass, r = 6, t1 = 5, t2 =7
m_eff
6
4
2
0
−2
0
5
10
15
t

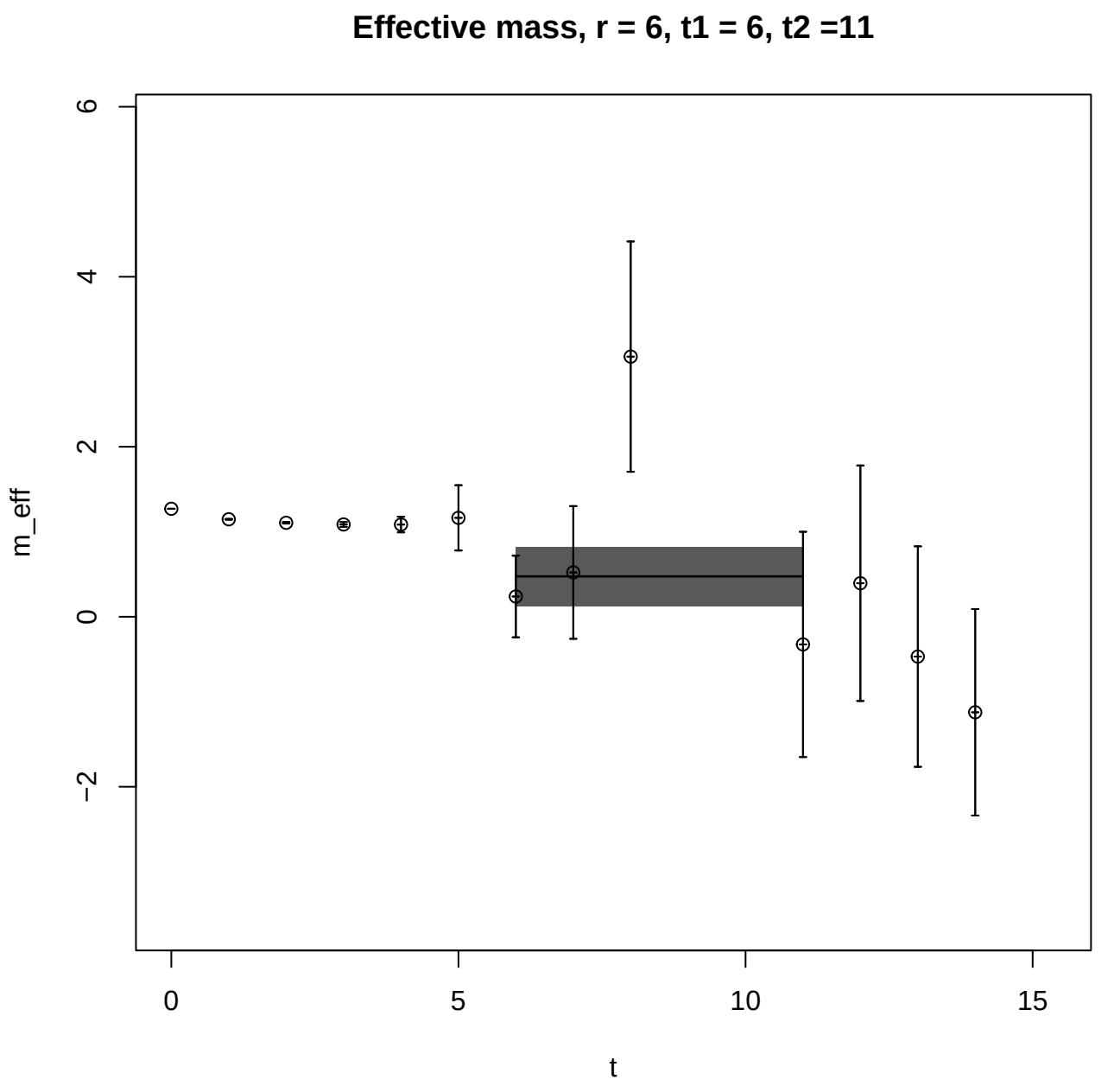

Effective mass, r = 6, t1 = 5, t2 =8
m_eff
t
6
4
2
0
−2
0
5
10
15

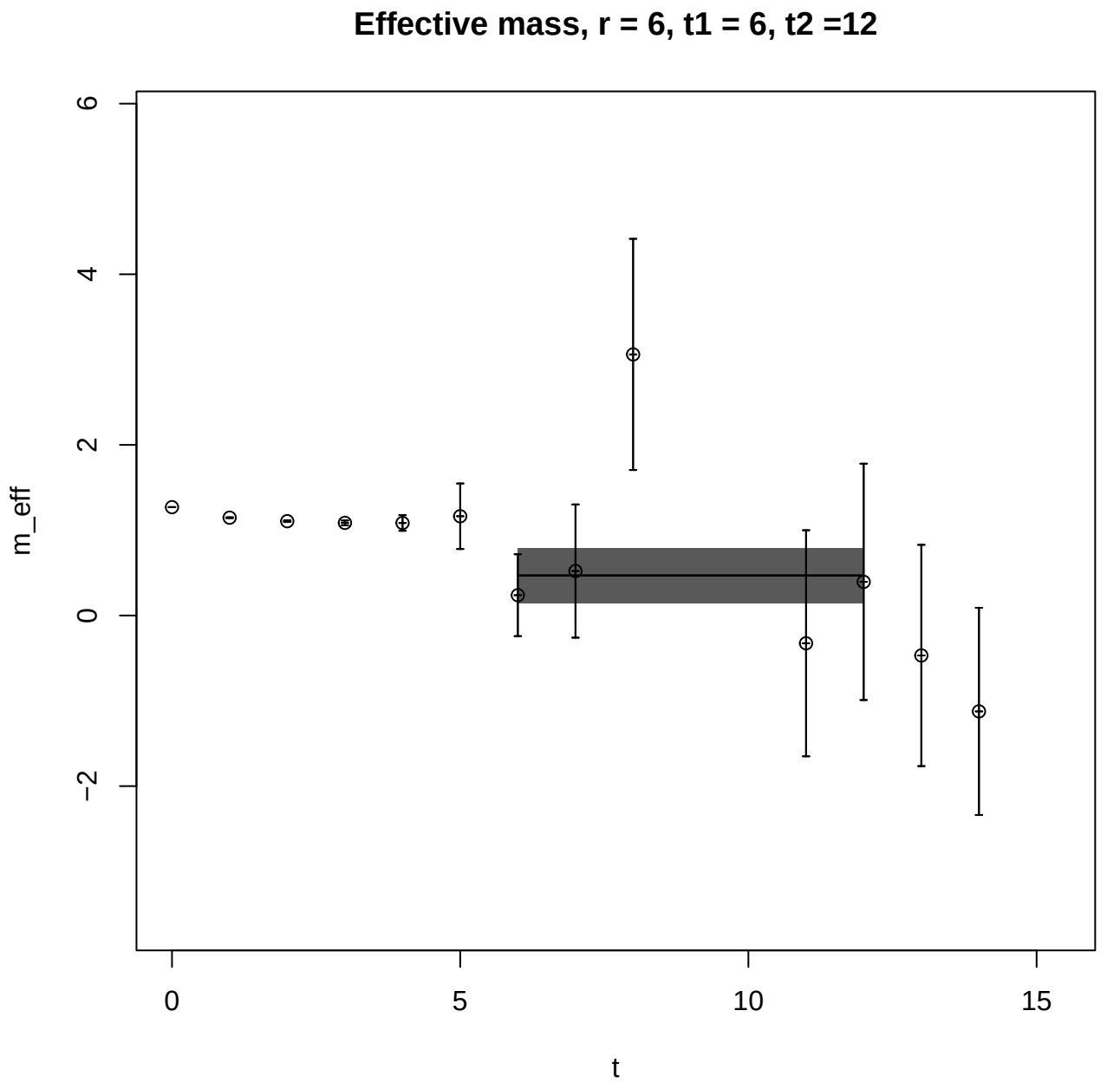

Effective mass, r = 6, t1 = 5, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

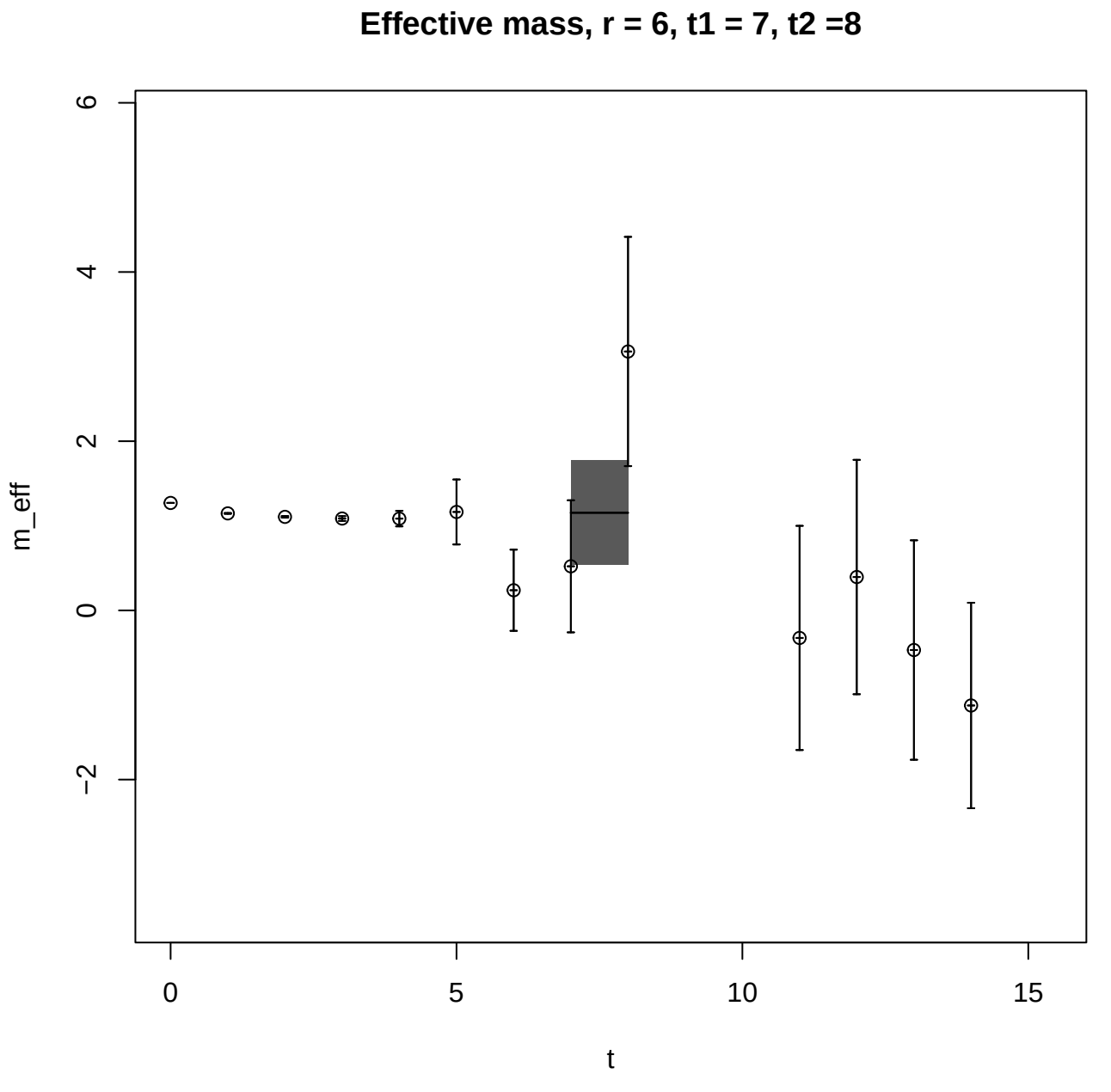

Effective mass, r = 6, t1 = 5, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

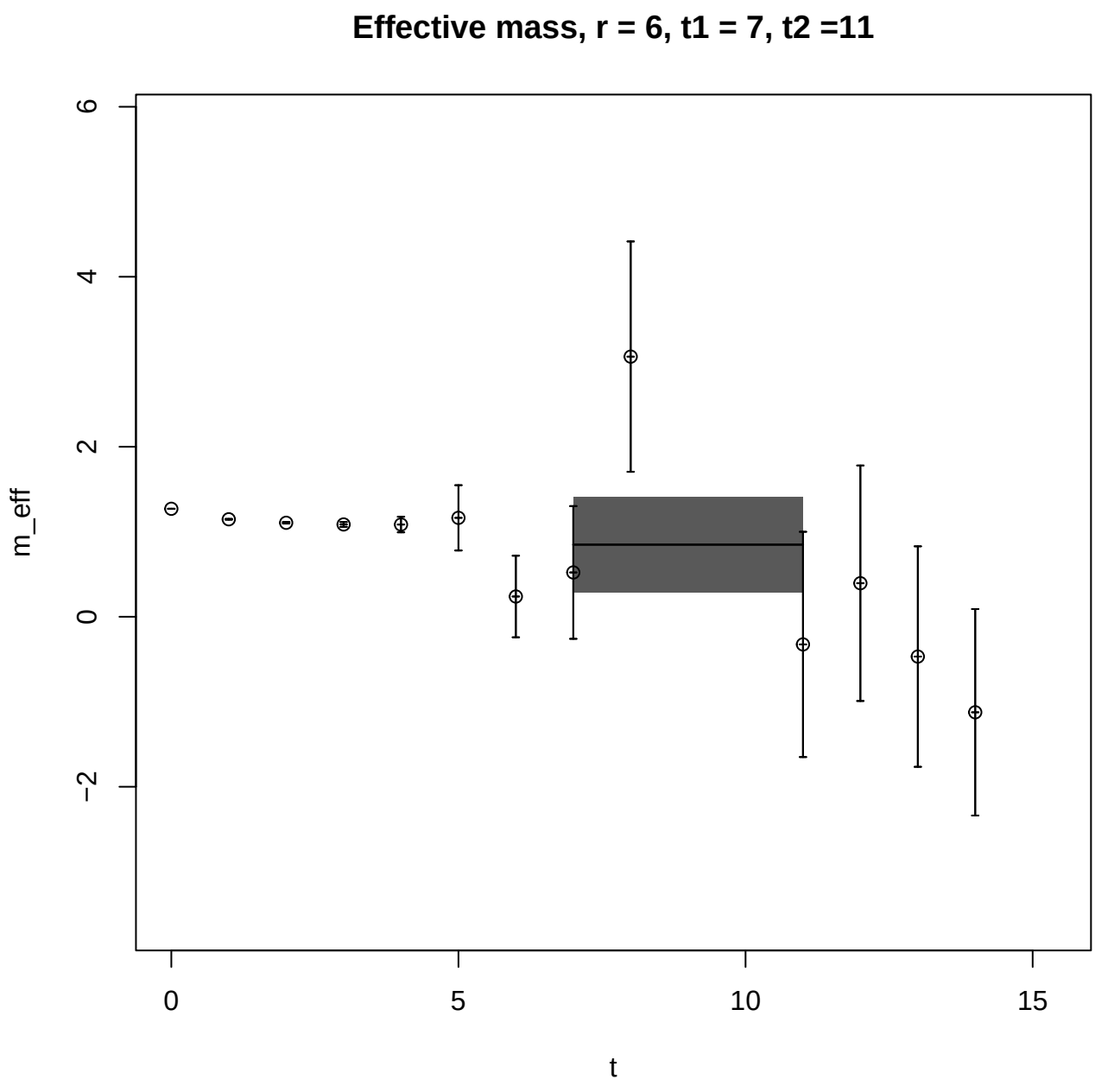

Effective mass, r = 6, t1 = 6, t2 =7
m_eff
6
4
2
0
−2
0
5
10
15
t

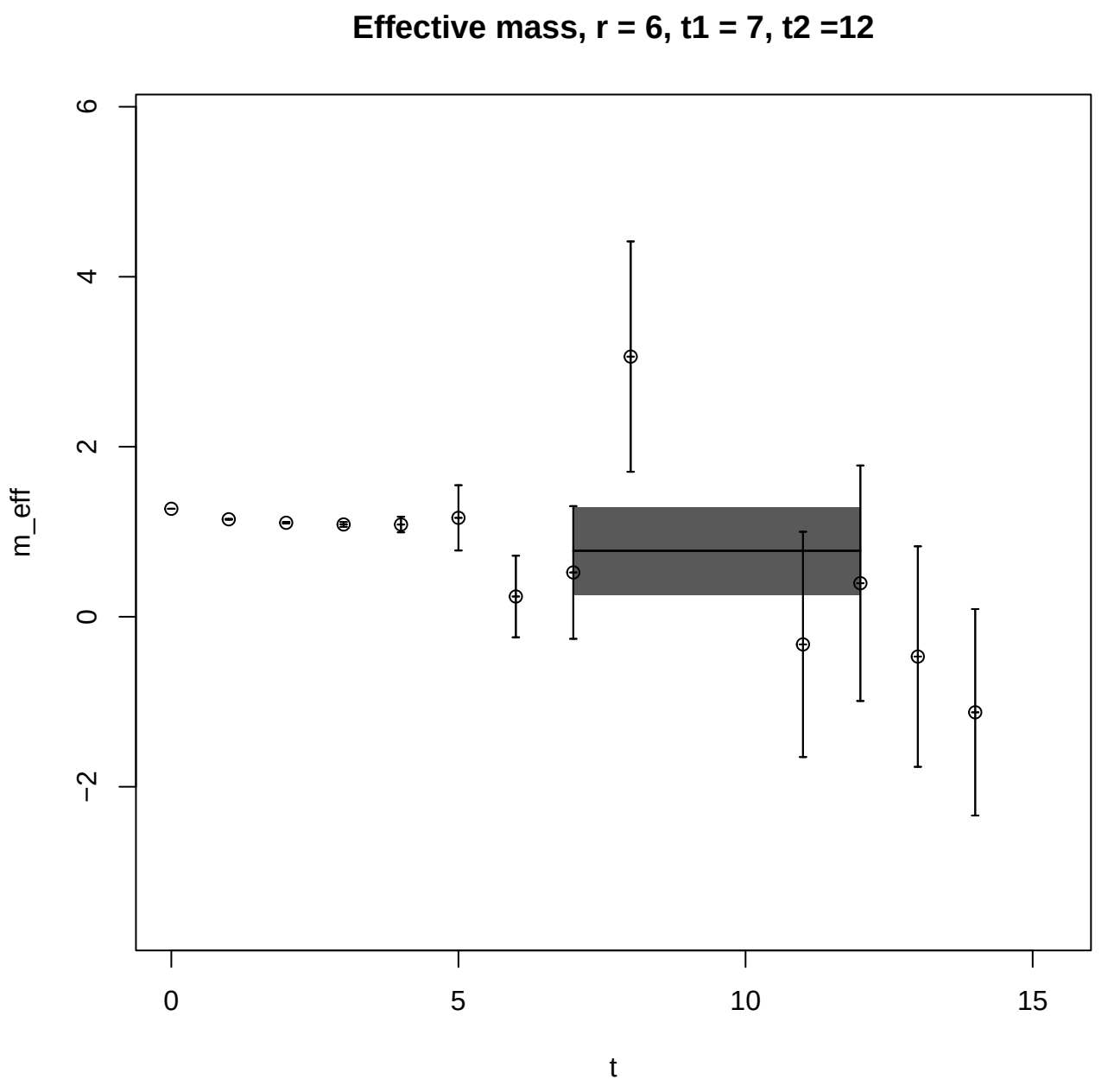

Effective mass, r = 6, t1 = 6, t2 =8
m_eff
t
6
4
2
0
−2
0
5
10
15

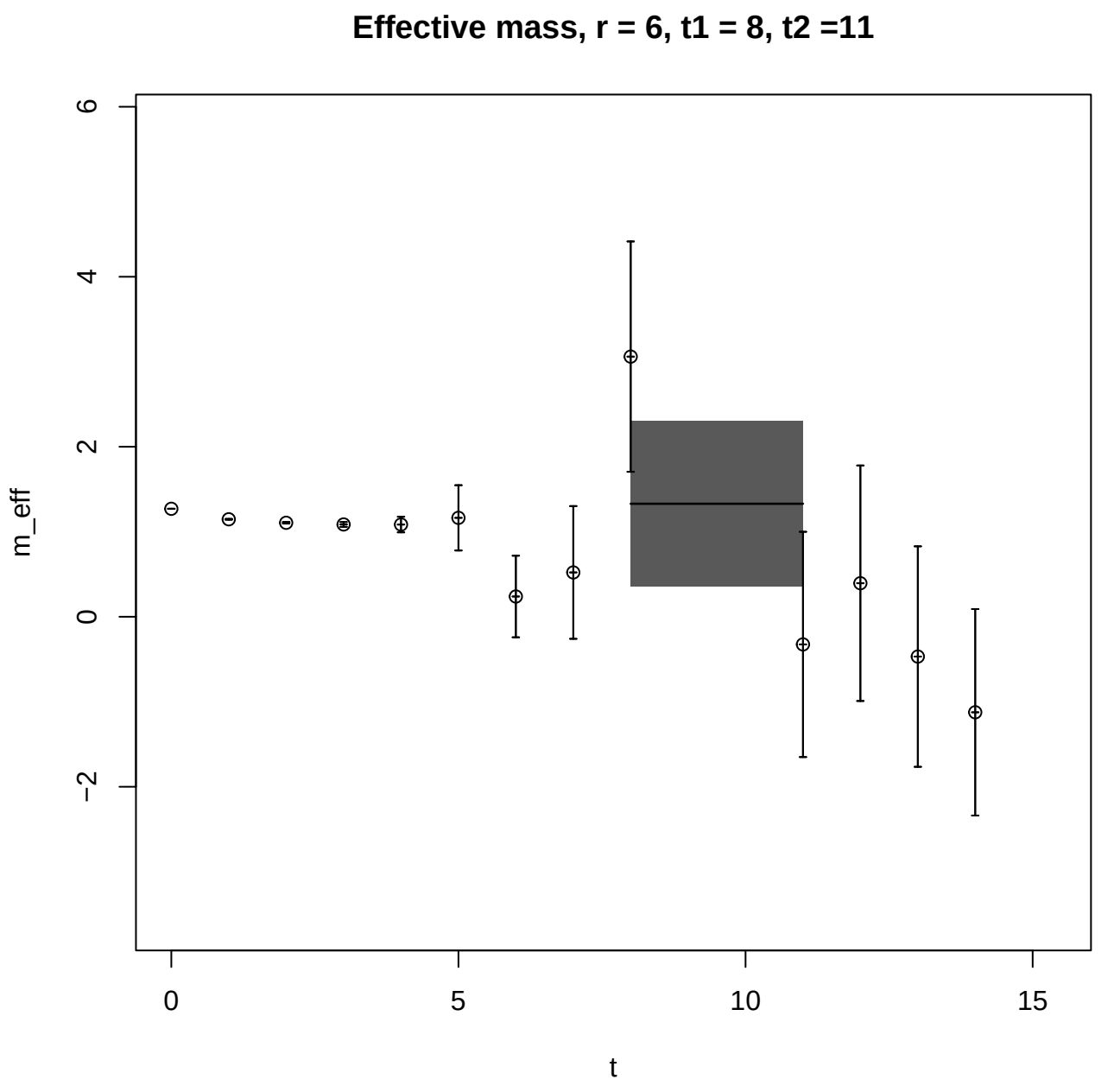

Effective mass, r = 6, t1 = 6, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

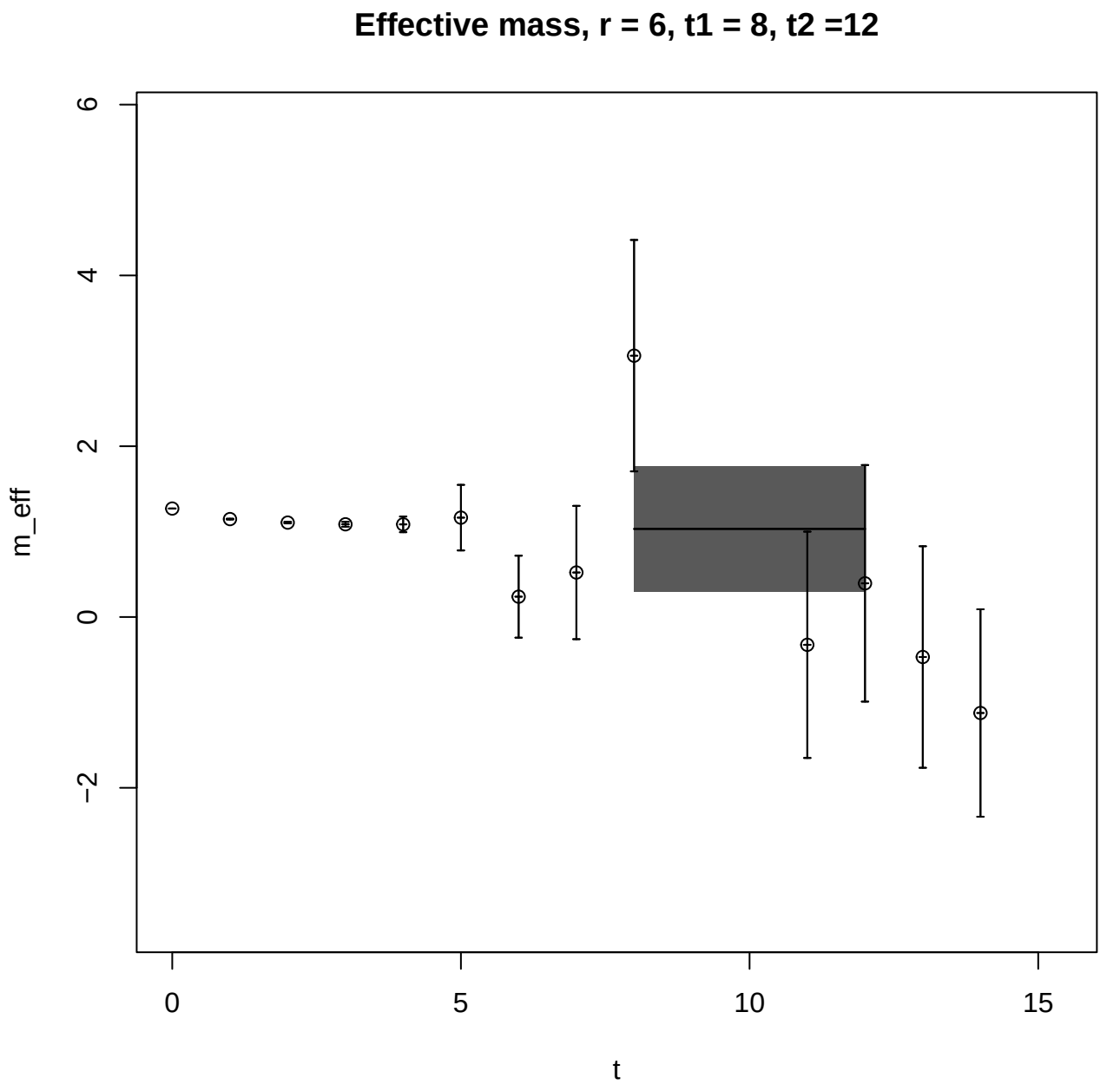

Effective mass, r = 6, t1 = 6, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

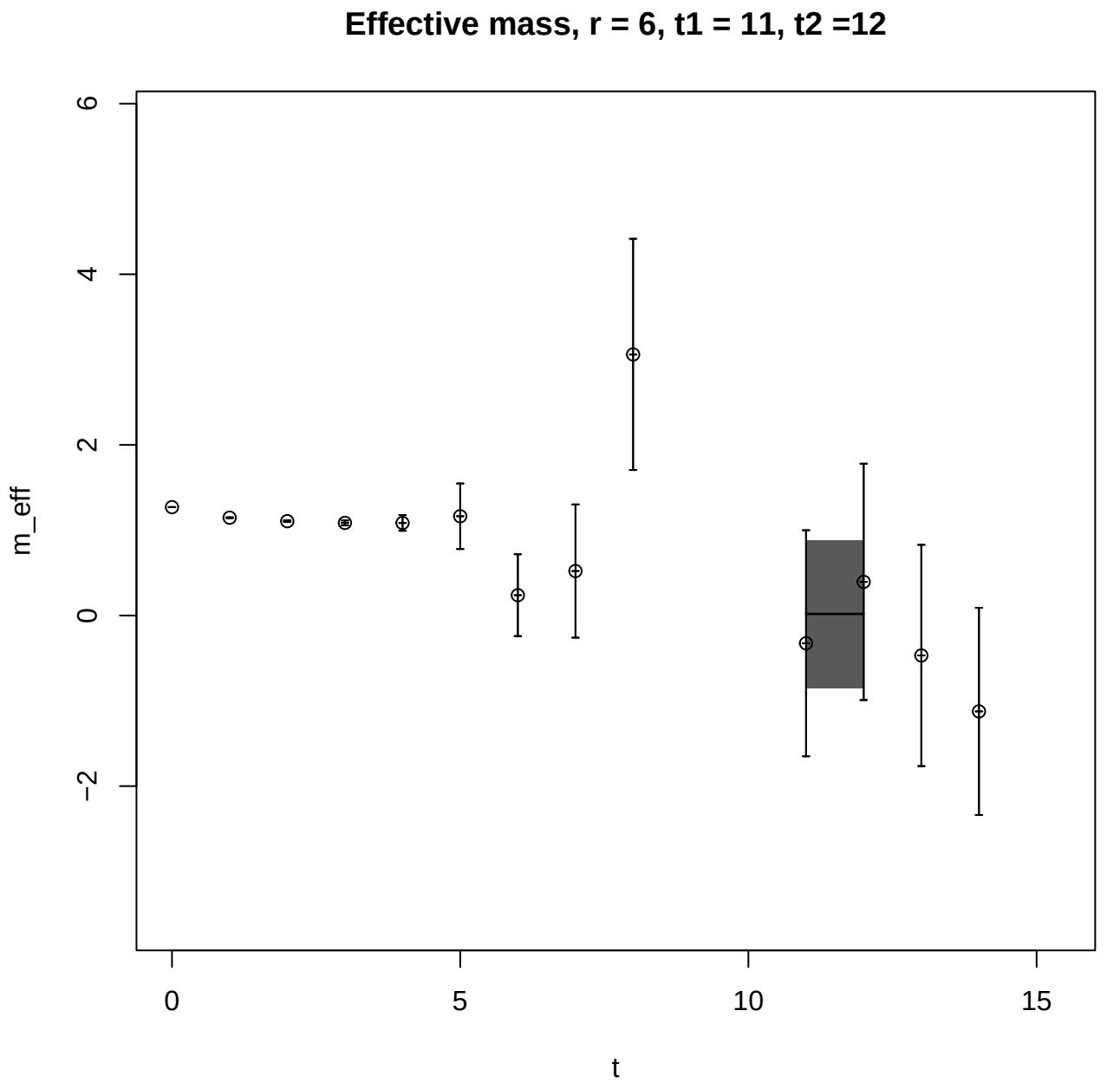

Effective mass, r = 6, t1 = 7, t2 =8
m_eff
t
6
4
2
0
−2
0
5
10
15

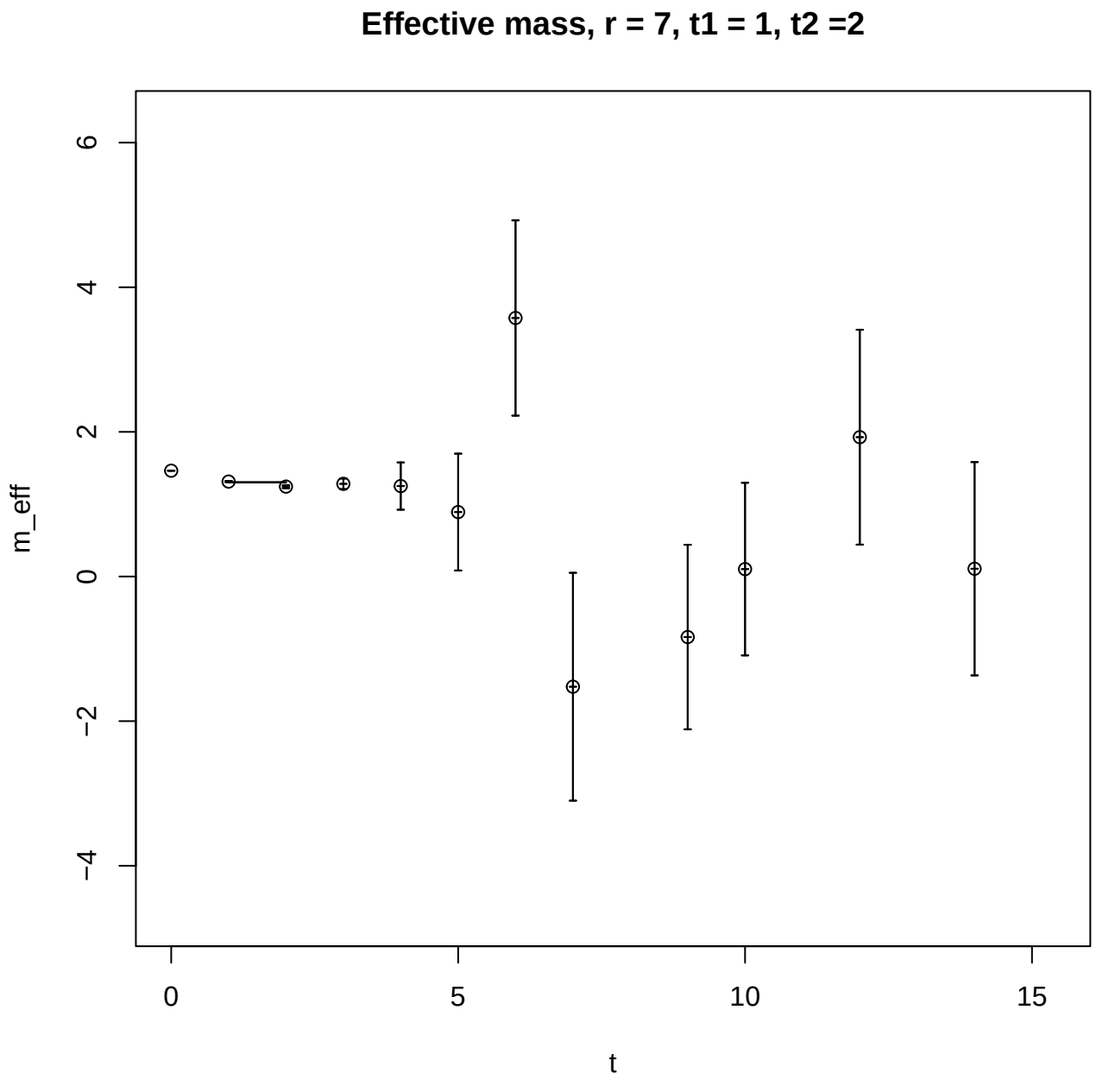

Effective mass, r = 6, t1 = 7, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

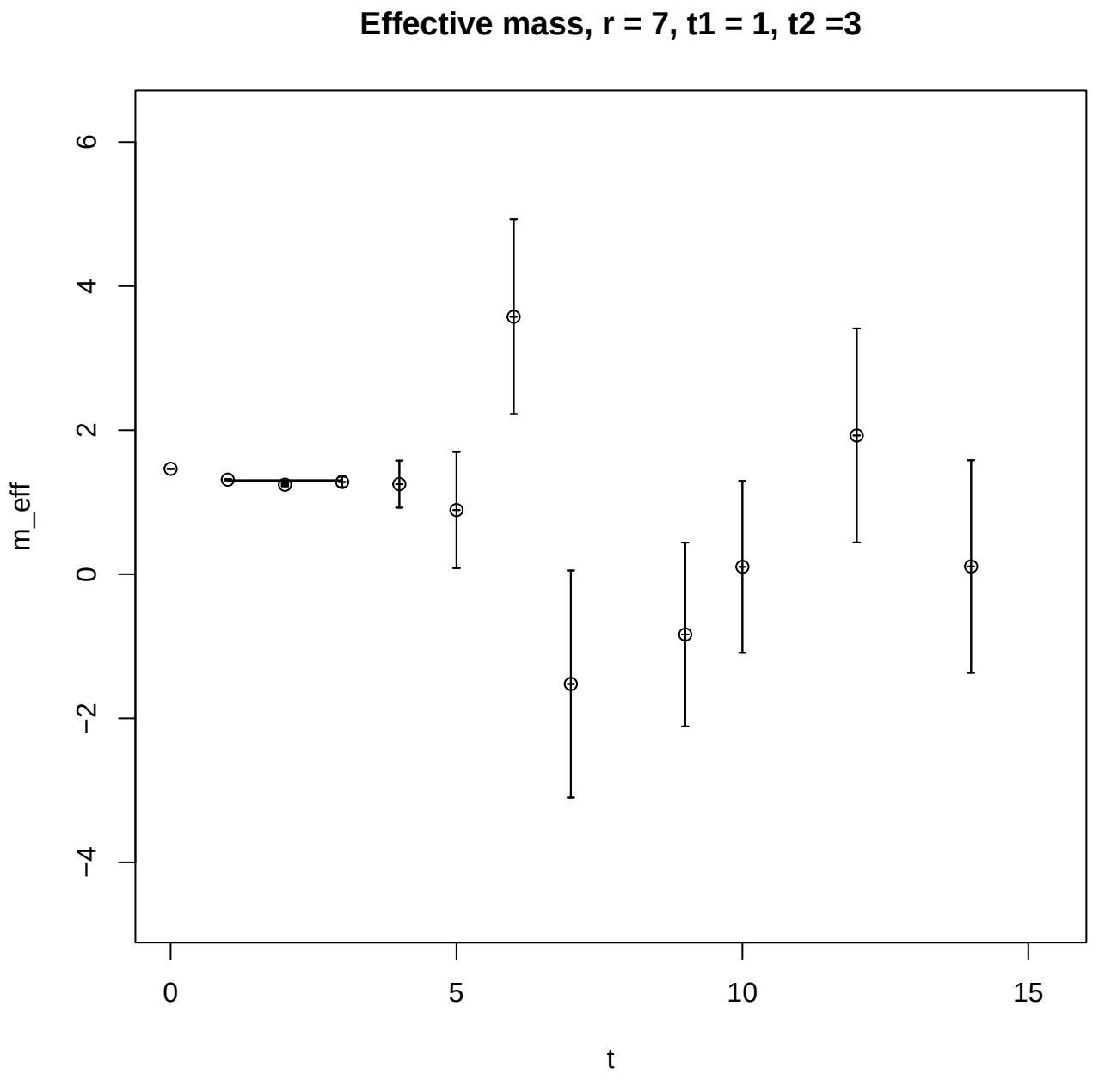

Effective mass, r = 6, t1 = 7, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

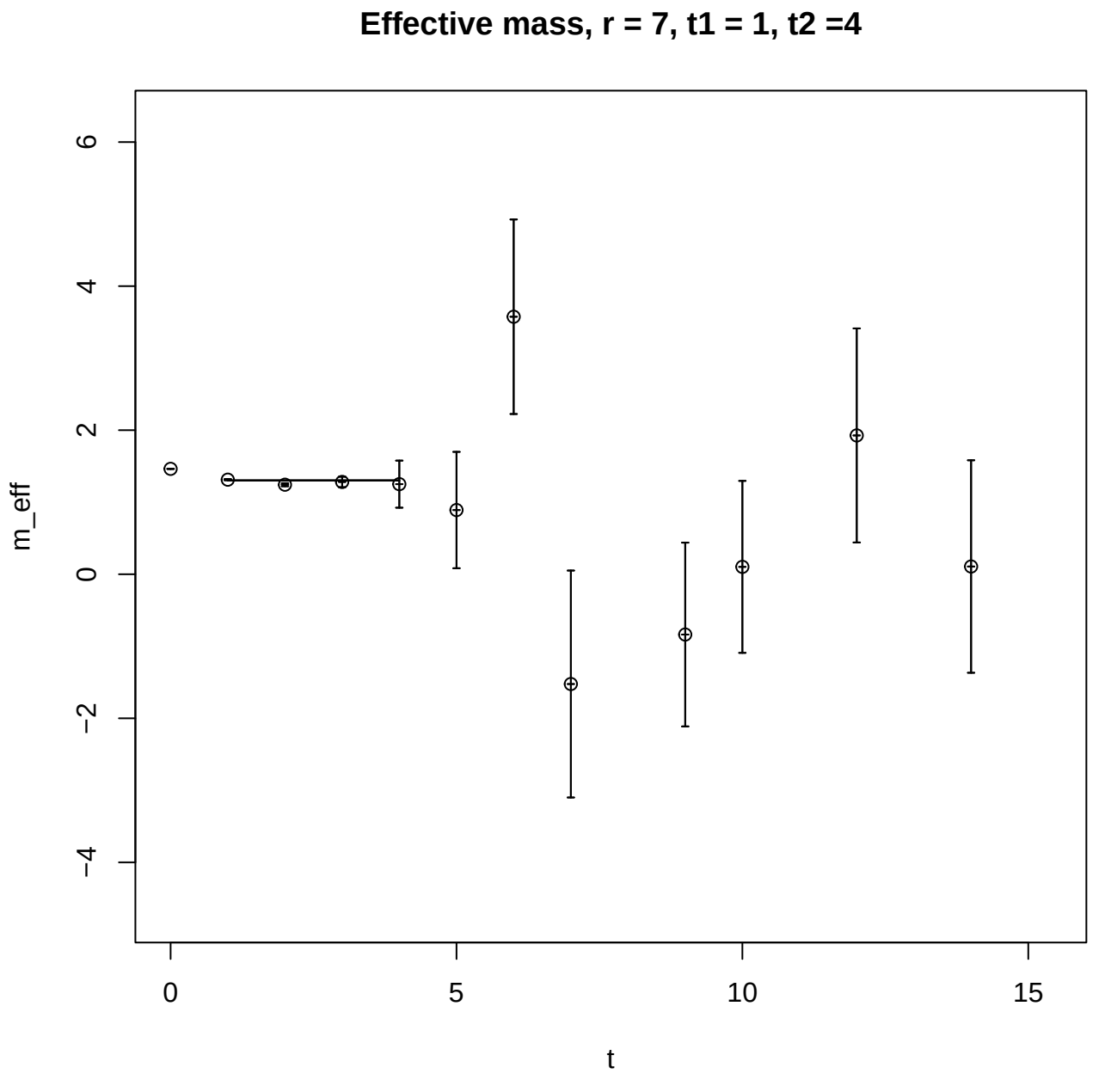

Effective mass, r = 6, t1 = 8, t2 =11
m_eff
t
6
4
2
0
−2
0
5
10
15

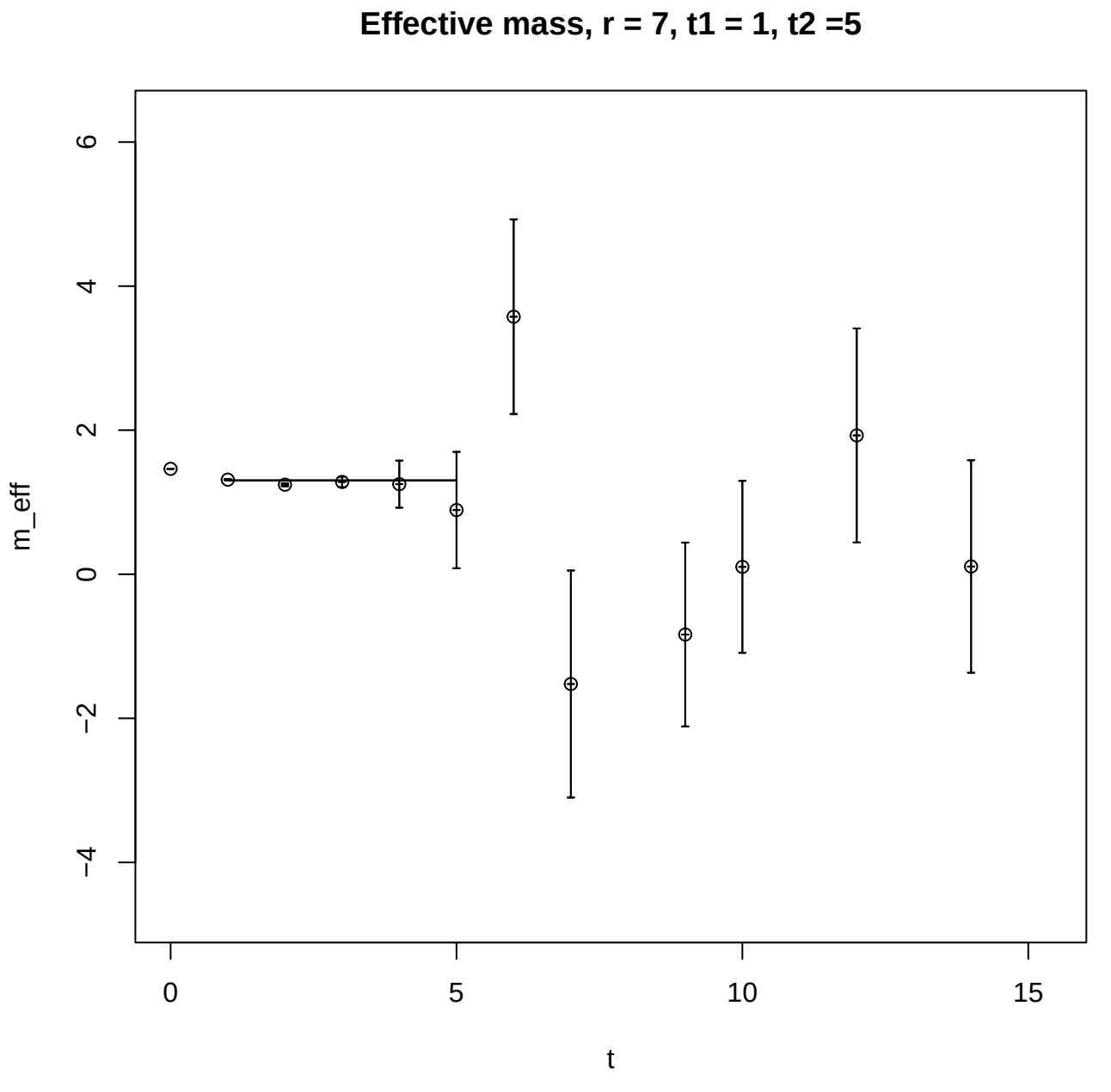

Effective mass, r = 6, t1 = 8, t2 =12
m_eff
t
6
4
2
0
−2
0
5
10
15

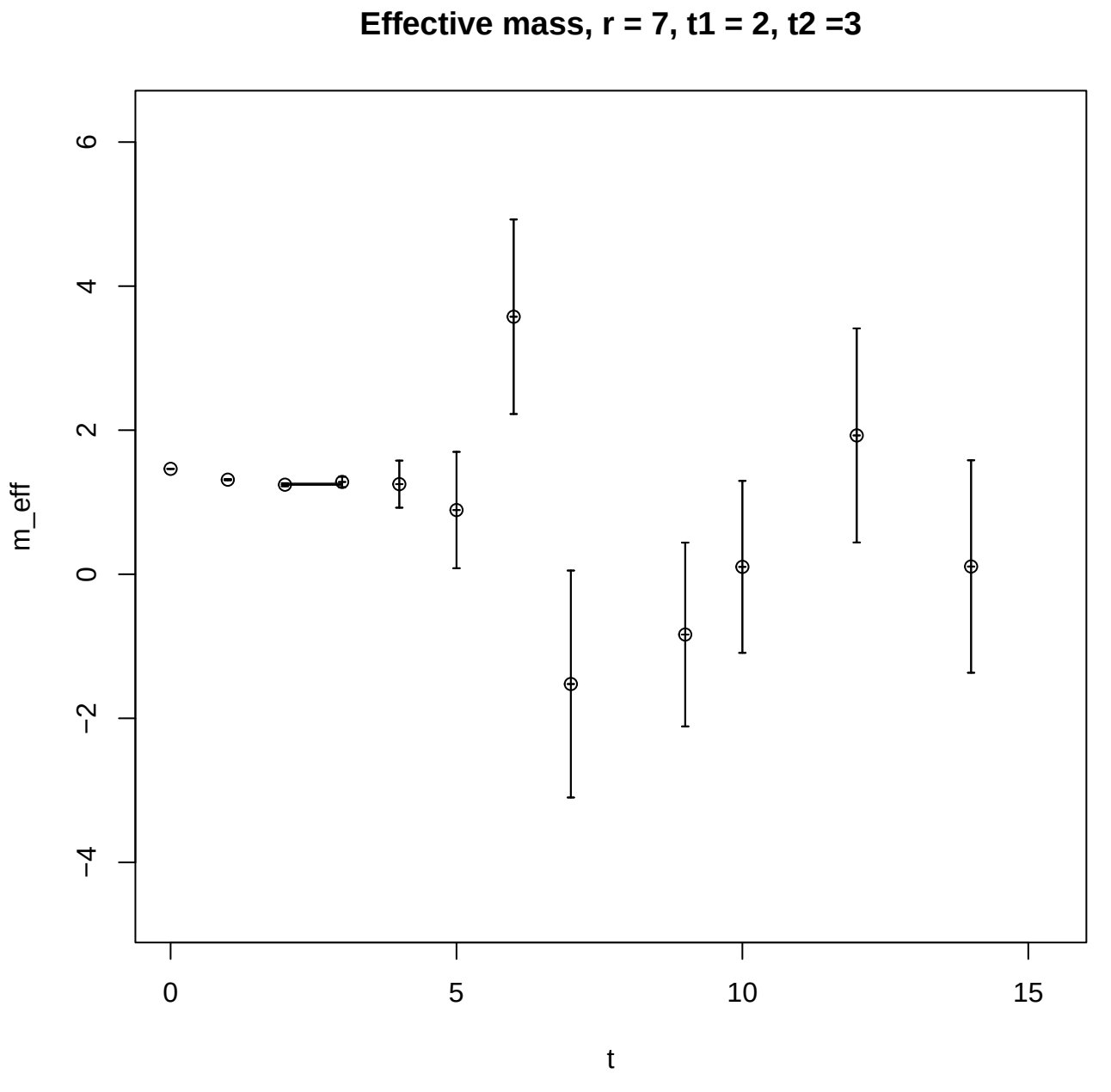

Effective mass, r = 6, t1 = 11, t2 =12
m_eff
6
4
2
0
−2
0
5
10
15
t

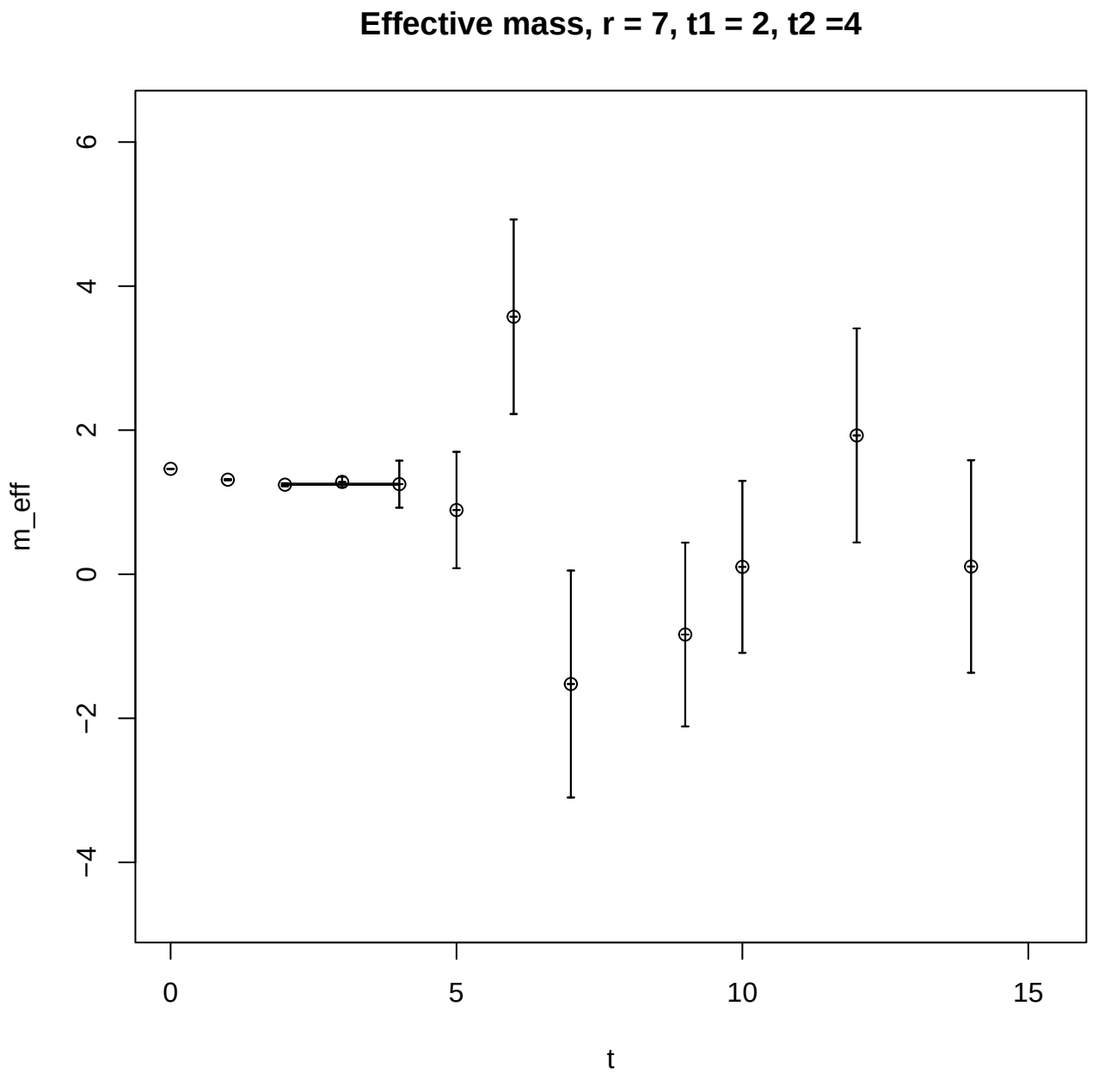

Effective mass, r = 7, t1 = 1, t2 =2
m_eff
6
4
2
0
−2
−4
0
5
10
15
t

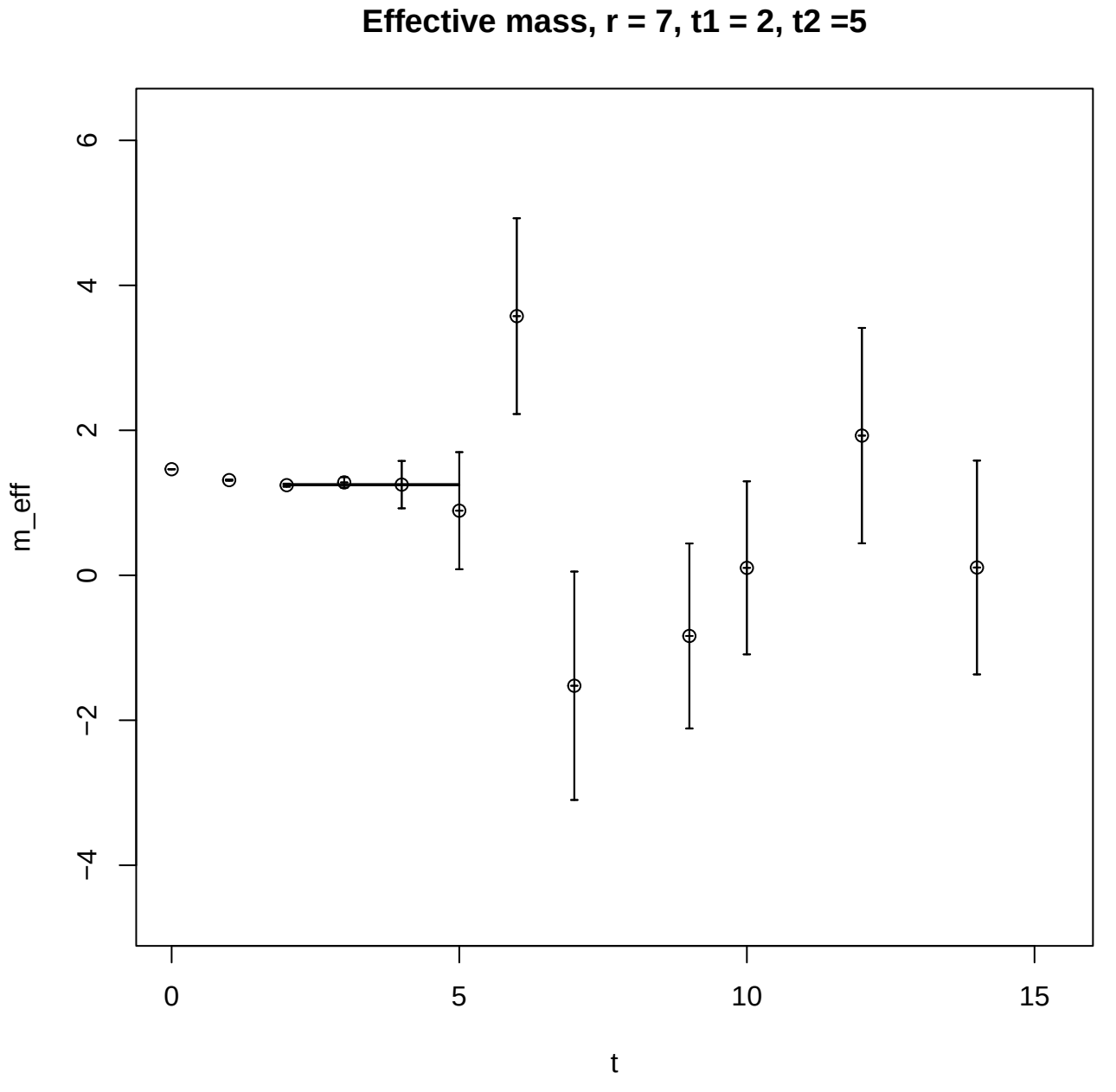

Effective mass, r = 7, t1 = 1, t2 =3
m_eff
6
4
2
0
−2
−4
0
5
10
15
t

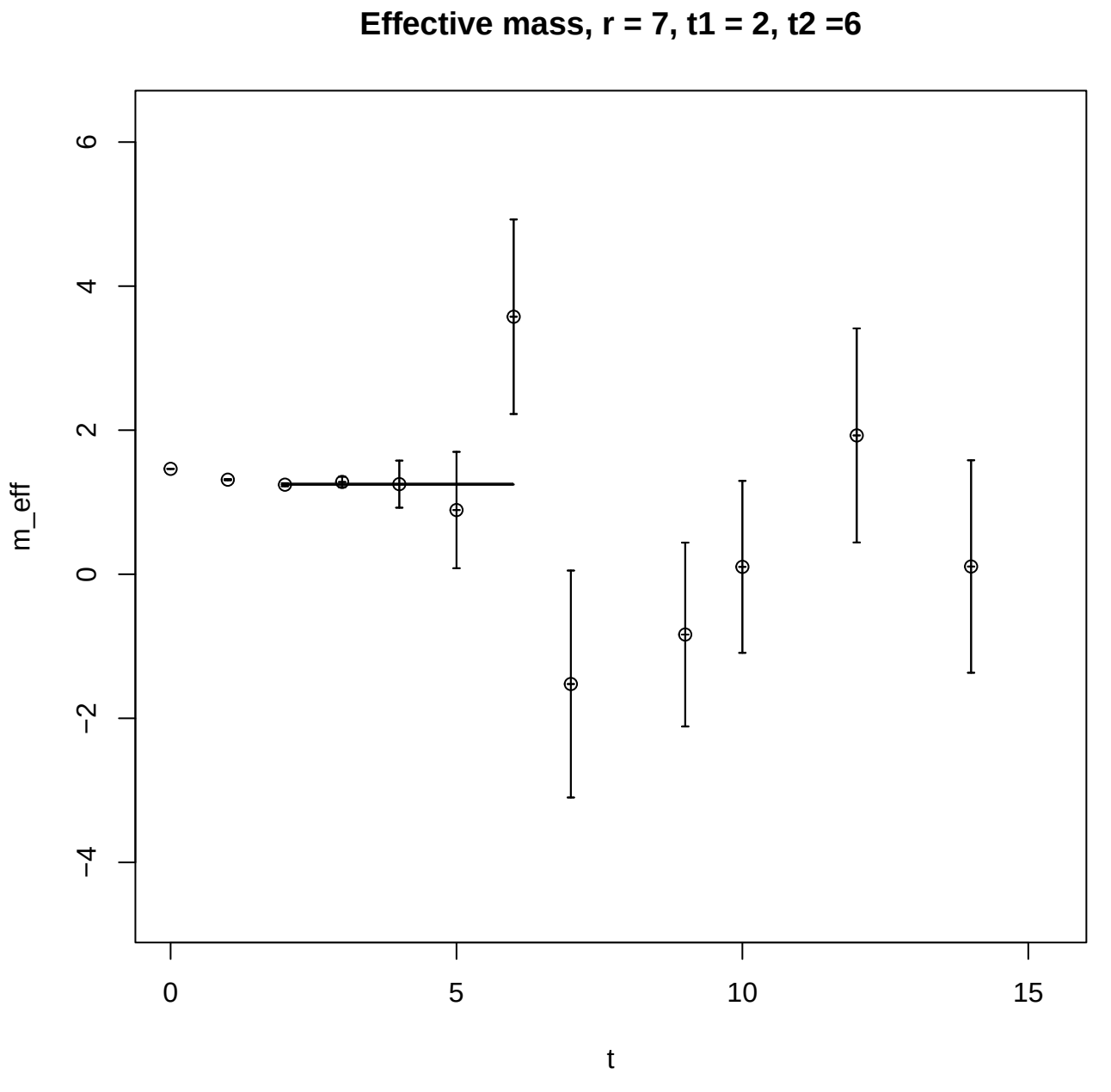

Effective mass, r = 7, t1 = 1, t2 =4
m_eff
6
4
2
0
−2
−4
0
5
10
15
t

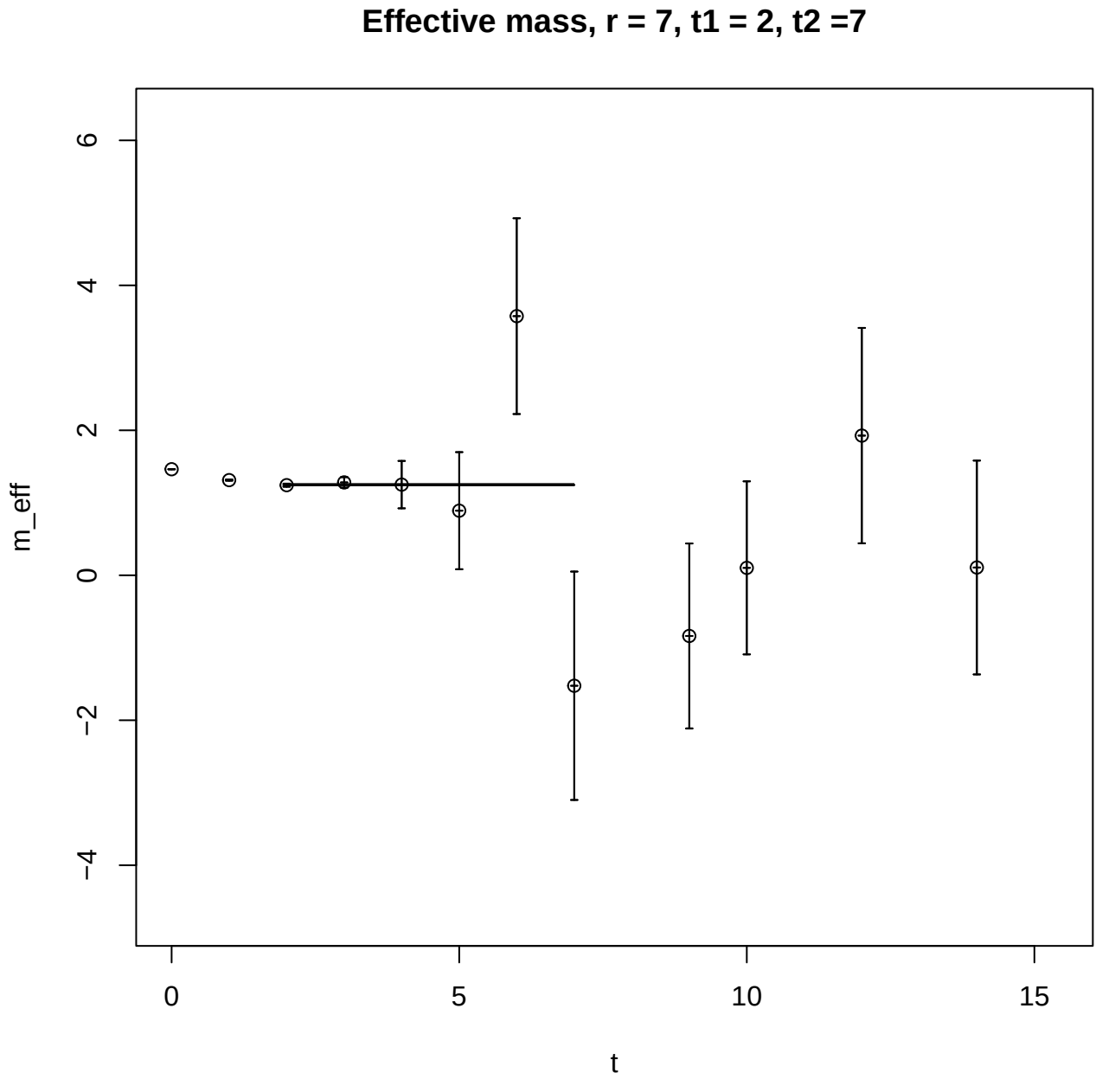

Effective mass, r = 7, t1 = 1, t2 =5
m_eff
t
6
4
2
0
−2
−4
0
5
10
15

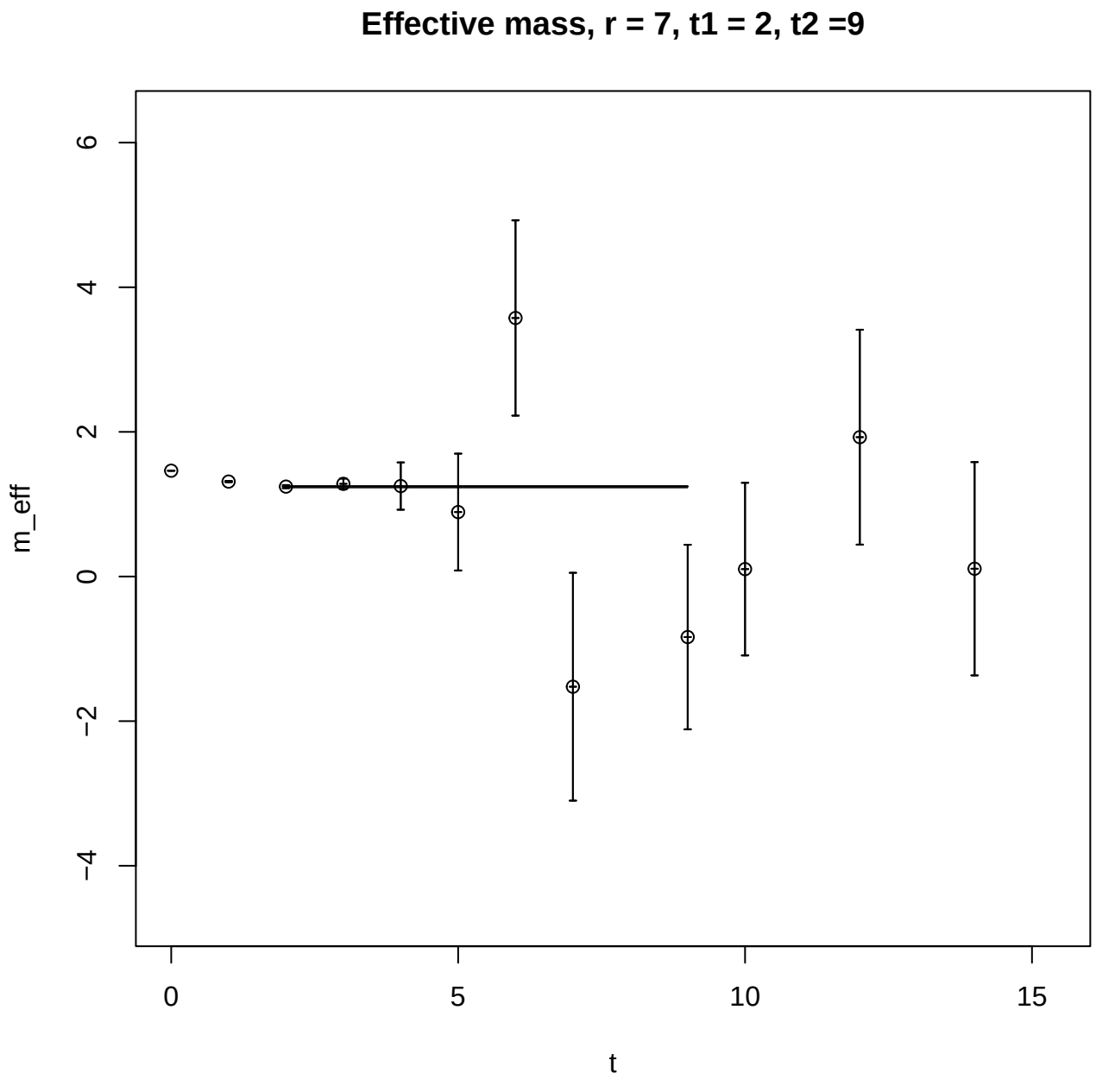

Effective mass, r = 7, t1 = 2, t2 =3
m_eff
6
4
2
0
−2
−4
0
5
10
15
t

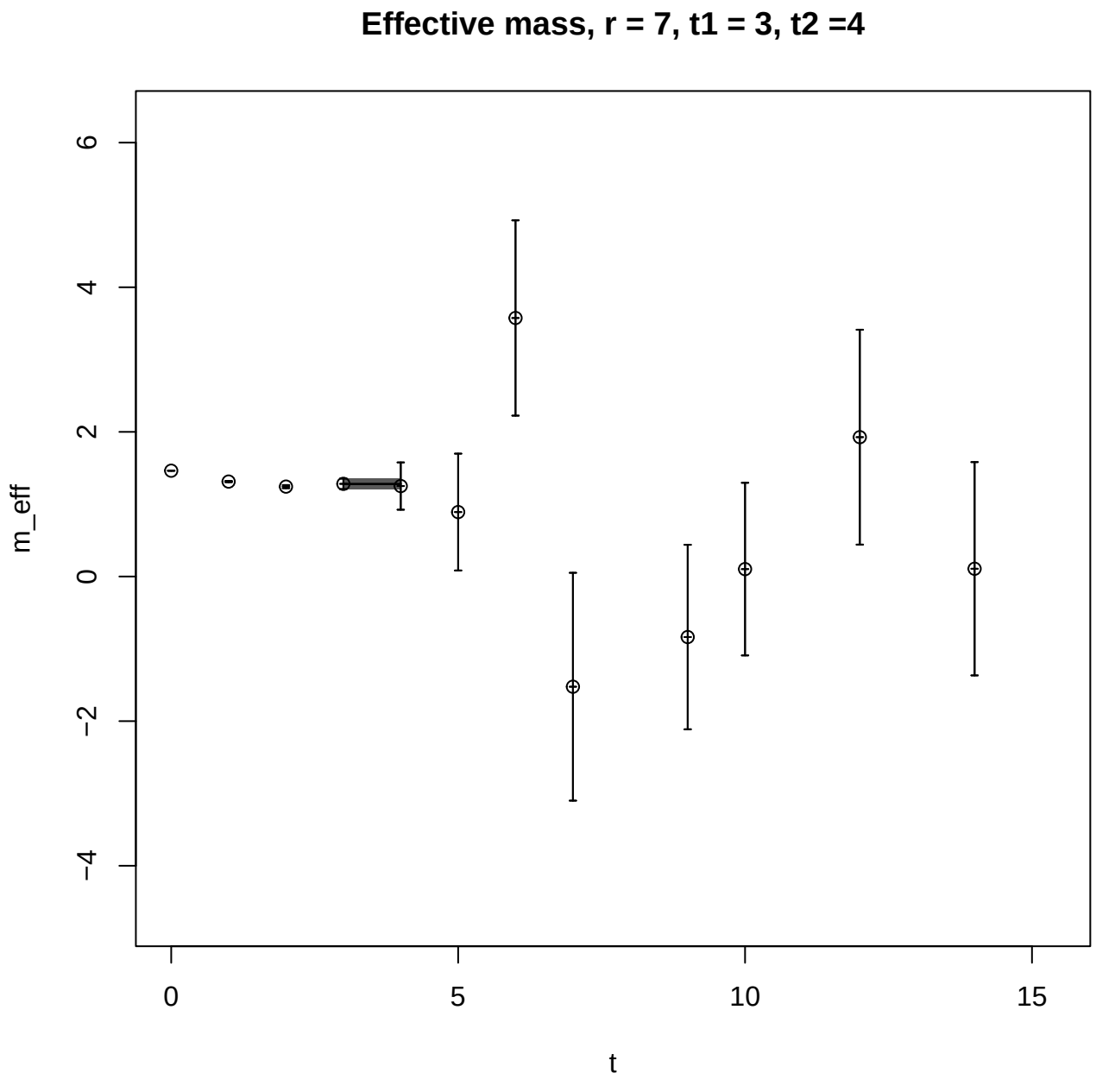

Effective mass, r = 7, t1 = 2, t2 =4
m_eff
t
6
4
2
0
−2
−4
0
5
10
15

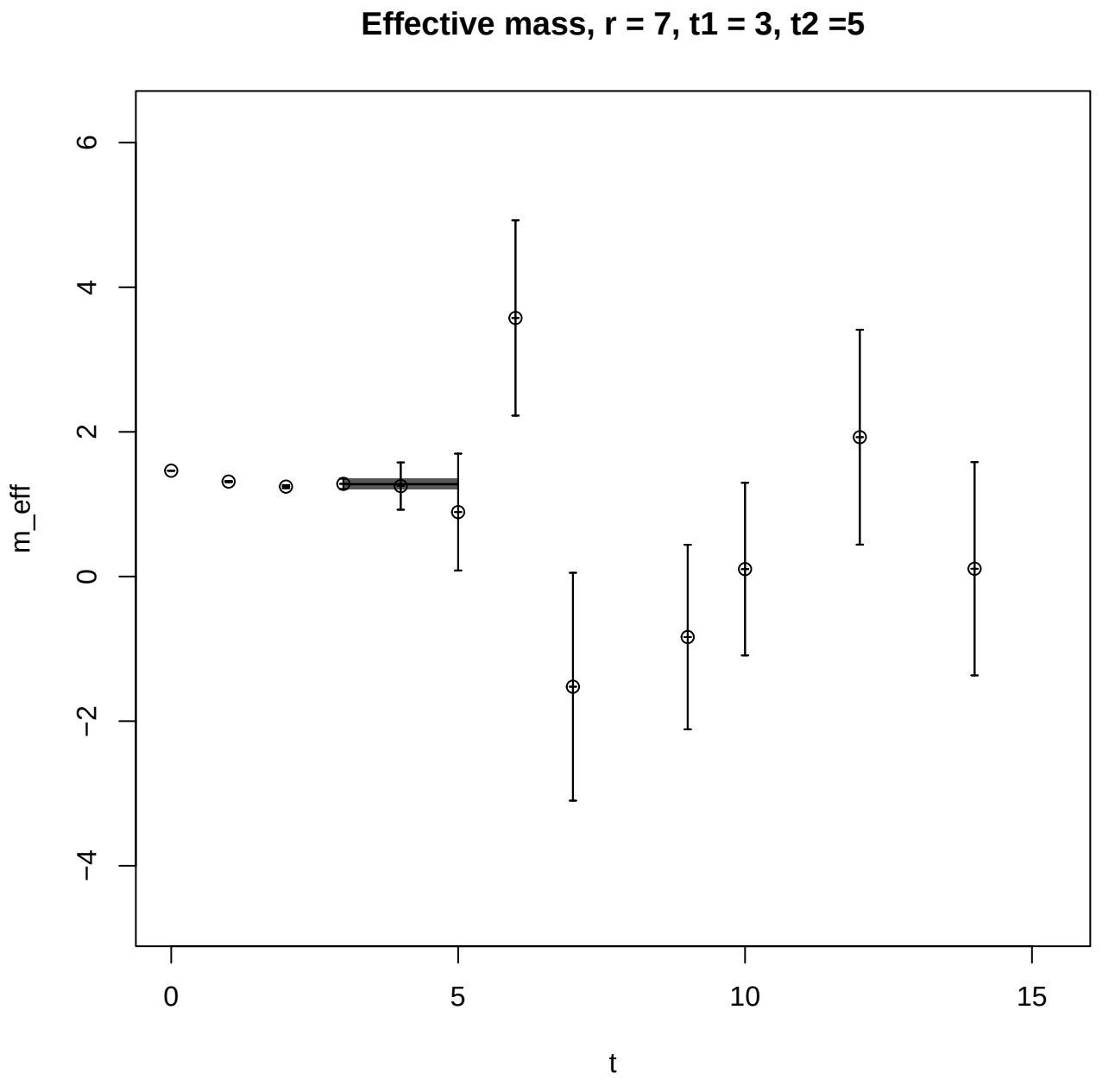

Effective mass, r = 7, t1 = 2, t2 =5
m_eff
t
6
4
2
0
−2
−4
0
5
10
15

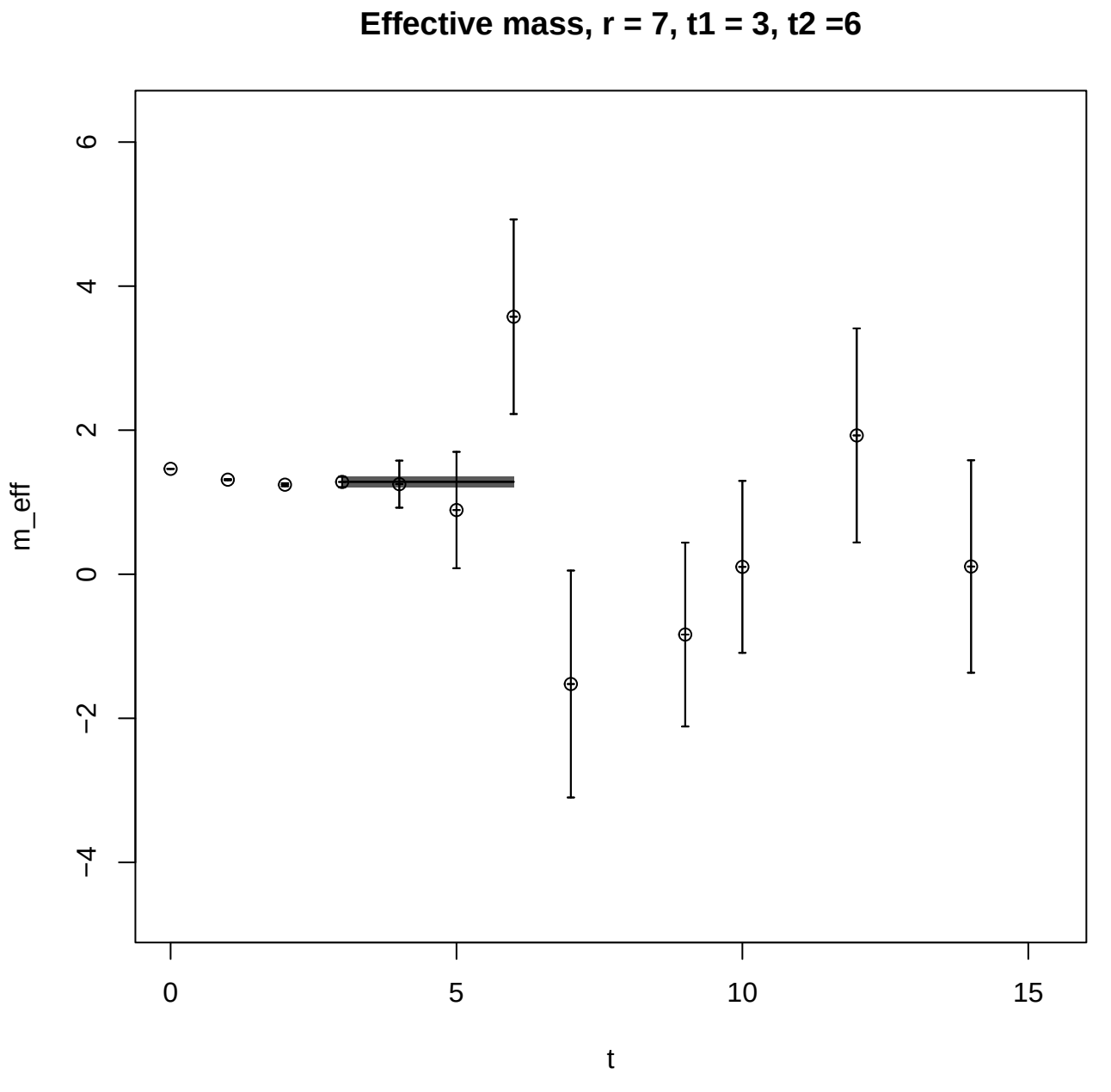

Effective mass, r = 7, t1 = 2, t2 =6
m_eff
t
6
4
2
0
−2
−4
0
5
10
15

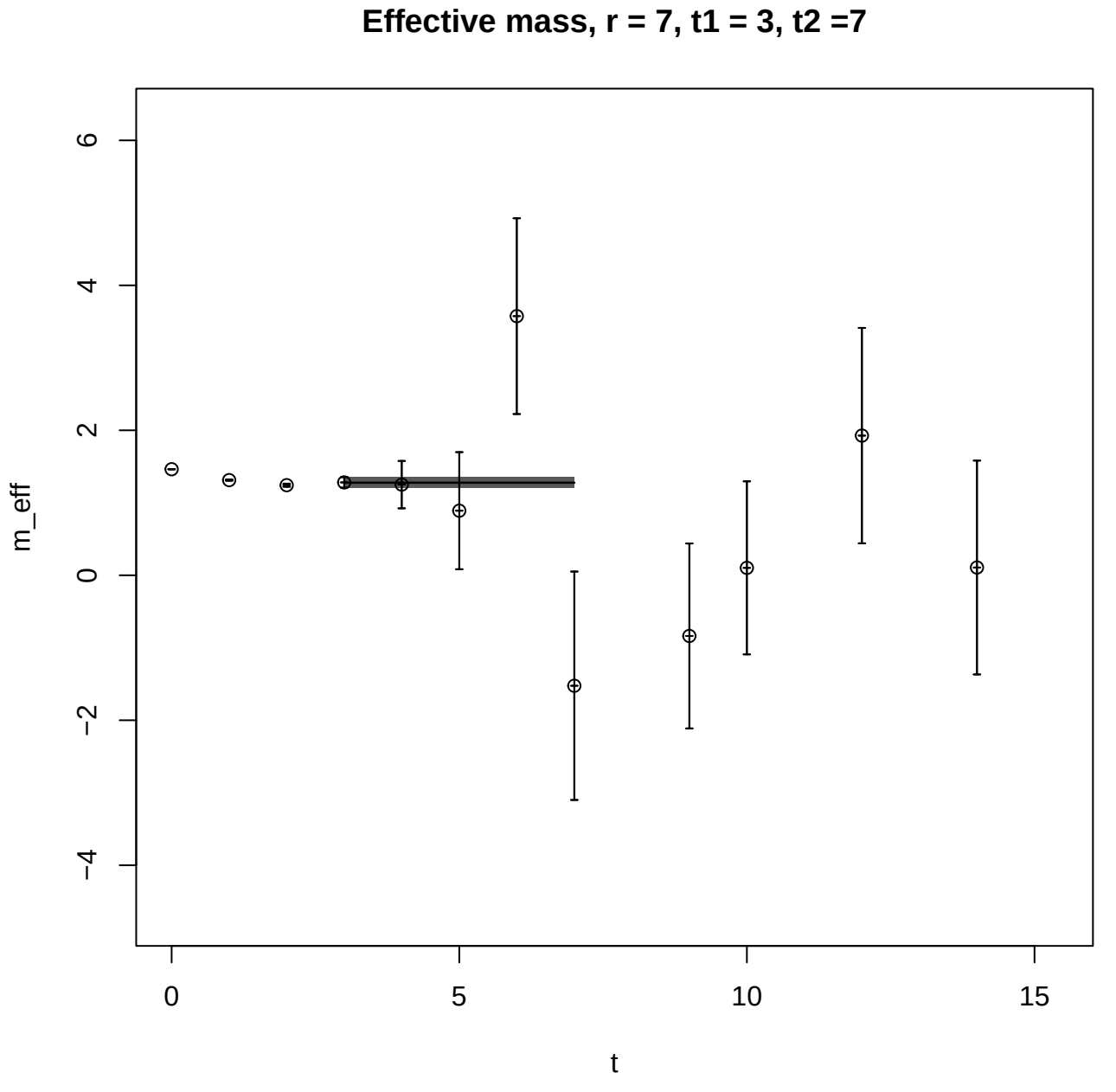

Effective mass, r = 7, t1 = 2, t2 =7
m_eff
t
6
4
2
0
−2
−4
0
5
10
15

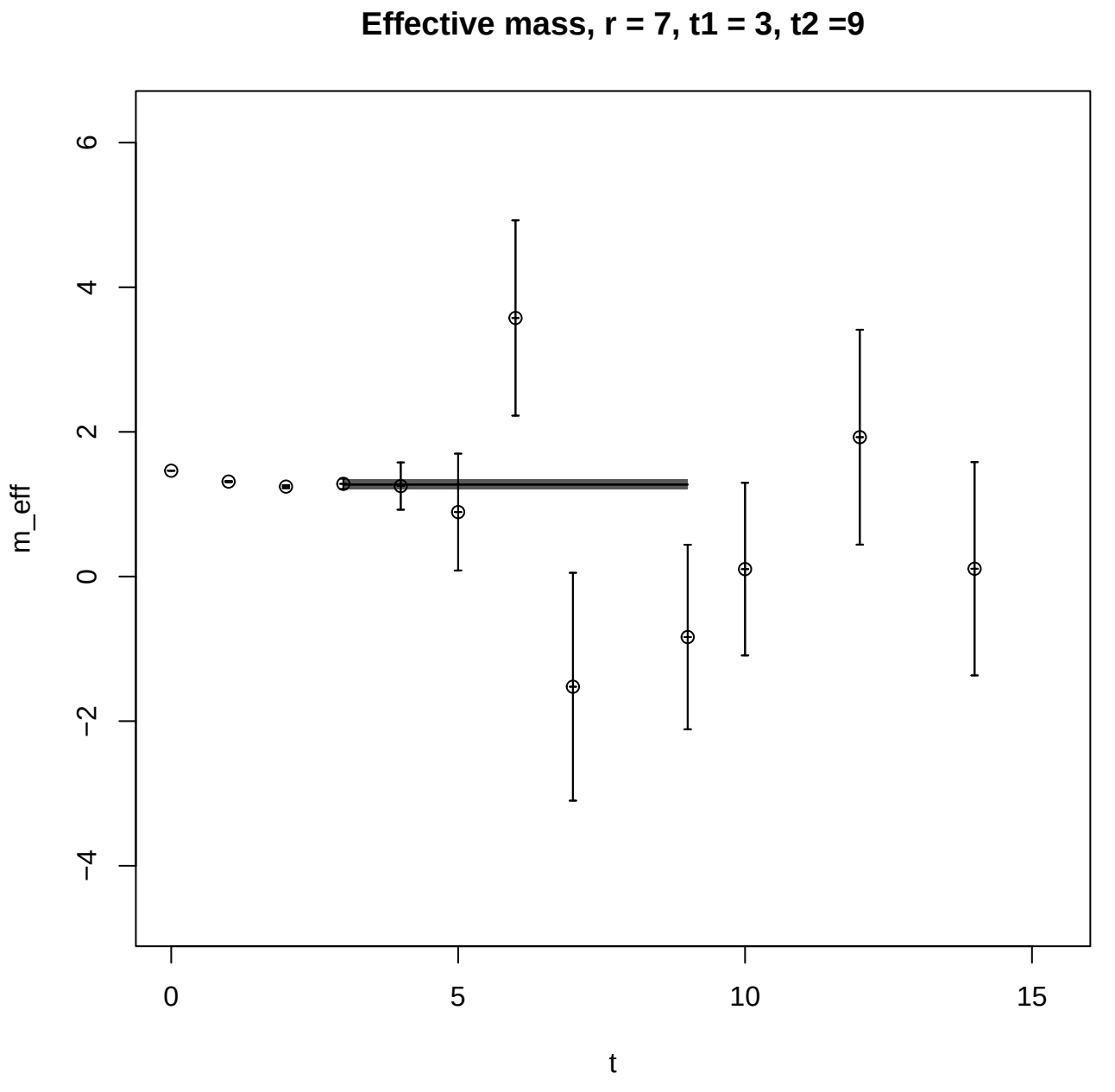

Effective mass, r = 7, t1 = 2, t2 =9
m_eff
t
6
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 3, t2 =4
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 3, t2 =5
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 3, t2 =6
m_eff
t
6
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 3, t2 =7
m_eff
t
6
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 3, t2 =9
m_eff
t
6
4
2
0
−2
−4
0
5
10
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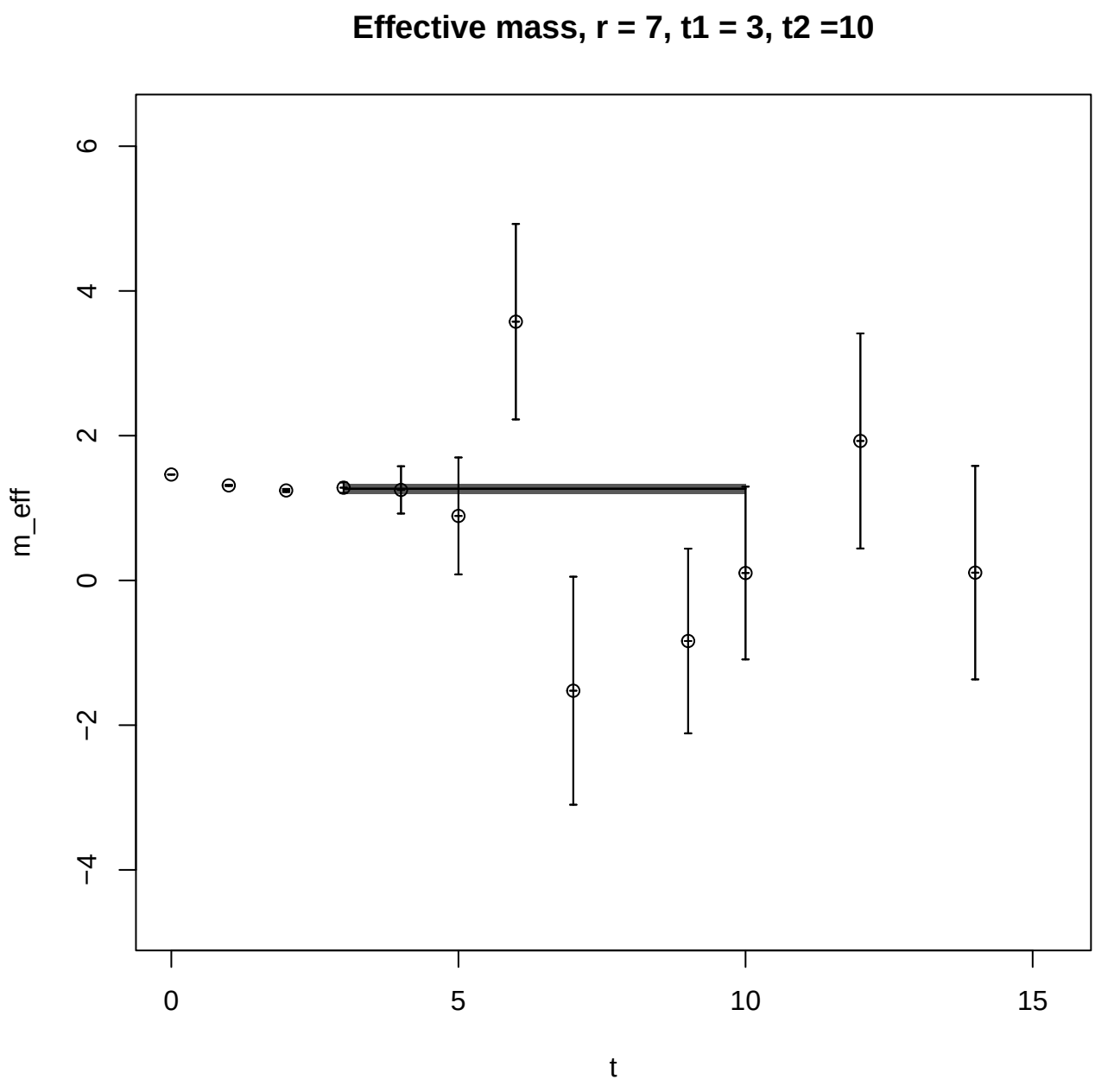
Effective mass, r = 7, t1 = 3, t2 =10
m_eff
t
6
4
2
0
−2
−4
0
5
10
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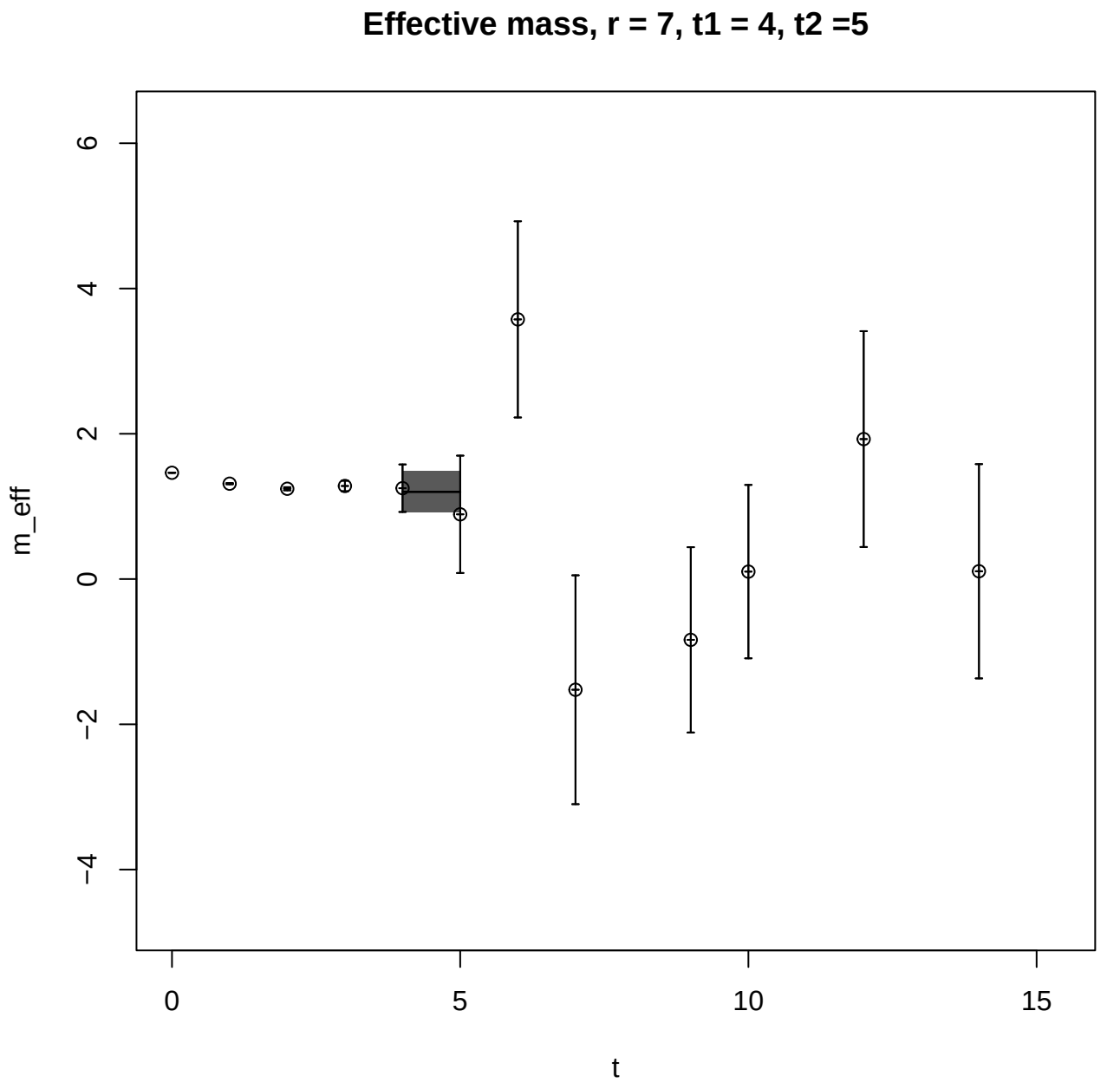
Effective mass, r = 7, t1 = 4, t2 =5
m_eff
t
6
4
2
0
−2
−4
0
5
10
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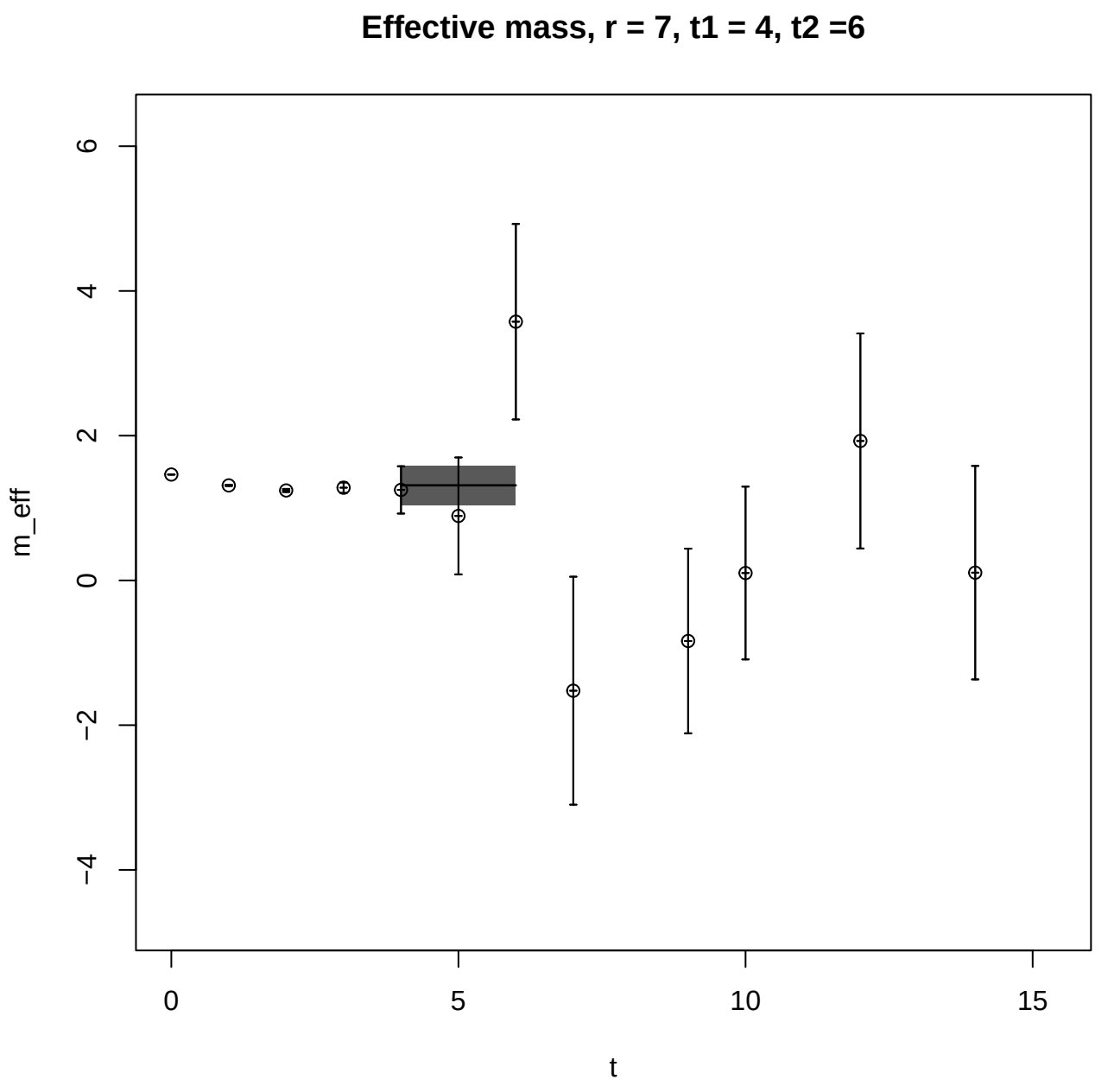
Effective mass, r = 7, t1 = 4, t2 =6
m_eff
t
6
4
2
0
−2
−4
0
5
10
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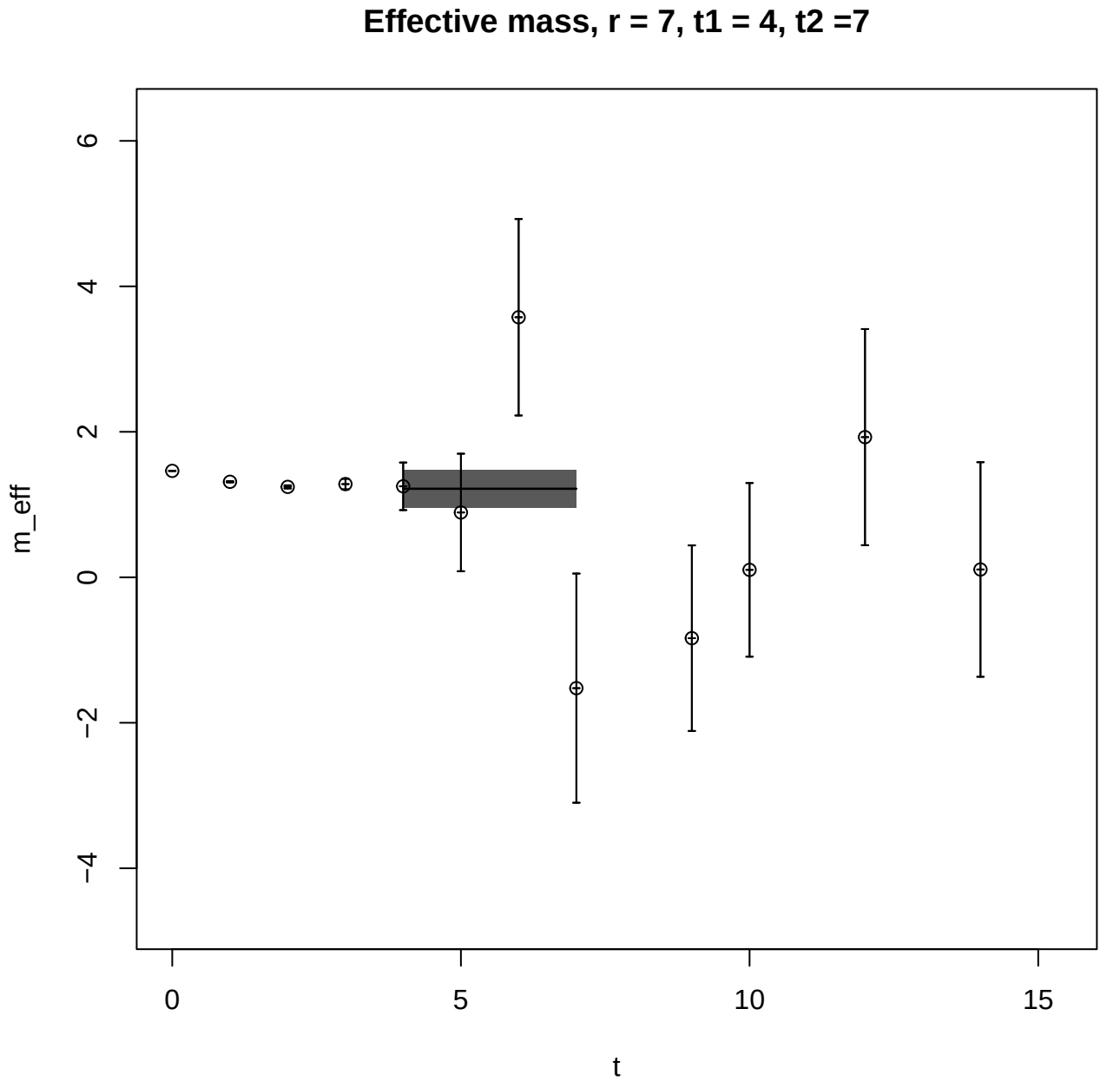
Effective mass, r = 7, t1 = 4, t2 =7
m_eff
6
4
2
0
−2
−4
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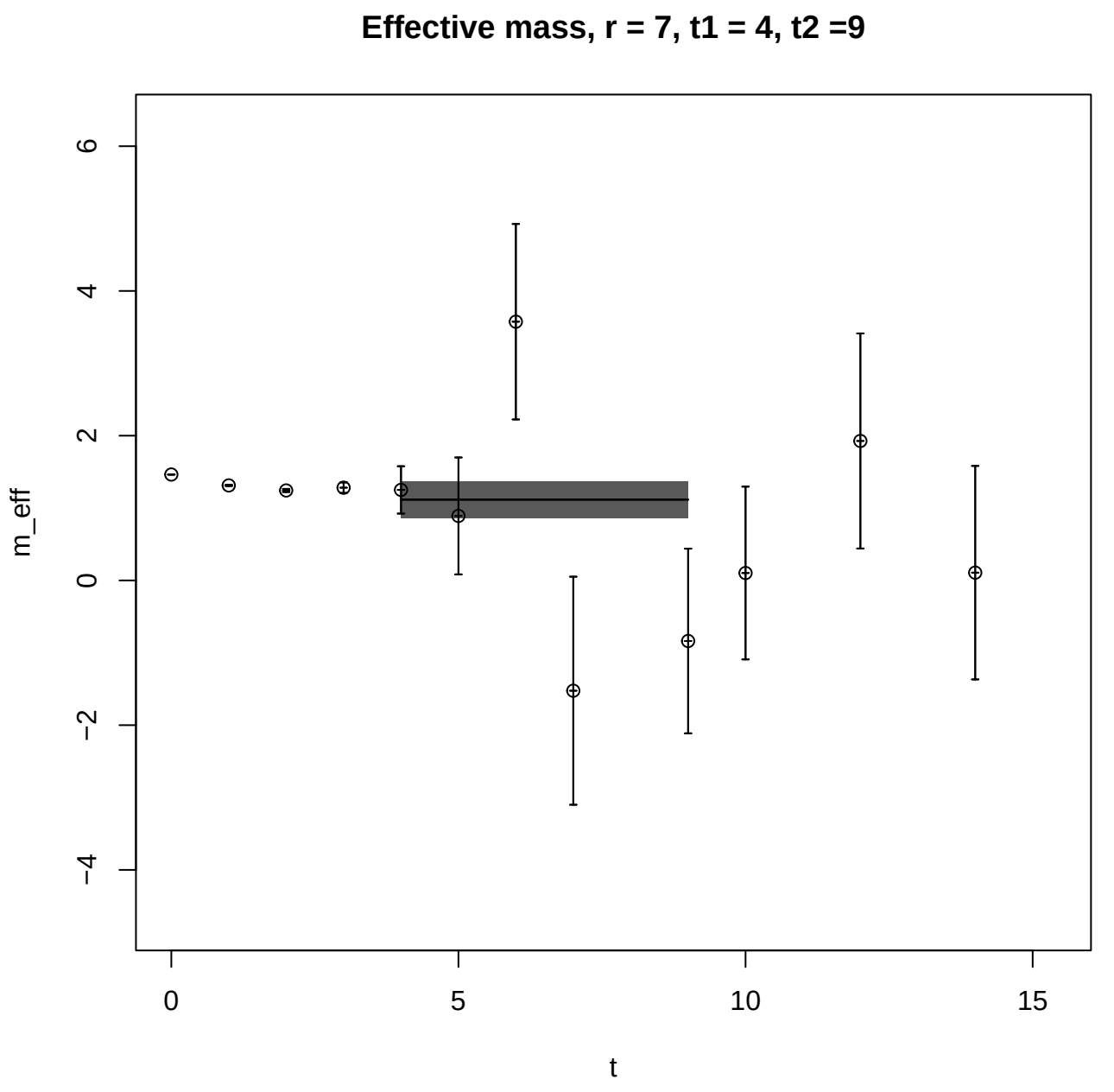
5
10
15
t


Effective mass, r = 7, t1 = 4, t2 =9
m_eff
t
6
4
2
0
−2
−4
0
5
10
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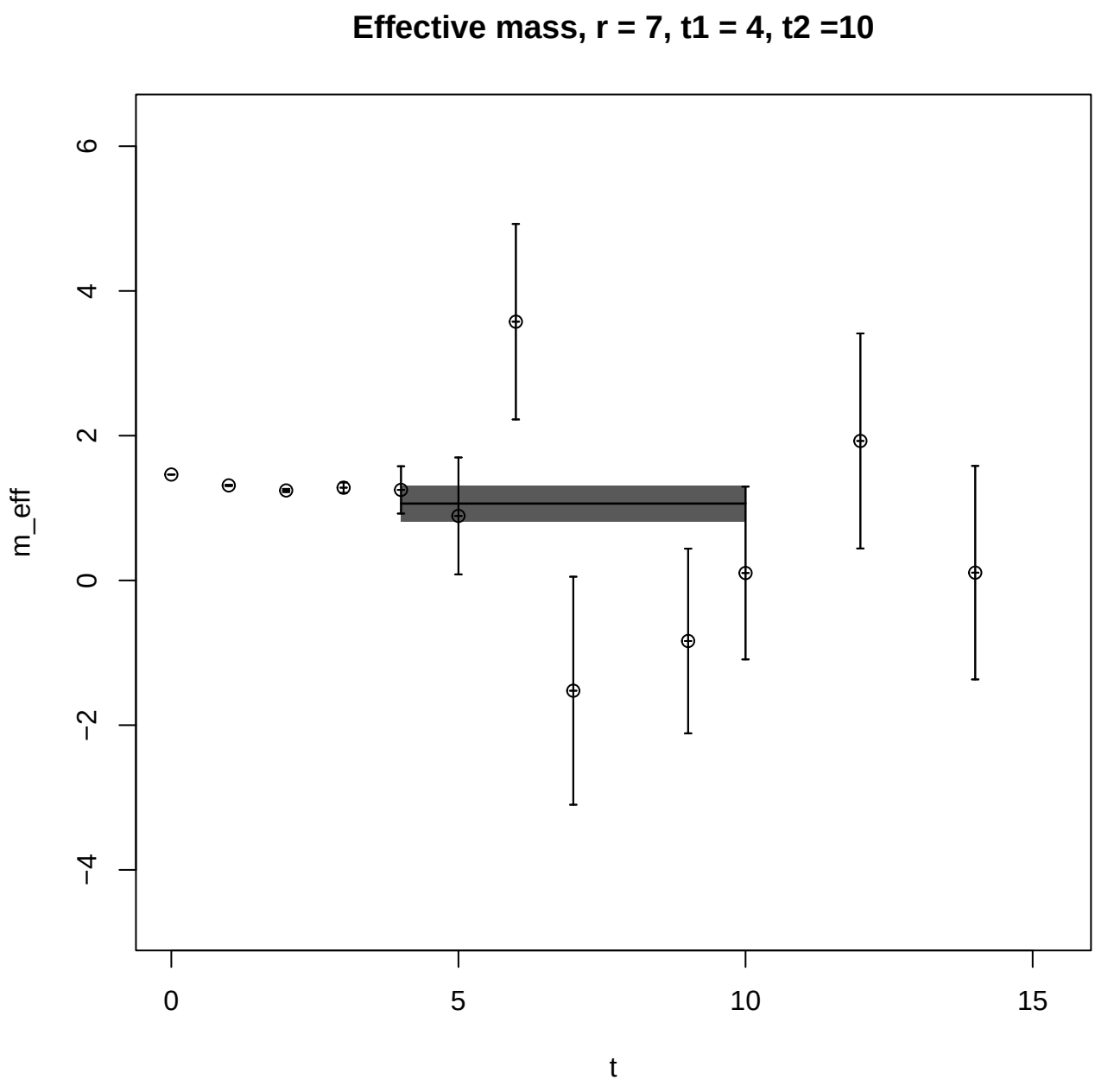
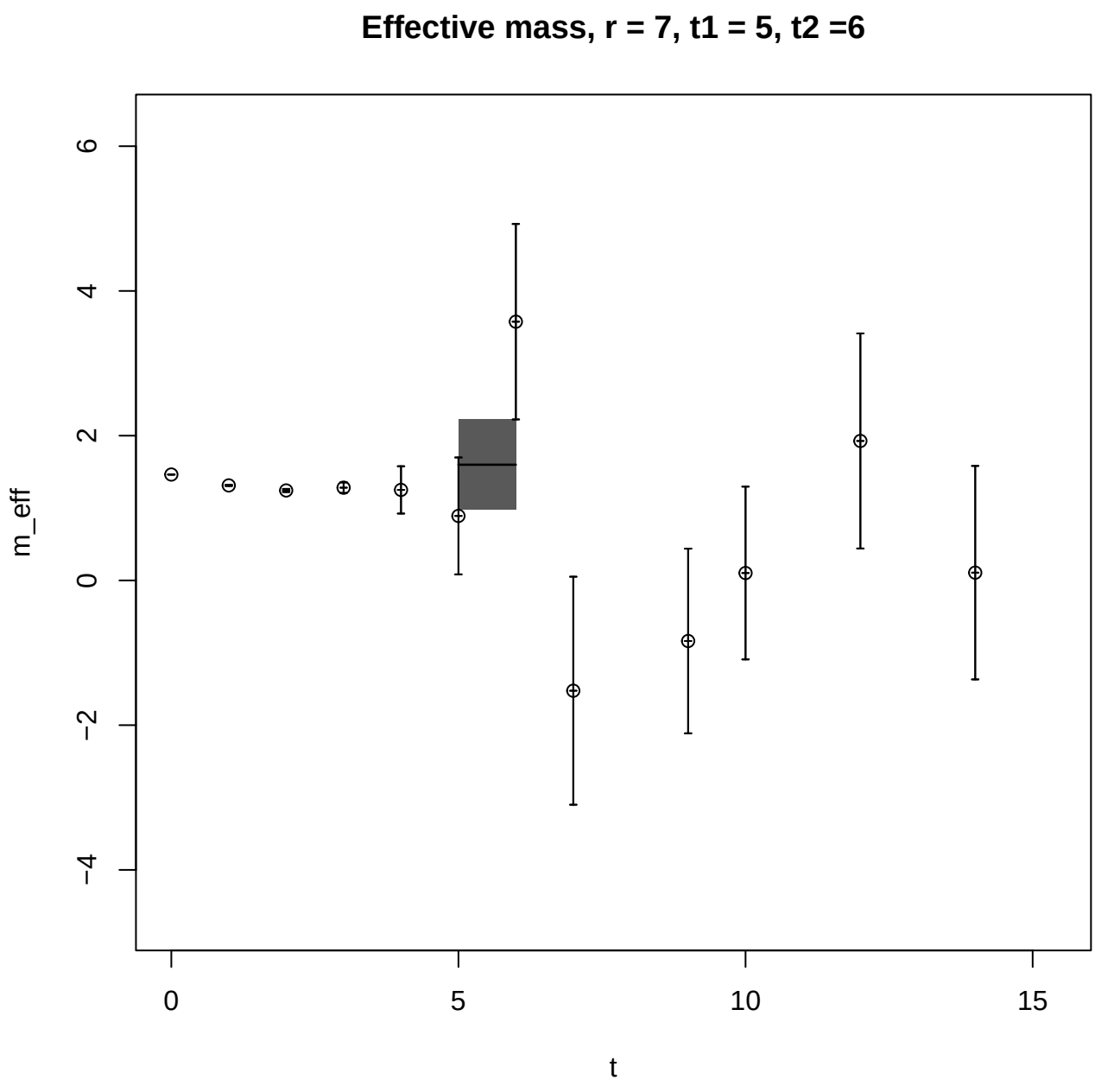

Effective mass, r = 7, t1 = 4, t2 =10
m_eff
t
6
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 7, t1 = 5, t2 =6
m_eff
6
4
2
0
−2
−4
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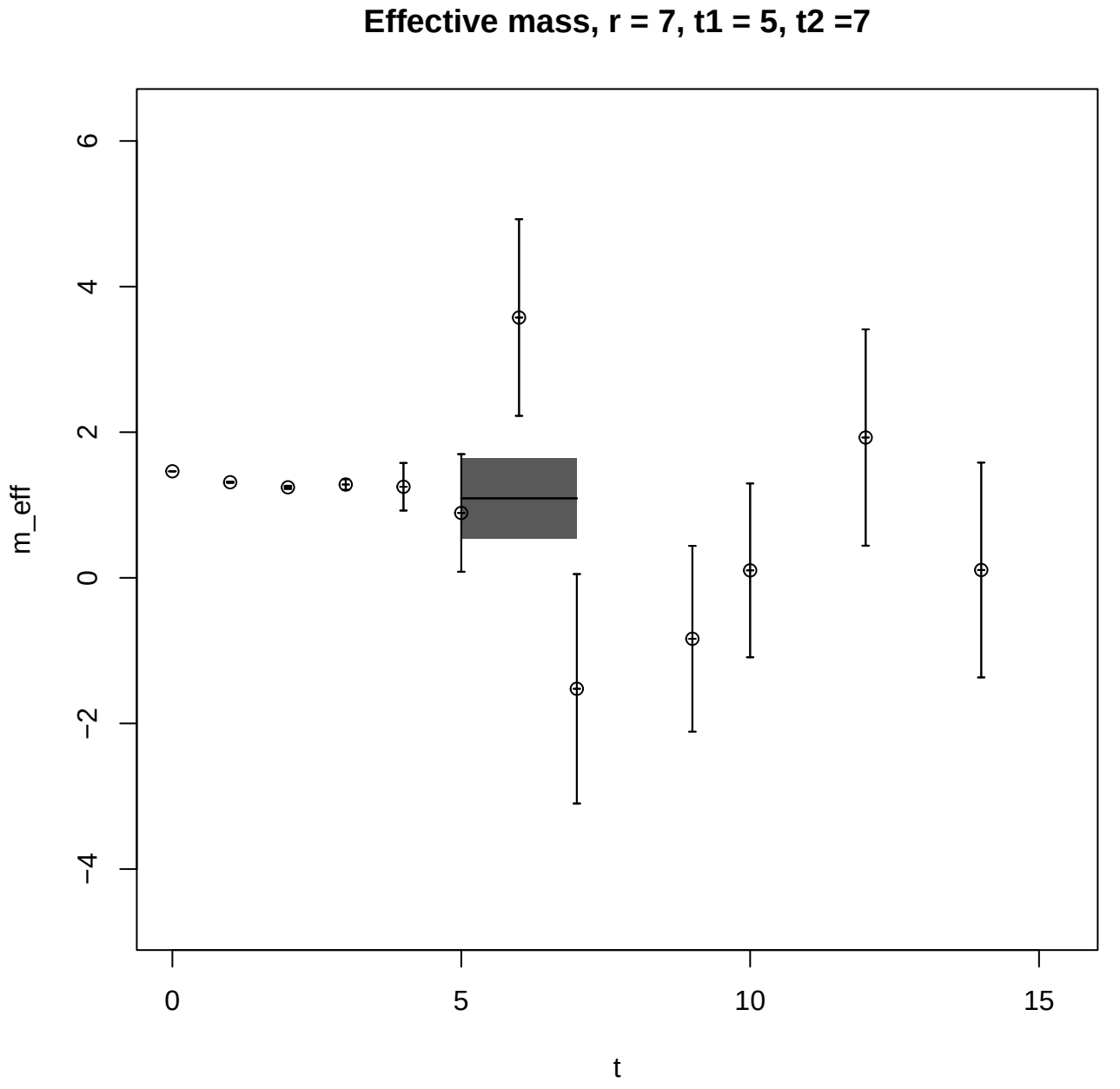
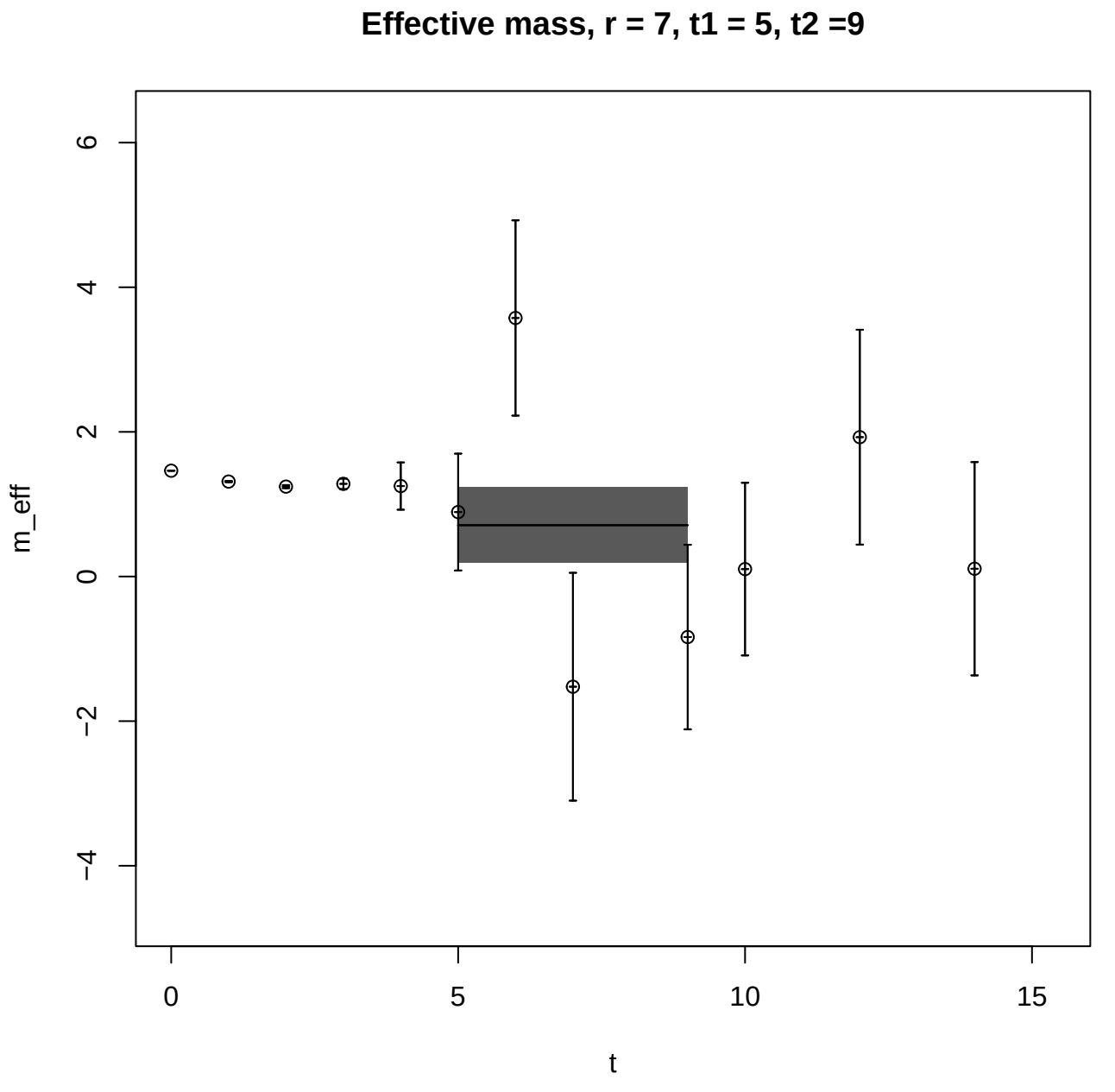
5
10
15
t


Effective mass, r = 7, t1 = 5, t2 =7
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 5, t2 =9
m_eff
6
4
2
0
−2
−4
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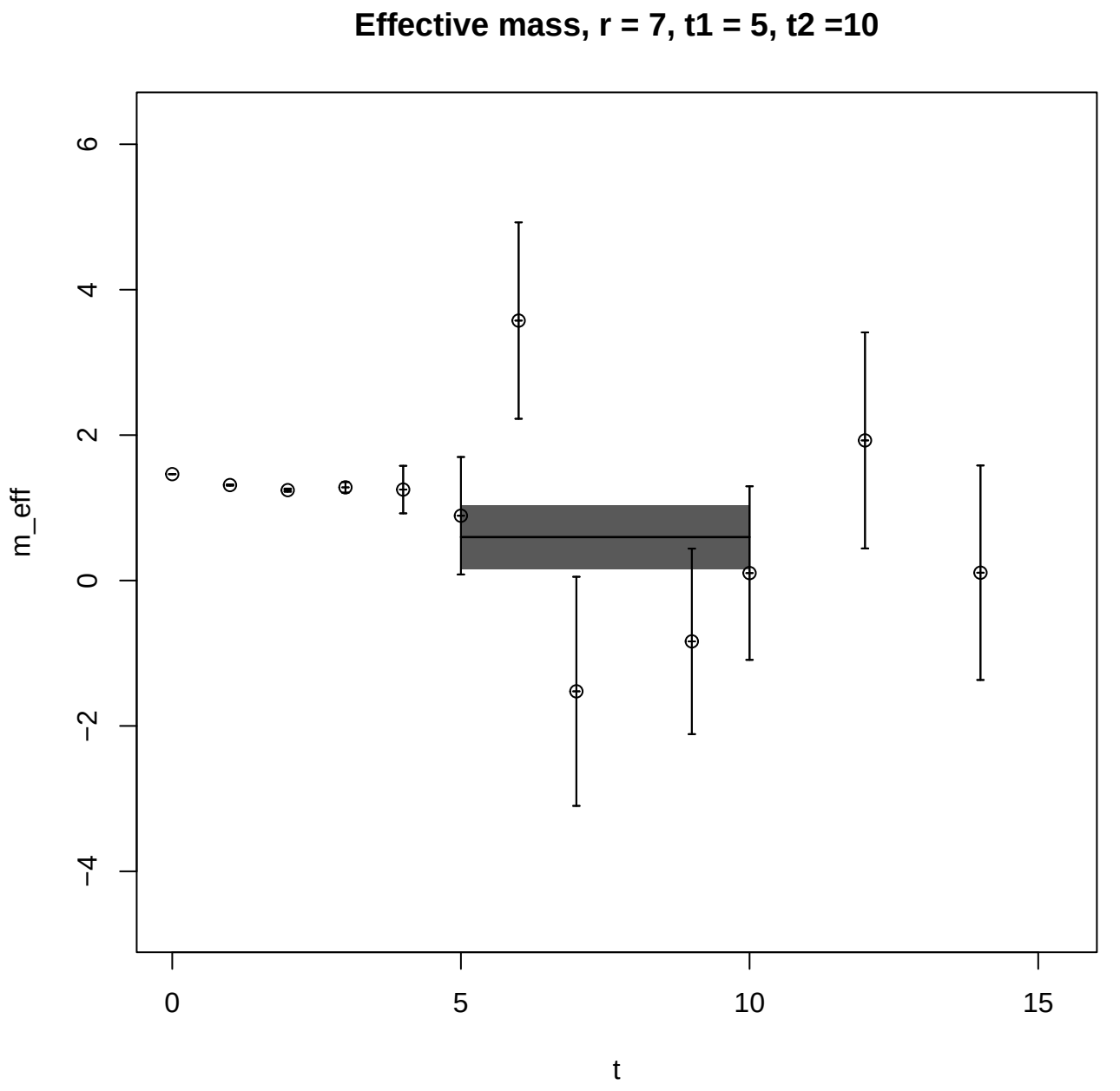
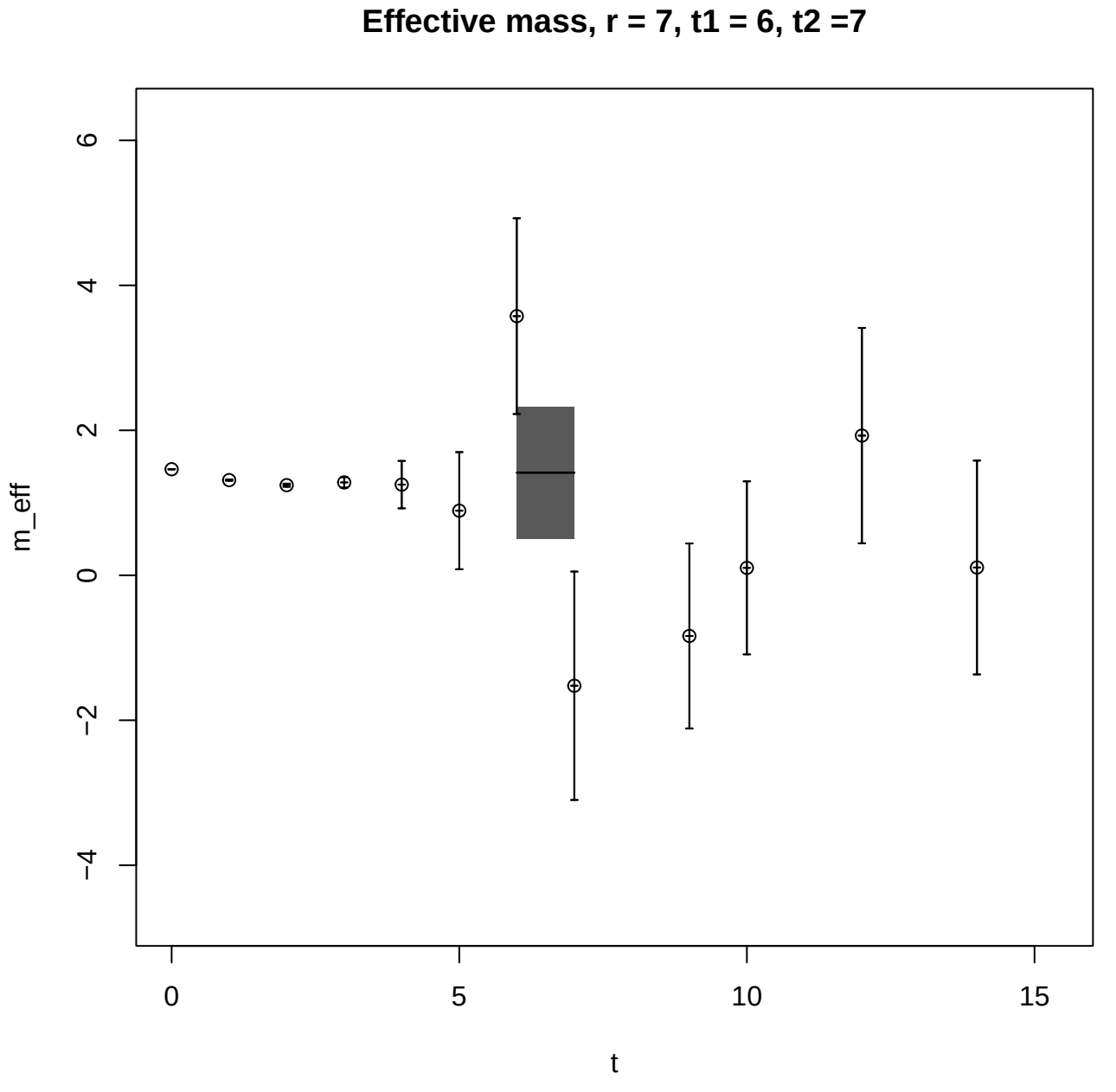
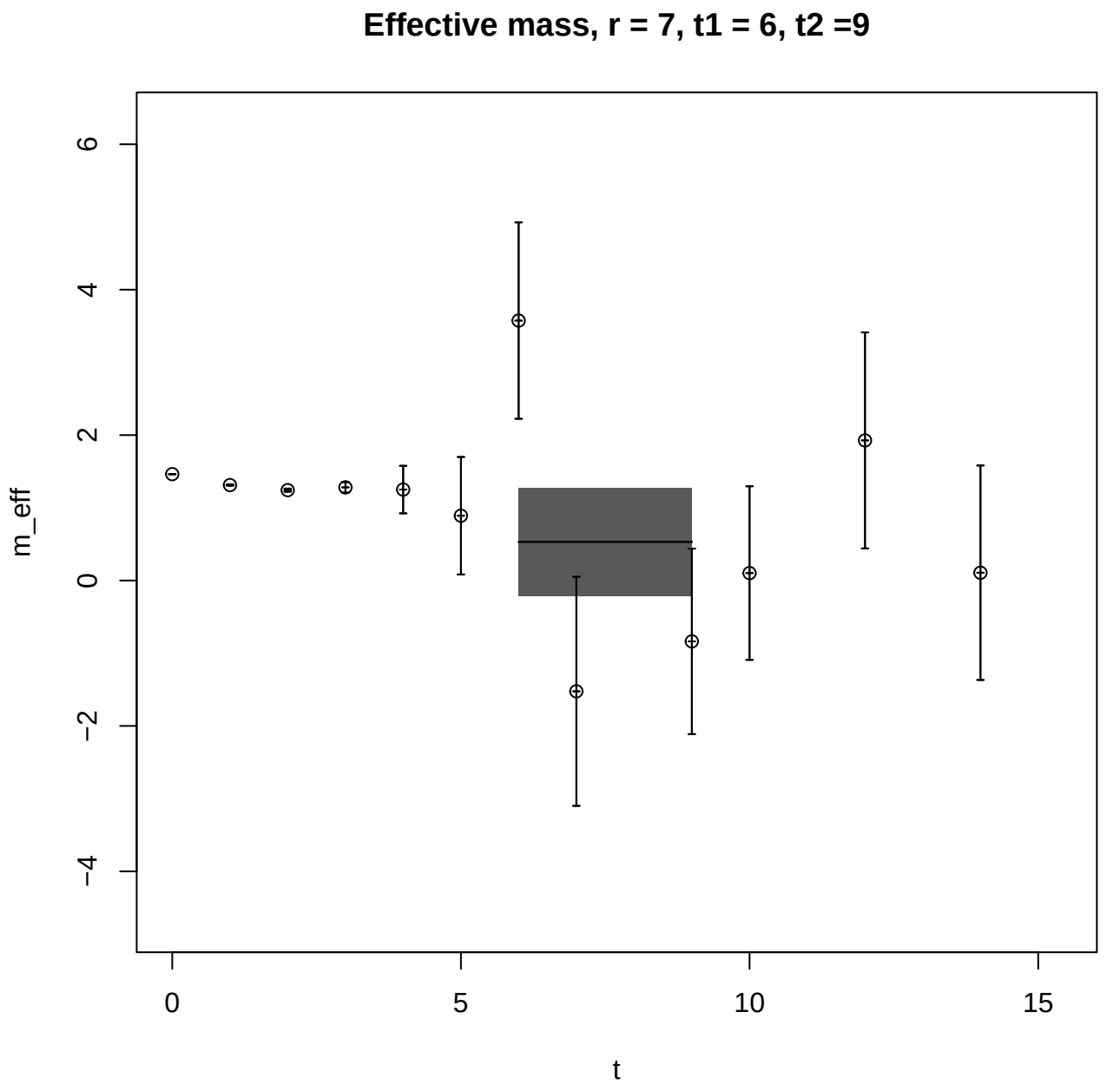
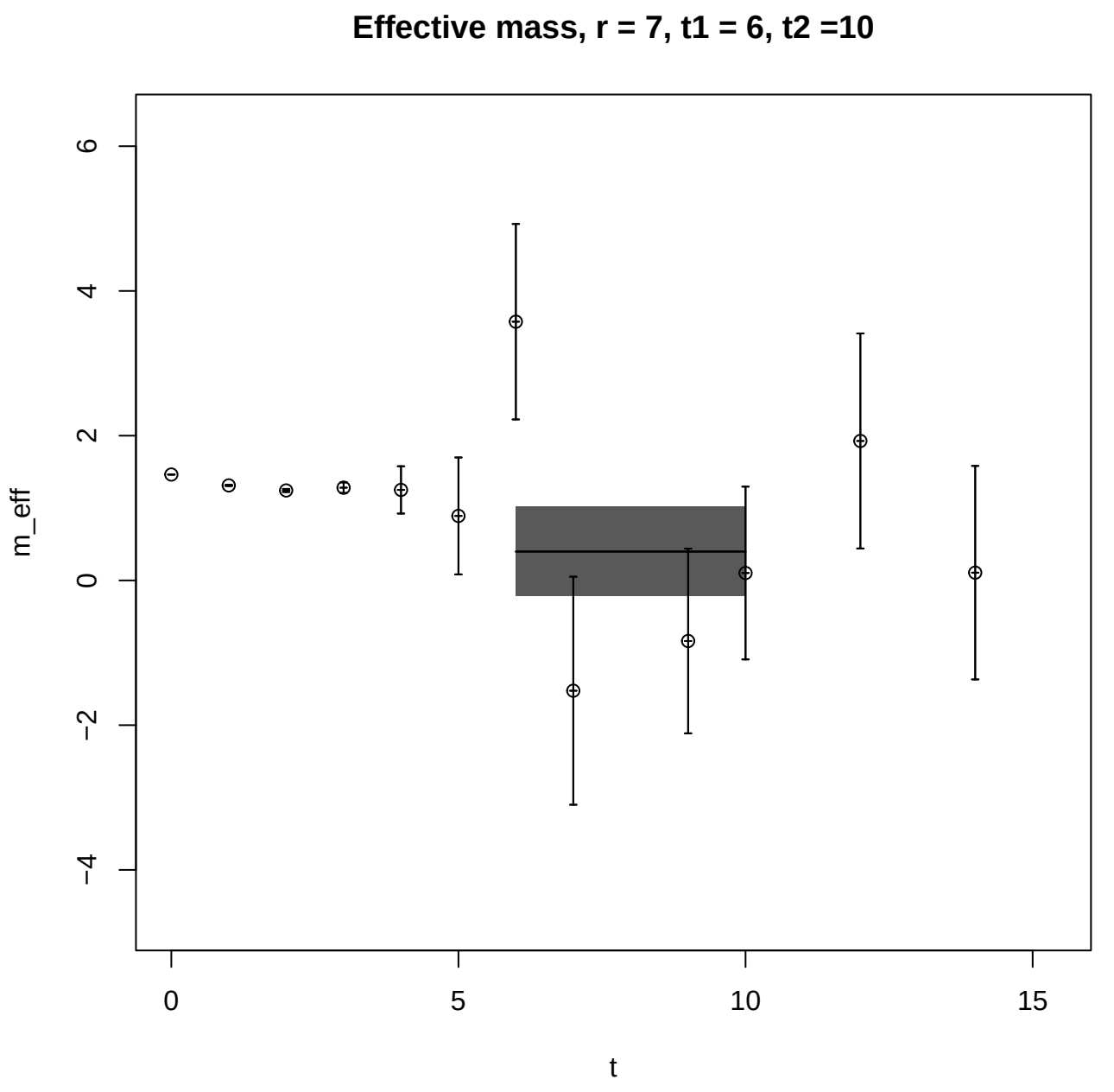
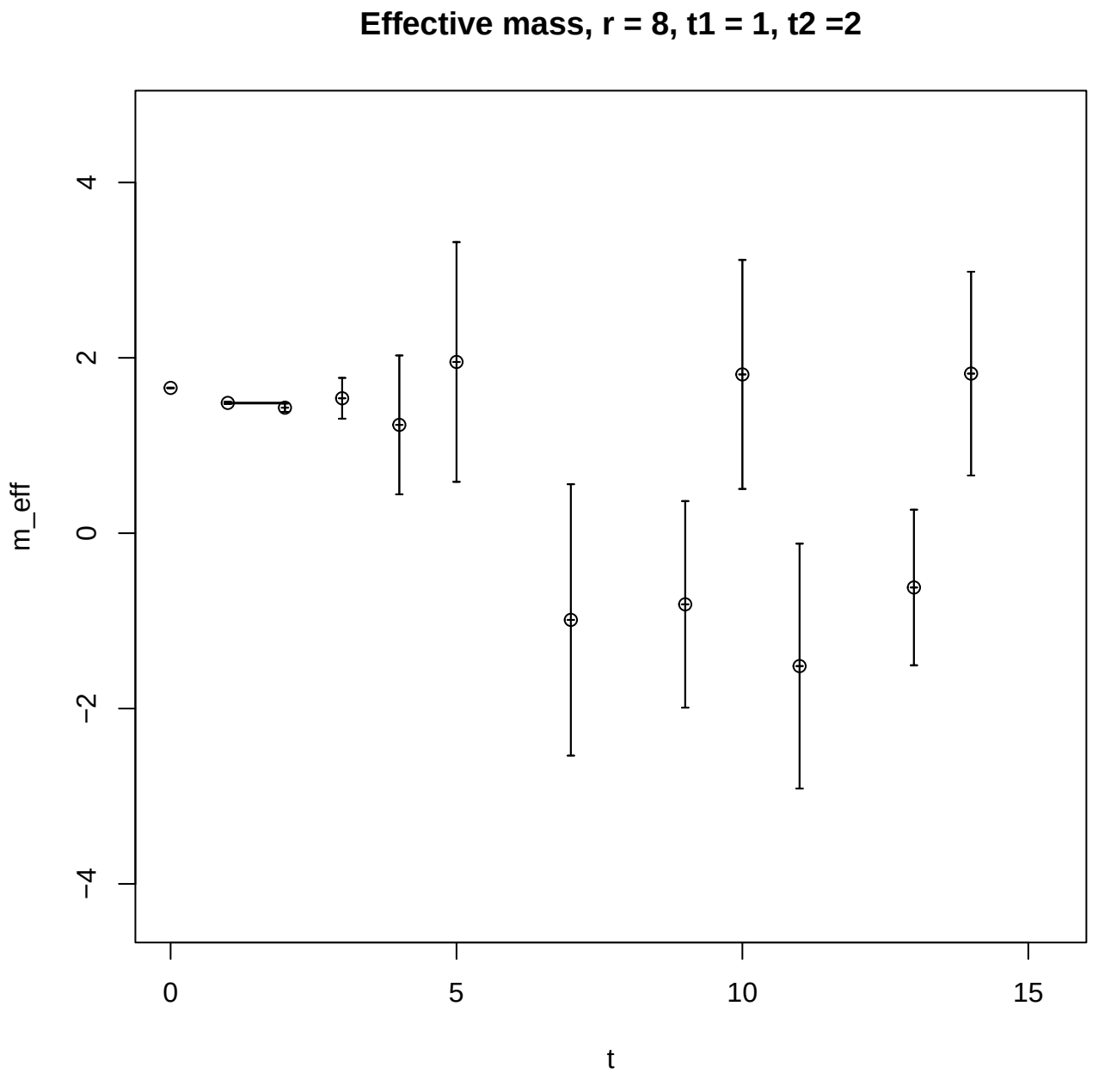
5
10
15
t


Effective mass, r = 7, t1 = 5, t2 =10
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 6, t2 =7
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 6, t2 =9
m_eff
6
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 7, t1 = 6, t2 =10
m_eff
t
6
4
2
0
−2
−4
0
5
10
15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m_eff
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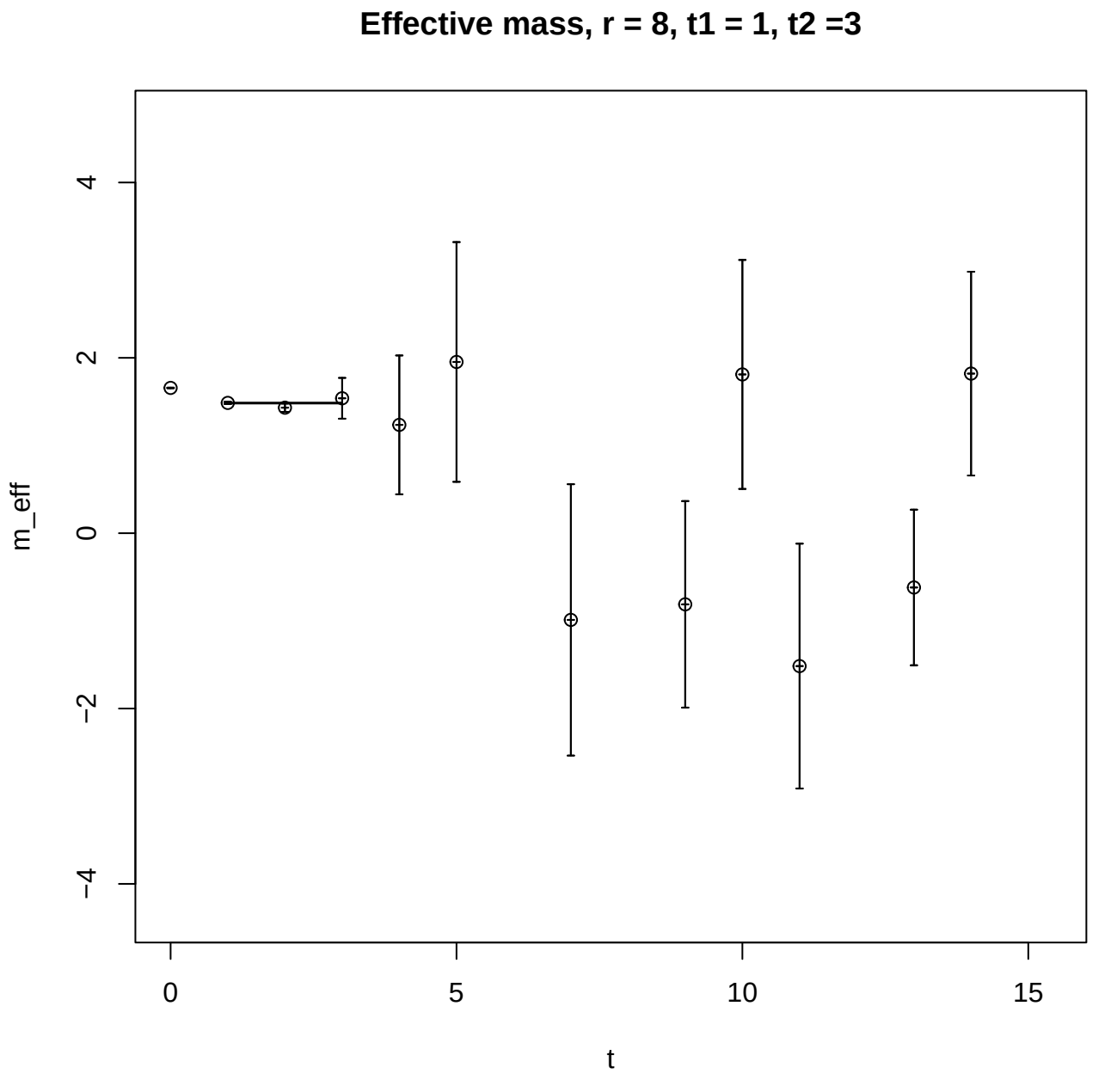
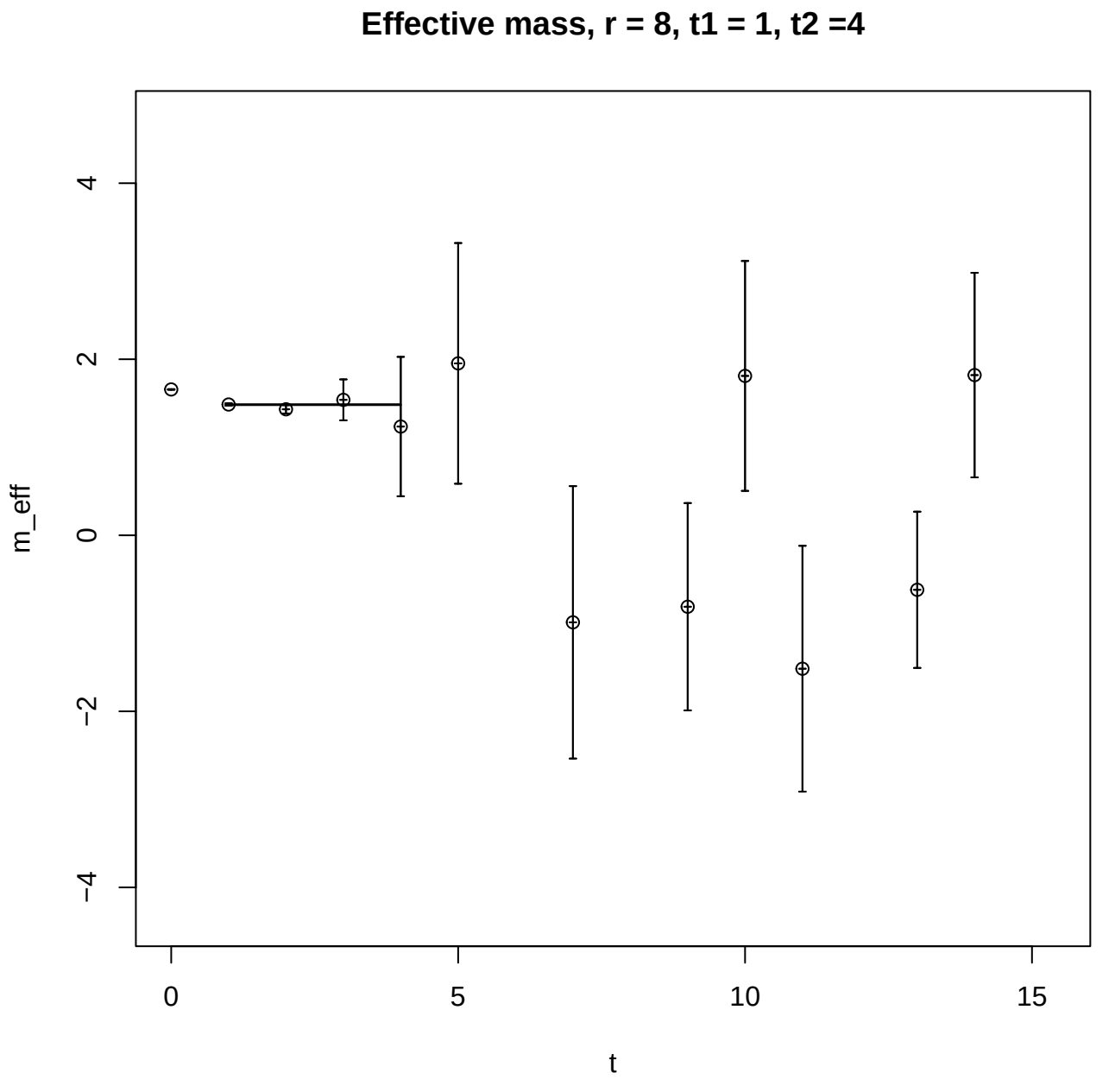
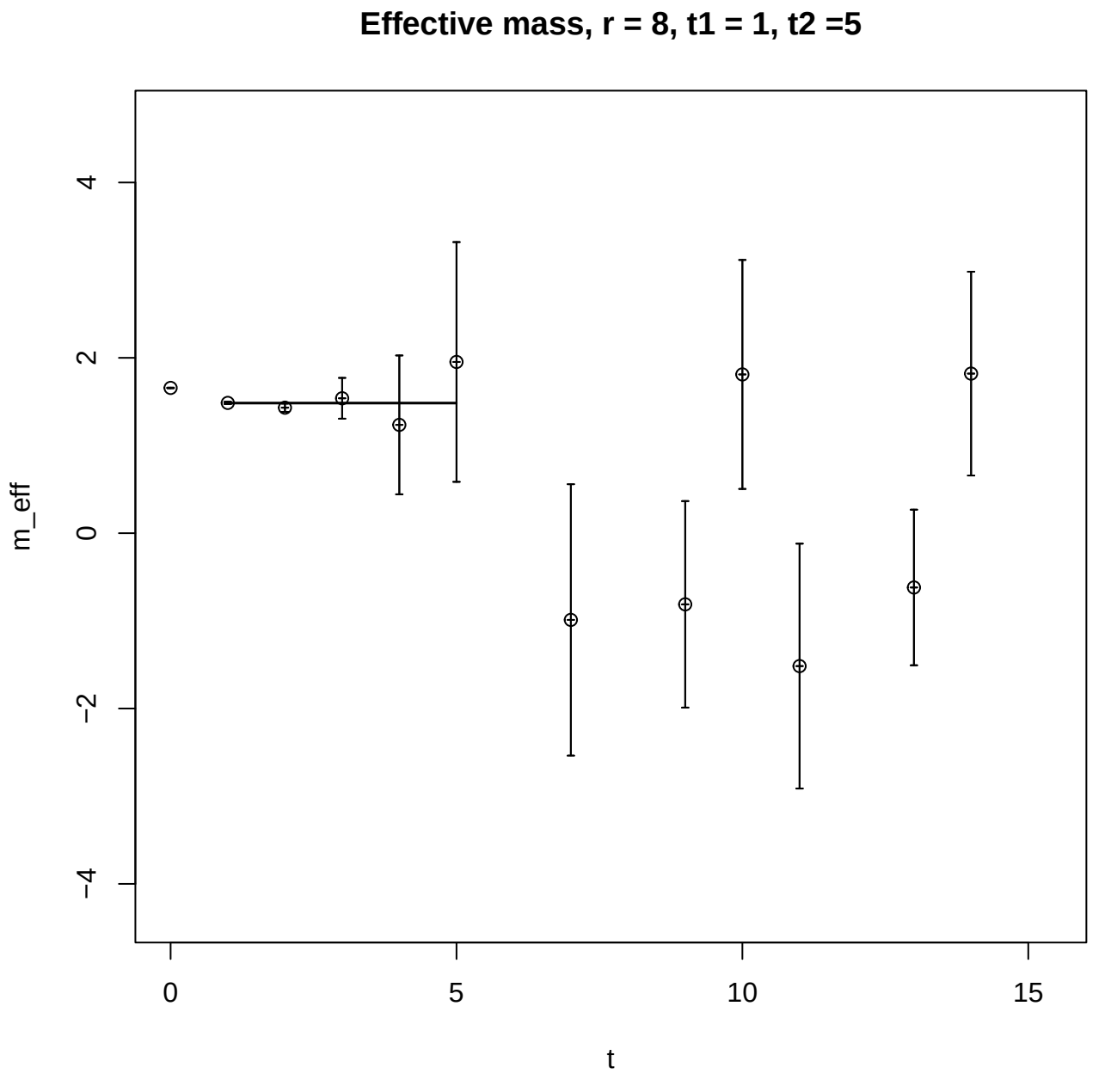
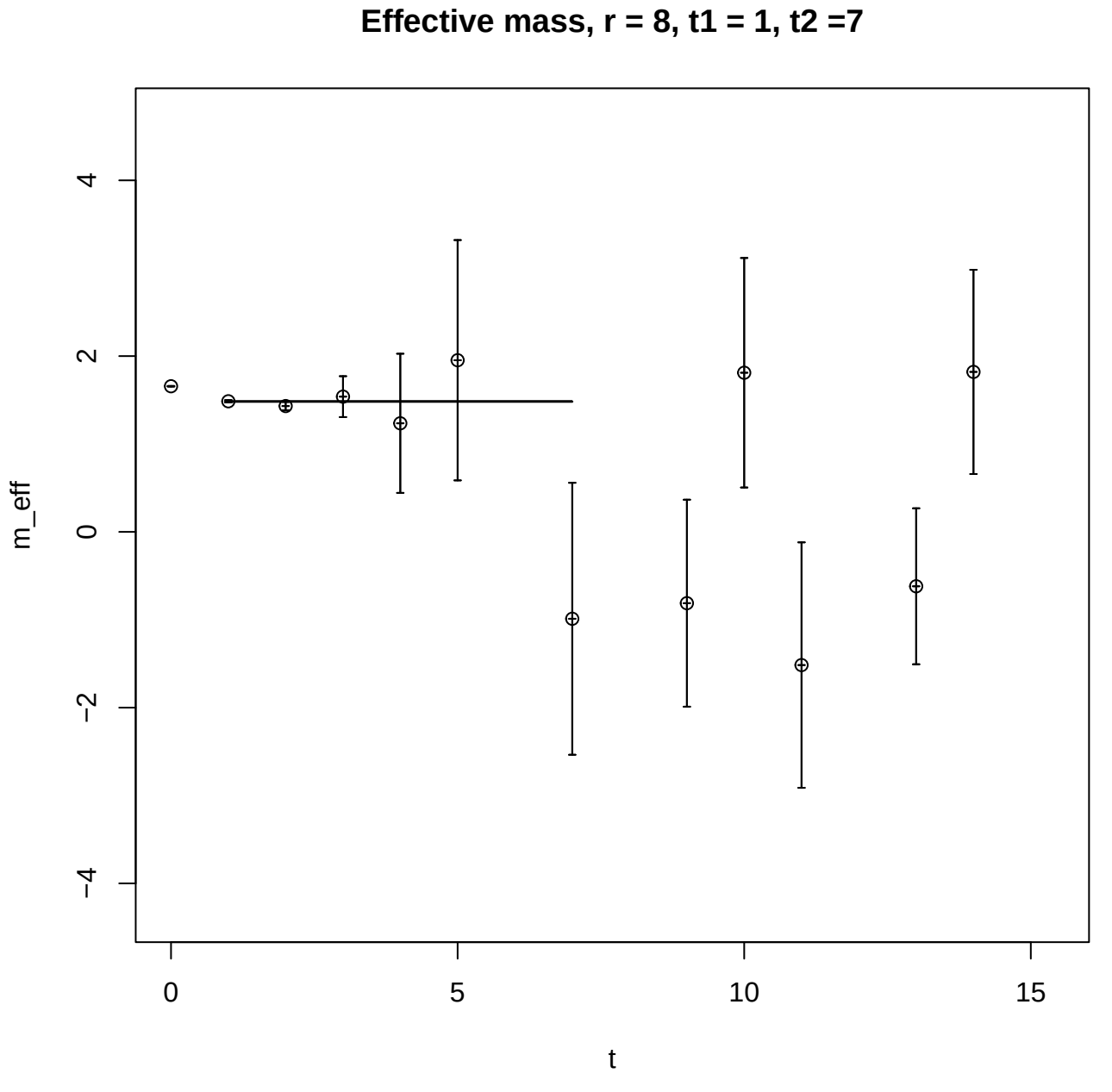
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =3
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 1, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 1, t2 =7
m_eff
t
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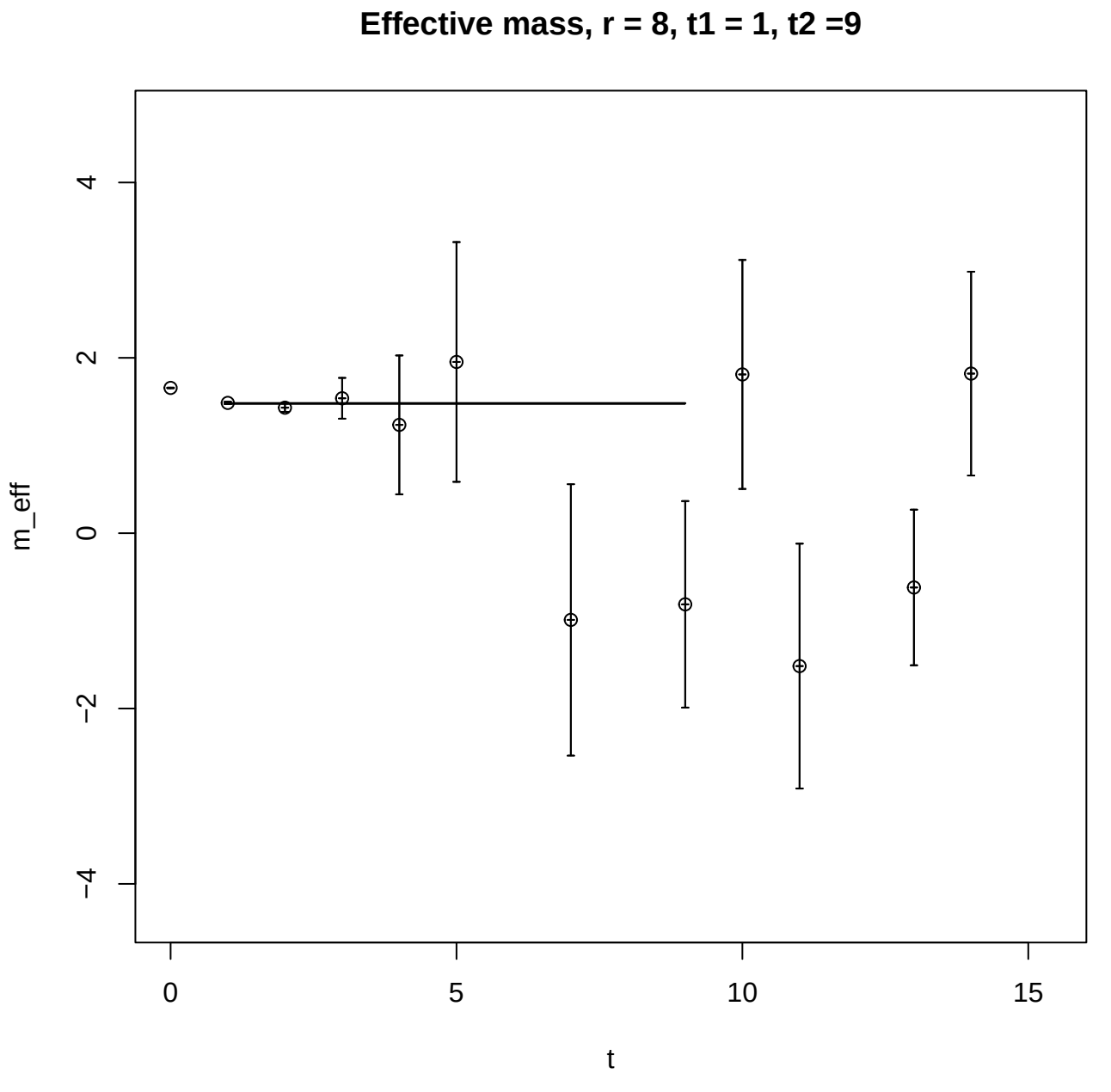
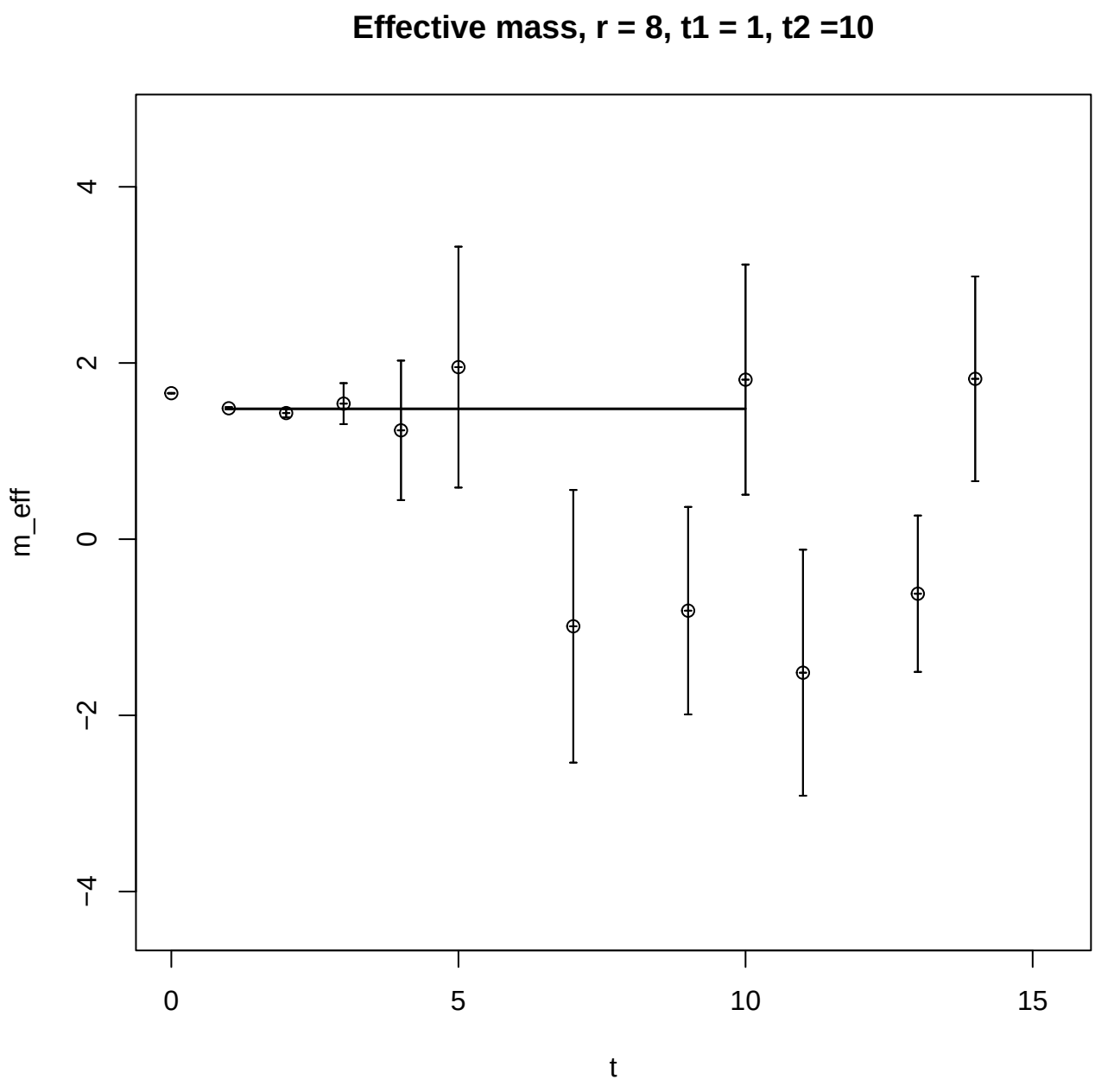
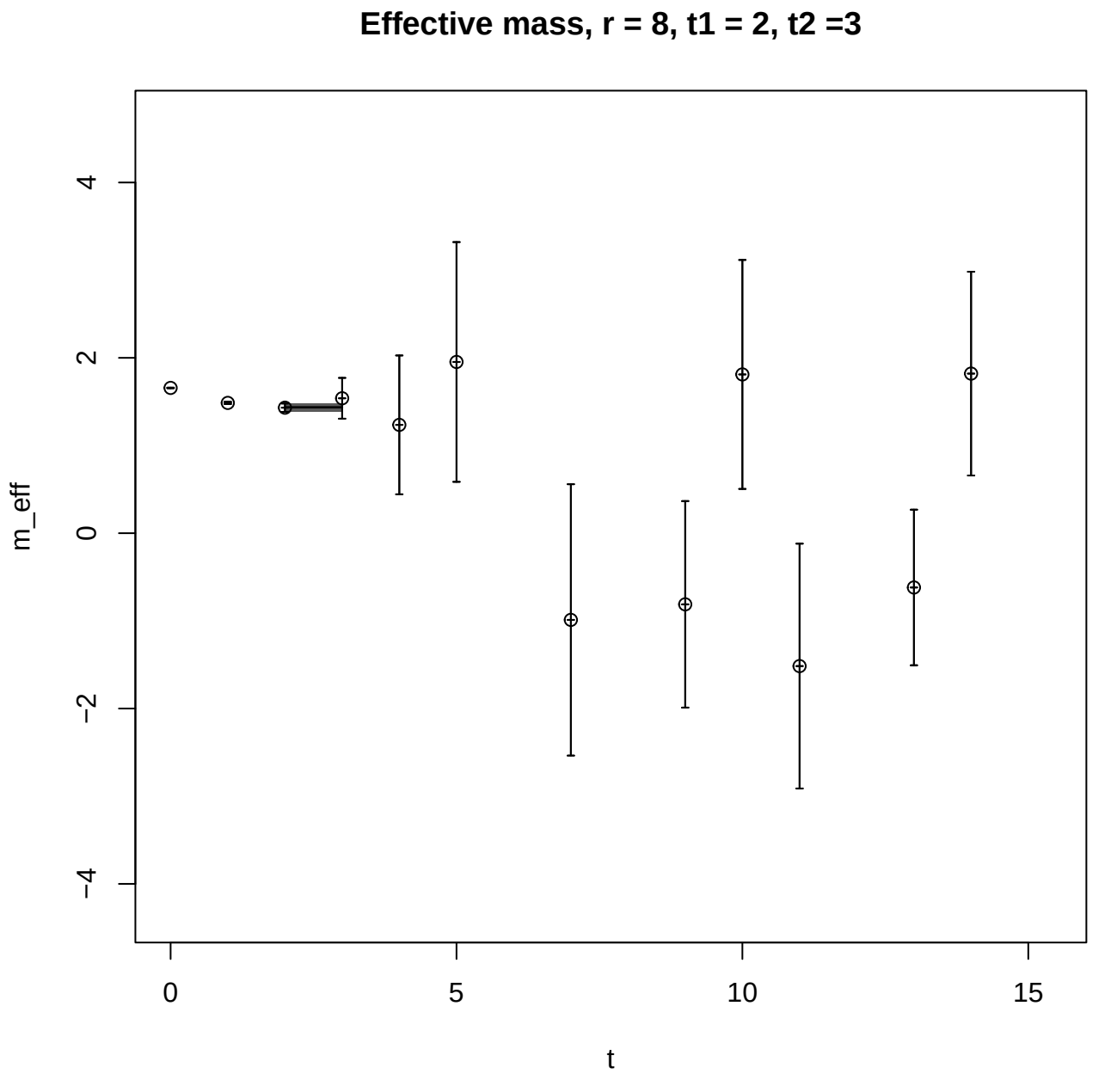
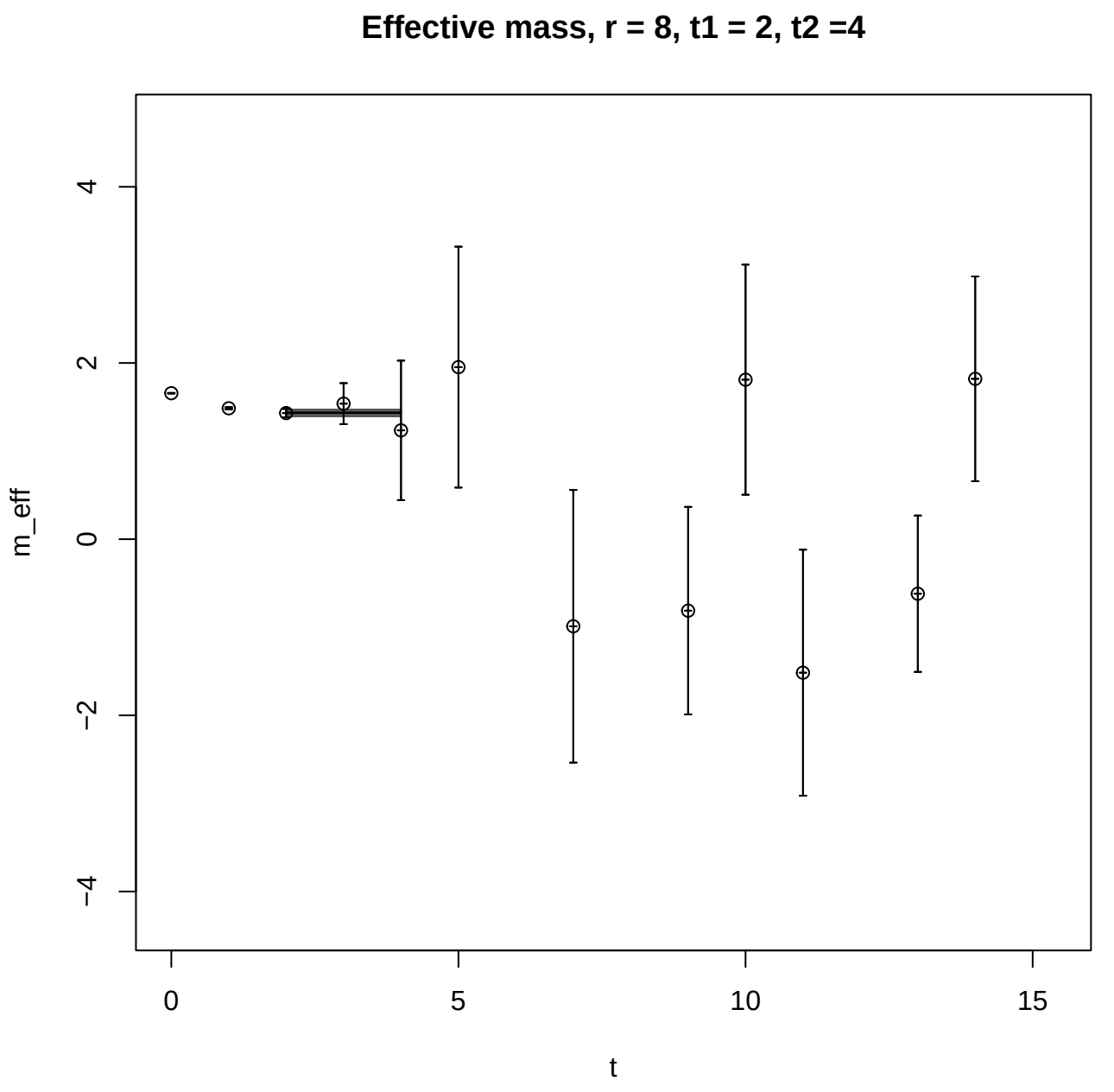
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 1, t2 =9
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 1, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15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m_eff
t
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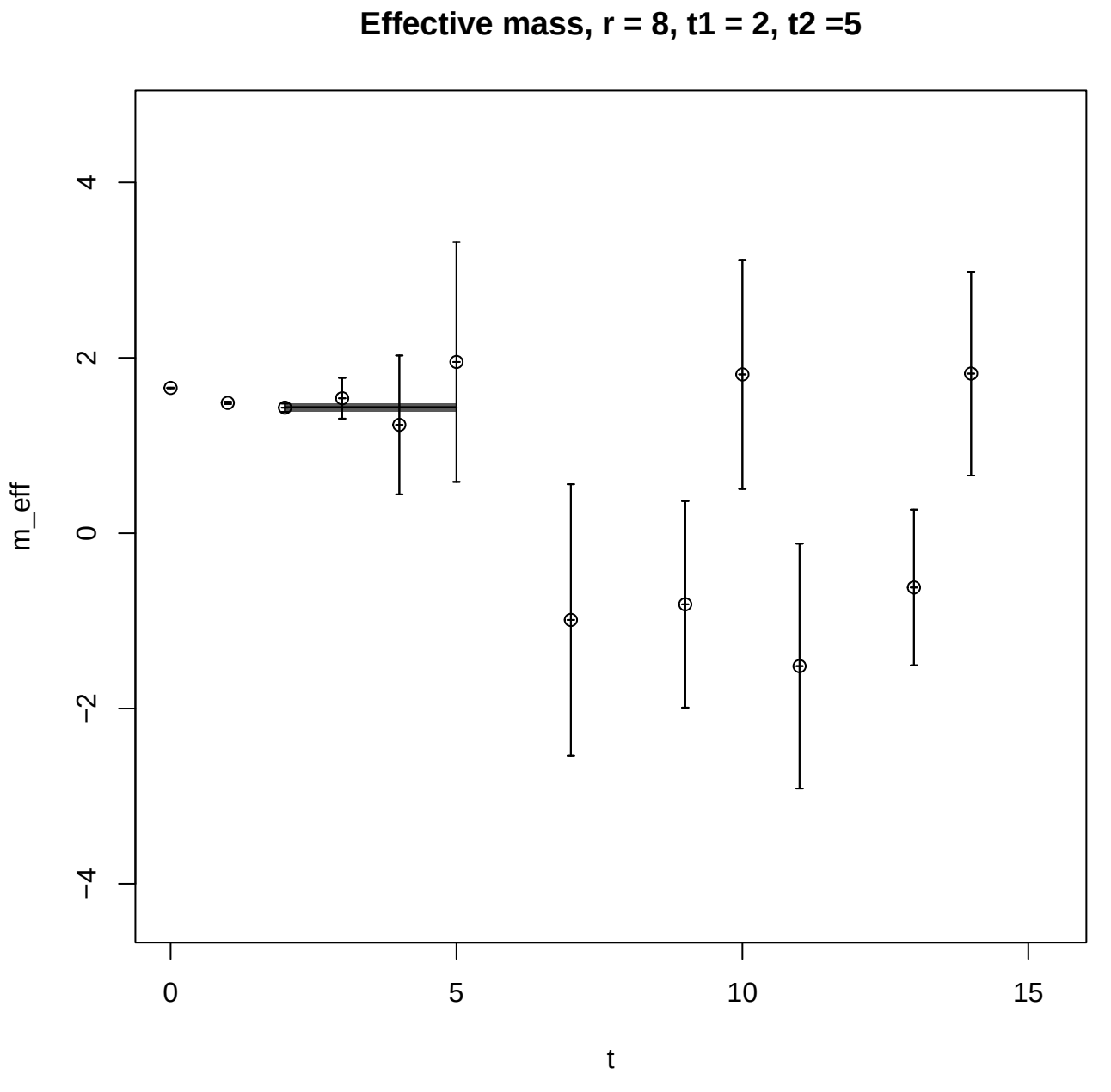
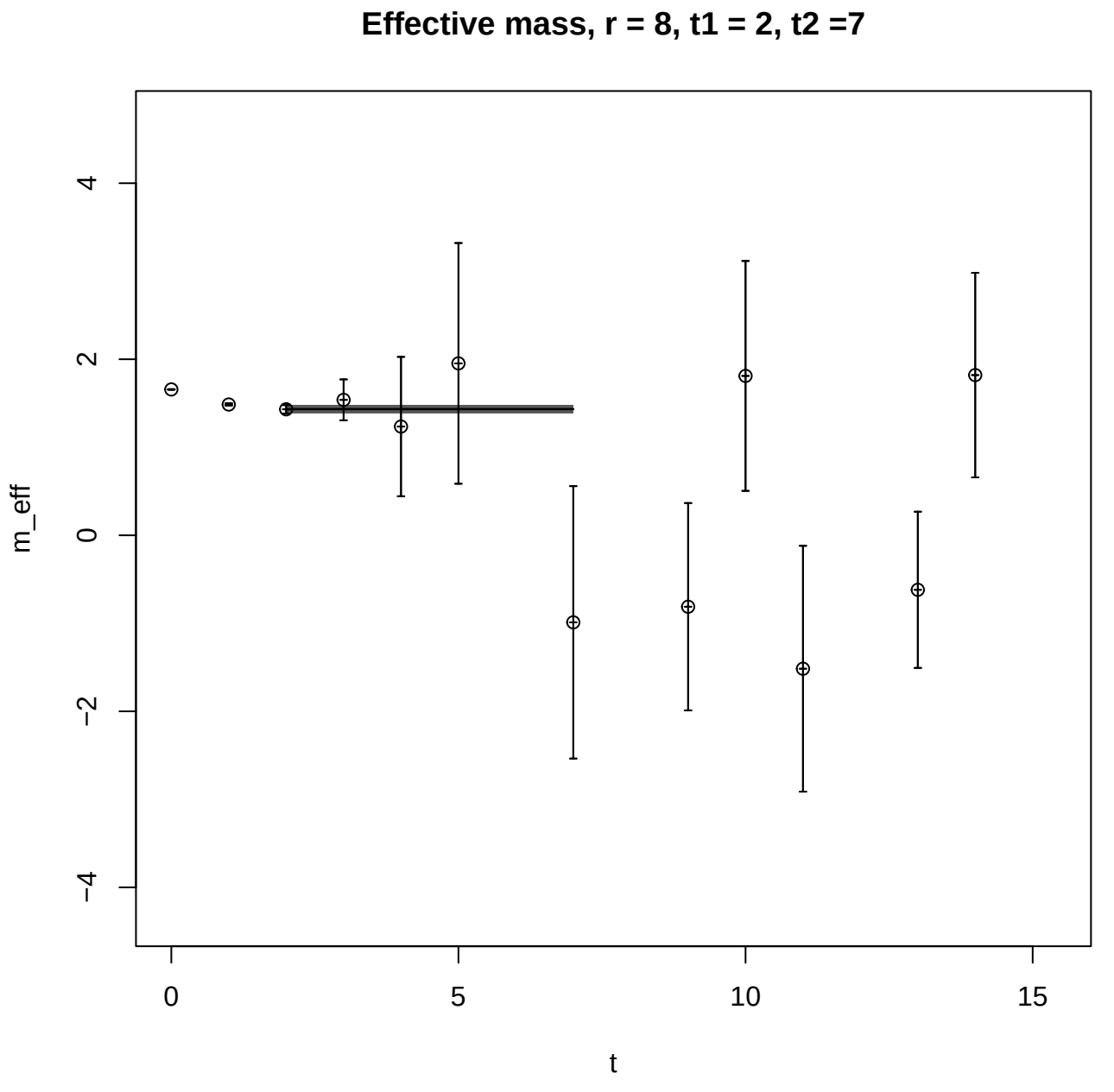
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 2, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 2, t2 =7
m_eff
t
4
2
0
−2
−4
0
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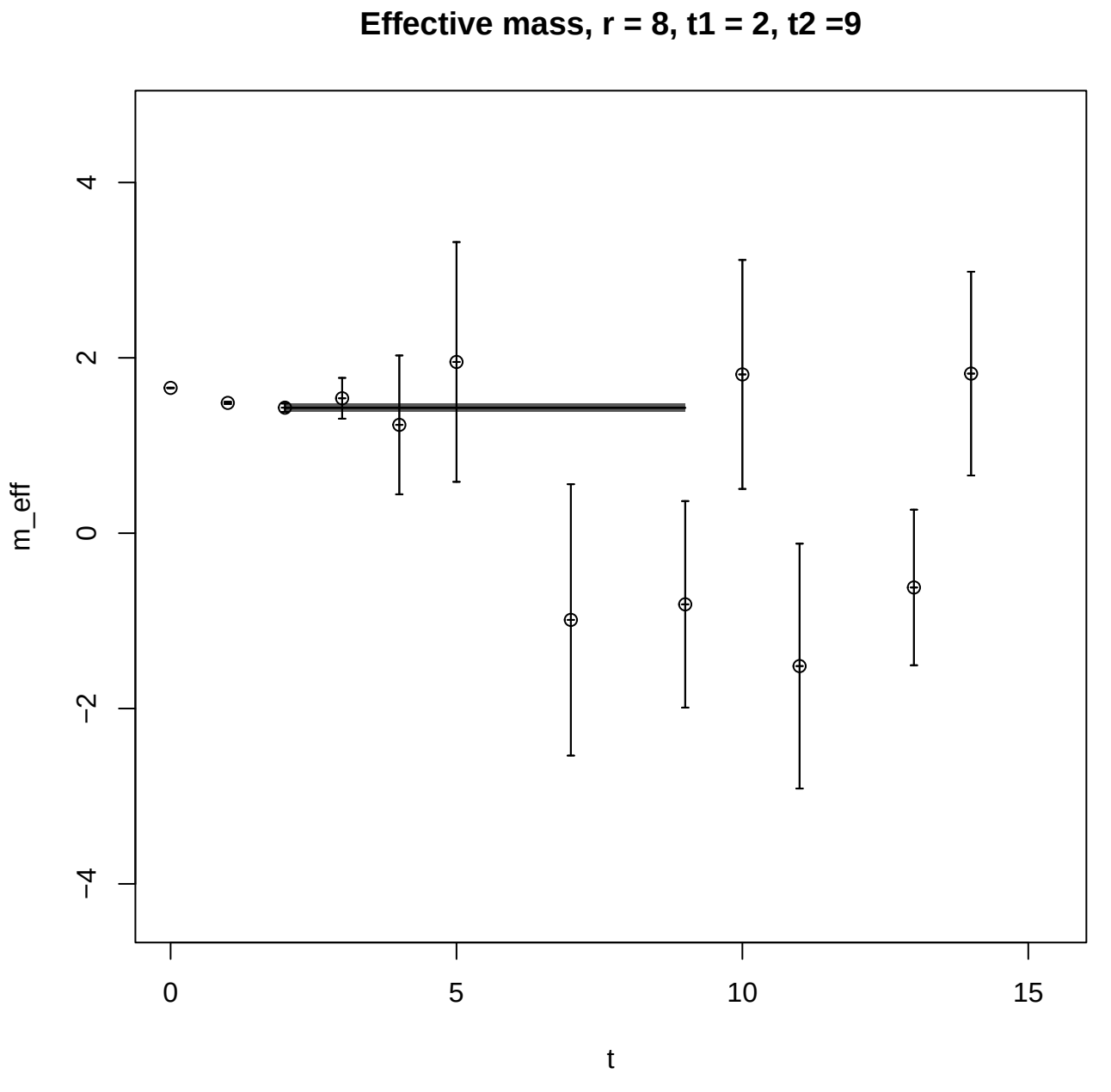
Effective mass, r = 8, t1 = 2, t2 =9
m_eff
t
4
2
0
−2
−4
0
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10
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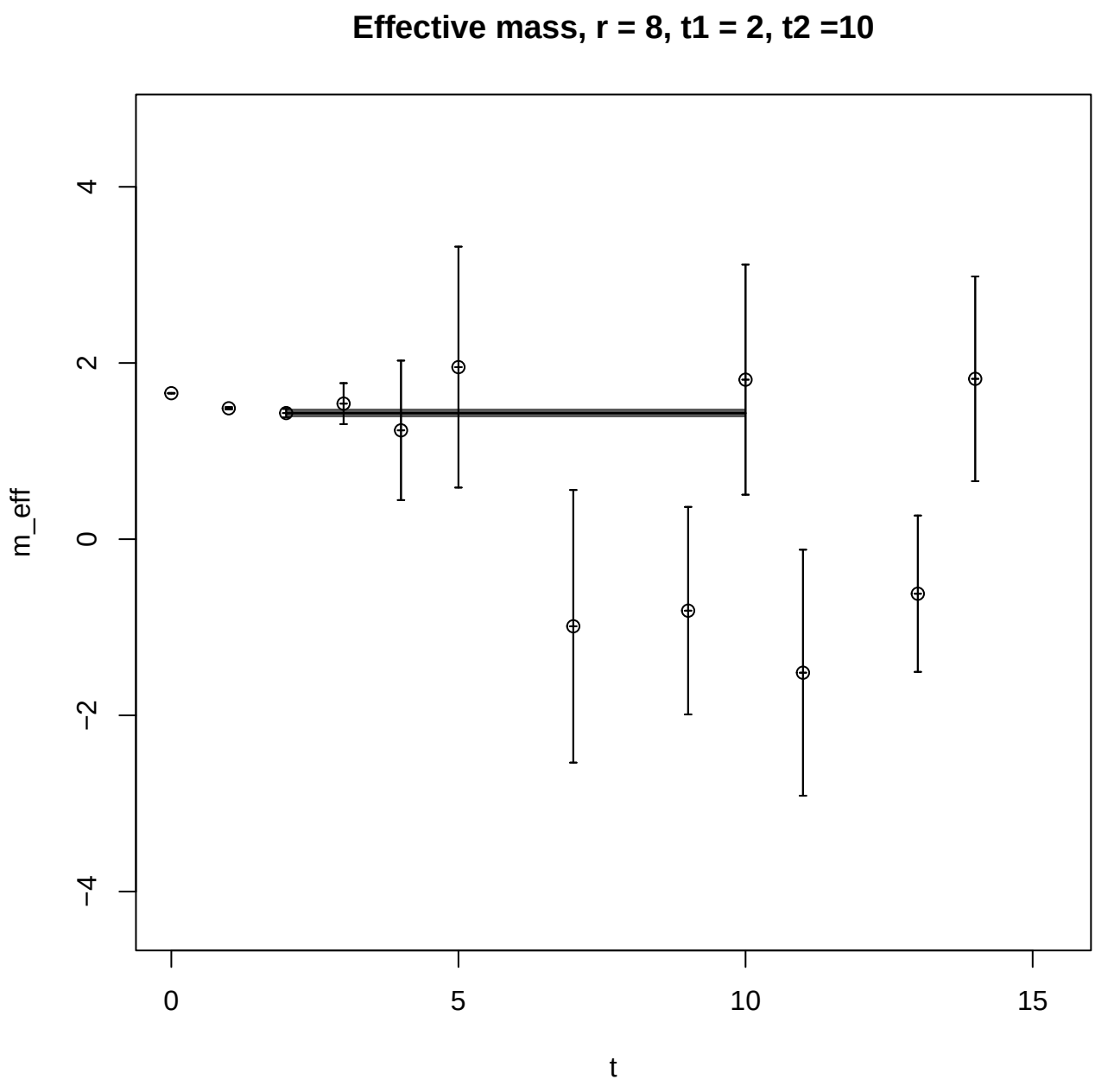
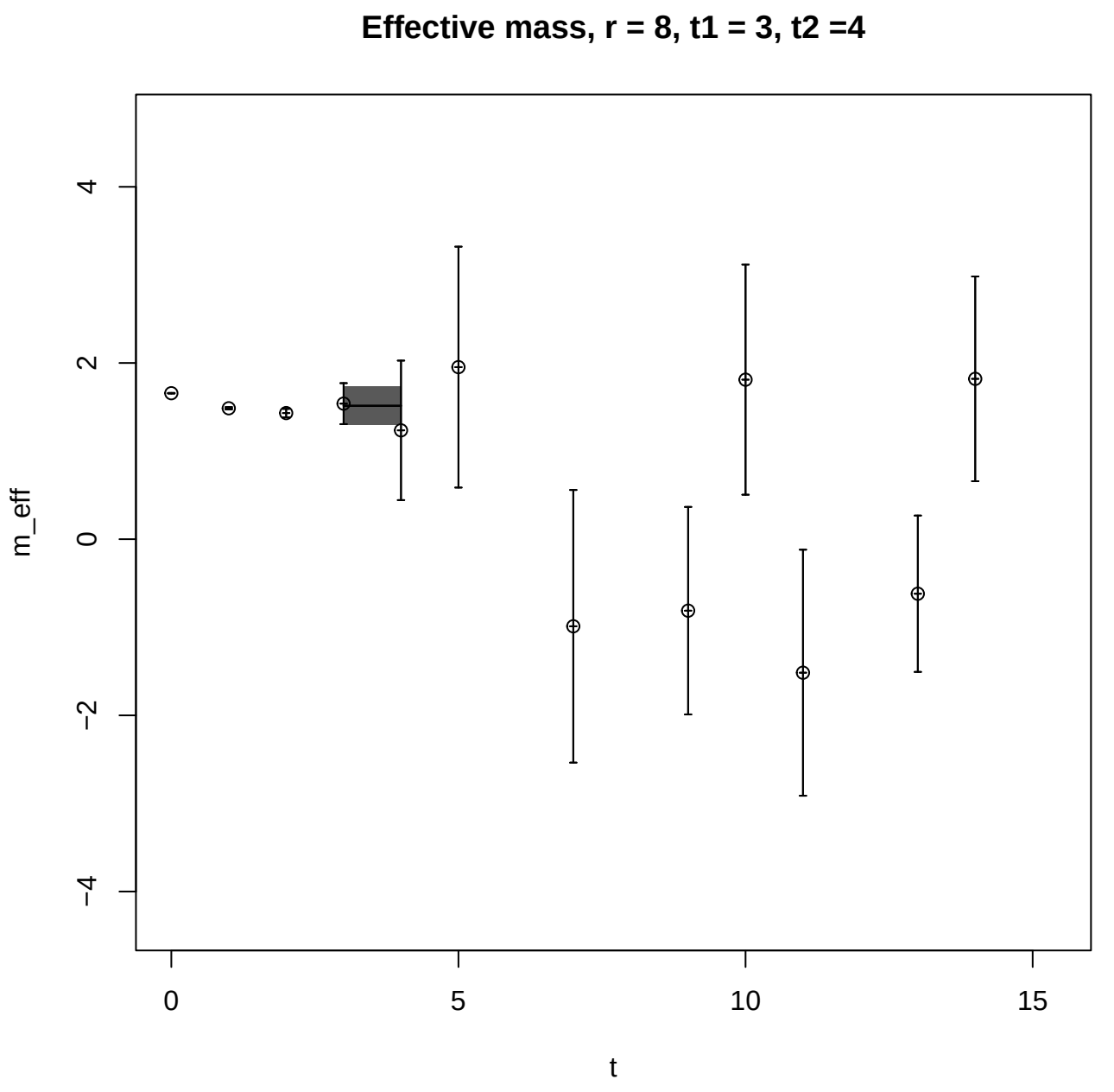
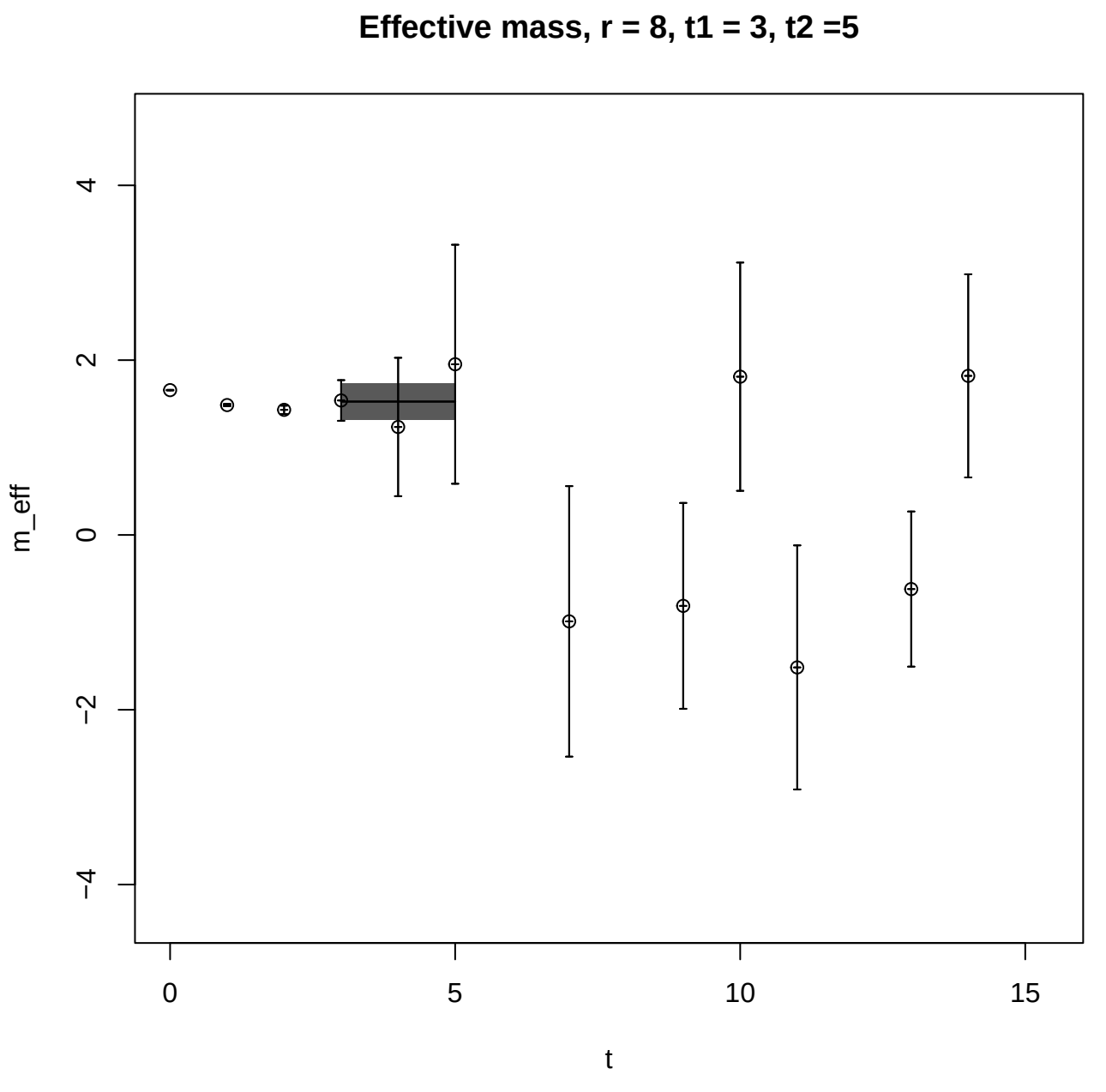
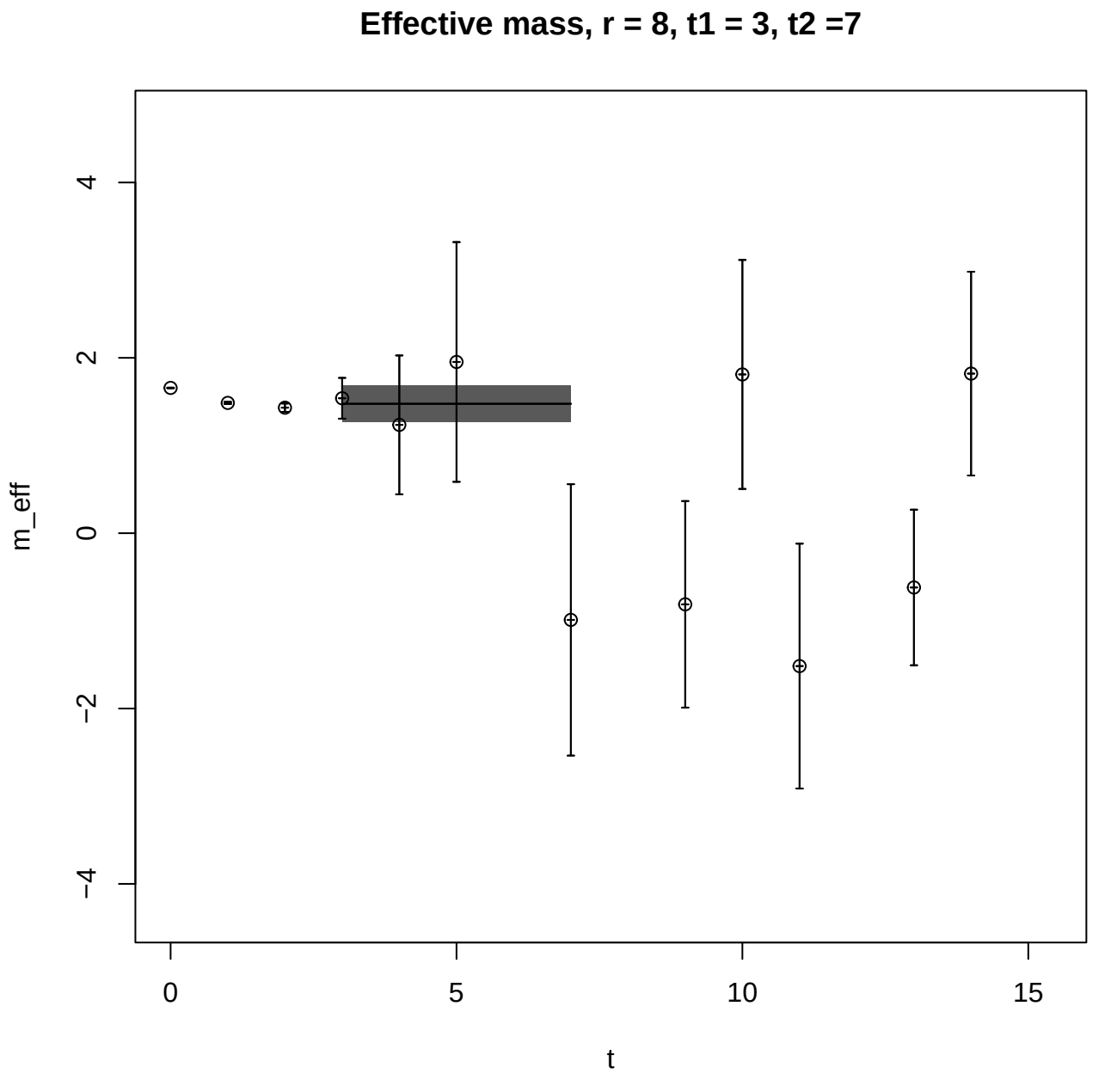

Effective mass, r = 8, t1 = 2, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 3, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 3, t2 =7
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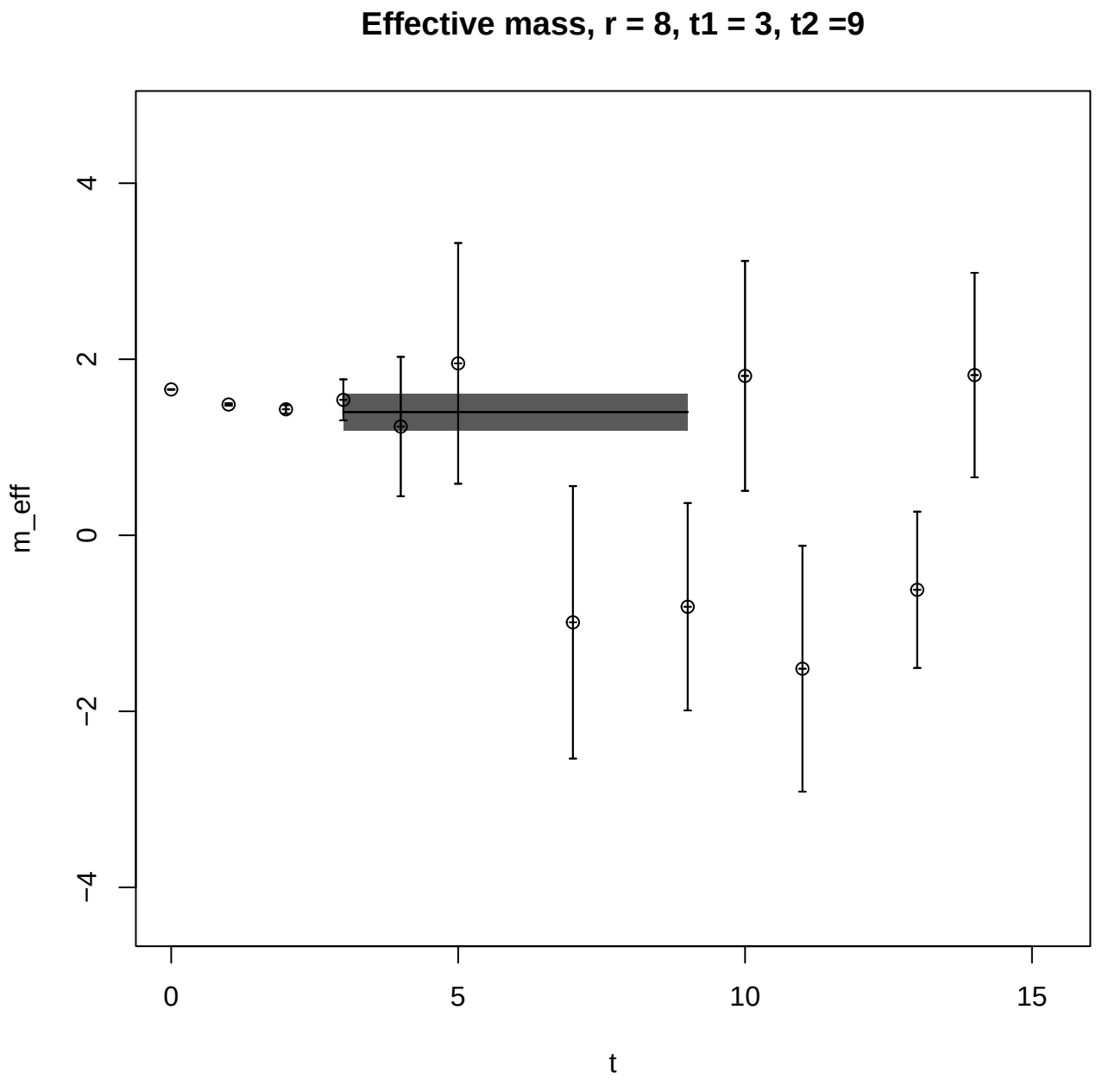
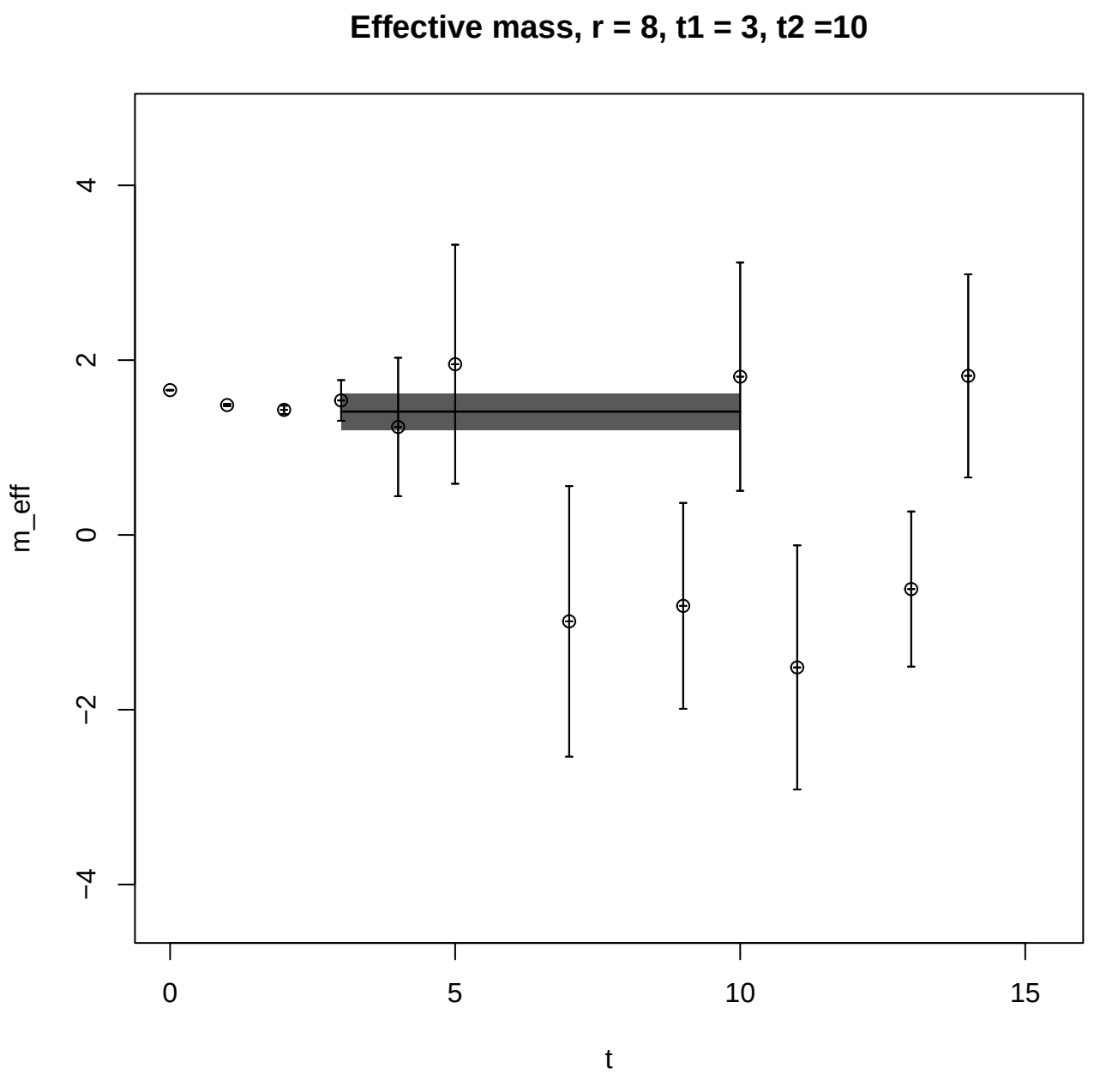
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 3, t2 =9
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 3, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
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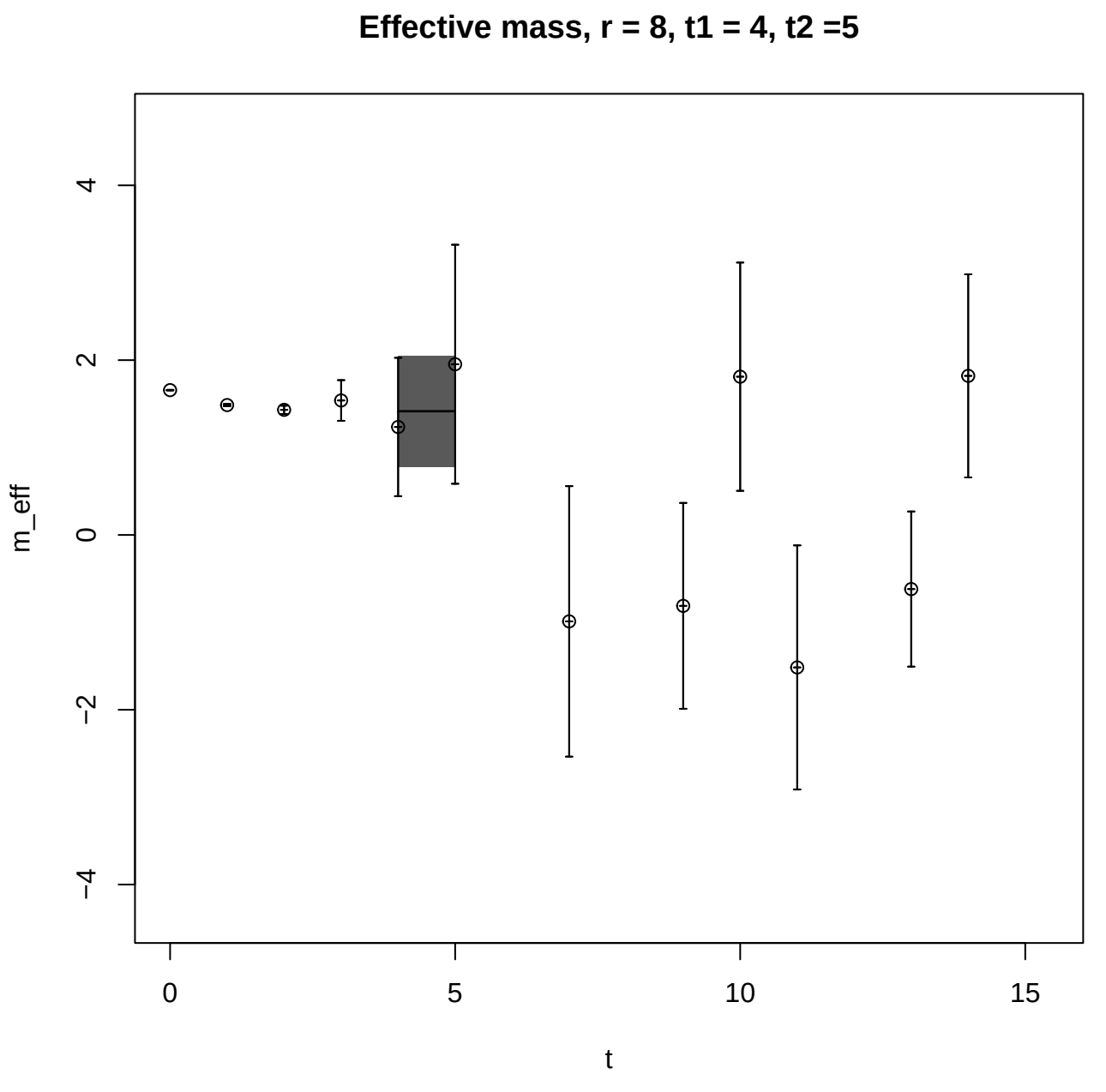
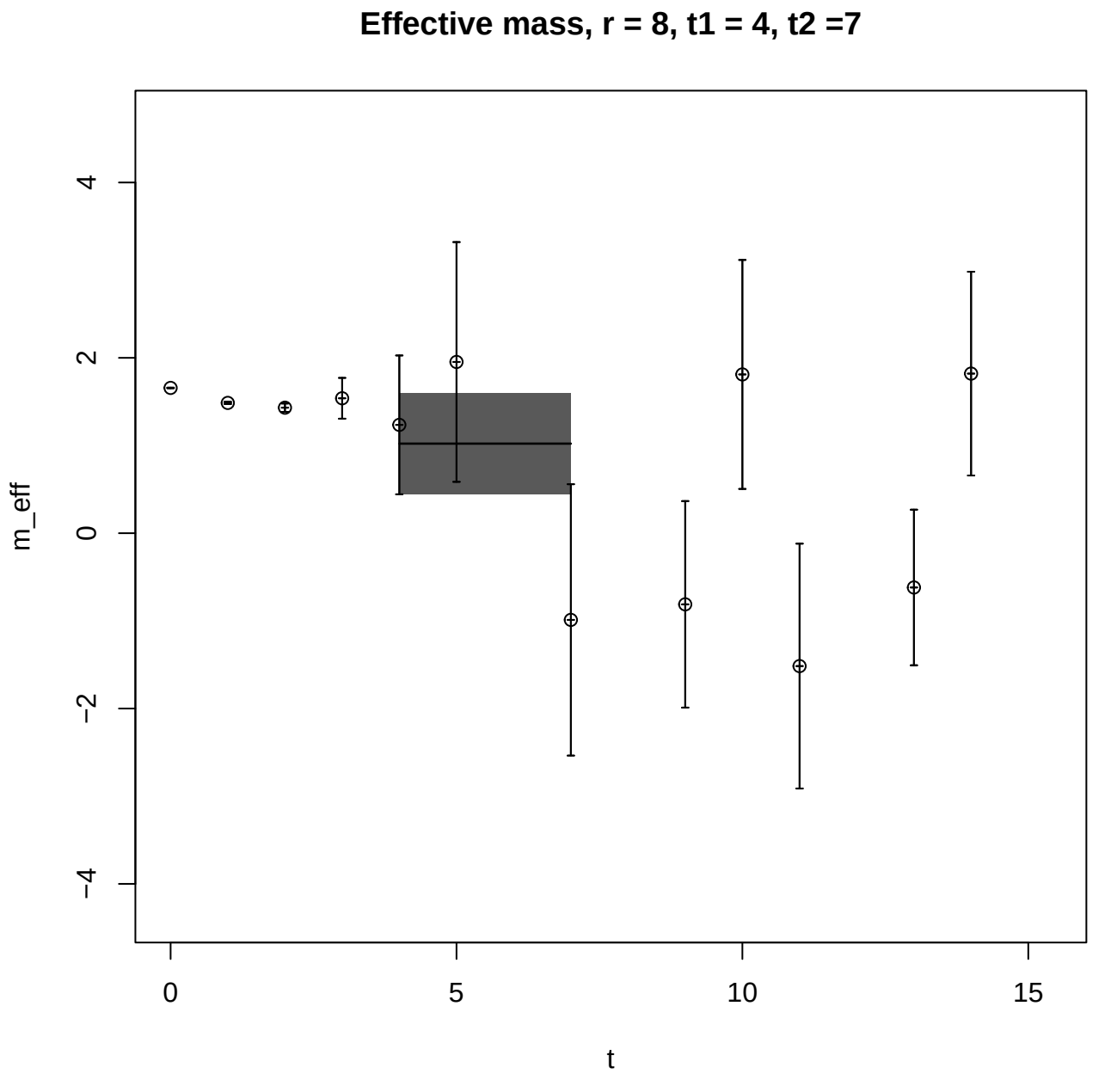
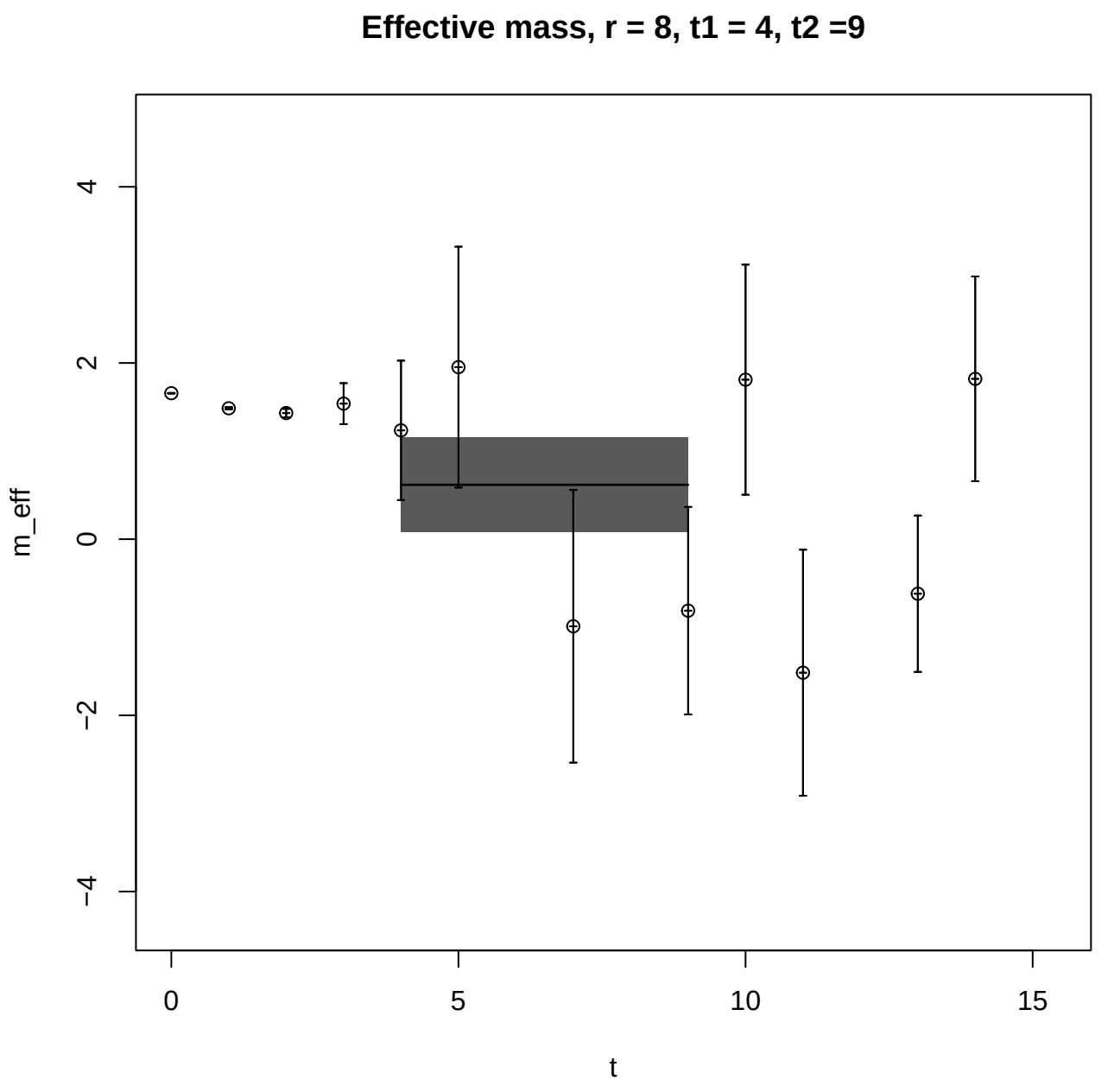

Effective mass, r = 8, t1 = 4, t2 =5
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 4, t2 =7
m_eff
4
2
0
−2
−4
0
5
10
15
t


Effective mass, r = 8, t1 = 4, t2 =9
m_eff
t
4
2
0
−2
−4
0
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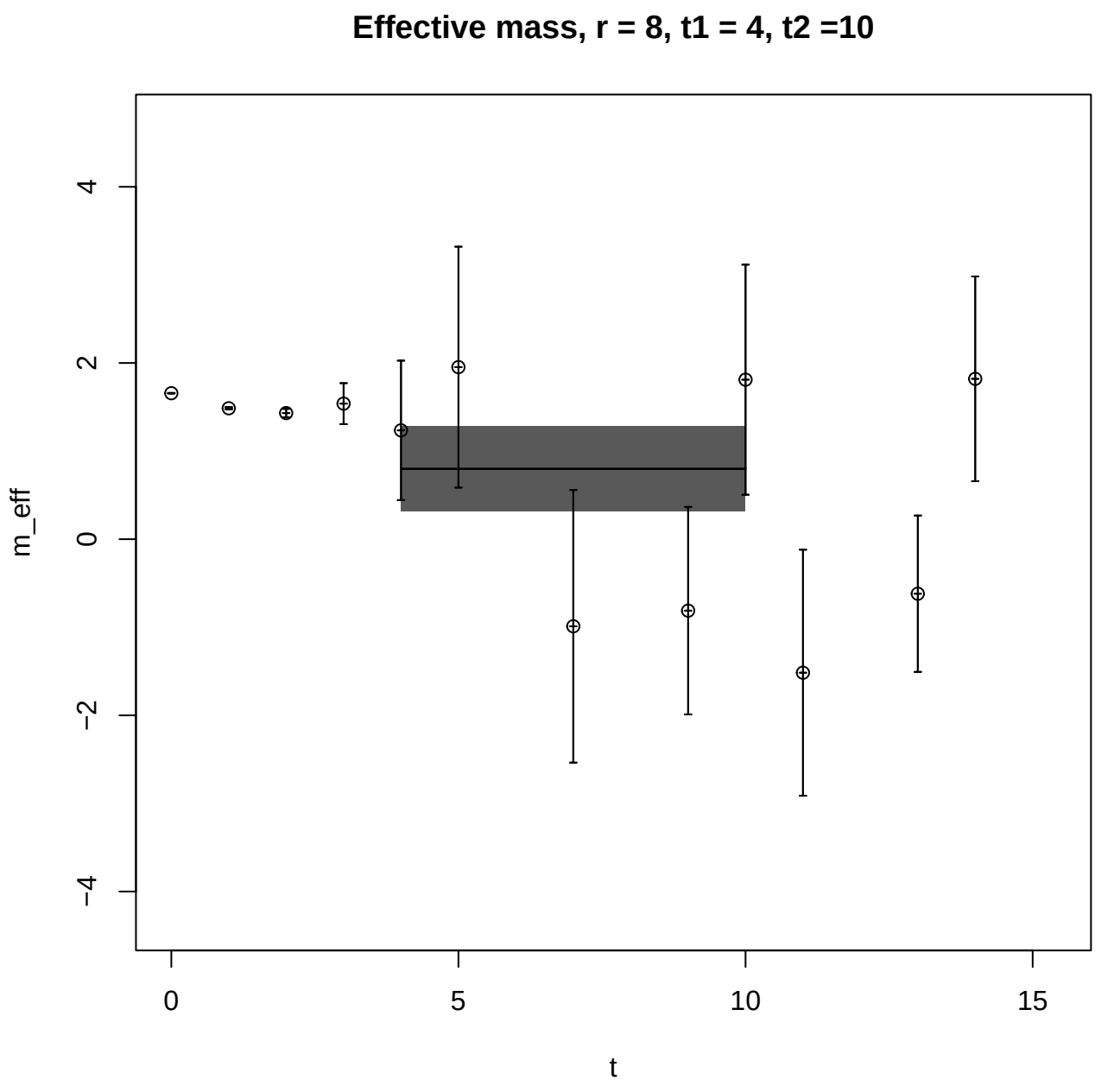
Effective mass, r = 8, t1 = 4, t2 =10
m_eff
t
4
2
0
−2
−4
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5
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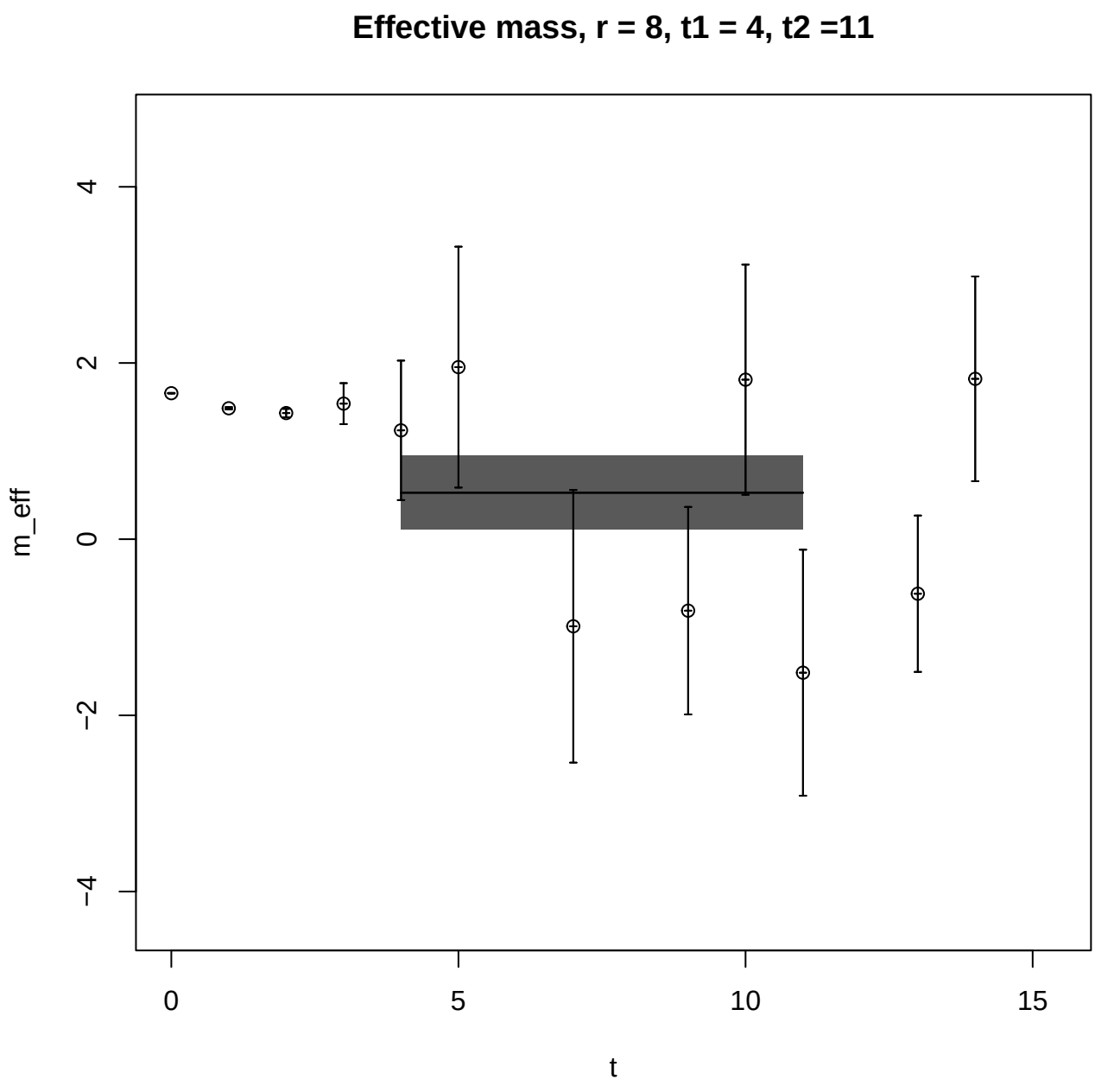
Effective mass, r = 8, t1 = 4, t2 =11
m_eff
t
4
2
0
−2
−4
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10
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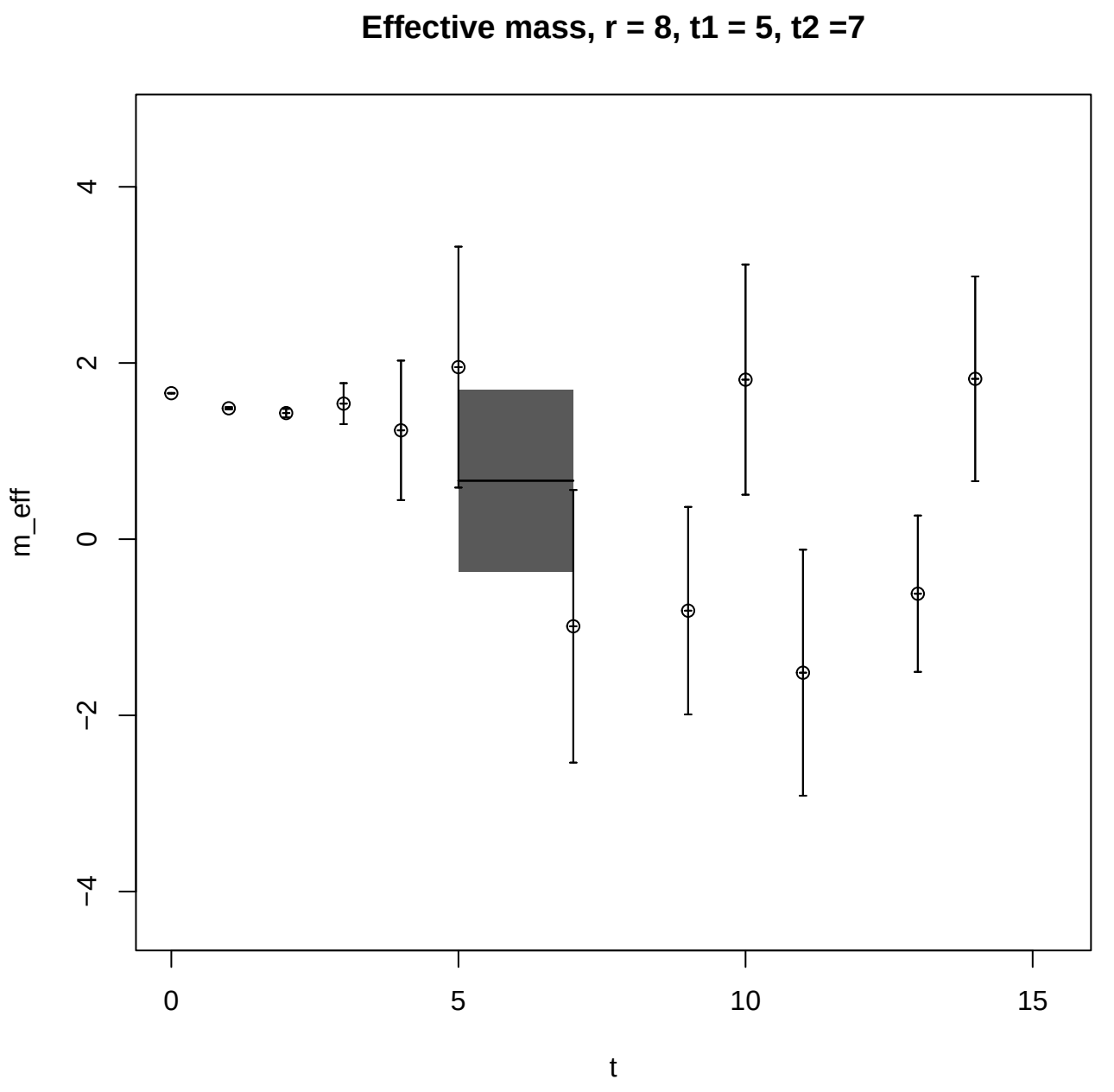
Effective mass, r = 8, t1 = 5, t2 =7
m_eff
t
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−2
−4
0
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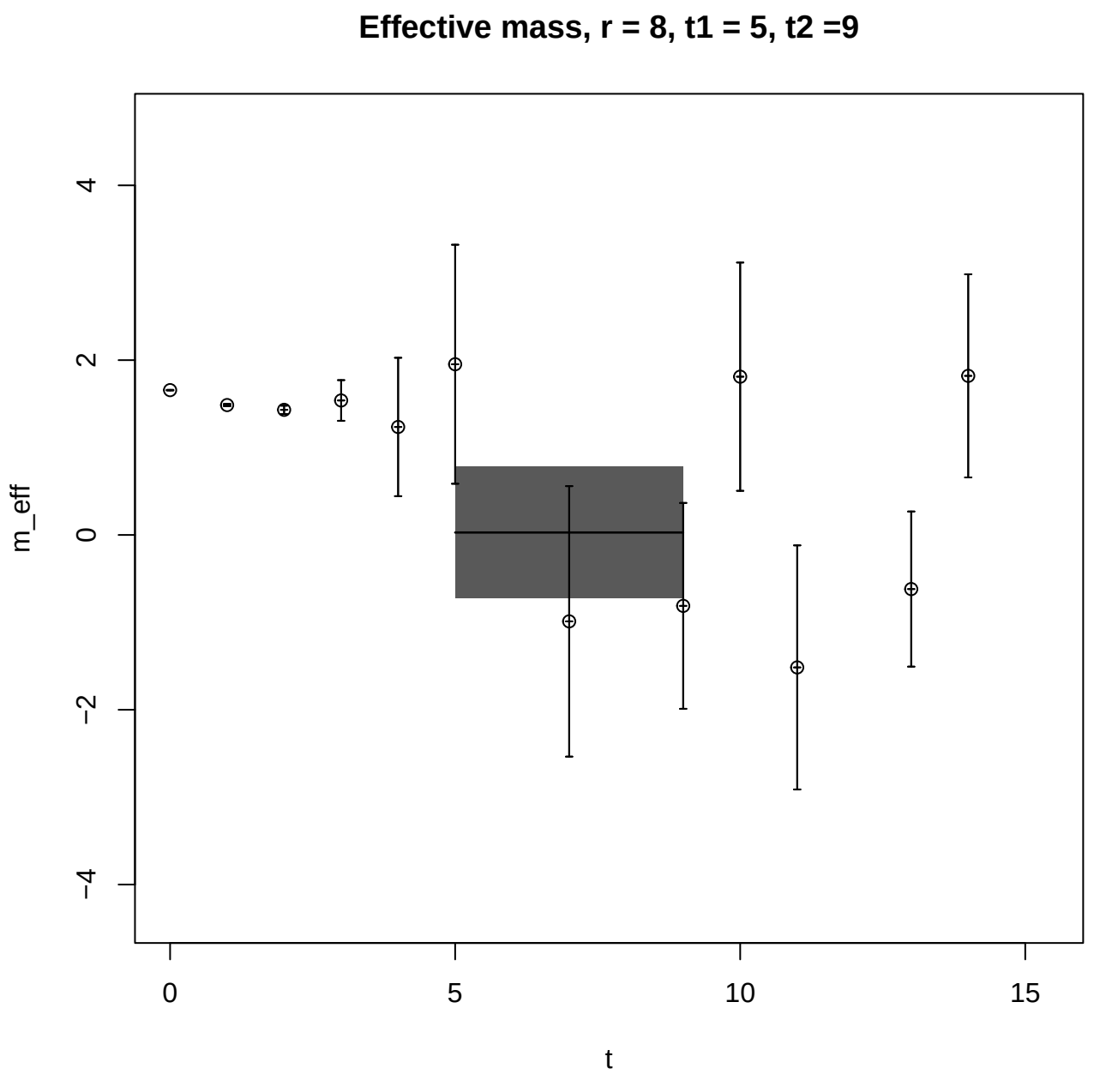
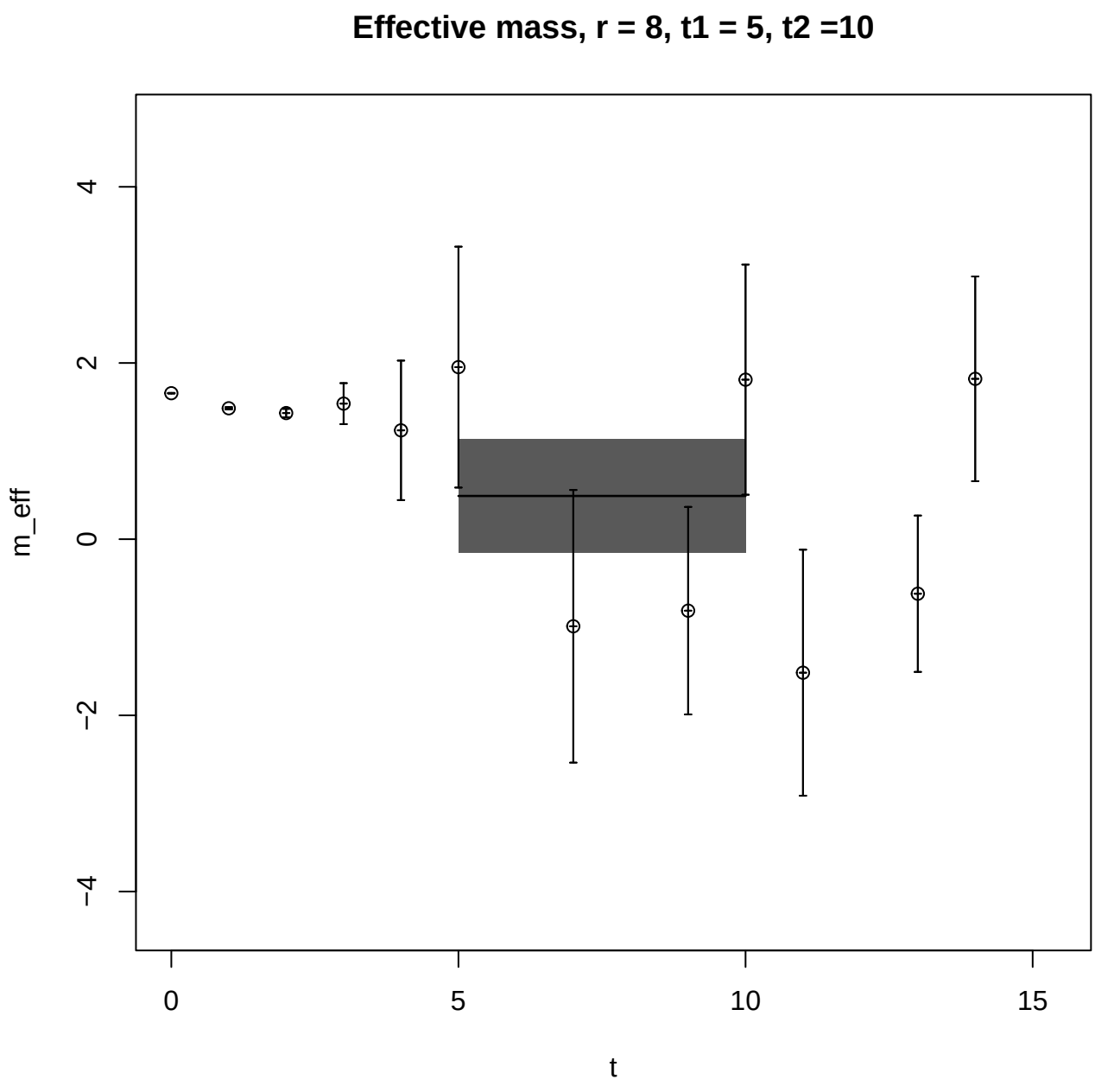
10
15


Effective mass, r = 8, t1 = 5, t2 =9
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 8, t1 = 5, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
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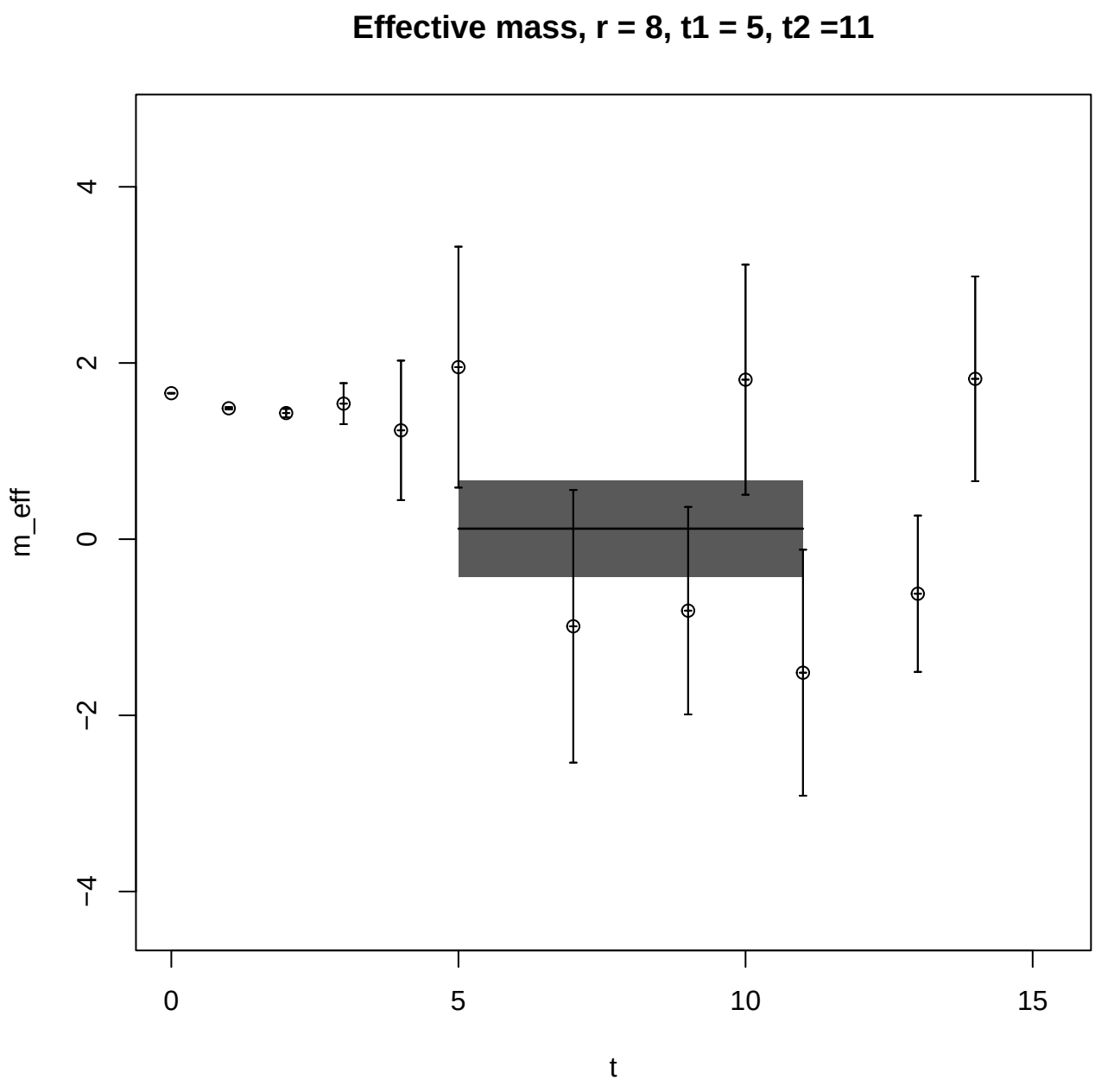
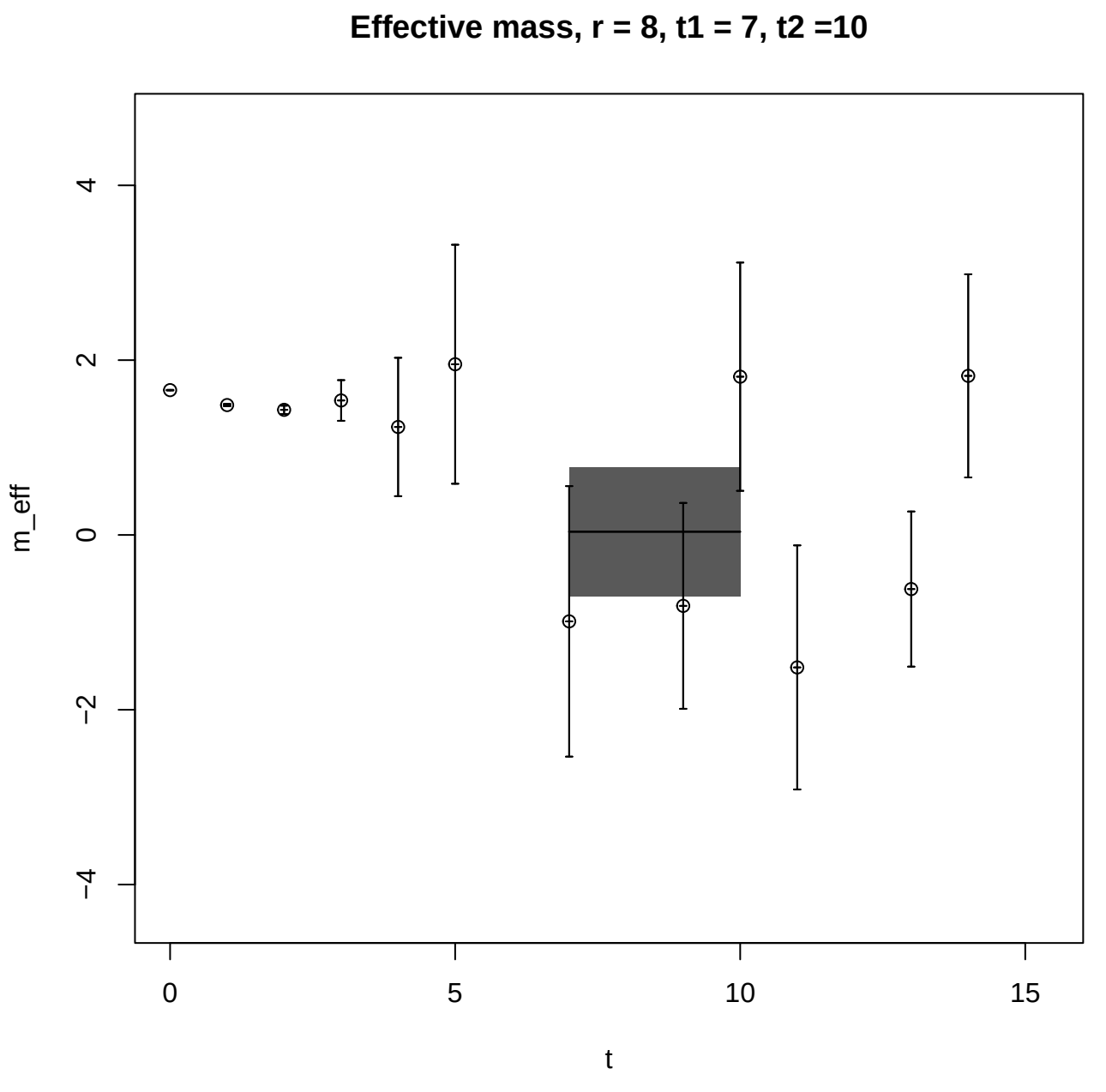
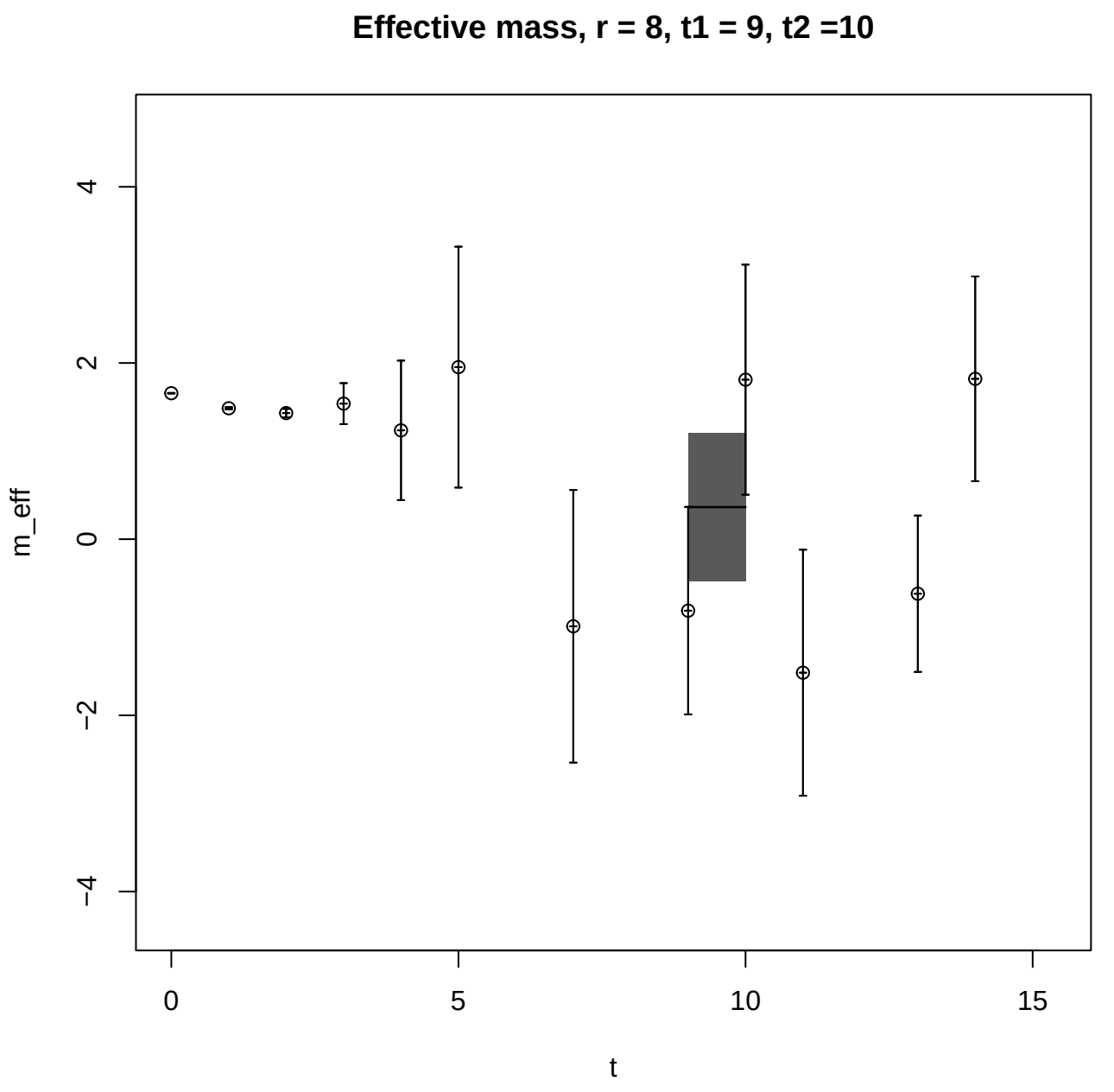
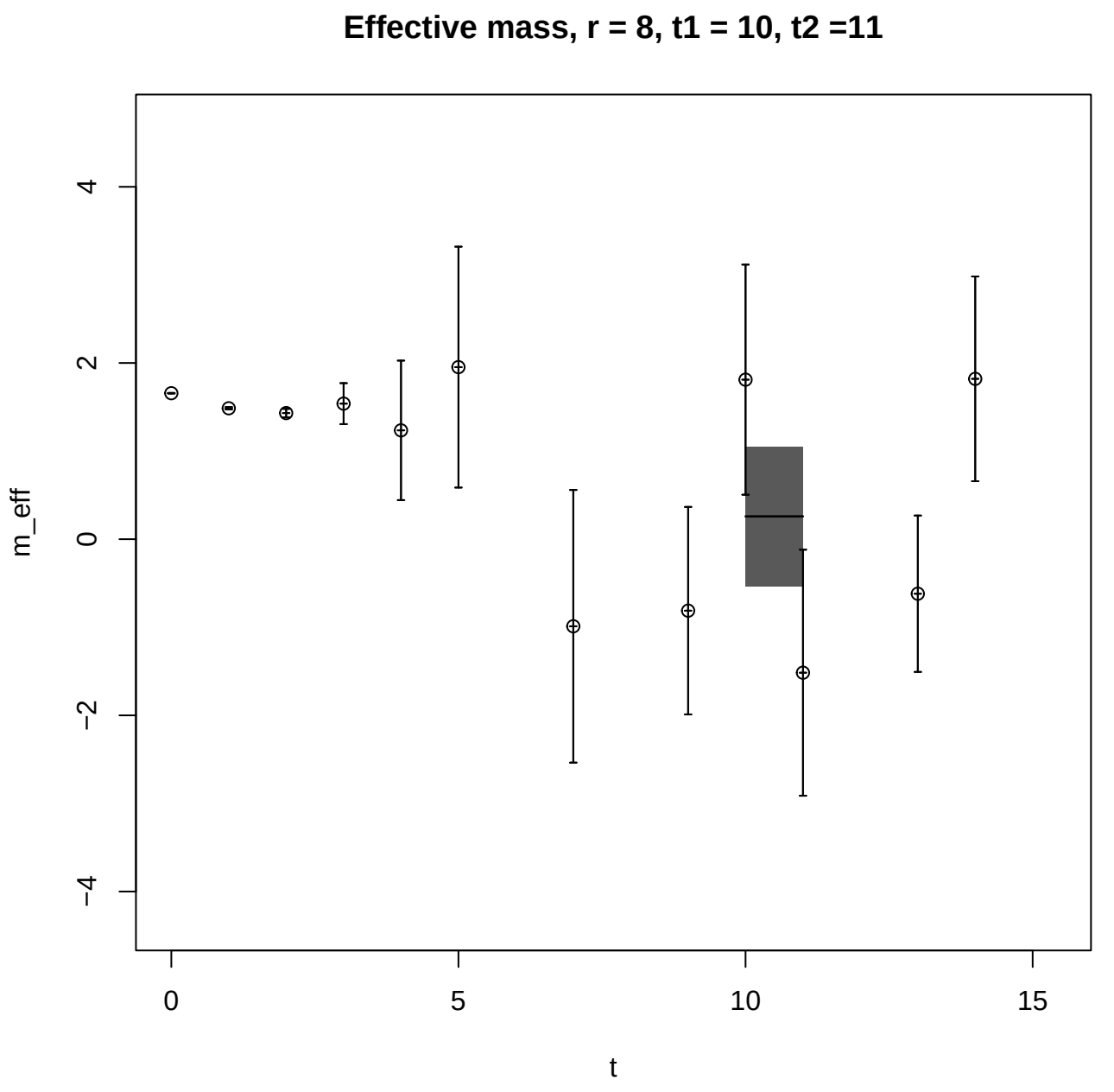
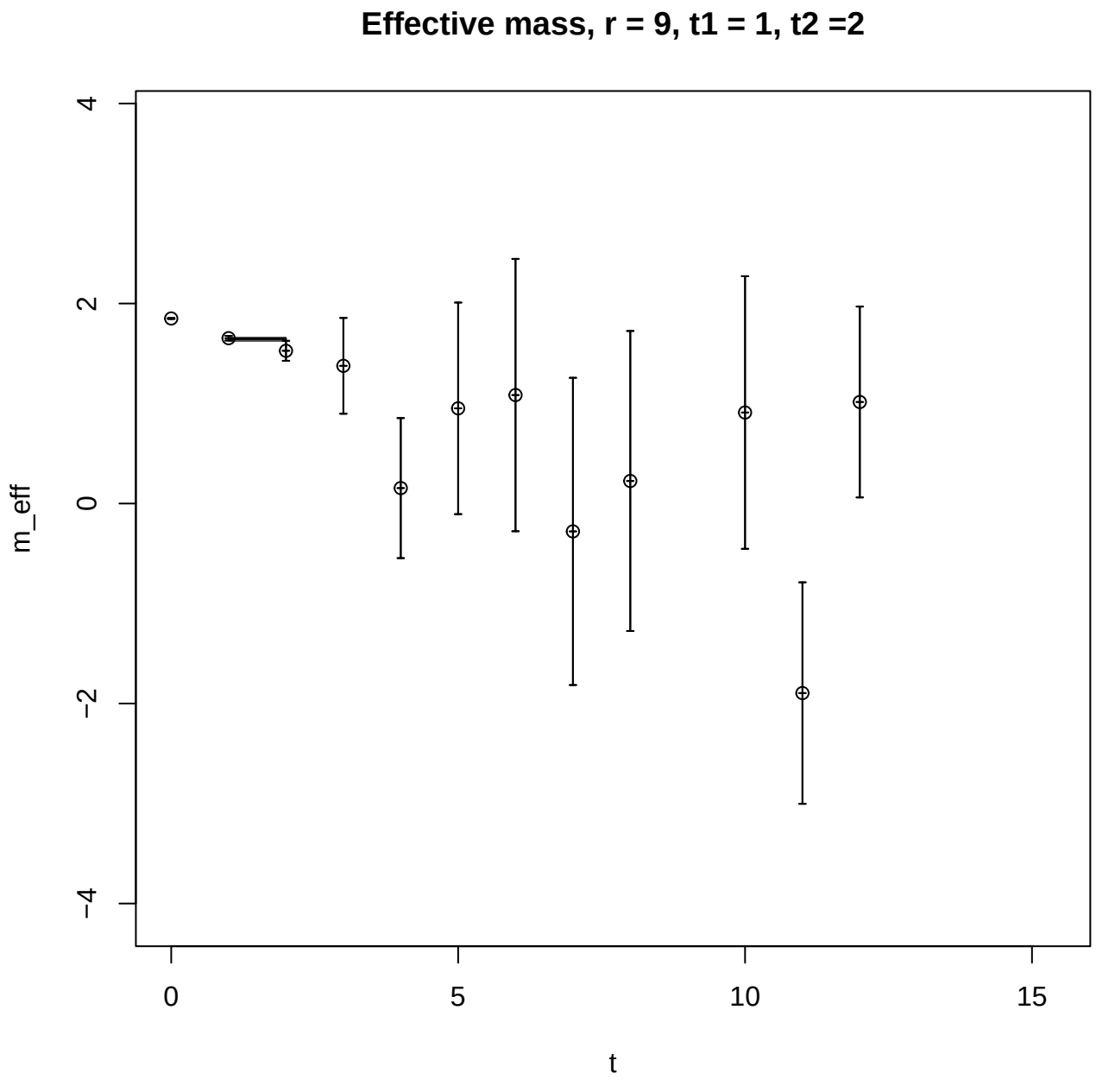
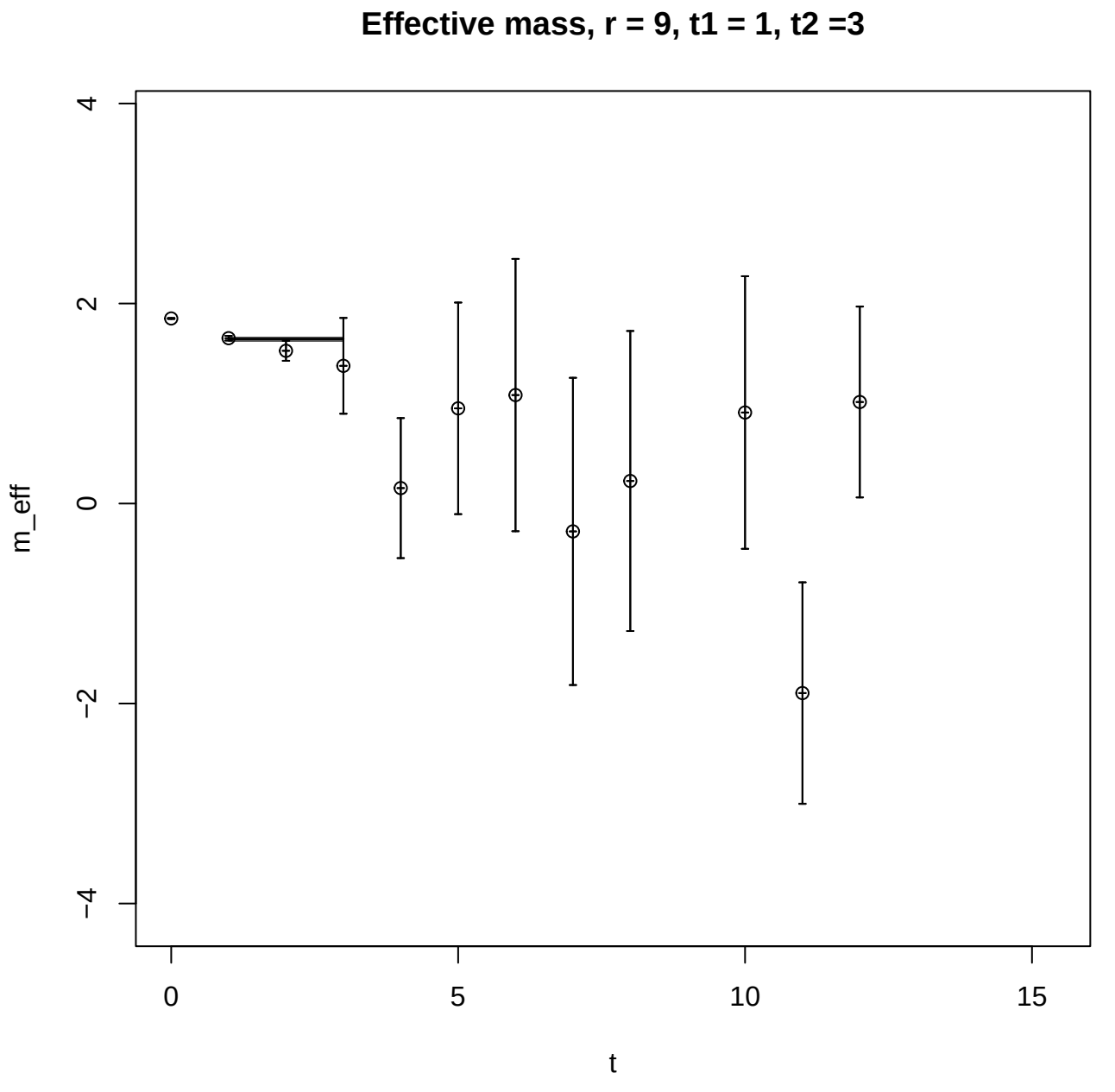

Effective mass, r = 8, t1 = 5, t2 =11
m_eff
t
4
2
0
−2
−4
0
5
10
15

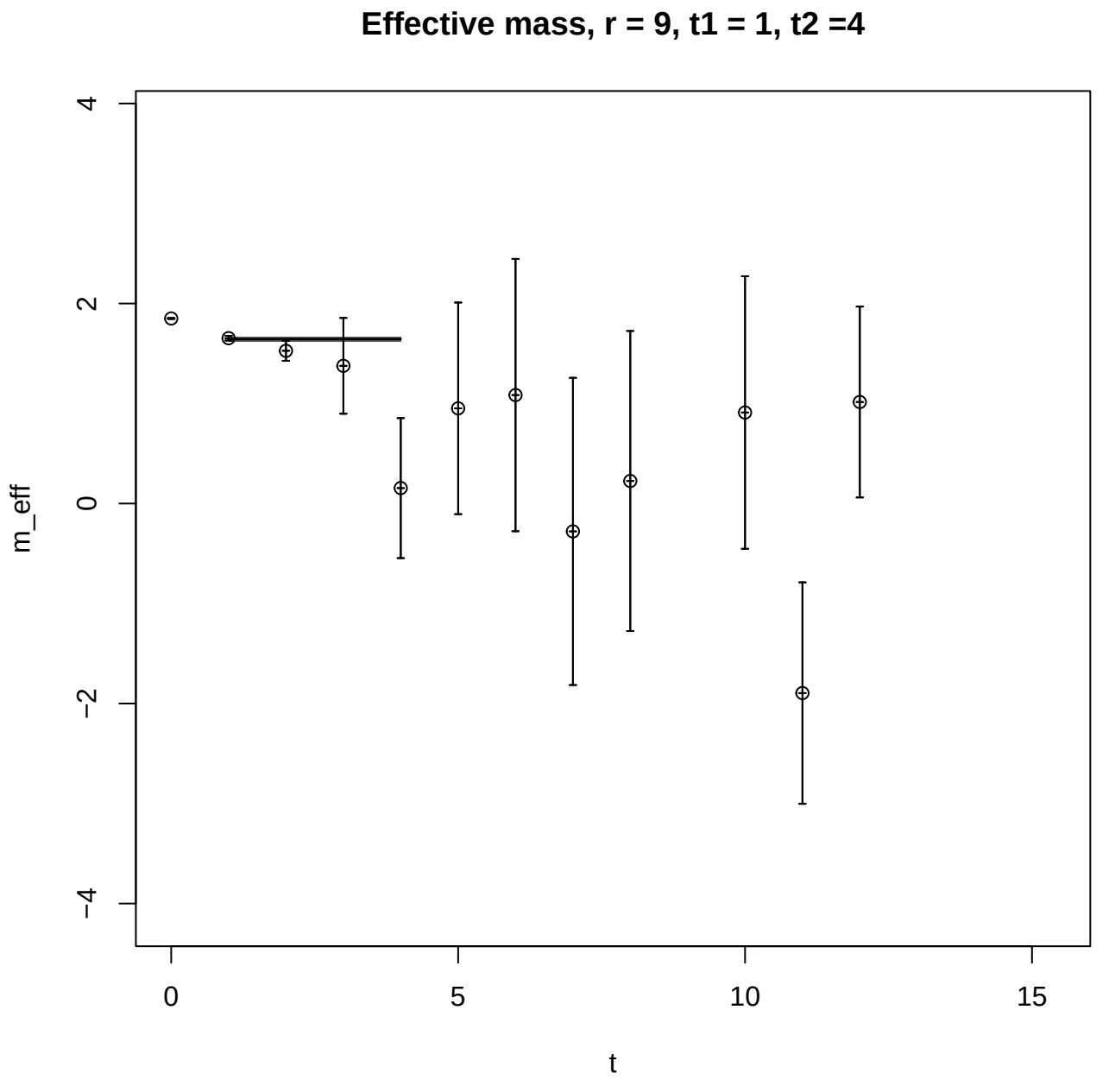

Effective mass, r = 8, t1 = 7, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

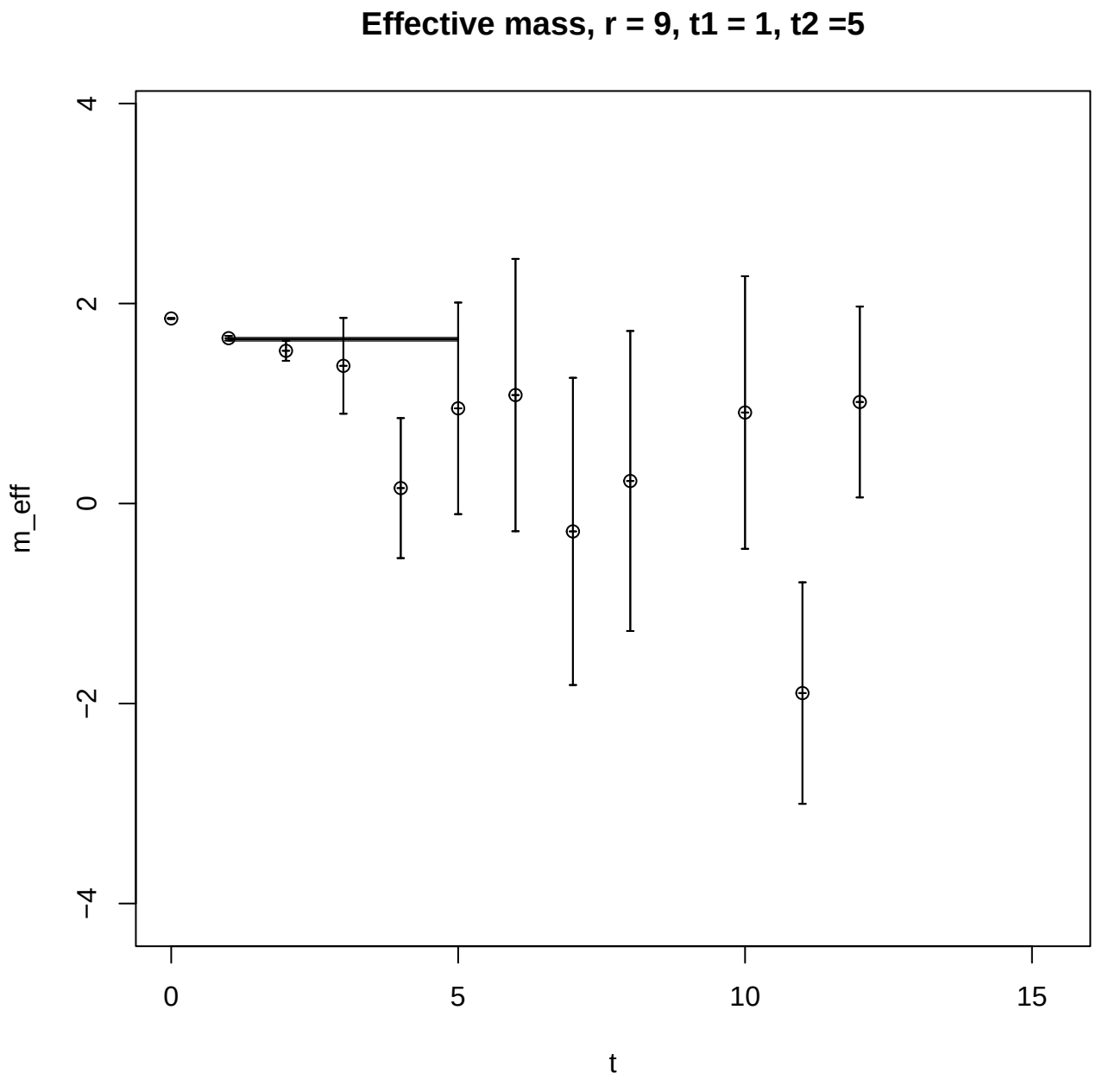

Effective mass, r = 8, t1 = 9, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

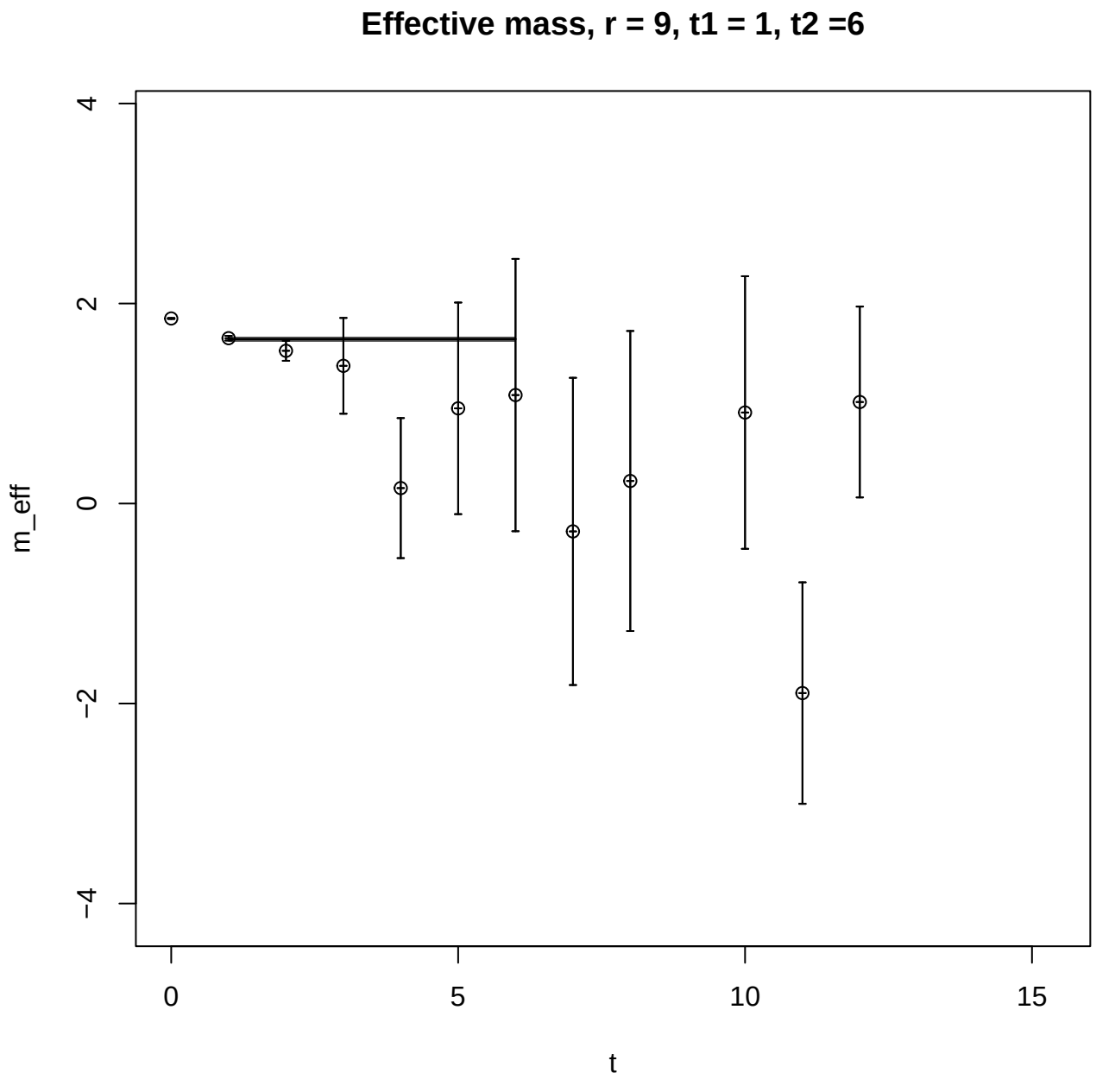

Effective mass, r = 8, t1 = 10, t2 =11
m_eff
4
2
0
−2
−4
0
5
10
15
t

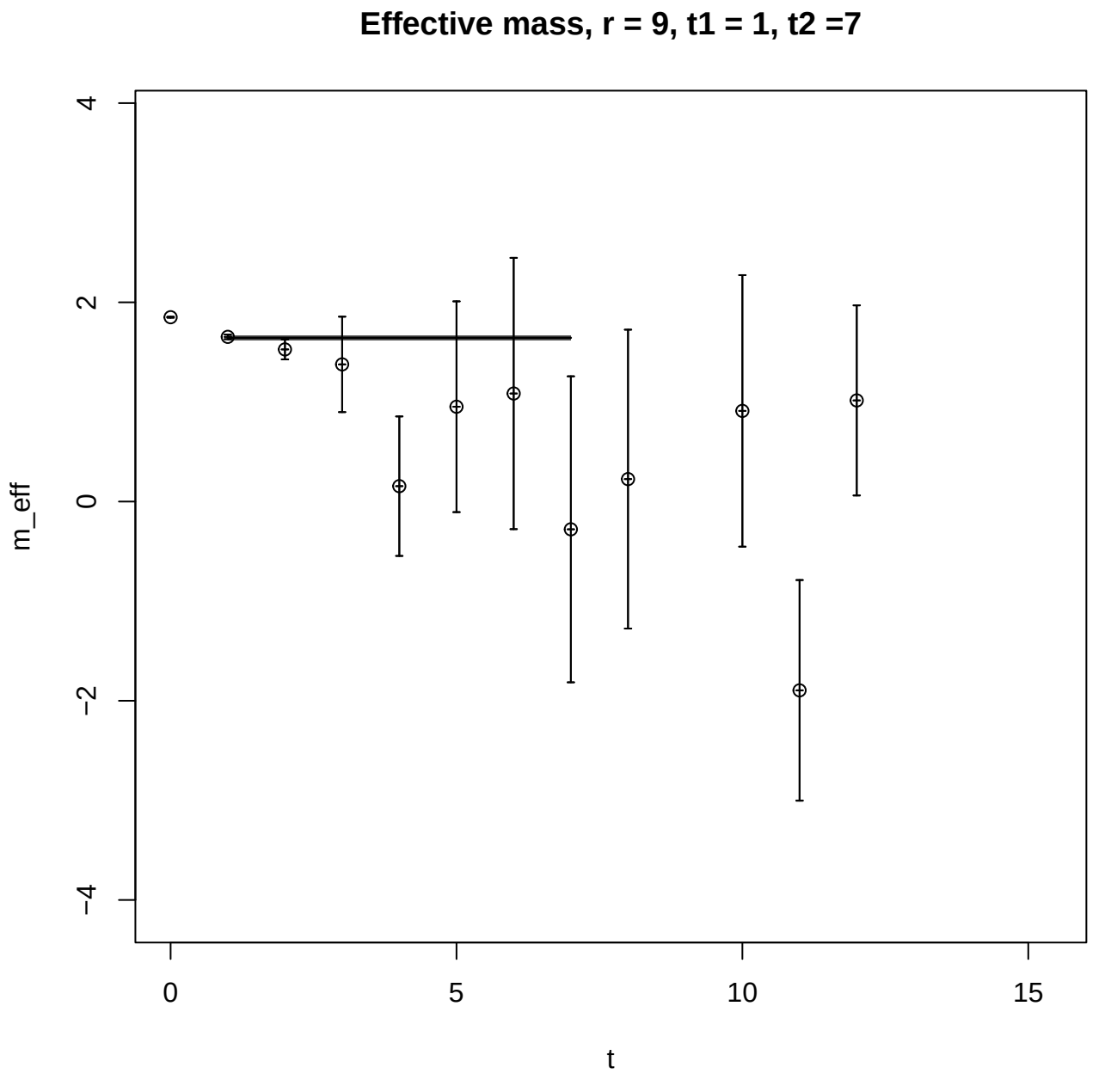

Effective mass, r = 9, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
0
5
10
15

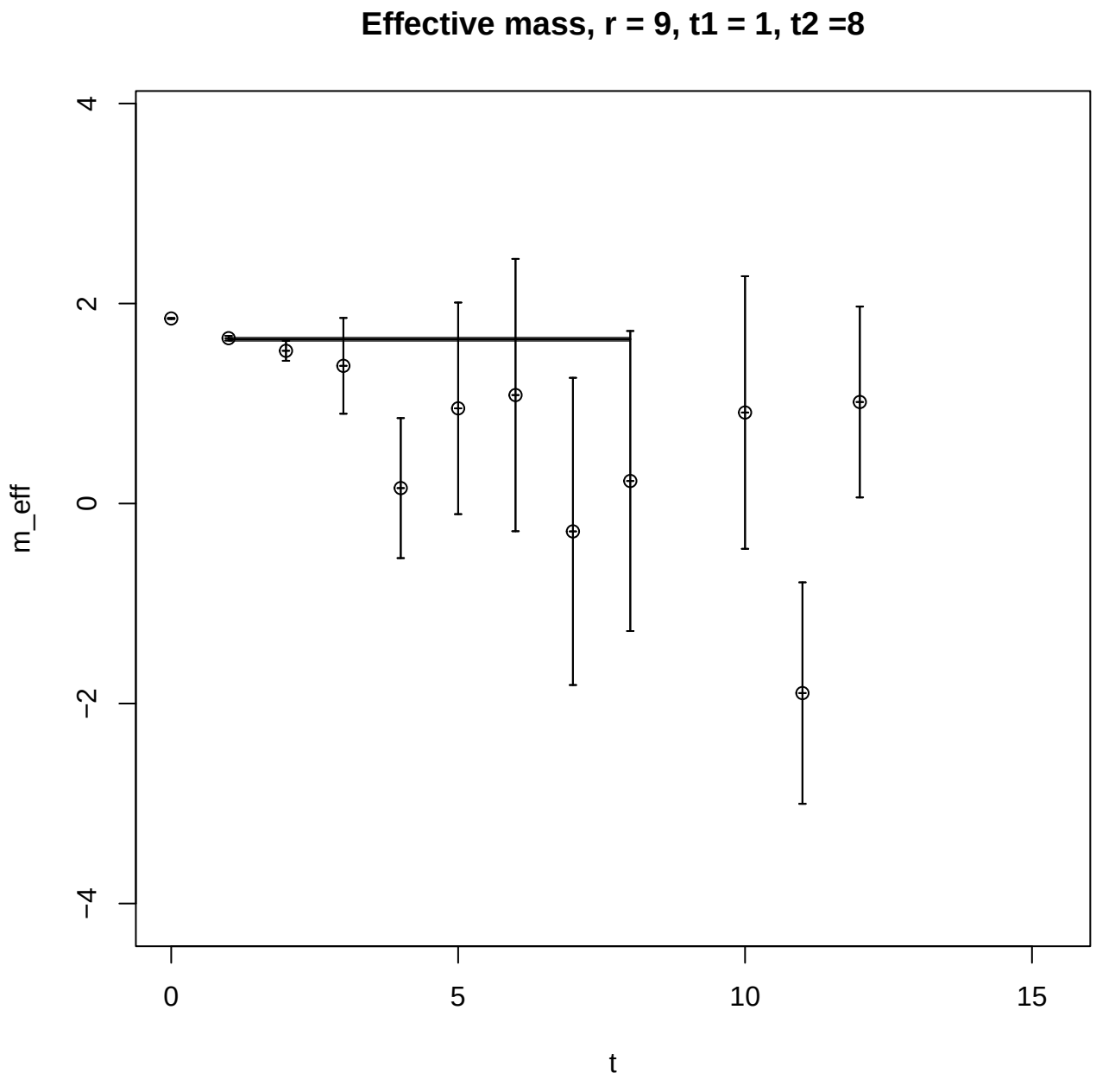

Effective mass, r = 9, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

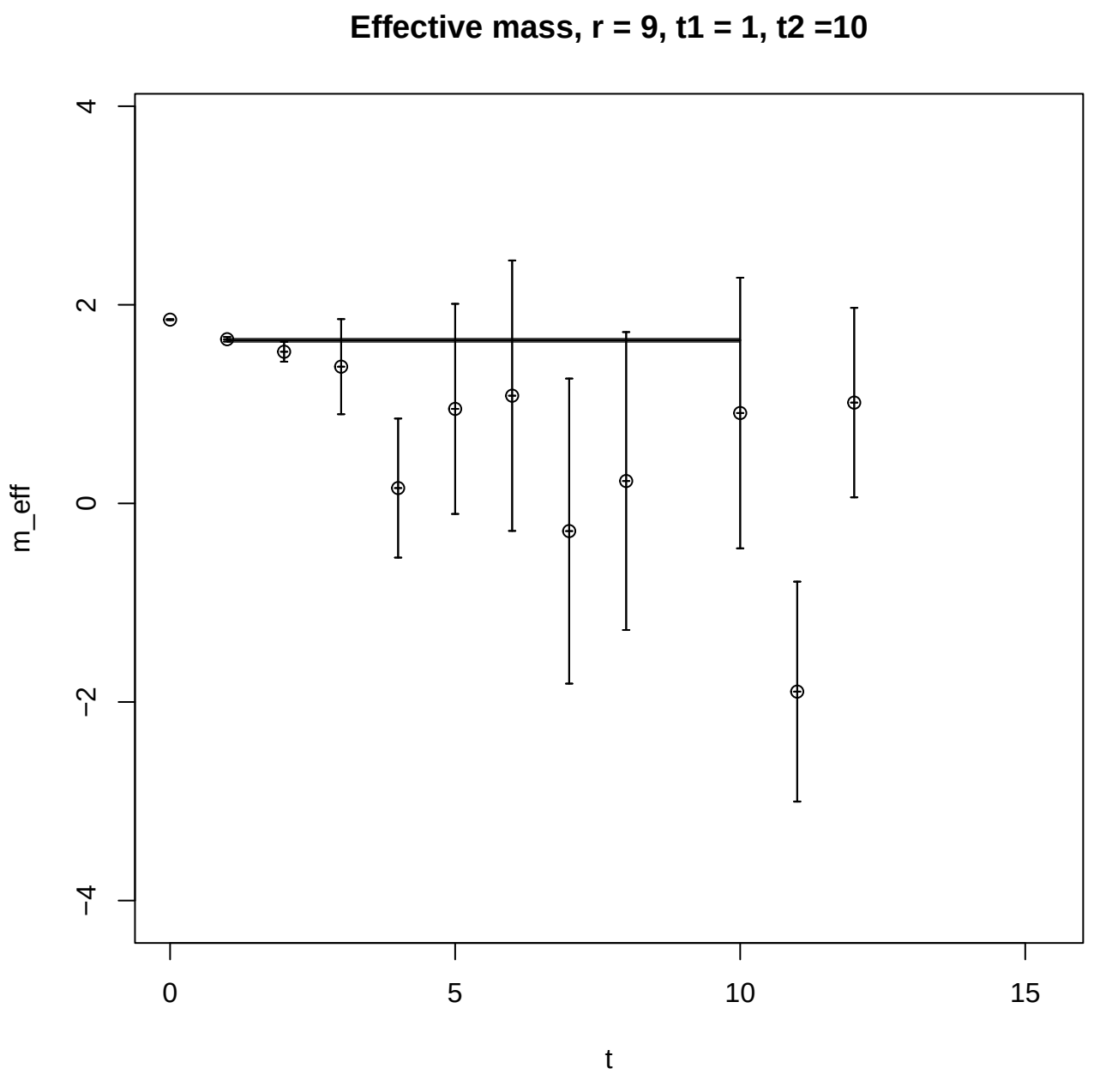

Effective mass, r = 9, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

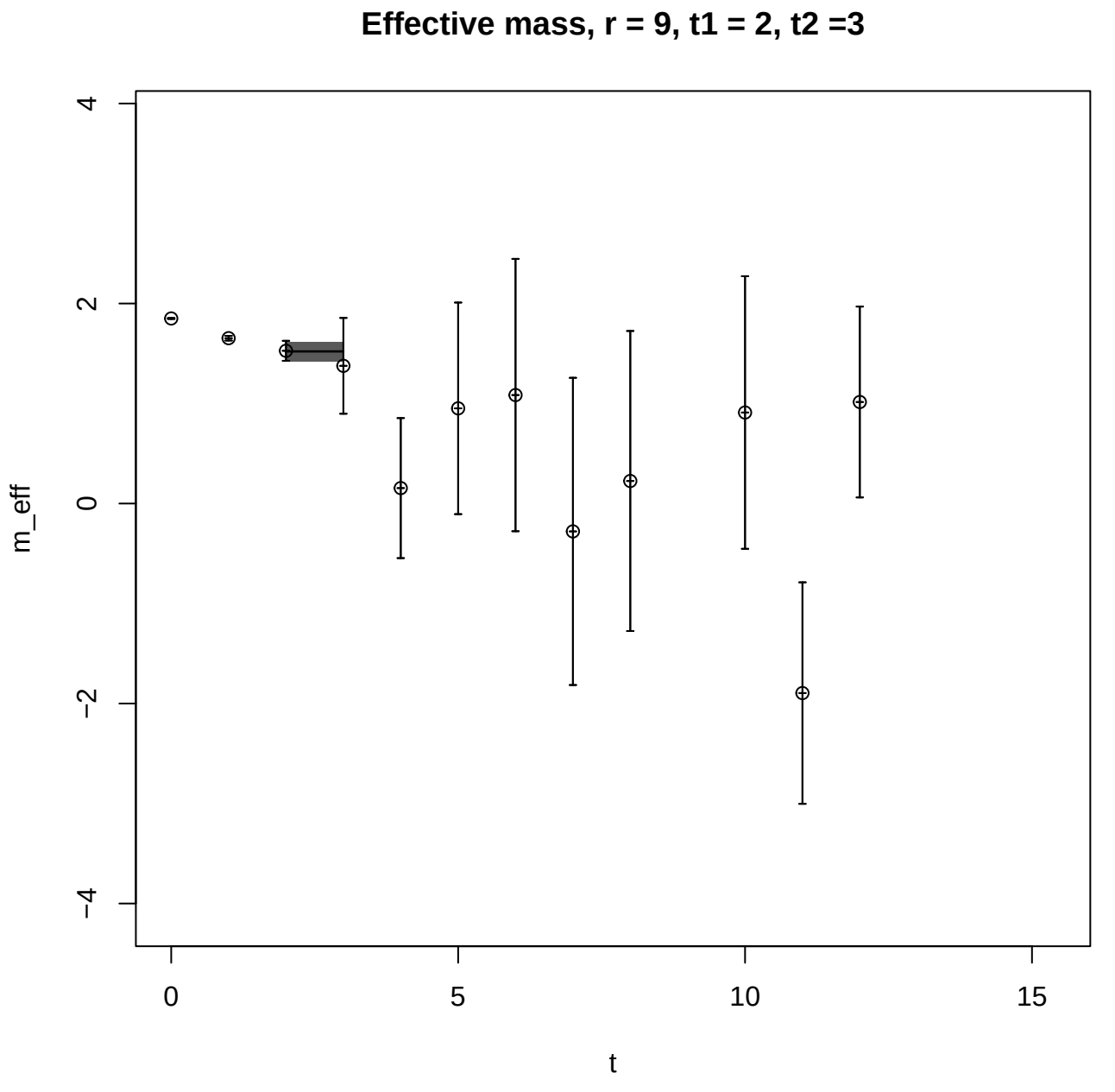

Effective mass, r = 9, t1 = 1, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

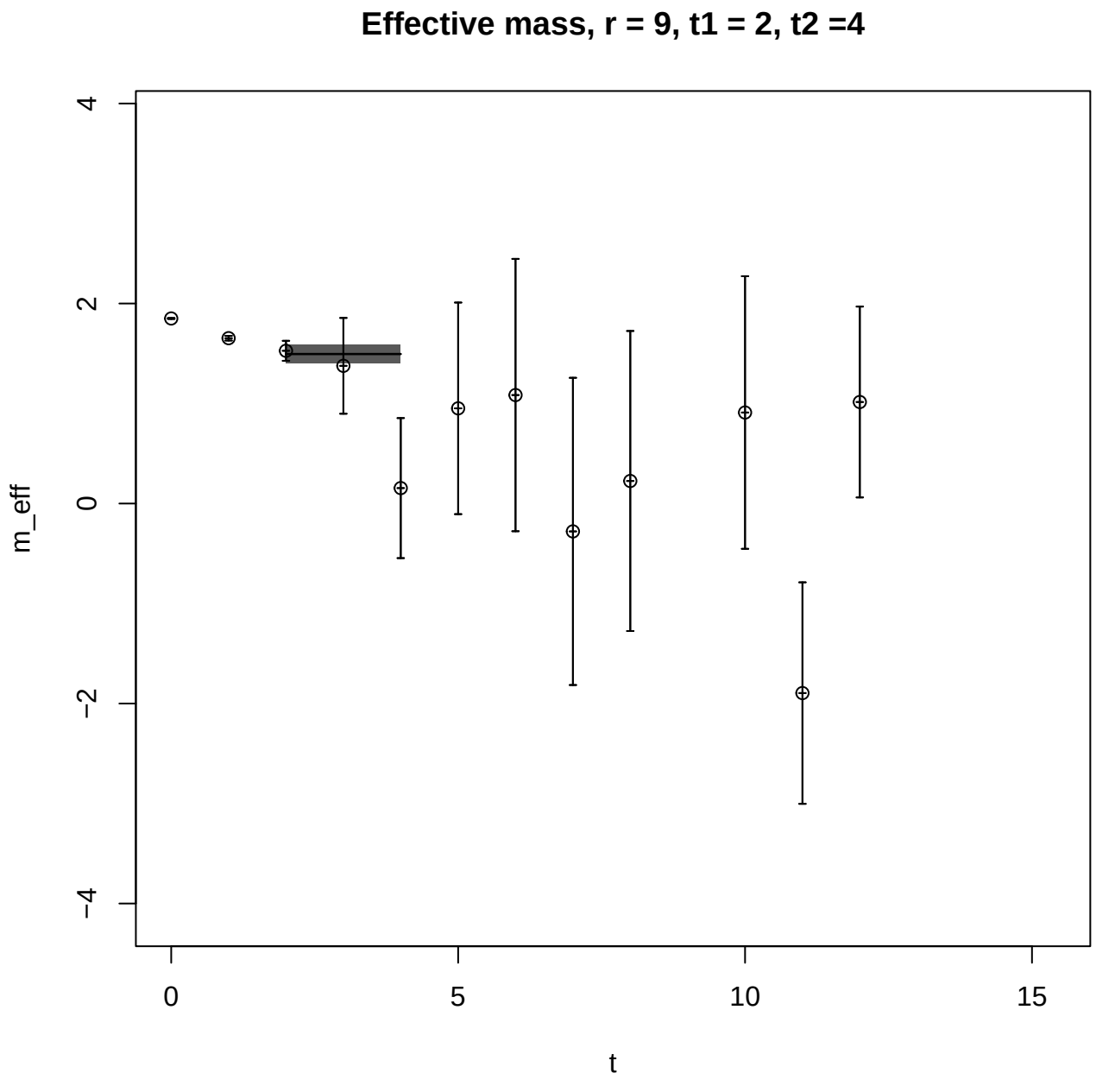

Effective mass, r = 9, t1 = 1, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15

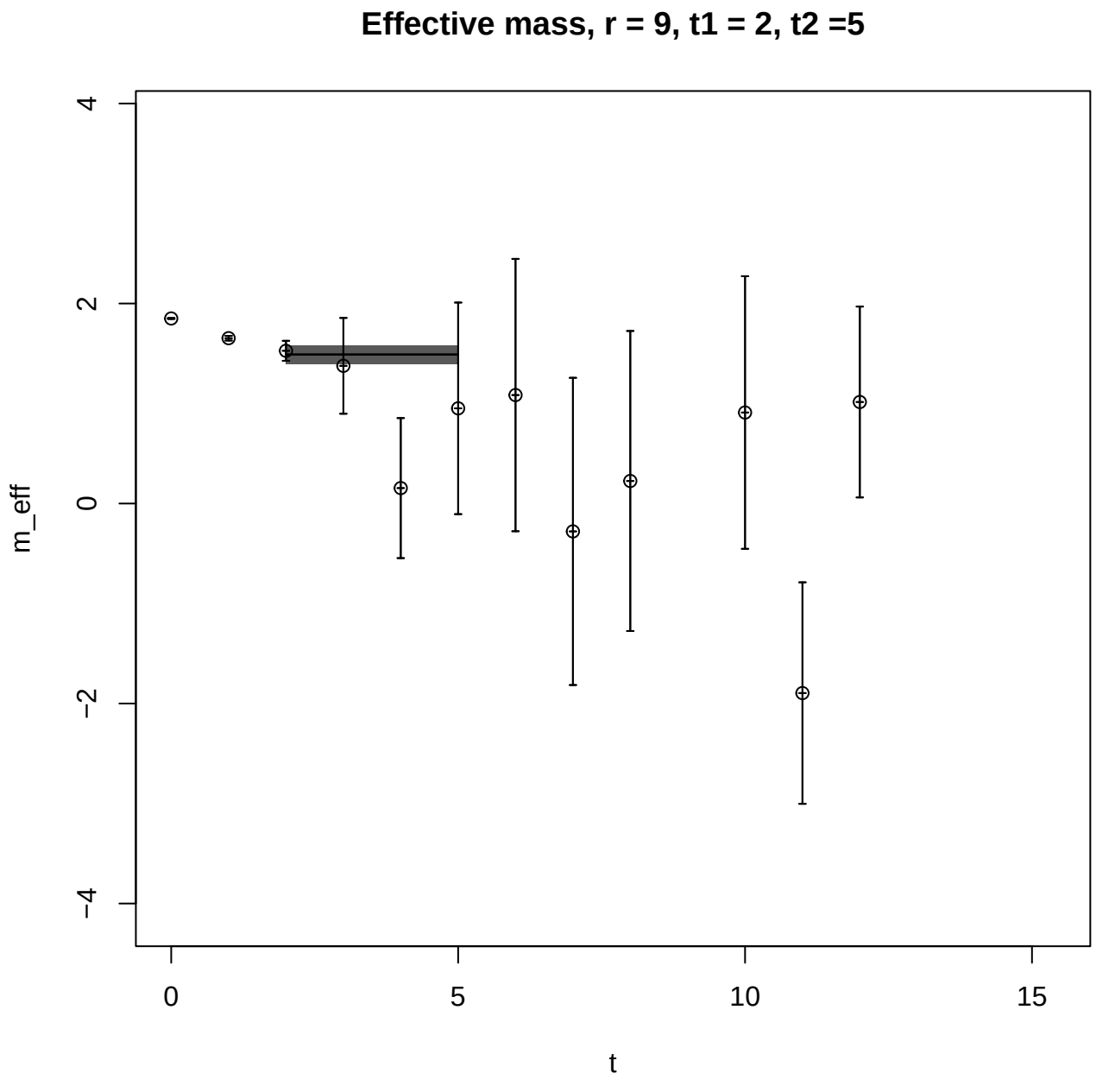

Effective mass, r = 9, t1 = 1, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

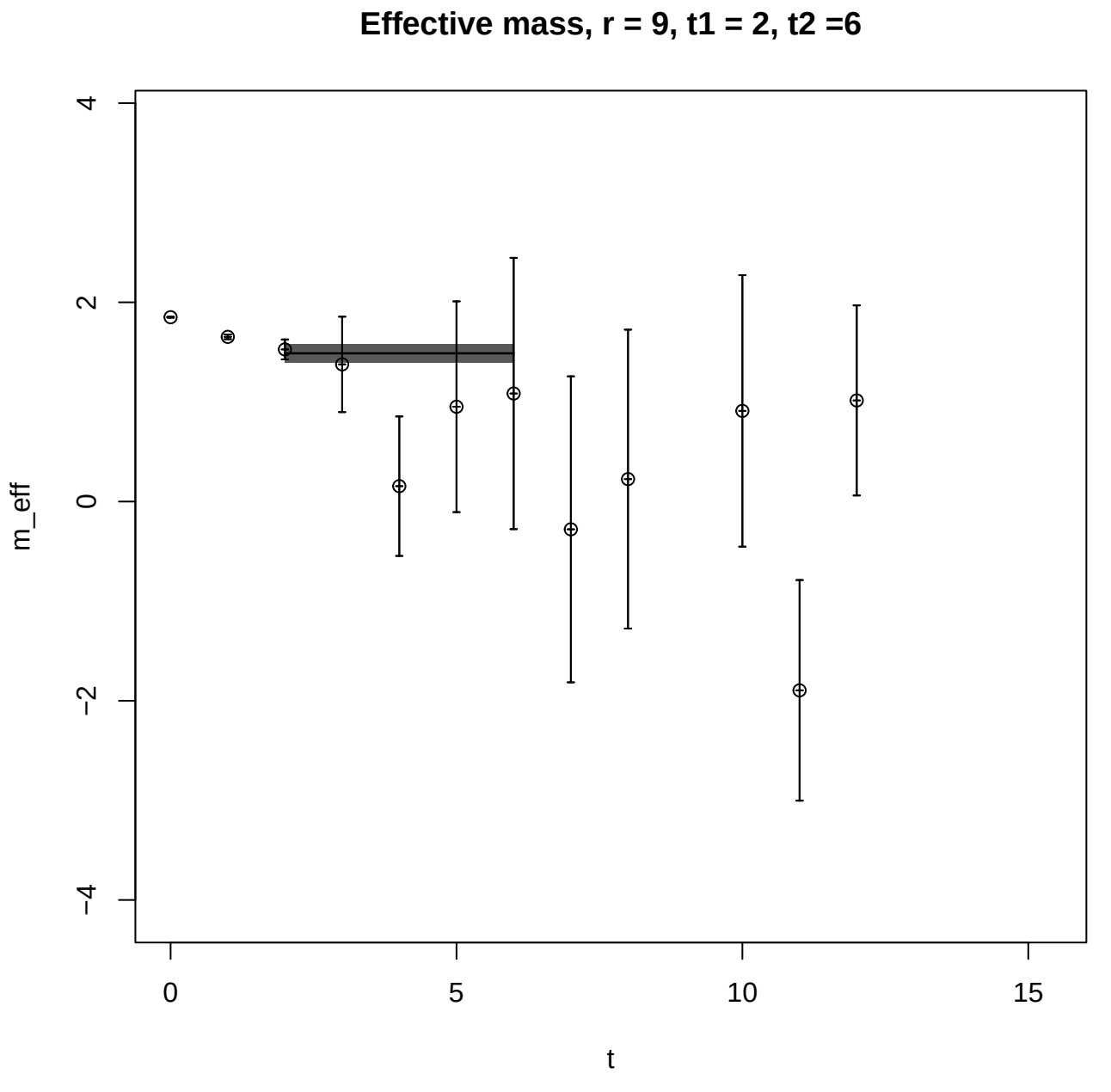

Effective mass, r = 9, t1 = 1, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

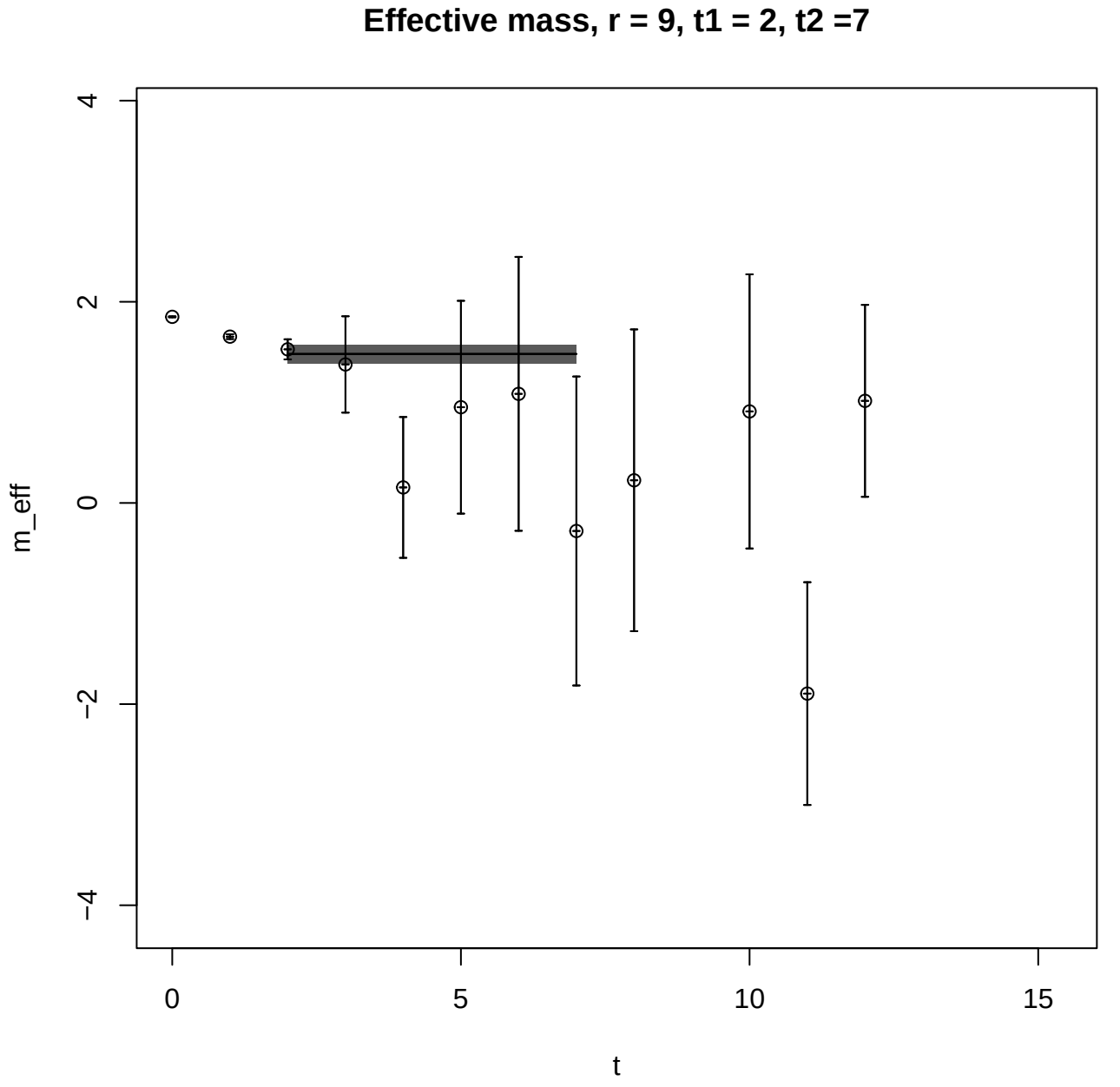

Effective mass, r = 9, t1 = 1, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

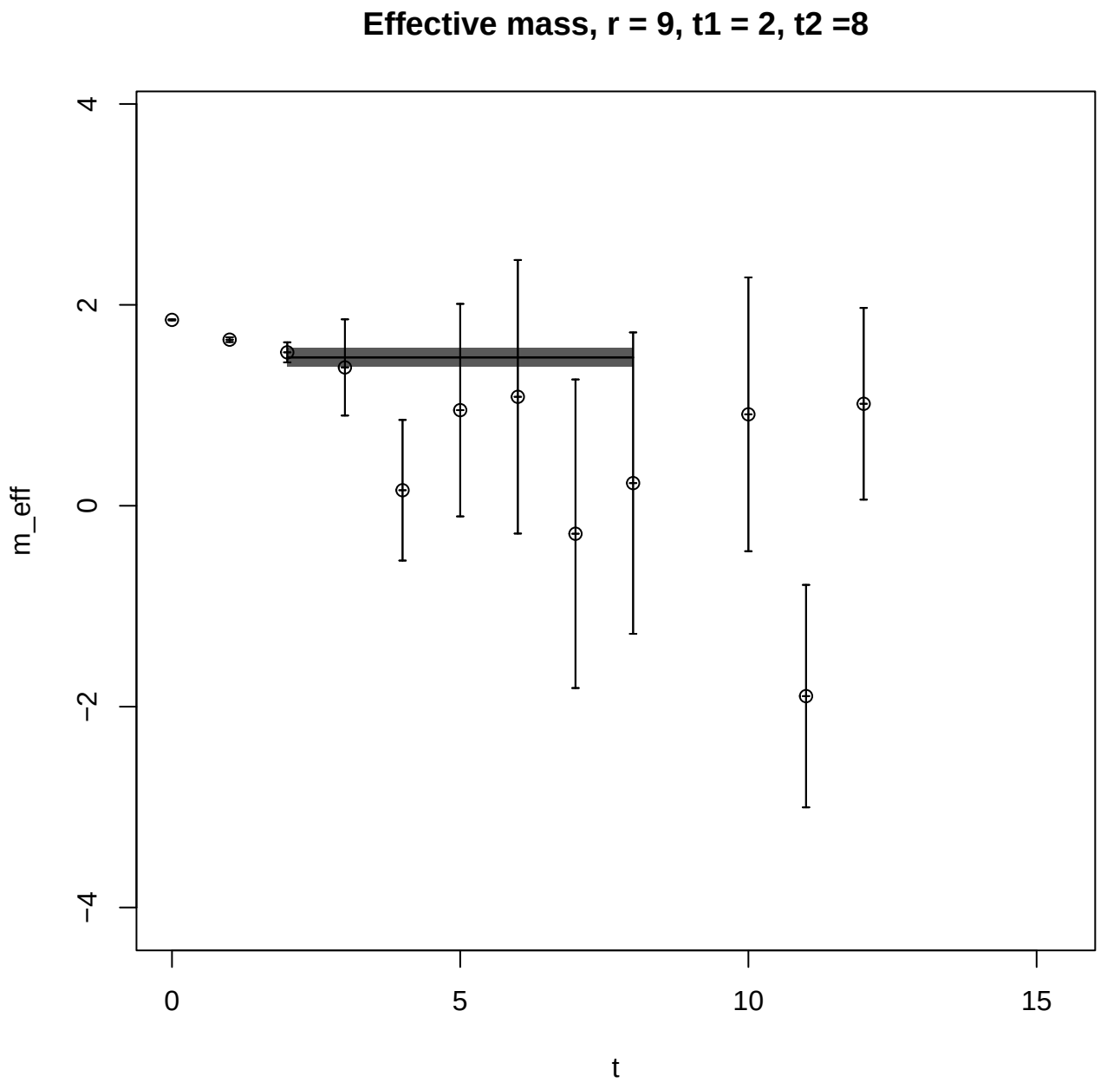

Effective mass, r = 9, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

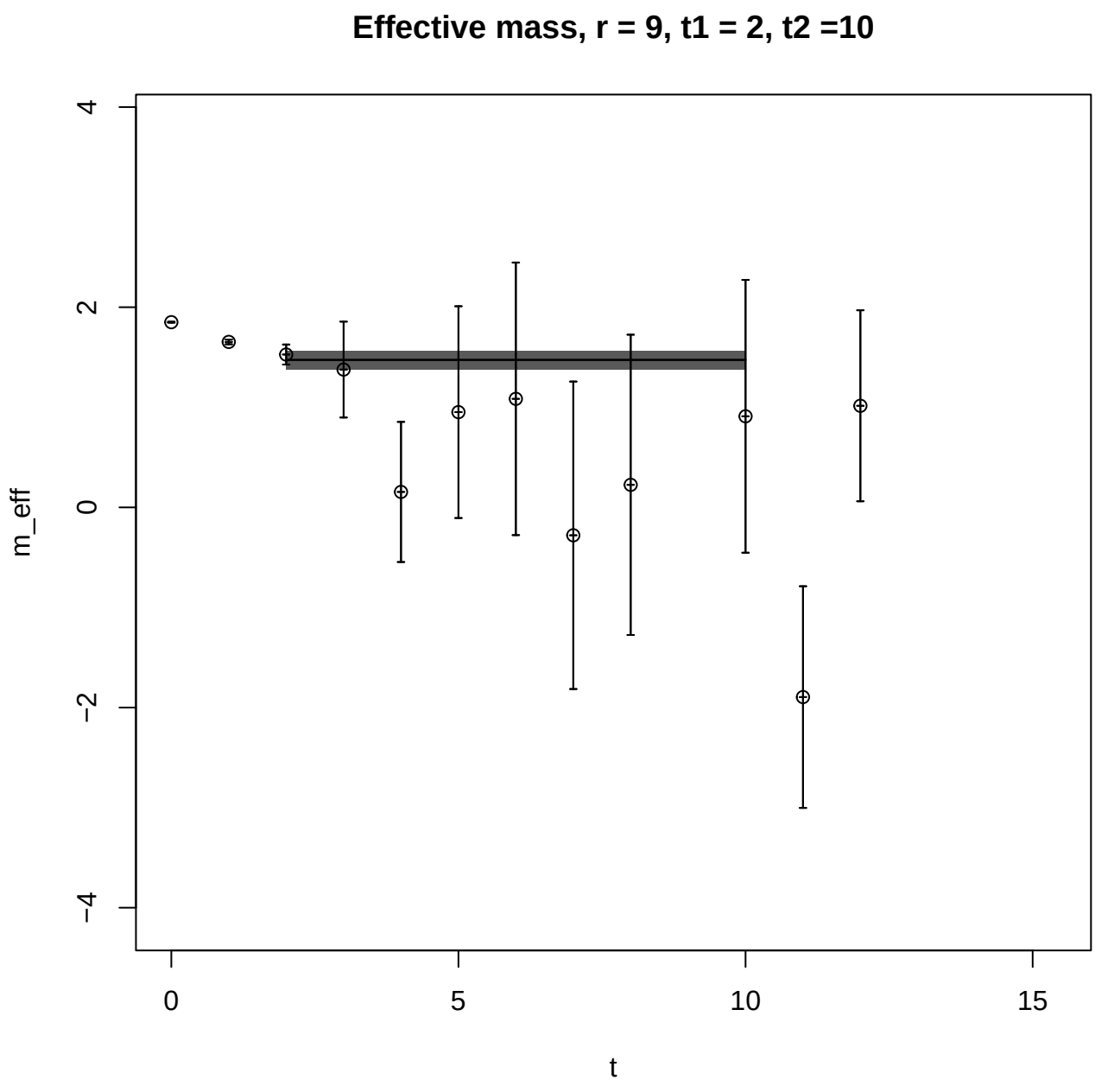

Effective mass, r = 9, t1 = 2, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

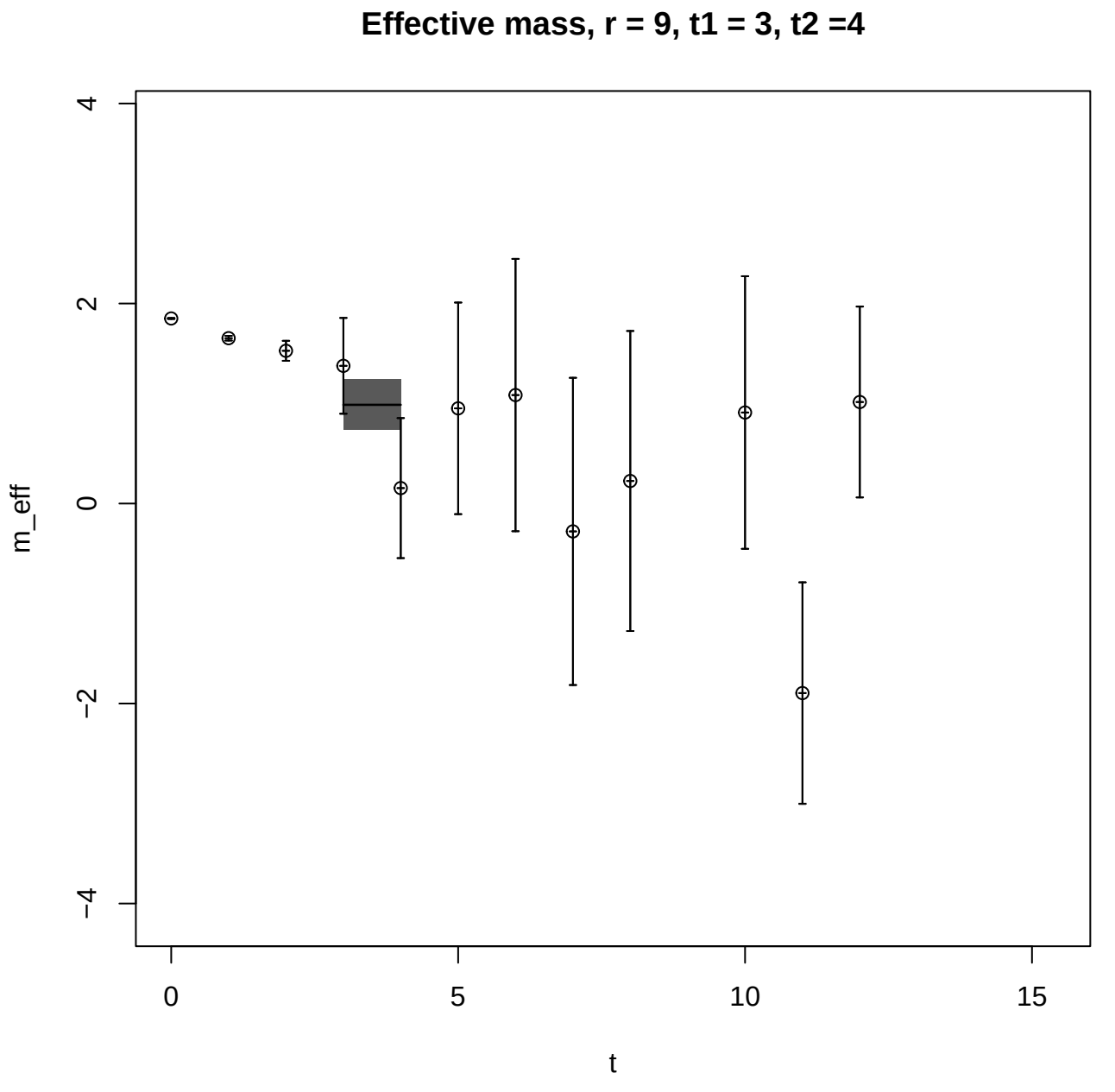

Effective mass, r = 9, t1 = 2, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

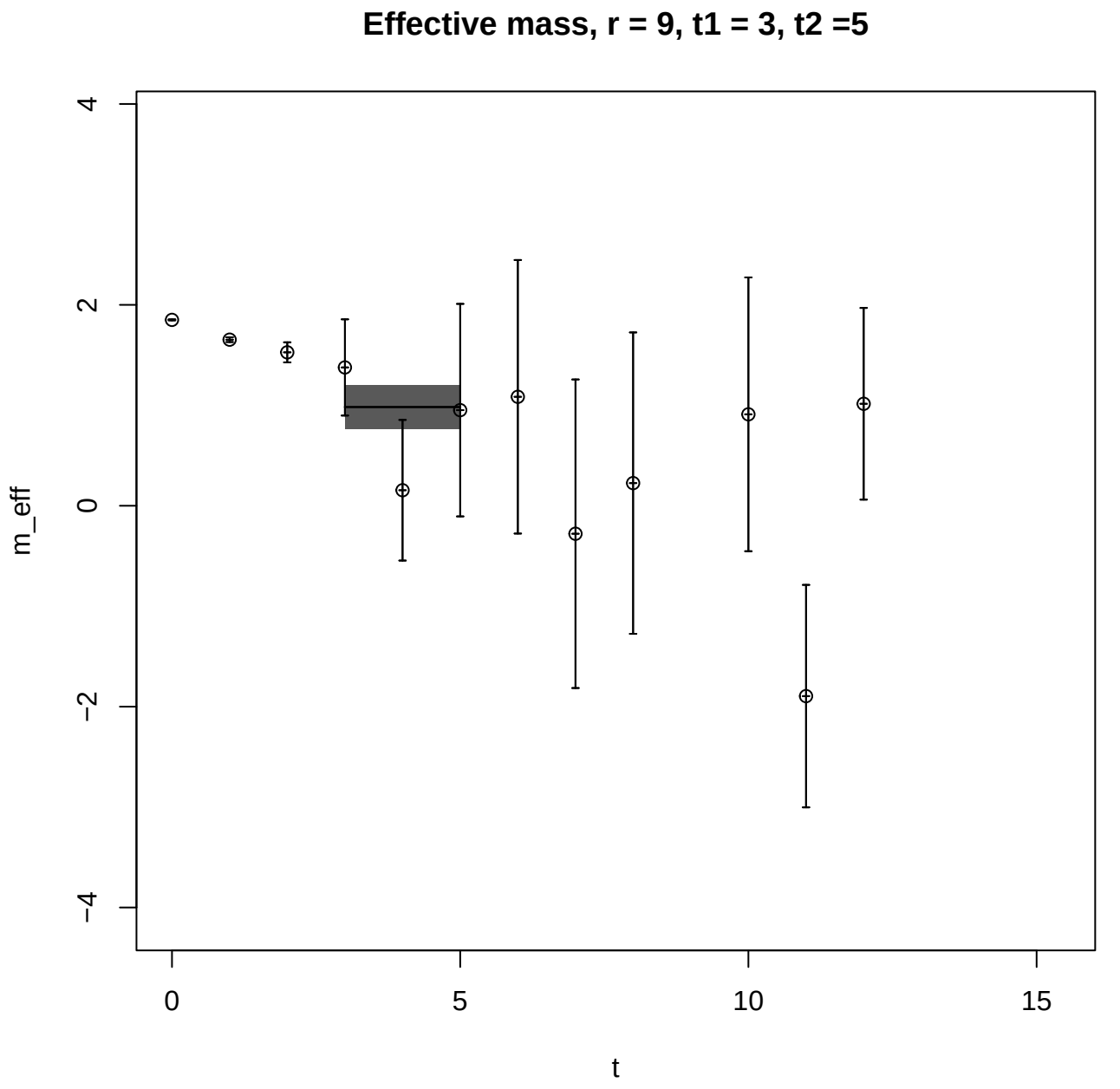

Effective mass, r = 9, t1 = 2, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15

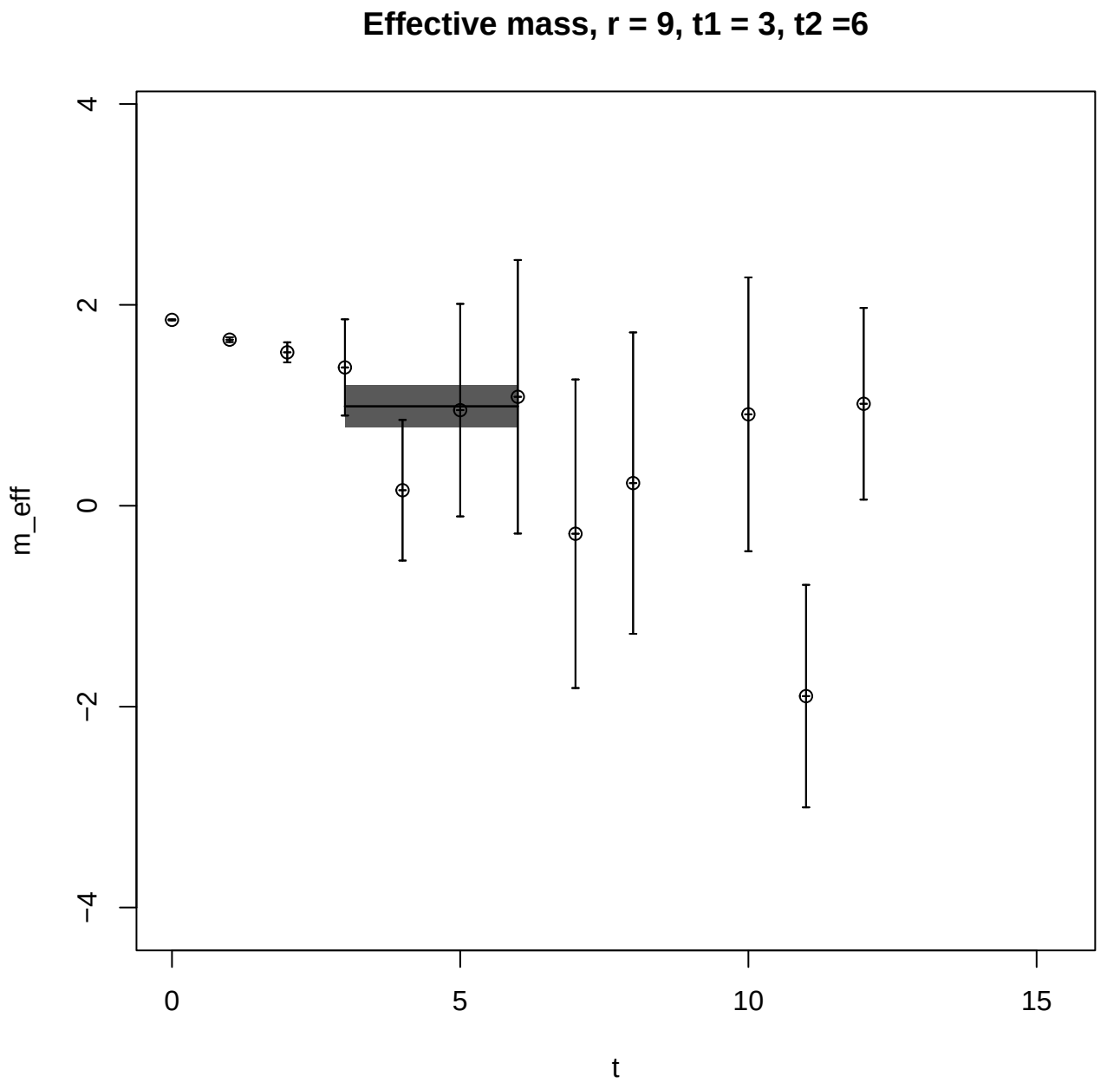

Effective mass, r = 9, t1 = 2, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

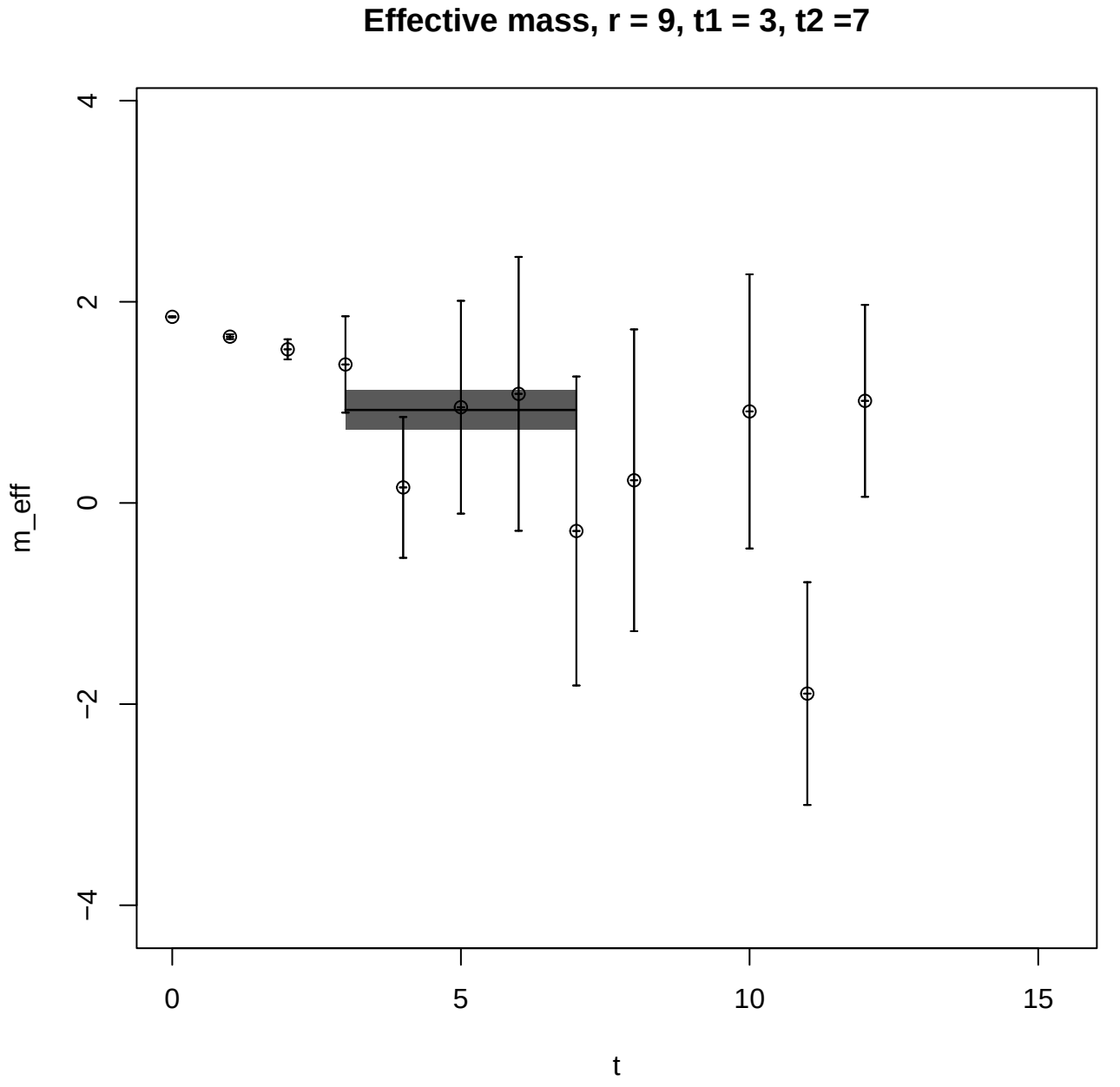

Effective mass, r = 9, t1 = 2, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

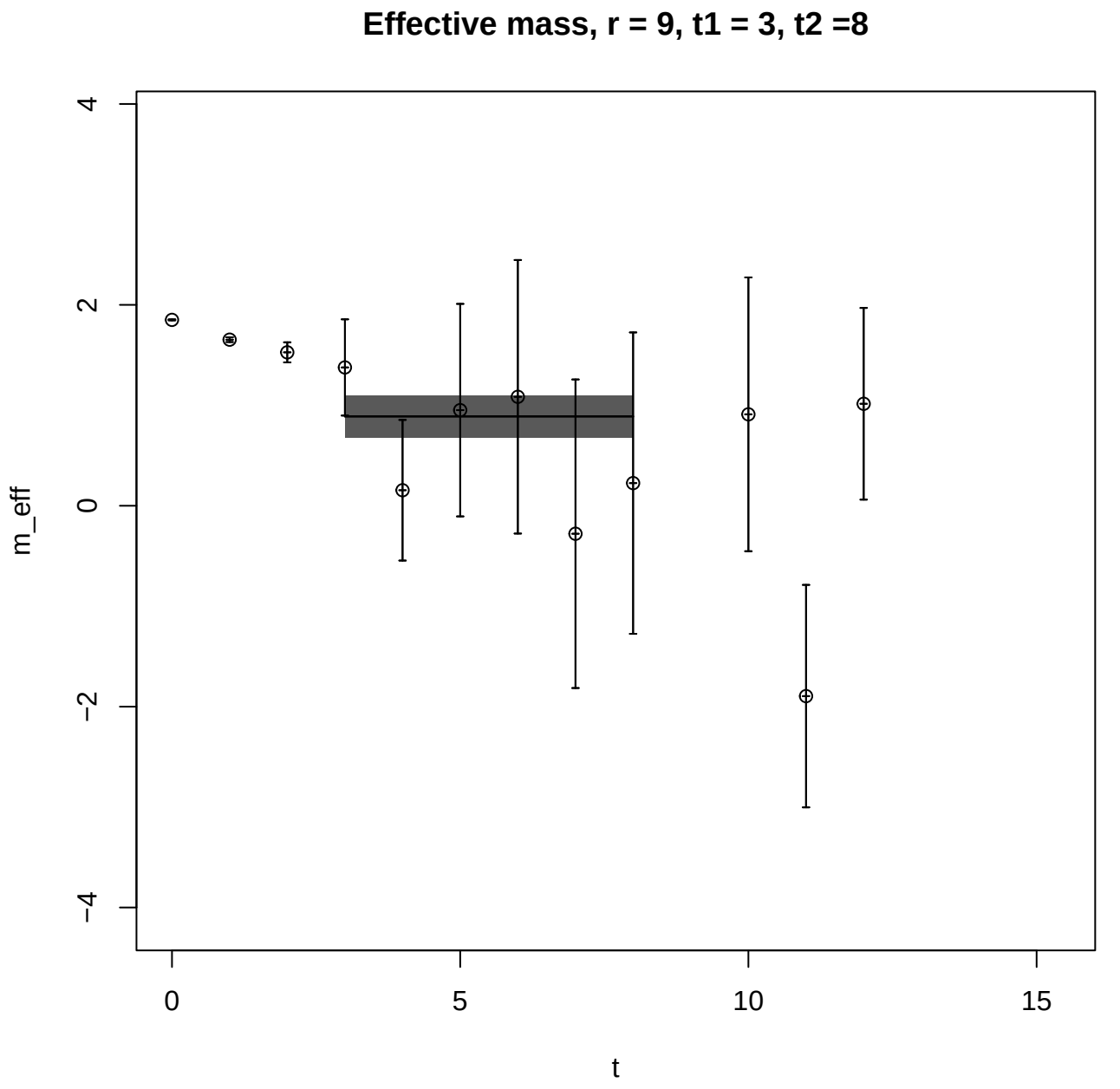

Effective mass, r = 9, t1 = 2, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

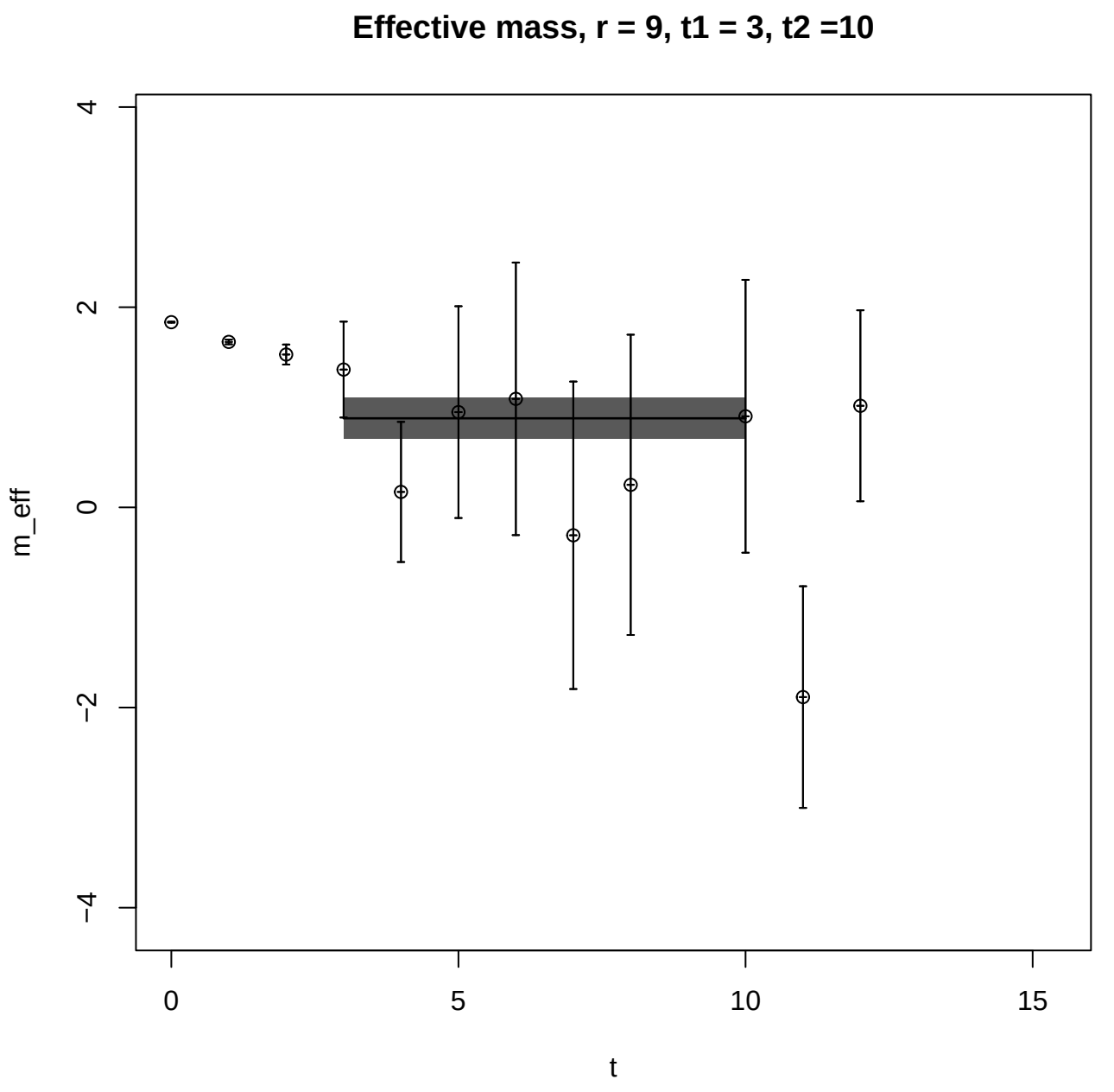

Effective mass, r = 9, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

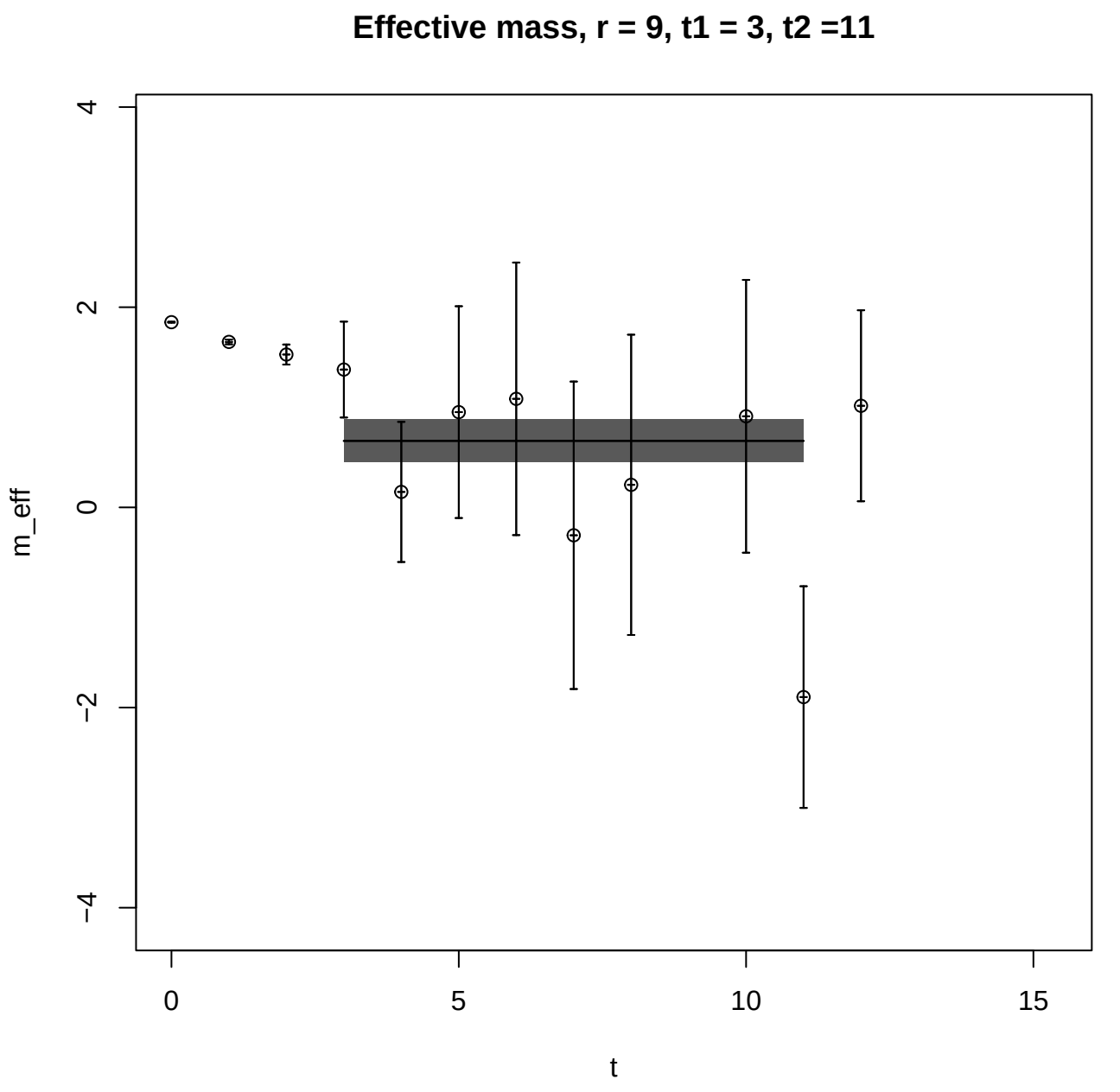

Effective mass, r = 9, t1 = 3, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

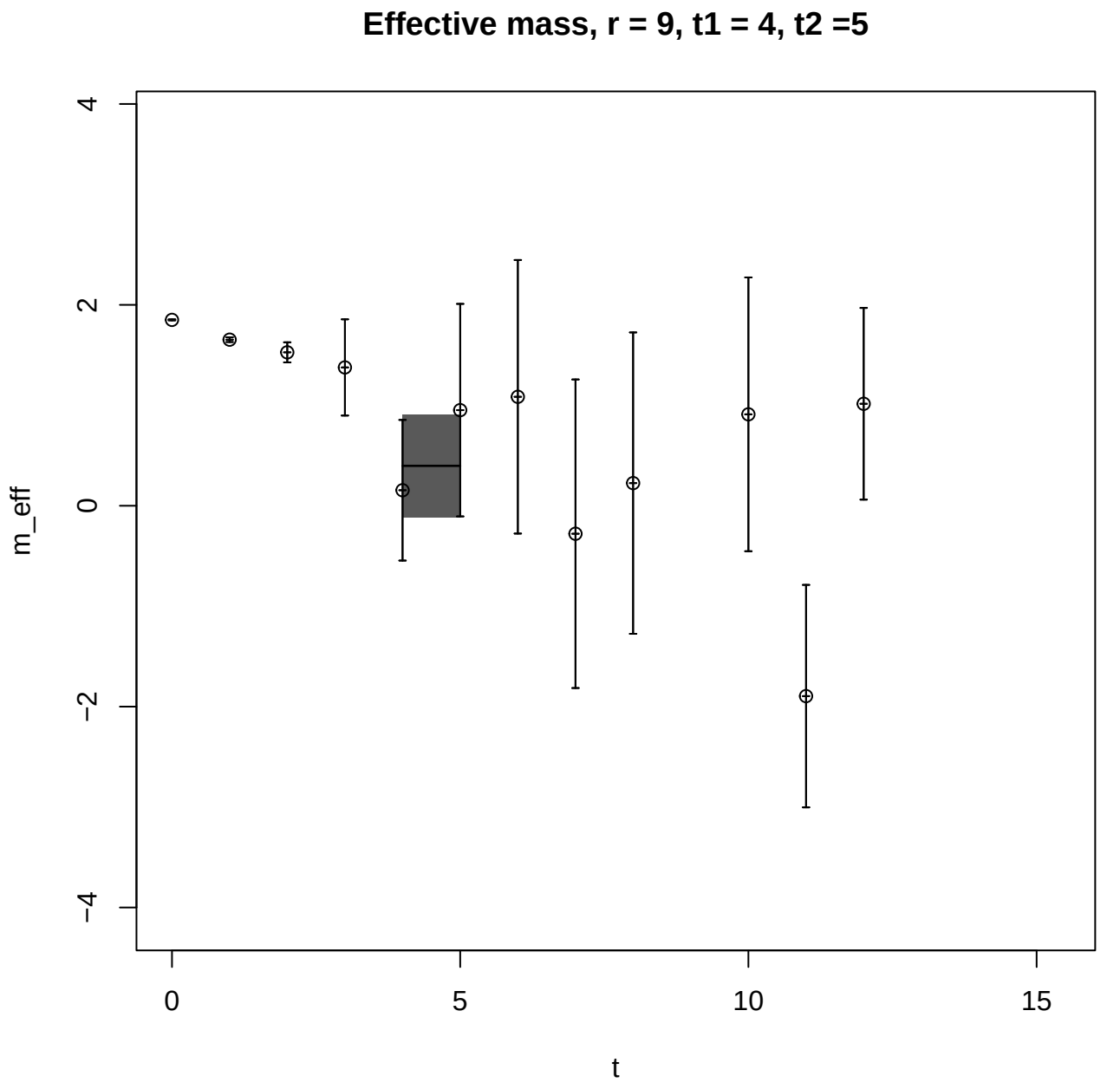

Effective mass, r = 9, t1 = 3, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 3, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 3, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 3, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 3, t2 =11
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 4, t2 =5
m_eff
t
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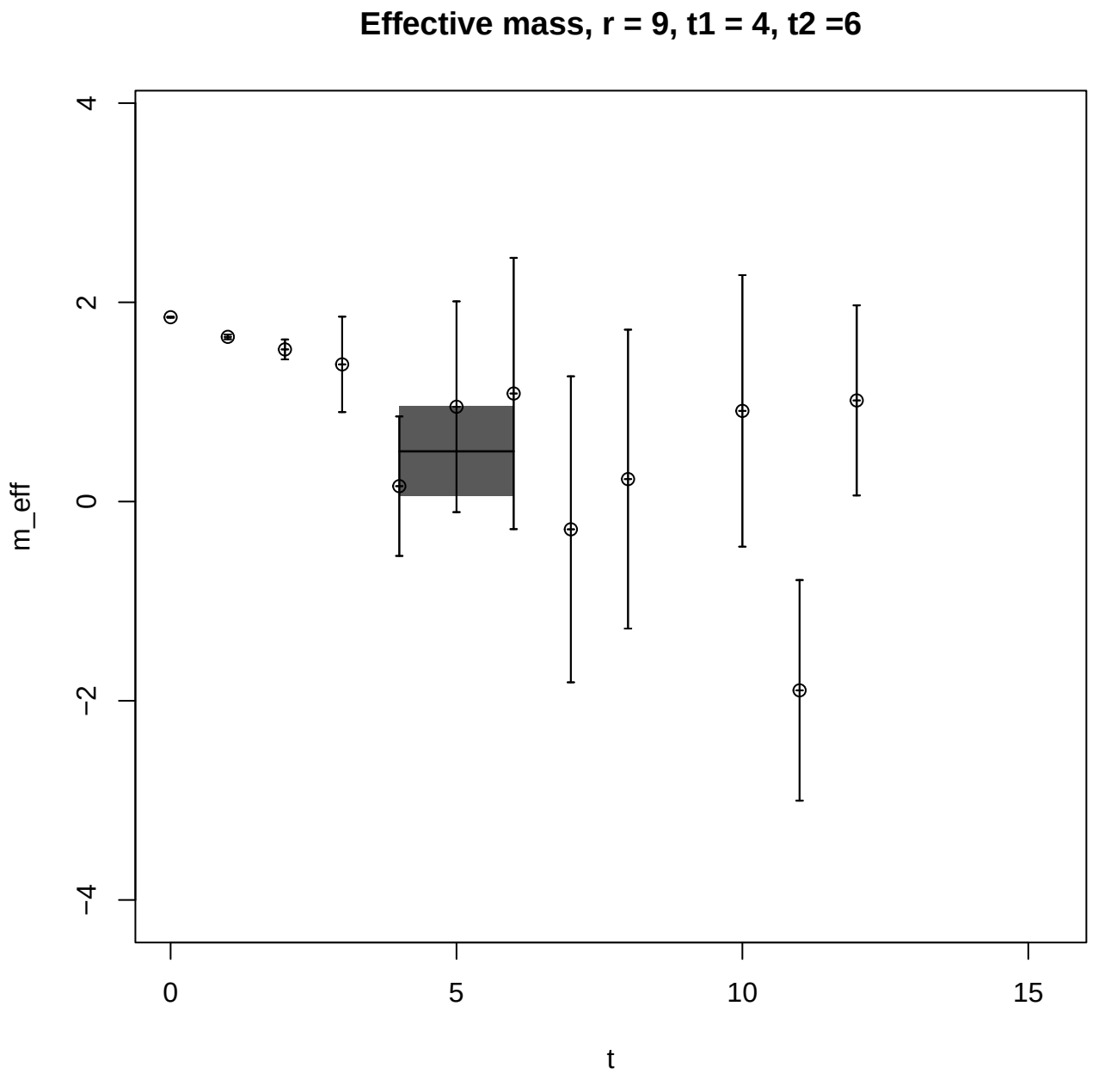
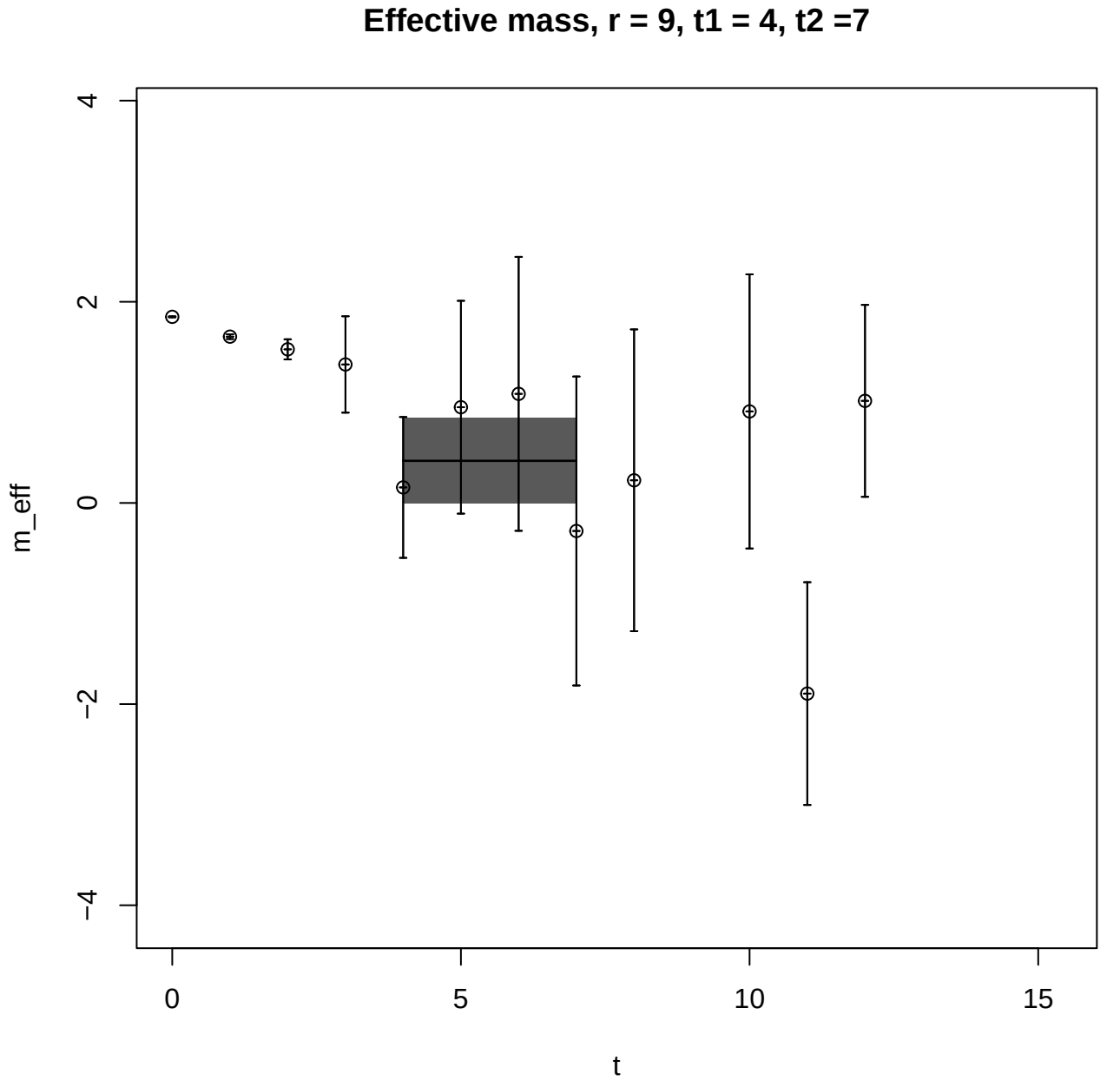
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 4, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 4, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
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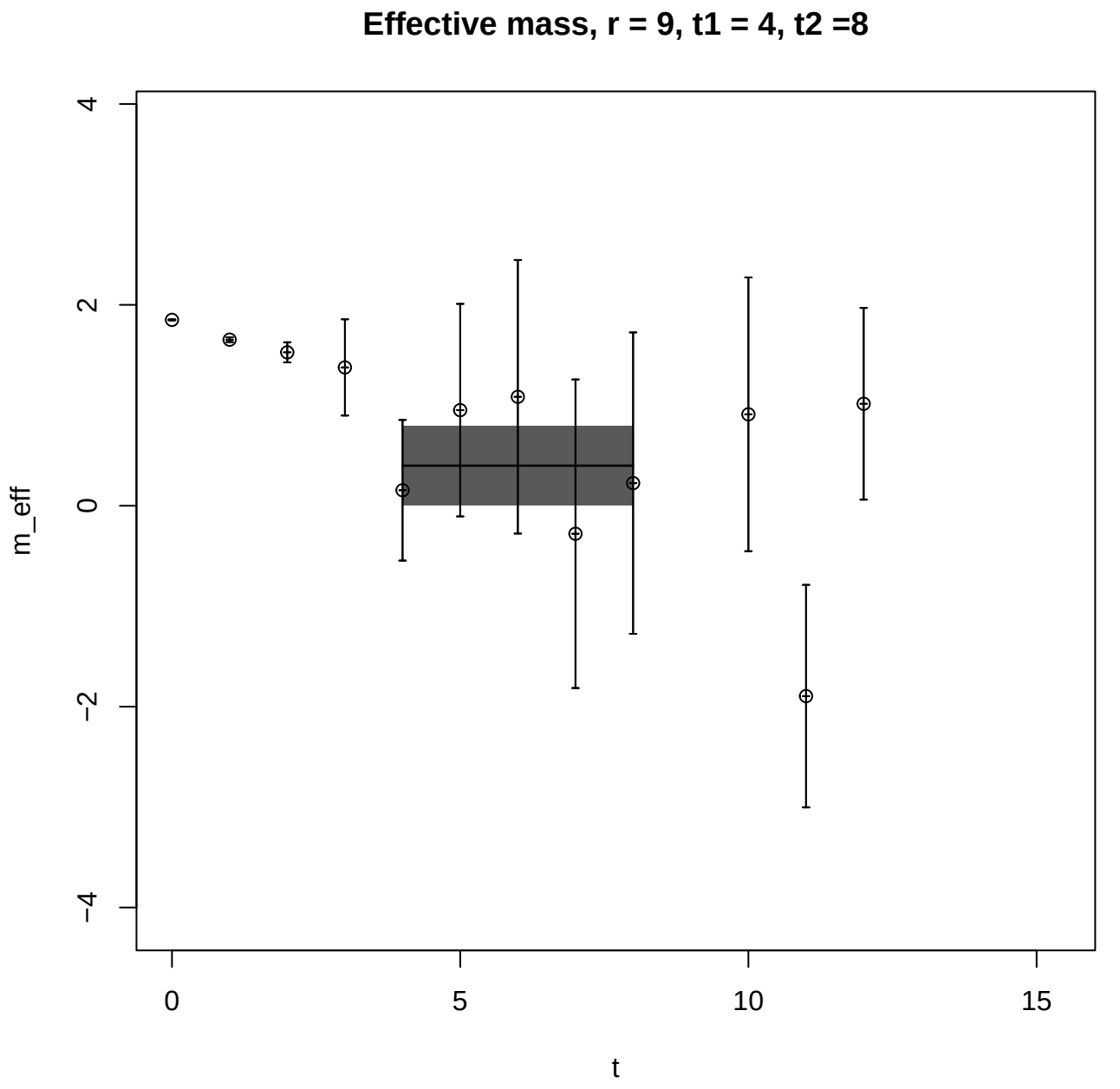
Effective mass, r = 9, t1 = 4, t2 =8
m_eff
t
4
2
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−2
−4
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10
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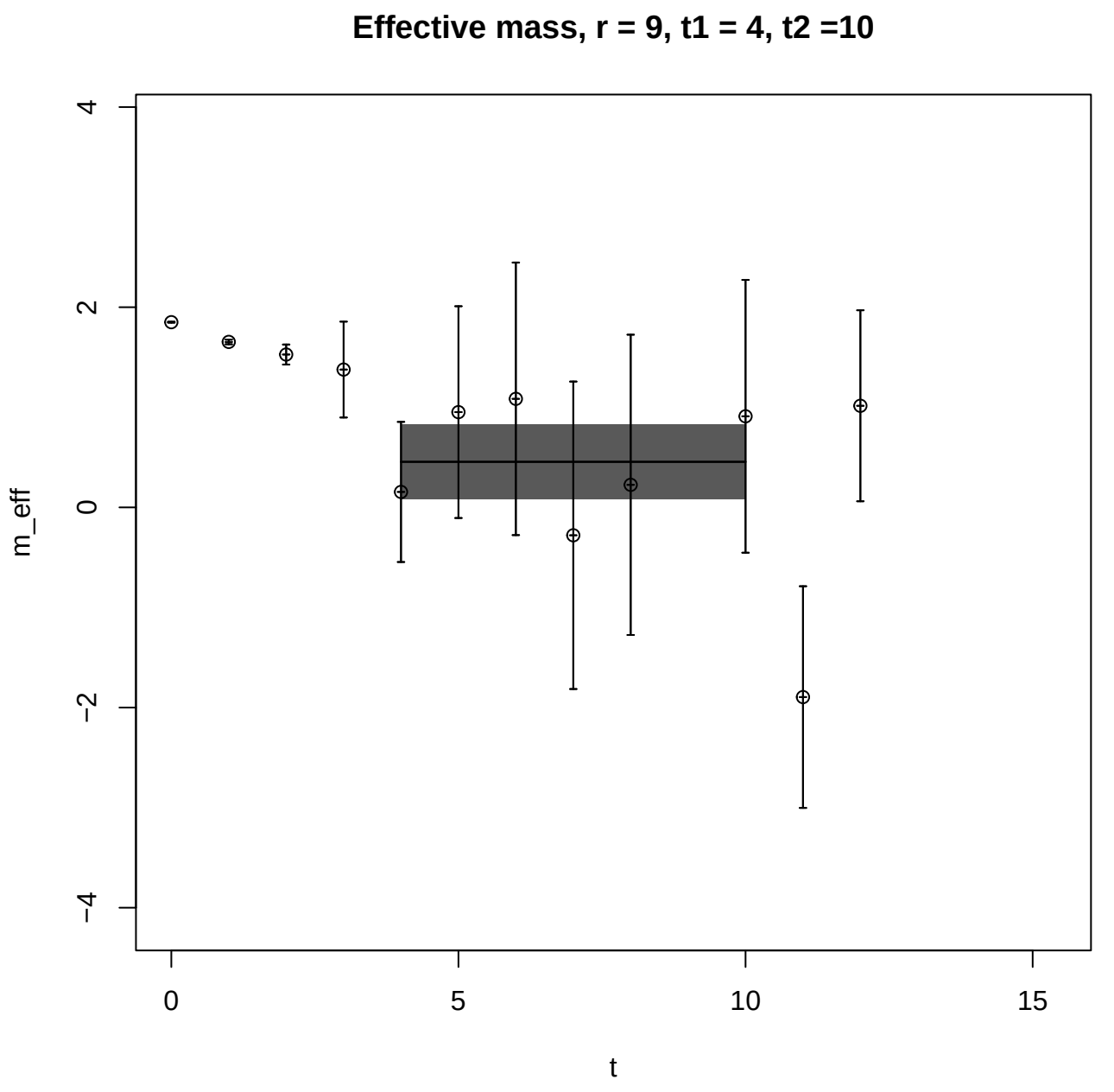
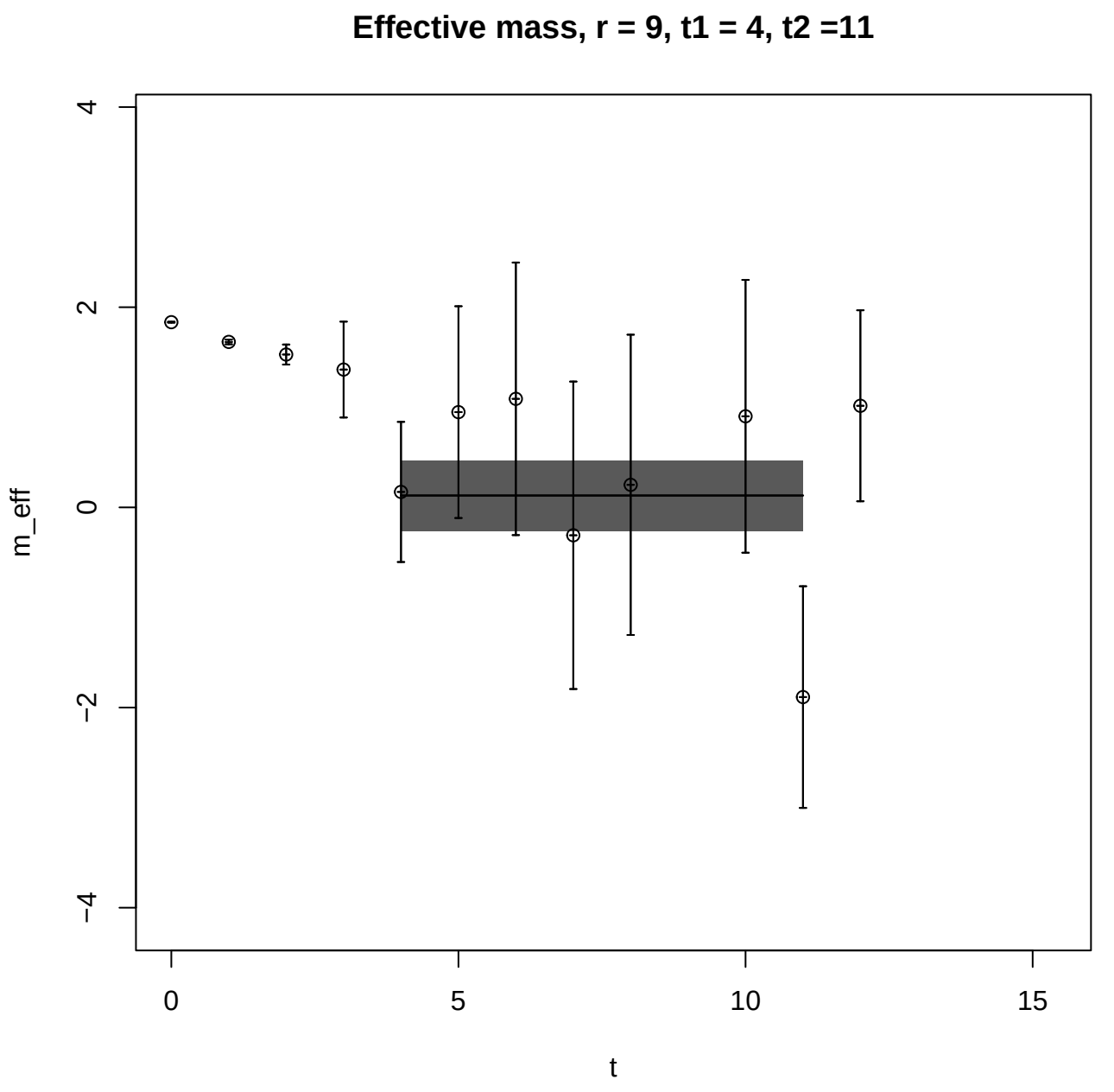

Effective mass, r = 9, t1 = 4, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 4, t2 =11
m_eff
t
4
2
0
−2
−4
0
5
10
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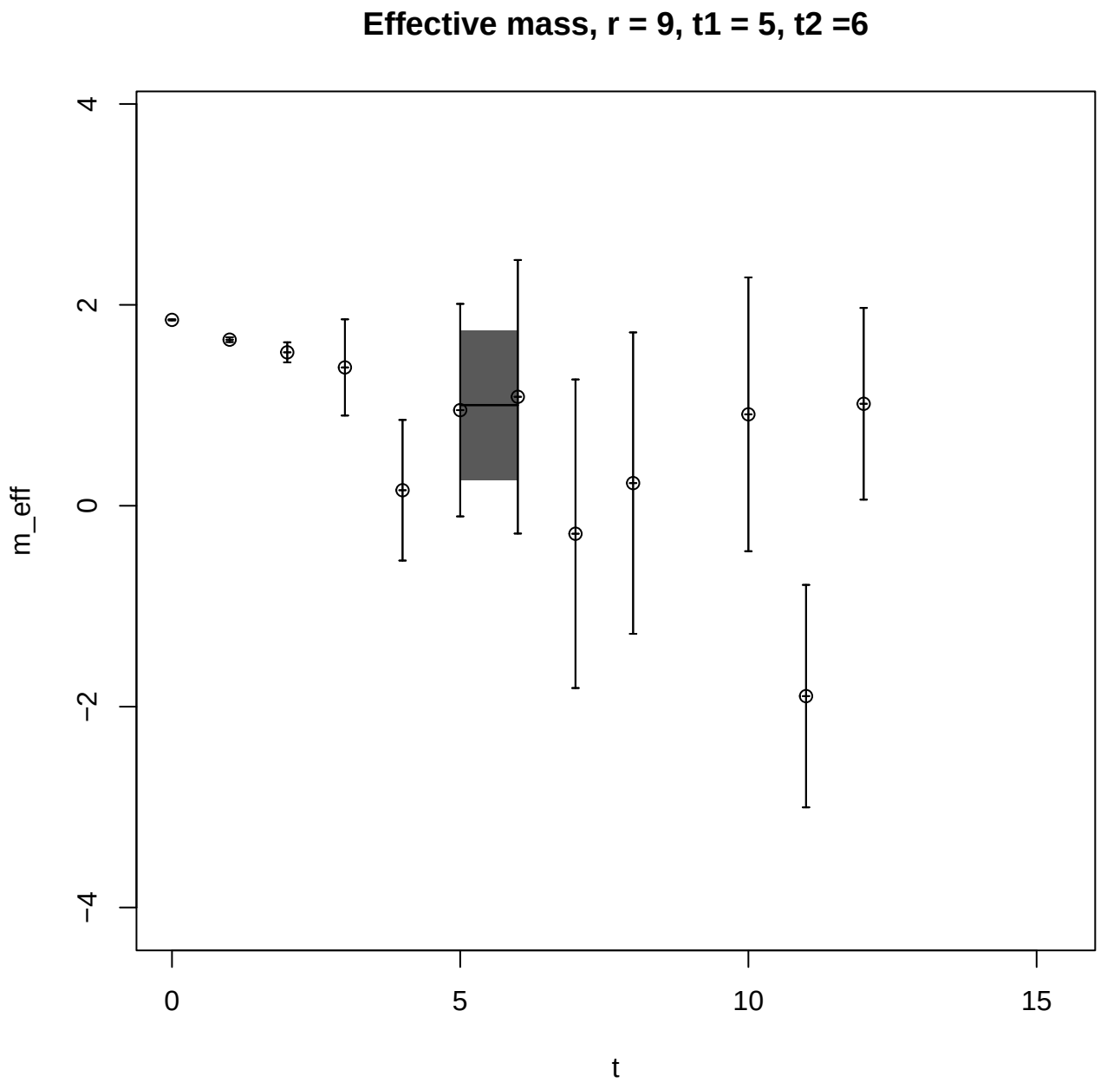
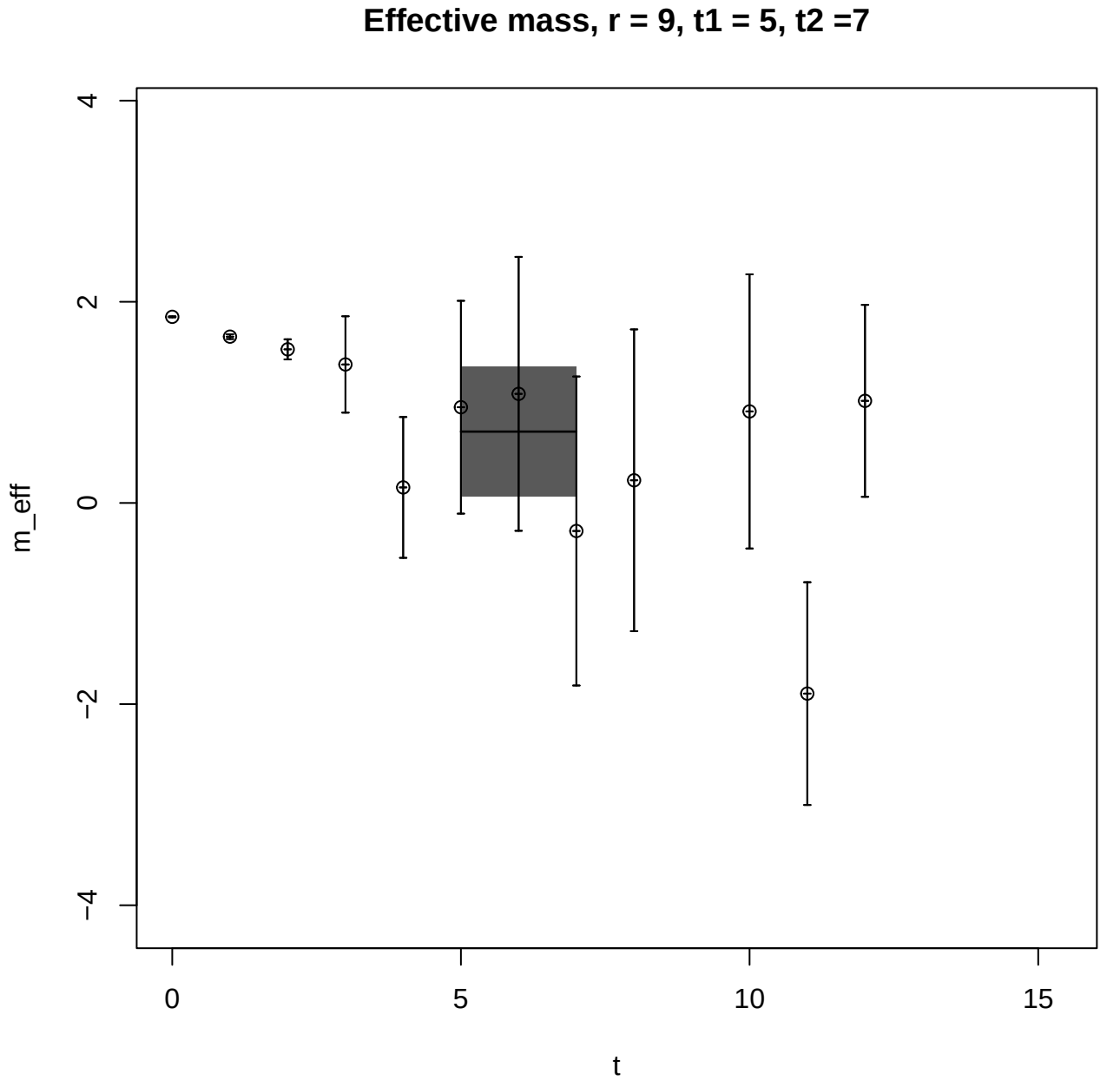
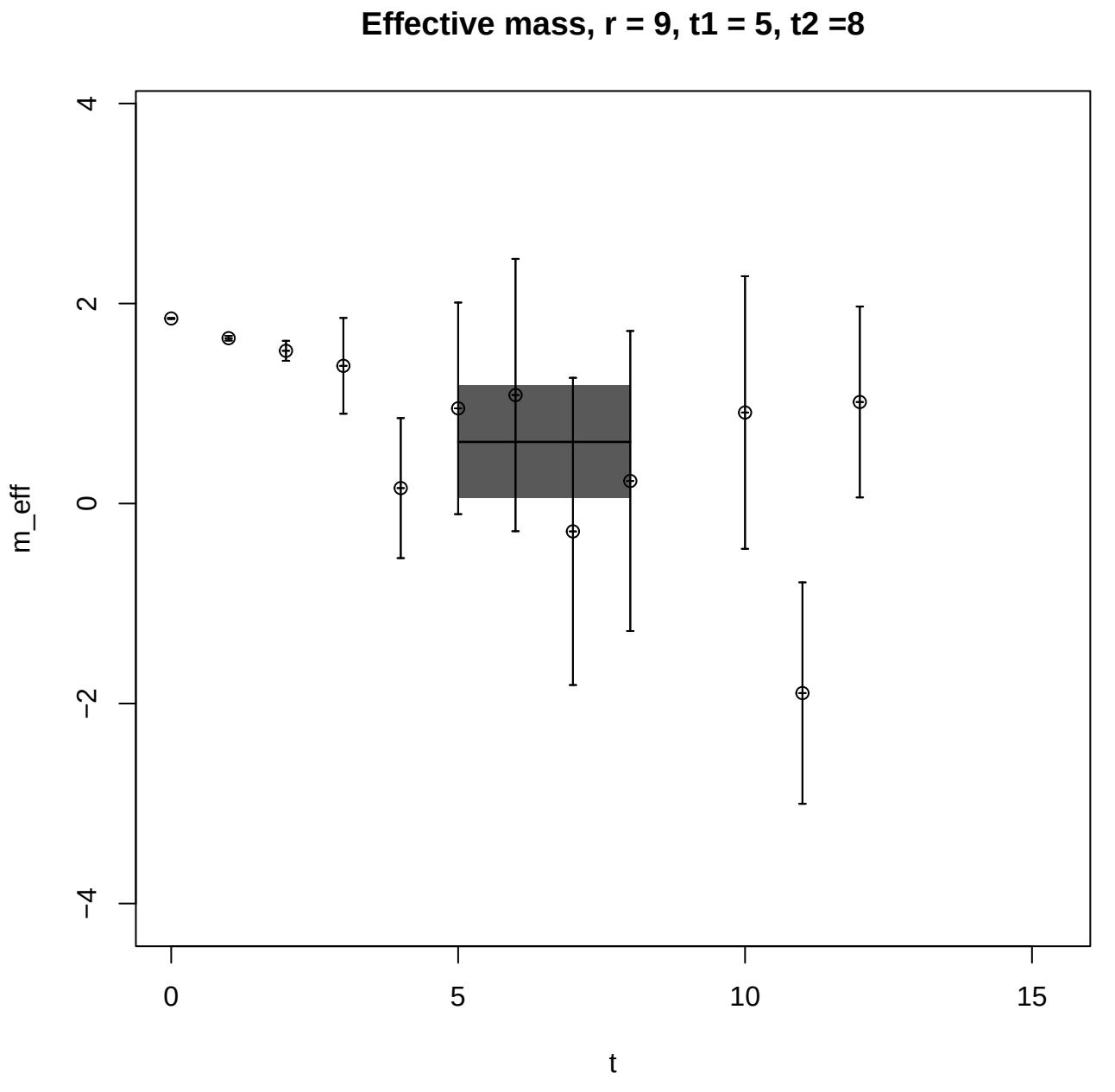

Effective mass, r = 9, t1 = 5, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 5, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 5, t2 =8
m_eff
t
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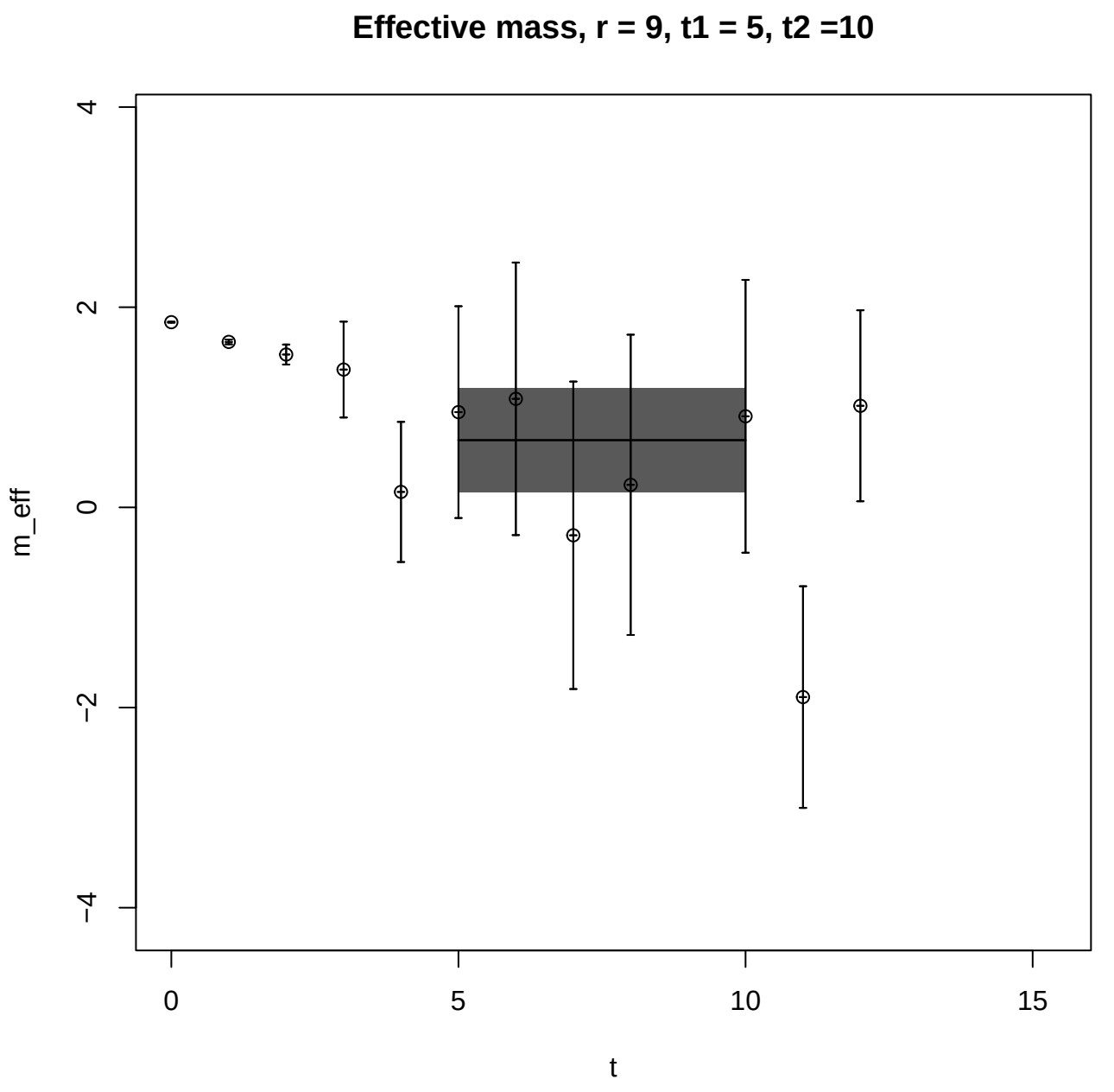
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 5, t2 =10
m_eff
t
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2
0
−2
−4
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10
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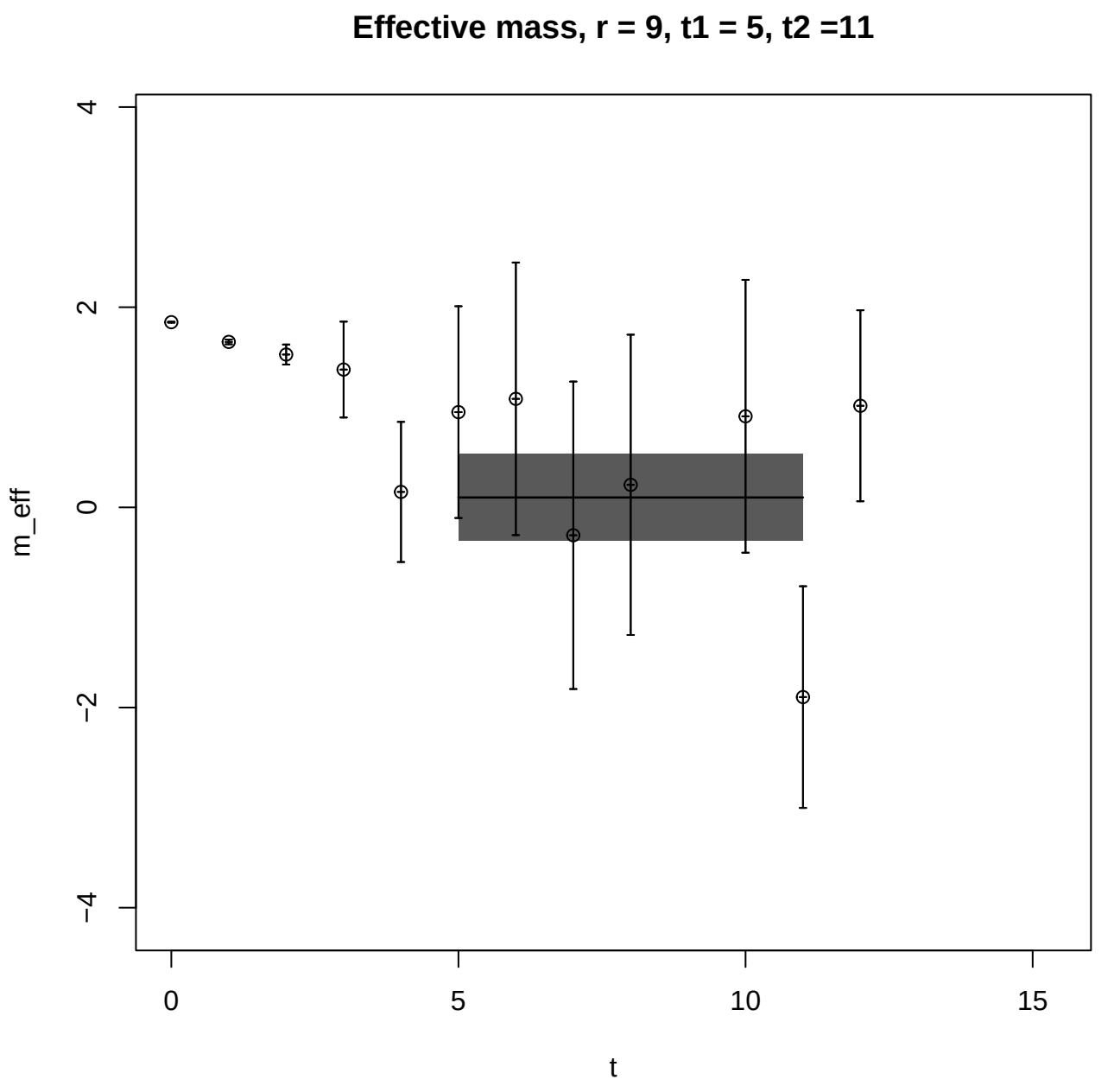
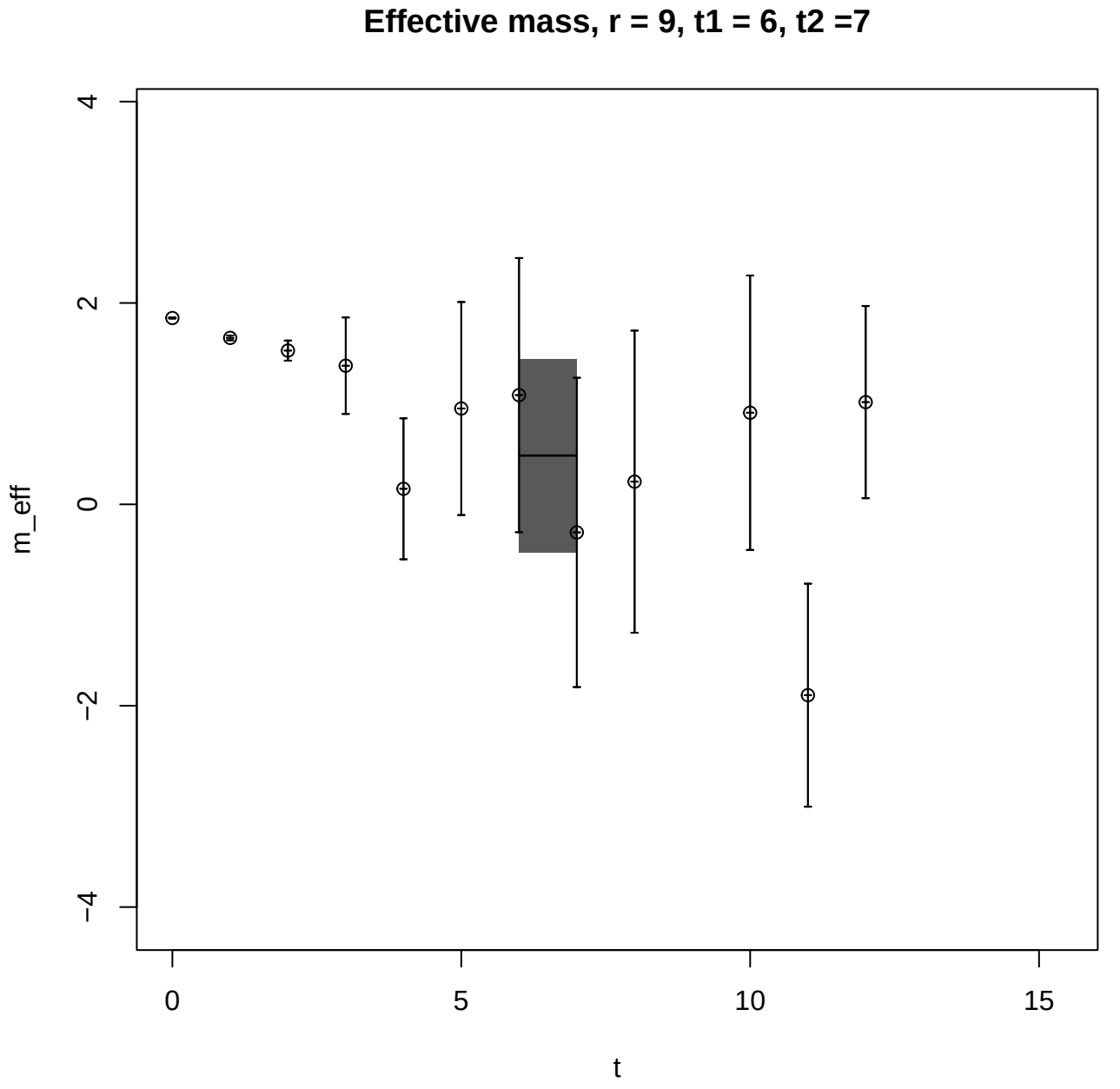

Effective mass, r = 9, t1 = 5, t2 =11
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 6, t2 =7
m_eff
t
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−4
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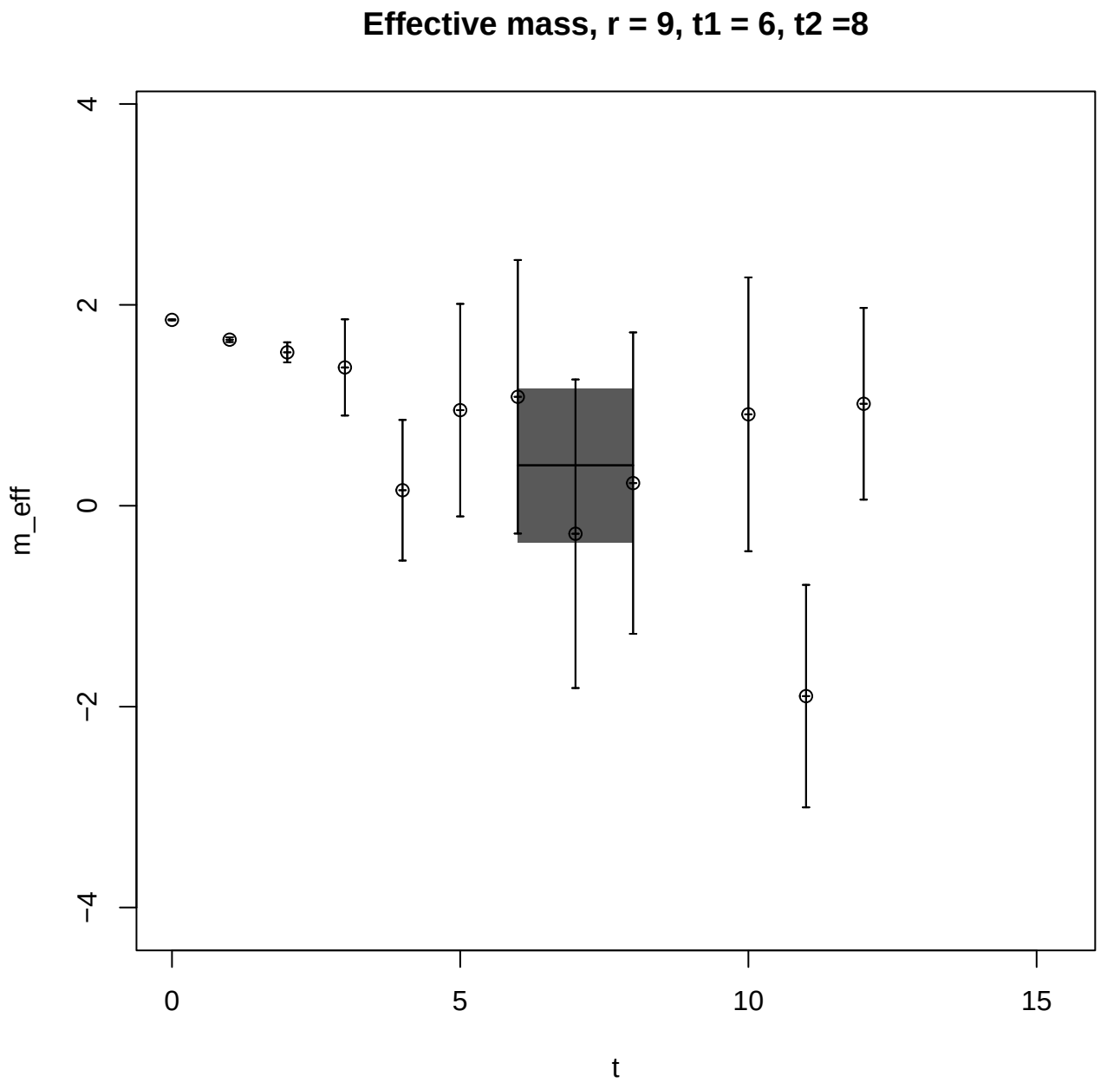
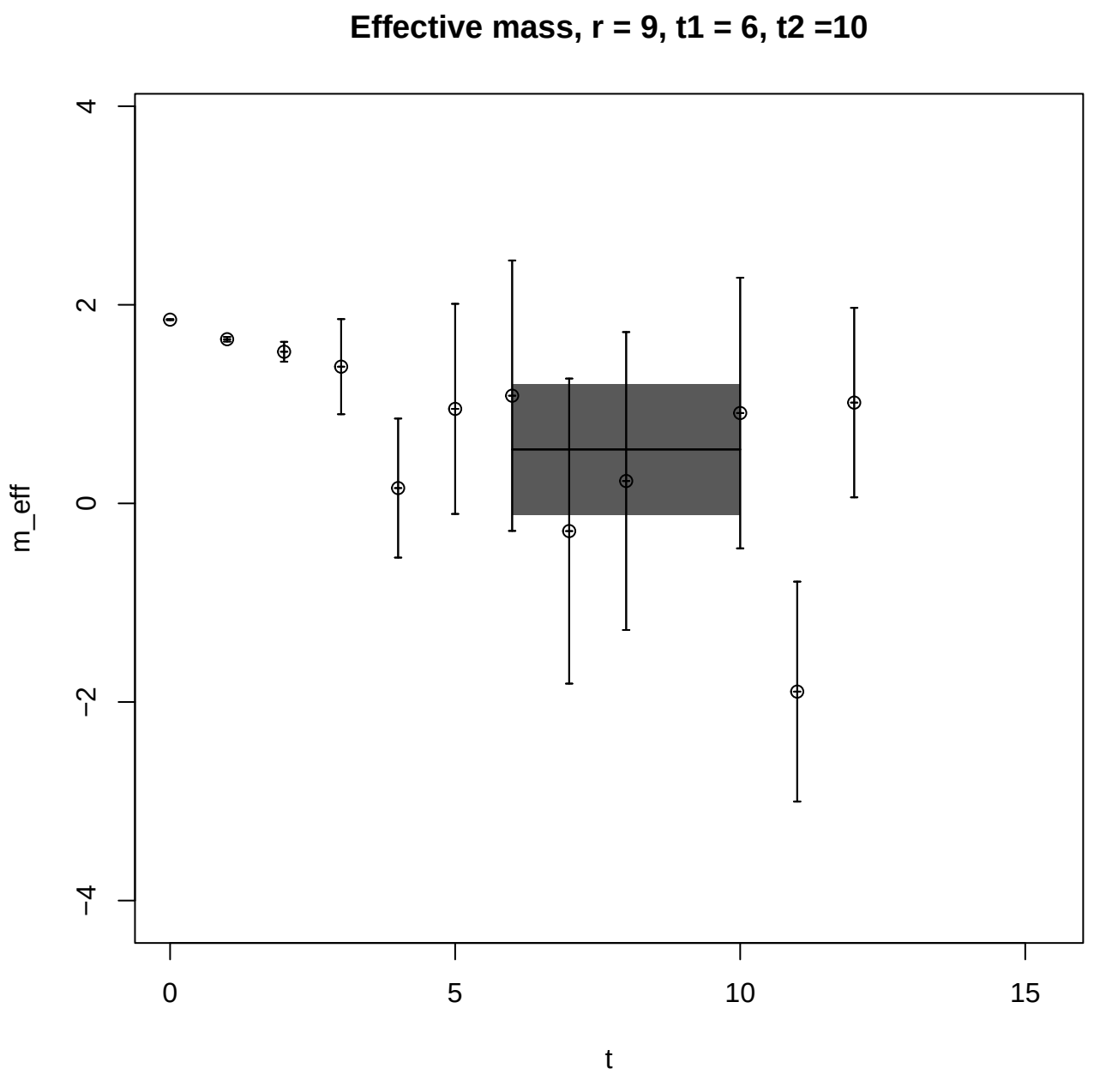

Effective mass, r = 9, t1 = 6, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15


Effective mass, r = 9, t1 = 6, t2 =10
m_eff
t
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−2
−4
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5
10
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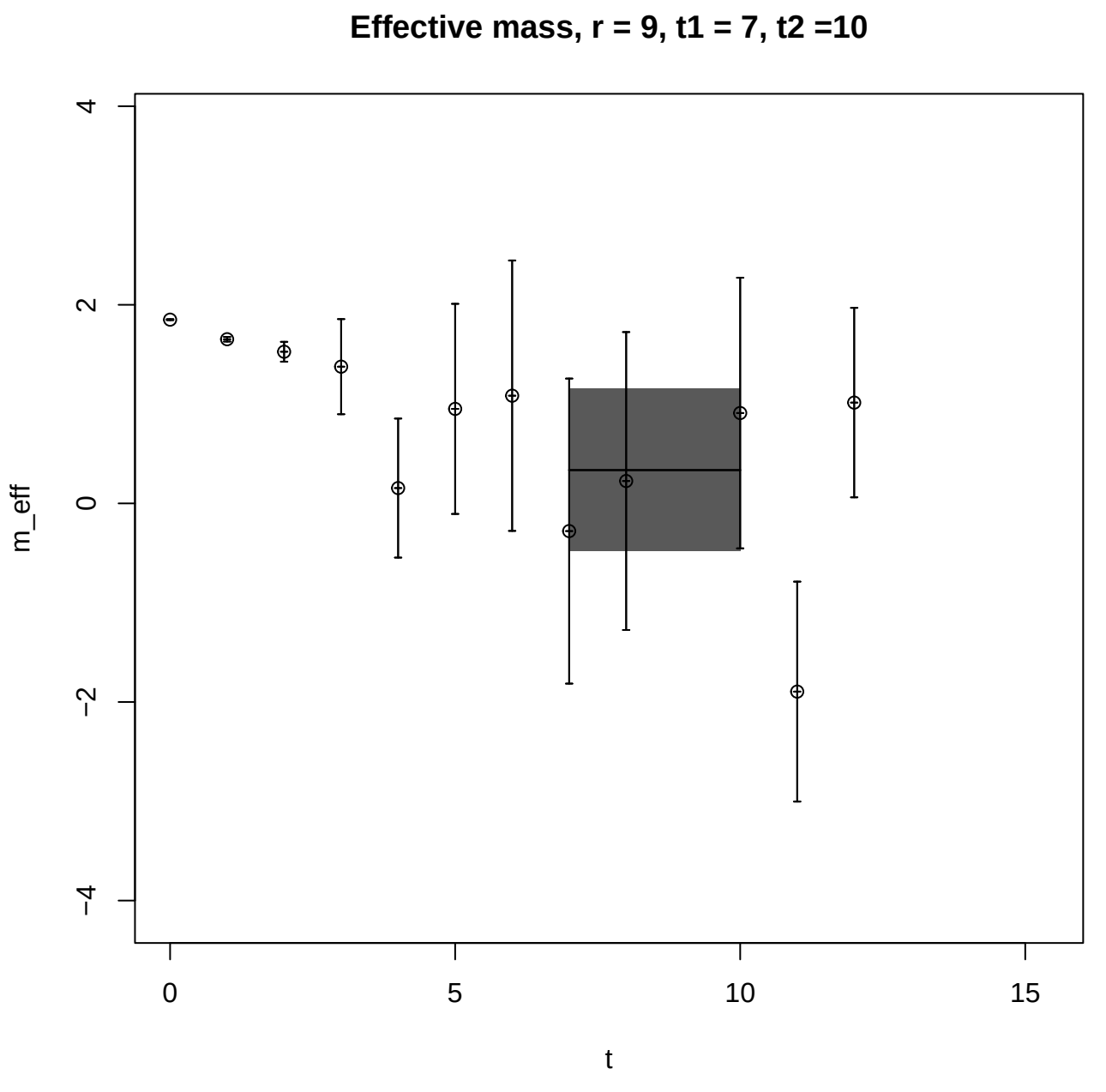
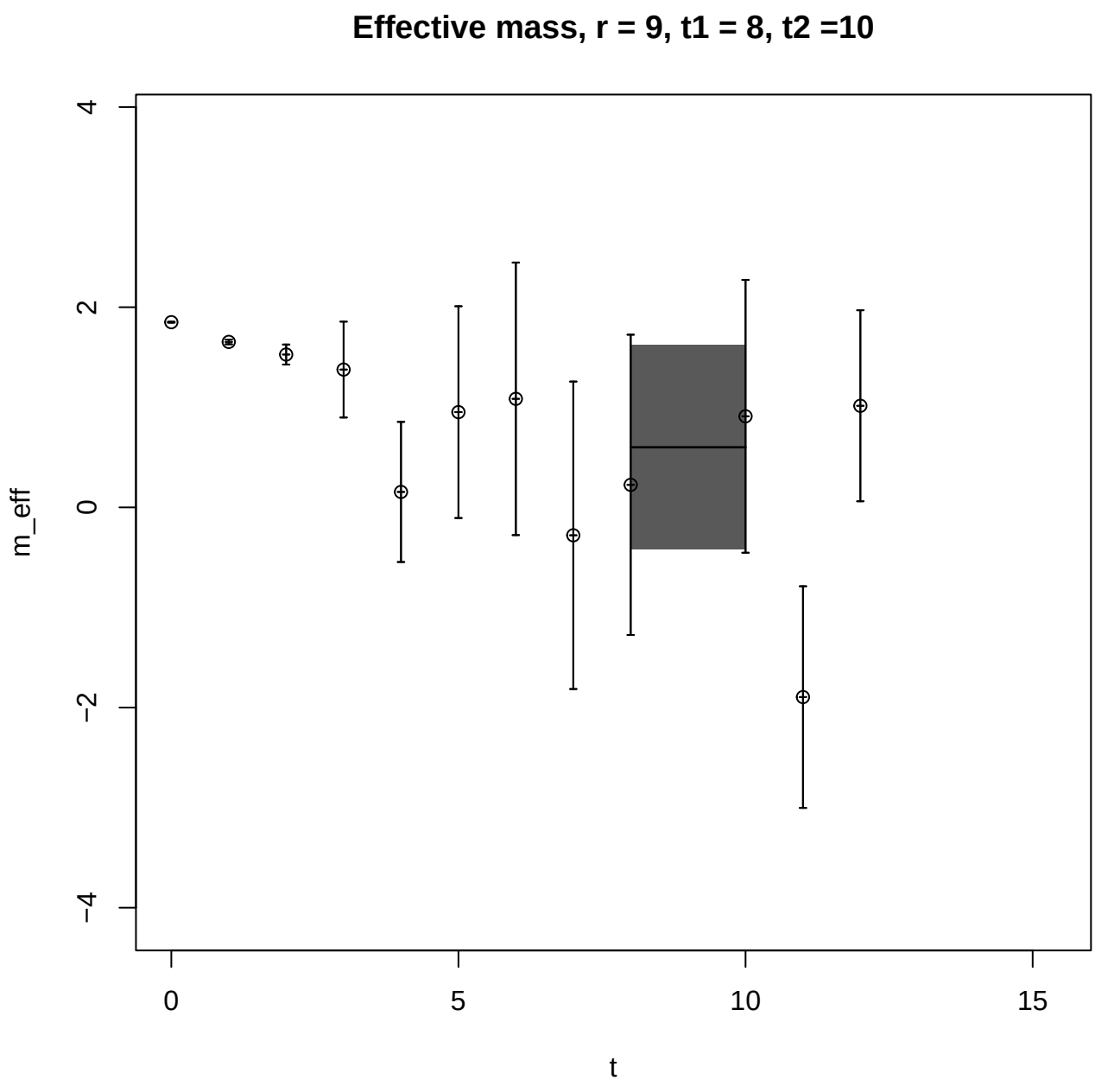
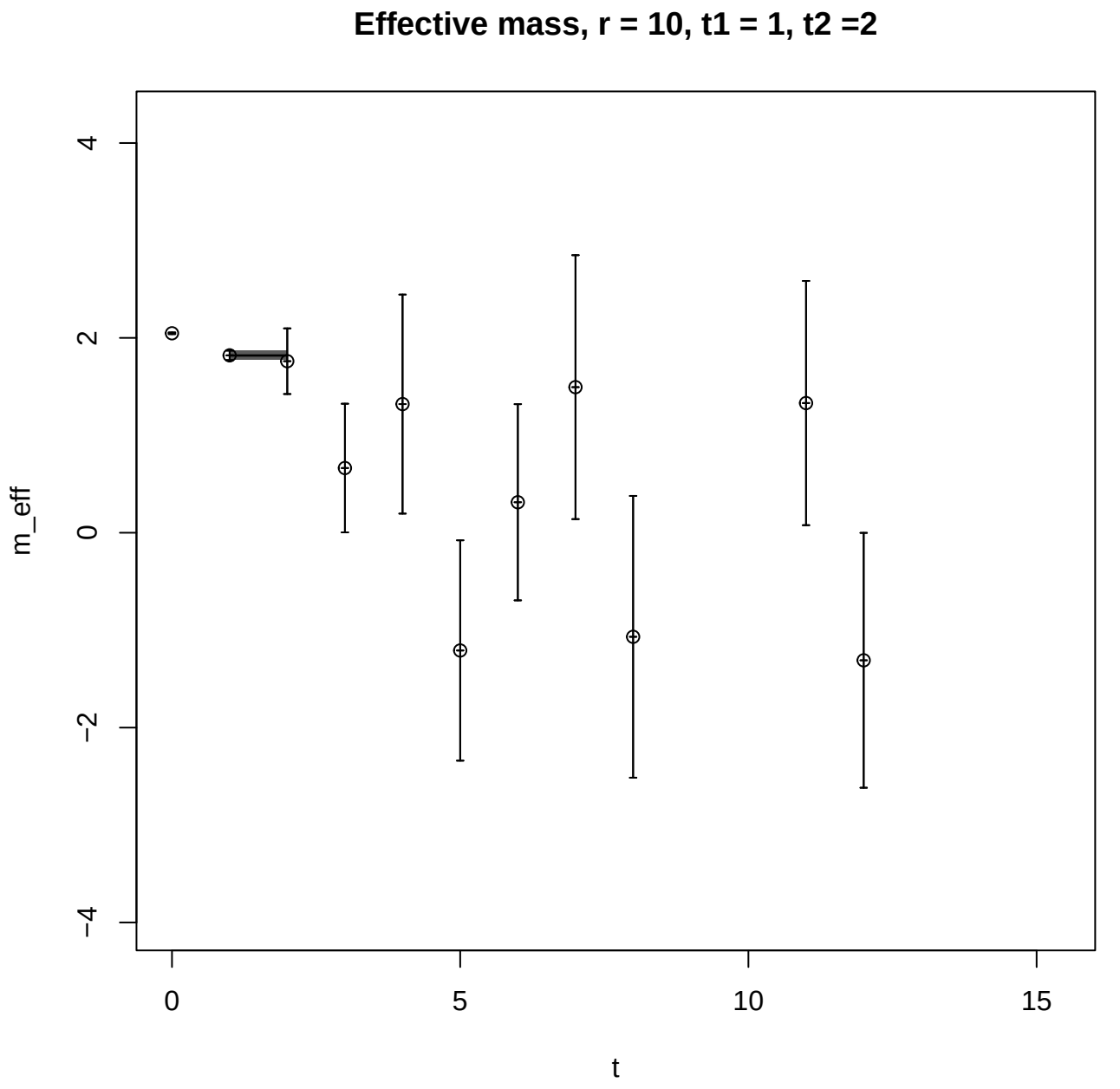
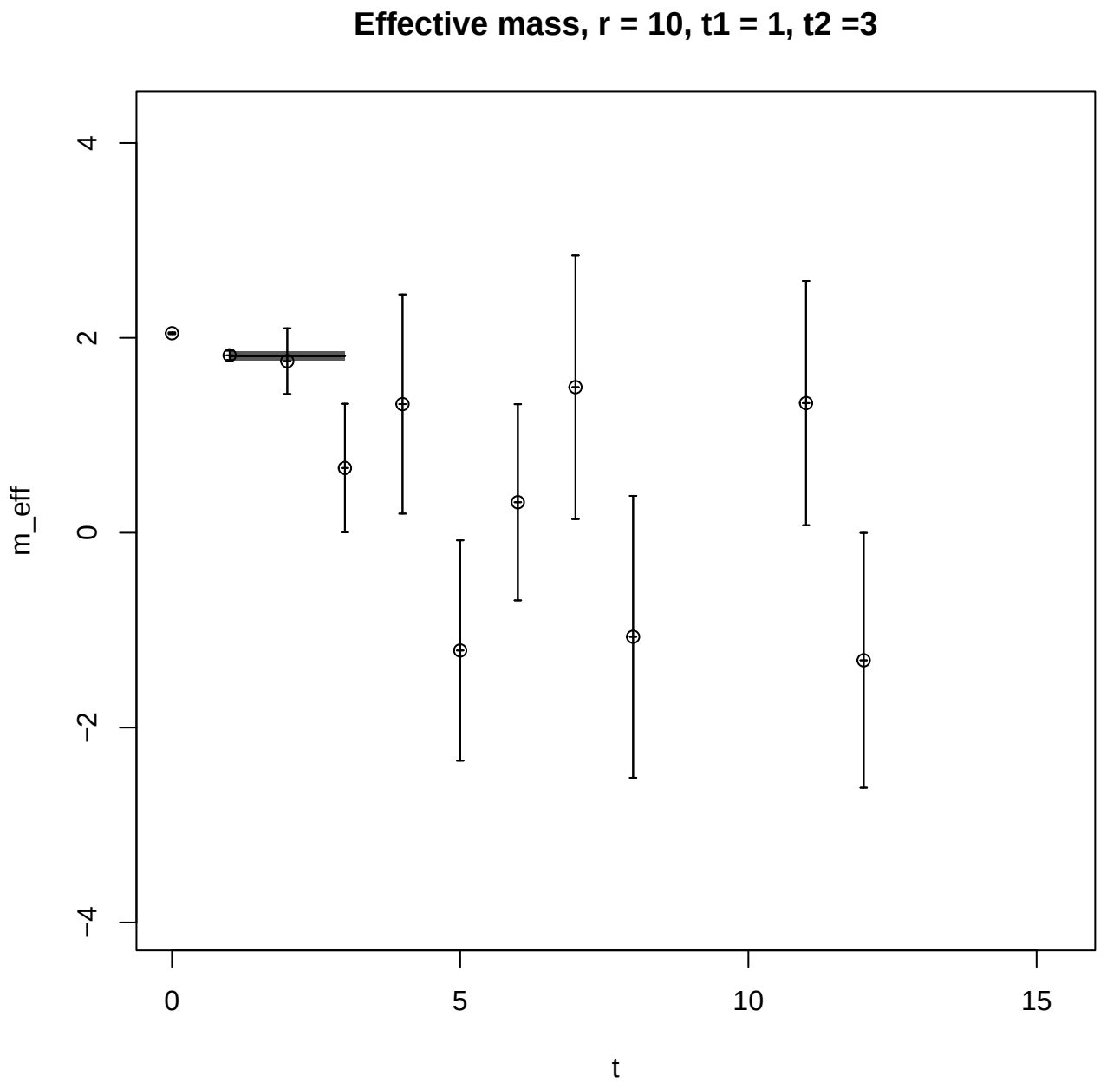
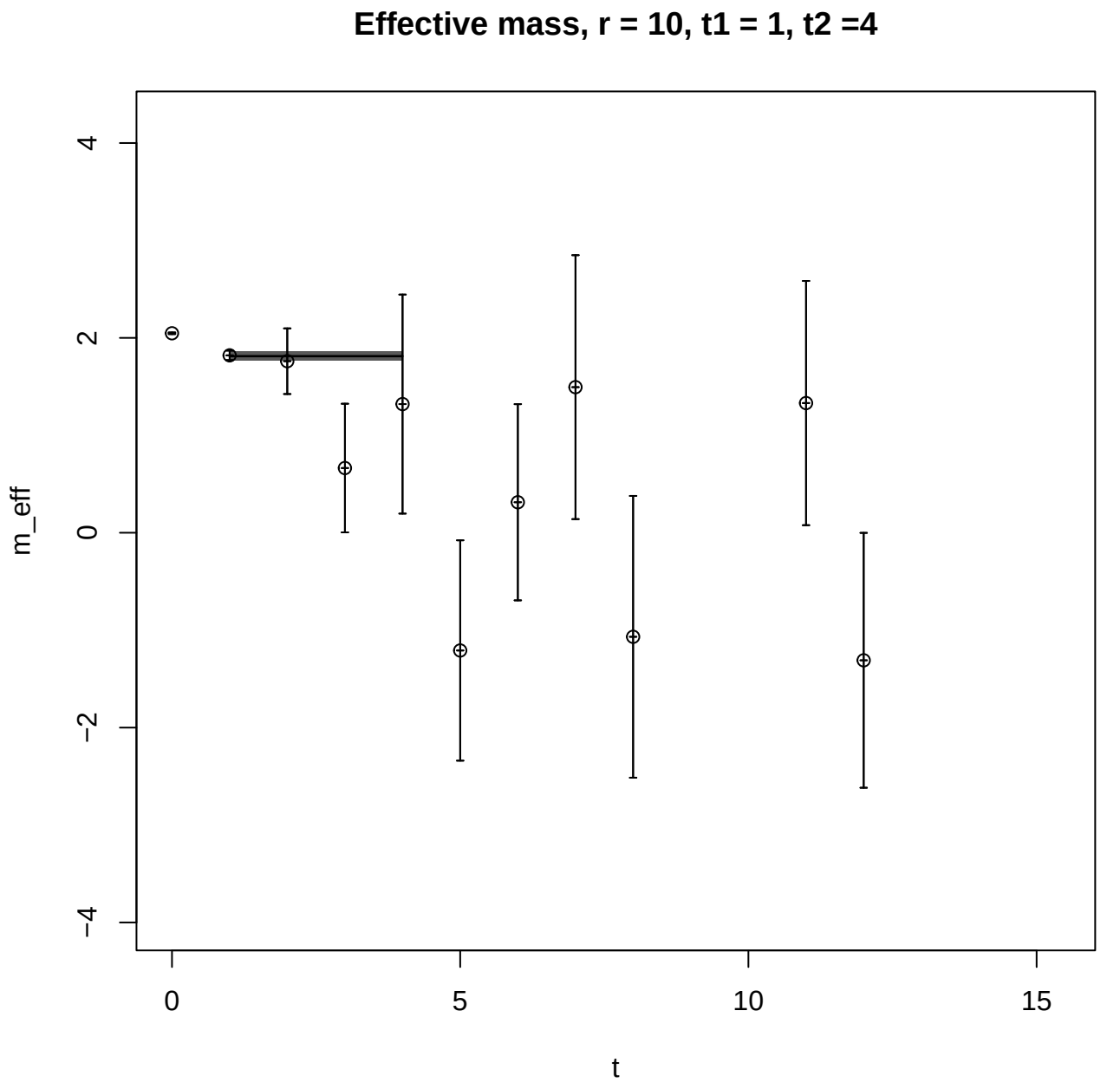
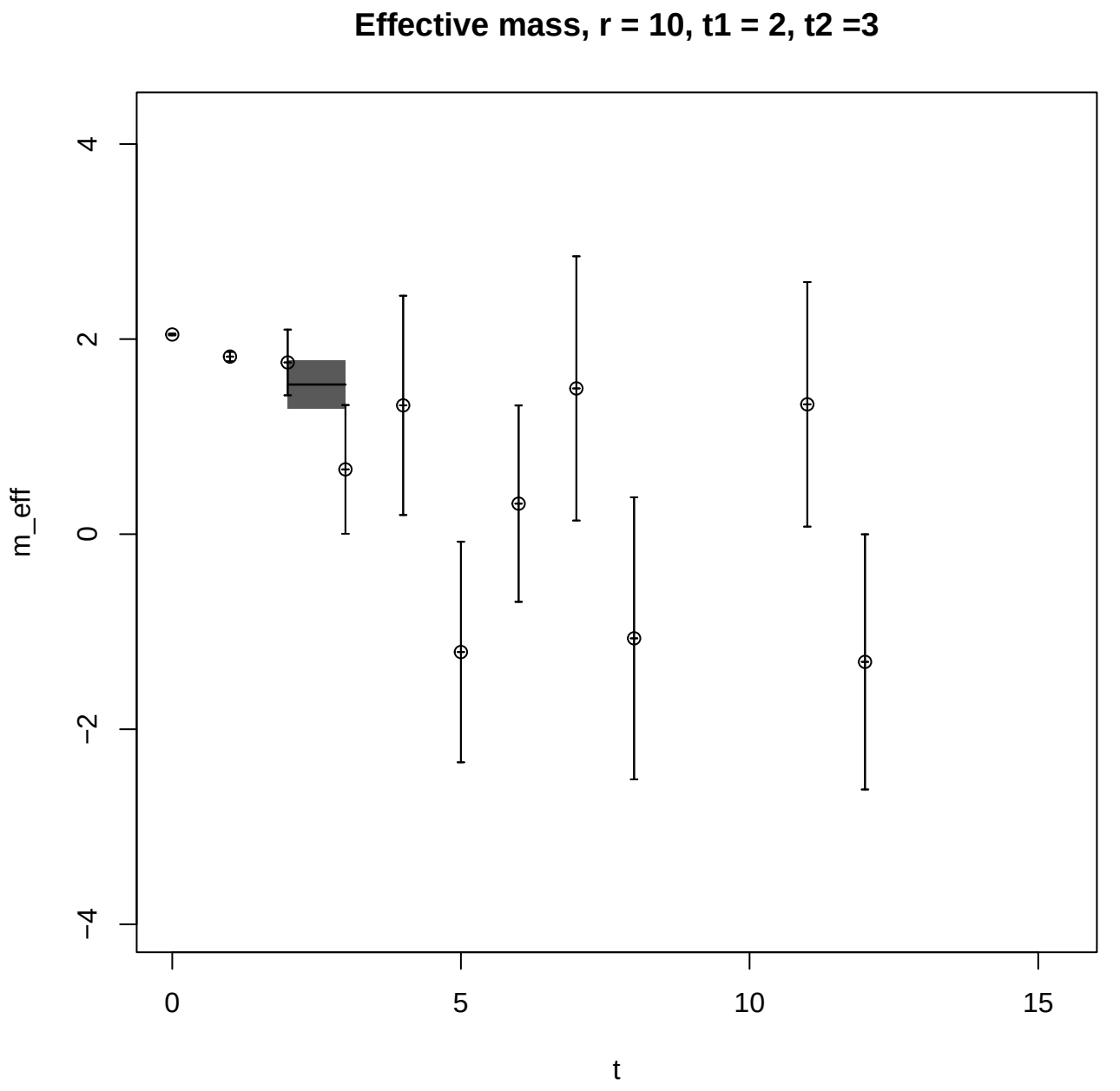

Effective mass, r = 9, t1 = 7, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

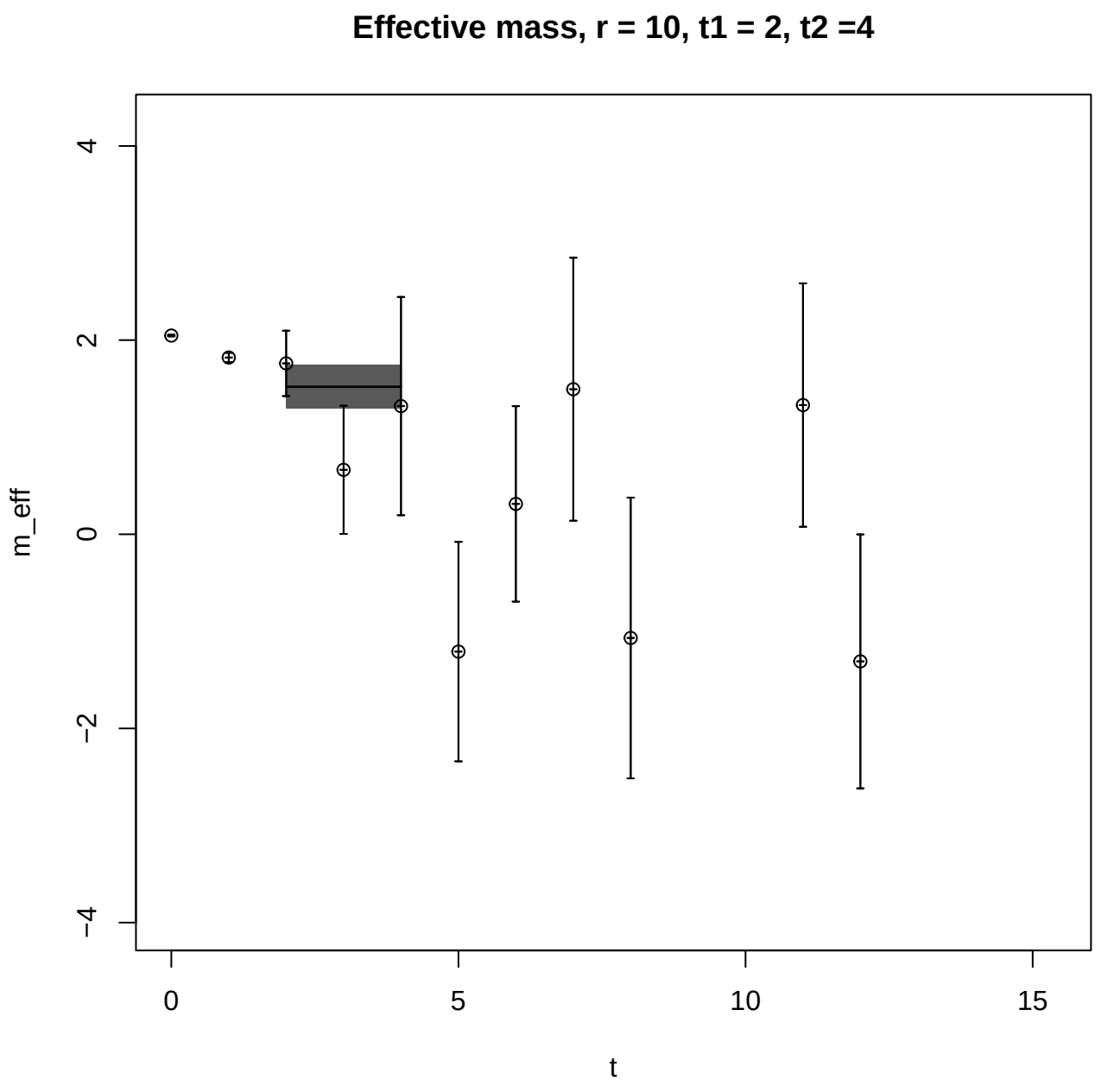

Effective mass, r = 9, t1 = 8, t2 =10
m_eff
t
4
2
0
−2
−4
0
5
10
15

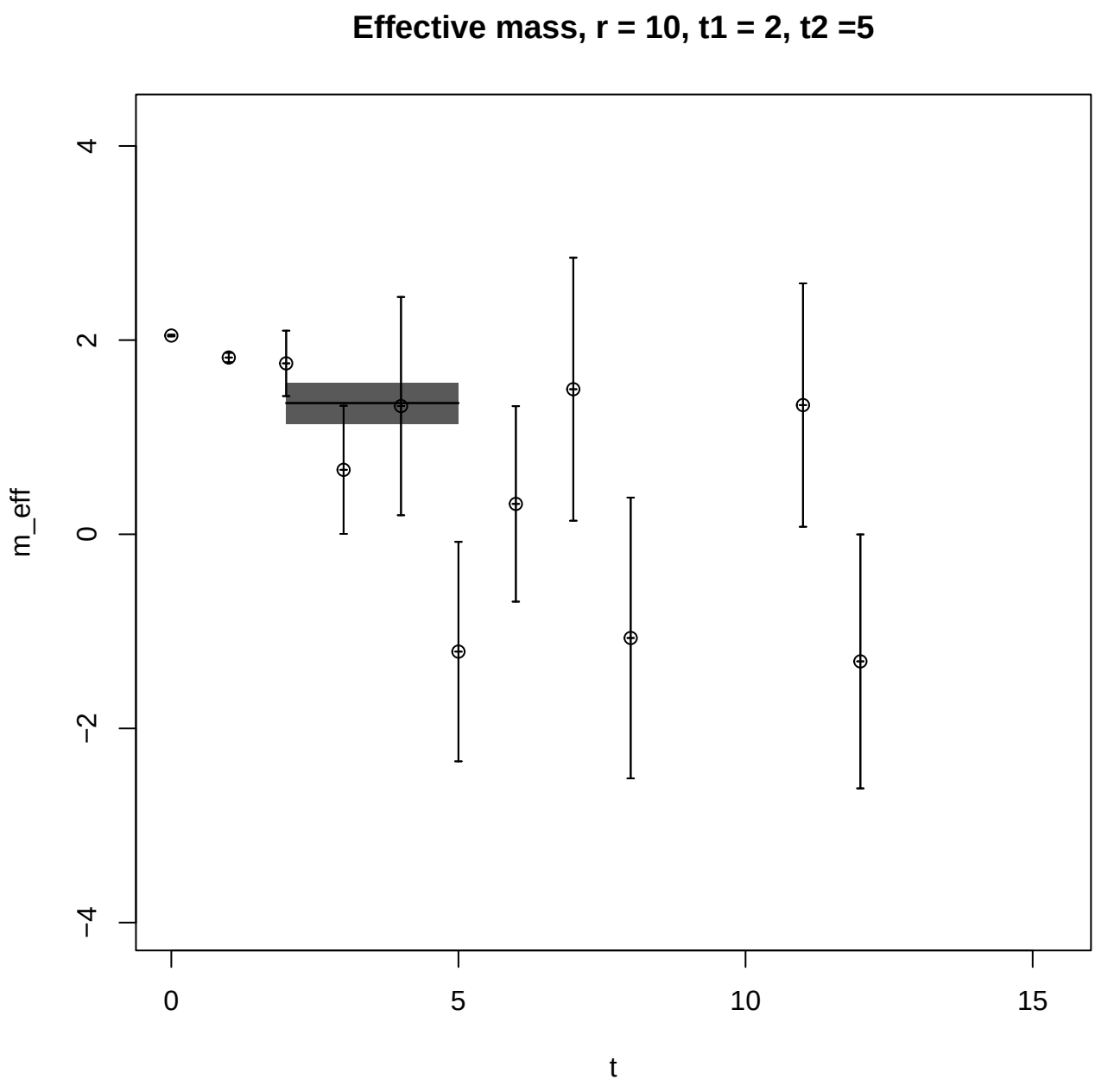

Effective mass, r = 10, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
0
5
10
15

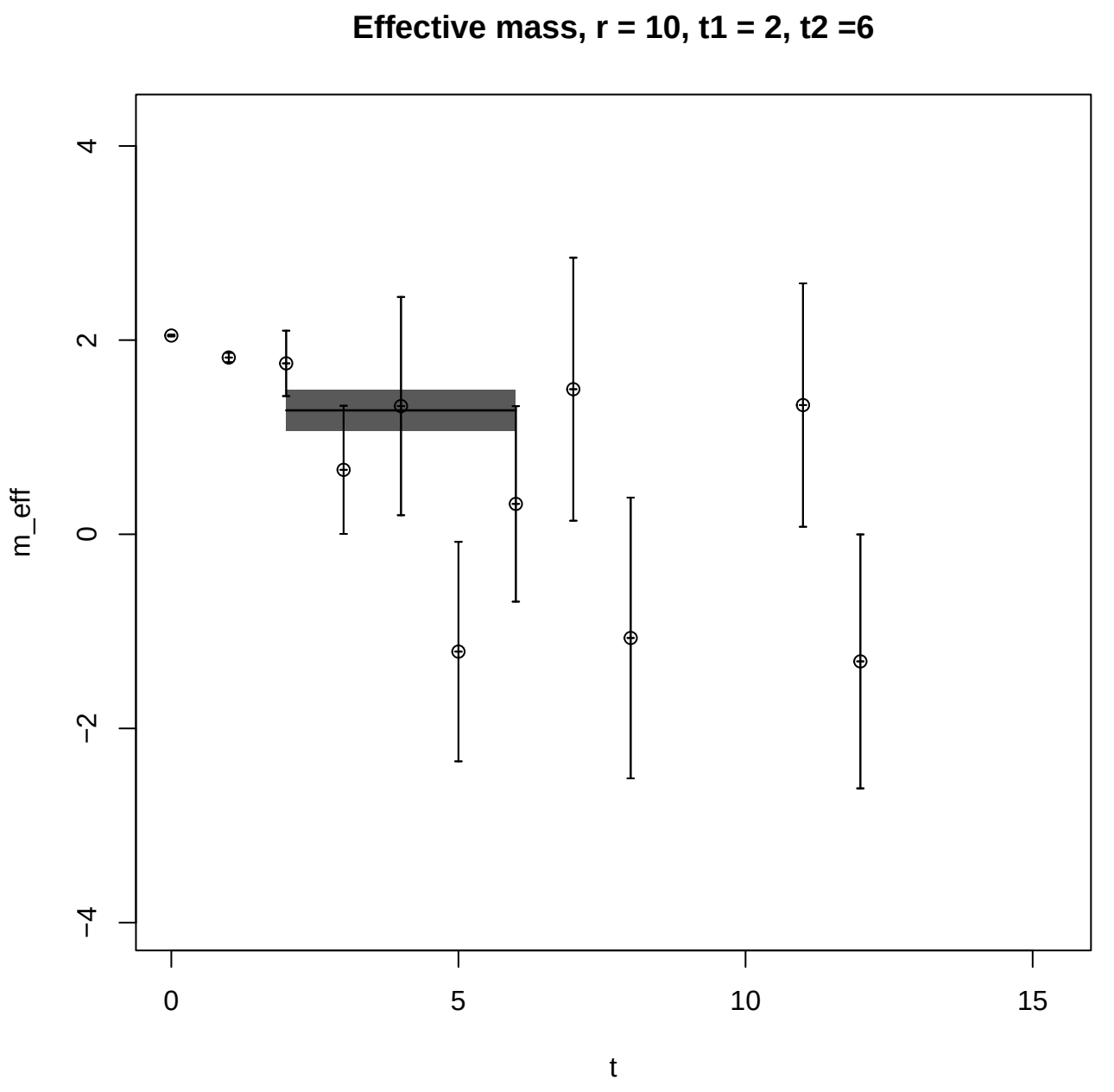

Effective mass, r = 10, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

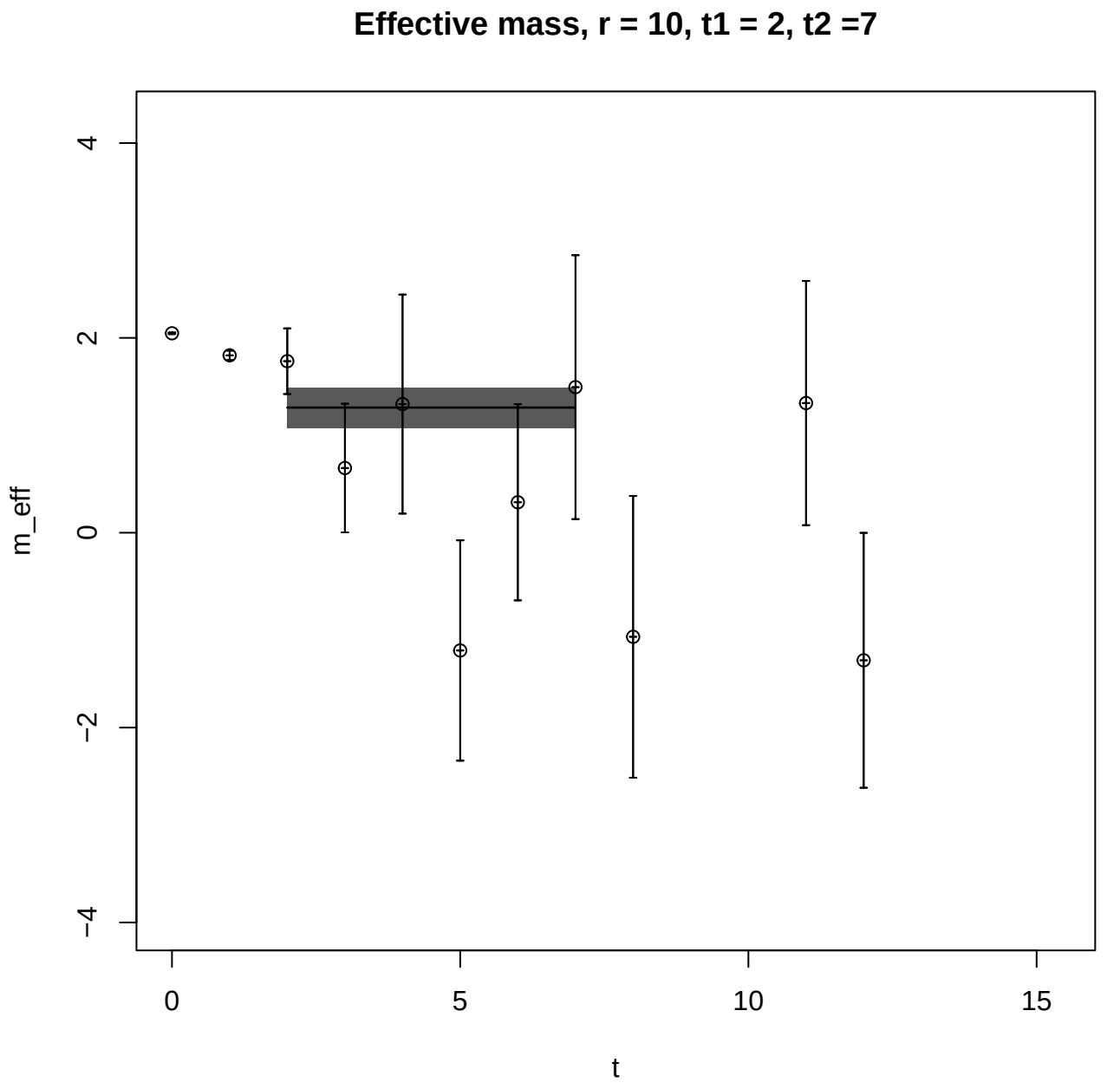

Effective mass, r = 10, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

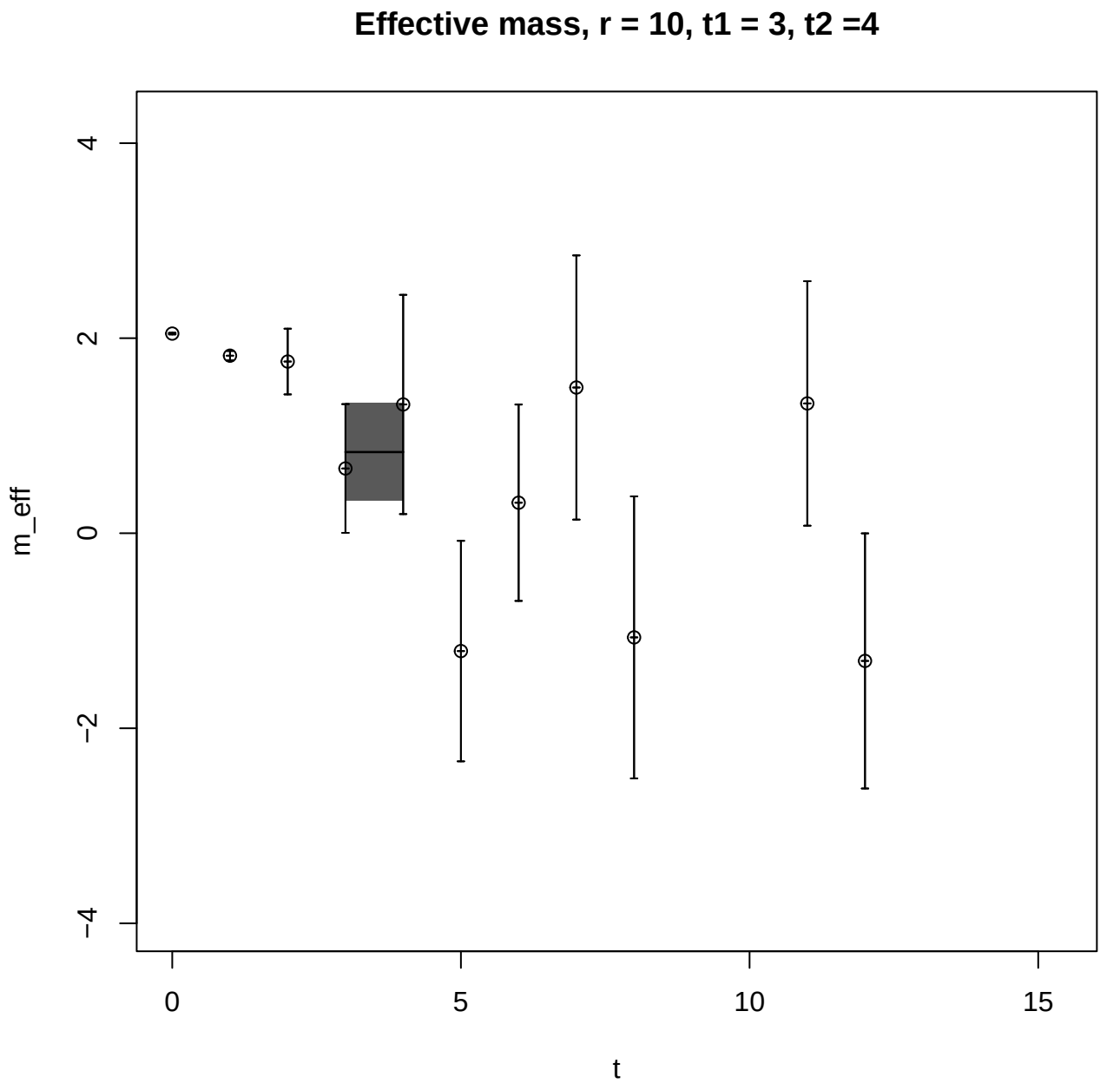

Effective mass, r = 10, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
0
5
10
15

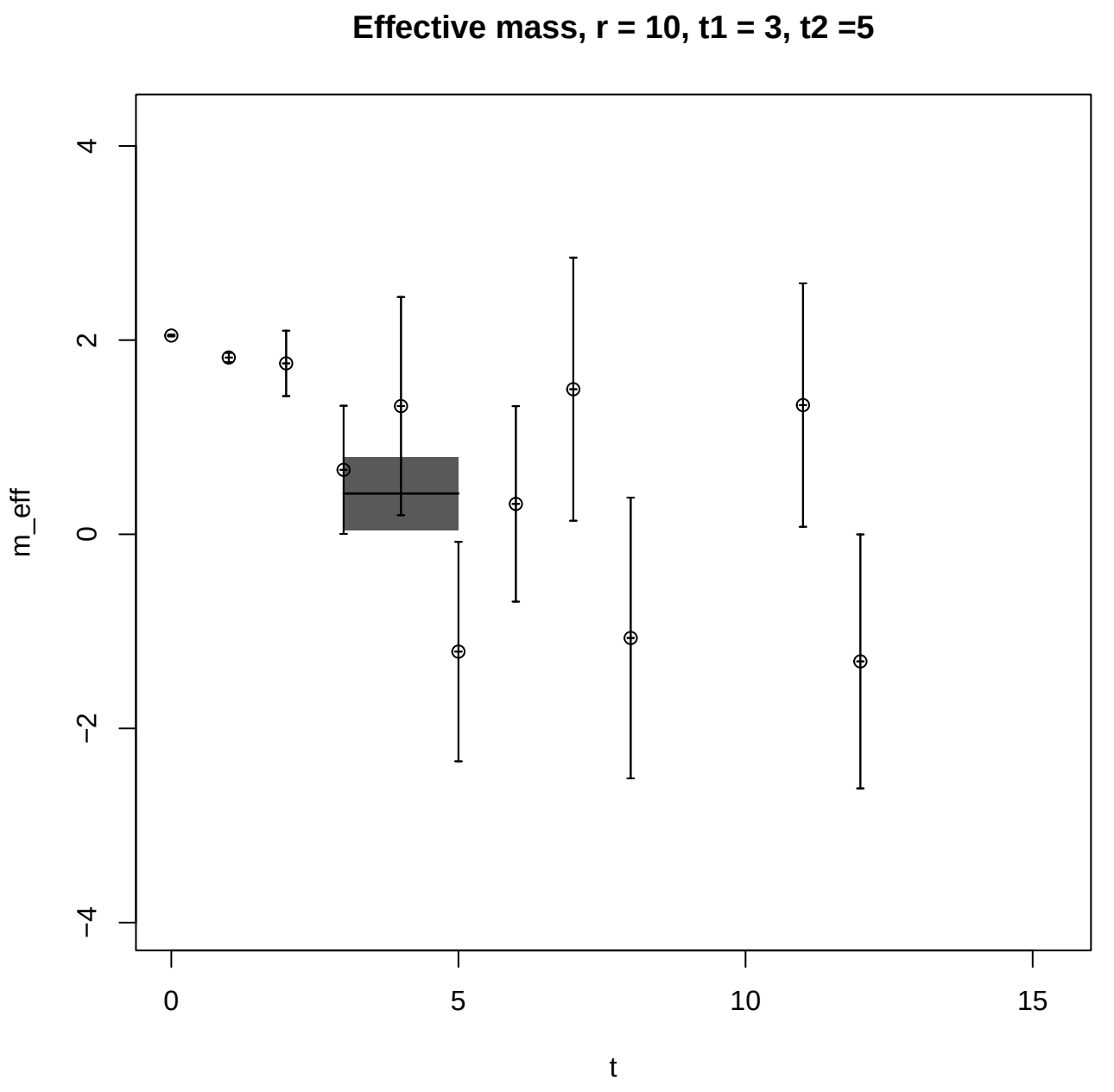

Effective mass, r = 10, t1 = 2, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

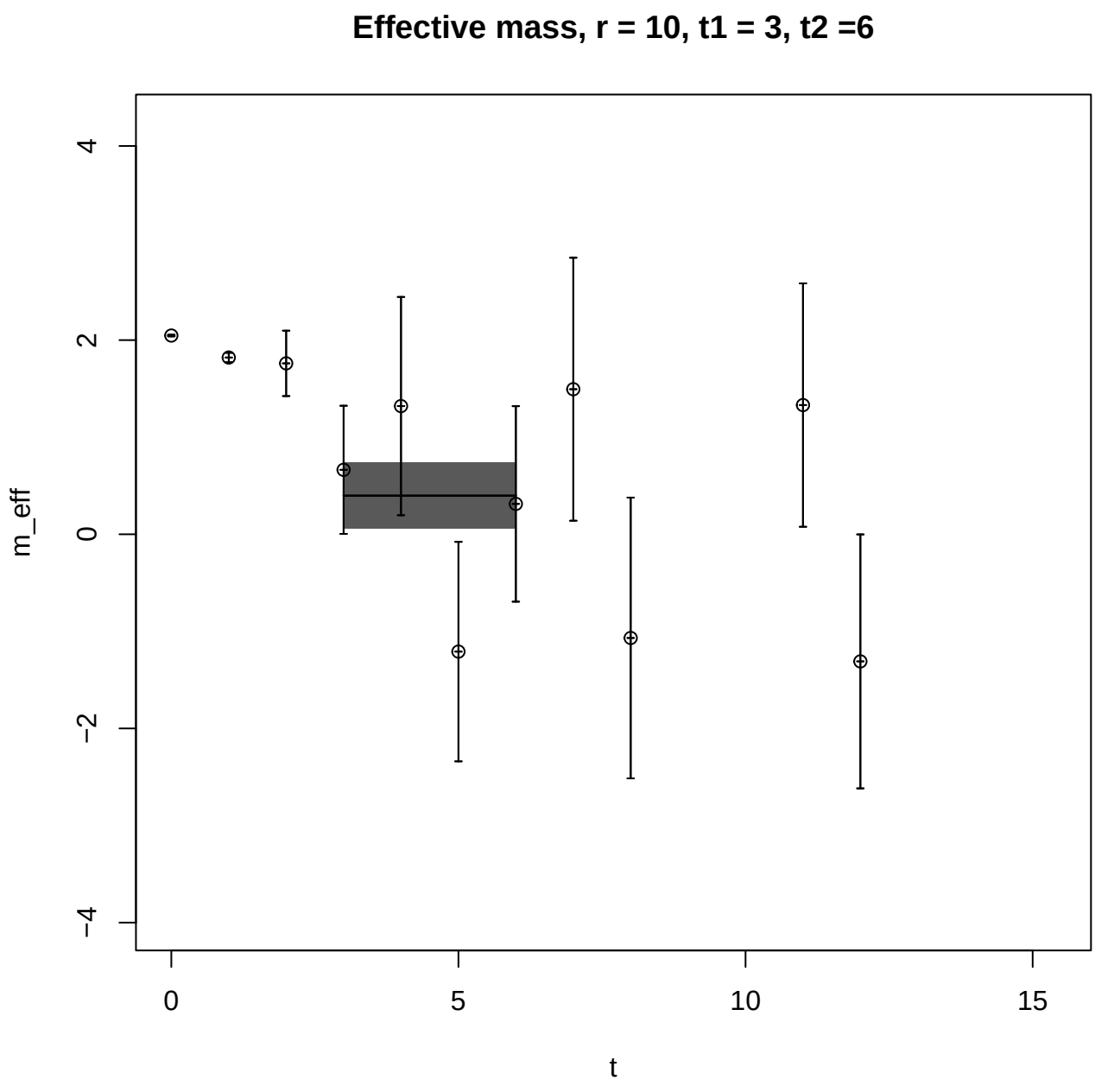

Effective mass, r = 10, t1 = 2, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

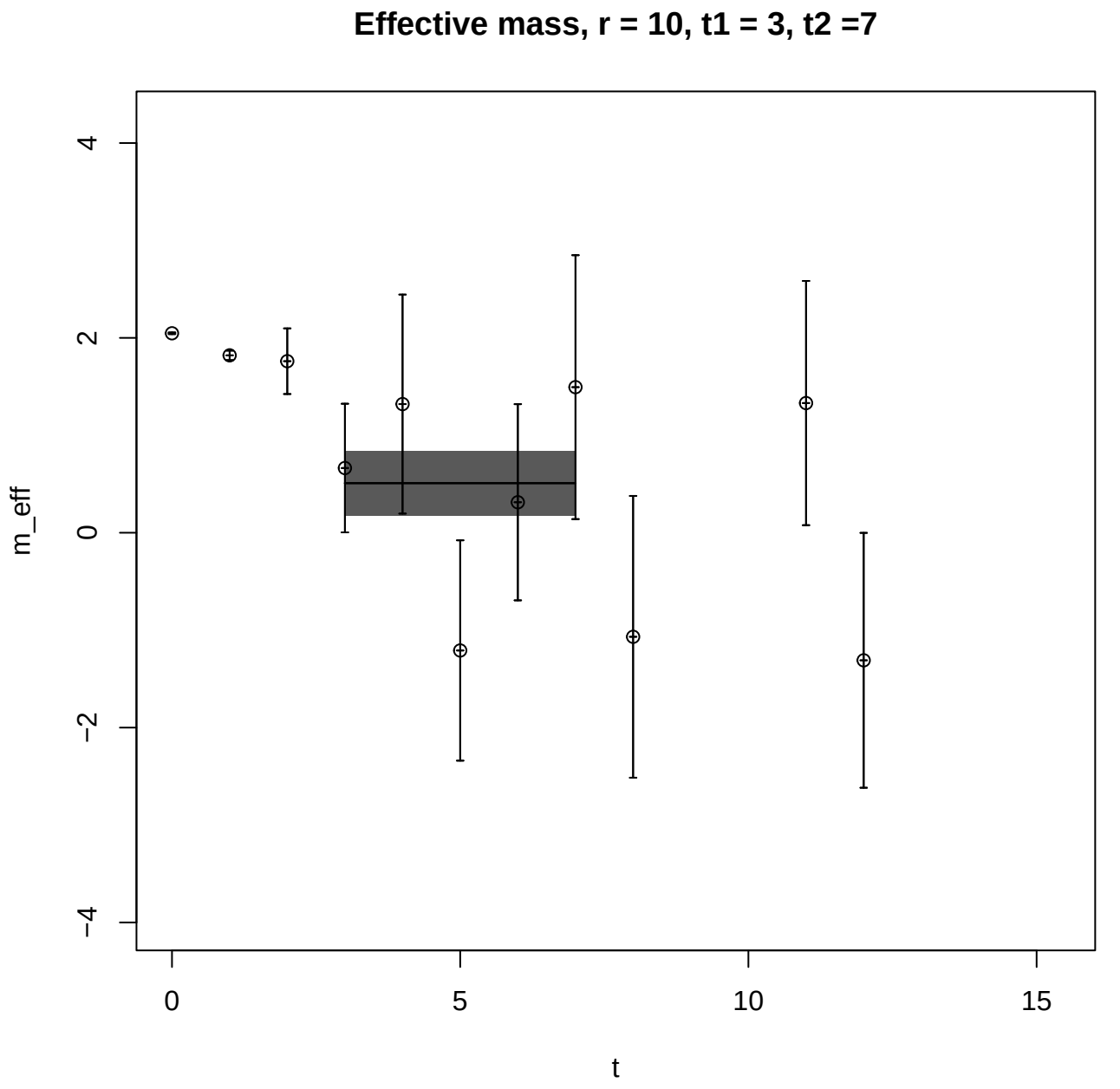

Effective mass, r = 10, t1 = 2, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15

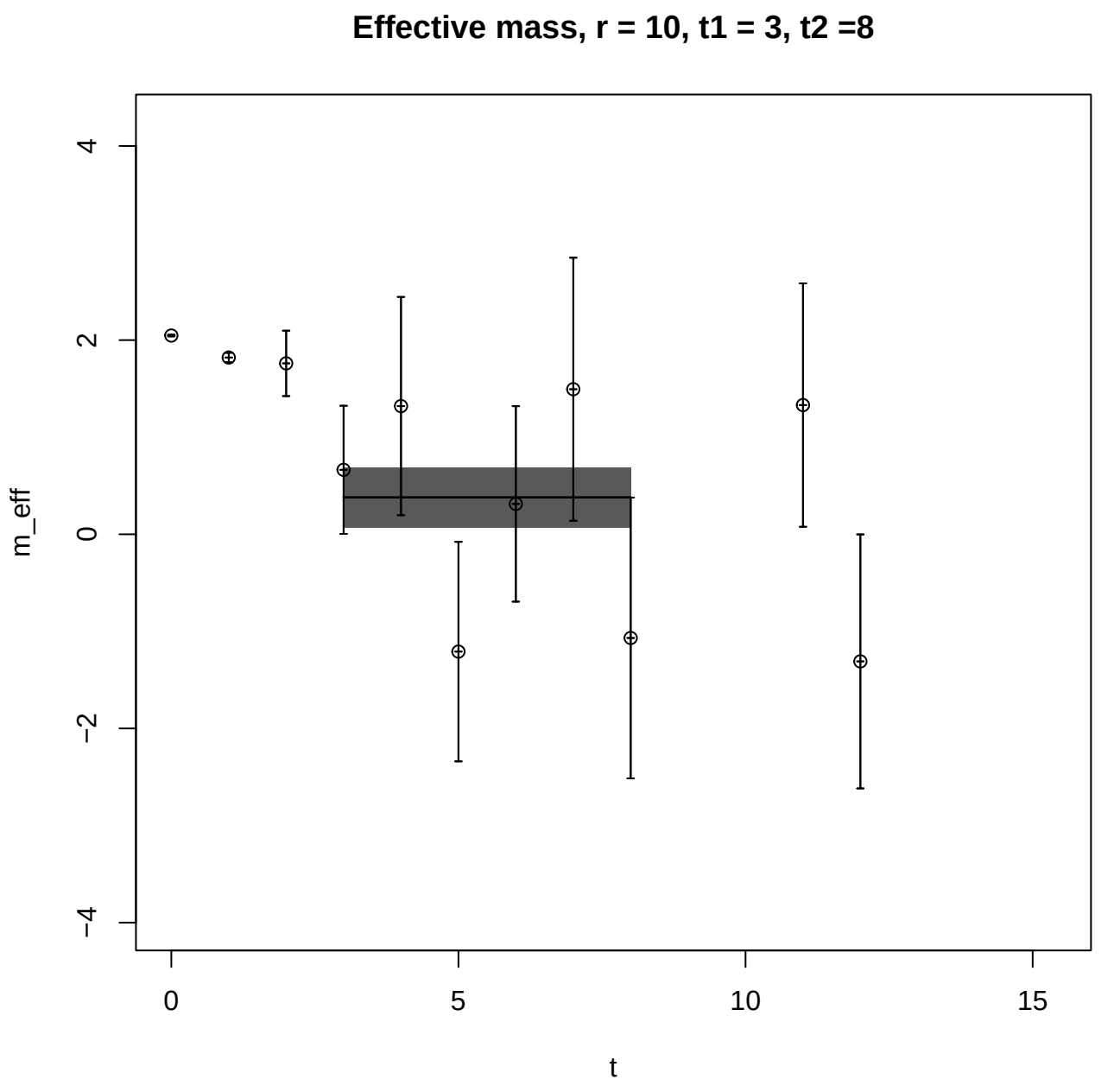

Effective mass, r = 10, t1 = 2, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

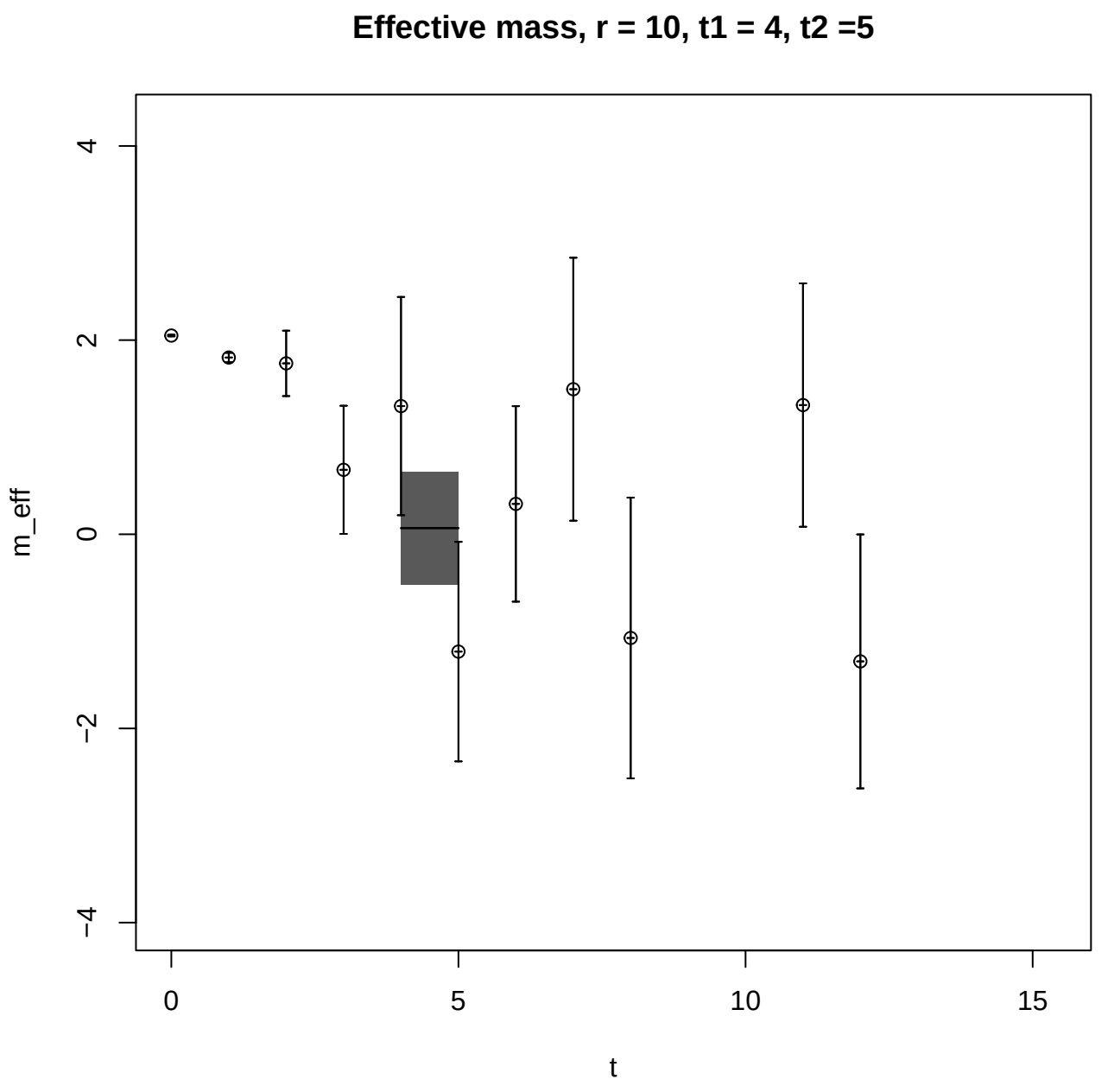

Effective mass, r = 10, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
0
5
10
15

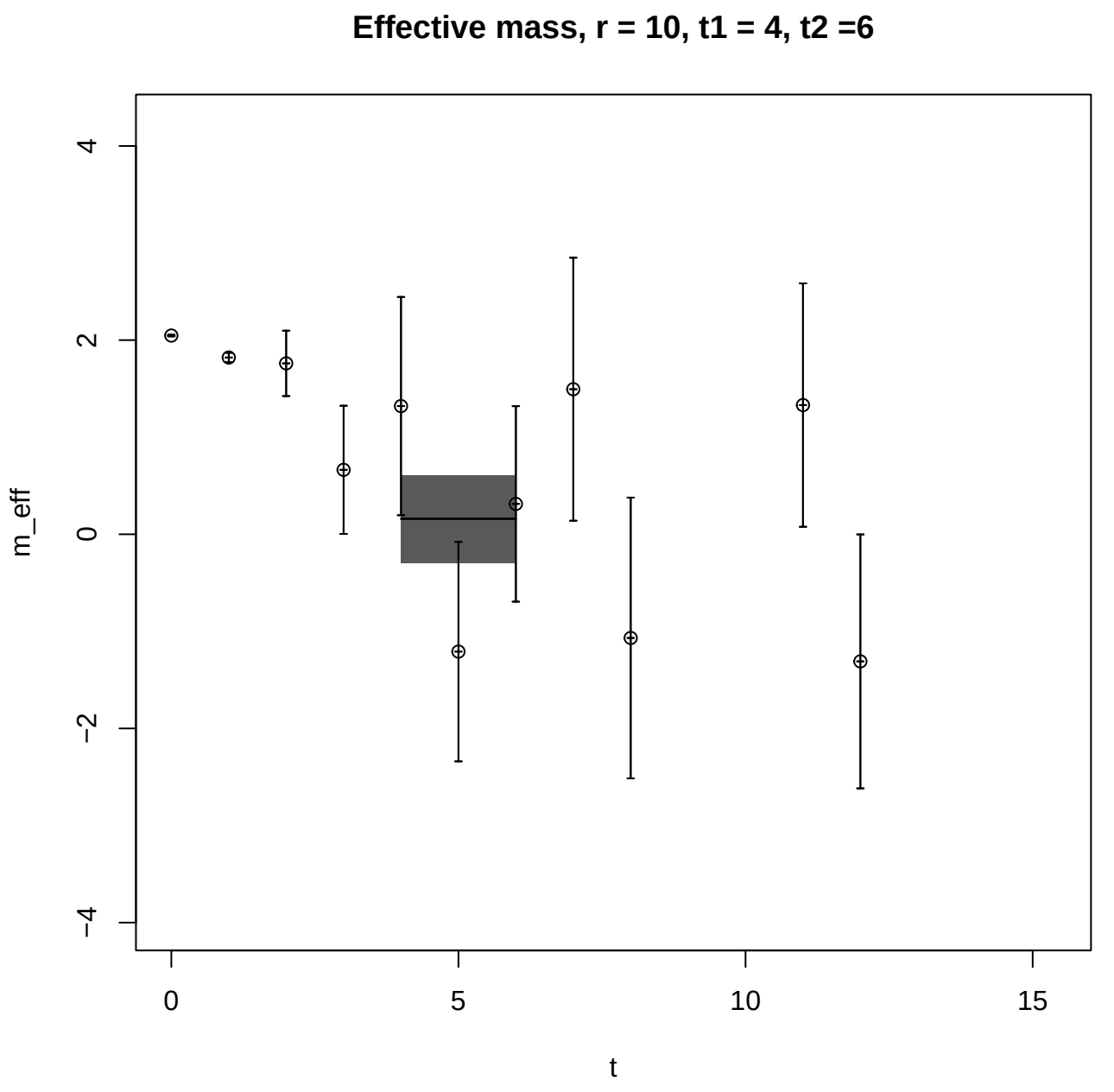

Effective mass, r = 10, t1 = 3, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

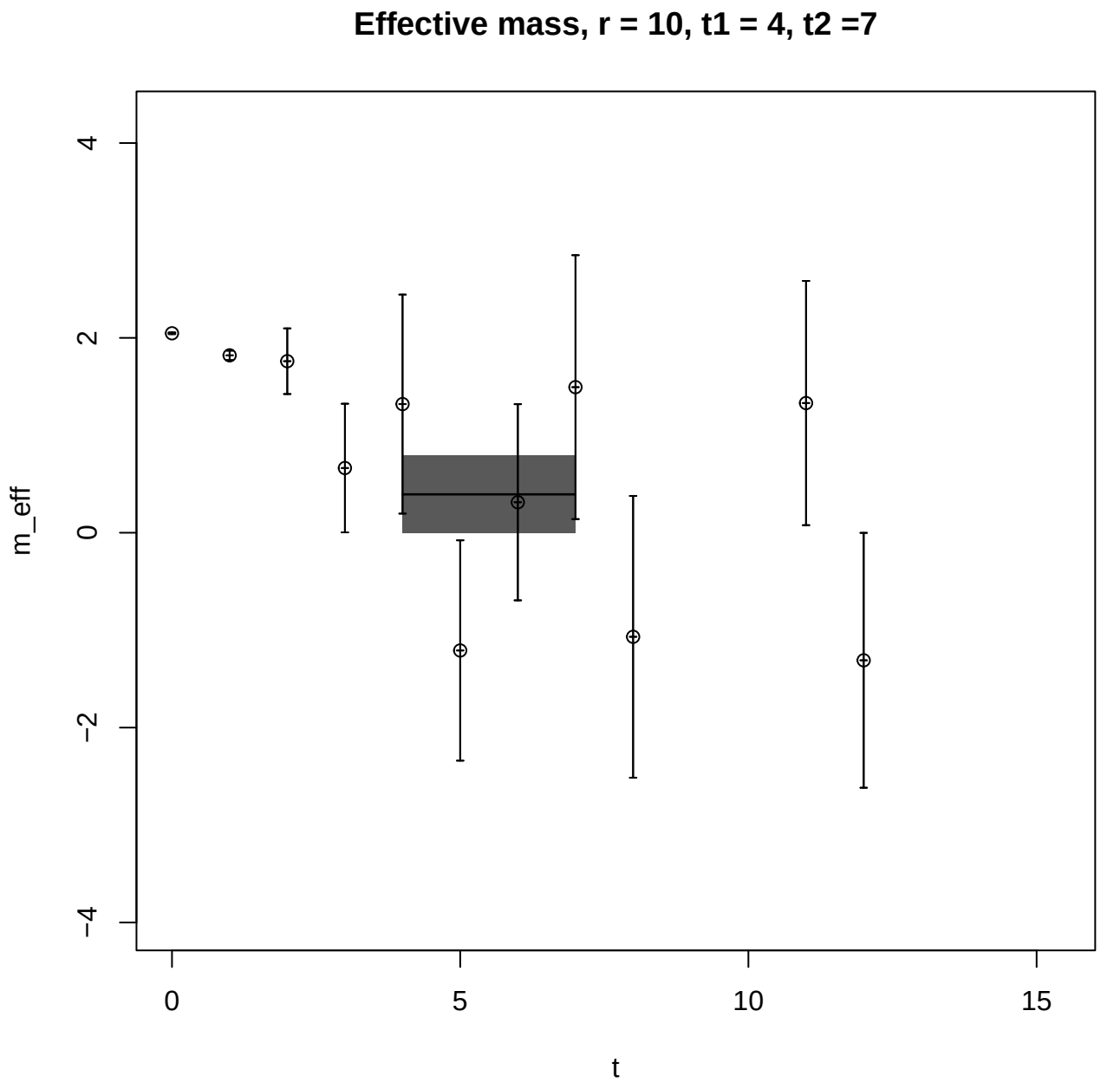

Effective mass, r = 10, t1 = 3, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15

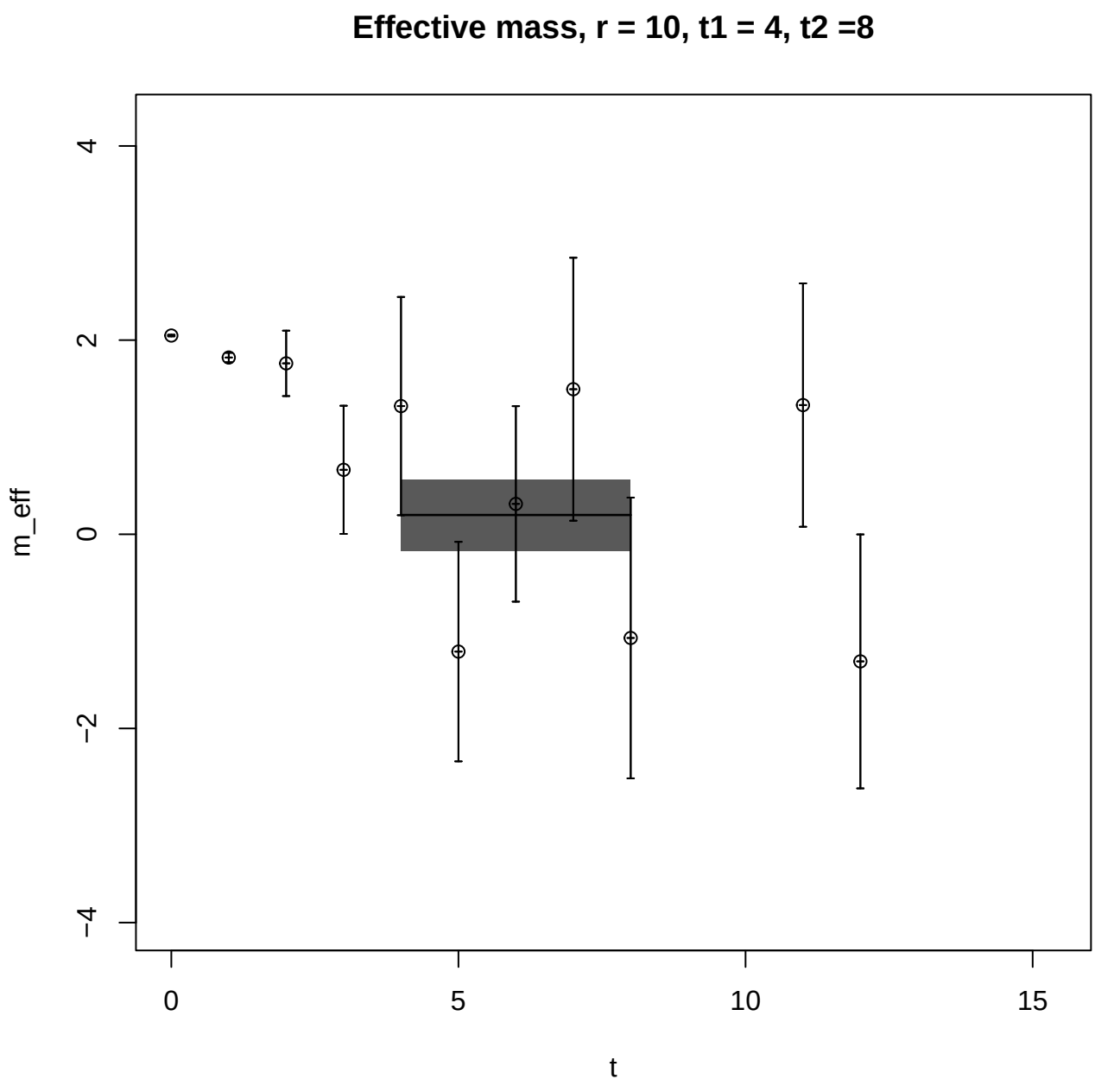

Effective mass, r = 10, t1 = 3, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

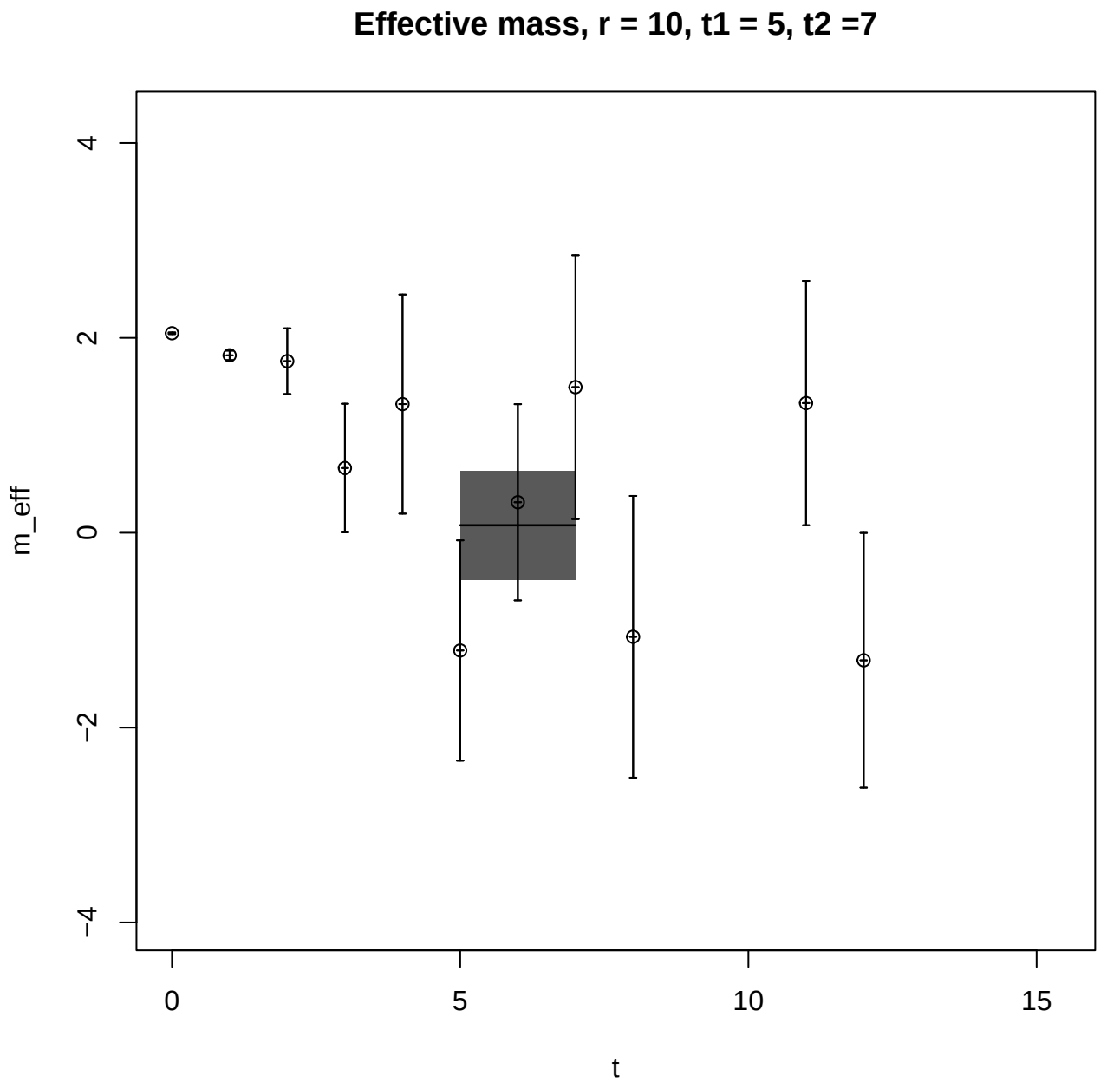

Effective mass, r = 10, t1 = 3, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

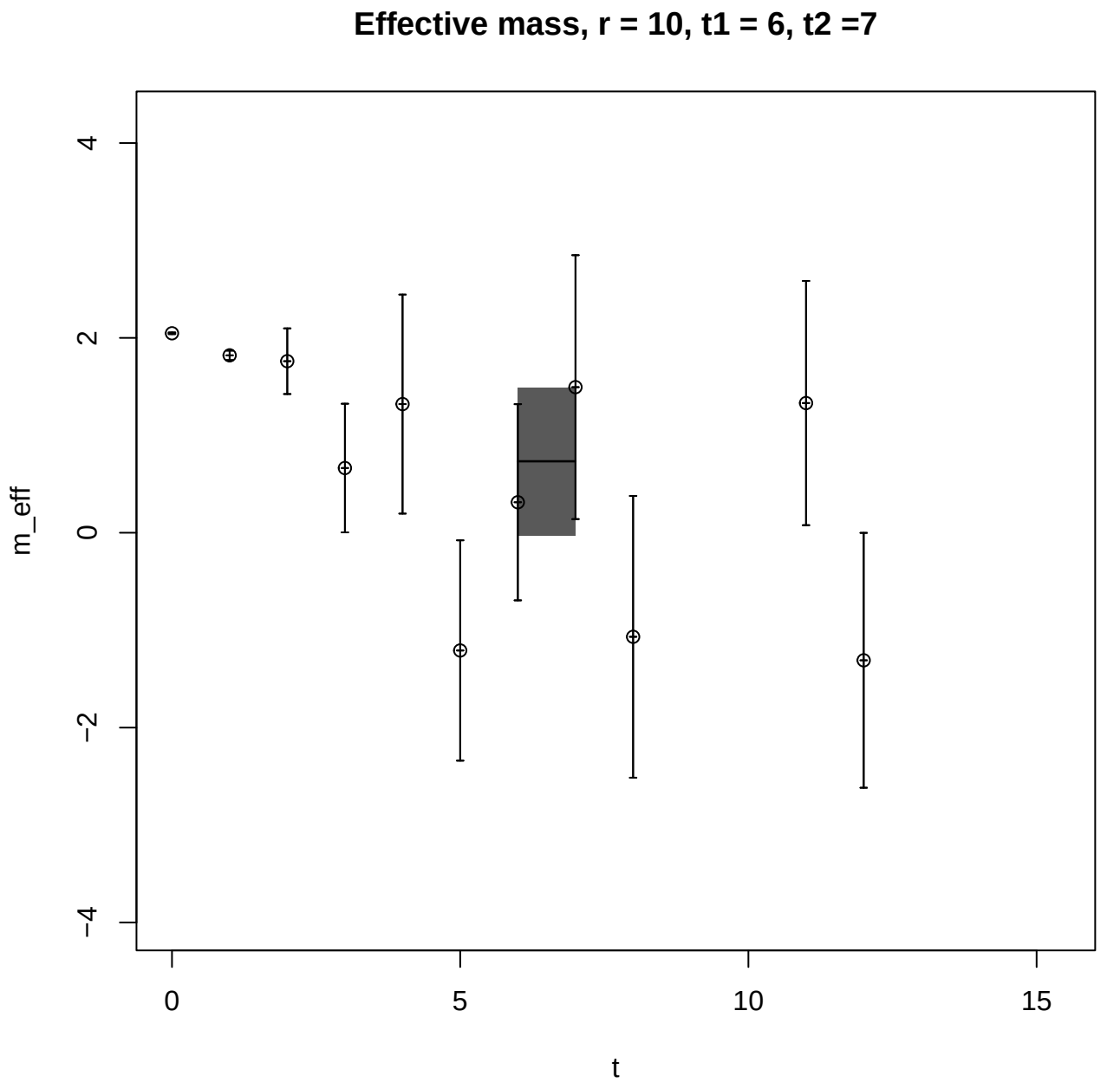

Effective mass, r = 10, t1 = 4, t2 =5
m_eff
t
4
2
0
−2
−4
0
5
10
15

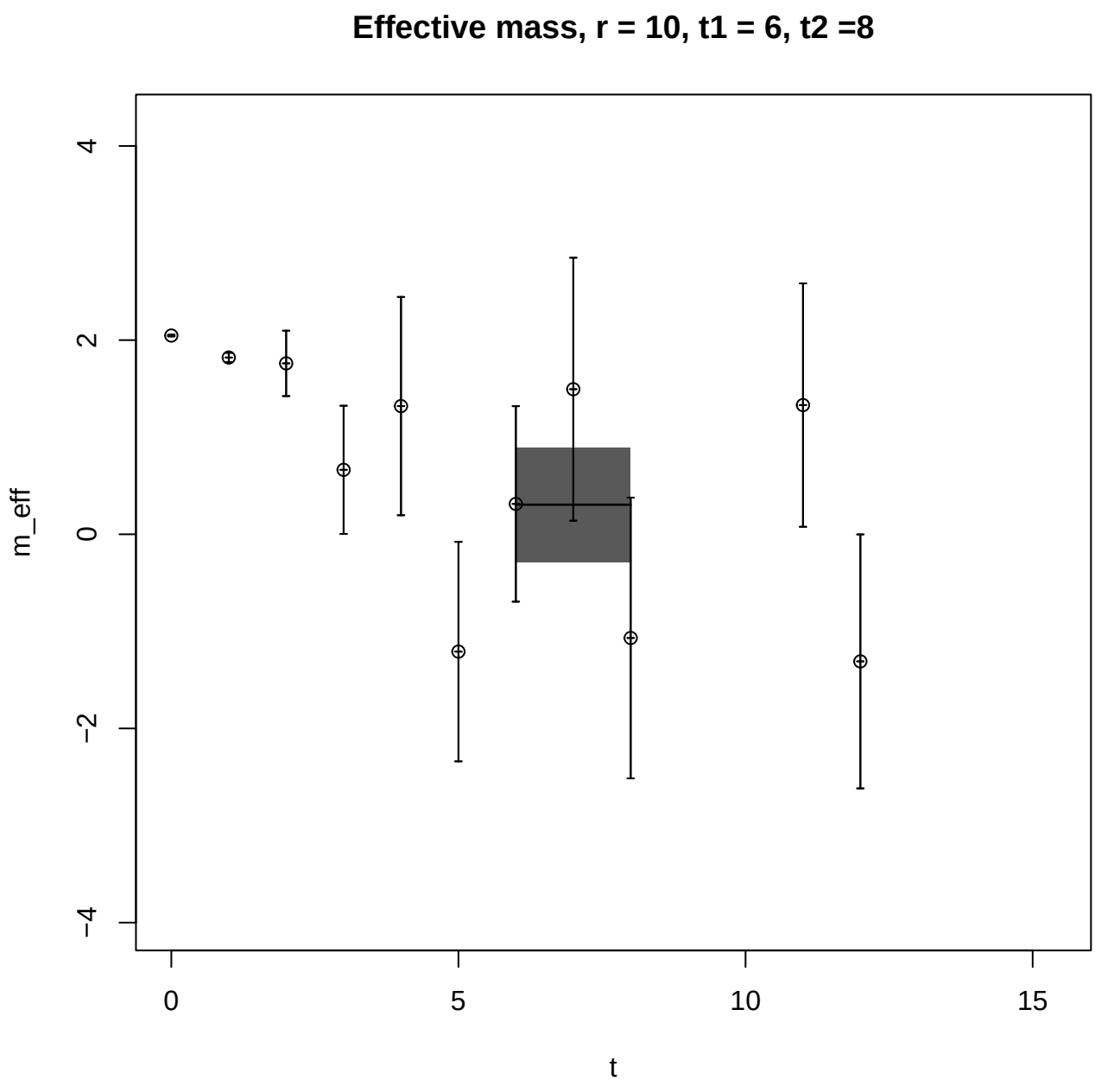

Effective mass, r = 10, t1 = 4, t2 =6
m_eff
t
4
2
0
−2
−4
0
5
10
15

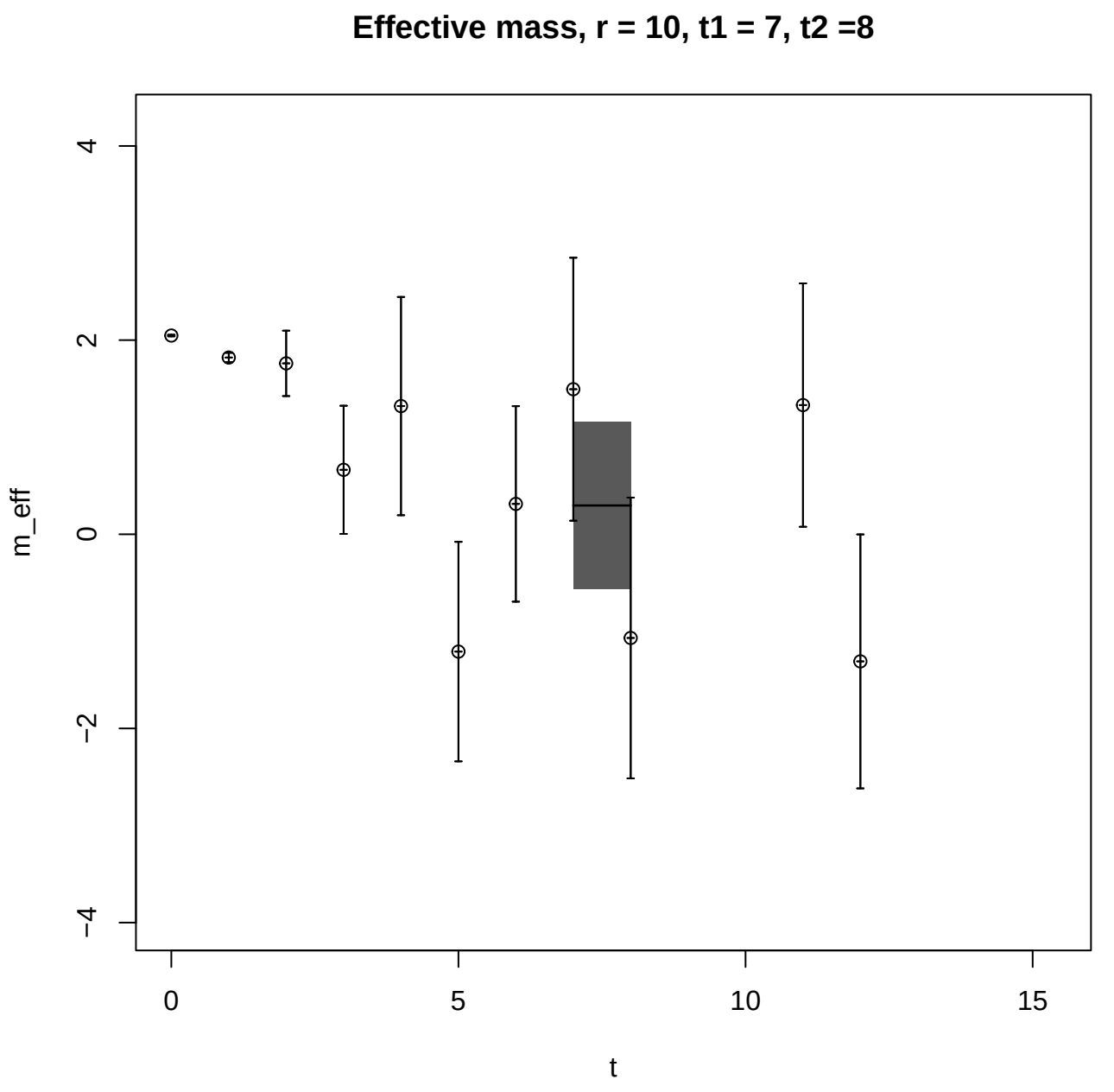

Effective mass, r = 10, t1 = 4, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

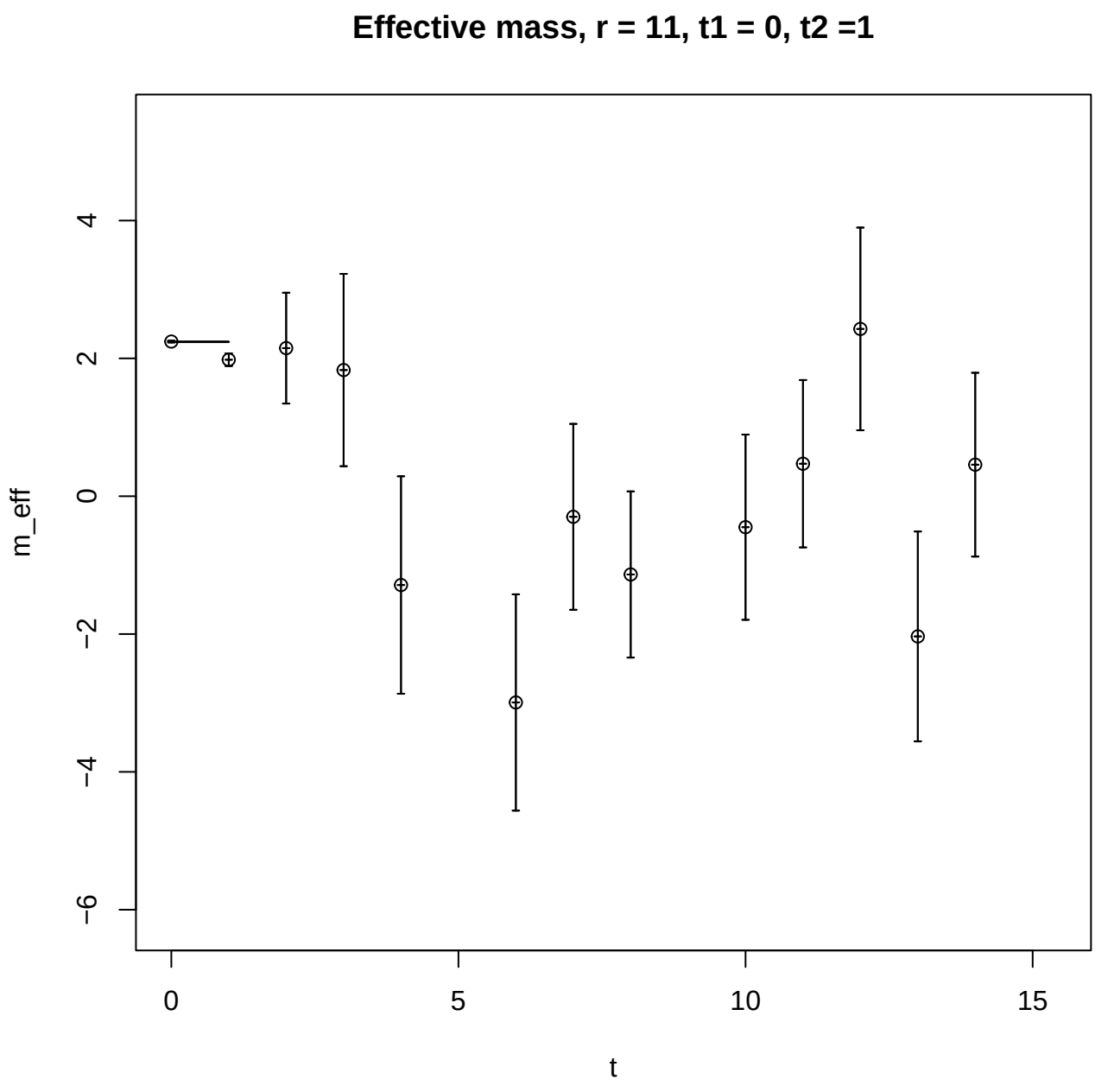

Effective mass, r = 10, t1 = 4, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

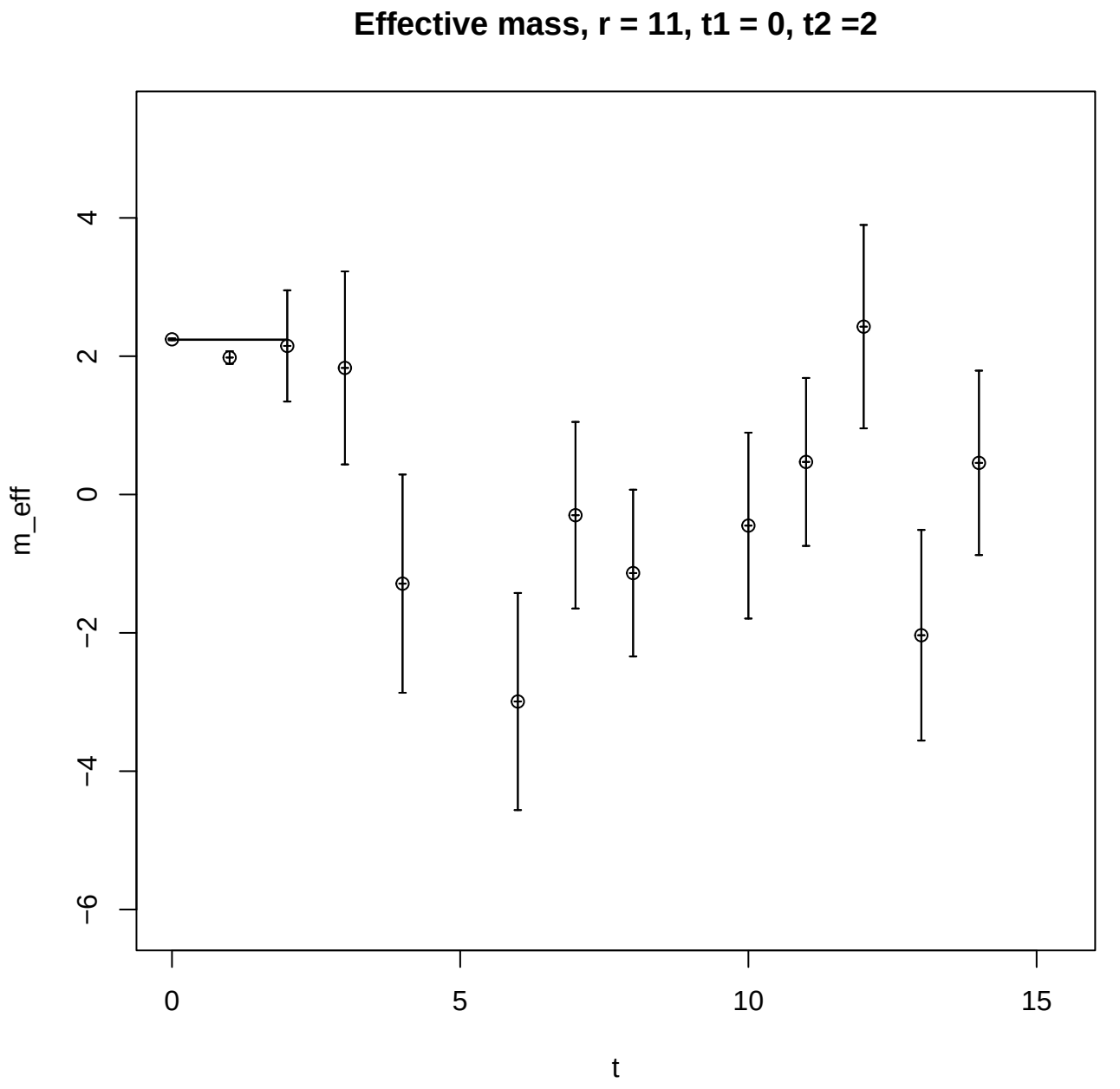

Effective mass, r = 10, t1 = 5, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

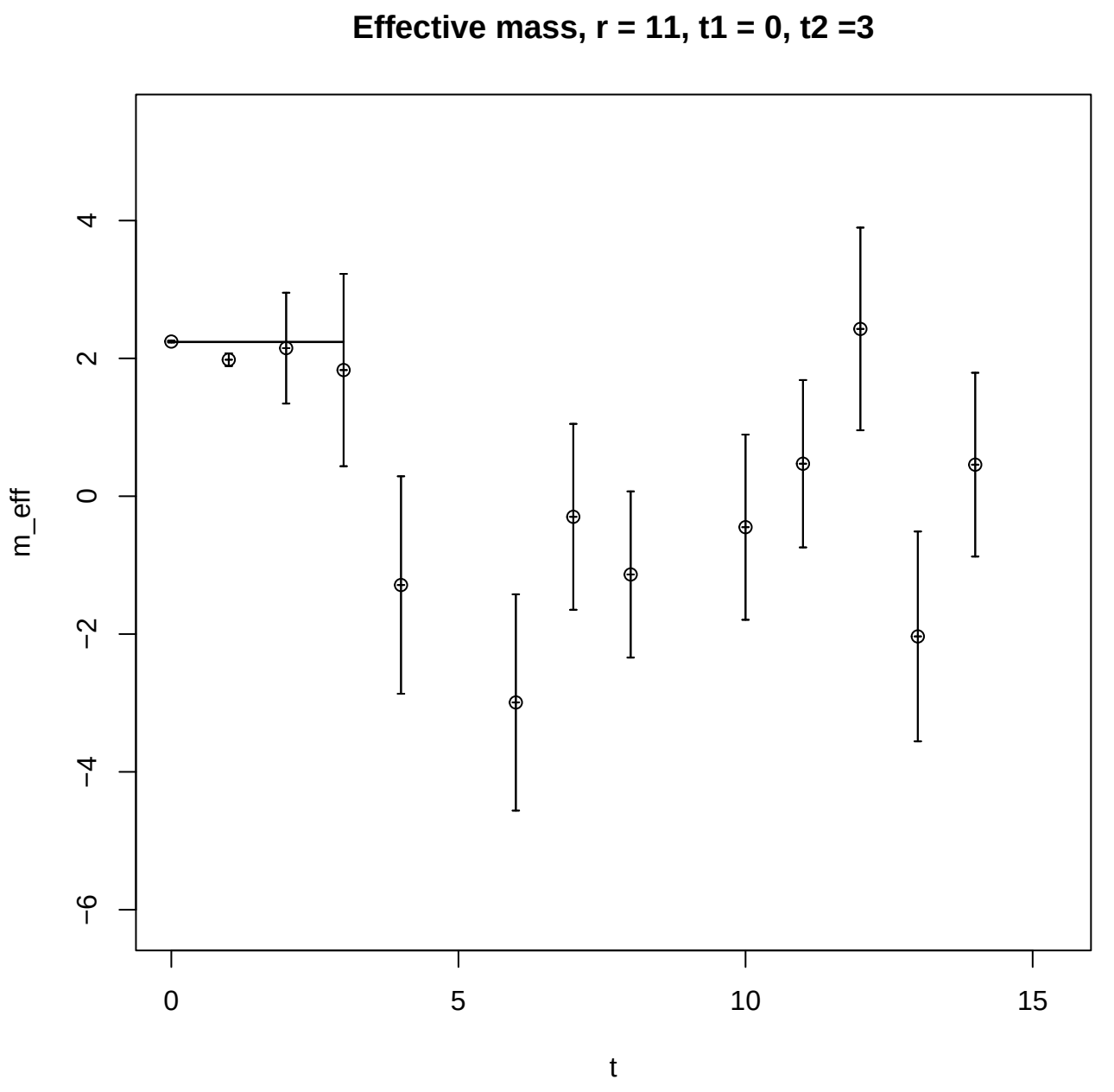

Effective mass, r = 10, t1 = 6, t2 =7
m_eff
t
4
2
0
−2
−4
0
5
10
15

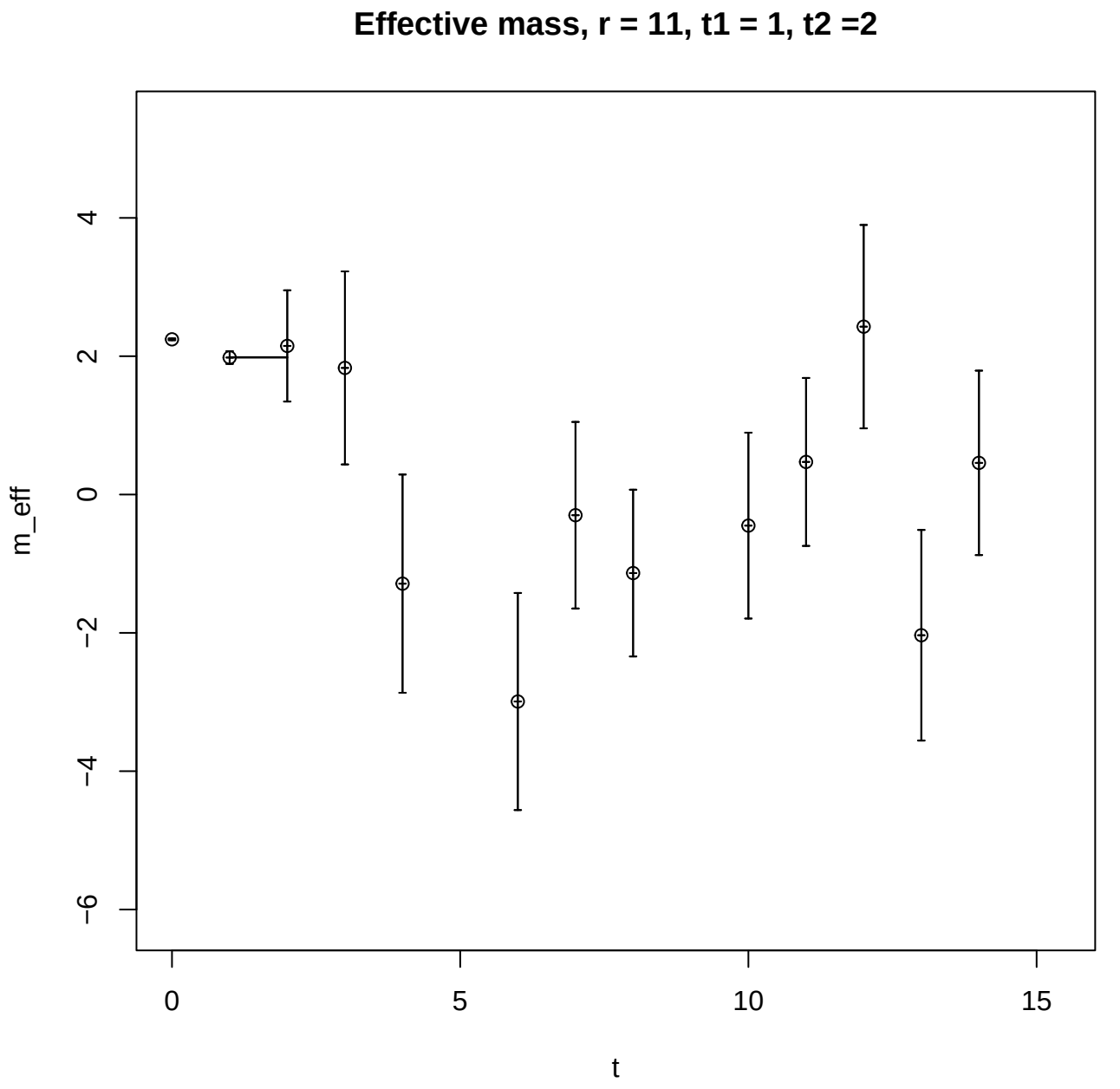

Effective mass, r = 10, t1 = 6, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

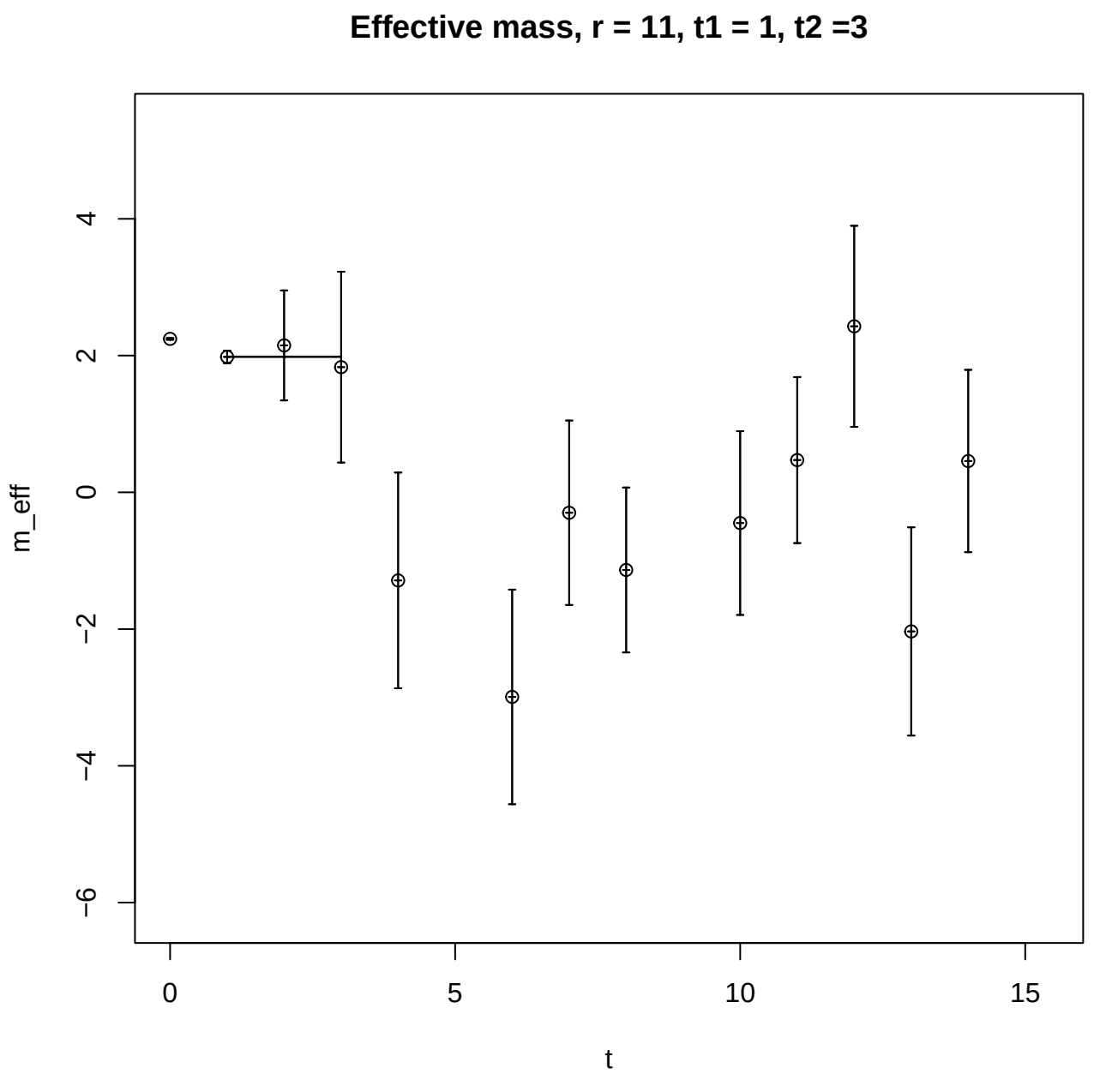

Effective mass, r = 10, t1 = 7, t2 =8
m_eff
t
4
2
0
−2
−4
0
5
10
15

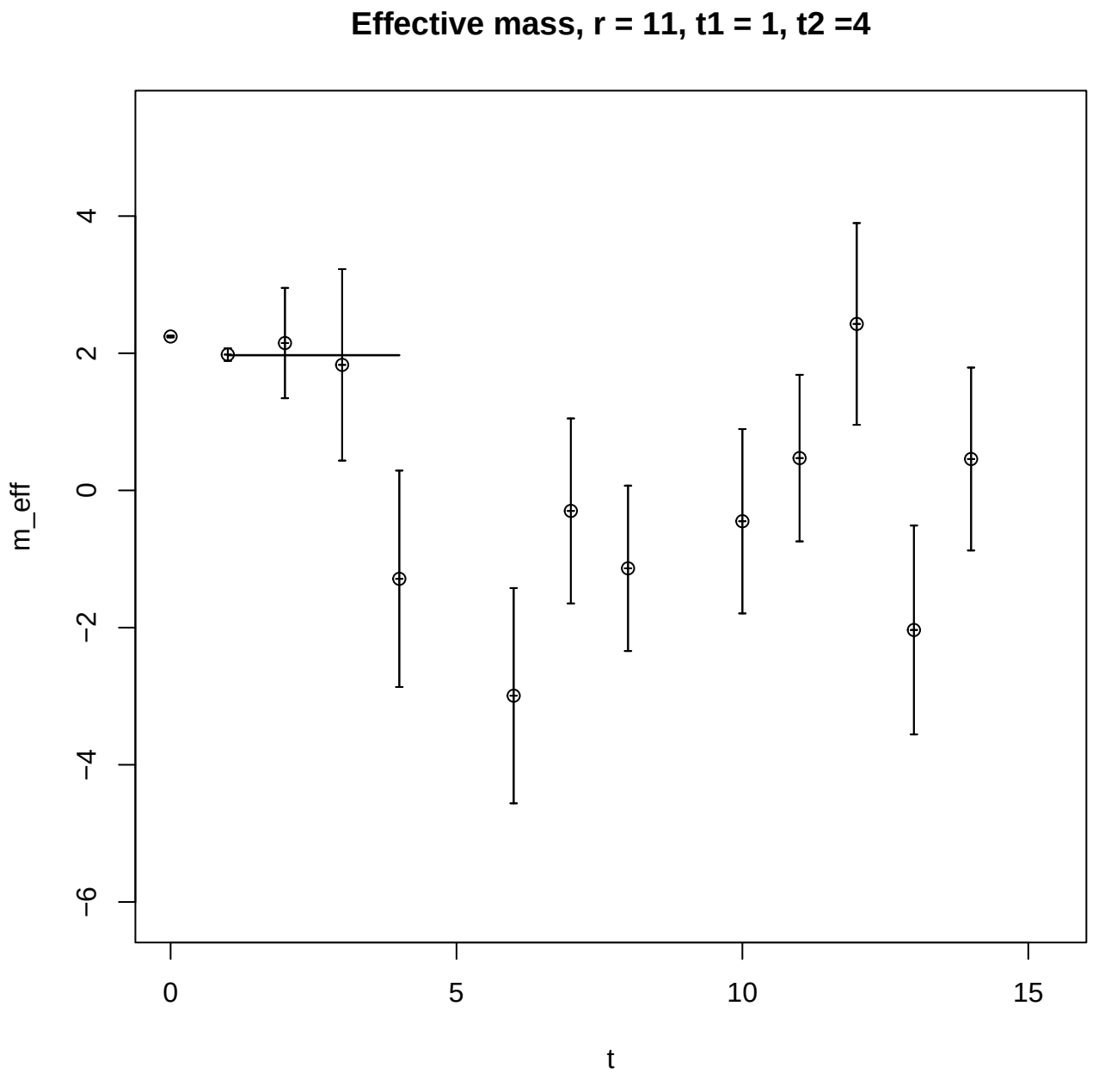

Effective mass, r = 11, t1 = 0, t2 =1
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

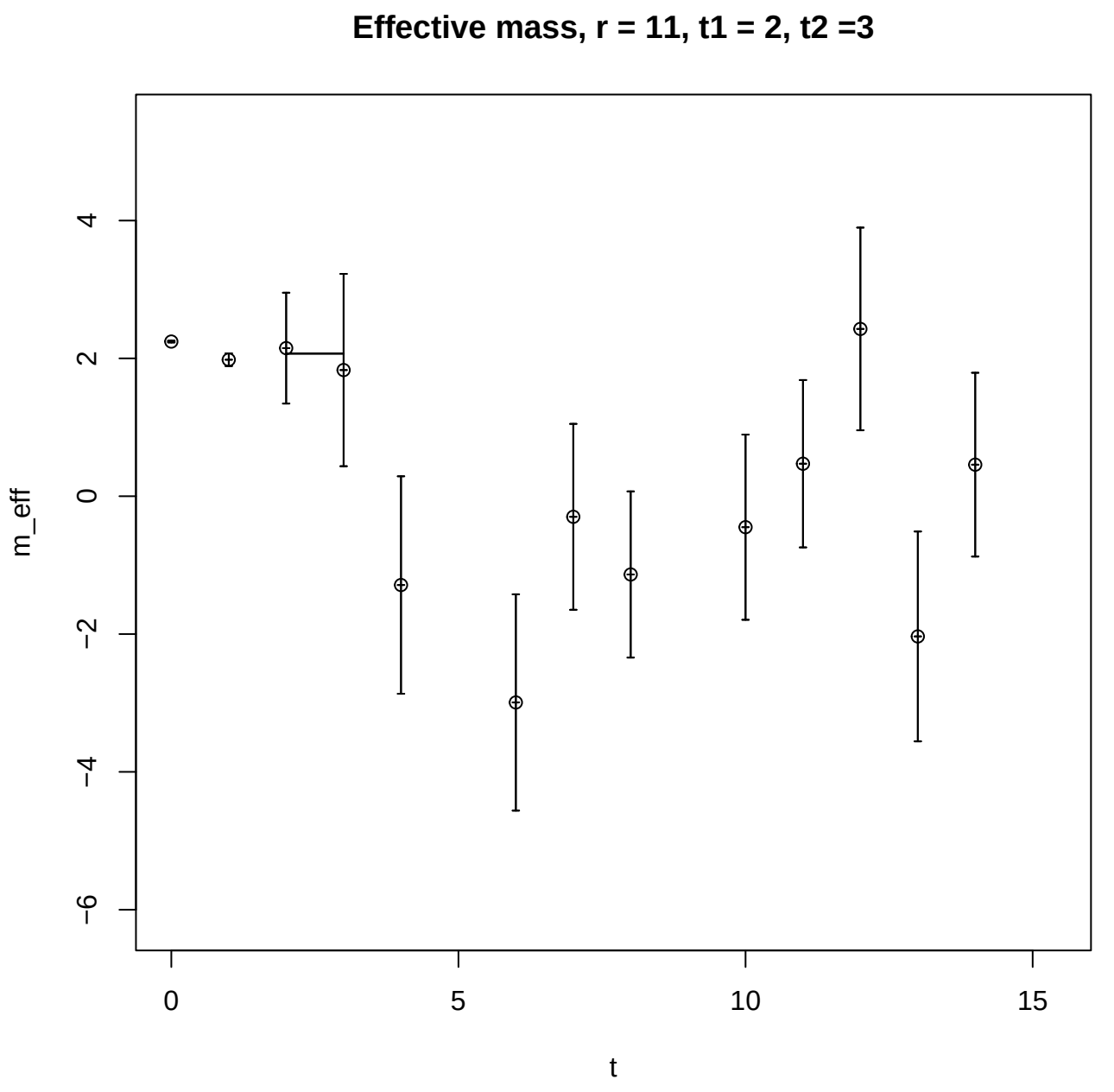

Effective mass, r = 11, t1 = 0, t2 =2
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

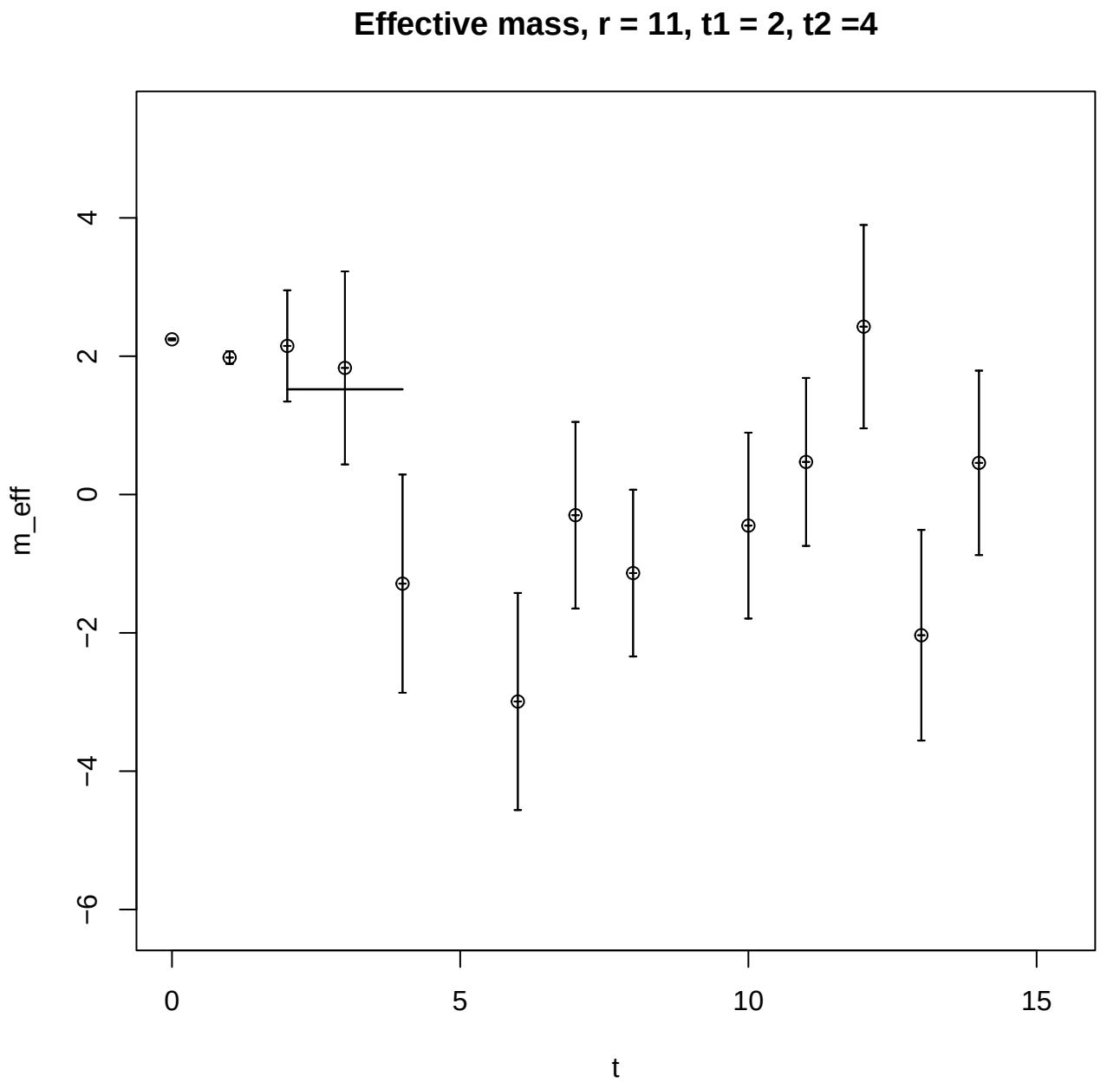

Effective mass, r = 11, t1 = 0, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

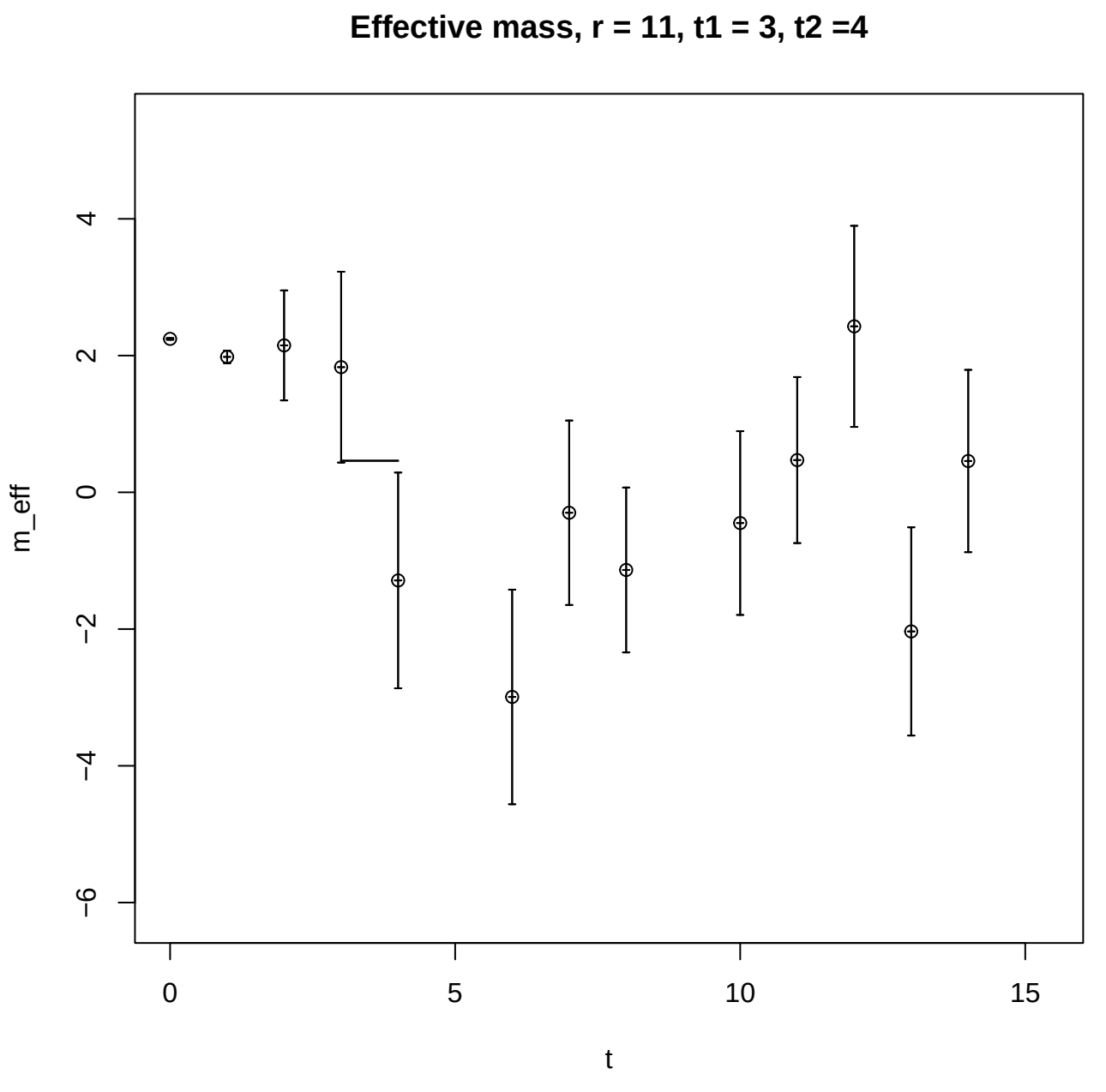

Effective mass, r = 11, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

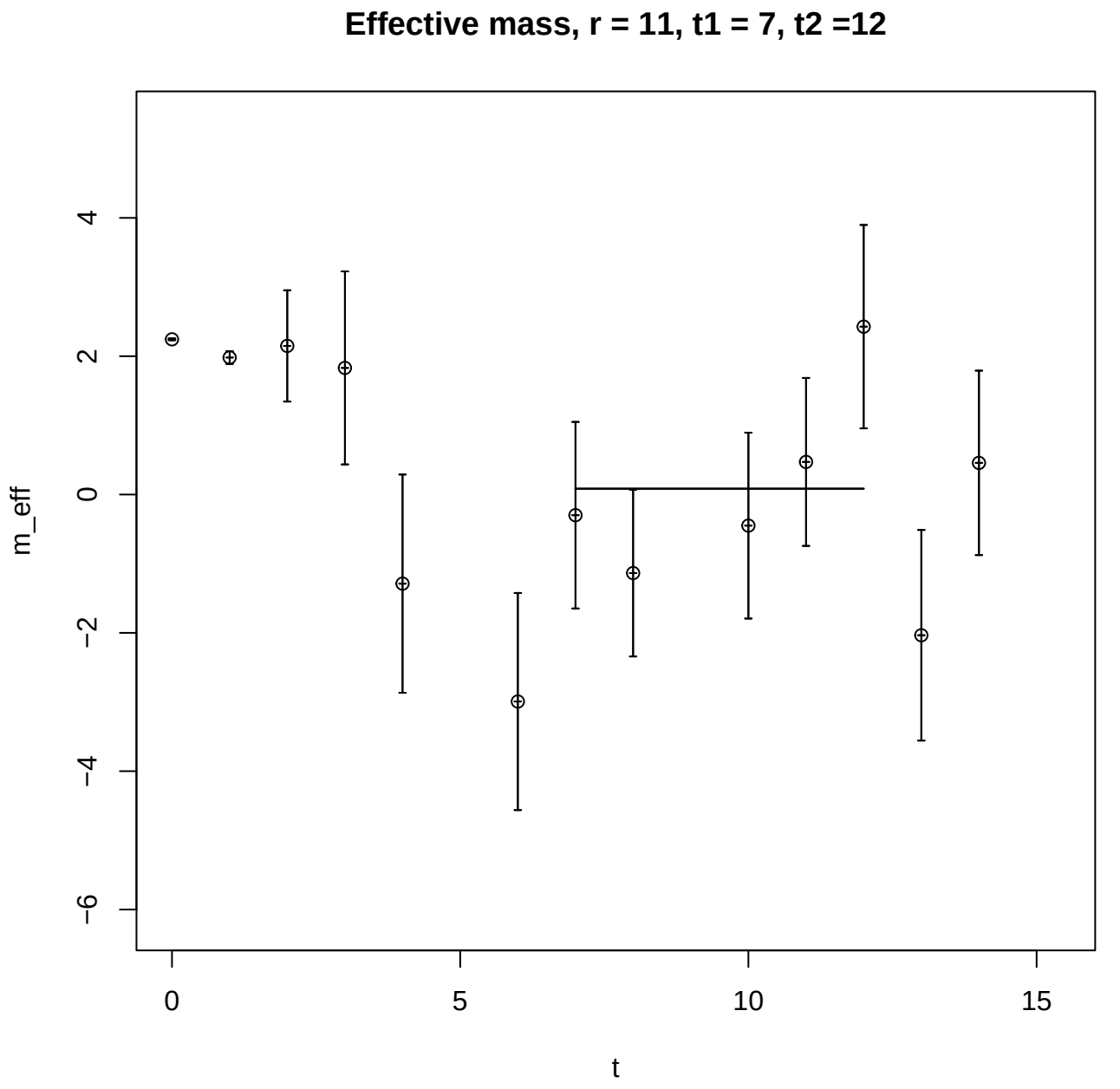

Effective mass, r = 11, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

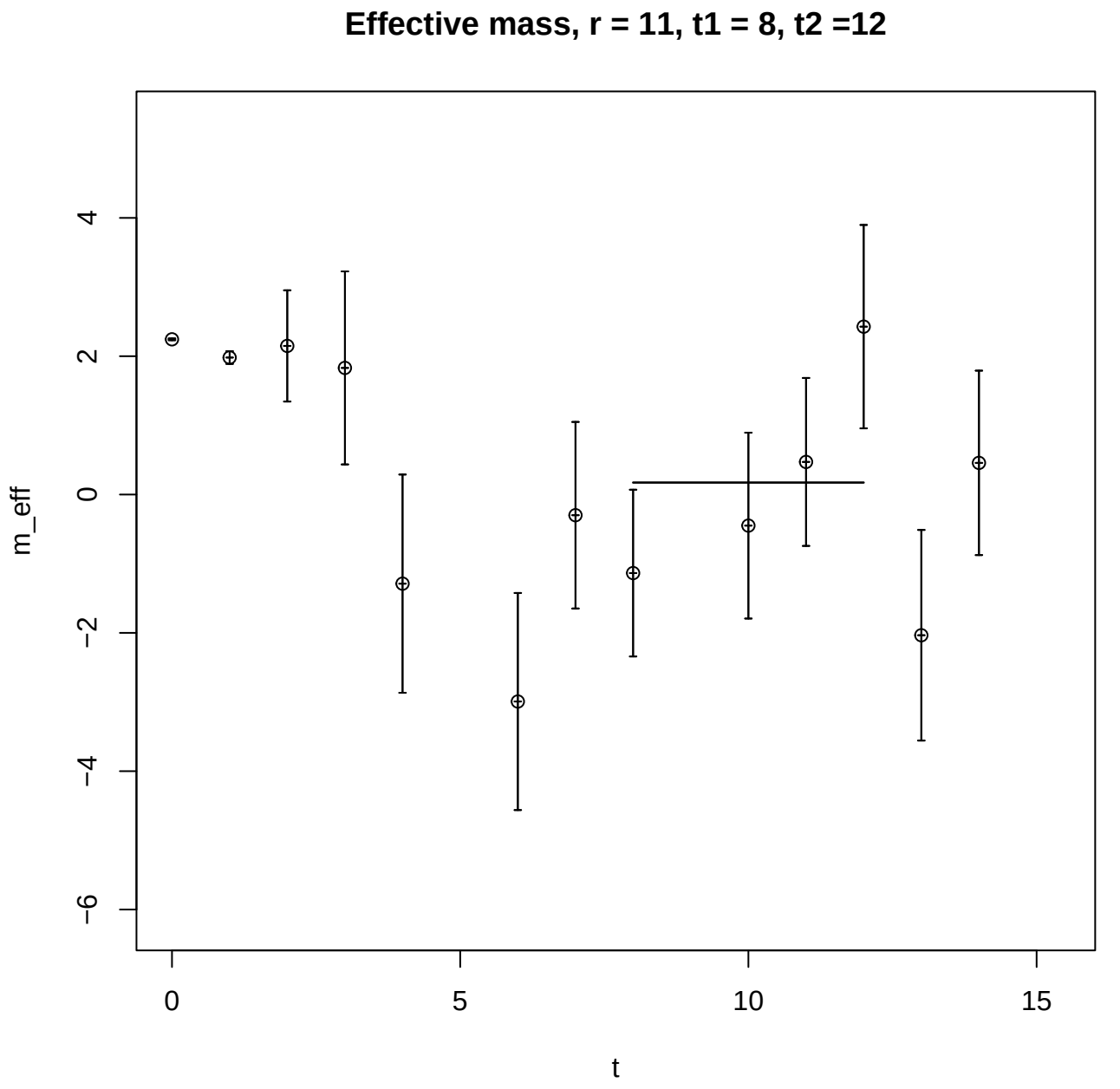

Effective mass, r = 11, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

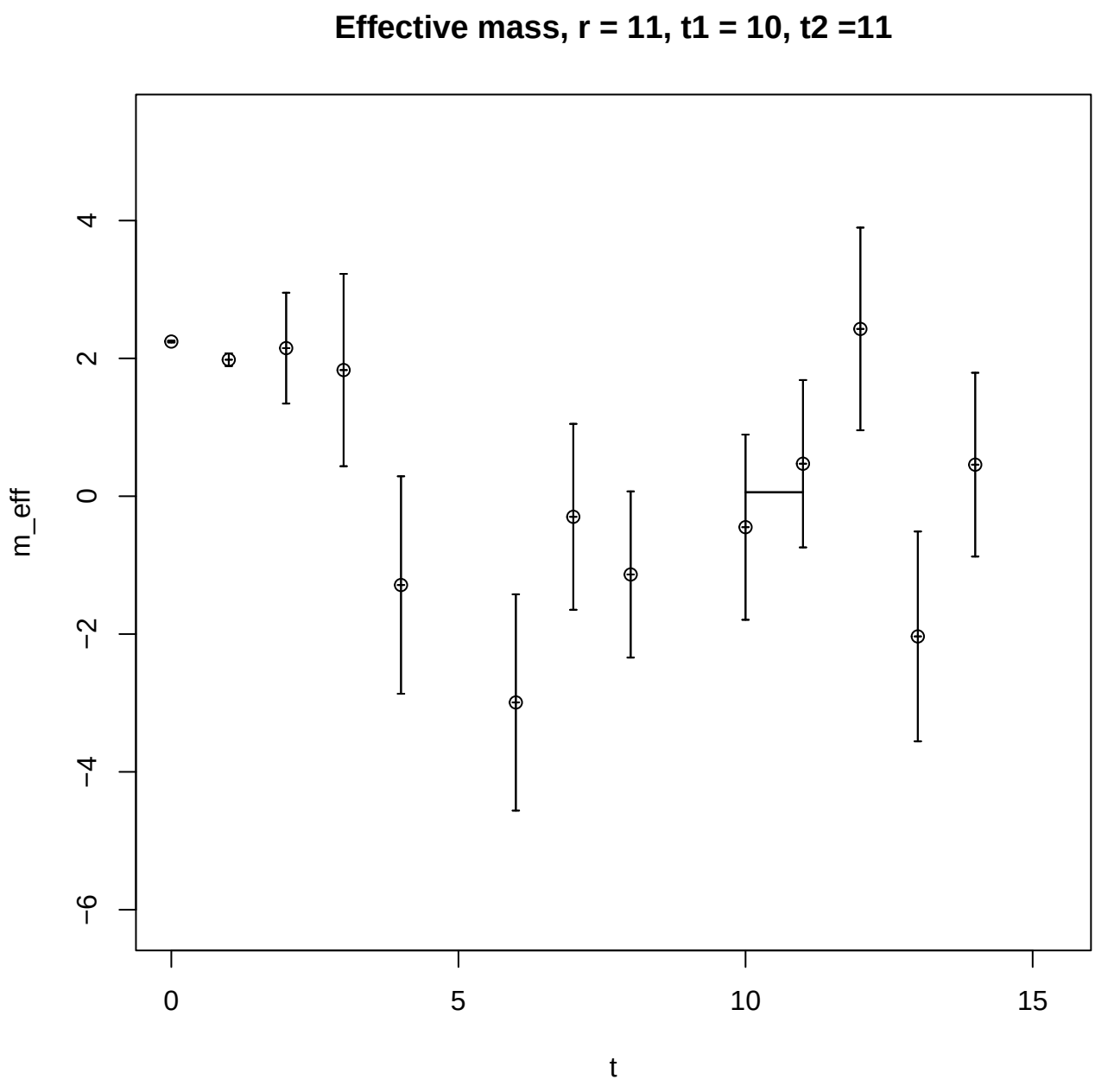

Effective mass, r = 11, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

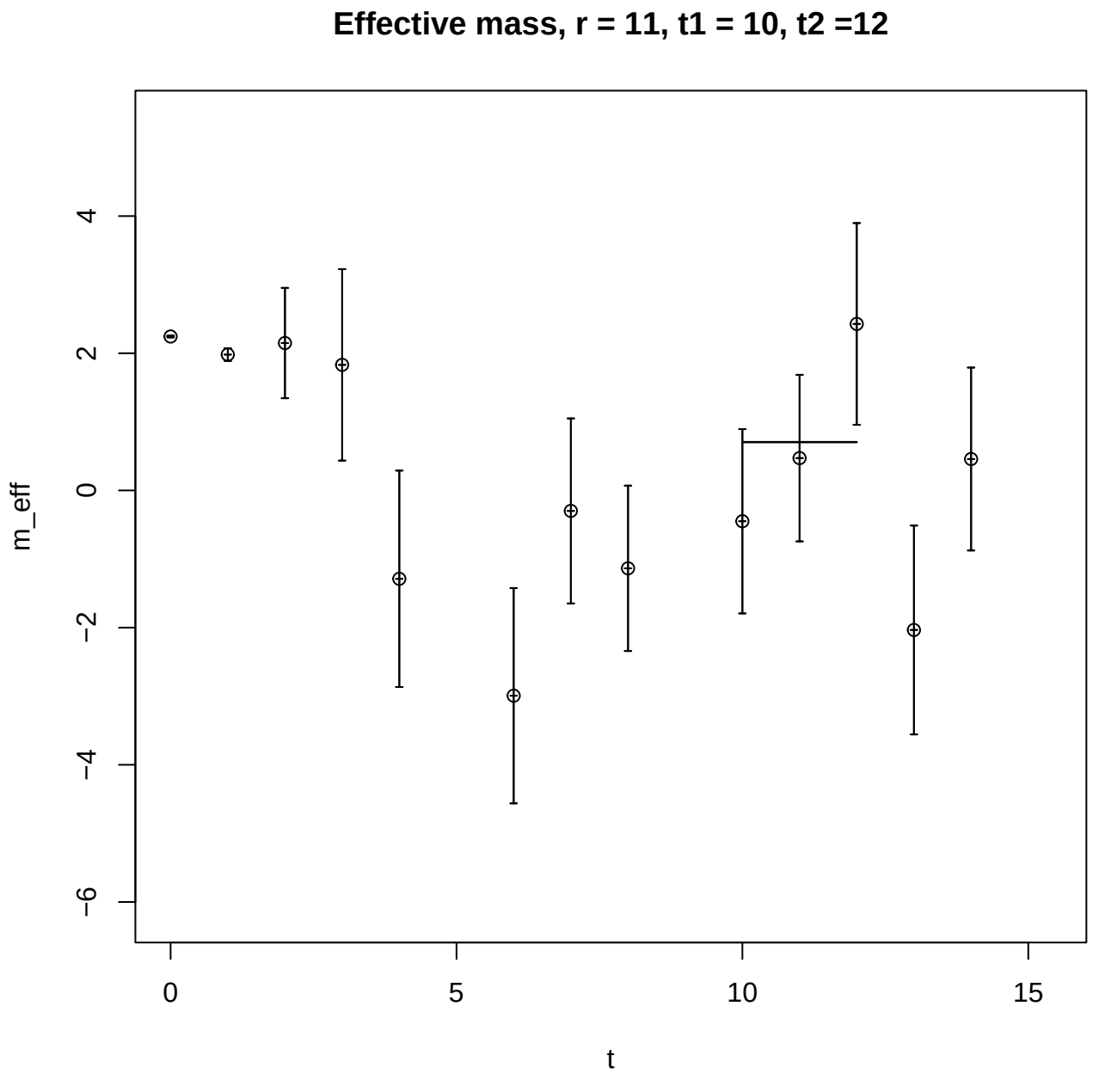

Effective mass, r = 11, t1 = 2, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

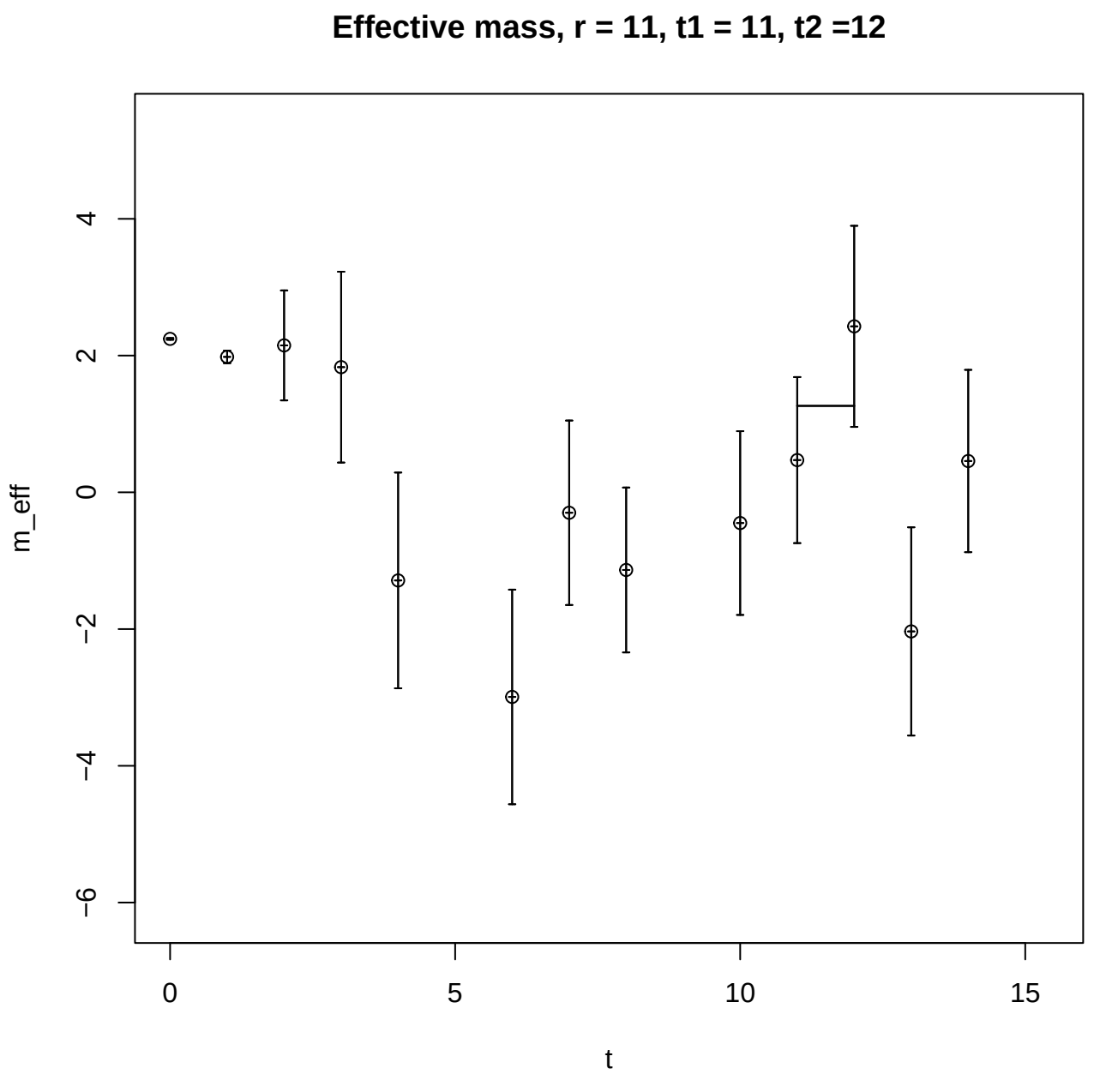

Effective mass, r = 11, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

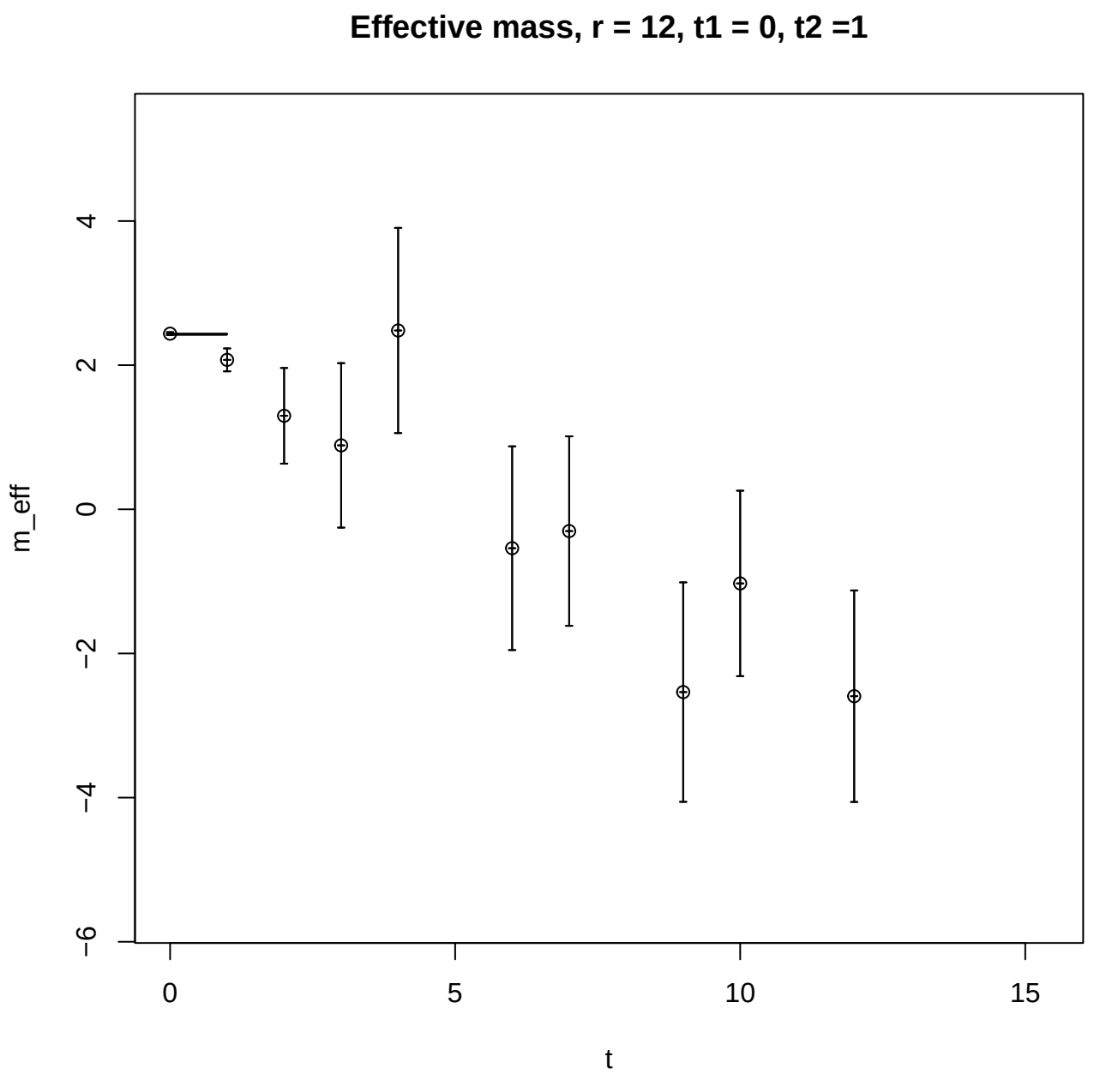

Effective mass, r = 11, t1 = 7, t2 =12
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

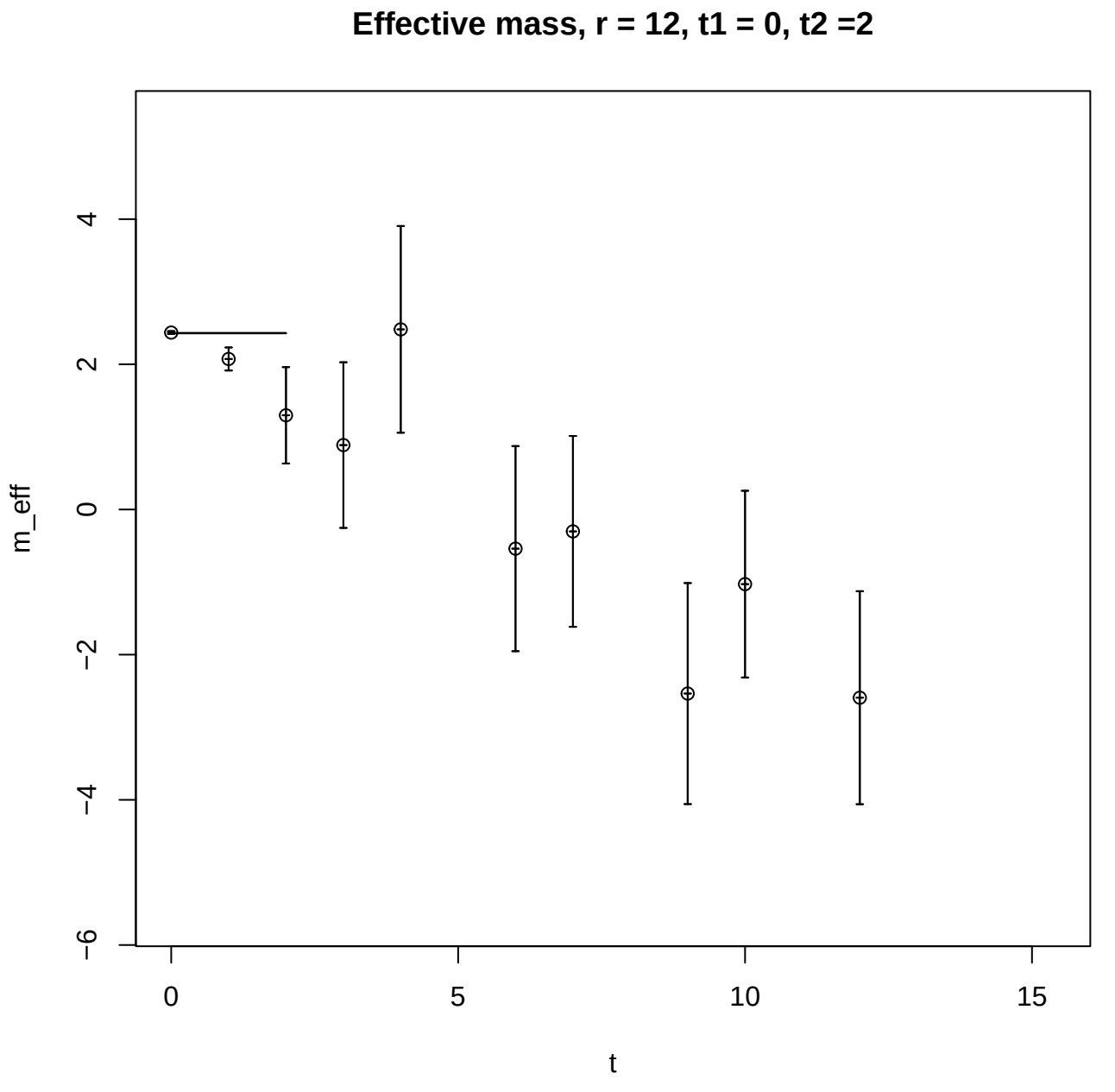

Effective mass, r = 11, t1 = 8, t2 =12
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

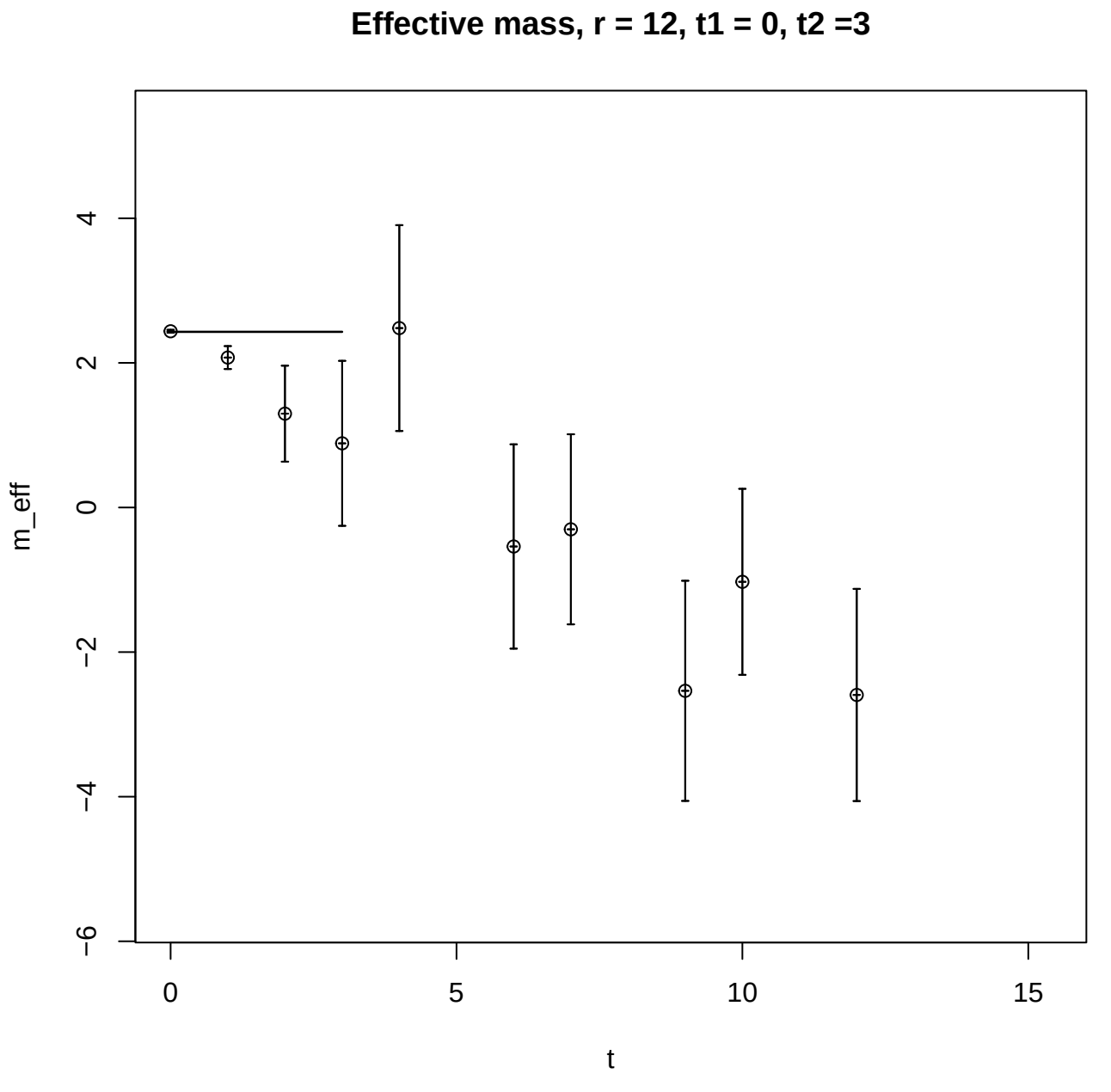

Effective mass, r = 11, t1 = 10, t2 =11
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

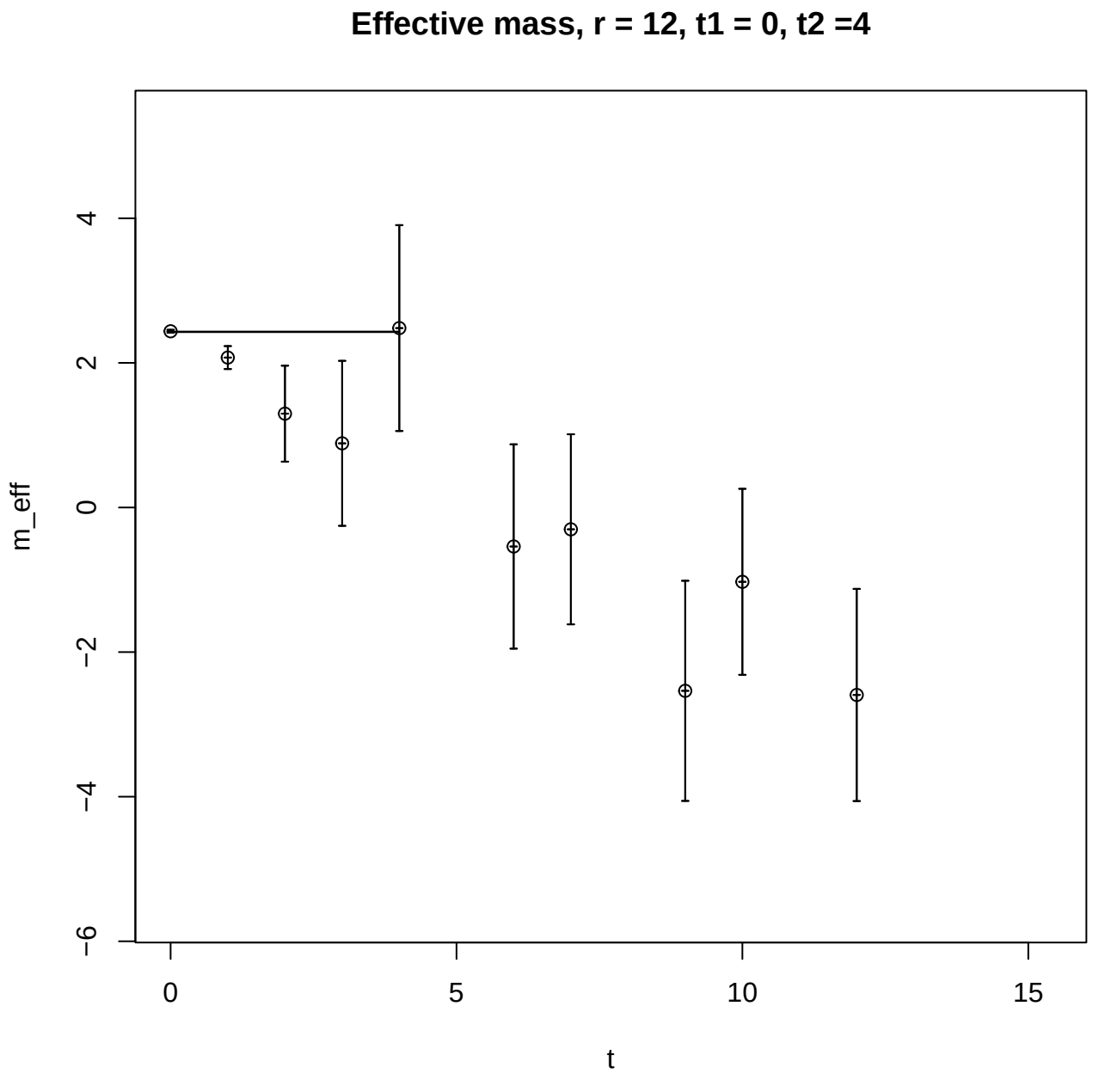

Effective mass, r = 11, t1 = 10, t2 =12
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

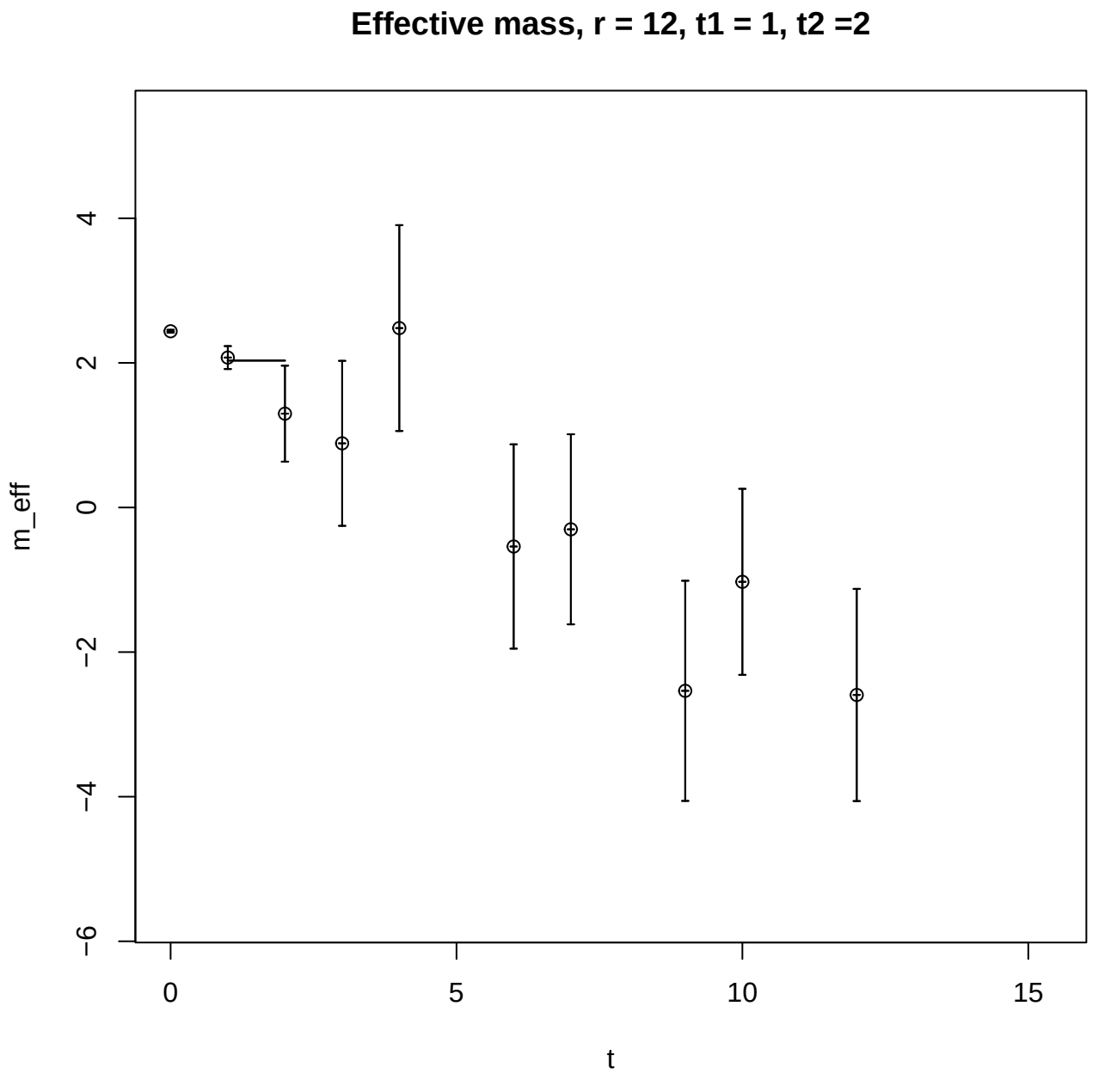

Effective mass, r = 11, t1 = 11, t2 =12
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

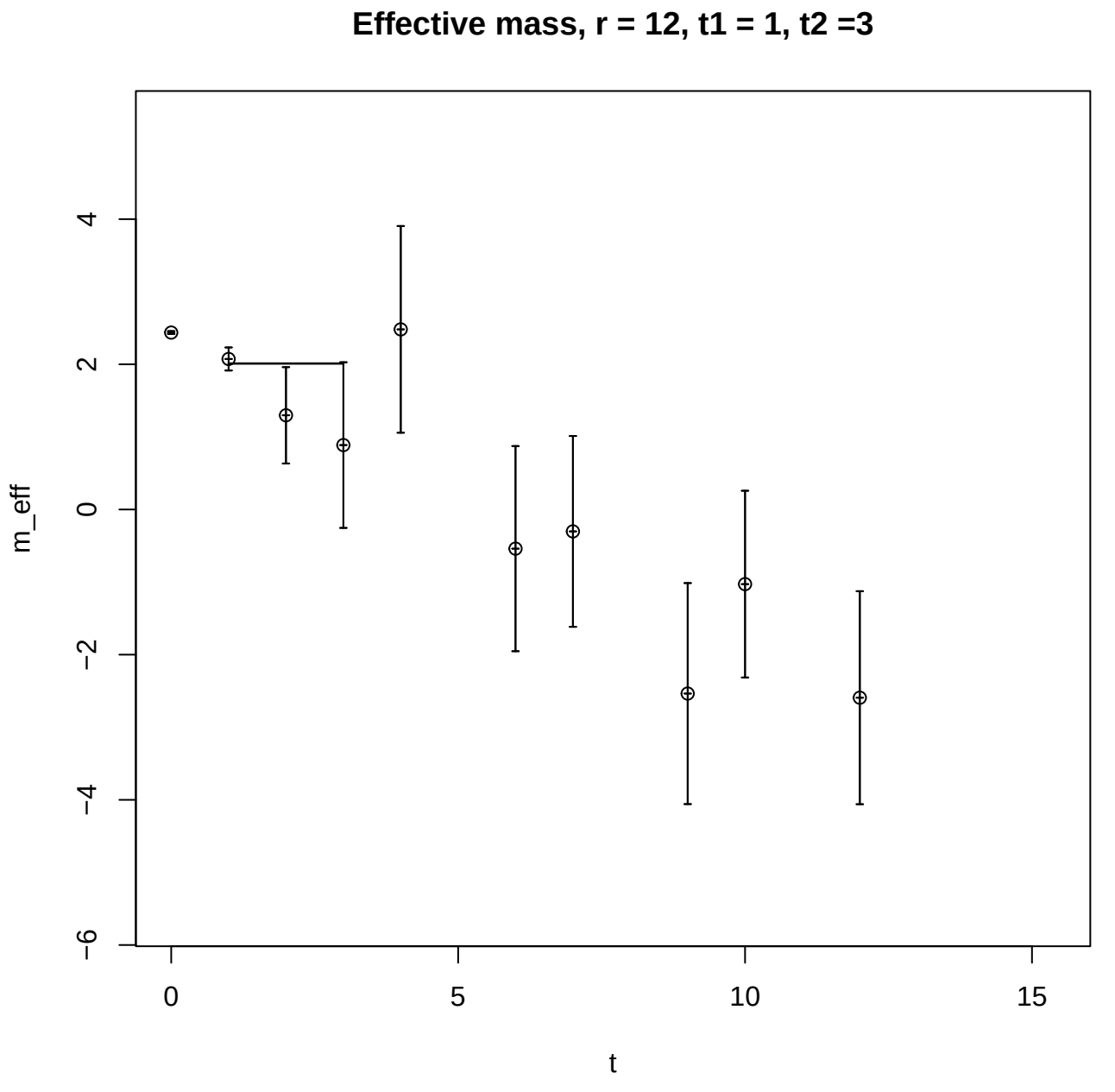

Effective mass, r = 12, t1 = 0, t2 =1
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

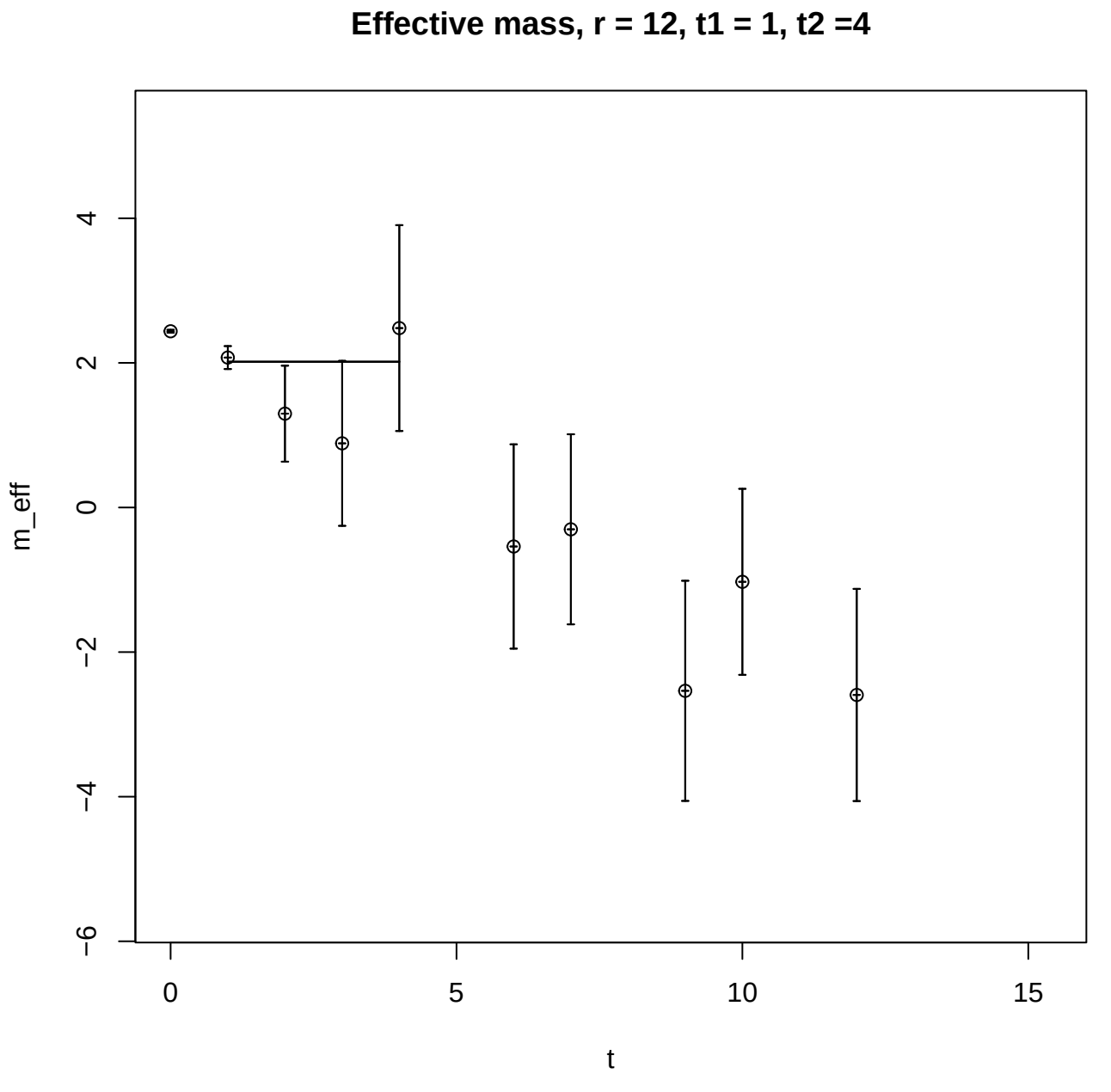

Effective mass, r = 12, t1 = 0, t2 =2
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

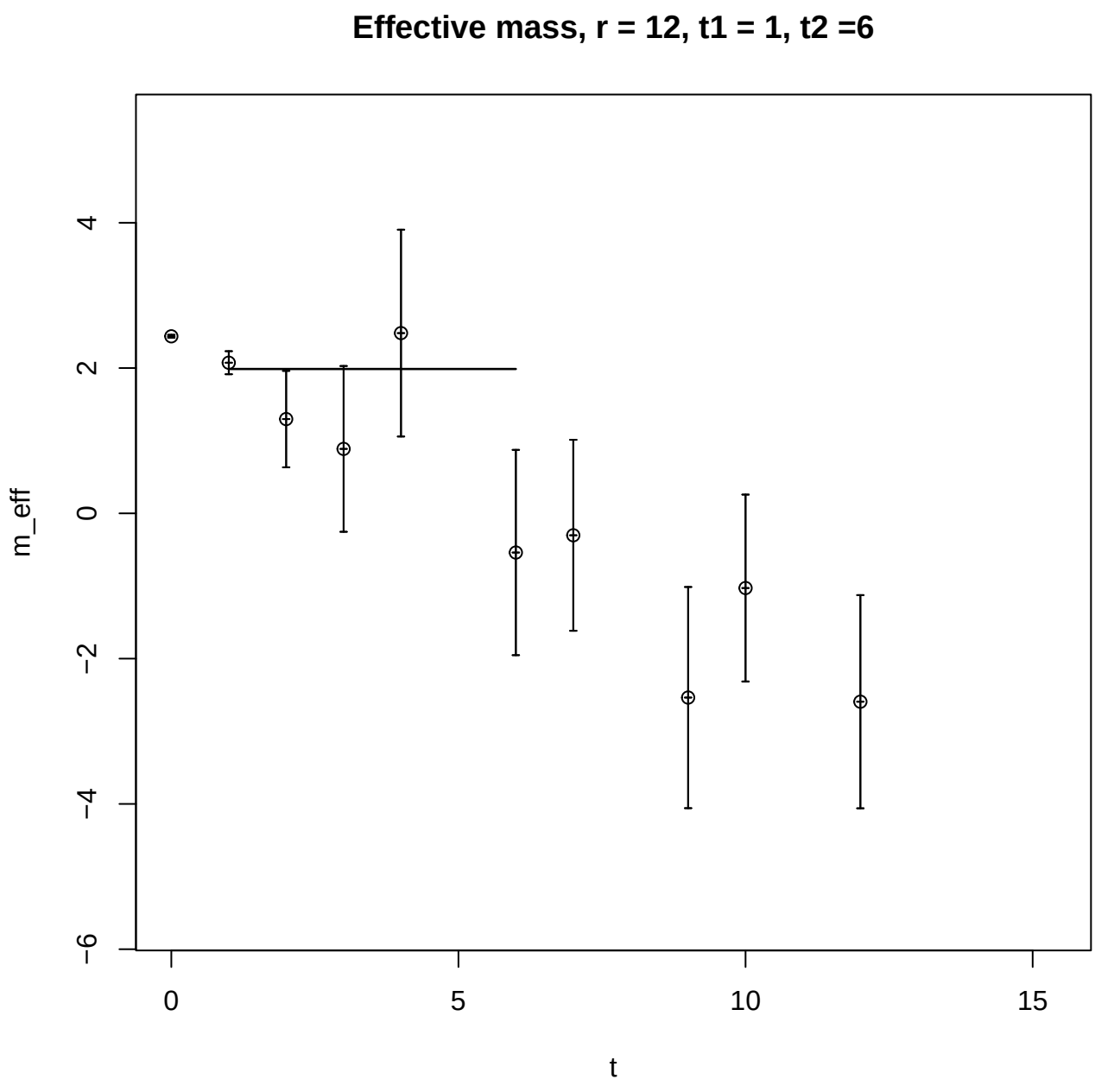

Effective mass, r = 12, t1 = 0, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

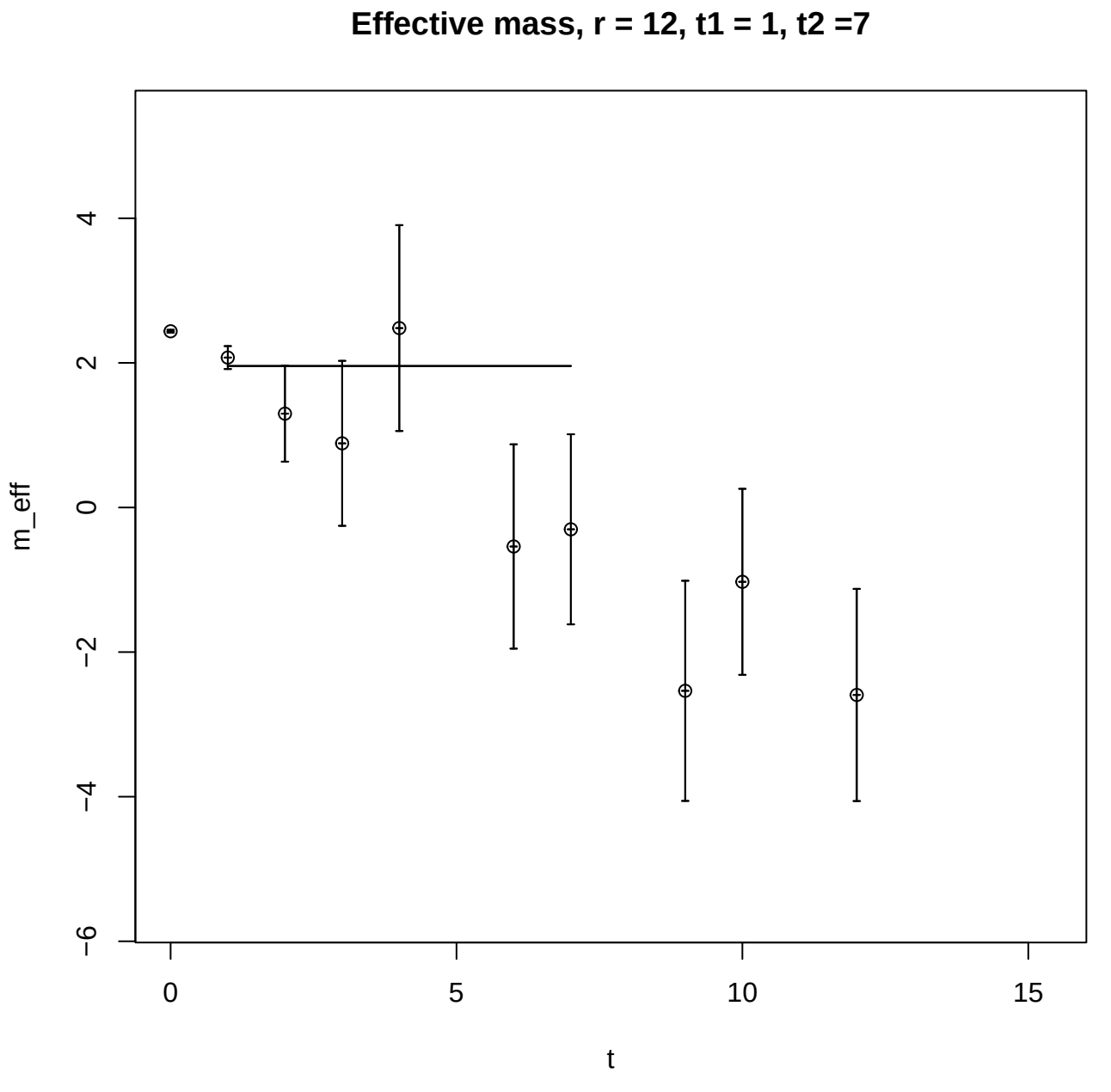

Effective mass, r = 12, t1 = 0, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

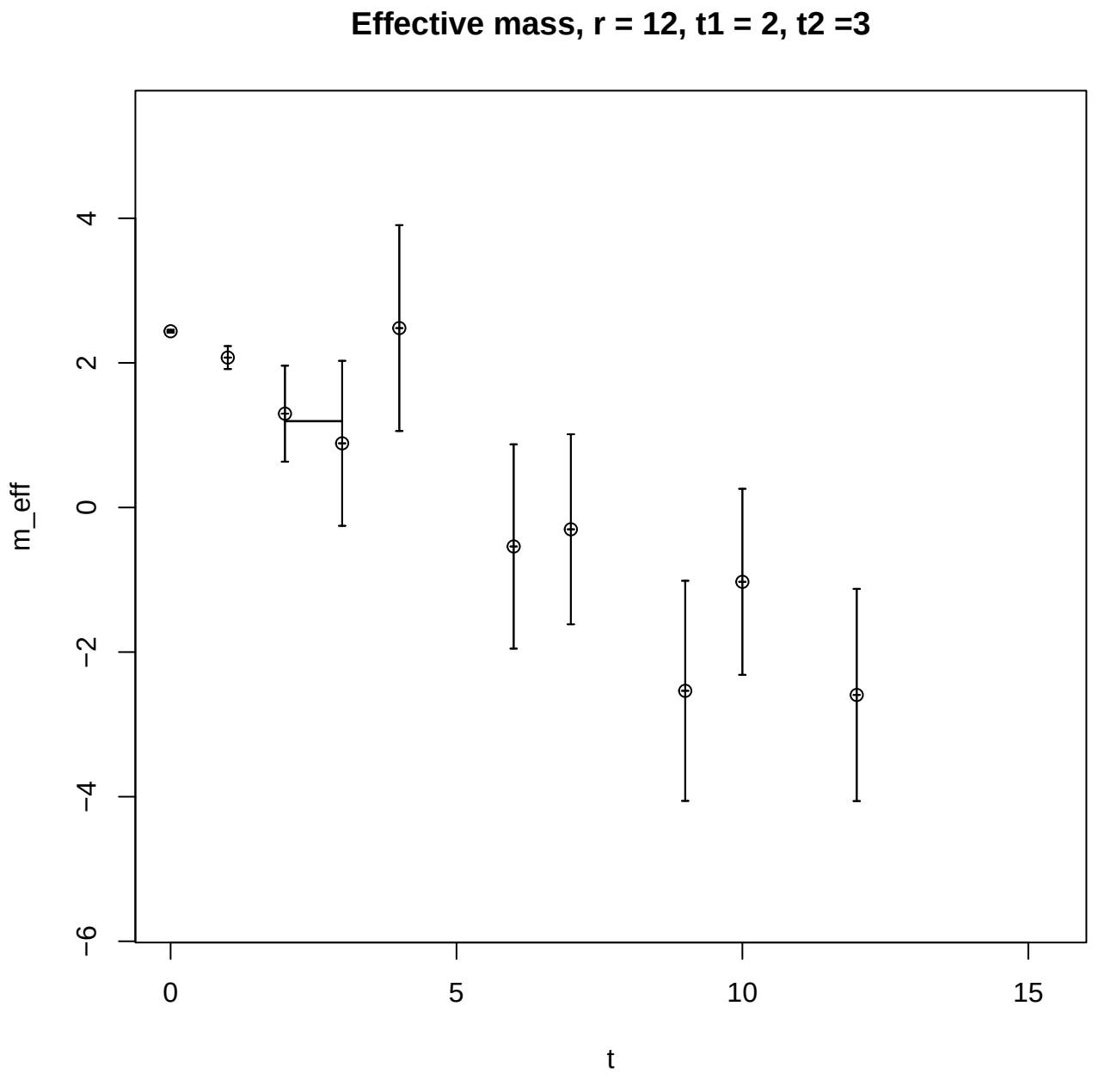

Effective mass, r = 12, t1 = 1, t2 =2
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

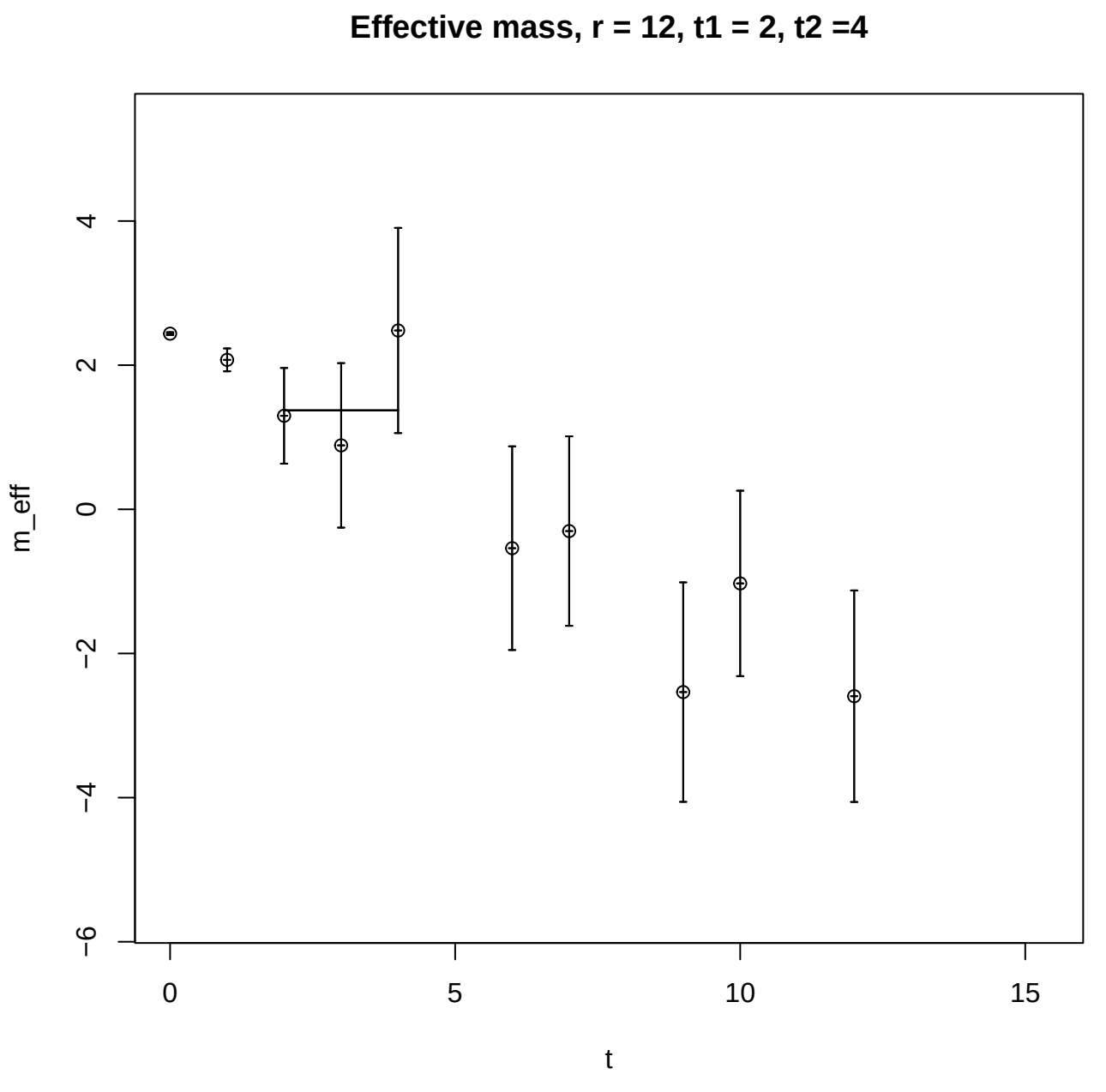

Effective mass, r = 12, t1 = 1, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

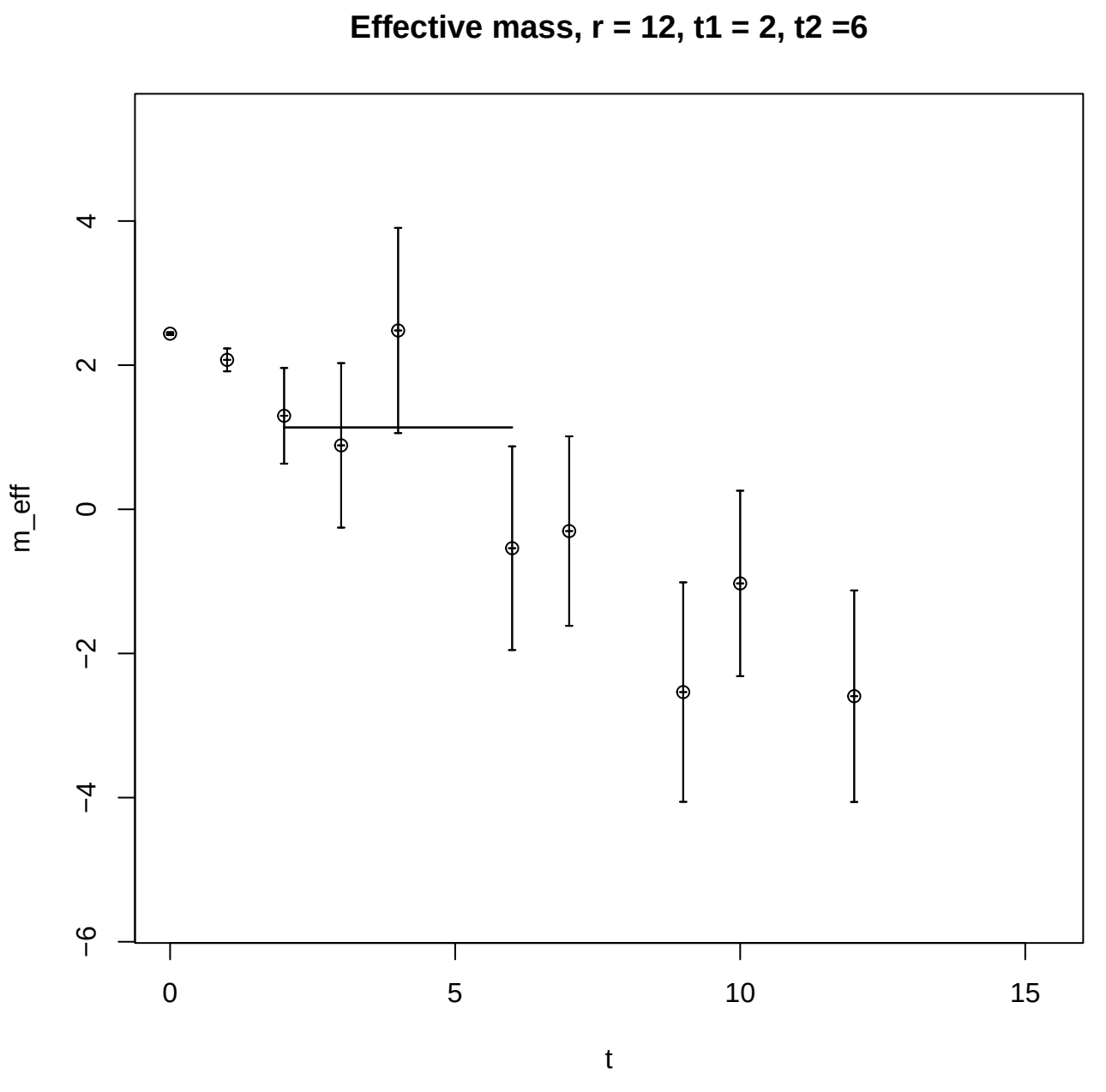

Effective mass, r = 12, t1 = 1, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

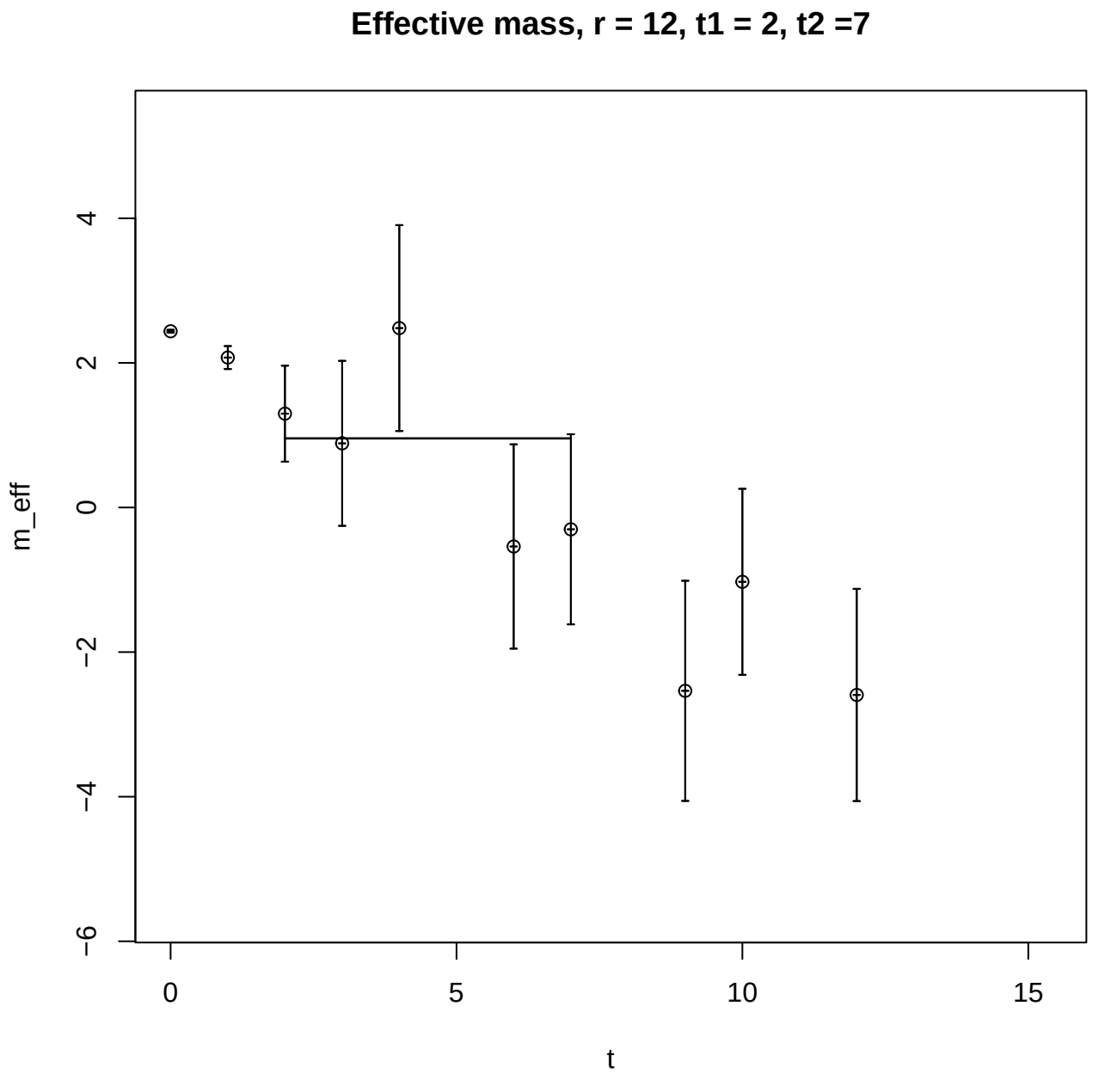

Effective mass, r = 12, t1 = 1, t2 =6
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

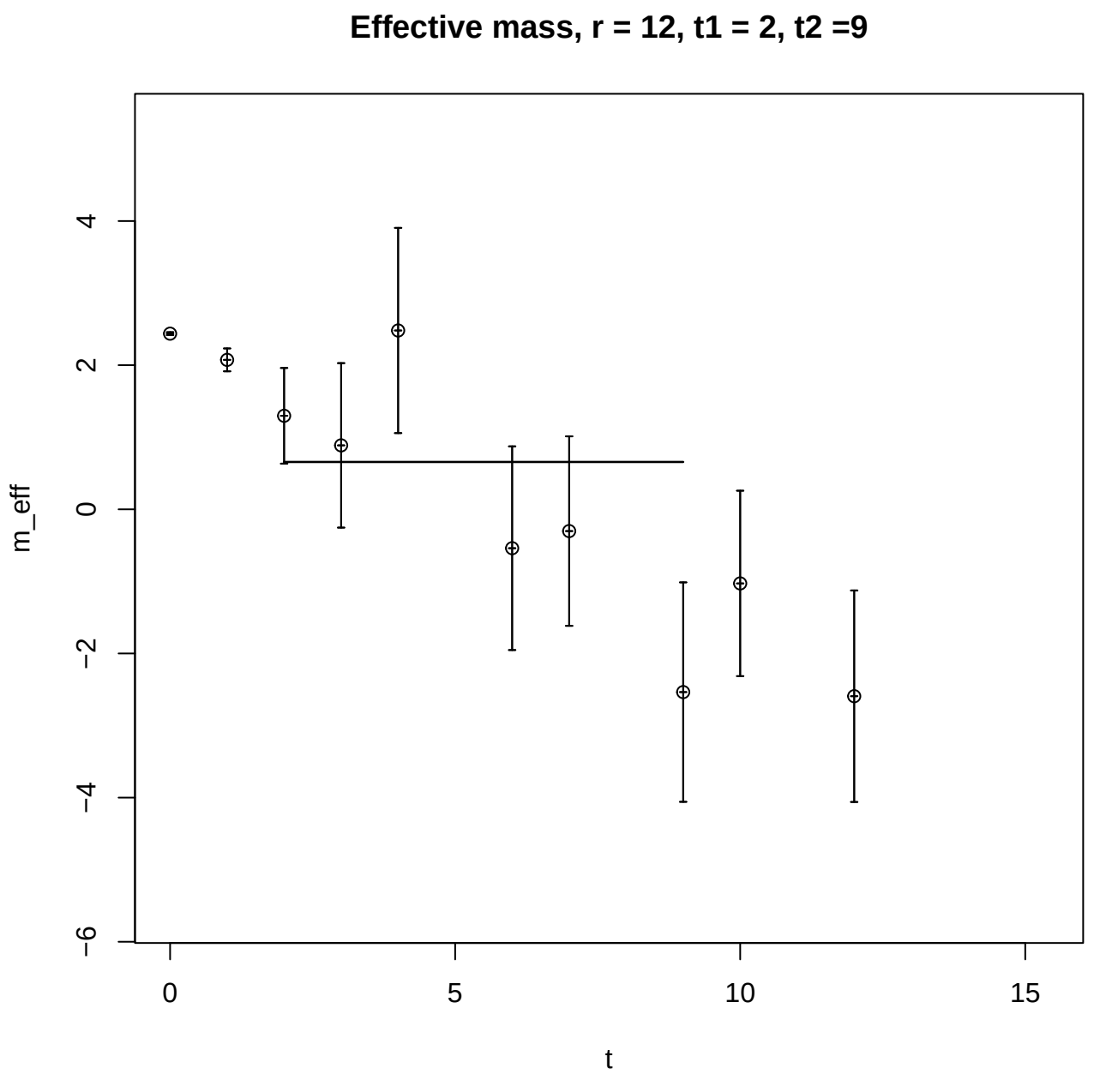

Effective mass, r = 12, t1 = 1, t2 =7
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

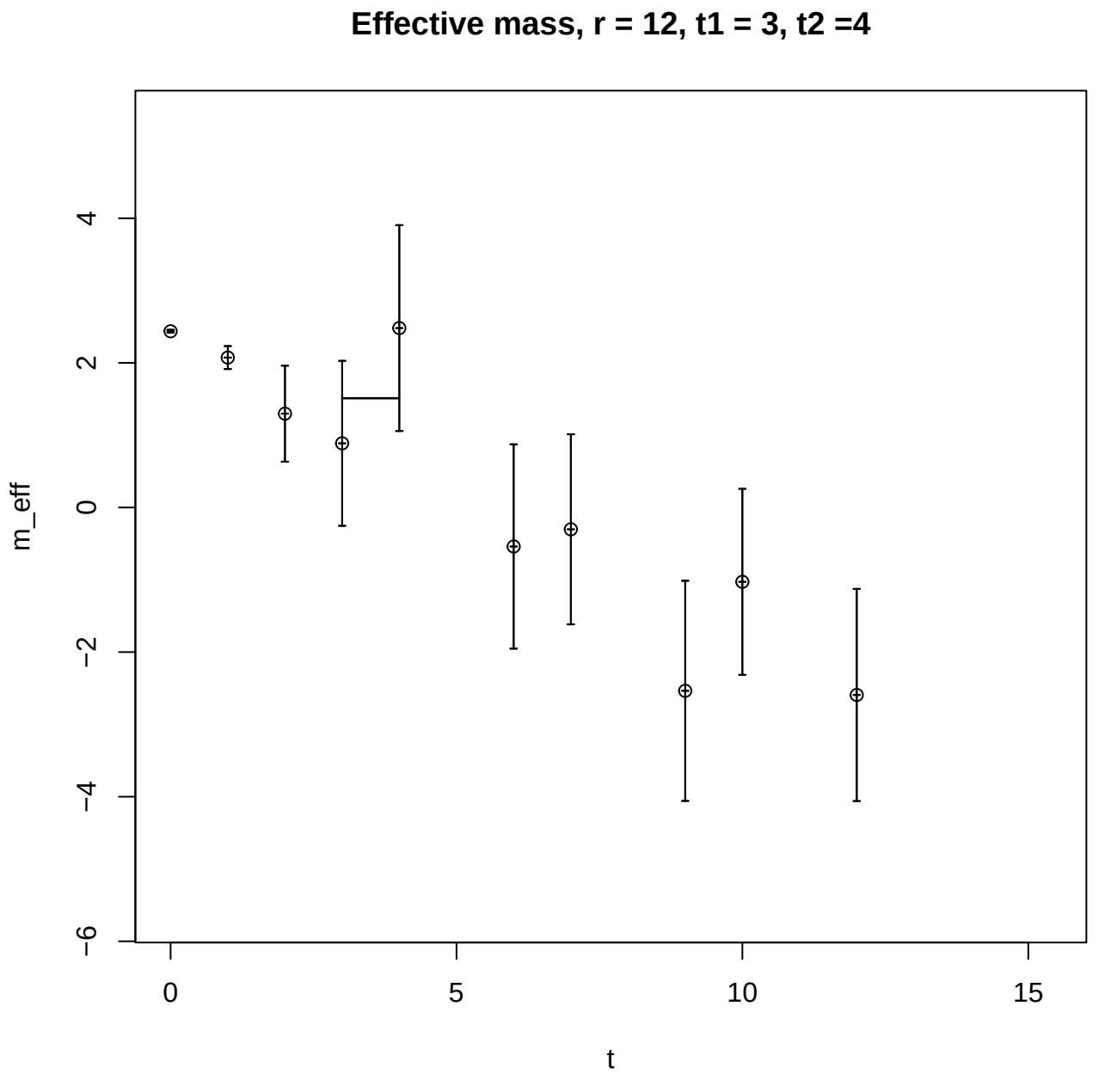

Effective mass, r = 12, t1 = 2, t2 =3
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

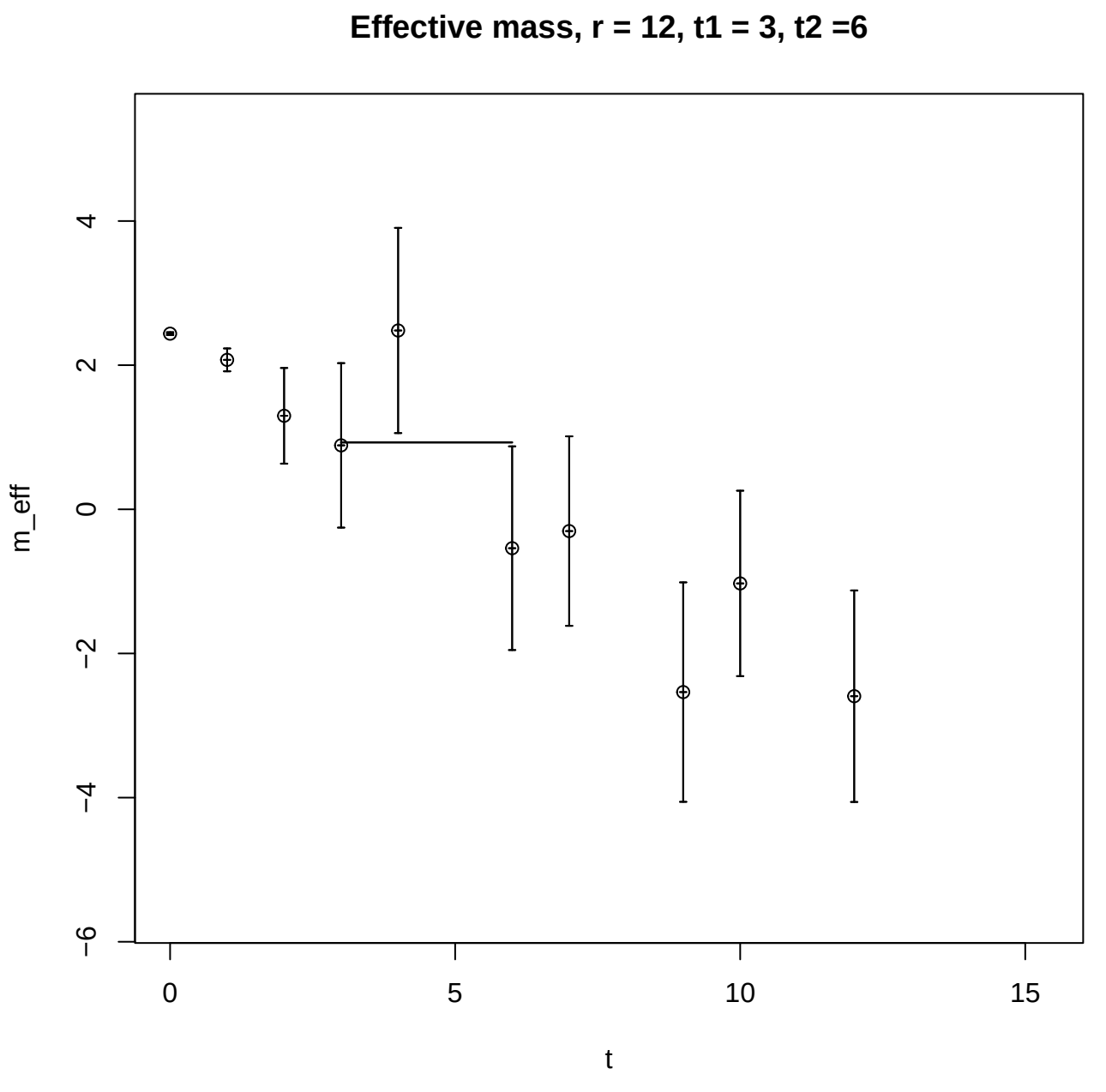

Effective mass, r = 12, t1 = 2, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

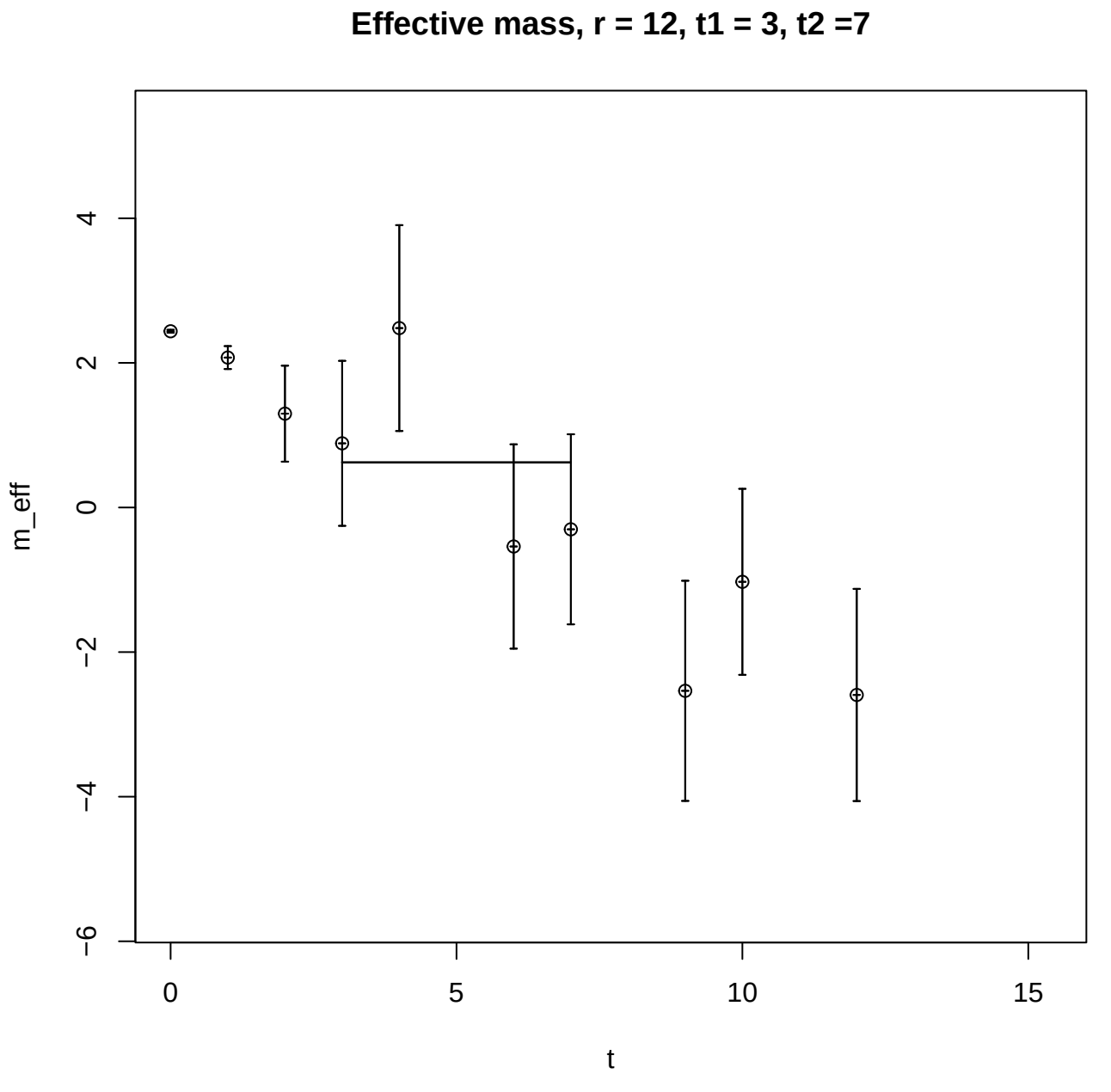

Effective mass, r = 12, t1 = 2, t2 =6
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

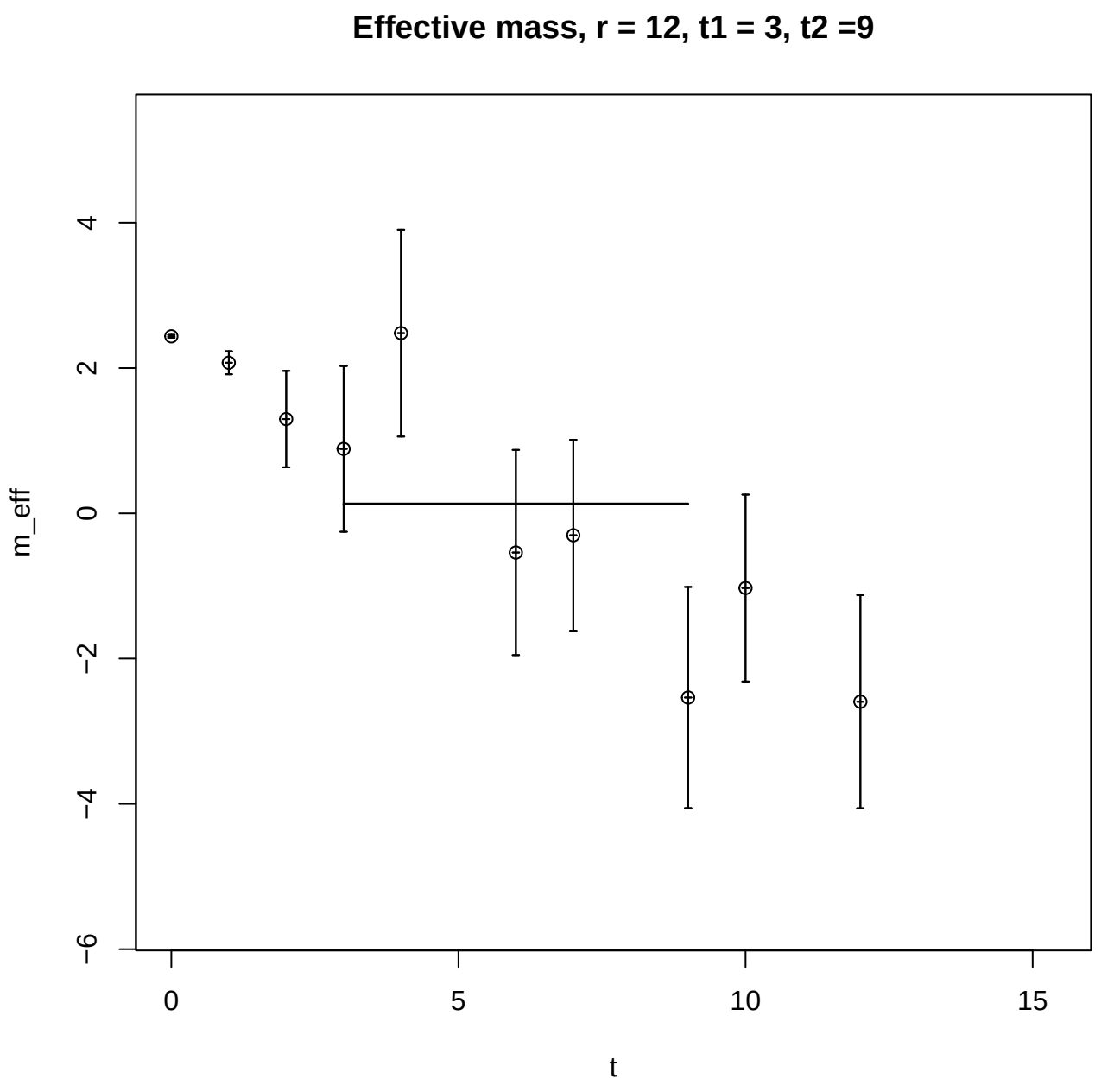

Effective mass, r = 12, t1 = 2, t2 =7
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

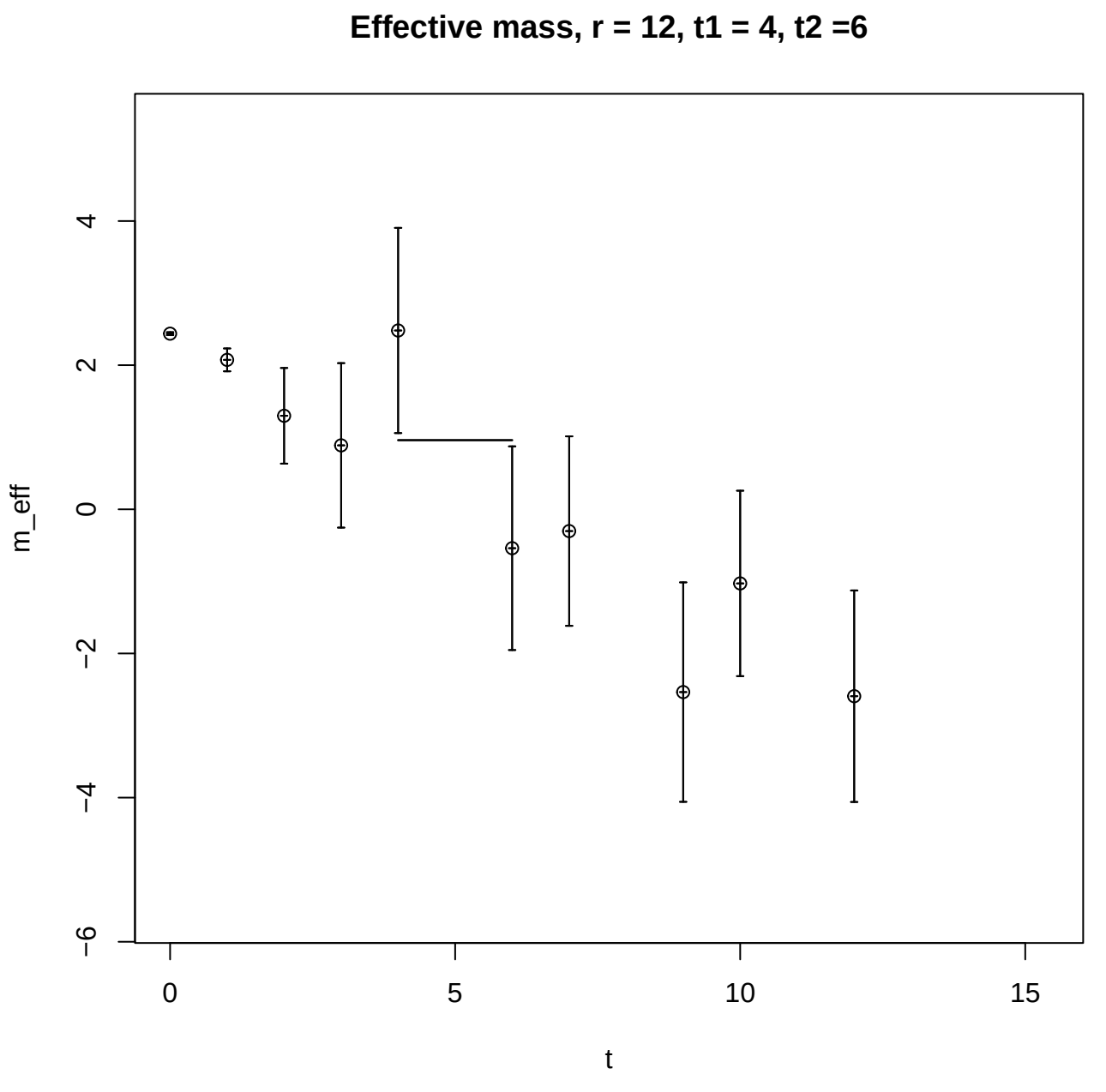

Effective mass, r = 12, t1 = 2, t2 =9
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15

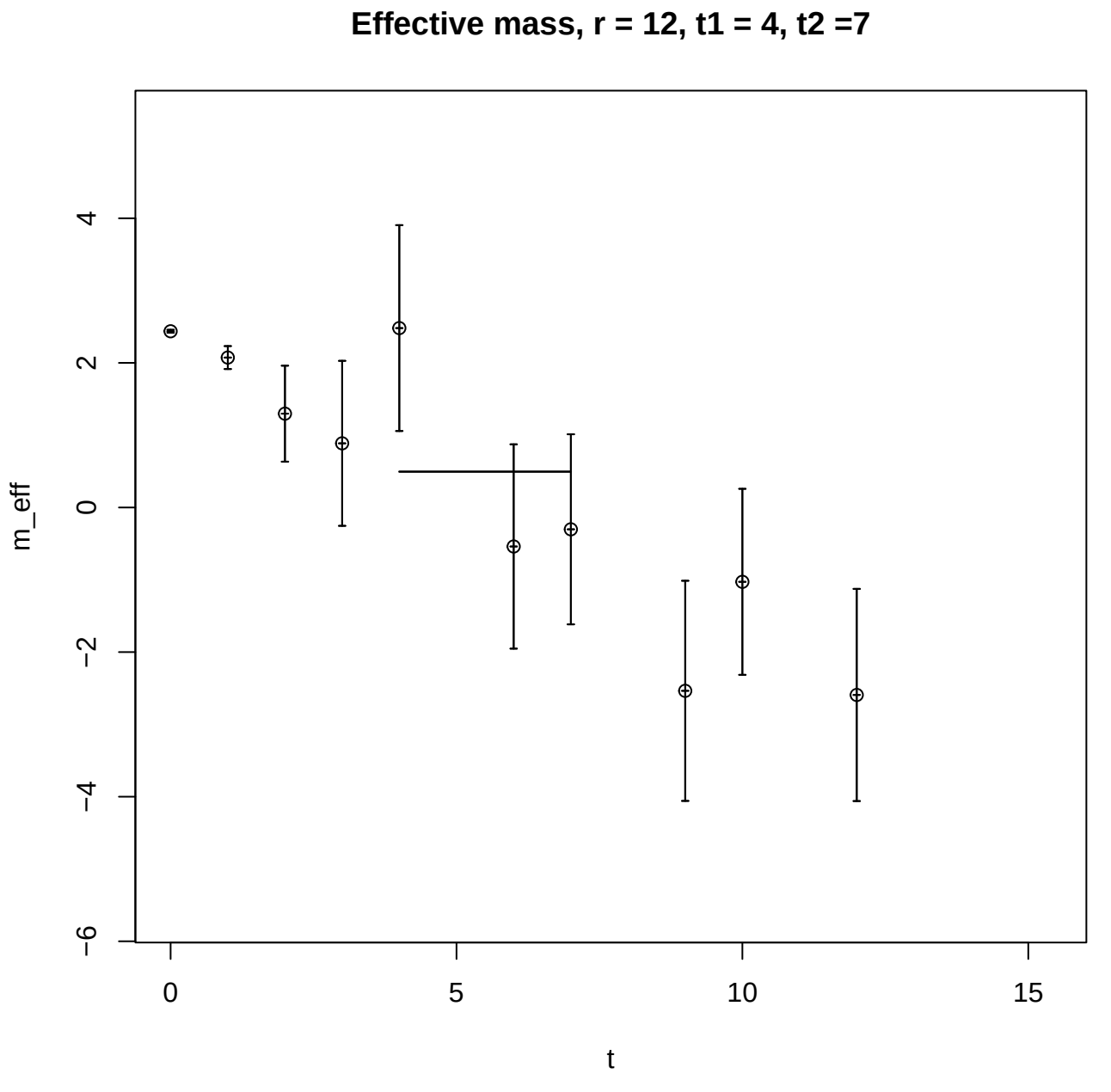

Effective mass, r = 12, t1 = 3, t2 =4
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15


Effective mass, r = 12, t1 = 3, t2 =6
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15


Effective mass, r = 12, t1 = 3, t2 =7
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15


Effective mass, r = 12, t1 = 3, t2 =9
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15


Effective mass, r = 12, t1 = 4, t2 =6
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15


Effective mass, r = 12, t1 = 4, t2 =7
m_eff
t
4
2
0
−2
−4
−6
0
5
10
15